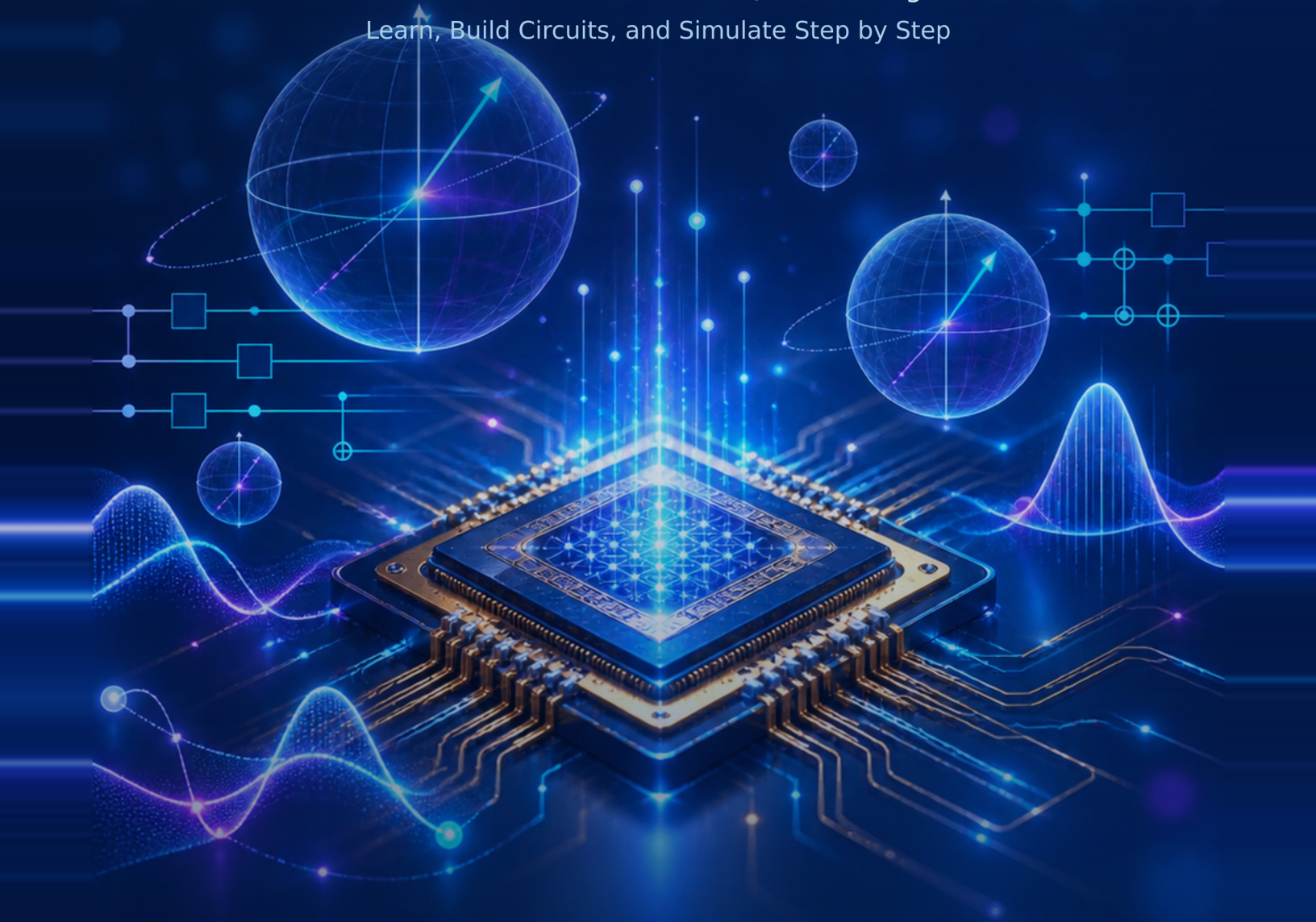


QUANTUM COMPUTING FOR EVERYONE

A Practical Guide from Basics to Quantum Algorithms
Learn, Build Circuits, and Simulate Step by Step



STUDENT EDITION

Accompanied by a separate Teacher's Guide

Dr. Ismael Khorshed Abdulrahman

Erbil Polytechnic University

FIRST EDITION · 2026

QUANTUM COMPUTING FOR EVERYONE

A Practical Guide from Basics to Quantum Algorithms

Learn, Build Circuits, and Simulate Step by Step

QCFE

STUDENT EDITION

Accompanied by a separate Teacher's Guide

First Edition — 2026

Quantum Computing for Everyone: A Practical Guide from Basics to Quantum Algorithms

From bits and complex numbers to real quantum algorithms — the companion book to the QCFE app

This book was built with the help of artificial intelligence (AI), under the direction and review of the author.

Dr. Ismael Khorshed Abdulrahman

Erbil Polytechnic University

PhD in Electrical and Computer Engineering

Tennessee Technological University, USA

Copyright © 2026 Ismael Khorshed Abdulrahman

All rights reserved.

No part of this publication may be reproduced, distributed, or transmitted in any form without prior written permission from the copyright holder, except for brief quotations used for purposes such as review, and scholarly citation.

This book and its companion app, Quantum Computing for Everyone (QCFE), are provided free of charge for anyone who wishes to learn. Voluntary donations in support of their continued development are warmly welcomed and appreciated.

How to Use This Book

This book is written in plain, simple English for students, teachers, engineers, and anyone who wants to learn quantum computing from the beginning. Short sentences and common words are used on purpose. Every technical word is explained the first time it is used, and also collected in the Glossary at the end of the book.

Quantum computing has a reputation for being mysterious and difficult. This book takes the opposite view: every idea in it can be understood with careful, honest, step-by-step mathematics — and all the background you need is taught from zero, inside the book. The story starts from familiar ground: Chapter 1 reviews bits, binary, and logic gates, and Chapter 2 gives the honest map of what quantum computers can and cannot do. The qubit itself arrives in Chapter 3. The two mathematical tools the whole book leans on — complex numbers, and vectors and matrices — are taught in full, from zero, in **Appendices A and B** at the back. If any chapter feels like it is using mathematics you have not met, go to the matching appendix first: Appendix A for complex numbers, Appendix B for vectors and matrices. Three more appendices serve as permanent companions: Appendix C teaches, from zero, the probability and statistics that this book's histograms and error bars run on; Appendix D collects every formula, gate, and rule of thumb onto a few reference pages; and Appendix E is a from-zero Python and NumPy quick start for the simulations and capstones. Teachers starting from zero should simply teach Appendices A–C right after Chapter 2, as the opening weeks of the course. If you can multiply and add carefully, you can learn everything here.

The book's eight parts climb steadily: bits and the quantum idea (Part One), the qubit and measurement (Part Two), gates and observables (Part Three), many qubits and entanglement (Part Four), protocols, programming, and quantum cryptography (Part Five), the famous algorithms (Part Six), quantum artificial intelligence and machine learning (Part Seven), and real hardware, noise, error correction, and capstone projects (Part Eight). A References and Further Reading section at the end lists the original papers and books behind every case study, so you can go to the sources.

Every chapter follows the same pattern: clear learning goals, plain explanations with figures and tables, important words with easy definitions, real equations with worked examples, practical activities, a real-life case study, common mistakes to avoid, a short summary, and at least ten practice problems at three levels — **[Basic]**, **[Standard]**, and **[Challenge]**.

This book is accompanied by a separate Teacher's Guide. The Teacher's Guide contains every chapter's self-test (the questions together with their answers and explanations), the complete step-by-step solutions to all of this book's practice problems, and other teaching-support materials. Students: try every practice problem honestly, on paper, before checking your work against the Teacher's Guide. Teachers: use the self-tests in class, as homework, or as revision.

This book is one part of the Quantum Computing for Everyone project. The same lessons, examples, circuits, and self-tests are also planned for the QCFE Android application and the QCFE Desktop application, with interactive circuit simulators, Bloch-sphere visualizers, measurement labs, and progress tracking. Wherever a lesson has a matching interactive tool, you will see an "Activity (App)" box and a blue "App tool" pointer. If you do not have the app yet, every activity can also be done on paper or with the free simulators described in Chapter 14.

A note on numbers: quantum hardware changes very quickly. Qubit counts, error rates, and company roadmaps in this book are clearly marked as example or approximate figures, correct to the best of our knowledge around 2026, and are used to teach you *how to read and judge* such numbers — not as a permanent record. Always check current sources before making a real technical decision. All mathematics, on the other hand, is timeless: every calculation in this book will still be exactly right in fifty years.

Table of Contents

How to Use This Book	4
FROM BITS TO THE QUANTUM IDEA	14
Bits and Classical Computing	15
1.1 Information Around Us	15
1.2 Bits: The Smallest Unit of Information	16
1.3 Binary Numbers: Counting with 0 and 1	18
1.4 Logic Gates and Classical Circuits.	20
1.5 What Classical Computers Do Well — and Where They Struggle	22
1.6 Error-Detecting Bits: The Parity Trick	24
Practice Problems	25
Why Quantum Computing Is Different.	27
2.1 A Short History of a Big Idea.	27
2.2 The Quantum World: Three Strange Rules	28
2.3 What Makes a Quantum Computer Different.	29
2.4 What Quantum Computers Are Good For (and Not Good For)	32
2.5 Hype vs. Reality: An Honest Look	33
2.6 The Double-Slit, With Numbers.	34
Practice Problems	35
THE QUBIT.	38
Qubits and Quantum States	39
3.1 From Bit to Qubit	39
3.2 Dirac Notation: Kets and Bras	40
3.3 The State Vector and Amplitudes	41
3.4 Normalization: The Rule Every State Obeys	42
3.5 Your First Quantum States	43
3.6 How Close Are Two States? Overlap as a Ruler.	44
3.7 Perfect Distinguishability: Orthogonal States.	45
Practice Problems	47
Superposition and Measurement	49
4.1 Superposition in Plain Language	49
4.2 The Born Rule: From Amplitudes to Probabilities	50
4.3 Measurement and Collapse	51
4.4 Shots and Histograms: Reading Quantum Statistics	52
4.5 Measuring in the $ +\rangle / -\rangle$ Basis.	54
4.6 Measuring Again: Sequences and the Quantum Zeno Effect	55
4.7 Certifying a State Factory: The Shot Budget.	56

Practice Problems	57
The Bloch Sphere	60
5.1 A Globe for One Qubit	60
5.2 The Angles θ and φ	61
5.3 Reading the Sphere: Latitude is Probability, Longitude is Phase	63
5.4 Global Phase vs. Relative Phase	64
5.5 Inside the Sphere: Mixed States	66
5.6 The Sphere as a Flight Log: Tracking a Circuit	67
Practice Problems	68
GATES AND MEASUREMENT	70
Single-Qubit Gates I: X, Z, and H	71
6.1 What a Quantum Gate Is	71
6.2 The X Gate: The Quantum NOT	72
6.3 The Z Gate: The Phase Flip	73
6.4 The Hadamard Gate: The Superposition Maker	74
6.5 Gates in Sequence: Interference in a Circuit	76
6.6 Two Reflections Make a Rotation: Composing X and Z	78
6.7 When Order Matters: A Commutation Field Guide	79
Practice Problems	80
Single-Qubit Gates II: Y, S, T, and Rotations	83
7.1 The Y Gate: The Third Pauli	83
7.2 The Phase Gates S and T	84
7.3 Rotation Gates: Any Angle You Like	86
7.4 Universality: Two Gates Are Enough	88
7.5 Compiling to the Menu: A Worked Gate Build	89
7.6 Euler's Recipe: Any Gate from Three Turns	90
Practice Problems	91
Observables and Expectation Values	94
8.1 Questions with Numbers: What an Observable Is	94
8.2 The Expectation Value $\langle Z \rangle$	95
8.3 Measuring $\langle X \rangle$ and $\langle Y \rangle$	96
8.4 The Bloch Connection: Averages Are Coordinates	97
8.5 The Matrix Formula and Many-Qubit Observables	98
8.6 Uncertainty, Made Quantitative	100
8.7 The Measurement Compiler: Rotate, Then Read	101
Practice Problems	103
MANY QUBITS AND ENTANGLEMENT	105
Multi-Qubit Systems and Tensor Products	106
9.1 Two Qubits: Four Possibilities at Once	106
9.2 The Tensor Product of States	107
9.3 Probability Tables and Marginals	108

9.4	Gates on Multi-Qubit Systems	109
9.5	Three Qubits and Beyond: The Space Explodes	110
9.6	Measuring Just One Qubit of a Pair	112
9.7	The W State: A Different Way to Share	113
	Practice Problems	114
	Controlled Gates	117
10.1	The CNOT Gate	117
10.2	Truth Table, Matrix, and a Copy Warning	118
10.3	Controlled-Z and SWAP	119
10.4	The Toffoli Gate: Classical Logic, Rebuilt Reversibly	120
10.5	Why Controlled Gates Complete the Toolbox	121
10.6	Building Oracles: From Logic to Circuits	122
10.7	Phase Kickback: The Control That Gets Painted	124
	Practice Problems	125
	Entanglement and Bell States	128
11.1	Making Entanglement: H + CNOT	128
11.2	The Four Bell States	129
11.3	Correlated Measurements: What Entanglement Does	130
11.4	What Entanglement Cannot Do	131
11.5	Bell-Basis Measurement	132
11.6	The CHSH Score: Putting a Number on Spookiness	133
11.7	Monogamy: Entanglement Cannot Be Shared Freely	135
	Practice Problems	136
	Quantum Circuits and Circuit Design	139
12.1	Reading a Full Circuit Diagram	139
12.2	From Circuit to State, Column by Column	140
12.3	Circuit Identities and Simplification	140
12.4	The No-Cloning Theorem	141
12.5	Designing Small Circuits: A Method	142
12.6	Scratch Qubits and Uncomputation	144
12.7	The Cost Sheet: Width, Size, and Depth	145
	Practice Problems	147
	PROTOCOLS, PROGRAMMING, AND CRYPTOGRAPHY	150
	Quantum Teleportation and Superdense Coding	151
13.1	The Problem: Sending a Qubit Without Sending It	151
13.2	The Protocol, Step by Step	152
13.3	Why Teleportation Breaks No Laws	153
13.4	Superdense Coding: The Mirror Protocol	154
13.5	Entanglement Swapping: Linking Qubits That Never Met	155
13.6	Teleporting Through Noise: The Fidelity of a Real Link	157

Practice Problems	158
Programming Quantum Computers	161
14.1 Simulators and Real Machines	161
14.2 Python and NumPy: This Book's Mathematics, Executable	162
14.3 Your First Qiskit Circuit	163
14.4 Running with Shots: Counts and Histograms	164
14.5 To the Cloud: Real Machines from Anywhere	165
14.6 Debugging Quantum Programs	166
14.7 From Counts to Truth: Correcting Readout Errors	168
Practice Problems	169
Quantum Cryptography and Communication	172
15.1 The Key Problem	172
15.2 Wiesner's Seed: Unforgeable States	173
15.3 BB84, Step by Step	173
15.4 Catching the Eavesdropper	174
15.5 From Lab to World: Networks, Satellites, and the Road Ahead	176
15.6 Entanglement-Based Keys: The E91 Idea	177
15.7 Two Defenses, One Deadline: PQC, QKD, and Mosca's Clock	179
Practice Problems	181
The Quantum Internet	184
16.1 The Enemy: Exponential Loss, and No Amplifiers	184
16.2 The Repeater Chain: Divide, Store, and Swap	185
16.3 Distillation: Trading Quantity for Quality	186
16.4 Fiber or Sky: The Satellite Alternative	186
Practice Problems	188
QUANTUM ALGORITHMS	190
First Algorithms: Deutsch–Jozsa and Bernstein–Vazirani	191
17.1 One Question, Two Answers: The Deutsch Problem	191
17.2 Why It Works: Kickback Meets Interference	192
17.3 Scaling Up: Deutsch–Jozsa	193
17.4 Reading a Hidden String: Bernstein–Vazirani	193
Practice Problems	195
Grover's Search Algorithm	197
18.1 The Search Problem	197
18.2 The Oracle: Marking Without Revealing	198
18.3 Amplitude Amplification: Inversion About the Mean	198
18.4 The Complete Algorithm — and a Perfect Small Case	200
18.5 How Many Iterations? $\sqrt{N}$, Overshoot, and Honest Limits	200
18.6 Several Needles at Once: Grover with Multiple Targets	202
18.7 Amplitude Amplification: Grover Beyond Search	204

Practice Problems	206
The QFT, Phase Estimation, and Shor's Algorithm	208
19.1 Waves, Periods, and Frequencies	208
19.2 The Quantum Fourier Transform	209
19.3 Quantum Phase Estimation	210
19.4 Shor's Algorithm: Factoring by Finding a Period	211
19.5 What Shor Means for Codes and Security	212
19.6 A Three-Qubit QFT, Gate by Gate	214
19.7 How Big Is the Machine That Breaks RSA?	215
Practice Problems	217
Variational Algorithms, Optimization, and Quantum Machine Learning	220
20.1 The Hybrid Idea: Quantum + Classical Loops	220
20.2 VQE: The Variational Quantum Eigensolver	221
20.3 Quantum Optimization and QAOA	223
20.4 Quantum Machine Learning: Promise and Caution	223
20.5 Measuring a Many-Term Energy — and Trusting the Bound	225
20.6 Optimizer Traps: Local Minima, Noise, and the Restart Discipline	226
Practice Problems	228
Quantum Simulation: The Original Killer App	230
21.1 The Problem Nature Solves Daily	230
21.2 Time Evolution as a Circuit Target	231
21.3 Trotter's Recipe: Slice Time Thin	231
21.4 The Prize List, Honestly Ranked	233
Practice Problems	234
QUANTUM AI AND MACHINE LEARNING	236
Encoding Data into Quantum States	237
22.1 The First Step of Every Quantum Model	237
22.2 Basis and Angle Encoding	238
22.3 Amplitude Encoding: Maximum Compression, Maximum Cost	240
22.4 Feature Maps: Nonlinearity for Free	240
22.5 The Encoding Bottleneck	241
22.6 How Many Qubits to Hold Your Data?	242
22.7 Feature Scaling: The Forgotten First Layer	243
Practice Problems	245
Variational Quantum Classifiers	248
23.1 Learning Machines in One Page	248
23.2 The Model: A Circuit as a Function	249
23.3 Loss and the Training Loop	249
23.4 A Complete, Real Training Run	250
23.5 Honest Evaluation	252
23.6 Decision Boundaries You Can Draw by Hand	254

23.7	From Scores to Decisions: Thresholds and Imbalance	255
	Practice Problems	256
	Quantum Kernels	259
24.1	The Similarity Trick	259
24.2	The Quantum Kernel	260
24.3	Measuring a Kernel on Hardware	260
24.4	A Working Kernel Machine	261
24.5	When Quantum Kernels Could Win — and the Fine Print	262
24.6	The Kernel Matrix Health Check	263
24.7	What the Classifier Remembers: Support Vectors	264
	Practice Problems	266
	Quantum Neural Networks and Training	269
25.1	From Classical Networks to Quantum Layers	269
25.2	Gradients: The Parameter-Shift Rule	270
25.3	Barren Plateaus: The Desert of Deep Random Circuits	271
25.4	Expressivity and Data Re-uploading	272
25.5	Quantum Data: Where the Edge Is Principled	273
25.6	Initialization, Depth, and Staying Out of the Desert	274
25.7	Overfitting: The Model That Memorized the Noise	276
	Practice Problems	277
	Quantum Generative Models and the Question of Advantage	280
26.1	Discriminate or Generate: Two Jobs for a Model	280
26.2	A Ruler Between Distributions	281
26.3	Quantum GANs: Two Models, One Duel	283
26.4	Where Could Quantum Advantage in AI Actually Live?	284
26.5	The Engineer's Compass	285
26.6	Judging a Generator: Beyond Eyeballing	286
	Practice Problems	287
	Quantum AI in Practice	289
27.1	The Honest Scoreboard	289
27.2	The Dequantization Story — and Its Permanent Lessons	290
27.3	Generative Models: Learning to Sample	290
27.4	Hybrid Workflows: Where Quantum Sits in a Real Pipeline	291
27.5	The Checklist — and the Road for Our Region	292
27.6	A Worked Audit, End to End	294
27.7	The Project Template: Six Steps from Idea to Report	295
	Practice Problems	297
	REAL MACHINES, REAL LIMITS, REAL PROJECTS	299
	Noise, Decoherence, and Errors	300
28.1	Why Qubits Are Fragile	300
28.2	T1 and T2: The Two Lifetimes	301

28.3	Gate Errors and Readout Errors.	303
28.4	Error Mitigation: Better Answers from Noisy Data	305
28.5	Measuring T_1 and T_2 Yourself	306
28.6	Zero-Noise Extrapolation, By Hand	308
	Practice Problems	309
	Quantum Error Correction	312
29.1	Why We Cannot Just Copy — and What We Can Do	312
29.2	The Three-Qubit Bit-Flip Code	313
29.3	Syndromes: Finding Errors Without Looking at Data	314
29.4	Toward the Surface Code: Distance and Threshold	315
29.5	Nine Qubits Against Everything: Shor's Code	318
29.6	The Logical Qubit Ledger: Overhead, Honestly Priced	320
	Practice Problems	322
	Quantum Hardware and Our Region	325
30.1	How to Build a Qubit: Five Technologies	325
30.2	Comparing Machines Honestly	326
30.3	Cold Machines: Inside the Refrigerator.	327
30.4	Cloud Quantum Computing from Anywhere	329
30.5	Building Quantum Skills in Iraq and the Kurdistan Region.	329
30.6	Reading a Real Datasheet	331
30.7	The Roadmap Game: Reading Vendor Promises.	333
	Practice Problems	334
	Capstone Projects	337
31.1	Project 1 — A Certified Quantum Coin	338
31.2	Project 2 — Build and Verify a Bell Pair.	338
31.3	Project 3 — The Teleportation Machine	339
31.4	Project 4 — A Secret Message with Superdense Coding.	339
31.5	Project 5 — Grover Search of an 8-Entry Database	339
31.6	Project 6 — Period Finding: Factor 15 End to End.	339
31.7	Project 7 — Ideal vs. Noisy: A Benchmark Report.	340
31.8	Project 8 — Protect One Qubit: A Repetition-Code Study.	340
31.9	Project 9 — Train a Quantum Classifier, Simulator to Hardware.	340
31.10	Project 10 — Quantum Kernel vs. Classical Baseline: The Fair Fight	341
31.11	Project 11 — The QML Claim Audit, in the Wild	341
31.12	Project 12 — A Realistic Briefing for Decision-Makers	341
	Practice Problems	342
	APPENDICES: THE MATHEMATICS AND TOOLS YOU NEED	343
	Appendix A — Complex Numbers	344
A.1	Why Quantum Computing Needs New Numbers	344
A.2	The Imaginary Unit i	345
A.3	Complex Numbers and the Complex Plane	346
A.4	Modulus, Conjugate, and $ z ^2$	348

A.5	The Unit Circle and $e^{i\theta}$ (Polar Form)	350
A.6	The Clock Numbers: Roots of Unity	353
	Practice Problems	354
	Appendix B — Vectors and Matrices	356
B.1	Vectors: Lists of Numbers with Meaning	356
B.2	Adding, Scaling, and the Inner Product	357
B.3	Matrices and Matrix-Vector Multiplication	359
B.4	Matrix $\times$ Matrix, the Identity, and the Dagger	360
B.5	Unitary Matrices: The Entrance Exam for Quantum Gates	362
B.6	Eigenvectors: The Directions a Matrix Keeps	363
	Practice Problems	364
	Appendix C — The Probability You Need	367
C.1	Probability: The Rules of Chance	367
C.2	Independence: When Chances Multiply	367
C.3	Expectation and Spread: One Number Each	368
C.4	Repetition and the Square-Root Law	368
C.5	Conditional Probability: Updating on Evidence	369
	Practice Problems	370
	Appendix D — Formula Sheet and Gate Reference	371
D.1	States and Probability	371
D.2	The Gate Menu	372
D.3	The Famous States and Protocol Scorecards	372
D.4	Noise, Errors, and Budgets	373
D.5	Quick Machine-Learning Reference	374
	Appendix E — Python and NumPy Quick Start	375
E.1	Installing Python (Once)	375
E.2	Python in Ten Minutes	375
E.3	NumPy Is the Quantum Part	376
E.4	A Complete First Simulation: The Bell Pair	376
E.5	Seeing Your Results	377
E.6	From NumPy to Qiskit	377
	References and Further Reading	379
	Glossary	385

PART ONE

FROM BITS TO THE QUANTUM IDEA

CHAPTER ONE

Bits and Classical Computing

Information, binary numbers, and logic — the world every computer lives in

IN THIS CHAPTER YOU WILL LEARN TO

- Explain what information is, and why every computer stores it as bits.
- Count how many different messages n bits can hold, using $M = 2^n$.
- Convert numbers between binary and decimal, in both directions.
- Read the truth tables of the NOT, AND, OR, and XOR logic gates, and trace a small logic circuit.
- Explain, with the example of exponential growth, why some problems defeat even the fastest classical computers — the door through which quantum computing enters.

1.1 Information Around Us

Look around you. A light switch is up or down. A door is open or closed. A phone battery is charged or empty. An exam answer is right or wrong. Each of these is a tiny piece of **information**: a choice between possibilities.

WORD TO KNOW

Information — Anything that answers a question or removes uncertainty. In computing, information is always stored as the physical state of something real — a voltage, a magnet, a pulse of light.

Here is the key idea of this whole chapter: **information is physical**. It never floats in the air as pure ideas. A message on your phone is really billions of tiny electrical charges inside a memory chip. A song is really a long list of numbers. A photo of Erbil's citadel is really a grid of numbers describing colors. To store, send, or process information, we must store, send, or process *physical states*.

The simplest possible piece of information is a choice between exactly two possibilities: yes/no, on/off, up/-down, true/false. Computer scientists call this smallest piece a **bit**. Everything your computer has ever done — every photo, video call, and game — is built from bits, processed billions of times per second.

WORD TO KNOW

Bit — The smallest unit of information: a single choice between two states, written 0 and 1. The word comes from “binary digit.”

CASE STUDY — FROM A ROOM-SIZED MACHINE TO YOUR POCKET • USA AND THE WORLD, 1946–TODAY

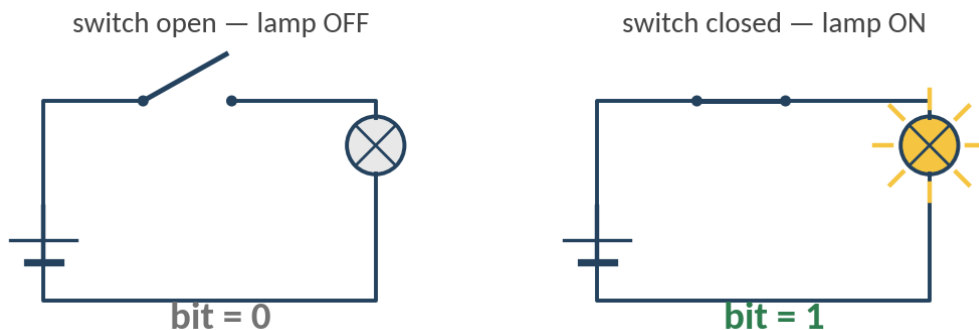
ENIAC, completed in 1946, was one of the first general-purpose electronic computers. It filled a large room, weighed about 27 tonnes, used about 150 kW of electric power (roughly as much as a whole neighborhood), and could do about 5,000 additions per second. A basic smartphone today is billions of times faster, runs on a battery, and fits in your pocket. This astonishing progress — driven by shrinking transistors — is the story of *classical* computing. In this book you will learn why, despite this progress, some problems still defeat every classical computer ever built, and what a completely different kind of computer can do about it.

1.2 Bits: The Smallest Unit of Information

A bit needs a physical home: some object with two reliably different states. Engineers have used many homes for bits over the years.

Physical homes for one bit

Technology	State for 0	State for 1	Where used
Switch or relay	open	closed	early machines
Voltage on a wire	low (≈ 0 V)	high (≈ 1 V)	processors
Magnetic region	magnetized one way	magnetized the other way	hard disks
Trapped charge	few electrons	many electrons	flash memory, SSD
Pulse of light	no pulse	pulse	fiber-optic cables



One bit, stored as a switch and shown by a lamp. The circuit has exactly two reliable states: lamp off (bit = 0) and lamp on (bit = 1).

Notice something important in every row of the table: the two states are *clearly different* and *stable*. A voltage of “about 0 V” and a voltage of “about 1 V” are easy to tell apart, even if electrical noise moves them a little. This is why classical computers almost never make hardware mistakes: their bits are rugged. Keep this in mind — in Chapter 28 you will see that quantum bits are exactly the opposite: fragile, and easily disturbed. That contrast explains much of what makes quantum computing hard to build.

One bit answers one yes/no question. More interesting messages need more bits, and here a beautiful pattern appears. With 1 bit you have 2 possible messages: 0 or 1. With 2 bits you have 4: 00, 01, 10, 11. With 3 bits you have 8: 000, 001, 010, 011, 100, 101, 110, 111. Each new bit *doubles* the count, because everything you had before can now appear once with the new bit at 0 and once with it at 1.

EQUATION: HOW MANY MESSAGES CAN N BITS HOLD?

$$M = 2^n$$

In words: Number of different messages = 2 multiplied by itself n times

Symbol	Meaning	Unit
M	the number of different messages (bit patterns) that can be stored	none — a count
n	the number of bits	bits

Each added bit doubles M . Powers of 2 grow very fast: $2^{10} = 1024$, 2^{20} is about a million, 2^{30} is about a billion.

WORKED EXAMPLE — HOW MANY BITS FOR THE KURDISH ALPHABET?

A student in Erbil wants to give every letter of a 34-letter alphabet its own bit pattern. How many bits does she need?

1. Write the equation: $M = 2^n$, and we need $M \geq 34$.
2. Try $n = 5$: $2^5 = 32$. Too few — only 32 patterns for 34 letters.
3. Try $n = 6$: $2^6 = 64$. Enough — 64 patterns for 34 letters, with 30 spare.

Answer: 6 bits per letter. (Real systems agree: text standards use groups of 8 bits, called bytes, with room to spare.)

WORD TO KNOW

Byte — A group of 8 bits, the everyday unit of computer storage. One byte holds $2^8 = 256$ different patterns. A kilobyte (kB) is about a thousand bytes; a megabyte (MB) about a million; a gigabyte (GB) about a billion.

COMMON MISTAKE

✗ Wrong: “A bit is a tiny amount of electricity.”

✓ Right: A bit is not a substance — it is a *choice* between two states. Electricity is just one popular way to

hold that choice. The same bit can move from a voltage (processor) to a magnet (disk) to light (fiber cable) without changing its value at all.

ACTIVITY (HOME) — BE A BIT DETECTIVE

1. Find five things in your home that hold exactly one bit of information (hint: switches, locks, indicator lamps).
2. For each one, write what physical state means 0 and what state means 1.
3. Then find one thing that holds *many* bits (a phone, a USB stick) and look up its capacity in gigabytes.
4. Convert that capacity to bits ($1 \text{ GB} \approx 8 \text{ billion bits}$). Write the number out fully once, to feel its size.

You need: Paper and pen

1.3 Binary Numbers: Counting with 0 and 1

We normally count in the **decimal** system, with ten digits (0–9). Computers count in the **binary** system, with two digits (0 and 1) — because two rugged states are what their hardware provides. Both systems use the same clever idea: *place value*. In decimal, the number 305 means $3 \times 100 + 0 \times 10 + 5 \times 1$. Each place is worth ten times the place to its right. In binary, each place is worth *two* times the place to its right: the places are worth 1, 2, 4, 8, 16, 32, ...

WORD TO KNOW

Binary number — A number written with only the digits 0 and 1, where each place is worth twice the place to its right. “1101” in binary means $8 + 4 + 0 + 1 = 13$.

EQUATION: THE VALUE OF A BINARY NUMBER

$$\text{value} = b_k 2^k + \dots + b_2 2^2 + b_1 2^1 + b_0 2^0$$

In words: Value = (each binary digit) $\times$ (the worth of its place), all added up

Symbol	Meaning	Unit
b_i	the binary digit (0 or 1) standing in place i , counting from the right and starting at 0	none — a digit
2^i	the worth of place i : the places are worth 1, 2, 4, 8, 16, ...	none — a count
k	the place number of the left-most digit	none

Digits that are 0 contribute nothing; you only add the place values under the 1s.

WORKED EXAMPLE — BINARY TO DECIMAL

What is the binary number 10110 in decimal?

1. Write the place values above the digits, from the right: 16, 8, 4, 2, 1.
2. Match them: $1 \rightarrow 16$, $0 \rightarrow 8$, $1 \rightarrow 4$, $1 \rightarrow 2$, $0 \rightarrow 1$.
3. Add the place values under the 1s: $16 + 4 + 2 = 22$.

Answer: 10110 (binary) = 22 (decimal)

WORKED EXAMPLE — DECIMAL TO BINARY

Write the decimal number 45 in binary.

1. Find the largest power of 2 that fits in 45: it is 32. Write a 1 in the 32s place. Remainder: $45 - 32 = 13$.
2. Does 16 fit in 13? No — write 0 in the 16s place.
3. Does 8 fit in 13? Yes — write 1. Remainder: $13 - 8 = 5$.
4. Does 4 fit in 5? Yes — write 1. Remainder: $5 - 4 = 1$.
5. Does 2 fit in 1? No — write 0. Does 1 fit? Yes — write 1. Remainder 0.

Answer: 45 (decimal) = 101101 (binary). Check: $32 + 8 + 4 + 1 = 45$ ✓

The numbers 0 to 15 in binary — worth memorizing slowly over this course

Decimal	Binary	Decimal	Binary
0	0000	8	1000
1	0001	9	1001
2	0010	10	1010
3	0011	11	1011
4	0100	12	1100
5	0101	13	1101
6	0110	14	1110
7	0111	15	1111

This table matters more than it looks. In quantum computing, the possible results of measuring a group of qubits are exactly these bit patterns. When Chapter 9 talks about “the state $|101\rangle$,” it will help enormously that you can read 101 instantly as “five.”

COMMON MISTAKE

X Wrong: “10 in binary is ten.”

✓ **Right:** 10 in binary is two ($1 \times 2 + 0 \times 1$). The same marks on paper mean different numbers in different systems. When there is any risk of confusion, say the digits aloud — “one-zero” — or write the system name after the number.

ACTIVITY (CLASSROOM) — BINARY ON YOUR FINGERS

1. Let each finger of one hand be a binary place: thumb = 1, index = 2, middle = 4, ring = 8, little = 16.
2. A raised finger is a 1; a folded finger is a 0. Show 5 (thumb + middle), then 11, then 21.
3. What is the largest number one hand can show? Predict it with $M = 2^n$, then check by raising all five fingers and adding the place values.
4. In pairs: one student shows a hand pattern, the other reads the number. Swap after five rounds.

You need: Your hands, Paper and pen for checking

► *App tool: Binary Converter — type a number and watch it become binary, one place at a time*

1.4 Logic Gates and Classical Circuits

Storing bits is only half the story — computers also *change* bits: they compute. The workers that do this are **logic gates**: tiny electronic devices that take one or two input bits and produce one output bit, following a fixed rule. The rule of a gate is written as its **truth table**: a complete list of every possible input and the output for each.

WORD TO KNOW

Logic gate — A device that turns one or two input bits into one output bit, following a fixed rule. Real gates are built from transistors; a modern processor contains billions of them.

WORD TO KNOW

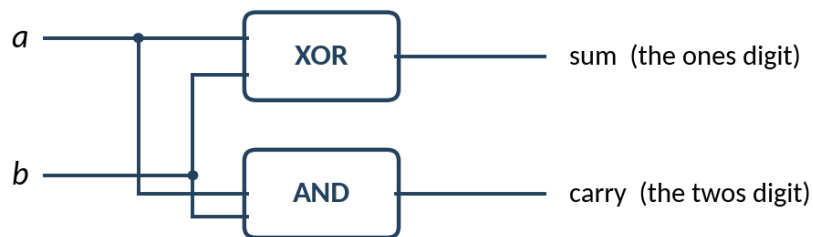
Truth table — A table listing every possible input combination of a gate or circuit, together with the output for each. Two inputs give $2^2 = 4$ rows; three inputs give 8 rows.

NOT		AND			OR			XOR		
in	out	a	b	out	a	b	out	a	b	out
0	1	0	0	0	0	0	0	0	0	0
1	0	0	1	0	0	1	1	0	1	1
		1	0	0	1	0	1	1	0	1
		1	1	1	1	1	1	1	1	0

The four most useful classical gates and their truth tables. NOT flips its input. AND outputs 1 only if both inputs are 1. OR outputs 1 if at least one input is 1. XOR (“exclusive or”) outputs 1 if the inputs differ.

Read each truth table slowly and say its rule in words. NOT: “the opposite.” AND: “both.” OR: “at least one.” XOR: “different.” These four rules, wired together in clever patterns, are enough to build everything a classical computer does — from adding numbers to streaming video.

Two of these gates deserve a special note for what comes later. First, **XOR** is exactly *addition of bits, keeping only the ones digit*: $1 + 1 = 10$ in binary, and XOR keeps the 0. In Chapter 10 you will meet the CNOT gate, the quantum cousin of XOR, and it will behave the same way. Second, notice that most gates *destroy information*: if an AND outputs 0, you cannot tell whether the inputs were 00, 01, or 10. Quantum gates, you will learn in Chapter 6, are never allowed to destroy information — one of the deepest differences between the two worlds.



A half-adder: the smallest possible adding machine. The XOR gate computes the ones digit of $a + b$; the AND gate computes the twos digit (the carry).

WORKED EXAMPLE — TRACING THE HALF-ADDER

Both inputs of the half-adder are 1 (we are adding $1 + 1$). What does the circuit output?

1. XOR rule (“different?”): inputs are 1 and 1 — not different — so sum = 0.
2. AND rule (“both?”): inputs are 1 and 1 — both are 1 — so carry = 1.
3. Read the answer as a two-bit binary number: carry then sum = 10.

Answer: sum = 0, carry = 1 — the circuit answers “ $1 + 1 = 10$,” which is binary for two. ✓

ACTIVITY (HOME) — BUILD A TRUTH TABLE

1. Draw the half-adder and its four possible inputs: 00, 01, 10, 11.
2. For each input, trace the circuit by hand and fill a truth table with columns *a*, *b*, *sum*, *carry*.
3. Check your table says: $0 + 0 = 00$, $0 + 1 = 01$, $1 + 0 = 01$, $1 + 1 = 10$.
4. Extra: how would you add three bits? (Real processors chain “full adders” — look up the idea if you are curious.)

You need: Paper and pen

► *App tool: Logic Lab — click gate inputs on and off and watch outputs change live*

1.5 What Classical Computers Do Well — and Where They Struggle

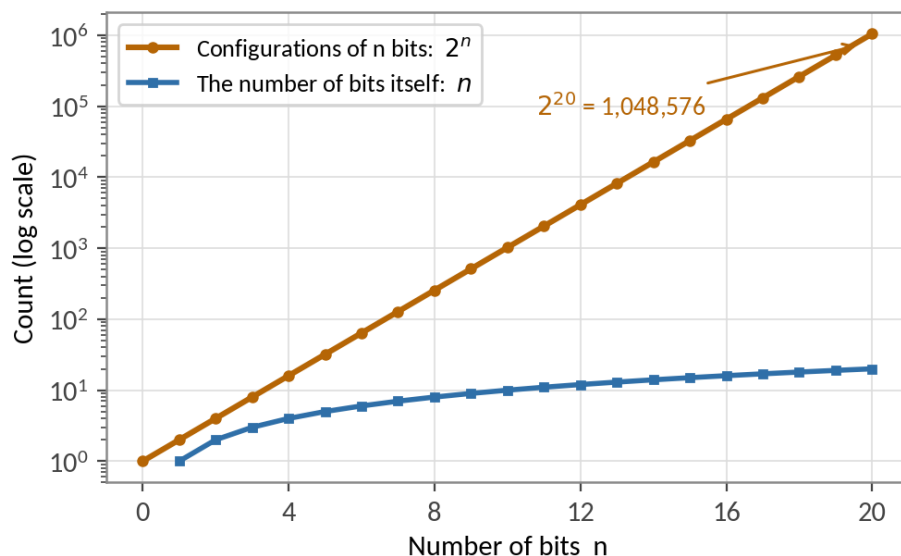
Classical computers are a triumph. They are fast (billions of operations per second), reliable (bit errors are astonishingly rare), cheap, and everywhere. For most tasks — documents, video, databases, controlling machines, even most scientific computing — they are, and will remain, the right tool. Any honest book about quantum computing must start by saying so.

But there is a family of problems where every classical computer, no matter how large, hits the same wall: problems whose number of possibilities grows *exponentially*.

WORD TO KNOW

Exponential growth — Growth by repeated multiplication, like 2^n : each step multiplies the total by a fixed factor. Exponential growth starts small and then explodes past anything that grows steadily.

You have already met the equation of this wall — it is $M = 2^n$ from Lesson 2, seen from the dark side. If a problem forces you to consider all 2^n configurations of n things, then each extra thing *doubles* the work. Chess-like games, molecule simulation, and code-breaking all have this flavor.



Steady growth versus exponential growth, on a logarithmic scale. The number of configurations 2^n (orange) races past the number of bits n (blue): at just $n = 20$ there are already more than a million configurations.

WORKED EXAMPLE — THE EXPONENTIAL WALL, WITH REAL NUMBERS

A search problem requires checking all configurations of n bits, and a fast computer checks one billion (10^9) configurations per second. How long does the full check take for $n = 30$, $n = 60$, and $n = 90$?

1. For $n = 30$: $2^{30} \approx 1.07 \times 10^9$ configurations $\div 10^9$ per second ≈ 1.1 seconds.
2. For $n = 60$: $2^{60} \approx 1.15 \times 10^{18}$, so time $\approx 1.15 \times 10^9$ seconds ≈ 37 years.
3. For $n = 90$: $2^{90} \approx 1.24 \times 10^{27}$, so time $\approx 1.24 \times 10^{18}$ seconds ≈ 39 billion years — about three times the age of the universe.

Answer: 1 second, 37 years, and 39 billion years. Adding 30 bits each time multiplied the work by about a billion. That is the exponential wall.

Engineers cannot beat this wall by building faster chips: going from 37 years to 37 minutes only moves the wall from $n = 60$ to about $n = 80$. The wall is mathematical, not technological.

REALITY CHECK

Exponential problems are not rare curiosities. Simulating the quantum behavior of molecules (for designing medicines and materials), optimizing over huge configuration spaces, and breaking certain codes are all exponential for classical machines. But be careful with the opposite claim, too: *most* everyday computing is NOT exponential, and a quantum computer offers it no advantage at all. Quantum computers are specialists, not faster laptops. Chapter 2 makes this precise.

CASE STUDY — SIMULATING NATURE, ATOM BY ATOM • THE WORLD'S SUPERCOMPUTERS

Modern supercomputers fill halls, use tens of megawatts of power, and perform more than 10^{18} operations per second. Yet when chemists ask them to simulate the exact quantum behavior of even a modest molecule — a few dozen interacting electrons — the exponential wall appears: each added electron roughly doubles the memory needed. Storing the exact quantum state of about 50 interacting particles would already exceed the memory of every computer on Earth combined. In 1981, the physicist Richard Feynman turned this frustration into a famous proposal: if nature is quantum, perhaps only a *quantum* machine can simulate it efficiently. That proposal is the seed this whole book grows from.

1.6 Error-Detecting Bits: The Parity Trick

Classical engineers solved a problem in the 1940s that will return, transformed, at this book's summit: bits get corrupted — by static, scratches, cosmic rays — and yet your files survive. The oldest defense is one extra bit.

WORD TO KNOW

Parity bit — An extra bit appended to a word, chosen so the total number of 1s is even. Any single flipped bit makes the count odd — announcing that an error happened, though not where.

WORKED EXAMPLE — ONE EXTRA BIT, ONE CAUGHT ERROR

Protect the 7-bit word 1011001 with a parity bit, then show how a corruption is caught.

1. Count the 1s: four — already even — so append parity 0: transmitted word 10110010.
2. Corruption strikes bit 3: received 10010010.
3. Receiver recounts: three 1s plus parity 0 = odd total. Alarm! The word is damaged (though the receiver cannot tell which bit lied).
4. Note the limit: two flips restore even parity — double errors slip through. Stronger codes (more parity bits, cleverly placed) locate and even fix errors: that is the *error-correcting* code inside every phone and hard drive.

Answer: transmitted 10110010; single flips are always detected, never located; double flips escape. Hold the idea gently — Chapter 29 rebuilds it for qubits, where the same “check the parity, not the data” philosophy has to work *without looking at the data at all*.

REALITY CHECK

Parity thinking is everywhere once you see it: the last digit of a credit-card number, a barcode's check digit, an ISBN's final character — all are parity's descendants, catching typing slips by arithmetic. Classical error correction is so cheap and perfected that we forget it exists; quantum error correction (Chapter 29) is the

same dream, priced a thousand times higher.

CHAPTER SUMMARY

- Information is physical: it is always stored as the state of something real. The smallest unit is the bit: one choice between two states, 0 and 1.
- n bits can hold $M = 2^n$ different patterns. Each added bit doubles the count.
- Binary numbers use place values 1, 2, 4, 8, 16, ... To read one, add the place values under the 1s.
- Logic gates (NOT, AND, OR, XOR) turn input bits into output bits by fixed truth-table rules; wired together, they perform all classical computation. Most classical gates destroy information — quantum gates will not.
- Classical bits are rugged and classical computers are superb at almost everything — but problems with 2^n configurations defeat them, because exponential growth outruns any buildable speed.
- The exponential cost of simulating quantum systems led Feynman (1981) to propose quantum computers in the first place.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] How many different patterns can 4 bits hold? List the first five patterns in binary counting order.

Problem 2 [Basic] Convert the binary number 1010 to decimal, showing the place values you added.

Problem 3 [Basic] Convert the decimal number 19 to binary, showing each step.

Problem 4 [Basic] Write the truth table of the AND gate, and state its rule in one short sentence.

Problem 5 [Basic] Append the correct parity bit to 1100110 and to 0000001, and state each transmitted 8-bit word.

Problem 6 [Basic] A received 8-bit word (parity included) is 10111010. Is it clean or corrupted? What can and cannot the receiver conclude?

Problem 7 [Standard] A remote control uses 7-bit codes. How many different commands can it send? A new model needs 200 commands — how many bits must its codes have?

Problem 8 [Standard] Convert 111111 (binary) to decimal *without adding six numbers*, by first finding the value of 1000000 (binary). Explain the trick you used.

Problem 9 [Standard] An XOR gate's inputs are $a = 1$ and $b = \text{unknown}$, and its output is 0. What is b ? Explain

using the rule “XOR asks: are the inputs different?”

Problem 10 [Standard] Trace the half-adder for inputs $a = 1$, $b = 0$: give sum and carry, and read the two-bit answer as a decimal number.

Problem 11 [Standard] A processor checks 10^9 configurations per second. Using $2^{10} \approx 10^3$, estimate how long it needs to check all configurations of 40 bits. Give the answer in minutes.

Problem 12 [Standard] Exactly two bits of a parity-protected word are flipped. Show with an example that the check passes, and compute the fraction of all possible double-flip patterns on an 8-bit word (there are $\binom{8}{2} = 28$) that parity misses — all of them, some, or none?

Problem 13 [Challenge] You need to label every seat in a stadium of 48,000 seats with a binary code. How many bits are needed? How many seats could the same number of bits label at most?

Problem 14 [Challenge] Build a truth table for the circuit “output = (a AND b) OR (NOT a).” There are four input rows. For which inputs is the output 0?

Problem 15 [Challenge] A team doubles its computer’s speed every two years. A problem currently needs 2^{80} steps and the computer does 10^9 steps per second. Roughly how many *years of doubling* are needed before the problem takes less than one day? (Use $2^{10} \approx 10^3$ and count carefully.) What does your answer say about fighting exponential walls with faster hardware?

Problem 16 [Challenge] Design a scheme with two parity bits over a 4-bit word arranged in a 2×2 grid (one parity per row): can it locate which row an error is in? Add column parities: what can be located now? (You have reinvented the rectangle code — the ancestor of Chapter 29’s syndrome tables.)

CHAPTER TWO

Why Quantum Computing Is Different

The ideas, the history, and an honest map of what quantum computers can and cannot do

IN THIS CHAPTER YOU WILL LEARN TO

- Tell the short history of quantum computing, from Planck's quanta to today's cloud machines.
- State the three quantum rules — superposition, measurement randomness, and entanglement — in plain language.
- Explain what really makes a quantum computer different: amplitudes that can cancel, and a state space that doubles with every qubit.
- Name the problem families where quantum computers promise real advantages — and the everyday tasks where they offer none.
- Judge a quantum-computing news claim calmly, using a short honesty checklist.

2.1 A Short History of a Big Idea

Quantum computing did not appear from nowhere. It sits on top of **quantum mechanics** — the physics of the very small — which is itself more than a century old, and the most precisely tested theory in all of science.

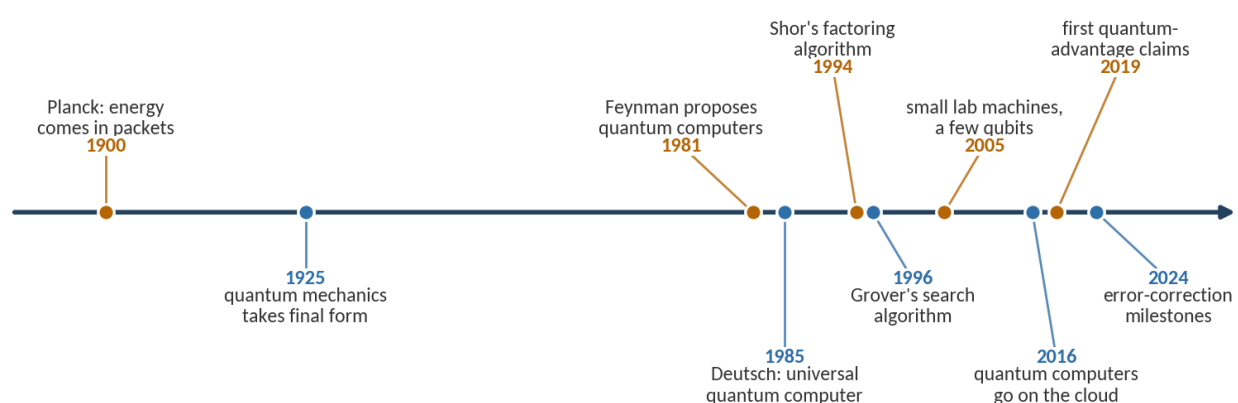
WORD TO KNOW

Quantum mechanics — The physics that describes atoms, electrons, and light. Its rules are different from everyday physics: quantities come in smallest packets (quanta), outcomes can be random, and objects can be in superposition.

The story begins in 1900, when Max Planck discovered that energy is exchanged in smallest packets, which he called *quanta*. By 1925–1927, physicists including Heisenberg, Schrödinger, Born, and Dirac had assembled the full mathematical machine of quantum mechanics. That machine explained atoms, chemistry, and light — and later gave us the transistor and the laser. Every classical computer is already, in this indirect sense, a child of quantum mechanics.

The idea of *computing with* quantum mechanics came much later. In 1981, Richard Feynman pointed at the

exponential wall you met in Chapter 1: simulating quantum systems seems to require exponential classical resources — so why not build a computer that is a quantum system? In 1985, David Deutsch made the idea precise, describing a universal quantum computer. For a decade it remained a beautiful curiosity. Then came the earthquake: in 1994, Peter Shor showed that a quantum computer could factor huge numbers efficiently — a task believed impossibly hard for classical machines, and the exact task protecting most encrypted communication (Chapter 19). Suddenly governments and companies cared. In 1996, Lov Grover added a general quantum search speedup (Chapter 18). Hardware crawled after theory: a few fragile qubits around 2000, dozens by the 2010s, and in 2016 IBM connected a small quantum processor to the internet, letting anyone — including students in Erbil, Duhok, or Sulaymaniyah — run real quantum circuits from a laptop. In 2019, Google claimed the first “quantum advantage” demonstration, and in the 2020s the frontier moved to the real bottleneck: quantum error correction (Chapters 28–29).



A century in one line: quantum mechanics matured for eighty years before anyone proposed computing with it, and the hardware era is younger than many students reading this book.

CASE STUDY — FEYNMAN'S CHALLENGE • MIT CONFERENCE, USA, 1981

At a physics-and-computation conference at MIT in 1981, Richard Feynman gave a talk that quantum scientists still quote. Classical computers, he argued, cannot efficiently simulate quantum nature: “Nature isn’t classical, dammit, and if you want to make a simulation of nature, you’d better make it quantum mechanical.” The talk set a research agenda decades long. It is worth noticing what Feynman did *not* say: he never promised a faster computer for everything. From its very first day, quantum computing was about a special class of problems — a point this book will repeat honestly, because much of today’s hype forgets it.

2.2 The Quantum World: Three Strange Rules

Everything new in this book flows from three experimental facts about the quantum world. You will meet each one mathematically later; here you only need them in plain language.

WORD TO KNOW

Superposition — A quantum object can be in a state that combines several classical possibilities at once, each with its own amplitude. A quantum coin can be in a genuine combination of heads and tails — not secretly one of them.

WORD TO KNOW

Measurement — The act of reading a quantum system. Measurement gives one definite, generally random result, and changes the state: the superposition collapses to the result you got.

WORD TO KNOW

Entanglement — A connection between two or more quantum objects in which their measurement results are correlated more strongly than any classical mechanism could arrange. Einstein called it “spooky”; experiments confirm it is real.

Rule 1 — Superposition. Small objects (electrons, photons, atoms) do not always have to be in one definite classical state. An electron can be in a combination of “here” and “there”; a photon can be in a combination of two polarizations. Each possibility in the combination carries a number called an *amplitude*. Amplitudes are the central objects of this entire book.

Rule 2 — Measurement randomness. When you measure, you get one answer, at random, with probabilities set by the amplitudes (the Born rule, Chapter 4). Before measurement, the superposition is not ignorance — it is a genuinely different kind of state, and experiments can tell the difference.

Rule 3 — Entanglement. Two quantum objects can share a joint state that cannot be described by describing each object separately. Measuring one instantly tells you about the other, no matter how far away — though, as you will prove in Chapter 11, this can never be used to send messages faster than light.

COMMON MISTAKE

✗ Wrong: “Quantum effects are unproven theory — maybe the strangeness will go away.”

✓ Right: Quantum mechanics is the most accurately tested theory in science, verified to better than parts per billion in some experiments. The 2022 Nobel Prize in Physics honored the experiments proving entanglement is real and cannot be explained by hidden classical information. The strangeness is measured fact; only our comfort with it is optional.

2.3 What Makes a Quantum Computer Different

A quantum computer is a machine whose bits obey the three rules above. Its basic unit is the **qubit**: a quantum system with two measurable states, labeled $|0\rangle$ and $|1\rangle$ (the funny brackets are Dirac notation, explained gently in Chapter 3).

WORD TO KNOW

Qubit — A quantum bit: any quantum object with two reliably measurable states, used to store quantum information. When measured, a qubit always answers 0 or 1 — but before measurement it can be in a superposition of both.



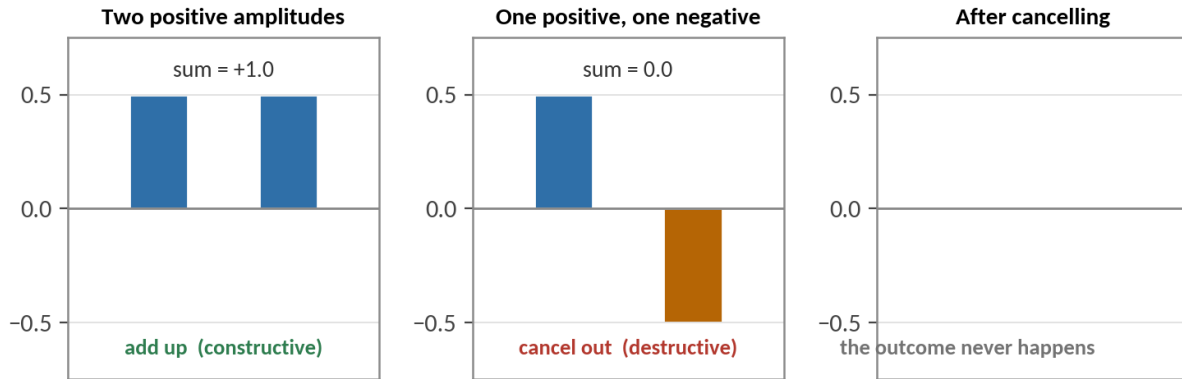
A bit is always one of two values. A qubit's state is described by two amplitudes, α and β — a richer object, even though measuring it still yields just 0 or 1.

Two consequences separate quantum computers from everything in Chapter 1.

Difference 1 — Amplitudes can cancel. Probabilities in ordinary life are never negative: adding a new way for something to happen can only make it more likely. Amplitudes can be negative (and even complex, Appendix A). When two paths to the same outcome have amplitudes $+0.5$ and -0.5 , the outcome *never happens at all*. This is **interference**, and it is the engine of every quantum algorithm: arrange the computation so that paths to wrong answers cancel and paths to the right answer reinforce.

WORD TO KNOW

Interference — The adding and cancelling of amplitudes. Amplitudes pointing the same way reinforce (constructive); amplitudes pointing opposite ways cancel (destructive). Quantum algorithms are interference machines.



The heart of quantum computing: amplitudes, unlike probabilities, can cancel. An outcome fed by amplitudes $+0.5$ and -0.5 has amplitude 0 — it never occurs.

Difference 2 — The description doubles with every qubit. One qubit is described by 2 amplitudes. Two qubits need 4 (one per pattern 00, 01, 10, 11). Three need 8. This is the 2^n of Chapter 1 again — but now working for us: the amplitudes are not just a list we must store; nature itself carries all 2^n of them at once, and a single gate operation updates them all.

EQUATION: THE SIZE OF A QUANTUM STATE

$$N_{\text{amp}} = 2^n$$

In words: A system of n qubits is described by 2^n amplitudes, one for each n -bit pattern

Symbol	Meaning	Unit
N_{amp}	the number of amplitudes in the state description	none — a count
n	the number of qubits	qubits

This is why classical simulation of quantum systems hits the exponential wall — and why Feynman wanted a quantum machine instead.

WORKED EXAMPLE — HOW BIG IS BIG?

How many amplitudes describe 10 qubits, 50 qubits, and 300 qubits? (Use $2^{10} = 1024 \approx 10^3$.)

1. For $n = 10$: $2^{10} = 1024$ amplitudes — easy for any laptop.
2. For $n = 50$: $2^{50} = (2^{10})^5 \approx (10^3)^5 = 10^{15}$ amplitudes. Stored at 16 bytes each, that is about 16,000 terabytes — roughly the limit of the world's biggest supercomputers.
3. For $n = 300$: $2^{300} \approx 10^{90}$ amplitudes — more than the estimated number of atoms in the observable universe (about 10^{80}).

Answer: $\sim 10^3$, $\sim 10^{15}$, and $\sim 10^{90}$. Somewhere between 50 and 300 qubits, exact classical simulation of arbitrary circuits becomes physically impossible — not just slow.

REALITY CHECK

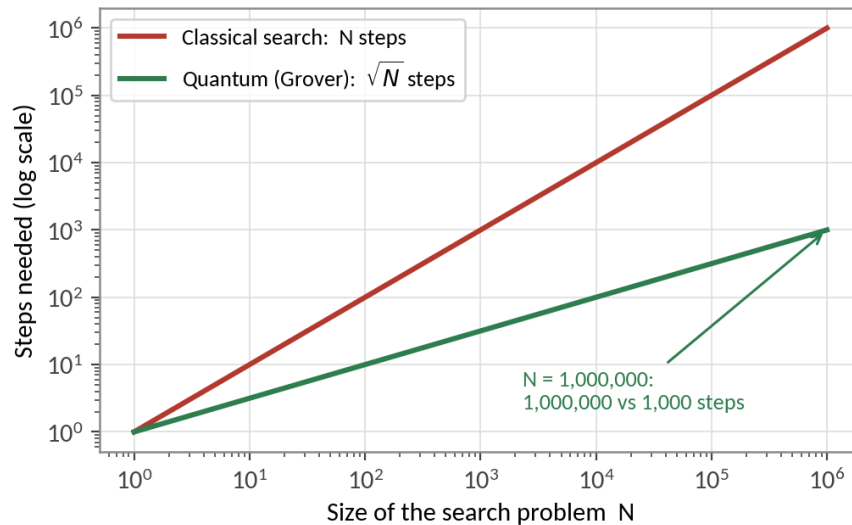
Careful with the most common myth in popular articles: a quantum computer does NOT “try all 2^n answers in parallel and pick the best.” It carries 2^n amplitudes, yes — but measurement returns *one random result*. Without interference cleverly concentrating amplitude on right answers, the machine is just an expensive random-number generator. Where such interference tricks exist (Chapters 18–20), quantum computers shine. Where they do not, quantum computers offer nothing. The number of known tricks is impressive but limited.

2.4 What Quantum Computers Are Good For (and Not Good For)

Sixty years of classical computer science plus forty years of quantum algorithms research give us a reasonably honest map.

Where quantum computers promise real advantage

Problem family	Why quantum helps	Expected speedup
Simulating molecules and materials	the simulated system and the computer obey the same physics (Feynman’s idea)	up to exponential
Factoring and related code-breaking	Shor’s algorithm finds hidden periods with the quantum Fourier transform	exponential (superpolynomial)
Unstructured search and many optimization subroutines	Grover’s amplitude amplification	quadratic ($N \rightarrow \sqrt{N}$)
Certified random numbers	measurement randomness is physical, not formula-based	new capability
Some machine-learning subroutines	linear-algebra speedups (with important caveats)	case by case; under research



What a quadratic speedup looks like. For a search over one million items, $\sqrt{N}$ means about a thousand quantum steps instead of a million classical ones. Valuable — but far less dramatic than the exponential speedups of simulation and factoring.

Just as important is the other half of the map. Quantum computers are *not* faster at: browsing, video, databases in everyday use, spreadsheets, ordinary arithmetic, or running your favorite game. Each quantum operation is currently far slower and far more error-prone than a classical one; a quantum computer pays off only when it needs *exponentially fewer* operations, and that happens only for special structures.

WORKED EXAMPLE — QUADRATIC VS EXPONENTIAL — WHY THE DIFFERENCE MATTERS

Problem A: search $N = 10^{12}$ items. Problem B: simulate a molecule needing $2^{60} \approx 10^{18}$ classical steps. A classical computer does 10^9 steps per second; suppose an ideal quantum computer does 10^6 quantum steps per second (quantum gates are slower). Compare honestly.

1. Problem A classical: $10^{12} \div 10^9 = 1000 \text{ s} \approx 17 \text{ minutes}$.
2. Problem A quantum (Grover): $\sqrt{10^{12}} = 10^6$ steps $\div 10^6 =$ about 1 second. Nice win — but the classical 17 minutes was already acceptable.
3. Problem B classical: $10^{18} \div 10^9 = 10^9 \text{ s} \approx 32 \text{ years}$ — unusable.
4. Problem B quantum: efficient simulation might need only $\sim 10^8$ quantum steps: $10^8 \div 10^6 = 100 \text{ s} \approx 2 \text{ minutes}$. From impossible to routine.

Answer: quadratic speedups improve the possible; exponential speedups make the impossible possible. That is why chemistry simulation and factoring dominate serious quantum road-maps.

2.5 Hype vs. Reality: An Honest Look

Quantum computing attracts money, headlines — and exaggeration. As a student of this field, you will be asked by friends and colleagues: “Is it true quantum computers will replace normal computers / break all security next year / solve climate change?” This lesson equips you to answer like a professional.

The honest current picture (around 2026) in one paragraph: today’s machines have from tens to a few hundred physical qubits (with some devices beyond a thousand); every qubit and gate is noisy; without error correction, circuits fail after a few hundred to a few thousand operations; error-corrected *logical* qubits exist only in small numbers as research milestones; no quantum computer has yet factored a number that threatens real encryption; and the most credible near-term value is in research — simulating small quantum systems and developing algorithms for the fault-tolerant machines to come.

A claim-checking checklist for quantum news

Ask this question	Why it matters
Physical qubits or logical (error-corrected) qubits?	1000 noisy physical qubits are far less useful than 50 good logical ones (Chapter 29).
What was actually computed?	Some demonstrations solve artificial problems chosen to be hard classically but useless practically.
Verified by whom?	Advantage claims are often narrowed or matched by better classical algorithms within months.
Speedup type: exponential or quadratic?	Quadratic advantages can vanish once error-correction overhead is included.
Timeline: demonstrated, or projected?	Road-maps are honest hopes, not deliveries.

CASE STUDY — THE “QUANTUM SUPREMACY” DEBATE · GOOGLE AND IBM, USA, 2019

In October 2019, Google announced that its 53-qubit Sycamore processor performed a sampling task in 200 seconds that, it estimated, would take the best supercomputer 10,000 years. Within days, IBM argued that a smarter classical method could do it in about 2.5 days, and in the following years improved classical algorithms shrank the gap dramatically. Who was right? Both, in parts: Sycamore really did something remarkable, *and* the “10,000 years” framing overstated the classical difficulty, *and* the task itself had no practical use. This episode is the perfect training example for the checklist above — and a preview of how science actually progresses: by claim, challenge, and correction.

ACTIVITY (HOME) — BE A CLAIM CHECKER

1. Find one recent news headline about quantum computing (any source).
2. Run it through the five checklist questions from this lesson. Write your answers.
3. Give the article an honesty grade from A (careful and precise) to D (pure hype), with one sentence of justification.
4. Rewrite its headline so that it would earn an A, without becoming boring.

You need: An internet-connected phone or computer, Paper and pen

► App tool: *Myth or Math quiz* — twenty quantum claims to sort into “myth” and “real,” with explanations

2.6 The Double-Slit, With Numbers

The famous two-slit experiment is usually told with pictures; this book can afford to tell it with amplitudes, because you now know the rule everything else in quantum computing runs on: *when a result can happen several indistinguishable ways, add the amplitudes first, then square.*

WORKED EXAMPLE — TWO PATHS, ONE DETECTOR

A particle can reach a detector by two paths of equal strength. At the screen's bright spot both path amplitudes are $+\frac{1}{2}$; at the dark spot they are $+\frac{1}{2}$ and $-\frac{1}{2}$. Compute the arrival probabilities, and compare with classical expectations.

1. Bright spot, quantum: add first, $\frac{1}{2} + \frac{1}{2} = 1$; then square: $P = 1$.
2. Dark spot, quantum: $\frac{1}{2} - \frac{1}{2} = 0$; square: $P = 0$ — the particle *never* arrives, though either path alone would deliver it.
3. Classical control: block one slit at a time and add the results: $P = |\frac{1}{2}|^2 + |\frac{1}{2}|^2 = \frac{1}{2}$ at *both* spots — no bright, no dark.
4. Now watch the slits (measure the path): the two ways become distinguishable, amplitudes stop adding, and the classical $\frac{1}{2}$ returns. Observation changes the arithmetic.

Answer: quantum 1 and 0; classical $\frac{1}{2}$ and $\frac{1}{2}$. Every quantum algorithm in this book is this table, scaled up: arrange computations so wrong answers meet like the dark spot (+ meets −) and right answers like the bright one. Interference is not a metaphor — it is addition, before squaring.

REALITY CHECK

The two-slit experiment is not a historical curiosity: it runs today with electrons, atoms, and even molecules of two thousand atoms, and the fringes appear exactly as amplitude arithmetic predicts. Feynman called it “the only mystery.” By the end of Part Three you will compute such patterns for a living, and Chapter 6 builds the smallest laboratory version — two Hadamard gates — where you can check every number by hand.

CHAPTER SUMMARY

- Quantum mechanics (1900–1927) explains the atomic world; quantum computing (1981–) is the young project of computing with its rules.
- Three rules power everything: superposition (states combine, with amplitudes), measurement (one random answer; the state collapses), and entanglement (correlations no classical story can produce).
- A qubit is described by two amplitudes; n qubits by 2^n . Amplitudes can be negative, so computational paths can cancel — interference is the engine of quantum algorithms.
- Real promise: simulating quantum systems, factoring (Shor), quadratic search speedups (Grover), certified randomness. No promise: everyday computing.
- Today's machines are noisy and small; error correction is the frontier. Judge every claim with the checklist: physical vs logical qubits, task usefulness, verification, speedup type, demonstrated vs projected.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] Put these events in time order and attach the right decade to each: Shor’s algorithm, Planck’s quanta, quantum computers on the cloud, Feynman’s proposal.

Problem 2 [Basic] State the three quantum rules in your own words, one sentence each.

Problem 3 [Basic] How many amplitudes describe a 4-qubit state? List the first four bit patterns they belong to.

Problem 4 [Basic] Two computational paths lead to the same outcome with amplitudes $+0.4$ and -0.4 . What is the total amplitude, and what does it mean physically?

Problem 5 [Basic] Two indistinguishable paths carry amplitudes 0.6 and -0.6 . Find the arrival probability, and the probability if the paths are made distinguishable.

Problem 6 [Basic] Now try amplitudes 0.6 and 0.8 toward one detector: $|0.6 + 0.8|^2 = 1.96$ — more than certainty, which is impossible! Resolve the paradox: why can two indistinguishable paths to the *same* outcome never carry amplitudes this large, and what legal probability results if both amplitudes are scaled by $\frac{1}{\sqrt{2}}$? (Hint: all outcomes’ probabilities must total 1.)

Problem 7 [Standard] Using $2^{10} \approx 10^3$, estimate how many qubits are needed before the number of amplitudes exceeds 10^{21} (the rough number of grains of sand on Earth).

Problem 8 [Standard] A friend says: “A 300-qubit computer stores 2^{300} numbers, so I can use it as an infinite hard disk.” Find the error, using Rule 2 (measurement).

Problem 9 [Standard] Classify each task as “likely quantum advantage” or “no advantage,” with one reason: (a) simulating a battery molecule, (b) streaming video, (c) factoring a 2048-bit number, (d) sorting a customer list.

Problem 10 [Standard] A search problem has $N = 4 \times 10^6$ possibilities. How many Grover steps ($\approx \sqrt{N}$) would a quantum computer need? How many classical checks on average ($\approx N/2$)?

Problem 11 [Standard] Explain in two sentences why “tries all answers at once” is a misleading description of quantum computing, and give the correct one-line description involving interference.

Problem 12 [Standard] At some screen position the amplitudes are 0.5 and $0.5e^{i120^\circ}$. Using $|a + b|^2 = |a|^2 + |b|^2 + 2|a||b|\cos(\Delta\varphi)$, compute the probability, and evaluate the same expression at $\Delta\varphi = 0^\circ$ and 180° to recover the bright and dark extremes.

Problem 13 [Challenge] A press release claims: “Our new 2000-qubit machine will design new medicines next year.” Write a five-line professional assessment using the claim-checking checklist.

Problem 14 [Challenge] Suppose future error correction requires 1000 physical qubits per logical qubit, and a useful chemistry simulation needs 100 logical qubits. How many physical qubits are required? Compare with today’s largest machines (order of 10^3), and state the conclusion in one sentence.

Problem 15 [Challenge] In the 2019 supremacy debate, Google said 10,000 years, IBM said 2.5 days. Take the ratio. If classical methods later improved by another factor of 100, what classical time results? What lesson does this teach about “quantum advantage” claims made against the *current best* classical algorithm?

Problem 16 [Challenge] A “which-path” detector of efficiency 50% tags half the particles (their amplitudes no longer interfere) and misses half (theirs do). For the dark-spot amplitudes $\pm \frac{1}{2}$, compute the arrival probability as a weighted mixture, and state the moral connecting measurement strength to fringe visibility.

PART TWO

THE QUBIT

CHAPTER THREE

Qubits and Quantum States

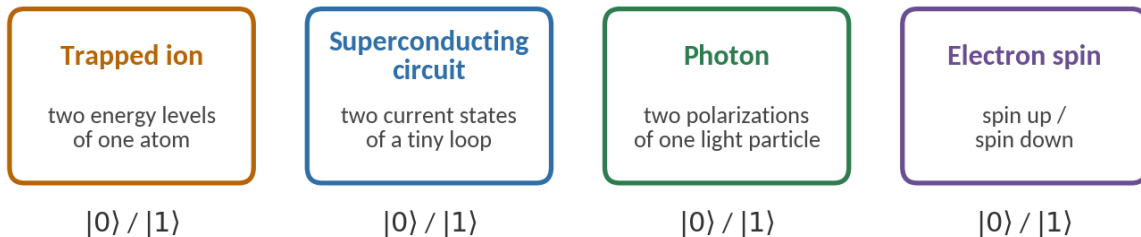
Dirac notation, amplitudes, and the rule every legal state obeys

IN THIS CHAPTER YOU WILL LEARN TO

- Describe what a qubit is physically, and name four real technologies that hold one.
- Read and write Dirac notation: $|0\rangle$, $|1\rangle$, $|\psi\rangle$, and $\langle\psi|$.
- Write any single-qubit state both as $\alpha|0\rangle + \beta|1\rangle$ and as a column vector.
- Check normalization ($|\alpha|^2 + |\beta|^2 = 1$) and normalize a state that fails it.
- Recognize the famous states $|+\rangle$, $|-\rangle$, and $|+i\rangle$, and compute with them.

3.1 From Bit to Qubit

A classical bit needs a physical object with two stable states (Chapter 1). A qubit needs more: a physical object with two states that can also hold *superpositions* of those states, undisturbed, long enough to compute. That is a far harder engineering demand — the object must be small enough for quantum rules to govern it and isolated enough for the outside world not to “measure” it by accident (Chapter 28 tells that struggle’s story).



Four real homes for a qubit. In each, two particular quantum states are chosen and labeled $|0\rangle$ and $|1\rangle$.

Whatever the hardware — an ion’s energy levels in a lab, a superconducting loop in a dilution refrigerator, a

photon’s polarization flying through fiber — the *mathematics* is identical. That is the power move of this book: master the 2-component arithmetic of one abstract qubit, and you have mastered every qubit technology on Earth.

WORD TO KNOW

Computational basis — The two agreed reference states of a qubit, written $|0\rangle$ and $|1\rangle$. They are the quantum versions of “bit = 0” and “bit = 1,” and they are orthogonal: $\langle 0|1\rangle = 0$.

3.2 Dirac Notation: Kets and Bras

Quantum mechanics has its own notation, invented by Paul Dirac, and at first sight it looks alien: $|0\rangle$, $|\psi\rangle$, $\langle\phi|$. Relax — it is nothing but a tidy costume for the vectors of Appendix B.

WORD TO KNOW

Ket — The symbol $|\rangle$ wrapped around a label, denoting a state vector. $|0\rangle$ is “the state labeled 0”; $|\psi\rangle$ is “the state labeled ψ ” (psi, a favorite Greek letter for states). Every ket IS a column vector.

EQUATION: THE BASIS KETS AS COLUMN VECTORS

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

In words: ket 0 = the column (1, 0); ket 1 = the column (0, 1)

Symbol	Meaning	Unit
$ 0\rangle$	the basis state “zero”: first component 1, second 0	none — a vector
$ 1\rangle$	the basis state “one”: first component 0, second 1	none — a vector

Careful: $|0\rangle$ is NOT the zero vector $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ — it is a perfectly healthy length-1 vector whose label is 0. The label names a state; it is not arithmetic.

Why bother with a new notation when columns worked fine? Three practical reasons. First, labels scale: for three qubits, $|101\rangle$ is instantly readable, while its 8-component column is not. Second, the notation separates *what the state means* (the label) from *how we compute* (the column). Third, it gives us a matching symbol for row vectors, which quantum mechanics uses constantly:

WORD TO KNOW

Bra — The symbol $\langle\psi|$: the dagger of the ket $|\psi\rangle$ — a row vector with conjugated entries. Bra and ket together make a “bra-ket” (bracket): $\langle\phi|\psi\rangle$ is exactly the inner product of Appendix B.

WORKED EXAMPLE — FROM KET TO BRA

Write the bra of $|\psi\rangle = \begin{pmatrix} 0.6 \\ 0.8i \end{pmatrix}$, and compute $\langle\psi|\psi\rangle$.

1. Dagger the column: transpose to a row and conjugate each entry.
2. $\langle\psi| = \begin{pmatrix} 0.6 & -0.8i \end{pmatrix}$ (the $0.8i$ became $-0.8i$).
3. Inner product: $\langle\psi|\psi\rangle = 0.6 \times 0.6 + (-0.8i)(0.8i) = 0.36 - 0.64i^2 = 0.36 + 0.64 = 1$.

Answer: $\langle\psi| = (0.6, -0.8i)$ and $\langle\psi|\psi\rangle = 1$ — this state has length 1, which the last lesson will reveal as the mark of every legal state.

3.3 The State Vector and Amplitudes

Here is the central definition of the entire book.

EQUATION: THE GENERAL STATE OF ONE QUBIT

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

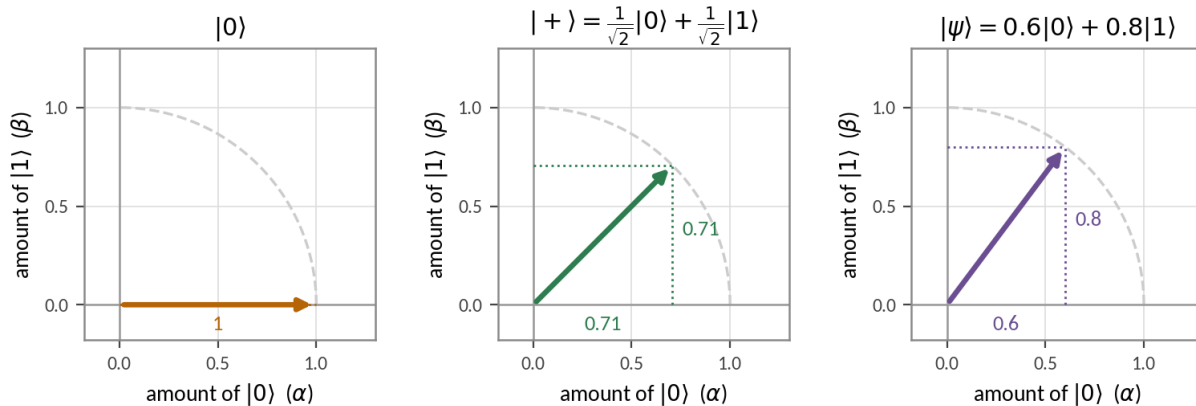
In words: Any qubit state = (amplitude α) $\times$ ket 0 + (amplitude β) $\times$ ket 1

Symbol	Meaning	Unit
$ \psi\rangle$	the qubit's state vector	none — a vector
α	the amplitude of $ 0\rangle$: a complex number ("how much $ 0\rangle$, with what phase")	none
β	the amplitude of $ 1\rangle$	none

This is vector scaling-and-adding from Appendix B, nothing more: $\alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$.

WORD TO KNOW

Amplitude — The complex number attached to each basis state in a superposition. Amplitudes are the working currency of quantum computing: gates transform them, interference adds and cancels them, and their squared moduli become measurement probabilities (Chapter 4).



Three states drawn as arrows in the $|0\rangle$ - $|1\rangle$ plane (possible when amplitudes are real). The components along each axis are the amplitudes. All three arrows have length 1.

Read a state like a recipe. The state $|\psi\rangle = 0.6|0\rangle + 0.8|1\rangle$ says: “mix 0.6 parts of $|0\rangle$ with 0.8 parts of $|1\rangle$.” The state $|\psi\rangle = |1\rangle$ says: “all $|1\rangle$, no $|0\rangle$ ” (amplitudes $\alpha = 0, \beta = 1$). And amplitudes may be complex: $|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle$ is a perfectly real physical state whose second amplitude carries a 90° phase — Chapter 5 will show it pointing along the y -axis of the Bloch sphere.

COMMON MISTAKE

✗ Wrong: “ α is the probability of getting 0.”

✓ Right: α is the *amplitude*; the probability is $|\alpha|^2$ (Chapter 4). The difference is not pedantry — amplitudes can be negative or complex and can cancel each other; probabilities cannot. Confusing the two makes interference impossible to understand.

3.4 Normalization: The Rule Every State Obeys

Measurement (Chapter 4) will turn amplitudes into probabilities via $|\alpha|^2$ and $|\beta|^2$. Probabilities of all outcomes must total exactly 1 — something must happen. This forces a law on every legal state vector:

EQUATION: THE NORMALIZATION CONDITION

$$|\alpha|^2 + |\beta|^2 = 1$$

In words: The squared sizes of the amplitudes must add to exactly 1

Symbol	Meaning	Unit
$ \alpha ^2$	squared modulus of the $ 0\rangle$ amplitude (use $a^2 + b^2$ from Appendix A if complex)	none
$ \beta ^2$	squared modulus of the $ 1\rangle$ amplitude	none

Geometrically: the state vector has length 1 — it lives on the unit circle (real amplitudes) or unit sphere (complex). This is exactly why gates must be unitary (Appendix B): they must keep states on the unit sphere.

WORKED EXAMPLE — CHECKING A STATE'S PAPERS

Is $|\psi\rangle = 0.6|0\rangle + 0.8|1\rangle$ a legal state? Is $|\phi\rangle = 0.5|0\rangle + 0.5|1\rangle$?

1. For $|\psi\rangle$: $|0.6|^2 + |0.8|^2 = 0.36 + 0.64 = 1.00$ ✓ — legal.
2. For $|\phi\rangle$: $|0.5|^2 + |0.5|^2 = 0.25 + 0.25 = 0.50 \neq 1$ — not legal as written.
3. Diagnosis: $|\phi\rangle$'s vector has length $\sqrt{0.5} \approx 0.707$, too short by that factor.

Answer: $|\psi\rangle$ is legal; $|\phi\rangle$ is not — but it can be repaired, as the next example shows.

WORKED EXAMPLE — NORMALIZING A STATE

Repair $|\phi\rangle = 0.5|0\rangle + 0.5|1\rangle$ by scaling it to length 1.

1. Find the current length: $L = \sqrt{|0.5|^2 + |0.5|^2} = \sqrt{0.5} = \frac{1}{\sqrt{2}} \approx 0.7071$.
2. Divide both amplitudes by L : new amplitudes $= 0.5 \div \frac{1}{\sqrt{2}} = 0.5\sqrt{2} = \frac{1}{\sqrt{2}} \approx 0.7071$.
3. Check: $\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2} + \frac{1}{2} = 1$ ✓

Answer: the normalized state is $\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$ — the equal superposition, star of the next lesson.

REALITY CHECK

Where do those strange $\frac{1}{\sqrt{2}}$ factors everywhere in quantum computing come from? Only from normalization. “Equal parts of two states” cannot use amplitudes $\frac{1}{2}$ and $\frac{1}{2}$ (squares total $\frac{1}{2}$) or 1 and 1 (squares total 2); equality plus normalization forces $\frac{1}{\sqrt{2}} \approx 0.707$ each. Whenever you see $\frac{1}{\sqrt{2}}$, read it as “equal shares, legally normalized” — not as a mysterious constant.

3.5 Your First Quantum States

A handful of specific states appear on nearly every page from here to the end of the book. Meet them now, learn them slowly, and the chapters ahead will feel like meeting old friends.

The single-qubit states you will meet everywhere

Name	Amplitude form	Column form	First impression
$ 0\rangle$	$1 \cdot 0\rangle + 0 \cdot 1\rangle$	$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$	“definitely 0”
$ 1\rangle$	$0 \cdot 0\rangle + 1 \cdot 1\rangle$	$\begin{pmatrix} 0 \\ 1 \end{pmatrix}$	“definitely 1”
$ +\rangle$	$\frac{1}{\sqrt{2}} 0\rangle + \frac{1}{\sqrt{2}} 1\rangle$	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	equal mix, phases aligned
$ -\rangle$	$\frac{1}{\sqrt{2}} 0\rangle - \frac{1}{\sqrt{2}} 1\rangle$	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$	equal mix, opposite phases
$ +i\rangle$	$\frac{1}{\sqrt{2}} 0\rangle + \frac{i}{\sqrt{2}} 1\rangle$	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix}$	equal mix, 90° phase

Notice what separates $|+\rangle$, $|-\rangle$, and $|+i\rangle$: nothing but phase. All three have $|\alpha|^2 = |\beta|^2 = \frac{1}{2}$, so all three answer a standard measurement with 50/50 randomness (Chapter 4) — yet they are physically different states,

distinguishable by other measurements, and gates transform them differently. *Phase is invisible to one measurement but not to physics.* Keeping that sentence in mind will carry you gracefully through Chapters 5–7.

WORKED EXAMPLE — VERIFYING $|-\rangle$ IS LEGAL AND ORTHOGONAL TO $|+\rangle$

Check that $|-\rangle$ is normalized, and compute $\langle + | - \rangle$.

1. Normalization: $\left|\frac{1}{\sqrt{2}}\right|^2 + \left|-\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2} + \frac{1}{2} = 1$ ✓ (a negative amplitude squares positive).

2. Bra of $|+\rangle$: all entries real, so $\langle + | = \frac{1}{\sqrt{2}}(1, 1)$.

3. Inner product: $\langle + | - \rangle = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \cdot \left(-\frac{1}{\sqrt{2}}\right) = \frac{1}{2} - \frac{1}{2} = 0$.

Answer: $|-\rangle$ is normalized, and $\langle + | - \rangle = 0$ — the pair $|+\rangle, |-\rangle$ is orthogonal, forming a second basis just as good as $|0\rangle, |1\rangle$. Chapter 4 measures in it.

CASE STUDY — THE FIRST TWO-LEVEL SYSTEMS PUT TO WORK • NIST, USA, 1995

The first controlled quantum logic on real qubits was demonstrated in 1995 at NIST in Boulder, USA, by Christopher Monroe, David Wineland, and colleagues, using a single beryllium ion held in an electromagnetic trap. Two energy levels of the ion served as $|0\rangle$ and $|1\rangle$; laser pulses steered its amplitudes with exquisite precision. Wineland shared the 2012 Nobel Prize in Physics for this line of work. Every state in this chapter — superpositions, phases, normalization — was realized in that laboratory on one single atom, decades before “quantum computing” became a headline word. Trapped ions remain among the highest-quality qubits in the world today (Chapter 30).

ACTIVITY (APP) — A GALLERY OF STATES

1. Open the *State Builder* in the QCFE app.
2. Type amplitudes $\alpha = 0.6, \beta = 0.8$. Confirm the app accepts the state (normalization bar shows exactly 1.00).
3. Try $\alpha = 0.5, \beta = 0.5$. Watch the normalization bar show 0.50 and use the “Normalize” button; compare the repaired amplitudes with the worked example.
4. Build $|+\rangle, |-\rangle$, and $|+i\rangle$ from the preset menu, and note how the phase dial (not the probability bars) is the only thing that changes.

You need: The QCFE app (Android or Windows)

► *App tool: State Builder — enter amplitudes, see the column vector, the normalization check, and the state drawn live*

3.6 How Close Are Two States? Overlap as a Ruler

One more tool completes the chapter’s kit — and quietly powers half the book ahead. Given two states, quantum mechanics has a standard ruler for “how alike are they?": the inner product of Appendix B, applied

to state vectors.

WORD TO KNOW

Fidelity (overlap) — The squared inner product $F = |\langle \phi | \psi \rangle|^2$ between two states: 1 for identical states, 0 for orthogonal ones, in between otherwise. The universal closeness score of quantum information — used to grade hardware gates, compare noisy states to ideals, and (Part Six) measure similarity between data points.

EQUATION: THE OVERLAP RULER

$$F(\phi, \psi) = |\langle \phi | \psi \rangle|^2$$

In words: Closeness of two states = squared modulus of their inner product

Symbol	Meaning	Unit
F	the fidelity: 1 = same state, 0 = perfectly distinguishable	none — 0 to 1
$\langle \phi \psi \rangle$	the inner product: conjugate the first state's amplitudes, multiply pairwise, add	none

Two meanings in one number: geometrically, how aligned the vectors are; physically (Chapter 4 preview), the probability that a $|\phi\rangle$ -versus-orthogonal measurement mistakes $|\psi\rangle$ for $|\phi\rangle$. Orthogonal states — $F = 0$ — are the only perfectly distinguishable pairs.

WORKED EXAMPLE — GRADING CLOSNESS ACROSS THE GALLERY

Compute the fidelity of $|\psi\rangle = 0.6|0\rangle + 0.8|1\rangle$ with each of $|0\rangle$, $|+\rangle$, and $|-\rangle$.

1. With $|0\rangle$: $\langle 0 | \psi \rangle = 0.6$, so $F = 0.36$ — mild resemblance.
2. With $|+\rangle$: $\langle + | \psi \rangle = \frac{0.6+0.8}{\sqrt{2}} = \frac{1.4}{\sqrt{2}} \approx 0.99$, so $F \approx 0.98$ — nearly the same state.
3. With $|-\rangle$: $\langle - | \psi \rangle = \frac{0.6-0.8}{\sqrt{2}} \approx -0.141$, so $F = 0.02$ — nearly orthogonal.

Answer: $F = 0.36, 0.98, 0.02$. The state is almost $|+\rangle$ and almost-not $|-\rangle$ — numbers you could already have guessed from the arrows figure, now made exact. When Chapter 28 reports “gate fidelity 99.5%,” it is exactly this ruler comparing what the gate made with what it should have made.

3.7 Perfect Distinguishability: Orthogonal States

Fidelity (previous lesson) measures how *alike* two states are; its zero point deserves its own spotlight. Two states with $\langle \phi | \psi \rangle = 0$ are called **orthogonal** — and they are the only pairs a single measurement can tell apart with certainty. Bits hide inside qubits precisely as orthogonal pairs: $|0\rangle / |1\rangle$, or equally well $|+\rangle / |-\rangle$.

EQUATION: THE ORTHOGONAL PARTNER

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad \implies \quad |\psi^\perp\rangle = \beta^*|0\rangle - \alpha^*|1\rangle$$

In words: Swap the amplitudes, conjugate them, flip one sign — the result is the unique state your state can never be mistaken for

Symbol	Meaning	Unit
$ \psi^\perp\rangle$	the orthogonal partner of $ \psi\rangle$	none
α^*, β^*	complex conjugates	none

Check: $\langle\psi^\perp|\psi\rangle = \beta\alpha - \alpha\beta = 0$ (real amplitudes shown; the conjugates make it work in general). On the Bloch sphere of Chapter 5, partners sit at antipodes — north pole and south pole of some axis.

WORKED EXAMPLE — BUILDING AND USING A PARTNER

Find the orthogonal partner of $|\psi\rangle = 0.6|0\rangle + 0.8|1\rangle$, verify it, and state what measurement distinguishes the pair perfectly.

1. Apply the recipe (real amplitudes): $|\psi^\perp\rangle = 0.8|0\rangle - 0.6|1\rangle$.
2. Verify orthogonality: $\langle\psi^\perp|\psi\rangle = 0.8 \times 0.6 + (-0.6) \times 0.8 = 0.48 - 0.48 = 0$ ✓
3. Verify it is a legal state: $0.64 + 0.36 = 1$ ✓
4. The distinguishing measurement: measure *in the basis* $\{|\psi\rangle, |\psi^\perp\rangle\}$ (Chapter 4 shows how any basis becomes a measurement): input $|\psi\rangle$ answers “first” with probability 1, input $|\psi^\perp\rangle$ answers “second” with probability 1.

Answer: $0.8|0\rangle - 0.6|1\rangle$. **Orthogonal pairs are the only perfectly readable alphabets — which is why non-orthogonal pairs (fidelity strictly between 0 and 1) become security features in Chapter 15: an eavesdropper cannot read what physics itself cannot distinguish.**

REALITY CHECK

This is the deep reason a qubit still carries only one readable bit despite its infinite dial settings: any measurement’s outcomes correspond to orthogonal states, and a two-dimensional space fits only two mutually orthogonal directions. Infinite states to *be*; two states to *be told apart*. Holevo’s theorem (References) is this observation grown up.

CHAPTER SUMMARY

- A qubit is any two-level quantum system; ions, superconducting circuits, photons, and spins all work, and all obey the same 2-component mathematics.
- Dirac notation: kets $|\psi\rangle$ are column vectors ($|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$); bras $\langle\psi|$ are daggered kets; $\langle\phi|\psi\rangle$ is the inner product.
- The general state is $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$, with complex amplitudes α, β .
- Every legal state is normalized: $|\alpha|^2 + |\beta|^2 = 1$. To repair a state, divide by its length. The $\frac{1}{\sqrt{2}}$ factors of

quantum computing are normalization at work.

- Know the gallery: $|0\rangle, |1\rangle, |+\rangle, |-\rangle, |+i\rangle$. The last three differ only in phase — same measurement probabilities, different physics.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] Write $|0\rangle$ and $|1\rangle$ as column vectors, and write the column vector of $|\psi\rangle = 0.28|0\rangle + 0.96|1\rangle$.

Problem 2 [Basic] Check whether $|\psi\rangle = 0.28|0\rangle + 0.96|1\rangle$ is normalized.

Problem 3 [Basic] Write the bra $\langle\psi|$ for $|\psi\rangle = \begin{pmatrix} 0.8 \\ 0.6 \end{pmatrix}$, and compute $\langle\psi|\psi\rangle$.

Problem 4 [Basic] Which of these are legal states? (a) $|a\rangle = |1\rangle$, (b) $|b\rangle = 0.9|0\rangle + 0.9|1\rangle$, (c) $|c\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$.

Problem 5 [Basic] Find the orthogonal partners of $|0\rangle$, of $|+\rangle$, and of $\frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$, and verify one of them by inner product.

Problem 6 [Basic] Where do a state and its partner sit on the Bloch sphere, relative to each other? Use $|0\rangle / |1\rangle$ and $|+\rangle / |-\rangle$ as checks.

Problem 7 [Standard] Normalize $|\phi\rangle = 3|0\rangle + 4|1\rangle$, and write the result with clean fractions.

Problem 8 [Standard] Compute $\langle 0|+\rangle$ and $\langle 1|+\rangle$. What do you notice about their values, and (Chapter 4 preview) what might their squares mean?

Problem 9 [Standard] For $|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle$: write the column vector, the bra, and verify normalization (remember $|i/\sqrt{2}|^2 = 1/2$).

Problem 10 [Standard] Show that $\langle +i | -i \rangle = 0$, where $|-i\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{i}{\sqrt{2}}|1\rangle$. (Careful with conjugates in the bra.)

Problem 11 [Standard] A student writes “ $|\psi\rangle = 0.7|0\rangle + 0.7|1\rangle$, so the qubit is 70% zero and 70% one.” Find both errors in this sentence.

Problem 12 [Standard] Compute the fidelity between $|+\rangle$ and $|+i\rangle$ (careful with the conjugate in the bra), and place the answer between the landmark values 0, $\frac{1}{2}$, and 1.

Problem 13 [Standard] Show that fidelity is symmetric — $|\langle\phi|\psi\rangle|^2 = |\langle\psi|\phi\rangle|^2$ — using the conjugate rule $(\langle u|v\rangle)^* = \langle v|u\rangle$.

Problem 14 [Standard] For the complex state $|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle$, apply the recipe with conjugation and verify $\langle\psi^\perp|\psi\rangle = 0$ carefully (this is where the conjugates earn their keep).

Problem 15 [Challenge] Find all real positive values a, b such that $|\psi\rangle = a|0\rangle + b|1\rangle$ is normalized and $b = 2a$. Then write the state.

Problem 16 [Challenge] The state $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{c}{2}|1\rangle$ is normalized, where c is a complex number. What is $|c|$? Give three different valid choices of c and explain in one sentence why they describe physically different states with identical measurement statistics in the standard basis.

Problem 17 [Challenge] Global phase: consider $|\psi\rangle$ and $|\psi'\rangle = e^{i\theta}|\psi\rangle$ (every amplitude multiplied by the same unit phase). Show $\langle\psi'|\psi'\rangle = \langle\psi|\psi\rangle$, and argue that no inner product with any bra can distinguish the squared moduli produced. (Full story in Chapter 5: global phase is unobservable; relative phase is everything.)

Problem 18 [Challenge] A noisy device aims to prepare $|+\rangle$ but delivers $|\psi\rangle = 0.74|0\rangle + 0.67|1\rangle$. Compute the preparation fidelity $|\langle+|\psi\rangle|^2$, and decide whether the device beats a “99% fidelity” specification.

Problem 19 [Challenge] Find all states with fidelity exactly $\frac{1}{2}$ to $|0\rangle$, describing the set in Bloch-sphere language (which latitude?). What familiar gallery states belong to it?

Problem 20 [Challenge] Alice sends one of $|0\rangle$ or $|+\rangle$ (not orthogonal: fidelity $\frac{1}{2}$). Show no single measurement identifies the state perfectly: for the standard-basis measurement, compute each input’s outcome distribution and exhibit the overlap; argue (using the partner construction) why every other basis fails somewhere too. What is the best single-shot success probability from the fidelity lesson — and why does Chapter 15 celebrate this failure?

CHAPTER FOUR

Superposition and Measurement

The Born rule, collapse, shots, and histograms — what happens when we look

IN THIS CHAPTER YOU WILL LEARN TO

- Explain superposition in plain, myth-free language.
- Use the Born rule to turn any state's amplitudes into measurement probabilities.
- Describe collapse: what the state is after a measurement.
- Predict measurement statistics for repeated runs (shots), and read a measurement histogram like an experimentalist.
- Compute probabilities for a measurement in the $|+\rangle / |-\rangle$ basis, and explain why phase becomes visible there.

4.1 Superposition in Plain Language

Chapter 3 gave you the mathematics of a state: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. This chapter asks the physical question: *what do we actually see* when we look at such a qubit?

First, let us say precisely what superposition is *not*. It is not “the qubit is both 0 and 1” (you never see both — a measurement always returns one digit). It is not “the qubit is either 0 or 1 and we just don’t know which” (that is classical ignorance; Bell tests, Chapter 11, prove nature does not work this way). Superposition is a third thing with no everyday analogue: a *definite* state, described exactly by two amplitudes, that produces *random* results when measured — with probabilities the amplitudes decide.

An honest picture: the state vector is like a perfectly definite recipe held by nature; measurement is an oven that can only output one of two dishes. Same recipe, same oven, honest randomness in which dish appears — but exact law in the long-run proportions.

WORD TO KNOW

Shot — One complete run of a quantum experiment: prepare the state, apply the circuit, measure once. Real quantum programs run thousands of shots and report the counts of each result.

4.2 The Born Rule: From Amplitudes to Probabilities

The bridge from amplitudes to observed randomness is one short equation — discovered by Max Born in 1926, and worth a Nobel Prize.

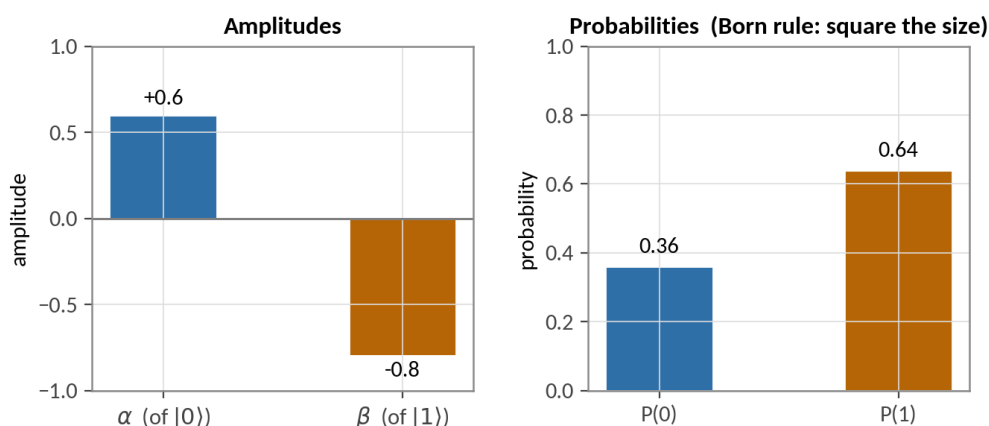
EQUATION: THE BORN RULE

$$P(0) = |\alpha|^2 \quad P(1) = |\beta|^2$$

In words: Probability of a result = squared modulus of its amplitude

Symbol	Meaning	Unit
$P(0), P(1)$	probabilities of measuring 0 and 1	none — numbers from 0 to 1
α, β	the amplitudes of $ 0\rangle$ and $ 1\rangle$ in the state	none — complex numbers
$ \cdot ^2$	squared modulus: $ a + bi ^2 = a^2 + b^2$ (Appendix A)	none

Normalization (Chapter 3) is exactly the guarantee that $P(0) + P(1) = 1$: something always happens.



From amplitudes to probabilities. Amplitudes may be negative (or complex); squaring their sizes produces honest probabilities. Note how $\beta = -0.8$ still gives $P(1) = 0.64$.

WORKED EXAMPLE — THE BORN RULE ON A GENERAL STATE

A qubit is in $|\psi\rangle = 0.6|0\rangle - 0.8|1\rangle$. Find the measurement probabilities.

1. Identify amplitudes: $\alpha = 0.6$, $\beta = -0.8$.
2. Born rule for 0: $P(0) = |0.6|^2 = 0.36$.
3. Born rule for 1: $P(1) = |-0.8|^2 = 0.64$ (the minus sign disappears in the squared modulus).
4. Check: $0.36 + 0.64 = 1.00$ ✓

Answer: $P(0) = 0.36$ (36%), $P(1) = 0.64$ (64%). The sign of β changed nothing here — but keep Chapter 2 in mind: in interference, before measurement, that sign can decide everything.

WORKED EXAMPLE — THE BORN RULE WITH A COMPLEX AMPLITUDE

Find the probabilities for $|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle$ (the state $|+i\rangle$ from Chapter 3).

1. $P(0) = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$.

2. For $\beta = \frac{i}{\sqrt{2}} = 0 + \frac{1}{\sqrt{2}}i$: $|\beta|^2 = 0^2 + \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}$.

3. Check: $\frac{1}{2} + \frac{1}{2} = 1$ ✓

Answer: 50% / 50% — identical statistics to $|+\rangle$ and $|-\rangle$ in this measurement. The phase i is invisible here; Lesson 5 shows the measurement that reveals it.

COMMON MISTAKE

X Wrong: “Probabilities can be added directly for amplitudes: if two paths lead to $|0\rangle$ with amplitudes 0.5 and -0.5 , the probability is $0.25 + 0.25 = 0.5$.”

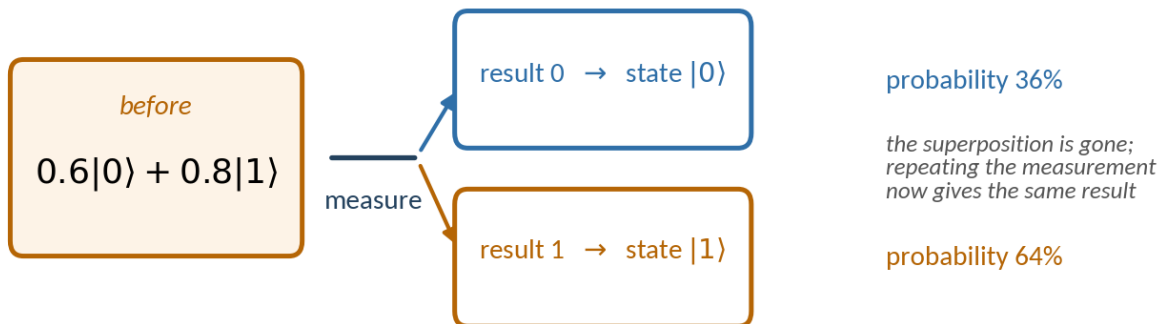
✓ Right: Add *amplitudes first*, then square: total amplitude = $0.5 + (-0.5) = 0$, so the probability is $|0|^2 = 0$. Amplitudes add, probabilities do not. This single ordering — add, then square — is the whole engine of quantum interference.

4.3 Measurement and Collapse

The Born rule tells you the odds. The second half of the measurement story tells you the aftermath: measurement *changes the state*.

WORD TO KNOW

Collapse — The change of state caused by measurement: after result 0, the qubit’s state is $|0\rangle$; after result 1, it is $|1\rangle$. The pre-measurement amplitudes are gone and cannot be recovered from the measured qubit.



Measurement in one picture: random result, probabilities from the Born rule, and collapse to the matching basis state. Measure the same qubit again immediately, and it repeats its answer with certainty.

Collapse has practical consequences that shape all of quantum computing. You cannot “peek” at a quantum computation midway without destroying the superposition it is building. You cannot read out the amplitudes α and β of one qubit — one shot yields one bit, and the recipe collapses. To learn the probabilities, you must prepare the *same* state many times and measure each copy fresh — which is exactly why quantum programs run in shots.

REALITY CHECK

A single measurement result tells you almost nothing. Getting “0” is consistent with $|0\rangle$, with $|+\rangle$, with $0.6|0\rangle + 0.8|1\rangle$ — with any state whose $\alpha \neq 0$. Quantum information is learned the way a casino learns its edge: statistically, over many trials. When someone says “the qubit was in superposition and we measured it,” always ask the professional’s question: *how many shots?*

4.4 Shots and Histograms: Reading Quantum Statistics

Run the same experiment N times and count the results: that is the daily work of quantum computing, and its standard picture is the **measurement histogram** — one bar per observed result, bar height = count.

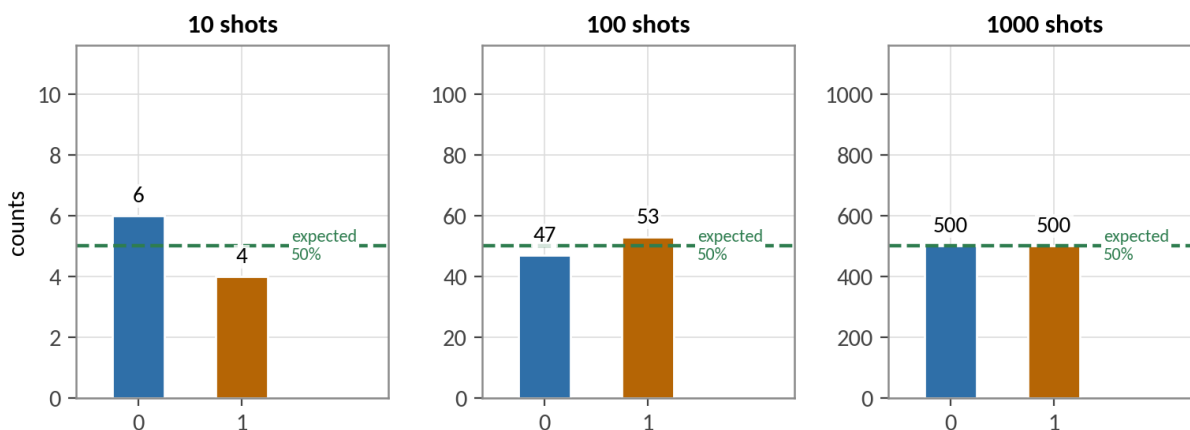
EQUATION: EXPECTED COUNTS FROM PROBABILITIES

$$\text{expected count of result } x = N_{\text{shots}} \times P(x)$$

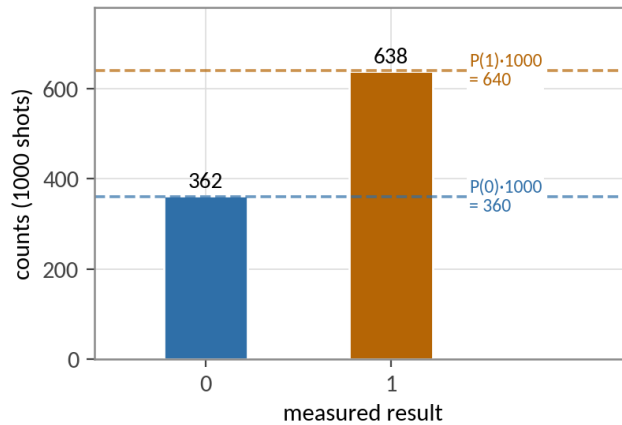
In words: Expected count = number of shots $\times$ probability of that result

Symbol	Meaning	Unit
N_{shots}	how many times the experiment is run	shots
$P(x)$	the Born-rule probability of result x	none

Real counts fluctuate around the expectation — more shots, smaller relative fluctuation. A useful rule of thumb: random wobble is about $\sqrt{N}$ counts.



The same 50/50 qubit measured 10, 100, and 1000 times (simulated). Small samples look lopsided; large samples settle toward the true probabilities. Never judge a qubit by ten shots.



A 1000-shot histogram for $|\psi\rangle = 0.6|0\rangle + 0.8|1\rangle$: counts 362 and 638 against expectations 360 and 640 — agreement well within the $\pm\sqrt{N} \approx \pm 32$ wobble.

WORKED EXAMPLE — PREDICTING A HISTOGRAM

The state $0.6|0\rangle - 0.8|1\rangle$ is measured with 500 shots. Predict the histogram, with a rough error bar.

1. Probabilities (Lesson 2): $P(0) = 0.36$, $P(1) = 0.64$.
2. Expected counts: $500 \times 0.36 = 180$ zeros and $500 \times 0.64 = 320$ ones.
3. Typical wobble: $\sqrt{500} \approx 22$ counts, so seeing anywhere around 180 ± 22 zeros is unremarkable.

Answer: about 180 zeros and 320 ones, give or take roughly 20 counts. If you saw 260 zeros, you should suspect the device or the state — that is 4 wobbles away.

WORKED EXAMPLE — READING A HISTOGRAM BACKWARD

An unknown qubit state gives 810 zeros in 1000 shots. Estimate $|\alpha|$ and propose a state consistent with the data.

1. Estimate the probability: $P(0) \approx 810/1000 = 0.81$.
2. Born rule backward: $|\alpha| = \sqrt{0.81} = 0.9$.
3. Then $|\beta| = \sqrt{1 - 0.81} = \sqrt{0.19} \approx 0.436$.

Answer: $|\alpha| = 0.9$, so one consistent state is $0.9|0\rangle + 0.436|1\rangle$. Note what the data can NOT tell you: the phases — $0.9|0\rangle - 0.436|1\rangle$ and $0.9|0\rangle + 0.436i|1\rangle$ fit equally well. Histograms in one basis never reveal phase.

ACTIVITY (APP) — FLIP A QUANTUM COIN 1000 TIMES

1. Open the *Measurement Lab* in the QCFE app and prepare $|+\rangle$ with the H gate.
2. Run 10 shots. Write the counts. Repeat the 10-shot run three times — notice how different the three small histograms look.

3. Now run 100 shots, then 1000. Watch the histogram settle toward 50/50.

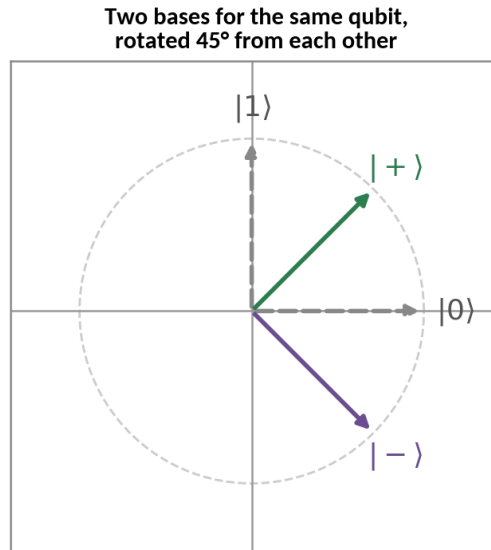
4. Change the state to $0.6|0\rangle + 0.8|1\rangle$ (amplitude sliders) and predict the 1000-shot histogram *before* running. Compare prediction and result.

You need: The QCFE app (Android or Windows), Paper and pen

► *App tool: Measurement Lab — prepare a state, choose the shots, and watch the histogram build up live, one shot at a time*

4.5 Measuring in the $|+\rangle / |-\rangle$ Basis

Standard measurement asks the qubit: “are you $|0\rangle$ or $|1\rangle$?” But that is a choice, not a law. Hardware can equally ask: “are you $|+\rangle$ or $|-\rangle$?” — a measurement in a rotated basis. (In practice, as Chapter 6 shows, one applies an H gate and then measures normally — same thing.)



The basis pair $|+\rangle, |-\rangle$ is the pair $|0\rangle, |1\rangle$ rotated by 45°. Both are legitimate orthogonal bases; a measurement can ask either question.

The Born rule generalizes with inner products (Appendix B): the amplitude for outcome $|\phi\rangle$ is the overlap $\langle\phi|\psi\rangle$.

EQUATION: THE BORN RULE IN ANY BASIS

$$P(\phi) = |\langle\phi|\psi\rangle|^2$$

In words: Probability of outcome ϕ = squared modulus of the overlap between the outcome state and the qubit's state

Symbol	Meaning	Unit
$P(\phi)$	probability that the measurement answers “ ϕ ”	none
$\langle\phi \psi\rangle$	the inner product (overlap) of $ \phi\rangle$ with $ \psi\rangle$	none — a complex number

The standard rule is the special case $\phi = 0, 1$: $\langle 0|\psi\rangle = \alpha$ and $\langle 1|\psi\rangle = \beta$. After the measurement, the state collapses to the outcome state $|\phi\rangle$.

WORKED EXAMPLE — PHASE BECOMES VISIBLE

Measure the state $|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$ in the $|+\rangle / |-\rangle$ basis. Find both probabilities.

1. Overlap with $|+\rangle$: $\langle +|-\rangle = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \cdot \left(-\frac{1}{\sqrt{2}}\right) = \frac{1}{2} - \frac{1}{2} = 0$.
2. So $P(+)=0$.
3. Overlap with $|-\rangle$: $\langle -|-\rangle = 1$ (a normalized state’s overlap with itself), so $P(-)=1$.

Answer: $P(+)=0$, $P(-)=1$ — certainty! In the standard basis, $|-\rangle$ looked like a fair coin; in its own basis it is completely predictable. The relative phase (the minus sign) was always physical — you just needed the right question to see it.

CASE STUDY — RANDOMNESS YOU CAN CERTIFY • NIST, USA, 2018

Classical computers cannot make true randomness: their “random” numbers come from formulas, and anyone knowing the formula and seed can predict them. Quantum measurement offers randomness guaranteed by physical law — and in 2018, NIST researchers went one step further, generating random bits from measurements on entangled photons arranged so that (via a Bell test, Chapter 11) any predictability would have produced impossible correlations. The result was randomness certified by physics itself, now feeding standards work and randomness-beacon services. Your Chapter 31 capstone builds a miniature version: a quantum coin whose fairness you verify with histograms exactly like this chapter’s.

4.6 Measuring Again: Sequences and the Quantum Zeno Effect

Collapse (Lesson 3) is not the end of a qubit’s story — it is a new beginning: the collapsed state is a perfectly good input for further gates and further measurements. Chaining measurements in *different* bases produces one of quantum mechanics’ most instructive surprises.

WORKED EXAMPLE — A MEASUREMENT SANDWICH: Z, THEN X, THEN Z

A qubit starts in $|0\rangle$. It is measured in the standard basis, then in the $|+\rangle / |-\rangle$ basis, then in the standard basis again. Find the probability that the final measurement gives 1.

1. First Z-measurement: the state is already $|0\rangle$, so the result is 0 with certainty and the state stays $|0\rangle$.

2. X-measurement on $|0\rangle$: overlaps with $|+\rangle$ and $|-\rangle$ are each $\frac{1}{\sqrt{2}}$ (Lesson 5), so the result is $+$ or $-$, each with probability $\frac{1}{2}$ — and the state collapses to $|+\rangle$ or $|-\rangle$ accordingly.
3. Final Z-measurement: whichever of $|\pm\rangle$ we now hold, $P(1) = \frac{1}{2}$.
4. Combine: $P(\text{final } 1) = 1 \times \frac{1}{2} = \frac{1}{2}$.

Answer: 50% — even though the qubit began as a certain $|0\rangle$ and nothing but *measurements* touched it. The middle measurement asked an incompatible question, and answering it re-randomized the first. Measurements are not passive lookups; each one rewrites the state in its own basis.

WORD TO KNOW

Quantum Zeno effect — The freezing of a quantum state by frequent measurement: a system repeatedly measured in the same basis keeps collapsing back to its last answer, and (in the limit of very frequent checks) barely evolves at all — “a watched quantum pot never boils.” Verified experimentally with trapped ions in 1990.

The same bookkeeping explains the Zeno effect. Suppose a slow process (say, a gentle rotation — Chapter 7) tries to carry $|0\rangle$ toward $|1\rangle$, but we measure in the standard basis after every tiny step. Each measurement almost certainly finds “still 0” (the rotation has barely begun) and collapses the state *exactly back* to $|0\rangle$, erasing the accumulated progress. Frequent watching pins the state in place — a real laboratory tool for holding qubits steady, and a vivid reminder that in the quantum world, checking is an action.

4.7 Certifying a State Factory: The Shot Budget

Laboratories buy and sell state preparation the way factories sell components: with specifications. “This source emits $|0\rangle$ with $|\alpha|^2 \geq 0.98$.” Accepting the shipment means designing a measurement plan whose statistics can actually see the spec — the everyday engineering face of this chapter’s wobble arithmetic.

EQUATION: THE SHOT-BUDGET RULE

$$\sigma \approx \sqrt{\frac{p(1-p)}{n}} \implies n \gtrsim p(1-p) \left(\frac{2}{\text{margin}} \right)^2$$

In words: To pin a probability down to a target margin at two-sigma confidence, invert the wobble formula for the shot count

Symbol	Meaning	Unit
p	the probability being estimated	none
n	number of shots	none
margin	the half-width you must resolve	none

Two useful extremes: near $p = 0.5$ the numerator is worst (0.25); near $p = 1$ it shrinks ($0.98 \times 0.02 \approx 0.02$), so certifying near-certainties is cheaper than measuring coin flips to the same margin.

WORKED EXAMPLE — ACCEPTING OR REJECTING THE SHIPMENT

Spec: $P(0) \geq 0.98$. Design a test resolving a margin of 0.01 at two sigma, and judge a run in which n shots give 969 zeros out of 1000.

1. Budget: $n \gtrsim 0.98 \times 0.02 \times (2/0.01)^2 = 0.0196 \times 40,000 \approx 784$ shots — the planned 1000 is comfortable.
2. Run the numbers: $\hat{p} = 0.969$, wobble $\sigma = \sqrt{0.969 \times 0.031/1000} \approx 0.0055$.
3. Compare with spec: the estimate sits $(0.98 - 0.969)/0.0055 = 2.0$ sigma *below* the promised 0.98 — right at the rejection line.
4. Professional verdict: flag the batch and re-test with $4\times$ the shots (doubling precision); a marginal reading deserves better data, not a coin-flip decision.

Answer: ≈ 800 shots suffice for the margin; the observed 96.9% sits two sigma under spec — evidence of a real shortfall, at the threshold where statistics stops excusing it. This accept/flag/reject discipline — spec, margin, budget, sigma-distance — is quality control for the quantum age, and Chapter 30's datasheets are read exactly this way.

REALITY CHECK

Beware the quiet assumption: shot statistics certify the *measured basis* only. A source could pass every standard-basis test while its phases wander drunkenly — invisible until an X- or Y-basis run (Chapter 8). Real certification (“tomography”) budgets shots in three bases, tripling the bill; entangled sources need Bell-test budgets on top (Chapter 11). The histogram you did not run is the spec you did not check.

CHAPTER SUMMARY

- Superposition is a definite state with random measurement results — not “both at once,” not hidden ignorance.
- Born rule: $P(0) = |\alpha|^2$, $P(1) = |\beta|^2$; in any basis, $P(\phi) = |\langle \phi | \psi \rangle|^2$. Amplitudes add *before* squaring — that ordering is interference.
- Measurement collapses the state to the observed outcome; amplitudes of one copy are unrecoverable, so quantum data is gathered over many shots.
- Expected count = shots $\times$ probability, with wobble $\approx \sqrt{N}$. Small samples mislead; histograms are read statistically.
- Standard-basis histograms never show phase; measuring in a rotated basis (e.g. $|+\rangle / |-\rangle$) turns phase into visible statistics.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] Find $P(0)$ and $P(1)$ for $|\psi\rangle = 0.8|0\rangle + 0.6|1\rangle$, and check they sum to 1.

Problem 2 [Basic] Find the probabilities for $|+\rangle$ and for $|-\rangle$ in the standard basis, and state in one sentence why they are identical.

Problem 3 [Basic] A state gives $P(1) = 0.25$. The state is measured once and the result is 1. What is the state immediately after this measurement, and what is the probability that an immediate second measurement also gives 1?

Problem 4 [Basic] With 2000 shots on $|+\rangle$, what are the expected counts, and what is the rough wobble $\sqrt{N}$?

Problem 5 [Basic] How many shots pin a fair-coin probability ($p \approx 0.5$) to a margin of ± 0.02 at two sigma? And to ± 0.01 ? State the scaling rule connecting the two answers.

Problem 6 [Basic] A 400-shot test of a claimed $P(0) = 0.98$ source returns 385 zeros. Compute $\hat{p}$ and its sigma-distance from spec, and give the verdict.

Problem 7 [Standard] Find the probabilities for $|\psi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}i|1\rangle$, checking normalization first.

Problem 8 [Standard] A 1000-shot histogram shows 495 zeros and 505 ones. A classmate concludes “the state must have been $|+\rangle$.” Name two other states from Chapter 3’s gallery producing the same expected histogram, and state what extra measurement would distinguish them.

Problem 9 [Standard] A qubit yields 0 in 160 of 1000 shots. Estimate $|\beta|$, and write one consistent state with real positive amplitudes.

Problem 10 [Standard] Compute $P(+) = |\langle +|\psi\rangle|^2$ for $|\psi\rangle = |0\rangle$. Interpret: what does a $|+\rangle / |-\rangle$ measurement answer when fed a definite $|0\rangle$?

Problem 11 [Standard] For $|\psi\rangle = 0.6|0\rangle + 0.8|1\rangle$, compute both $P(+)$ and $P(-)$, and check they sum to 1. (Work the inner products carefully: $\langle +|\psi\rangle = \frac{0.6+0.8}{\sqrt{2}}$, $\langle -|\psi\rangle = \frac{0.6-0.8}{\sqrt{2}}$.)

Problem 12 [Standard] Redo the measurement sandwich starting from $|1\rangle$: give the probability tree and the final $P(0)$.

Problem 13 [Standard] In the sandwich, replace the middle X-measurement by a second Z-measurement. Now what is $P(\text{final } 1)$, and what general rule about repeated same-basis measurements does this confirm?

Problem 14 [Standard] Show that certifying $p = 0.98$ to margin 0.01 is about 12 times cheaper in shots than certifying $p = 0.5$ to the same margin, using the budget formula.

Problem 15 [Challenge] Two states, $|a\rangle = 0.6|0\rangle + 0.8|1\rangle$ and $|b\rangle = 0.6|0\rangle - 0.8|1\rangle$, give identical standard-basis histograms. Compute $P(+)$ for each, and design a two-histogram experimental procedure that identifies which state a mystery machine is producing.

Problem 16 [Challenge] A device claims to produce $|+\rangle$, but in 10,000 shots you count 5350 zeros. The wobble is $\sqrt{10000} = 100$. How many wobbles from expectation is this? Is the device credible? What $|\alpha|$ does the data suggest instead?

Problem 17 [Challenge] (Preview of quantum key distribution.) Alice sends a qubit that is either $|0\rangle$ or $|+\rangle$, chosen secretly. Eve measures it in the standard basis and forwards the collapsed state. If the qubit was $|+\rangle$, what does Eve forward, with what probabilities — and show that Bob, measuring in the $|+\rangle / |-\rangle$ basis, now gets $|-\rangle$ (an “impossible” result had Eve not interfered) with probability $\frac{1}{2}$. Moral: in the quantum world, eavesdropping leaves fingerprints.

Problem 18 [Challenge] Zeno arithmetic: a rotation moves $|0\rangle$ by a tiny angle $\theta = 1^\circ$ toward $|1\rangle$ between checks, so each check finds 1 with probability $\sin^2(0.5^\circ) \approx 7.6 \times 10^{-5}$. Over 90 check-steps (a would-be 90° rotation), estimate the total probability the watched qubit EVER leaves $|0\rangle$, and compare with the unwatched outcome.

Problem 19 [Challenge] A mystery machine hands you two independent copies of its state, promised to be either $|0\rangle$ or $|+\rangle$. Measure copy 1 in the standard basis and copy 2 in the $\pm$ basis. Compute all four outcome probabilities for each candidate state, and show that two of the outcomes identify the state *with certainty* (which ones, and why can they never lie?). What fraction of runs ends in a certain identification, and what overall success probability results if you guess on the ambiguous runs? (Single-copy measurements can reach about 85% accuracy but can *never* say “certain” — you have discovered unambiguous state discrimination.)

Problem 20 [Challenge] Design the full acceptance test for a claimed $|+\rangle$ source with specs $\langle X \rangle \geq 0.96$: which basis rotation precedes measurement, what p does the spec translate to, what shot budget resolves margin 0.01 — and which second test (with its own budget) would catch an impostor source emitting the *mixture* of Chapter 5’s mixed-state lesson?

CHAPTER FIVE

The Bloch Sphere

One globe for every qubit state — angles, poles, and phases made visible

IN THIS CHAPTER YOU WILL LEARN TO

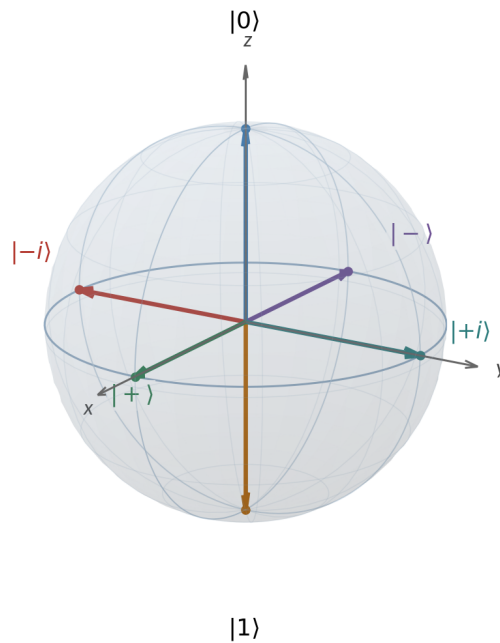
- Explain why every single-qubit state fits on the surface of one sphere.
- Convert between a state's angles (θ, φ) and its amplitude form.
- Place the gallery states $|0\rangle$, $|1\rangle$, $|+\rangle$, $|-\rangle$, $|+i\rangle$, $|-i\rangle$ on the sphere from memory.
- Read probabilities from latitude and phase from longitude.
- Distinguish global phase (unphysical) from relative phase (physical), and explain why the sphere ignores the former.

5.1 A Globe for One Qubit

A single-qubit state $\alpha|0\rangle + \beta|1\rangle$ contains two complex numbers — four real numbers in total. That sounds impossible to draw. But two of the four are not really free: normalization removes one ($|\alpha|^2 + |\beta|^2 = 1$), and global phase (Lesson 4) removes another. Exactly two meaningful real parameters remain — and any two-parameter surface can be drawn. Felix Bloch's beautiful discovery (1946): the natural surface is a *sphere*, and every possible qubit state is exactly one point on it.

WORD TO KNOW

Bloch sphere — A unit sphere on which every pure single-qubit state corresponds to exactly one surface point. North pole = $|0\rangle$, south pole = $|1\rangle$, equator = equal superpositions. The standard map of one-qubit quantum computing.



The Bloch sphere with the six landmark states. The poles are the basis states; the equator carries the equal superpositions, distinguished only by phase (longitude).

Learn the landmarks the way sailors learn stars. North pole: $|0\rangle$. South pole: $|1\rangle$. On the equator, four familiar faces from Chapter 3: $|+\rangle$ on the $+x$ axis, $|-\rangle$ opposite it on $-x$, $|+i\rangle$ on $+y$, $|-i\rangle$ on $-y$. Two warnings as you build intuition. First, *orthogonal states sit at opposite poles of a diameter, not at right angles*: $|0\rangle$ and $|1\rangle$ are 180° apart on the sphere. The 90° of vector space becomes 180° of sphere — angles on the Bloch sphere are doubled. Second, the sphere is an abstract map, not a picture of the physical device: no electron is literally spinning along that arrow (though for real spins the map is delightfully literal).

5.2 The Angles θ and φ

Two numbers locate any point on a globe: latitude and longitude. The Bloch sphere's version:

WORD TO KNOW

Polar angle θ — The angle down from the north pole, from 0° (north, $|0\rangle$) through 90° (equator) to 180° (south, $|1\rangle$). It controls the balance between $|0\rangle$ and $|1\rangle$ — the latitude.

WORD TO KNOW

Azimuthal angle φ — The angle around the vertical axis, measured from the $+x$ direction: the longitude. It records the relative phase between the amplitudes.

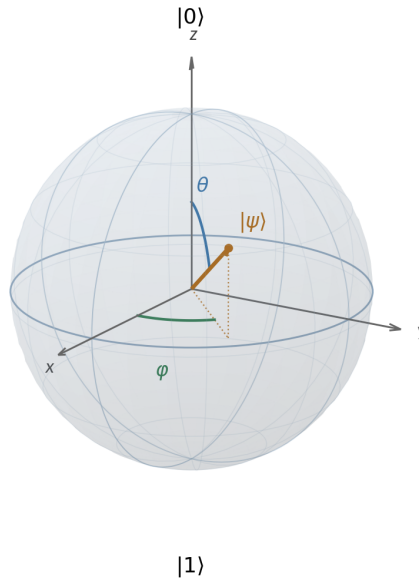
EQUATION: THE STATE AT ANGLES (θ, φ)

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\varphi} \sin\left(\frac{\theta}{2}\right) |1\rangle$$

In words: Amplitude of $|0\rangle = \cos(\text{half the polar angle})$; amplitude of $|1\rangle = (\text{phase factor}) \times \sin(\text{half the polar angle})$

Symbol	Meaning	Unit
θ	polar angle: 0° at $ 0\rangle$, 180° at $ 1\rangle$	degrees or radians
φ	azimuthal angle: relative phase of the $ 1\rangle$ amplitude	degrees or radians
$e^{i\varphi}$	the unit-circle phase factor from Appendix A	none

The half-angles $\theta/2$ are the angle-doubling in reverse: walking 180° on the sphere ($|0\rangle$ to $|1\rangle$) turns the vector by 90° . Normalization is automatic: $\cos^2 + \sin^2 = 1$.



A general state located by its two angles: θ down from the pole, φ around from the $+x$ axis. Here $\theta = 60^\circ$, $\varphi = 45^\circ$.

WORKED EXAMPLE — FROM ANGLES TO AMPLITUDES

Write the state at $\theta = 60^\circ$, $\varphi = 90^\circ$ in amplitude form, and give its measurement probabilities.

1. Half-angle: $\theta/2 = 30^\circ$, so $\cos 30^\circ \approx 0.866$ and $\sin 30^\circ = 0.5$.
2. Phase factor: $e^{i90^\circ} = i$ (Appendix A landmark).
3. Assemble: $|\psi\rangle = 0.866 |0\rangle + 0.5 i |1\rangle$.
4. Born rule: $P(0) = 0.866^2 = 0.75$; $P(1) = |0.5 i|^2 = 0.25$.

Answer: $|\psi\rangle \approx 0.866|0\rangle + 0.5i|1\rangle$, measuring 0 with 75% and 1 with 25%. High on the sphere = mostly $|0\rangle$, exactly as intuition wants.

WORKED EXAMPLE — FROM AMPLITUDES TO ANGLES

Locate $|\psi\rangle = 0.6|0\rangle + 0.8|1\rangle$ on the sphere.

1. Match to the formula: $\cos(\theta/2) = 0.6 \Rightarrow \theta/2 \approx 53.1^\circ$, so $\theta \approx 106.3^\circ$.

2. Consistency check: $\sin(53.1^\circ) \approx 0.8$ ✓ — matches the $|1\rangle$ amplitude.

3. Phase: the $|1\rangle$ amplitude is real and positive, so $e^{i\varphi} = 1 \Rightarrow \varphi = 0^\circ$.

Answer: $(\theta, \varphi) \approx (106^\circ, 0^\circ)$ — slightly below the equator (more $|1\rangle$ than $|0\rangle$), on the x -side of the sphere.

5.3 Reading the Sphere: Latitude is Probability, Longitude is Phase

The sphere is not just an address system — it is a nomogram you can read physics from.

How to read a Bloch point

Where the point is	What it tells you
Near the north pole (θ small)	$P(0)$ close to 1 — almost certainly measures 0
On the equator ($\theta = 90^\circ$)	perfect balance: $P(0) = P(1) = 0.5$
Near the south pole	$P(1)$ close to 1
Longitude φ	the relative phase — invisible to a standard measurement, decisive for interference and for gates (Chapters 6–7)
Antipodal (opposite) points	orthogonal states — a measurement basis, like $ 0\rangle / 1\rangle$ or $ +\rangle / -\rangle$

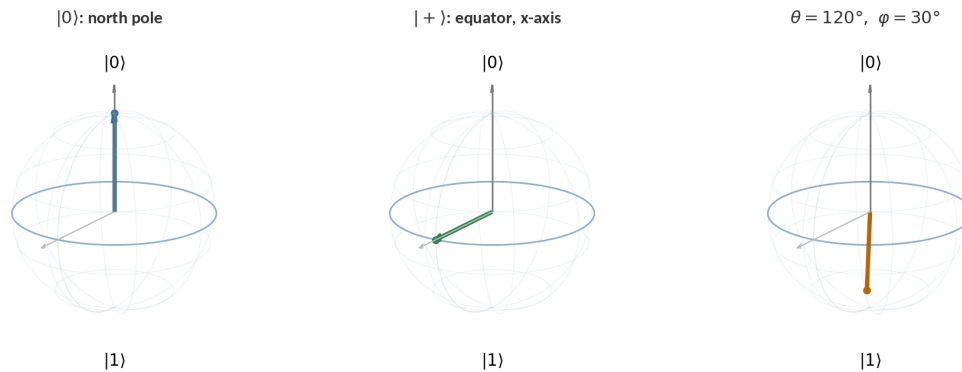
EQUATION: PROBABILITIES FROM LATITUDE

$$P(0) = \cos^2\left(\frac{\theta}{2}\right) \qquad P(1) = \sin^2\left(\frac{\theta}{2}\right)$$

In words: Probability of 0 = $\cos^2$ of half the polar angle; probability of 1 = $\sin^2$ of half the polar angle

Symbol	Meaning	Unit
θ	the polar angle of the state's Bloch point	degrees or radians

φ appears nowhere in this equation — the mathematical statement of “standard measurements cannot see phase.”



Reading latitude: at the pole, certainty; on the equator, a fair coin; at $\theta = 120^\circ$ (below the equator),
 $P(1) = \sin^2 60^\circ = 0.75$.

COMMON MISTAKE

✗ Wrong: “ $|+\rangle$ points along x , so measuring it must give some new result called ‘plus’.”

✓ Right: A standard measurement projects onto the *poles*: any equator state gives 0 or 1, each with 50%. The sphere direction tells you the state, not the readout dial. (To make $|+\rangle / |-\rangle$ the readout dial, you measure along the x -axis instead — Chapter 4, Lesson 5.)

5.4 Global Phase vs. Relative Phase

Chapter 3’s challenge problem hinted at something strange: the states $|\psi\rangle$ and $e^{i\gamma}|\psi\rangle$ — every amplitude multiplied by the same unit factor — give identical probabilities in every basis, respond identically to every gate, and are physically the *same state*. That overall factor is called **global phase**, and no experiment in the universe can detect it.

WORD TO KNOW

Global phase — An overall factor $e^{i\gamma}$ multiplying an entire state. It changes no probability and no physical prediction; states differing only by global phase are the same physical state. The Bloch sphere automatically ignores it.

WORD TO KNOW

Relative phase — The phase *difference* between the $|0\rangle$ and $|1\rangle$ amplitudes — the angle φ . Fully physical: it moves the state around the equator direction, changes interference, and distinguishes $|+\rangle$ from $|-\rangle$ from $|+i\rangle$.

Global phase: physically identical

same physics

$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = e^{i\theta} \cdot \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$|\psi\rangle$ and $e^{i\theta}|\psi\rangle$: same probabilities,
same gate behavior — same state

Relative phase: physically different

different physics

$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \neq \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$|+\rangle$ vs $|-\rangle$: same probabilities now,
but gates treat them differently

The rule that confuses everyone once and then never again: an overall phase on the whole state changes nothing; a phase between the amplitudes changes the state.

This is why the Bloch sphere works at all. Four real numbers, minus one for normalization, minus one for global phase, leaves two: θ and φ . The sphere is not a lossy sketch of the state — it is a *complete, exact* description of everything physical about one qubit.

WORKED EXAMPLE — SAME STATE IN DISGUISE

Show that $|\psi_1\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$ and $|\psi_2\rangle = -\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$ are the same physical state, while $|+\rangle$ is different from both.

1. Factor $|\psi_2\rangle$: $|\psi_2\rangle = (-1) \left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle \right) = e^{i180^\circ} |\psi_1\rangle$.

2. The factor e^{i180° multiplies the *whole* state: a global phase. So $|\psi_2\rangle$ and $|\psi_1\rangle$ share one Bloch point ($|-\rangle$, on $-x$).

3. Compare with $|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$: relative to $|\psi_1\rangle$, only the $|1\rangle$ amplitude changed sign — a *relative* phase of 180° , moving the point from $-x$ to $+x$.

Answer: $|\psi_1\rangle = |\psi_2\rangle$ physically (global phase -1); $|+\rangle$ sits at the opposite equator point (relative phase). Same-looking minus signs, completely different meanings — location is everything.

CASE STUDY — FROM NUCLEAR MAGNETS TO HOSPITAL SCANNERS • FELIX BLOCH, 1946

Felix Bloch introduced his sphere in 1946 to describe nuclear spins precessing in magnetic fields — pure physics, decades before quantum computing existed. The same mathematics became the working language of nuclear magnetic resonance (NMR), where technicians literally speak in Bloch-sphere terms: a “ 90° pulse” tips spins from the pole to the equator; a “ 180° pulse” flips them pole to pole. NMR grew into MRI, the hospital scanner that images the human body by steering billions of hydrogen-nuclei qubits around Bloch spheres — and, in the 1990s, NMR machines hosted the first small quantum computations ever run. When you learn the Bloch sphere, you are learning the control language of real machines, from clinics in Erbil to quantum labs worldwide.

ACTIVITY (APP) — FLY AROUND THE SPHERE

1. Open the *Bloch Explorer* in the QCFE app. Drag the sphere to view it from all sides.
2. Use the θ slider (with $\varphi = 0$): watch the state slide from $|0\rangle$ down through $|+\rangle$ to $|1\rangle$ while the live probability bars update. Verify $P(0) = 0.75$ at $\theta = 60^\circ$.
3. Set $\theta = 90^\circ$ and sweep φ from 0° to 360° : the point orbits the equator through $|+\rangle, |+i\rangle, |-\rangle, |-i\rangle$ — while the probability bars never move.
4. Type the amplitudes 0.6 and $0.8i$ and read off (θ, φ) from the display. Check θ against the formula $\cos(\theta/2) = 0.6$.

You need: The QCFE app (Android or Windows), A calculator

► App tool: *Bloch Explorer* — drag a state around the sphere and watch amplitudes and probabilities respond in real time

5.5 Inside the Sphere: Mixed States

Every state in this chapter has lived on the sphere's surface. Real laboratories constantly meet a second kind of state that lives *inside* it — and telling the two apart is a professional skill.

WORD TO KNOW

Mixed state — A statistical mixture of pure states — for example, “ $|0\rangle$ with probability 75%, $|1\rangle$ with probability 25%,” produced by classical ignorance or by noise. On the Bloch sphere, a mixture sits at the weighted average of its ingredients' points: strictly inside the ball. The extreme case, the maximally mixed state at the very center, gives 50/50 for every measurement in every basis.

The crucial contrast: the superposition $|\psi\rangle = \sqrt{0.75}|0\rangle + \sqrt{0.25}|1\rangle$ and the mixture “75% $|0\rangle$, 25% $|1\rangle$ ” give *identical* standard-basis statistics ($P(0) = 0.75$) — yet they are physically different, and one measurement family exposes them.

WORKED EXAMPLE — SUPERPOSITION OR MIXTURE? THE X-MEASUREMENT VERDICT

Both candidate states above are handed to you, unlabeled. Predict each one's statistics in the $|+\rangle / |-\rangle$ basis.

1. The superposition (real, positive amplitudes): $P(+)=|\langle+|\psi\rangle|^2=\frac{(\sqrt{0.75}+\sqrt{0.25})^2}{2}=\frac{(0.866+0.5)^2}{2}\approx 0.933$ — strongly biased toward $+$.
2. The mixture: each ingredient is measured separately and the results pooled. $|0\rangle$ gives $P(+)=\frac{1}{2}$; $|1\rangle$ gives $P(+)=\frac{1}{2}$; pooled: $0.75\times\frac{1}{2}+0.25\times\frac{1}{2}=\frac{1}{2}$ exactly.
3. Verdict rule: run the X-basis histogram. Bias $\approx 93/7$: superposition. Flat 50/50: mixture.

Answer: the superposition shows interference-grade bias in the rotated basis; the mixture cannot — its Bloch arrow (length $0.75 - 0.25 = 0.5$, pointing at the north pole's latitude equivalent) has no equatorial

component at all. “Same probabilities in one basis” never means “same state”; phase coherence lives in the other bases, and noise (Chapter 28) is precisely the process that erodes it, dragging arrows off the surface toward the center.

The arrow-length rule from Chapter 8 completes the picture: pure states have Bloch length 1; mixtures have length < 1 ; total ignorance sits at length 0. Tomography that returns a short arrow is not failing — it is measuring your state’s purity.

5.6 The Sphere as a Flight Log: Tracking a Circuit

The Bloch sphere’s daily job in laboratories is not philosophy — it is *navigation*: plotting where a circuit takes the state, gate by gate, the way a pilot logs waypoints. One worked flight makes the method yours.

WORKED EXAMPLE — LOGGING THE FLIGHT H, THEN T, THEN H

Start at $|0\rangle$ and log the journey through H, then T, then H — then predict the final measurement.

1. Waypoint 0: $|0\rangle$ — north pole.
2. After H: $|+\rangle$ — equator, longitude 0° (the $+x$ axis).
3. After T: the T gate turns 45° about the z-axis (Chapter 7): still on the equator, longitude 45° ; state $\frac{1}{\sqrt{2}}(|0\rangle + e^{i45^\circ}|1\rangle)$. Note: *latitude unchanged* — a z-turn never changes $P(0)$, which is why the standard histogram cannot see it.
4. After H: H maps the equator point at longitude φ to a point at *latitude* determined by φ — the phase becomes population. Compute: amplitudes $(\frac{1+e^{i45^\circ}}{2}, \frac{1-e^{i45^\circ}}{2})$, so $P(0) = \frac{1+\cos 45^\circ}{2} = 0.854$.

Answer: the flight lands at a point with $P(0) = 85.4\%$ — the invisible 45° phase, made visible by the final H. This three-gate pattern (H, phase, H) is the *interferometer*: the measuring instrument for phases, and the primitive inside Ramsey experiments (Chapter 28), phase estimation (Chapter 19), and every quantum sensor.

WORD TO KNOW

Interferometer — The H-phase-H sandwich (or its laboratory equivalent): a circuit converting an invisible relative phase φ into a visible probability $P(0) = (1 + \cos \varphi)/2$. Quantum computing’s voltmeter.

REALITY CHECK

Two flight-log habits professionals keep. First: z-rotations (S , T , R_z) never move latitude — if your histogram should have changed and did not, suspect that your circuit’s action was “only phase” and add the final H that reads it. Second: any single-qubit circuit, however long, lands somewhere reachable by *one* rotation — the whole flight compresses (Chapter 7’s Euler lesson), which is exactly what compilers exploit. The sphere is not just a picture of states; it is a picture of *computations*.

CHAPTER SUMMARY

- Every single-qubit state is one point on the Bloch sphere: $|0\rangle$ north, $|1\rangle$ south, equal superpositions on the equator ($|+\rangle$ at $+x$, $|+i\rangle$ at $+y$, $|-\rangle$ at $-x$, $| -i\rangle$ at $-y$).
- The state at angles (θ, φ) is $\cos(\theta/2) |0\rangle + e^{i\varphi} \sin(\theta/2) |1\rangle$ — note the half-angles.
- Latitude sets probabilities: $P(0) = \cos^2(\theta/2)$, $P(1) = \sin^2(\theta/2)$. Longitude is relative phase, invisible to standard measurement but fully physical.
- Orthogonal states are antipodal (sphere angles are doubled vector angles).
- Global phase is undetectable and the sphere ignores it; relative phase moves you around the sphere. Two parameters, one sphere, zero information lost.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] Give the Bloch coordinates (θ, φ) of: (a) $|0\rangle$, (b) $|1\rangle$, (c) $|+\rangle$, (d) $|+i\rangle$.

Problem 2 [Basic] What are $P(0)$ and $P(1)$ for the state at $\theta = 90^\circ$, $\varphi = 137^\circ$? What role did φ play?

Problem 3 [Basic] Write the state at $\theta = 180^\circ$, $\varphi = 0^\circ$ using the angle formula, and simplify.

Problem 4 [Basic] Which pairs are antipodal on the sphere: $(|0\rangle, |1\rangle)$? $(|+\rangle, |-\rangle)$? $(|0\rangle, |+\rangle)$? What does antipodal mean physically?

Problem 5 [Basic] Log the flight $|0\rangle \rightarrow H \rightarrow S \rightarrow H$: waypoints (pole/equator, longitudes) and the final $P(0)$ using $(1 + \cos \varphi)/2$ with $\varphi = 90^\circ$.

Problem 6 [Basic] Why can no sequence of S and T gates alone ever change $P(0)$ of any starting state? Answer in one sentence about axes.

Problem 7 [Standard] Write the state at $\theta = 120^\circ$, $\varphi = 180^\circ$ in amplitude form (use $\cos 60^\circ = 0.5$, $\sin 60^\circ \approx 0.866$, $e^{i180^\circ} = -1$), and give its probabilities.

Problem 8 [Standard] Locate $|\psi\rangle = 0.28 |0\rangle + 0.96 |1\rangle$: find θ (from $\cos(\theta/2) = 0.28$) and φ .

Problem 9 [Standard] Two states sit at the same $\theta = 70^\circ$ but $\varphi = 0^\circ$ and $\varphi = 90^\circ$. Write both in amplitude form, and name one experiment from Chapter 4 that would distinguish them and one that would not.

Problem 10 [Standard] Simplify and locate on the sphere: $|\psi\rangle = e^{i\pi/3} \left(\frac{1}{\sqrt{2}} |0\rangle + \frac{i}{\sqrt{2}} |1\rangle \right)$. What kind of phase is $e^{i\pi/3}$ here?

Problem 11 [Standard] A state has $P(1) = 0.9$. Find θ (from $\sin^2(\theta/2) = 0.9$), and say in words where the point sits.

Problem 12 [Standard] Locate the mixture “50% $|0\rangle$, 50% $|1\rangle$ ” on the Bloch ball, and explain why NO measurement basis can distinguish it from “50% $|+\rangle$, 50% $|-\rangle$.”

Problem 13 [Standard] A state’s tomography returns Bloch coordinates $(0.4, 0, 0.4)$. Compute the arrow length, classify the state (pure or mixed), and give its purity ratio relative to a surface state in the same direction.

Problem 14 [Standard] $\text{Log } |0\rangle \rightarrow \text{H} \rightarrow \text{Z} \rightarrow \text{H}$ and confirm the interferometer formula at $\varphi = 180^\circ$ predicts your endpoint. Which classic identity from Chapter 6 did the flight just prove geographically?

Problem 15 [Challenge] Show that the states at (θ, φ) and $(180^\circ - \theta, \varphi + 180^\circ)$... are they orthogonal? Test with $\theta = 60^\circ, \varphi = 0^\circ$: write both states and compute their inner product. Generalize in one sentence about antipodal points.

Problem 16 [Challenge] Using the angle formula, verify that the six landmark states pairwise: $|+i\rangle$ has $(\theta, \varphi) = (90^\circ, 90^\circ)$ and $|-i\rangle$ has $(90^\circ, 270^\circ)$; compute $\langle +i | -i \rangle$ from the amplitude forms and confirm antipodal $\Rightarrow$ orthogonal.

Problem 17 [Challenge] A colleague proposes storing information in global phase: “encode 0 as $|+\rangle$ and 1 as $-|+\rangle$.” Using this chapter, explain in three sentences why no receiver could ever read this message — and propose the minimal change (which phase to use instead) that makes the scheme work, naming the two resulting states.

Problem 18 [Challenge] An imperfect preparation produces $|+\rangle$ with probability 0.9 and $|-\rangle$ with probability 0.1. Find the resulting Bloch arrow (average the two points), predict $P(+)$ in an X-basis measurement, and compare with the pure state whose arrow points the same way at full length.

Problem 19 [Challenge] Explain in four sentences why decoherence (Chapter 28) is well pictured as “arrows shrinking toward the center,” connecting: environment measurement (Chapter 4), the mixture verdict of this lesson, and the T2 fringe decay you will meet in the Ramsey experiment.

Problem 20 [Challenge] A mystery box applies an unknown phase φ . You run the H-box-H interferometer with 4000 shots and count 3415 zeros. Estimate φ (two values are consistent — why?), give the wobble-based uncertainty on P , and design the one extra experiment (hint: insert an S) that breaks the tie between the two candidate angles.

PART THREE

GATES AND MEASUREMENT

CHAPTER SIX

Single-Qubit Gates I: X, Z, and H

X, Z, and H — the first three machines of quantum computing, in matrices, circuits, and spheres

IN THIS CHAPTER YOU WILL LEARN TO

- Define a quantum gate as a unitary matrix acting on the state vector.
- Apply the X, Z, and H gates to any state by matrix arithmetic — and check the results on the Bloch sphere.
- Read and write single-qubit circuit diagrams, including the order rule: circuits read left to right, matrices multiply right to left.
- Compute the output of gate sequences like HH and HZH, and explain the interference inside them.
- Use the identities $XX = ZZ = HH = I$ and $HZH = X$ fluently.

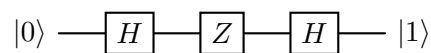
6.1 What a Quantum Gate Is

Classical gates (Chapter 1) turn bits into bits by truth tables. Quantum gates turn *states* into *states* — and since states are vectors, gates are matrices. Appendix B already interviewed the candidates and posted the entrance requirement on the door: a gate must be **unitary** ($U^\dagger U = I$), so that legal states go to legal states and nothing is ever lost.

WORD TO KNOW

Quantum gate — A unitary matrix applied to the state of one or more qubits: $|\psi_{\text{out}}\rangle = U |\psi_{\text{in}}\rangle$. The quantum analogue of a logic gate — but always reversible, and able to create and steer superpositions.

In circuit diagrams, a gate is a labeled box on a horizontal *wire*. The wire is not a physical cable — it is the timeline of one qubit, read **left to right**:



Anatomy of a one-qubit circuit: the input state enters at the left; boxes apply gates in time order; the right side is the output. This particular circuit is computed, step by step, in Lesson 5.

One rule causes more student errors than any other, so let us nail it immediately. The circuit above applies H first. But the matrix acting first must stand *next to the vector* (Appendix B), so the output is

$$|\psi_{\text{out}}\rangle = H Z H |0\rangle$$

— the leftmost gate in the diagram is the *rightmost* matrix in the product. Circuits read like English, matrix products like Arabic; both describe the same sequence of events.

COMMON MISTAKE

✗ Wrong: Circuit “ H then Z then H ” means the matrix product HZH read as “apply the left H first” — so any order works, it’s all the same letters.

✓ Right: The first gate applied is the matrix nearest the ket: circuit $H \rightarrow Z \rightarrow H$ means $H(Z(H|\psi\rangle))$. Here the product is symmetric so the risk hides; but circuit $X \rightarrow Z$ is the matrix ZX , and $ZX \neq XZ$ (Appendix B, Problem 6). Always build products from the ket outward.

6.2 The X Gate: The Quantum NOT

EQUATION: THE X (NOT) GATE

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad X|0\rangle = |1\rangle \quad X|1\rangle = |0\rangle$$

In words: X swaps the two amplitudes: what belonged to $|0\rangle$ now belongs to $|1\rangle$ and vice versa

Symbol	Meaning	Unit
X	the bit-flip (NOT) gate	none — a unitary matrix

On the Bloch sphere: a 180° rotation about the x -axis. Applied twice: $XX = I$ — flipping twice does nothing.

On basis states, X is exactly the classical NOT. Its quantum personality appears on superpositions: it swaps the amplitudes, both at once.

WORKED EXAMPLE — X ON A SUPERPOSITION

Apply X to $|\psi\rangle = 0.6|0\rangle + 0.8|1\rangle$, and compare measurement statistics before and after.

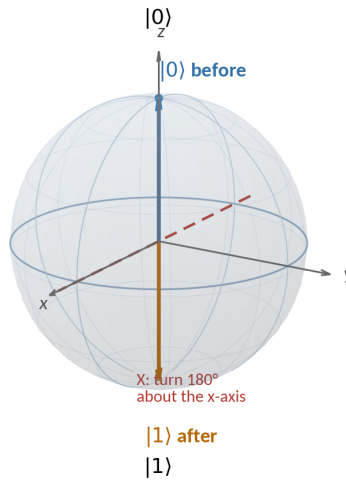
1. Write the input as a column: $\begin{pmatrix} 0.6 \\ 0.8 \end{pmatrix}$.

2. Multiply: $X \begin{pmatrix} 0.6 \\ 0.8 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0.6 \\ 0.8 \end{pmatrix} = \begin{pmatrix} 0 \cdot 0.6 + 1 \cdot 0.8 \\ 1 \cdot 0.6 + 0 \cdot 0.8 \end{pmatrix} = \begin{pmatrix} 0.8 \\ 0.6 \end{pmatrix}$.

3. Read the output: $X|\psi\rangle = 0.8|0\rangle + 0.6|1\rangle$.

4. Statistics: before, $P(0) = 0.36$; after, $P(0) = 0.64$ — the probabilities traded places, as a NOT should arrange.

Answer: $X|\psi\rangle = 0.8|0\rangle + 0.6|1\rangle$. **One gate moved both amplitudes — no classical gate ever touches a probability distribution this way.**



X as geometry: a half-turn about the x -axis carries $|0\rangle$ to $|1\rangle$ (and the whole sphere with it). States on the axis, like $|+\rangle$, do not move at all: $X|+\rangle = |+\rangle$.

6.3 The Z Gate: The Phase Flip

EQUATION: THE Z (PHASE-FLIP) GATE

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad Z|0\rangle = |0\rangle \quad Z|1\rangle = -|1\rangle$$

In words: Z leaves the $|0\rangle$ amplitude alone and flips the sign of the $|1\rangle$ amplitude

Symbol	Meaning	Unit
Z	the phase-flip gate	none — a unitary matrix

On the Bloch sphere: a 180° rotation about the z -axis. Again $ZZ = I$.

To classical eyes, Z looks useless: it never changes a 0 into a 1, and after Z, standard measurement probabilities are exactly what they were ($|\alpha|^2 = |\beta|^2$). But you know from Chapters 4 and 5 what a sign between amplitudes is: a *relative phase* — invisible in one basis, decisive in another.

WORKED EXAMPLE — Z TURNS $|+\rangle$ INTO $|-\rangle$

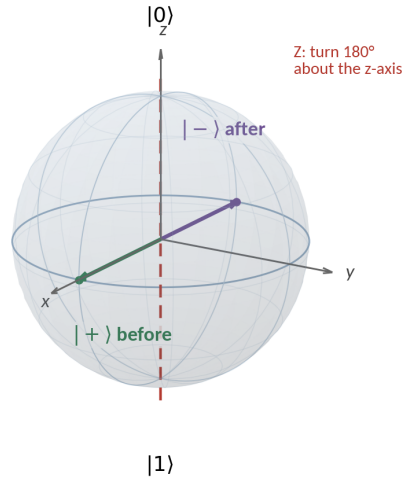
Apply Z to $|+\rangle$ and identify the output.

1. Input column: $|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

2. Multiply: $Z|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

3. Recognize the output from Chapter 3's gallery: this is exactly $|-\rangle$.

Answer: $Z|+\rangle = |-\rangle$ — on the sphere, a half-turn about the vertical axis carries $+x$ to $-x$. Standard-basis statistics unchanged (50/50 both), yet the state is probably different: measure in the $|+\rangle / |-\rangle$ basis (Chapter 4, Lesson 5) and the answer flips from certain- $+$ to certain- $-$.



Z as geometry: a half-turn about the z-axis. Poles stay (up to global phase); the equator rotates by 180°: $|+\rangle \leftrightarrow |-\rangle$, and $|+i\rangle \leftrightarrow |-i\rangle$.

X and Z are two of the three *Pauli gates*; their sibling Y waits in Chapter 7. A cultural note you will need when reading anything about quantum error correction (Chapter 29): every error a qubit can suffer can be written as some mixture of “an X happened” (bit flip) and “a Z happened” (phase flip). Master these two gates and you have vocabulary for quantum noise itself.

6.4 The Hadamard Gate: The Superposition Maker

The star of the show. Neither X nor Z can create a superposition from a basis state — check: they only shuffle or re-sign amplitudes. The Hadamard gate is the standard tool that makes superpositions from nothing, and (being its own inverse) also folds them back.

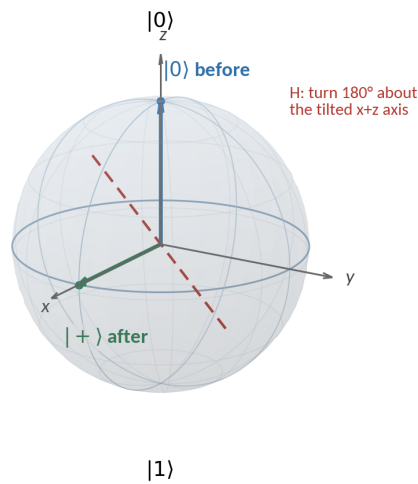
EQUATION: THE HADAMARD GATE

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad H|0\rangle = |+\rangle \quad H|1\rangle = |-\rangle$$

In words: H mixes the amplitudes: new top = (top + bottom)/√2, new bottom = (top - bottom)/√2

Symbol	Meaning	Unit
H	the Hadamard gate	none — a unitary matrix
$\frac{1}{\sqrt{2}}$	normalization's fingerprint (Chapter 3)	none

On the Bloch sphere: a 180° rotation about the diagonal axis halfway between x and z — which is exactly why it exchanges the z -pole states with the x -axis states. $HH = I$ (Appendix B proved it).



H as geometry: a half-turn about the tilted $(x+z)$ axis swaps north pole and $+x$: $|0\rangle \leftrightarrow |+\rangle$, and likewise $|1\rangle \leftrightarrow |-\rangle$.

WORKED EXAMPLE — H ON EACH BASIS STATE — MEMORIZE BOTH

Compute $H|0\rangle$ and $H|1\rangle$.

$$1. H|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = |+\rangle.$$

$$2. H|1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = |-\rangle.$$

3. Notice the difference: both outputs are 50/50 states, but $H|1\rangle$ carries the minus — the input bit survives, hidden in the phase.

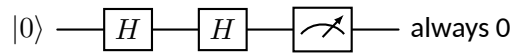
Answer: $H|0\rangle = |+\rangle$, $H|1\rangle = |-\rangle$. **H converts bit information into phase information — and, being self-inverse, phase information back into bit information. Algorithms exploit both directions constantly.**

REALITY CHECK

“Apply H and the qubit does everything at once” is hype’s favorite sentence. Truth: $H|0\rangle = |+\rangle$, and measuring $|+\rangle$ gives a fair coin — as useful as flipping dinars. The power is never in creating a superposition (one gate, trivial); it is in what you *do between* H and the measurement: steering phases so that interference concentrates probability on answers (Lesson 5 shows the smallest example; Grover in Chapter 18 shows the famous one).

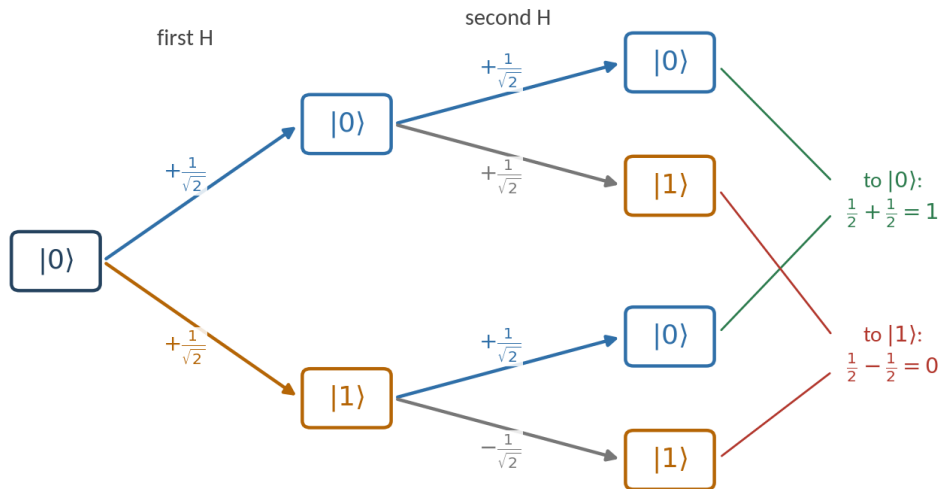
6.5 Gates in Sequence: Interference in a Circuit

Time to compute a full circuit. Two gates first:



H twice. Algebra says $HH = I$; the interesting question is what happens inside.

Algebraically the answer is instant: $HH|0\rangle = I|0\rangle = |0\rangle$; the measurement gives 0 with certainty. But watch the amplitudes flow — this is the smallest interference experiment in quantum computing:



Inside HH: after the first H, two branches. The second H splits each branch again. Two paths arrive at $|0\rangle$ with amplitudes $+\frac{1}{2}, +\frac{1}{2}$: they add to 1. Two paths arrive at $|1\rangle$ with $+\frac{1}{2}, -\frac{1}{2}$: they cancel to 0. The minus sign in H’s matrix single-handedly deletes the wrong outcome.

If measurement had happened between the two H gates, each branch would have collapsed (Chapter 4), the phases would have been destroyed, and the final result would be 50/50. Interference needs the superposition to survive untouched — the deep reason quantum computers must isolate their qubits so fiercely (Chapter 28).

WORKED EXAMPLE — THE CIRCUIT $H \rightarrow Z \rightarrow H$, COMPLETELY

Compute the output of the chapter-opening circuit: input $|0\rangle$, gates H then Z then H.

1. After the first H: $|0\rangle \mapsto |+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.
2. After Z (Lesson 3's example): $|+\rangle \mapsto |-\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$.
3. After the second H: $H|-\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1-1 \\ 1+1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$.
4. Conclusion as one matrix: $HZH|0\rangle = |1\rangle = X|0\rangle$.

Answer: the circuit outputs $|1\rangle$, with certainty. In fact $HZH = X$ exactly (Appendix B, Problem 12): a phase flip, dressed between two Hadamards, becomes a bit flip. Sandwiching a gate in H's moves it between the z -language and the x -language — a trick used from error correction to algorithm design.

Your gate toolbox so far

Gate	Matrix	Action on basis	Bloch picture
X	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	$ 0\rangle \leftrightarrow 1\rangle$	180° about x
Z	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	$ 1\rangle \rightarrow - 1\rangle$	180° about z
H	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	$ 0\rangle \rightarrow +\rangle, 1\rangle \rightarrow -\rangle$	180° about $x+z$ diagonal
I	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	nothing	no turn

CASE STUDY — FIVE QUBITS FOR THE WHOLE WORLD • IBM QUANTUM EXPERIENCE, 2016

In May 2016, IBM did something no one had done: it connected a real quantum processor — five superconducting qubits in a dilution refrigerator in New York — to the public internet, free. Anyone could drag gates onto wires in a web editor, press run, and receive genuine quantum measurement histograms. Within two years, users had run millions of experiments, classrooms worldwide replaced “imagine a quantum computer” with “log in and try,” and research papers appeared whose experiments were run entirely through the cloud — including by students in countries with no quantum lab of their own. The circuits you drew in this chapter, X's and H's on wires, are exactly the language of those cloud editors; in Chapter 14 you will write them in Python and, if you wish, run them on real hardware from anywhere — Kurdistan included.

ACTIVITY (APP) — BUILD YOUR FIRST CIRCUITS

1. Open the *Circuit Builder* in the QCFE app. Drag an X onto the wire; input $|0\rangle$; run 100 shots. Histogram: all 1s.

2. Replace X with H. Run 100 shots: roughly 50/50. Run again — slightly different 50/50, as Chapter 4 promised.
3. Build H-H. Before running, predict the histogram. Run: all 0s — interference deleted the 1s. Toggle the app's "show amplitudes" view and find the $+\frac{1}{2}, -\frac{1}{2}$ cancellation from this lesson's figure.
4. Build H-Z-H and verify the output is always 1, matching the worked example.

You need: The QCFE app (Android or Windows)

► App tool: Circuit Builder — drag X, Z, H onto a wire; see live state, Bloch point, amplitudes, and histograms

6.6 Two Reflections Make a Rotation: Composing X and Z

A closing geometric gem ties the chapter's gates together — and hands you a tool professionals use daily to simplify circuits without touching a matrix.

Each of X, Z, H is a 180° turn about its own axis. A classical fact of geometry (true on any sphere): *two 180° turns about different axes compose into a single rotation about the axis perpendicular to both, by twice the angle between them.* The x - and z -axes sit 90° apart, so Z-then-X should equal a 180° rotation about the y -axis — which Chapter 7 will name the Y gate.

WORKED EXAMPLE — VERIFYING THE GEOMETRY WITH ALGEBRA

Compute the matrix product XZ , compare it with the Y gate ($Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, previewed here), and reconcile with the rotation picture.

1. Multiply (right-to-left order — Z acts first): $XZ = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$

2. Compare with Y: $-iY = -i \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \checkmark$ — so $XZ = -iY$.

3. Interpret the $-i$: a global phase (Chapter 5) — physically invisible. As operations on states, Z-then-X and Y are the same machine.

4. Cross-check on a state: $XZ|+\rangle = X|-\rangle = -|-\rangle \dots$ and $Y|+\rangle$? Chapter 7's worked example will give $-i|-\rangle$ — again the same state, phases aside.

Answer: $XZ = -iY$: two half-turns about perpendicular axes make the third Pauli, up to global phase. The practical harvest: any string of X's and Z's collapses, by geometry, to at most one Pauli — circuits full of flips are never as long as they look.

WORD TO KNOW

Pauli group (working notion) — The family $\{I, X, Y, Z\}$ with phases: closed under multiplication — any product of Paulis is again a Pauli, up to a factor $\pm 1, \pm i$. The algebraic backbone of error analysis (Chapter 28) and error correction (Chapter 29).

6.7 When Order Matters: A Commutation Field Guide

Multiplying the gate alphabet raises a question silent in classical circuits: does the *order* of gates matter? Sometimes no — and when it does, the difference is measurable. The field guide: z-family gates (Z , S , T , R_z) commute among themselves; x-family likewise; *across* families, order matters, and the previous lesson's $XZ = -ZX$ was the first sighting.

WORKED EXAMPLE — SAME GATES, DIFFERENT ORDER, DIFFERENT HISTOGRAM

Compare the circuits T-then-H and H-then-T acting on $|0\rangle$ (reading left to right), measured in the standard basis after a final H in both cases is *not* added — measure directly. Then explain the general lesson.

1. Circuit A (T first): $T|0\rangle = |0\rangle$ (T only phases $|1\rangle$; the pole is unmoved). Then H: $|+\rangle$. Measured: 50/50.
2. Circuit B (H first): $H|0\rangle = |+\rangle$. Then T: equator, longitude 45° . Measured: still 50/50 — *the standard histogram ties!*
3. Break the tie by measuring in the X basis (append H, Chapter 4): circuit A's $|+\rangle$ gives $P(+) = 1$; circuit B's 45° state gives $P(+) = \frac{1+\cos 45^\circ}{2} = 0.854$.
4. Conclusion: $TH \neq HT$ — the states differ (longitudes 0° vs 45°), and a phase-sensitive measurement exposes it even when the lazy histogram cannot.

Answer: the two orders produce genuinely different states, distinguishable at 100% vs 85.4% in the right basis. Commutation is not algebraic pedantry: compilers may only swap gates that commute, and every “optimization” that reorders non-commuting gates is a bug with a histogram to prove it.

WORD TO KNOW

Commuting gates — Gates with $AB = BA$: applying them in either order gives the same circuit. Same-axis rotations commute; different-axis rotations generally do not — the algebraic face of the uncertainty trade-offs of Chapter 8.

REALITY CHECK

A practical rule saves hours of debugging: *diagonal gates commute*. Z , S , T , R_z , CZ — all diagonal in the standard basis — can be freely reordered among themselves, merged, and slid past each other, which is why compilers love pushing them together into one combined phase. The moment an H or X enters, the sliding stops. Check any proposed reorder with a 2×2 multiplication — twenty seconds that Chapter 14's debugging lesson will thank you for.

ACTIVITY (CLASS) — THE INTERFERENCE DIAL, ON PAPER

1. Everyone computes H then H on $|0\rangle$ by hand (four amplitudes, two cancellations) and confirms $P(0) = 1$.
2. Now insert Z between the H's; recompute. The cancellation flips: $P(1) = 1$.

3. Split the class: half insert S, half insert T. Using the interferometer formula $P(0) = (1 + \cos \varphi)/2$, each half predicts its number — S gives 50% (a right-angle phase hides from this detector completely!), T gives 85.4% — then checks in the QCFE app.

4. Discuss: one inserted gate, four different histograms — the dial is the relative phase, and interference is the meter that reads it.

You need: Paper and pen, The QCFE app for checking

CHAPTER SUMMARY

- A quantum gate is a unitary matrix; a circuit applies gates left to right, and the matching matrix product is written right to left (first gate nearest the ket).
- $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ swaps amplitudes (bit flip; 180° about x). $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ flips the $|1\rangle$ sign (phase flip; 180° about z): invisible to standard statistics, physical nonetheless.
- $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ makes and unmakes superpositions: $|0\rangle \leftrightarrow |+\rangle$, $|1\rangle \leftrightarrow |-\rangle$; it converts bit information to phase information and back.
- $XX = ZZ = HH = I$; and $HZH = X$ — the Hadamard sandwich exchanges bit-language and phase-language.
- HH on $|0\rangle$ is the smallest interference experiment: paths to $|1\rangle$ cancel ($+\frac{1}{2} - \frac{1}{2}$), paths to $|0\rangle$ reinforce. Measuring mid-circuit destroys the effect.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] Apply X to $|1\rangle$, to $|+\rangle$, and to $0.28|0\rangle + 0.96|1\rangle$ (matrix arithmetic, all three).

Problem 2 [Basic] Apply Z to $|0\rangle$, to $|1\rangle$, and to $|-\rangle$. Which outputs are physically new states, and which differ only by global phase?

Problem 3 [Basic] Verify by matrix multiplication that $ZZ = I$.

Problem 4 [Basic] Draw the circuit that applies, in time order, H then X to input $|0\rangle$, and write its output as a matrix product (which letter stands next to $|0\rangle$?).

Problem 5 [Basic] Do X and Z commute? Compute both products for $|+\rangle$ as input and exhibit the difference (recall $XZ = -iY$ from the previous lesson — what is ZX ?).

Problem 6 [Basic] Show S and T commute by multiplying their matrices both ways. What single gate equals ST?

Problem 7 [Standard] Compute the output of circuit $X \rightarrow H$ on input $|0\rangle$ (i.e. the product $HX|0\rangle$), and locate the result on the Bloch sphere.

Problem 8 [Standard] Compute $H(0.6|0\rangle + 0.8|1\rangle)$ and give the output's measurement probabilities. Note how unequal amplitudes give partial — not total — interference.

Problem 9 [Standard] Show by explicit matrix multiplication that $XZ \neq ZX$, then compute both $XZ|+\rangle$ and $ZX|+\rangle$ and identify each output in gallery language (up to global phase).

Problem 10 [Standard] Circuit: $|0\rangle \rightarrow H \rightarrow (\text{measure}) \rightarrow H \rightarrow \text{measure}$. Unlike the HH circuit, the middle measurement collapses the state. Compute the probability tree and show the final result is 50/50 — then explain in one sentence what the middle measurement destroyed.

Problem 11 [Standard] Using $HZH = X$, predict HXH without computing — then verify by matrix multiplication. (The sandwich works both directions.)

Problem 12 [Standard] Compute ZX and express it as (phase) $\times$ (Pauli). Comparing with $XZ = -iY$: by what factor do the two orders differ, and why does that factor never affect measurement statistics?

Problem 13 [Standard] Reduce the string $XZXZ$ to a single phase times a single Pauli, working two letters at a time.

Problem 14 [Standard] Verify that H commutes with neither Z nor X , but that $HZH = X$ (the sandwich identity): compute all three products. Why is a sandwich not a swap?

Problem 15 [Challenge] Find the output of circuit $H \rightarrow Z \rightarrow H \rightarrow Z \rightarrow H \rightarrow Z \rightarrow H$ on $|0\rangle$ by pairing sandwiches: rewrite the product using $HZH = X$ repeatedly rather than multiplying seven matrices. (Answer in gallery form, global phase allowed.)

Problem 16 [Challenge] The gate $M = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ looks like a shuffled H . Check whether M is unitary; compute $M|0\rangle$ and $MM|0\rangle$; and explain why M , unlike H , is *not* its own inverse (what did the moved minus sign change?).

Problem 17 [Challenge] Alice encodes one secret bit b as follows: she prepares $|0\rangle$, applies H , then applies Z if $b = 1$ (nothing if $b = 0$), then applies H again, and hands Bob the qubit. What state does Bob receive in each case, and how does he read b with one standard measurement? Explain in two sentences how this protocol hides a bit in a phase mid-flight — and why an eavesdropper measuring in the standard basis mid-protocol would scramble it.

Problem 18 [Challenge] Using the two-reflections rule, predict — before any algebra — what single rotation the product HZ is (axes of H and Z are 45° apart; twice that is 90° , about which axis?). Then test your prediction on $|0\rangle$: compute $HZ|0\rangle$ and check the result sits where a 90° turn about your predicted axis would put it.

Problem 19 [Challenge] The look-ahead trick: show that $XH = HZ$ by multiplying both sides out — “pushing

an X leftward through an H turns it into a Z.” Then use this rewriting to simplify the circuit H, X, H, X (as matrices: $XHXH$) to a single named gate times a phase.

Problem 20 [Challenge] A compiler wants to move an S gate from before an H to after it: $HS \rightarrow S'H$ for some replacement S' . Using the sandwich identities, find the correct S' (hint: it is a rotation about a different axis — which?), and verify on the input $|0\rangle$ that $HS|0\rangle = S'H|0\rangle$.

CHAPTER SEVEN

Single-Qubit Gates II: Y, S, T, and Rotations

Y, S, T, and the rotation gates — fine control of the sphere, and why it is enough

IN THIS CHAPTER YOU WILL LEARN TO

- Apply the Y gate and place it in the Pauli family alongside X and Z.
- Use the phase gates S and T, and track the phases they attach to $|1\rangle$.
- Read and apply the rotation gates $R_x(\theta)$, $R_y(\theta)$, $R_z(\theta)$ as turns of the Bloch sphere.
- Chain phase gates: verify $T^2 = S$, $S^2 = Z$, and locate each on the unit circle.
- Explain single-qubit universality: why H and T alone can build (approximately) any one-qubit gate.

7.1 The Y Gate: The Third Pauli

Chapter 6 introduced two of the three Pauli gates. The third, Y, flips the bit *and* the phase at once — with a pair of i 's in exactly the right places to keep it unitary.

EQUATION: THE Y GATE

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad Y|0\rangle = i|1\rangle \quad Y|1\rangle = -i|0\rangle$$

In words: Y swaps the amplitudes AND attaches phases: a bit flip and a phase flip in one move

Symbol	Meaning	Unit
Y	the third Pauli gate	none — a unitary matrix
$i, -i$	unit phases (Appendix A): $ \pm i ^2 = 1$, so probabilities are unaffected by them	none

On the Bloch sphere: a 180° rotation about the y -axis. Like its siblings, $YY = I$. Up to global phase, Y acts as “Z then X”: $XZ = -iY$.

WORKED EXAMPLE — Y ON A BASIS STATE AND ON $|+\rangle$

Compute $Y|0\rangle$ and $Y|+\rangle$, and interpret each.

1. $Y|0\rangle = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ i \end{pmatrix} = i|1\rangle$. Global phase i aside, this is $|1\rangle$: a bit flip.

2. $Y|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} -i \\ i \end{pmatrix} = \frac{-i}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

3. Factor the global phase $-i$: the remaining state is $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = |-\rangle$.

Answer: $Y|0\rangle = i|1\rangle$ (physically $|1\rangle$) and $Y|+\rangle = -i|-\rangle$ (physically $|-\rangle$). Both agree with geometry: a half-turn about y maps pole to pole and $+x$ to $-x$.

The Pauli family — one gate per axis

Gate	Matrix	Plain-language action	Bloch rotation
X	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	bit flip	180° about x
Y	$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$	bit flip + phase flip	180° about y
Z	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	phase flip	180° about z

7.2 The Phase Gates S and T

Z flips the phase of $|1\rangle$ by 180° . Quantum algorithms need gentler touches — quarter turns, eighth turns — and two standard gates provide them.

EQUATION: THE S AND T GATES

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \quad T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

In words: S attaches a 90° phase to $|1\rangle$; T attaches a 45° phase to $|1\rangle$

Symbol	Meaning	Unit
S	the quarter-turn phase gate ("phase gate")	none
T	the eighth-turn phase gate	none
$e^{i\pi/4}$	the 45° unit phase = $\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$ (Appendix A's worked example)	none

Both leave $|0\rangle$ untouched and leave all standard-basis probabilities unchanged — pure phase machinery. On the sphere: rotations about z by 90° (S) and 45° (T).

The Z -family forms a ladder set of turns about the same axis, and chaining them adds the angles — Appendix A's unit circle at work:

EQUATION: THE PHASE LADDER

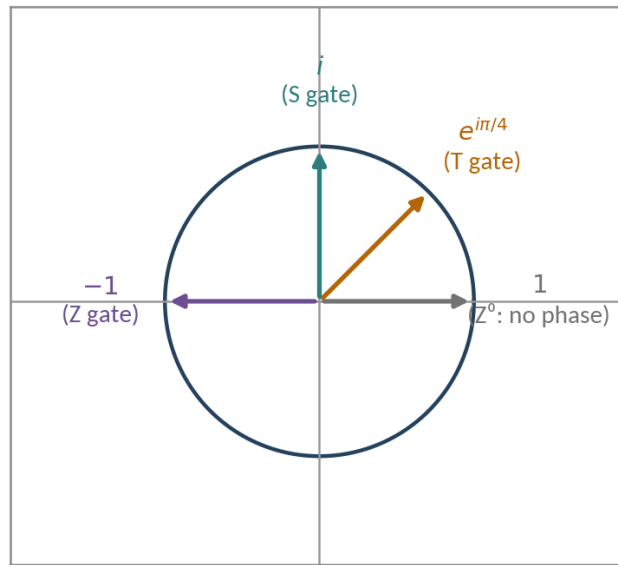
$$T^2 = S \quad S^2 = Z \quad T^4 = Z \quad Z^2 = S^4 = T^8 = I$$

In words: Two eighth-turns make a quarter-turn; two quarter-turns make a half-turn; eight eighth-turns bring you home

Symbol	Meaning	Unit
T^2	T applied twice (matrix product TT)	none

Check one rung by pure phase arithmetic: $e^{i\pi/4} \cdot e^{i\pi/4} = e^{i\pi/2} = i$ — which is exactly S 's phase.

The phases the Z-family gates put on $|1\rangle$



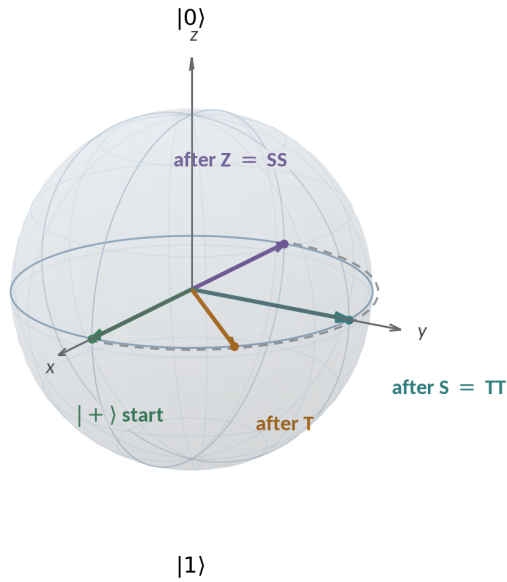
The phases the Z-family attaches to $|1\rangle$, living on Appendix A's unit circle: T at 45° , S at 90° , Z at 180° .

WORKED EXAMPLE — WALKING $|+\rangle$ AROUND THE EQUATOR

Start from $|+\rangle$ and apply T , then T again (making S in total), then S (making Z in total). Name each intermediate state.

1. After T : $\frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi/4} |1\rangle)$ — the equator point at longitude $\varphi = 45^\circ$.
2. After the second T : phases add, $\frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi/2} |1\rangle) = \frac{1}{\sqrt{2}} (|0\rangle + i |1\rangle) = |+\rangle$, at $\varphi = 90^\circ$.
3. After S : add another 90° : $\frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi} |1\rangle) = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) = |-\rangle$, at $\varphi = 180^\circ$.

Answer: $|+\rangle \xrightarrow{T} (\varphi=45^\circ) \xrightarrow{T} |+\rangle \xrightarrow{S} |-\rangle$. Phase gates march equator states around the sphere, one fixed step per gate — and standard-basis histograms never notice a thing.



The worked example drawn: T, then T, then S walk $|+\rangle$ around the equator to $|-\rangle$. Latitude (probabilities) never changes; longitude (phase) does all the moving.

COMMON MISTAKE

X Wrong: “S and T do nothing measurable — the histogram is identical before and after, so they are decorative.”

✓ Right: They do nothing measurable *in the standard basis, applied once, alone*. Sandwich them between Hadamards, or use them on interfering paths, and the phase becomes hard statistics: for example $H S H |0\rangle$ gives $P(0) = P(1) = \frac{1}{2}$ with a 90° internal phase, while $H Z H |0\rangle$ gives $P(1) = 1$. Phase gates are the steering wheel of interference — Chapters 18 and 19 are driven with them.

7.3 Rotation Gates: Any Angle You Like

X, Y, Z, S, T are fixed turns. Hardware, however, can rotate a qubit by *any* angle — a laser pulse twice as long turns the state twice as far. The general gates come in three families, one per axis.

EQUATION: THE ROTATION GATES

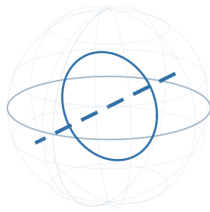
$$R_z(\theta) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix} \quad R_y(\theta) = \begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix} \quad R_x(\theta) = \begin{pmatrix} \cos \frac{\theta}{2} & -i \sin \frac{\theta}{2} \\ i \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}$$

In words: Each rotates the Bloch sphere by angle θ about its own axis

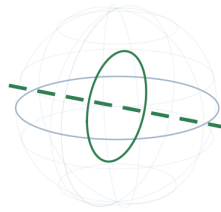
Symbol	Meaning	Unit
θ	the rotation angle on the Bloch sphere	radians or degrees
$\theta/2$	the half-angle appearing in the matrices — the sphere's angle-doubling again (Chapter 5)	radians or degrees

Special cases (up to global phase): $R_z(180^\circ) = Z$, $R_z(90^\circ) = S$, $R_z(45^\circ) = T$; $R_x(180^\circ) = X$; $R_y(180^\circ) = Y$. The fixed gates are just marked stops on three continuous dials. (This book writes R_z in its phase-gate form; some texts multiply by a global phase $e^{-i\theta/2}$ — physically identical.)

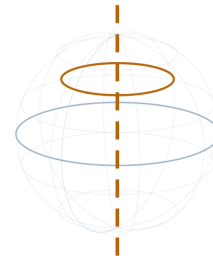
Rx: turn around x



Ry: turn around y



Rz: turn around z



Three dials, one sphere: R_x , R_y , and R_z each turn the state about their own axis by any chosen angle.

WORKED EXAMPLE — AN R_y ROTATION BY 60°

Apply $R_y(60^\circ)$ to $|0\rangle$ and give the resulting state and probabilities.

1. Half-angle: $\theta/2 = 30^\circ$: $\cos 30^\circ \approx 0.866$, $\sin 30^\circ = 0.5$.
2. Multiply: $R_y(60^\circ) |0\rangle = \begin{pmatrix} 0.866 & -0.5 \\ 0.5 & 0.866 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.866 \\ 0.5 \end{pmatrix}$.
3. Read the state: $0.866 |0\rangle + 0.5 |1\rangle$ — real amplitudes, on the sphere's xz -meridian at $\theta = 60^\circ$.
4. Born rule: $P(0) = 0.75$, $P(1) = 0.25$.

Answer: $R_y(60^\circ) |0\rangle = 0.866 |0\rangle + 0.5 |1\rangle$, with a 75/25 split. R_y is the workhorse for preparing any real-amplitude state: dial θ , get $\cos(\theta/2) |0\rangle + \sin(\theta/2) |1\rangle$.

WORKED EXAMPLE — DESIGNING A STATE TO ORDER

A protocol needs a qubit with exactly $P(1) = 0.10$. Which R_y rotation on $|0\rangle$ prepares it?

1. Target: $\sin^2(\theta/2) = 0.10 \Rightarrow \sin(\theta/2) = \sqrt{0.10} \approx 0.316$.
2. Inverse sine: $\theta/2 \approx 18.4^\circ$, so $\theta \approx 36.9^\circ$.

3. Check: $\cos(18.4^\circ) \approx 0.949$, and $0.949^2 \approx 0.90$ ✓

Answer: $R_y(36.9^\circ)|0\rangle \approx 0.949|0\rangle + 0.316|1\rangle$, giving $P(1) = 0.10$. Real quantum programs (Chapter 20's variational circuits) are built almost entirely from such dialed rotations.

7.4 Universality: Two Gates Are Enough

A remarkable theorem closes this part of the book. You now hold infinitely many gates (R_x, R_y, R_z at every angle) — but hardware wants a *small, fixed* menu it can calibrate to perfection. How small can the menu be?

WORD TO KNOW

Universal gate set — A small set of gates whose combinations can build (to any desired accuracy) every possible quantum gate. For one qubit, $\{H, T\}$ is universal; adding CNOT (Chapter 10) makes $\{H, T, \text{CNOT}\}$ universal for *all* quantum computing.

The idea in three steps, no proof needed. First, any single-qubit unitary is some rotation of the sphere (about some axis, by some angle). Second, rotations about two *different* axes can be combined to approximate a rotation about *any* axis — and H and T give you two different axes (T turns about z ; HTH turns about x , by the sandwich rule of Chapter 6). Third, the tiny 45° step of T is “incommensurate” enough that repeating and mixing these turns fills the sphere of rotations densely — like a clock hand that jumps a never-repeating step and eventually visits every minute mark as closely as you please. (The Solovay-Kitaev theorem sharpens “eventually” into “efficiently.”)

REALITY CHECK

Universality is why real machines survive on tiny gate menus: a few calibrated pulses per qubit, plus one entangling gate. But approximation has a price — hitting a target gate to accuracy ε costs a string of H 's and T 's that grows as the accuracy demand tightens. Compilers for future error-corrected machines (Chapter 29) spend much of their effort “ T -counting”: the T gate is by far the most expensive to protect, so algorithms are judged by how few T 's they need. The humble eighth-turn is, economically, the most precious gate in quantum computing.

CASE STUDY — TUNING PHASES FOR A LIVING • RAMSEY INTERFEROMETRY AND ATOMIC CLOCKS

The chain “ H — controlled phase — H ” is not just a textbook pattern; it is the daily bread of the world's best measuring devices. In Ramsey interferometry (Nobel Prize 1989), an atom is put into superposition (an H -like pulse), allowed to accumulate phase $e^{i\theta}$ at a rate set by its energy (an R_z whose angle records elapsed time), then folded back (second H -like pulse) so the phase becomes a population difference — a histogram. Cesium atomic clocks run exactly this circuit billions of times to define the world's second; GPS satellites carry such clocks, which is why your phone knows where you are. Every time you navigate a street in Erbil, a Bloch-sphere phase measurement made it possible.

ACTIVITY (APP) — THE ROTATION DIALS

1. Open the *Gate Dials* tool in the QCFE app. Start from $|0\rangle$.
2. Sweep the R_y dial slowly from 0° to 180° and watch the Bloch arrow swing from north pole to south pole through $|+\rangle$; watch $P(1)$ climb from 0 to 1.
3. Set $R_y(36.9^\circ)$ and confirm the app reports $P(1) \approx 0.10$, matching the worked example.
4. Start again from $|+\rangle$: apply T eight times, watching the arrow hop 45° per press around the equator and return home — the phase ladder $T^8 = I$, live.

You need: The QCFE app (Android or Windows)

► App tool: *Gate Dials* — continuous R_x , R_y , R_z sliders plus one-press X, Y, Z, H, S, T buttons, with the Bloch arrow and histogram always live

7.5 Compiling to the Menu: A Worked Gate Build

Universality (Lesson 5) promises that H and T suffice; here is the promise kept, on a job small enough to do by hand — the daily bread of a quantum compiler.

WORKED EXAMPLE — BUILD $R_z(135^\circ)$ FROM THE HARDWARE MENU

A device offers only {H, S, T} (plus their daggers). Compile the rotation $R_z(135^\circ)$, exactly.

1. Decompose the angle in the menu's currency: $135^\circ = 90^\circ + 45^\circ$.
2. Match menu items: $S = R_z(90^\circ)$ and $T = R_z(45^\circ)$ (Lesson 2's ladder, read backwards).
3. Same-axis rotations add angles (Appendix A's multiply-the-phases rule), so $R_z(135^\circ) = ST$ — order irrelevant here, since both turn about z .
4. Cost accounting, compiler-style: 1 T-gate, 1 Clifford (S). An alternative expansion TTT also works ($3 \times 45^\circ$) but spends three T's — and Lesson 5 warned that T is the expensive gate under error correction. The compiler picks ST .

Answer: $R_z(135^\circ) = ST$, at T-count 1. Exact compilations exist whenever the target angle is a multiple of 45° ; all other angles (say $R_z(10^\circ)$) are approximated by longer H/T strings — the Solovay-Kitaev territory of Lesson 5, where T-count grows as accuracy demands tighten.

WORD TO KNOW

T-count — The number of T gates in a compiled circuit — the standard cost metric for future fault-tolerant machines, where each T requires an expensive distilled resource state (Chapter 29) while Clifford gates (H, S, CNOT) are comparatively free.

WORKED EXAMPLE — AN APPROXIMATE COMPILER, GRADED WITH FIDELITY

The target is $R_z(50^\circ)$; the menu forces a choice between the exact-angle neighbors $R_z(45^\circ) = T$ and $R_z(90^\circ) = S$. Which single-gate approximation is better, and how good is it?

1. The approximation error is a phase misalignment of 5° (choosing T) versus 40° (choosing S).
2. Grade with the overlap ruler (Chapter 3): acting on $|+\rangle$, an angle error δ leaves fidelity $\cos^2(\delta/2)$.
3. T's grade: $\cos^2(2.5^\circ) \approx 0.998$. S's grade: $\cos^2(20^\circ) \approx 0.883$.

Answer: choose T — 99.8% fidelity from one gate. Squeezing out the last 0.2% is what longer H/T strings buy, at rising T-count: compiling is always this trade, precision against price.

7.6 Euler's Recipe: Any Gate from Three Turns

The rotation gates promise “any angle about any axis”; Euler's theorem delivers something stronger — *any* single-qubit gate whatsoever, exotic or plain, factors into just three turns about two fixed axes:

EQUATION: THE ZYZ DECOMPOSITION

$$U = e^{i\gamma} R_z(\alpha) R_y(\beta) R_z(\delta)$$

In words: Any one-qubit gate = a z-turn, then a y-turn, then a z-turn (read right to left), times an invisible global phase

Symbol	Meaning	Unit
α, β, δ	the three Euler angles	rad or $^\circ$
$e^{i\gamma}$	global phase — physically irrelevant	none
R_z, R_y	the rotation gates of this chapter	none

Why it works, in Bloch language: the first z-turn picks a longitude, the y-turn sets the latitude change, the last z-turn fixes the final longitude — three dials cover the whole sphere of possibilities. Hardware loves this: many machines physically implement only R_z and one fixed-axis pulse, and compile everything else through exactly this recipe.

WORKED EXAMPLE — DECOMPOSING THE HADAMARD

Find Euler angles for H, and verify the claim by matrix multiplication.

1. Guess from geography: H exchanges the z-axis and x-axis — a 90° turn about y *almost* does this, but sends $|0\rangle \rightarrow |+\rangle$ while wrecking the equator's orientation; a 180° z-turn first fixes that.
2. Claim: $H = e^{i90^\circ} R_y(90^\circ) R_z(180^\circ)$ — that is $\alpha = 0^\circ, \beta = 90^\circ, \delta = 180^\circ$.
3. Verify: $R_z(180^\circ) = \begin{pmatrix} e^{-i90^\circ} & 0 \\ 0 & e^{i90^\circ} \end{pmatrix} = -i \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$; multiply by $R_y(90^\circ) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ and by $e^{i90^\circ} = i$: the product is $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = H$ ✓
4. Sanity check on a state: $R_z(180^\circ) |0\rangle \sim |0\rangle$ (poles ignore z-turns), then $R_y(90^\circ)$ carries it to $|+\rangle$ — and indeed $H |0\rangle = |+\rangle$.

Answer: $H = e^{i90^\circ} R_y(90^\circ) R_z(180^\circ)$. One glamorous gate, three plain turns. Every compiler you will ever use (Chapter 14) runs this factorization on every single-qubit gate in your circuit — usually into the hardware’s native R_z - $\sqrt{X}$ - R_z dialect, a cousin of ZYZ.

REALITY CHECK

Euler’s recipe also explains a compiler magic trick: any *run* of consecutive single-qubit gates on one wire — ten, fifty, a thousand — multiplies into a single 2×2 unitary, which then re-factors into just three turns. Long single-qubit stretches cost almost nothing after compilation; it is the CNOTs between wires that resist all such compression (Chapter 10) — one more reason the two-qubit gate count rules the budgets of Chapter 30.

CHAPTER SUMMARY

- $Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ completes the Pauli trio: bit-and-phase flip, 180° about y ; $XX = YY = ZZ = I$.
- S and T attach phases i (90°) and $e^{i\pi/4}$ (45°) to $|1\rangle$: $T^2 = S$, $S^2 = Z$, $T^8 = I$. Alone they change no standard-basis statistics; inside interference they change everything.
- $R_x(\theta)$, $R_y(\theta)$, $R_z(\theta)$ rotate the sphere by any angle about their axis; matrices carry half-angles. R_y prepares any real-amplitude state to order.
- Fixed gates are marked stops on the rotation dials: Z, S, T on the z -dial; X on x ; Y on y .
- Universality: $\{H, T\}$ builds any one-qubit gate to any accuracy (add CNOT for everything). Hardware thrives on small menus; T is the costly gate that compilers count.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher’s Guide.

Problem 1 [Basic] Compute $Y|1\rangle$ and identify the physical state (name the global phase you factored out).

Problem 2 [Basic] Apply S to $|+\rangle$ and name the output from the Chapter 3 gallery.

Problem 3 [Basic] Using phase arithmetic only (add the angles), verify $S^2 = Z$ and $T^4 = Z$.

Problem 4 [Basic] What rotation gate and angle turn $|0\rangle$ into $|+\rangle$? (Hint: which axis moves pole to equator along the xz -meridian — and check against the R_y matrix at that angle.)

Problem 5 [Basic] Give Euler angles (α, β, δ) for the gates Z , S , and $R_y(30^\circ)$ themselves (no computation needed — read them off).

Problem 6 [Basic] Why can the global phase $e^{i\gamma}$ be ignored for any circuit that ends in a measurement? One sentence, citing the Born rule.

Problem 7 [Standard] Compute $T|-\rangle$, writing the output in the form $\frac{1}{\sqrt{2}}(|0\rangle + e^{i\varphi}|1\rangle)$ and giving φ in degrees. (Recall $|-\rangle$ has $\varphi = 180^\circ$.)

Problem 8 [Standard] Apply $R_y(90^\circ)$ to $|0\rangle$: compute the matrix at $\theta = 90^\circ$ ($\cos 45^\circ = \sin 45^\circ = \frac{1}{\sqrt{2}}$), multiply, and name the output state.

Problem 9 [Standard] Find the R_y angle that prepares $P(1) = 0.36$ from $|0\rangle$, and write the prepared state.

Problem 10 [Standard] Show that $SS^\dagger = I$ by writing $S^\dagger$ explicitly and multiplying. What gate is $S^\dagger$ on the phase circle (which angle does it attach to $|1\rangle$)?

Problem 11 [Standard] Compute the circuit $H \rightarrow T \rightarrow H$ on $|0\rangle$ far enough to find $P(0)$ and $P(1)$. (Multiply carefully; you will need $|1 + e^{i\pi/4}|^2 = 2 + \sqrt{2}$, which you can verify from $e^{i\pi/4} = \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$.) Compare with $H \rightarrow Z \rightarrow H$ and $H \rightarrow H$ to see the T-phase steering the split.

Problem 12 [Standard] Compile $R_z(225^\circ)$ from $\{S, T\}$ at minimum T-count, and give one more expansion with a worse T-count.

Problem 13 [Standard] Using the daggers: express $R_z(-45^\circ)$ and $R_z(315^\circ)$ each as a single menu gate, and explain why they are the same physical operation.

Problem 14 [Standard] Compute the matrix $R_y(90^\circ)R_z(180^\circ)$ explicitly (as in the worked example) and confirm that without the e^{i90° it equals $-iH$. Then confirm $(-iH)(-iH) = -I$: does the doubled circuit still act as identity on measurement statistics?

Problem 15 [Challenge] Prove $XZ = -iY$ by matrix multiplication, and conclude how Y relates physically to “Z then X.” Why does the factor $-i$ not matter?

Problem 16 [Challenge] Using the sandwich rule ($H \cdot (z\text{-turn}) \cdot H = \text{same turn about } x$), write the gate $R_x(45^\circ)$ as a three-gate circuit using only H and T. Verify at one checkpoint: apply your circuit to $|+\rangle$ and check the state does not move (why must a state on the x -axis be unmoved by any x -rotation?).

Problem 17 [Challenge] A hardware menu offers only $\{H, S\}$ — Hadamard plus quarter-turn. Explain why this menu is NOT universal for one qubit, by finding the (finite!) set of equator longitudes reachable from $|+\rangle$ using z -turns of 90° and sphere-swaps of H. What does the T gate’s 45° provide that 90° steps never can? (One honest sentence about “fine-grained angles” earns full credit; the full theory is called the Clifford hierarchy.)

Problem 18 [Challenge] Compile $R_x(90^\circ)$ using the sandwich rule (H turns z -rotations into x -rotations): write the three-gate sequence, state its T-count, and verify your compile sends $|0\rangle$ to a state on the y -meridian (compute it).

Problem 19 [Challenge] A target $R_z(60^\circ)$ must reach fidelity ≥ 0.999 on $|+\rangle$. Show the best single menu gate fails the specification (compute its fidelity), and determine the smallest angle error δ that passes — the accuracy a longer H/T string would need to deliver.

Problem 20 [Challenge] X is a 180° turn about the x-axis — almost. Compute the matrix $R_x(180^\circ)$ from $R_x(\theta) = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} X$, and show $X = i R_x(180^\circ)$: equal only up to a global phase i . Verify both send $|0\rangle \rightarrow |1\rangle$ (up to phase). Explain why hardware implementing X as $R_x(180^\circ)$ is correct for every circuit that ends in measurement — and in what situation (building a *controlled*- X from a controlled- $R_x(180^\circ)$, Chapter 10) the leftover i becomes a measurable relative phase.

CHAPTER EIGHT

Observables and Expectation Values

Turning histograms into numbers — the averages that power chemistry, optimization, and quantum

AI

IN THIS CHAPTER YOU WILL LEARN TO

- Explain what an observable is: a measurable question whose answers carry numerical values.
- Compute the expectation value $\langle Z \rangle$ from probabilities, and estimate it from a real histogram.
- Measure $\langle X \rangle$ and $\langle Y \rangle$ by rotating the basis before the meter — and know which circuit does which.
- Read $\langle X \rangle$, $\langle Y \rangle$, $\langle Z \rangle$ as the Bloch coordinates of the state, and reconstruct a state from measured averages (tomography).
- Use the matrix formula $\langle A \rangle = \langle \psi | A | \psi \rangle$, extend it to multi-qubit observables like $Z \otimes Z$, and budget the shots an accurate average needs.

8.1 Questions with Numbers: What an Observable Is

So far, measurement has answered with *labels*: 0 or 1, pattern by pattern (Chapter 4). Science, though, usually wants *quantities*: the energy of a molecule, the magnetization of a material, the cost of a delivery route, the score of a classifier. The bridge between quantum measurements and quantities is the **observable**: a measurement whose outcomes are assigned numerical values, so that repeated shots produce not just a histogram but an *average number*.

WORD TO KNOW

Observable — A measurable quantity: a choice of measurement together with a numerical value for each outcome. The standard single-qubit observable Z assigns the value $+1$ to result 0 and -1 to result 1. In the mathematics, every observable is a matrix whose eigenvalues are the possible values.

Why $+1$ and -1 rather than 0 and 1? Convenience with meaning: the average then runs from $+1$ (“always 0”) through 0 (“perfectly balanced”) to -1 (“always 1”), making balance, bias, and correlation instantly readable — exactly like a survey scored from -1 to $+1$. This convention is universal across quantum computing, and

it is the Pauli matrices X , Y , Z of Chapters 6–7 that play the starring role — the same matrices, now cast as *questions* rather than *operations*. (One matrix, two jobs: applied to a state, Z is a gate; asked of a state, Z is an observable. The context — circuit box versus measured average — always tells you which is meant.)

8.2 The Expectation Value $\langle Z \rangle$

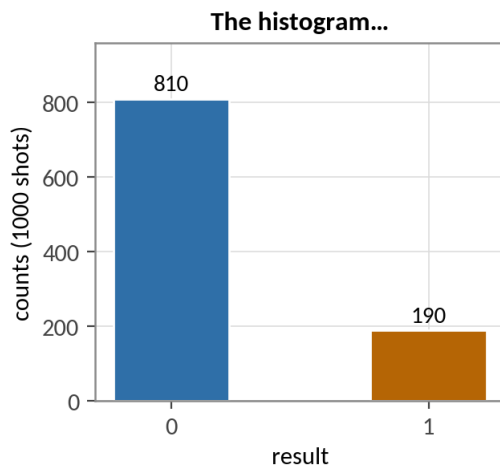
EQUATION: THE EXPECTATION VALUE OF Z

$$\langle Z \rangle = (+1)P(0) + (-1)P(1) = P(0) - P(1)$$

In words: Average of Z = probability of 0 minus probability of 1

Symbol	Meaning	Unit
$\langle Z \rangle$	the expectation (average) value of the observable Z	none — between -1 and $+1$
$P(0), P(1)$	the Born-rule probabilities of Chapter 4	none

For the general state $\alpha|0\rangle + \beta|1\rangle$: $\langle Z \rangle = |\alpha|^2 - |\beta|^2$. Landmarks: $\langle Z \rangle = +1$ for $|0\rangle$, -1 for $|1\rangle$, and 0 for every equator state ($|+\rangle$, $|-\rangle$, $|+i\rangle$, $\dots$).



...becomes one number

$$\begin{aligned}\langle Z \rangle &= P(0) - P(1) \\ &= 0.81 - 0.19 = +0.62\end{aligned}$$

*value +1 for result 0, value -1 for result 1,
averaged over all shots*

From counts to a quantity: score each shot (+1 for 0, -1 for 1) and average. A 1000-shot histogram with 810 zeros compresses into the single number $\langle Z \rangle = +0.62$.

WORKED EXAMPLE — $\langle Z \rangle$ THREE WAYS

A qubit is prepared in $|\psi\rangle = 0.6|0\rangle + 0.8|1\rangle$. Find $\langle Z \rangle$ exactly, then estimate it from a run of 1000 shots that returned 372 zeros.

1. Exact, from amplitudes: $P(0) = 0.36$, $P(1) = 0.64$, so $\langle Z \rangle = 0.36 - 0.64 = -0.28$.
2. From the experiment: $\hat{P}(0) = 0.372$, so $\widehat{\langle Z \rangle} = 0.372 - 0.628 = -0.256$.
3. Judge the gap with Chapter 4's wobble: the uncertainty of a 1000-shot average is about $1/\sqrt{1000} \approx 0.032$,

and $|-0.256 - (-0.28)| = 0.024$ sits comfortably inside it.

Answer: $\langle Z \rangle = -0.28$ exactly; the experiment estimated -0.26 ± 0.03 — agreement, once you speak the language of shot noise. Every expectation value from real hardware comes with such an error bar; professionals never quote one without it.

COMMON MISTAKE

X Wrong: “ $\langle Z \rangle = -0.28$ means each shot returns the number -0.28 .”

✓ Right: No single shot ever returns -0.28 — each shot gives $+1$ or -1 , nothing else. The expectation value is a property of the *state*, revealed by averaging many shots, exactly as a die’s average of 3.5 is never the face you roll. Confusing the average with an outcome is the most common expectation-value error in early lab reports.

8.3 Measuring $\langle X \rangle$ and $\langle Y \rangle$

Z asks “0 or 1?” — the poles question. The observable X asks the equator question “ $|+\rangle$ or $|-\rangle$?” (values $+1$ and -1), and Y asks “ $|+i\rangle$ or $|-i\rangle$?”. Hardware meters, however, only read the standard basis. The trick you learned in Chapter 4, Lesson 5 — rotate the state, then measure — becomes a standard recipe:

Basis-change recipes for the three Pauli observables

To measure	Apply before the meter	Then
$\langle Z \rangle$	nothing	$\langle Z \rangle = P(0) - P(1)$
$\langle X \rangle$	H	$\langle X \rangle = P(0) - P(1)$ of the rotated run
$\langle Y \rangle$	$S^\dagger$, then H	$\langle Y \rangle = P(0) - P(1)$ of the rotated run

Why does the H recipe work? Because H maps the X -question onto the Z -question: it sends $|+\rangle \rightarrow |0\rangle$ and $|-\rangle \rightarrow |1\rangle$ (Chapter 6), so “measure Z after H” answers exactly “was it $|+\rangle$ or $|-\rangle$?”. The Y recipe adds $S^\dagger$ to first turn the y -axis onto the x -axis. One important bookkeeping note: the three averages need three *separate* runs — a shot measured for X is collapsed and cannot also be measured for Z . Expectation-value experiments always come as a small family of circuits differing only in their final rotations.

WORKED EXAMPLE — ALL THREE AVERAGES OF $|+\rangle$

Predict $\langle Z \rangle$, $\langle X \rangle$, and $\langle Y \rangle$ for the state $|+\rangle$.

- $\langle Z \rangle$: $|+\rangle$ has $P(0) = P(1) = \frac{1}{2}$, so $\langle Z \rangle = 0$.
- $\langle X \rangle$: apply H first: $H|+\rangle = |0\rangle$, so the rotated run gives 0 always — $\langle X \rangle = P(0) - P(1) = 1 - 0 = +1$.
- $\langle Y \rangle$: apply $S^\dagger$ then H. $S^\dagger|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle) = |-i\rangle$, and H maps it to $\frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$ -rotated form with $P(0) = P(1) = \frac{1}{2}$ — so $\langle Y \rangle = 0$. (Compute: $H|-i\rangle = \frac{1}{2}\begin{pmatrix} 1-i \\ 1+i \end{pmatrix}$, and $|1 \pm i|^2/4 = \frac{1}{2}$ each.)

Answer: $(\langle X \rangle, \langle Y \rangle, \langle Z \rangle) = (+1, 0, 0)$ — three numbers that should look familiar: they are exactly the coordinates of $|+\rangle$'s point on the Bloch sphere. That is no coincidence, as the next lesson reveals.

8.4 The Bloch Connection: Averages Are Coordinates

Here is one of the most elegant facts in single-qubit physics, and the payoff of Chapter 5:

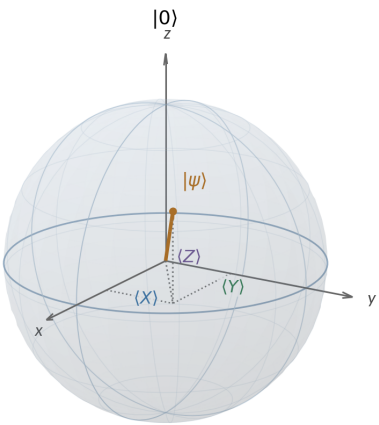
EQUATION: EXPECTATION VALUES = BLOCH COORDINATES

$$(\langle X \rangle, \langle Y \rangle, \langle Z \rangle) = (x, y, z)_{\text{Bloch}}$$

In words: The three Pauli averages of a state are exactly the coordinates of its point on the Bloch sphere

Symbol	Meaning	Unit
$\langle X \rangle, \langle Y \rangle, \langle Z \rangle$	the three measured Pauli averages	each between -1 and $+1$
$(x, y, z)_{\text{Bloch}}$	the state's position on the unit sphere (Chapter 5)	none

Check the landmarks: $|0\rangle$ at the north pole has $(0, 0, +1)$; $|+\rangle$ on the x -axis has $(+1, 0, 0)$ — matching the worked example. For pure states $\langle X \rangle^2 + \langle Y \rangle^2 + \langle Z \rangle^2 = 1$; noisy (mixed) states fall inside the sphere, and the shortfall from 1 is itself a practical purity meter.



$|1\rangle$

The measurement meaning of the Bloch sphere: a state's three Pauli averages are literally its coordinates. Measuring all three (over many shots) is called tomography — experimentally locating the state.

WORD TO KNOW

State tomography — Reconstructing an unknown state from measured averages: for one qubit, estimate $\langle X \rangle, \langle Y \rangle, \langle Z \rangle$ over many shots and read off the Bloch point. The standard way labs verify “we really prepared the state we claim.”

WORKED EXAMPLE — TOMOGRAPHY OF A MYSTERY QUBIT

Three runs of 2000 shots each on identically prepared mystery qubits give: Z -run 84.5% zeros; X -run (H first) 74.9% zeros; Y -run ($S^\dagger$ H first) 50.6% zeros. Locate and identify the state.

1. Convert each to an average: $\langle Z \rangle = 0.845 - 0.155 = 0.69$; $\langle X \rangle = 0.749 - 0.251 = 0.50$; $\langle Y \rangle = 0.506 - 0.494 = 0.01 \approx 0$.

2. Purity check: $0.50^2 + 0.01^2 + 0.69^2 = 0.25 + 0.00 + 0.476 = 0.726...$ noticeably below 1 — either noise, or rounding; the shot-noise bar per average is $1/\sqrt{2000} \approx 0.022$, too small to explain the gap alone, so allow for slight decoherence (Chapter 28) and proceed with the direction.

3. Direction on the sphere: in the xz -plane ($y \approx 0$), at angle $\theta = \arccos(z/r)$ with $r = \sqrt{0.726} \approx 0.852$: $\theta \approx \arccos(0.69/0.852) \approx 36^\circ$ from the pole, $\varphi = 0$.

Answer: the state points 36° down from $|0\rangle$ along the $+x$ meridian — consistent with the preparation $R_y(36^\circ)|0\rangle \approx 0.95|0\rangle + 0.31|1\rangle$, slightly dulled by noise (its Bloch arrow has length 0.85 rather than 1). This is a genuine lab workflow, at Chapter-7 mathematics.

8.5 The Matrix Formula and Many-Qubit Observables

Everything above has a one-line algebraic engine, and it is the formula that runs quantum chemistry and quantum machine learning alike:

EQUATION: THE EXPECTATION-VALUE FORMULA

$$\langle A \rangle = \langle \psi | A | \psi \rangle = \langle \psi | (A | \psi \rangle)$$

In words: Average of observable A = bra of the state, times A applied to the state

Symbol	Meaning	Unit
A	the observable's matrix (e.g. X, Y, Z , or sums of their products)	none
$\langle \psi A \psi \rangle$	inner product of $ \psi\rangle$ with $A \psi\rangle$ (Appendix B arithmetic)	none — always real for observables

Two-step recipe: apply the matrix to the ket, then inner-product with the original bra. For $A = Z$ this reproduces $|\alpha|^2 - |\beta|^2$ — try it once and the general machine is yours.

WORKED EXAMPLE — THE FORMULA IN ACTION

Compute $\langle X \rangle$ for $|\psi\rangle = 0.6|0\rangle + 0.8|1\rangle$ using $\langle\psi|X|\psi\rangle$.

1. Apply the matrix: $X|\psi\rangle = X \begin{pmatrix} 0.6 \\ 0.8 \end{pmatrix} = \begin{pmatrix} 0.8 \\ 0.6 \end{pmatrix}$ (the amplitude swap of Chapter 6).
2. Inner product with the bra: $\langle\psi|X|\psi\rangle = (0.6)(0.8) + (0.8)(0.6) = 0.48 + 0.48 = 0.96$.
3. Sanity: this state sits near the $+x$ side of its meridian, so a large positive $\langle X \rangle$ is exactly right; and the recipe view agrees — H-then-measure on this state gives $P(0) = 0.98$ (Chapter 4, Problem 9), so $\langle X \rangle = 0.98 - 0.02 = 0.96$ ✓

Answer: $\langle X \rangle = 0.96$, by matrix arithmetic and by circuit recipe alike. Two roads, one number — when your two roads disagree in the lab, you have found a bug or a noise source.

Observables scale to registers by the tensor logic of Chapter 9. The workhorse two-qubit observable is $Z \otimes Z$ (“ZZ”): value $+1$ when the two measured bits *agree*, -1 when they *disagree* — the correlation meter. For a Bell state $|\Phi^+\rangle$, every shot agrees: $\langle ZZ \rangle = +1$; for independent coins, agreements and disagreements balance: $\langle ZZ \rangle = 0$. Sums of such Pauli products are how real problems enter the machine: a molecule’s energy, a MaxCut cost (Chapter 20), a classifier’s decision score (Part Six) — each is a recipe like $E = 0.4 \langle ZZ \rangle - 1.1 \langle XI \rangle + \dots$, measured term by term with the basis-change recipes of Lesson 3 and assembled classically.

REALITY CHECK

Averages cost shots, and the bill is the field’s quiet tax. One expectation value to precision ε needs about $1/\varepsilon^2$ shots: precision 0.01 costs $\sim 10,000$ shots *per term, per circuit setting*. A modest chemistry Hamiltonian with a few hundred Pauli terms, evaluated at every step of a variational optimization (Chapter 20), multiplies into millions of shots — often the true bottleneck, ahead of qubit counts. When Part Six discusses training quantum models, remember: every gradient the optimizer sees is bought, shot by shot, with this arithmetic.

CASE STUDY — THE AVERAGES THAT MEASURE REALITY’S STRANGENESS · CHSH, 1969–TODAY

The most consequential expectation values in physics history are four of this chapter’s correlation meters. The CHSH version of Bell’s test (Clauser, Horne, Shimony, Holt, 1969) asks two distant parties to measure Pauli-style observables along tilted axes and combine four measured correlations into one score, $S = \langle A_1 B_1 \rangle + \langle A_2 B_1 \rangle + \langle A_2 B_2 \rangle - \langle A_1 B_2 \rangle$. Every classical hidden-answer theory obeys $|S| \leq 2$; quantum mechanics predicts up to $2\sqrt{2} \approx 2.83$ for Bell pairs at the optimal angles. The Nobel-recognized experiments of Chapter 11 are, operationally, exactly this chapter’s workflow — basis rotations, shot counting, averages with error bars — pushed to heroic precision. When you compute a $\langle ZZ \rangle$ on the app or on cloud hardware tonight, you are handling the same experimental currency that settled a seventy-year argument about the nature of reality.

ACTIVITY (APP) — THE AVERAGE-O-METER

1. Open the *Observable Lab* in the QCFE app. Prepare $R_y(60^\circ)|0\rangle$ and read the exact $(\langle X \rangle, \langle Y \rangle, \langle Z \rangle)$ from the display; check $\langle Z \rangle = \cos 60^\circ = 0.5$ by hand.
2. Run 100-shot estimates of each average five times; watch the estimates scatter around the exact values with spread $\approx 1/\sqrt{100} = 0.1$.
3. Increase to 10,000 shots and watch the spread shrink tenfold — the $1/\varepsilon^2$ tax in action.
4. Prepare $|\Phi^+\rangle$ in the two-qubit panel and measure $\langle ZZ \rangle$ and $\langle XX \rangle$ (the app applies the H's for you). Both near +1? You have verified entangled correlations in two bases — the heart of the CHSH test.

You need: The QCFE app (Android or Windows), A calculator

► App tool: *Observable Lab* — live $(\langle X \rangle, \langle Y \rangle, \langle Z \rangle)$ readouts with shot-noise simulation, plus two-qubit correlation meters

8.6 Uncertainty, Made Quantitative

“You cannot know Z and X at once” has been this book’s slogan since Chapter 4. Observables let us upgrade the slogan to a theorem with numbers in it. Every observable carries, in a given state, not just an average but a **spread** — and quantum mechanics sets a floor under certain *products* of spreads.

EQUATION: SPREAD, AND THE UNCERTAINTY FLOOR

$$\Delta A = \sqrt{\langle A^2 \rangle - \langle A \rangle^2}, \quad \Delta Z \Delta X \geq |\langle Y \rangle|$$

In words: Spread = the histogram’s standard deviation. For the Pauli pair Z and X , the spreads’ product can never fall below the size of Y ’s average

Symbol	Meaning	Unit
ΔA	standard deviation of A’s outcomes	none
$\langle A^2 \rangle$	average of the <i>squared</i> outcomes	none
$ \langle Y \rangle $	the floor, set by the third Pauli	none

For any Pauli P , outcomes are ± 1 , so $\langle P^2 \rangle = 1$ and $\Delta P = \sqrt{1 - \langle P \rangle^2}$ — spread and average trade off along a circle. The floor is Robertson’s 1929 inequality specialized to qubits.

WORD TO KNOW

Uncertainty relation — A provable floor under the product of two observables’ spreads, holding for every state — a property of the state itself, not of clumsy measuring devices. Zero spread in one incompatible observable forces full spread in the other.

WORKED EXAMPLE — THE EXTREMES, THEN THE EQUATOR

Compute ΔZ and ΔX for $|0\rangle$, for $|+\rangle$, and check the floor in both cases.

1. $|0\rangle$: $\langle Z \rangle = 1 \Rightarrow \Delta Z = \sqrt{1 - 1} = 0$; $\langle X \rangle = 0 \Rightarrow \Delta X = 1$. Product: 0. Floor: $\langle Y \rangle = 0$. $0 \geq 0$ ✓
2. $|+\rangle$: mirror image — $\Delta Z = 1$, $\Delta X = 0$, product 0, floor 0 ✓
3. Moral so far: certainty in one of the pair is legal, but only where the floor happens to vanish.

Answer: the poles and the $\pm$ axis are the lucky places where one spread can be zero. The floor becomes interesting — and binding — between them, as the next example shows.

WORKED EXAMPLE — A STATE THAT SITS EXACTLY ON THE FLOOR

Take the equator state $\frac{1}{\sqrt{2}}(|0\rangle + e^{i45^\circ}|1\rangle)$, whose Bloch coordinates are $\langle X \rangle = \langle Y \rangle = 0.707$, $\langle Z \rangle = 0$. Test the uncertainty relation.

1. $\Delta Z = \sqrt{1 - 0^2} = 1$.
2. $\Delta X = \sqrt{1 - 0.707^2} = \sqrt{0.5} = 0.707$.
3. Product: $1 \times 0.707 = 0.707$. Floor: $|\langle Y \rangle| = 0.707$.

Answer: $\Delta Z \Delta X = 0.707 = |\langle Y \rangle|$ — the inequality holds *with equality*: this state is exactly as certain about Z and X together as nature permits. States on the floor are called minimum-uncertainty states, and lasers, atomic clocks, and gravitational-wave detectors are engineered to live there.

REALITY CHECK

Uncertainty is a property of the *state*, not a flaw of the *apparatus*: even a perfect machine measuring perfect copies shows the spreads, because each copy genuinely lacks a definite pre-answer (Chapter 11 closed that loophole). Practical corollary: to know a qubit’s state you must estimate all three averages $\langle X \rangle$, $\langle Y \rangle$, $\langle Z \rangle$ from three separate histogram runs — the “tomography” of this chapter’s Bloch-coordinates lesson, and the reason state estimation costs shots in triplicate.

8.7 The Measurement Compiler: Rotate, Then Read

Hardware measures one thing: Z, qubit by qubit. Yet this book keeps saying “measure X” or “estimate $\langle Y \rangle$.” The bridge is a two-line compiler: *rotate the state so the desired axis becomes the z-axis, then read Z as usual.*

EQUATION: THE ROTATE-THEN-READ RULES

measure X : apply H , read Z measure Y : apply $S^\dagger$ then H , read Z
In words: One (or two) fixed gates convert any Pauli measurement into the standard one

Symbol	Meaning	Unit
H	swaps the x- and z-axes	none
$S^\dagger$	the -90° z-turn (undoes S)	none
read Z	the ordinary standard-basis measurement	none

Why it works: $HXH = Z$, so X 's eigenstates $|\pm\rangle$ land on $|0\rangle, |1\rangle$; for Y , the $S^\dagger$ first turns Y 's equator points onto X 's, and H finishes. Every "measurement setting" in Chapters 20–27 is exactly such a prefix.

WORKED EXAMPLE — COMPILING A Y-MEASUREMENT

The equator state $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{i45^\circ}|1\rangle)$ has $\langle Y \rangle = \sin 45^\circ = 0.707$. Compile the Y -measurement and predict the compiled histogram.

1. Prefix the circuit with $S^\dagger$, then H ; then measure normally.
2. Prediction from the expectation value: $\langle Y \rangle = P(\text{outcome } +) - P(\text{outcome } -)$, so $P(0_{\text{after prefix}}) = \frac{1+\langle Y \rangle}{2} = \frac{1+0.707}{2} = 0.854$.
3. Check the plumbing on a landmark: for $|+i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$ (the $+y$ pole), $S^\dagger$ gives $|+\rangle$ and H gives $|0\rangle$ — the $+y$ pole reads "0" with certainty, as it must ✓

Answer: prefix $S^\dagger, H$; expect 85.4% zeros. Rotate-then-read is why a machine with one hard-wired measurement can still estimate every Bloch coordinate, every Pauli term of a molecule's energy, and every CHSH correlator — the entire observable menu, served by two little gates.

REALITY CHECK

The compiler view also prices tomography honestly: the three Bloch coordinates need three differently-prefixed histogram runs — no circuit measures $\langle X \rangle, \langle Y \rangle, \langle Z \rangle$ simultaneously, and the uncertainty lesson explains why nature forbids it. When Chapter 20 batches Hamiltonian terms into "measurement settings," it is literally grouping terms whose prefixes agree qubit-by-qubit. One idea, half the field's plumbing.

CHAPTER SUMMARY

- An observable is a measurement with numerical values attached; the Pauli matrices double as the standard observables, with values ± 1 .
- $\langle Z \rangle = P(0) - P(1)$: histograms compress into averages; estimates carry shot-noise bars $\approx 1/\sqrt{N}$.
- $\langle X \rangle$ and $\langle Y \rangle$ are measured by rotating first (H for X ; $S^\dagger$ then H for Y) — separate runs for separate observables.
- $(\langle X \rangle, \langle Y \rangle, \langle Z \rangle)$ are the state's Bloch coordinates; measuring all three is tomography, and arrow length < 1 diagnoses noise.
- The algebraic engine: $\langle A \rangle = \langle \psi | A | \psi \rangle$. Multi-qubit products like $Z \otimes Z$ meter correlations; real applica-

tions measure sums of Pauli terms — at $\sim 1/\varepsilon^2$ shots per term, the working tax of variational algorithms and quantum machine learning.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] Compute $\langle Z \rangle$ for: (a) $|0\rangle$, (b) $|1\rangle$, (c) $|-\rangle$, (d) $0.8|0\rangle + 0.6|1\rangle$.

Problem 2 [Basic] A 500-shot run returns 335 zeros. Estimate $\langle Z \rangle$ and its shot-noise bar ($\approx 1/\sqrt{500}$).

Problem 3 [Basic] Which circuit measures $\langle X \rangle$? Draw it (one qubit, one extra gate, one meter), and state the conversion from counts to the average.

Problem 4 [Basic] Read the coordinates: a state has $(\langle X \rangle, \langle Y \rangle, \langle Z \rangle) = (0, -1, 0)$. Which gallery state is it (Chapter 3)?

Problem 5 [Basic] Compute ΔZ for the state with $P(0) = 0.9$: first find $\langle Z \rangle$, then the spread. Sketch where on the Bloch sphere such states live.

Problem 6 [Basic] Using $\Delta P = \sqrt{1 - \langle P \rangle^2}$, explain why no qubit state can have $\Delta Z = \Delta X = \Delta Y = 0$ — what would its Bloch vector have to be, and what rule forbids that?

Problem 7 [Basic] Write the compiled circuits (prefix + read) for estimating $\langle X \rangle$ and $\langle Y \rangle$ of any state, and state how the counts convert to the estimate.

Problem 8 [Basic] What histogram does the compiled X-measurement give for input $|0\rangle$? Compute it and reconcile with $\langle X \rangle_{|0\rangle} = 0$.

Problem 9 [Standard] Using the matrix formula, compute $\langle Z \rangle$ and $\langle X \rangle$ for $|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle$, and confirm the pair $(\langle X \rangle, \langle Z \rangle) = (0, 0)$ places the state on the y -axis. (Its $\langle Y \rangle$, by elimination and purity, must be what?)

Problem 10 [Standard] For the state $R_y(\theta)|0\rangle = \cos \frac{\theta}{2}|0\rangle + \sin \frac{\theta}{2}|1\rangle$, show $\langle Z \rangle = \cos \theta$ using $\cos^2 - \sin^2$ of half-angles — the identity quoted in Chapter 20's VQE example.

Problem 11 [Standard] Tomography drill: runs give $\langle X \rangle = -0.71$, $\langle Y \rangle = 0.00$, $\langle Z \rangle = +0.70$. Locate the state on the sphere (quadrant and angles, roughly), check purity, and propose the R_y preparation angle that fits.

Problem 12 [Standard] Two-qubit correlations: compute $\langle ZZ \rangle$ for (a) $|\Phi^+\rangle$, (b) $|\Psi^+\rangle$, (c) $|+\rangle \otimes |+\rangle$, using “+1 if bits agree, -1 if they disagree” on each state's probability table.

Problem 13 [Standard] Shot budgeting: a Hamiltonian has 40 Pauli terms, each needing precision 0.02. Estimate the shots per energy evaluation, and the total for a 200-step VQE optimization. Express the total in

hours at 10,000 shots per second.

Problem 14 [Standard] For $|\psi\rangle = 0.8|0\rangle + 0.6|1\rangle$: compute $\langle Z \rangle$, $\langle X \rangle$, $\langle Y \rangle$, all three spreads, and verify $\Delta Z \Delta X \geq |\langle Y \rangle|$.

Problem 15 [Standard] Show that for every pure qubit state, $\Delta X^2 + \Delta Y^2 + \Delta Z^2 = 2$, using $r_x^2 + r_y^2 + r_z^2 = 1$. (A conservation law for ignorance: squeezing spread out of one observable pumps it into the others.)

Problem 16 [Standard] A tomography budget: with n shots per basis, each average carries wobble $\approx 1/\sqrt{n}$. How many total shots (three bases) are needed to pin each Bloch coordinate to ± 0.03 ?

Problem 17 [Standard] For $|\psi\rangle = 0.6|0\rangle + 0.8|1\rangle$: predict the compiled X-measurement's $P(0)$ from $\langle X \rangle = 2 \times 0.6 \times 0.8 = 0.96$, and verify by computing $|\langle +|\psi \rangle|^2$ directly.

Problem 18 [Challenge] Prove that $\langle A \rangle$ is always real when $A^\dagger = A$ (true for X, Y, Z): take the complex conjugate of $\langle \psi|A|\psi \rangle$, use $(\langle u|v \rangle)^* = \langle v|u \rangle$ and the dagger rules of Appendix B, and conclude. Why must a physical average be real?

Problem 19 [Challenge] The uncertainty seesaw: compute $\langle Z \rangle$ and $\langle X \rangle$ for the family $R_y(\theta)|0\rangle$ at $\theta = 0^\circ, 45^\circ, 90^\circ$, and verify $\langle X \rangle^2 + \langle Z \rangle^2 = 1$ throughout. What does this identity say about knowing both averages sharply at once — and which famous physics principle is peeking through?

Problem 20 [Challenge] CHSH by the numbers: for a Bell pair, quantum mechanics gives correlations $\langle AB \rangle = \cos(\text{angle between the two measurement axes})$. Evaluate $S = \langle A_1 B_1 \rangle + \langle A_2 B_1 \rangle + \langle A_2 B_2 \rangle - \langle A_1 B_2 \rangle$ for the optimal settings where consecutive axes are 45° apart (angles: $A_1 = 0^\circ, A_2 = 90^\circ, B_1 = 45^\circ, B_2 = 135^\circ$), and confirm $S = 2\sqrt{2} \approx 2.83 > 2$. You have just reproduced, on paper, the number behind the 2022 Nobel Prize.

Problem 21 [Challenge] Find the equator state (phase φ) that *maximizes* the product $\Delta Z \Delta X$, compute that maximum, and compare it against the floor $|\langle Y \rangle|$ at the same φ . Where is the gap between product and floor largest?

Problem 22 [Challenge] Design the compiled measurement for the diagonal observable $\frac{1}{\sqrt{2}}(X + Z)$ (the “H-axis”): which single rotation gate maps its eigenstates onto $|0\rangle, |1\rangle$? (Hint: Euler’s lesson — a 45° turn about y.) Predict $P(0)$ for input $|0\rangle$, and check against $\langle \frac{X+Z}{\sqrt{2}} \rangle_{|0\rangle} = \frac{1}{\sqrt{2}}$.

PART FOUR

MANY QUBITS AND ENTANGLEMENT

CHAPTER NINE

Multi-Qubit Systems and Tensor Products

Two qubits, four amplitudes — the tensor product and the exponential state space

IN THIS CHAPTER YOU WILL LEARN TO

- Write two-qubit basis states $|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$ and read them as binary numbers.
- Combine single-qubit states into multi-qubit states with the tensor product $\otimes$.
- Build probability tables for multi-qubit measurements, and compute single-qubit (marginal) probabilities from them.
- Apply a gate to one qubit of a multi-qubit system (e.g. $H \otimes I$) and to both at once.
- Explain why n qubits need 2^n amplitudes, and what that means for simulation.

9.1 Two Qubits: Four Possibilities at Once

One qubit gave us a two-dimensional world: amplitudes for $|0\rangle$ and $|1\rangle$. Add a second qubit and the possible measurement outcomes become the two-bit patterns of Chapter 1: 00, 01, 10, and 11. Quantum mechanics attaches one complex amplitude to *each* pattern:

EQUATION: THE GENERAL TWO-QUBIT STATE

$$|\psi\rangle = a_{00}|00\rangle + a_{01}|01\rangle + a_{10}|10\rangle + a_{11}|11\rangle = \begin{pmatrix} a_{00} \\ a_{01} \\ a_{10} \\ a_{11} \end{pmatrix}$$

In words: A two-qubit state = one amplitude for each of the four bit patterns, stacked into a 4-component vector

Symbol	Meaning	Unit
$ xy\rangle$	the basis state “first qubit shows x , second shows y ”	none
a_{xy}	the amplitude of pattern xy	none — complex

Normalization extends naturally: $|a_{00}|^2 + |a_{01}|^2 + |a_{10}|^2 + |a_{11}|^2 = 1$. The Born rule too: $P(xy) = |a_{xy}|^2$.

The notation $|01\rangle$ is shorthand for “qubit A in $|0\rangle$ and qubit B in $|1\rangle$.” Read the labels as binary numbers and the four basis states are simply $|0\rangle, |1\rangle, |2\rangle, |3\rangle$ in counting order — some books even write them that way. (Ordering convention: in this book, the left symbol is qubit A = the top wire of a circuit. When you meet Qiskit in Chapter 14, note it numbers the other way — always check the convention before comparing outputs.)

WORD TO KNOW

Product state — A multi-qubit state that can be written as “qubit A in some state” $\otimes$ “qubit B in some state” — each qubit has its own story. States that cannot be split this way are entangled (Chapter 11).

9.2 The Tensor Product of States

If qubit A is in $|\phi_A\rangle = a|0\rangle + b|1\rangle$ and qubit B is independently in $|\phi_B\rangle = c|0\rangle + d|1\rangle$, what is the joint state? The combining rule is the **tensor product**: every amplitude of A multiplies every amplitude of B, and the labels concatenate.

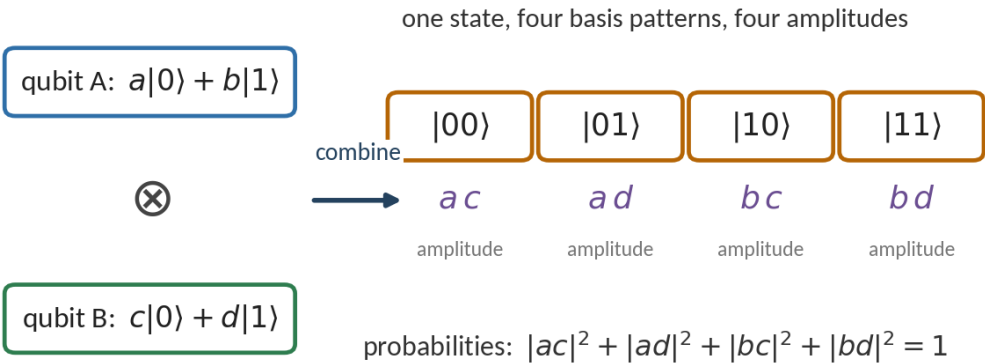
EQUATION: TENSOR PRODUCT OF TWO QUBIT STATES

$$(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$$

In words: Multiply out the brackets; amplitudes multiply, labels join up

Symbol	Meaning	Unit
$\otimes$	the tensor product (“combine independent systems”)	none
a, b	amplitudes of qubit A	none
c, d	amplitudes of qubit B	none

It is ordinary bracket expansion — the same skill as $(a + b)(c + d)$ in algebra — with kets glued side by side. Column-vector version: $\begin{pmatrix} a \\ b \end{pmatrix} \otimes \begin{pmatrix} c \\ d \end{pmatrix} = (ac, ad, bc, bd)^T$.



The tensor product visualized: each amplitude of A pairs with each amplitude of B, producing the four amplitudes of the joint state.

WORKED EXAMPLE — COMBINING $|+\rangle$ WITH $|0\rangle$

Qubit A is in $|+\rangle$; qubit B is in $|0\rangle$. Find the joint state.

1. Write the factors: $|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$ and $|0\rangle = 1 \cdot |0\rangle + 0 \cdot |1\rangle$.

2. Expand: $\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) \otimes |0\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|10\rangle$.

3. Check normalization: $\frac{1}{2} + \frac{1}{2} = 1$ ✓

Answer: $|+\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|10\rangle$: qubit A undecided, qubit B certainly 0 — exactly what independence should look like. Patterns 01 and 11 (where B shows 1) have amplitude zero.

WORKED EXAMPLE — COMBINING TWO SUPERPOSITIONS

Both qubits are in $|+\rangle$. Find the joint state and its probability table.

1. Expand: $|+\rangle \otimes |+\rangle = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$.

2. Born rule on each pattern: $P = \left|\frac{1}{2}\right|^2 = \frac{1}{4}$ for all four.

3. Sanity check: $4 \times \frac{1}{4} = 1$ ✓

Answer: the uniform superposition over all four patterns, each measured with probability 25%. In general, H on every one of n qubits makes the uniform superposition over all 2^n patterns — the standard opening move of the algorithms in Part Five.

COMMON MISTAKE

✗ Wrong: $|0\rangle \otimes |1\rangle$ and $|1\rangle \otimes |0\rangle$ are the same thing — “one qubit of each.”

✓ Right: Order is physical: $|01\rangle$ says A shows 0 and B shows 1; $|10\rangle$ says the opposite. They are different (orthogonal!) basis states, as different as the numbers 1 and 2 whose binary names they carry.

9.3 Probability Tables and Marginals

With four (or more) outcomes, the professional habit is a **probability table** — and from it, answers to part-questions like “what is the chance qubit A shows 1, regardless of B?”

Probability table for $|\psi\rangle = 0.8|00\rangle + 0.6|11\rangle$ (a preview of entanglement)

Pattern (A, B)	Amplitude	Probability
00	0.8	0.64
01	0	0
10	0	0
11	0.6	0.36

EQUATION: MARGINAL (SINGLE-QUBIT) PROBABILITY

$$P(A=1) = |a_{10}|^2 + |a_{11}|^2$$

In words: Probability that qubit A shows 1 = add the probabilities of every pattern whose first bit is 1

Symbol	Meaning	Unit
$P(A=1)$	probability that qubit A alone is measured as 1	none
a_{10}, a_{11}	amplitudes of the patterns with first bit 1	none

Add probabilities across patterns here, not amplitudes — the patterns are distinguishable outcomes, not interfering paths to the same outcome.

WORKED EXAMPLE — MARGINALS FROM THE TABLE

For $|\psi\rangle = 0.8|00\rangle + 0.6|11\rangle$: find $P(A=1)$, $P(B=1)$, and the probability both show the same digit.

1. $P(A=1)$: patterns 10 and 11 give $0 + 0.36 = 0.36$.
2. $P(B=1)$: patterns 01 and 11 give $0 + 0.36 = 0.36$.
3. Same digit: patterns 00 and 11 give $0.64 + 0.36 = 1.00$.

Answer: each qubit alone shows 1 with probability 0.36 — but they agree with probability 1. Individually random, jointly locked: hold that thought two chapters.

9.4 Gates on Multi-Qubit Systems

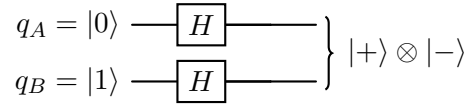
Gates extend by the same $\otimes$ logic. To apply H to qubit A while leaving B alone, the joint operation is $H \otimes I$ — “H here, nothing there.” You rarely need its 4×4 matrix: work term by term on the state, gate-ing the matching slot of each ket.

WORKED EXAMPLE — H ON ONE QUBIT OF A PAIR

Apply H to qubit A of $|01\rangle$ (i.e. compute $(H \otimes I)|01\rangle$).

1. Split the ket: $|01\rangle = |0\rangle_A \otimes |1\rangle_B$.
2. Gate the A slot: $H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$; B's slot rides along untouched.
3. Recombine: $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |1\rangle = \frac{1}{\sqrt{2}}|01\rangle + \frac{1}{\sqrt{2}}|11\rangle$.

Answer: $\frac{1}{\sqrt{2}} |01\rangle + \frac{1}{\sqrt{2}} |11\rangle$ — qubit A went into superposition; qubit B stayed a definite 1. On a circuit diagram this is simply an H box on A's wire and empty wire for B.



Parallel gates: one H per wire in the same time step realizes $H \otimes H$. Independent gates on independent qubits — nothing communicates (that begins in Chapter 10).

WORKED EXAMPLE — TWO GATES AT ONCE

Compute $(H \otimes H) |01\rangle$ — the circuit above.

1. Gate each slot: $H |0\rangle = |+\rangle$ and $H |1\rangle = |-\rangle$.
2. Combine: $|+\rangle \otimes |-\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$.
3. Expand: $\frac{1}{2} (|00\rangle - |01\rangle + |10\rangle - |11\rangle)$.

Answer: $\frac{1}{2} (|00\rangle - |01\rangle + |10\rangle - |11\rangle)$: all four patterns equally likely (each $P = \frac{1}{4}$), with a pattern of signs. Those signs are relative phases — data for interference, as always.

9.5 Three Qubits and Beyond: The Space Explodes

Everything scales by the same rules. Three qubits: eight basis states $|000\rangle$ through $|111\rangle$, eight amplitudes. And in general — the equation you met as a promise in Chapter 2, now as arithmetic you own:

EQUATION: DIMENSION OF THE STATE SPACE

$$\dim = 2^n$$

In words: n qubits are described by 2^n complex amplitudes, one per n -bit pattern

Symbol	Meaning	Unit
n	the number of qubits	qubits
dim	the number of basis patterns = number of amplitudes	none

Growth by doubling: every added qubit doubles the vector. 10 qubits: 1024 amplitudes. 40 qubits: about 10^{12} . 300 qubits: more amplitudes than atoms in the observable universe.

WORKED EXAMPLE — THE MEMORY BILL FOR SIMULATION

A laptop stores one amplitude in 16 bytes. How much memory does the full state vector of 30, 40, and 50 qubits need?

1. $n = 30$: $2^{30} \approx 1.07 \times 10^9$ amplitudes $\times 16 \text{ B} \approx 17 \text{ GB}$ — a strong laptop just manages.
2. $n = 40$: $2^{40} \approx 1.1 \times 10^{12}$ amplitudes $\rightarrow$ about 17.6 TB — a server cluster.
3. $n = 50$: $2^{50} \approx 1.1 \times 10^{15}$ amplitudes $\rightarrow$ about 18 PB (petabytes) — roughly the total memory of a top national supercomputer.

Answer: 17 GB $\rightarrow$ 17.6 TB $\rightarrow$ 18 PB for just 30 $\rightarrow$ 40 $\rightarrow$ 50 qubits. Each extra qubit doubles the bill — this is Feynman’s wall from Chapter 1, now computed by you personally.

REALITY CHECK

The 2^n explosion cuts both ways, and honesty requires saying both. *For* quantum computing: a 300-qubit machine manipulates a state no classical memory could even store — genuine, physical, exponential parallelism of amplitudes. *Against* the hype: you still cannot *read* 2^n amplitudes — measuring n qubits yields just n classical bits (one pattern), and the state collapses (Chapter 4). The amplitudes are working capital, not output. Algorithms must funnel that exponential interior into a few readable bits — which is why good quantum algorithms are rare and precious.

CASE STUDY — THE 50-QUBIT FRONTIER • SUMMIT SUPERCOMPUTER VS SYCAMORE, 2019

The memory bill you just computed is not hypothetical — it decided a famous scientific duel. Google’s 2019 quantum-advantage experiment (Chapter 2) used 53 qubits precisely because a 53-qubit state vector ($2^{53} \approx 9 \times 10^{15}$ amplitudes, over a hundred petabytes at full precision) sat just beyond the world’s largest classical memories, including the Summit supercomputer’s. IBM’s rebuttal — “2.5 days, not 10,000 years” — worked by *not* storing the full vector, trading memory for disk space and cleverness. The whole debate, on both sides, is Lesson 5 of this chapter with a budget: whoever understands 2^n understands where classical simulation ends.

ACTIVITY (APP) — TWO QUBITS, FOUR BARS

1. Open the *Two-Qubit Lab* in the QCFE app. Prepare $|0\rangle \otimes |0\rangle$ and run 1000 shots: one bar, all 00.
2. Apply H to qubit A only. Predict the four probabilities, then run and check (50% 00, 50% 10).
3. Apply H to both qubits. Predict, run, check: four equal bars of about 250.
4. Use the amplitude view: set A to $0.6|0\rangle + 0.8|1\rangle$ and B to $|+\rangle$, and verify one displayed amplitude against your own hand calculation of $a_{10} = 0.8 \times \frac{1}{\sqrt{2}} \approx 0.566$.

You need: The QCFE app (Android or Windows), Paper and pen

► *App tool: Two-Qubit Lab — prepare each qubit, see all four amplitudes and the live histogram, and pull the marginals apart*

9.6 Measuring Just One Qubit of a Pair

Nothing forces us to measure every qubit at once. A real program often reads out one qubit now and keeps computing with the rest. The rule is a natural extension of the Born rule, in two moves: *add up* the probabilities of all patterns consistent with the observed digit, then *keep only* those patterns' amplitudes, renormalized so the survivors' probabilities total 100%.

EQUATION: PARTIAL MEASUREMENT (MEASURING QUBIT A ALONE)

$$P(A=0) = |a_{00}|^2 + |a_{01}|^2, \quad |\psi_{\text{after}}\rangle = \frac{a_{00}|00\rangle + a_{01}|01\rangle}{\sqrt{P(A=0)}}$$

In words: Probability = sum over consistent patterns; new state = the consistent amplitudes, rescaled to length 1

Symbol	Meaning	Unit
$P(A=0)$	probability qubit A reads 0	none
a_{0y}	amplitudes of patterns whose first digit is 0	none
$\sqrt{P(A=0)}$	renormalization factor	none

The same recipe with digits 1, or with qubit B, works identically. Qubit B's amplitudes are not erased — they are filtered: only the branches consistent with what A showed survive.

WORD TO KNOW

Partial measurement — A measurement of some (not all) qubits of a multi-qubit state. The unmeasured qubits remain quantum, in a state filtered and renormalized according to the observed result.

WORKED EXAMPLE — MEASURING HALF OF A THREE-PATTERN STATE

Let $|\psi\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{\sqrt{2}}{2}|11\rangle$. Qubit A is measured. Find both outcomes' probabilities and the state of B afterward in each case.

1. Check normalization: $\frac{1}{4} + \frac{1}{4} + \frac{1}{2} = 1$ ✓
2. $P(A=0)$: patterns 00 and 01 give $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$.
3. Collapse for $A=0$: keep $\frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle$, divide by $\sqrt{1/2}$: $\frac{1}{\sqrt{2}}(|00\rangle + |01\rangle) = |0\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.
4. $P(A=1) = \frac{1}{2}$; the only consistent pattern is 11, so the state snaps to $|11\rangle$.

Answer: each result has probability $\frac{1}{2}$. If A shows 0, qubit B is left in $|+\rangle$ — still genuinely undecided, a fair coin. If A shows 1, qubit B is already certain: it *will* read 1. One measurement on A changed what B can do — that is correlation, and in Chapter 11 it becomes the main character.

WORKED EXAMPLE — ONE AT A TIME OR ALL AT ONCE — SAME ANSWER

For the same $|\psi\rangle$, compute $P(\text{pattern } 01)$ two ways: directly, and as a chain (first A, then B).

1. Direct Born rule: $P(01) = |\frac{1}{2}|^2 = \frac{1}{4}$.

2. Chain: $P(A=0) = \frac{1}{2}$; given that, B is in $|+\rangle$, so $P(B=1 | A=0) = \frac{1}{2}$.

3. Multiply: $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ ✓

Answer: $\frac{1}{4}$ both ways. Measuring qubit by qubit or all at once gives the same statistics — quantum mechanics is consistent about it, and this consistency is what lets real hardware read out its qubits in any order.

REALITY CHECK

Partial measurement is why entanglement cannot send messages. When A is measured, B's *conditional* state changes — but someone holding only B, ignorant of A's result, still sees the old marginal probabilities, exactly as computed by the marginal rule of this chapter. Only when A's result arrives by ordinary (light-speed-limited) communication can B's owner make use of the sharper prediction. Chapter 13 builds its entire protocol on this fine print.

9.7 The W State: A Different Way to Share

GHZ is not the only celebrity among three-qubit states. Its rival distributes entanglement in a fundamentally different style:

EQUATION: THE W STATE

$$|W\rangle = \frac{1}{\sqrt{3}}(|001\rangle + |010\rangle + |100\rangle)$$

In words: Exactly one excitation, shared equally among three qubits — nobody knows who holds it

Symbol	Meaning	Unit
$ W\rangle$	the three-qubit W state	none
$\frac{1}{\sqrt{3}}$	equal amplitudes for the three placements	none

Named after Wolfgang Dür, who proved (2000) that W and GHZ are inequivalent: no local operations convert one into the other. Three qubits can be entangled in two genuinely different ways.

WORKED EXAMPLE — W VS GHZ: THE ROBUSTNESS TEST

Compute each qubit's marginal $P(1)$ in $|W\rangle$, then compare how W and GHZ survive losing their third qubit.

1. Marginals: qubit C shows 1 only in pattern 001: $P(C=1) = \frac{1}{3}$; by symmetry each qubit's marginal is $\frac{1}{3}$.

2. Lose qubit C from GHZ ($\frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$): the partial-measurement rule says C's result (0 or 1, half/half) collapses A,B to $|00\rangle$ or $|11\rangle$ — definite states, entanglement *annihilated*.

3. Lose qubit C from W: if C reads 1 (probability $\frac{1}{3}$), A,B collapse to $|00\rangle$; but if C reads 0 (probability $\frac{2}{3}$), A,B collapse to $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ — a perfect *Bell state*.

Answer: marginals $\frac{1}{3}$ each; after losing one qubit, GHZ keeps nothing while W keeps full two-qubit entanglement two times in three. GHZ is the all-or-nothing pact, W the resilient collective — and the choice between such structures is a real design decision in quantum networks, where losing a photon is Tuesday.

WORD TO KNOW

W state — The three-qubit state sharing one excitation equally. Entangled in a class of its own (inequivalent to GHZ), and famously robust: losing any one qubit usually leaves the other two still entangled.

REALITY CHECK

The GHZ/W split is the first glimpse of a wild landscape: with more qubits, the number of inequivalent entanglement classes explodes, and classifying them remains research mathematics. For this book's purposes one moral suffices — “entangled” is not one thing. When a paper says a machine “created entanglement among 20 qubits,” the professional's follow-up is Chapter 30's favorite: *which* entanglement, verified *how*?

CHAPTER SUMMARY

- Two qubits have four basis patterns $|00\rangle, |01\rangle, |10\rangle, |11\rangle$; a state assigns each an amplitude; Born rule and normalization extend pattern-by-pattern.
- Independent qubits combine by the tensor product: expand the brackets, multiply amplitudes, join labels. $|+\rangle \otimes |+\rangle =$ the uniform superposition.
- Probability tables organize outcomes; marginals add probabilities over the patterns that match the part-question.
- Gates on one qubit of many: gate the matching slot, carry the rest ($H \otimes I$). Parallel boxes in one circuit column = tensor product of gates.
- n qubits $\Rightarrow 2^n$ amplitudes: the exponential space that makes simulation crushing and quantum hardware promising — while measurement still returns only n bits.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] Write the four two-qubit basis states as 4-component columns (in the order 00, 01, 10, 11).

Problem 2 [Basic] Compute $|1\rangle \otimes |+\rangle$, expanded into basis kets with amplitudes.

Problem 3 [Basic] For $|\psi\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$: what is $P(01)$? What is $P(A=0)$?

Problem 4 [Basic] How many amplitudes describe 5 qubits? Write the first three basis patterns and the last

one.

Problem 5 [Basic] For $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$, qubit A is measured. Give both probabilities and B's state after each outcome. What does the answer say about whether A and B were correlated?

Problem 6 [Basic] For $|\psi\rangle = 0.8|00\rangle + 0.6|11\rangle$, qubit B is measured and shows 1. What was the probability of that, and what is the full two-qubit state afterward?

Problem 7 [Basic] Verify $|W\rangle$ is normalized, and compute $P(\text{exactly one } 1)$ and $P(\text{all zeros})$ for a full three-qubit measurement.

Problem 8 [Basic] Using the partial-measurement rule, confirm the two collapse outcomes when qubit C of $|W\rangle$ is measured, with their probabilities.

Problem 9 [Standard] Compute $(0.6|0\rangle + 0.8|1\rangle) \otimes (0.8|0\rangle + 0.6|1\rangle)$ and verify the four probabilities sum to 1.

Problem 10 [Standard] Build the probability table of $|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle - \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle$, then find $P(A=1)$ and $P(B=0)$.

Problem 11 [Standard] Apply $X \otimes I$ to $|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|01\rangle$ (flip slot A in every term). Then apply $I \otimes X$ to the original instead. Compare the two outputs in words.

Problem 12 [Standard] Compute $(H \otimes I)|10\rangle$, then apply $(H \otimes I)$ again to your result and confirm you return to $|10\rangle$ (the identity $HH = I$ working slot-wise).

Problem 13 [Standard] A 3-qubit state is $|\psi\rangle = \frac{1}{\sqrt{2}}|000\rangle + \frac{1}{\sqrt{2}}|111\rangle$. Find $P(\text{pattern } 000)$, $P(\text{middle qubit} = 1)$, and $P(\text{all three agree})$.

Problem 14 [Standard] For the product state $|+\rangle \otimes |+\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$, show by the partial-measurement rule that measuring A (either result) leaves B in exactly $|+\rangle$ — unchanged. State the general moral about product states.

Problem 15 [Standard] Let $|\psi\rangle = \frac{1}{\sqrt{8}}(\sqrt{3}|00\rangle + |01\rangle + 2|11\rangle)$. Check normalization, then find $P(A=0)$, the collapsed state if A shows 0, and $P(B=0 | A=0)$.

Problem 16 [Standard] For $|\psi\rangle = 0.8|00\rangle + 0.6|11\rangle$, compute the probability of every measurement history when the qubits are measured A-first, and again B-first. Confirm the joint statistics agree.

Problem 17 [Standard] Generalize: write $|W_4\rangle$ for four qubits, give each marginal, and the probability that losing one qubit (measuring it as 0) leaves the rest in $|W_3\rangle$.

Problem 18 [Challenge] Show that $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ can NOT be written as $(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle)$: assume it can, match all four amplitudes ($ac = \frac{1}{\sqrt{2}}$, $ad = 0$, $bc = 0$, $bd = \frac{1}{\sqrt{2}}$), and derive a contradiction. (You have just proven your first entangled state entangled — two chapters early.)

Problem 19 [Challenge] Using $2^{10} \approx 10^3$: a simulator stores amplitudes at 16 bytes each and a data center offers 1 exabyte (10^{18} bytes). What is the largest n it can hold? What does your answer say about the marketing phrase “we simulate any quantum computer classically”?

Problem 20 [Challenge] The uniform 3-qubit superposition $\frac{1}{\sqrt{8}} \sum_{x=000}^{111} |x\rangle$ is measured. (a) What is the probability the result, read as binary, is an even number? (b) A colleague claims “measuring all three qubits gives three bits, so eight patterns of information per shot.” Correct the claim: how many *patterns* appear per shot, and how much information is that?

Problem 21 [Challenge] Alice and Bob share $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. Before any measurement, compute Bob’s $P(B=1)$. After Alice measures (result unknown to Bob), what is Bob’s $P(B=1)$, averaged over Alice’s outcomes? Explain carefully why these agree, and why this is precisely the “no faster-than-light messages” guarantee.

Problem 22 [Challenge] Show that $|W\rangle$ cannot be written as any tensor product $|a\rangle \otimes |bc\rangle$: assume it could, and derive a contradiction from the zero amplitude of $|011\rangle$ together with the nonzero amplitudes of $|001\rangle$ and $|010\rangle$. (You have proven genuine three-party entanglement — no shortcut through pairing.)

CHAPTER TEN

Controlled Gates

CNOT, controlled-Z, SWAP, and Toffoli — the gates that make qubits talk

IN THIS CHAPTER YOU WILL LEARN TO

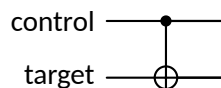
- Apply the CNOT gate to basis states and superpositions, and write its 4×4 matrix.
- Read control dots and target symbols in circuit diagrams.
- Use the controlled-Z and SWAP gates, and recognize their circuit symbols.
- Trace the Toffoli gate and explain how it rebuilds classical logic reversibly.
- Explain why controlled gates are the ingredient that single-qubit gates can never replace.

10.1 The CNOT Gate

Every gate so far acts on one qubit alone; a register of such loners could never compute anything interesting together. The missing ingredient is a gate where one qubit's state *conditions* what happens to another. The workhorse is the controlled-NOT:

WORD TO KNOW

CNOT — The controlled-NOT (also written CX): a two-qubit gate that flips the target qubit if and only if the control qubit is $|1\rangle$. The standard entangling gate of quantum computing.



The CNOT symbol: a solid dot on the control wire, connected to a $\oplus$ on the target wire. Memory aid: $\oplus$ is the XOR symbol of Chapter 1 — and on basis states CNOT computes exactly target $\oplus$ control.

EQUATION: THE CNOT GATE

$$\text{CNOT} : \quad |00\rangle \rightarrow |00\rangle \quad |01\rangle \rightarrow |01\rangle \quad |10\rangle \rightarrow |11\rangle \quad |11\rangle \rightarrow |10\rangle \quad \text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

In words: If the control (first) bit is 1, flip the target (second) bit; otherwise do nothing

Symbol	Meaning	Unit
control	the first qubit: decides, never changes	none
target	the second qubit: gets flipped when control is 1	none

The matrix is the identity with the last two rows swapped — “swap the amplitudes of $|10\rangle$ and $|11\rangle$.” It is unitary, and its own inverse: CNOT twice = nothing.

On basis states, CNOT is classical: control stays, target becomes XOR. The quantum personality appears when the control is in superposition — then the gate acts on every branch at once:

WORKED EXAMPLE — CNOT WITH A SUPERPOSED CONTROL

Compute CNOT applied to $|+\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|10\rangle$.

1. Apply the rule branch by branch (linearity: gates act on each term).
2. Branch $|00\rangle$: control is 0 — unchanged: $|00\rangle$.
3. Branch $|10\rangle$: control is 1 — flip the target: $|11\rangle$.
4. Collect: $\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$.

Answer: $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ — the state Chapter 9’s challenge problem proved impossible to split into two independent qubits. One CNOT on a superposed control has produced *entanglement*. Chapter 11 is entirely about this output.

COMMON MISTAKE

X Wrong: “The control qubit is never affected by a CNOT.”

✓ Right: True on basis states, false in general. Feed the control a superposition in the $|+\rangle / |-\rangle$ sense and phases can flow *backward* from target to control (the famous “phase kickback,” Chapter 19). Safe statement: in the standard basis, control decides and target flips; in rotated bases, influence runs both ways. Quantum gates have no one-way doors.

10.2 Truth Table, Matrix, and a Copy Warning

CNOT on the four basis states

In: control	In: target	Out: control	Out: target
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

Look at the table with Chapter 1 eyes: out-target = in-control XOR in-target, and *nothing is erased* — from the output you can always reconstruct the input (apply CNOT again!). Compare the classical AND, which crushed three input pairs into one output. Controlled gates are how quantum computing rebuilds logic *without* deleting information — unitarity in action.

One tempting misreading: on inputs $|x\rangle|0\rangle$, CNOT gives $|x\rangle|x\rangle$ — it looks like a copying machine. For basis states, it is. For superpositions, it is not: as the worked example showed, $|+\rangle|0\rangle$ became an entangled pair, *not* two independent copies $|+\rangle|+\rangle$ (expand the latter — it contains $|01\rangle$ and $|10\rangle$ terms the entangled state lacks). Chapter 12 sharpens this into a theorem: perfect copying of unknown quantum states is impossible for *any* machine.

10.3 Controlled-Z and SWAP

Any gate can be promoted to a controlled version: “if control is $|1\rangle$, apply the gate to the target.” After CNOT, the most useful is controlled-Z:

EQUATION: THE CONTROLLED-Z GATE

$$\text{CZ } |11\rangle = -|11\rangle, \quad \text{all other basis states unchanged} \quad \text{CZ} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

In words: CZ flips the sign of the $|11\rangle$ amplitude only — a phase flip that needs both qubits to be 1

Symbol	Meaning	Unit
CZ	the controlled-Z gate	none

Beautifully symmetric: “control” and “target” are interchangeable (both must be 1). Its circuit symbol is honest about this: two dots, no $\oplus$. Hardware loves CZ — for several platforms it is the native two-qubit gate, with CNOT built as *H-on-target, CZ, H-on-target* (the sandwich rule again).



Left: controlled-Z — two dots, perfectly symmetric. Right: SWAP — two crosses, exchanging the qubits' states.

WORD TO KNOW

SWAP — The two-qubit gate that exchanges the states of its qubits: $|xy\rangle \rightarrow |yx\rangle$. Buildable from three CNOTs in alternating directions — a classic exercise you will do below.

SWAP matters in real hardware because most chips only allow two-qubit gates between *neighboring* qubits; compilers insert SWAP chains to bring distant qubits together, and every inserted SWAP costs three CNOTs of precious fidelity (Chapter 28).

WORKED EXAMPLE — CZ IN ACTION — PHASES NEED NO FLIPS

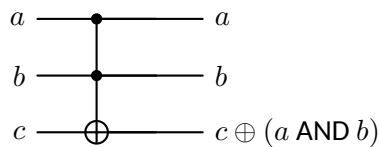
Apply CZ to $|+\rangle \otimes |+\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$.

1. Only the $|11\rangle$ amplitude changes sign.
2. Output: $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$.
3. Standard-basis probabilities: still $\frac{1}{4}$ each — yet this state is now entangled (it cannot be factored; try the Chapter 9 method).

Answer: $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$. One quiet minus sign, invisible to a histogram, and the two qubits are inseparably linked. Entanglement can hide entirely in phase.

10.4 The Toffoli Gate: Classical Logic, Rebuilt Reversibly

Two controls, one target: flip the target only if *both* controls are 1. That is the Toffoli gate (CCX), and it carries classical computing into the quantum world.



The Toffoli (CCX) gate: two control dots, one target. With the target prepared as $c = 0$, the output bottom wire is exactly $a \text{ AND } b$ — with the inputs still intact above it.

WORKED EXAMPLE — AND, WITHOUT BURNING THE EVIDENCE

Feed the Toffoli gate $a = 1, b = 1, c = 0$, then $a = 1, b = 0, c = 0$. Outputs?

1. Case 1: both controls are 1 $\Rightarrow$ flip target: output $(1, 1, 0 \oplus 1) = (1, 1, 1)$. Bottom wire reads $1 = 1 \text{ AND } 1$ ✓
2. Case 2: controls 1, 0 — not both 1 $\Rightarrow$ target unchanged: output $(1, 0, 0)$. Bottom wire $0 = 1 \text{ AND } 0$ ✓
3. Note what survived: in both cases a and b exit unchanged — the inputs are still there, so the operation can be undone (apply Toffoli again).

Answer: the Toffoli computes AND onto a fresh qubit while preserving its inputs — reversible logic. With

Toffoli (plus X), every classical circuit — adders, comparators, entire arithmetic units — can be rebuilt inside a quantum computer. Shor’s algorithm (Chapter 19) contains exactly such reversible arithmetic.

REALITY CHECK

“Reversible AND” is not a quantum luxury — it answers a deep classical question too. Landauer (1961) showed that *erasing* information has an unavoidable energy cost; Bennett (1973) showed computation can, in principle, avoid erasure entirely — using exactly such reversible gates. Quantum computers are forced to this discipline by unitarity; the bonus is that Toffoli-style arithmetic can then run on *superposed* inputs: one circuit evaluating a function on many arguments’ amplitudes at once. That trick powers the oracles of Grover and Shor in Part Five.

10.5 Why Controlled Gates Complete the Toolbox

Single-qubit gates rotate each Bloch sphere separately: n lonely globes. Any circuit made only of them keeps every qubit forever independent — a product state at all times, describable by just $2n$ numbers instead of 2^n . No entanglement, and (a theorem makes it precise) nothing a classical computer couldn’t simulate with ease.

Add one entangling gate — CNOT is the standard choice — and the picture transforms:

WORD TO KNOW

Universal gate set (full) — $\{H, T, \text{CNOT}\}$: with these three gates alone, any quantum computation on any number of qubits can be built to any accuracy. H and T steer single spheres (Chapter 7); CNOT ties spheres together.

The two-qubit repertoire at a glance

Gate	Symbol	Basis action	Special role
CNOT (CX)	dot + $\oplus$	target $\rightarrow$ target $\oplus$ control	the standard entangler
CZ	dot + dot	sign flip on $ 11\rangle$	symmetric; hardware-native on several platforms
SWAP	cross + cross	$ xy\rangle \rightarrow yx\rangle$	routing on real chips; = 3 CNOTs
Toffoli (CCX)	dot + dot + $\oplus$	target $\rightarrow$ target $\oplus$ (a AND b)	reversible classical logic; oracle building

CASE STUDY — THE FIRST TWO-QUBIT LOGIC GATE • NIST, BOULDER, USA, 1995

Proposing CNOT on paper (Cirac and Zoller, spring 1995) and making one are different sports. By December

1995, Christopher Monroe and David Wineland’s NIST group had coaxed a single trapped beryllium ion into performing a working controlled-NOT — control and target encoded in two different degrees of freedom of the *same* ion, steered by precisely timed laser pulses. Fidelity was about 90% — heroic then, hopeless for algorithms, and the ancestor of every entangling gate since. Thirty years later, the best trapped-ion two-qubit gates exceed 99.9% fidelity, and the CNOT count of an algorithm remains the first number any hardware engineer asks about. When Chapter 28 discusses gate errors, it is overwhelmingly *this* gate family being graded.

ACTIVITY (APP) — CONTROL AND TARGET

1. In the *Two-Qubit Lab*, build control = $|1\rangle$, target = $|0\rangle$, add a CNOT, run: histogram all 11.
2. Change the control to $|0\rangle$: all 00 — the gate really is conditional.
3. Now set the control to $|+\rangle$ (H before the CNOT). Predict the histogram from this chapter’s worked example, run 1000 shots, and check: about half 00, half 11, *never* 01 or 10.
4. Replace CNOT by CZ with both qubits in $|+\rangle$. Confirm the histogram shows four equal bars — then switch on the amplitude view and find the lone minus sign.

You need: The QCFE app (Android or Windows)

► *App tool: Two-Qubit Lab — drag CNOT, CZ, SWAP and Toffoli onto the wires; watch amplitudes, phases, and histograms respond*

10.6 Building Oracles: From Logic to Circuits

Every algorithm in Part Six leans on an **oracle** — a circuit that knows the problem’s yes/no function $f(x)$. Students often imagine the oracle as a mysterious black box handed down from above. It is not. *You* build it, from the problem’s own logic, using exactly the controlled gates of this chapter. There are two standard styles.

EQUATION: THE TWO ORACLE STYLES

$$\text{bit oracle: } U_f |x\rangle |a\rangle = |x\rangle |a \oplus f(x)\rangle \qquad \text{phase oracle: } O_f |x\rangle = (-1)^{f(x)} |x\rangle$$

In words: Bit oracle: write the answer into a helper qubit. Phase oracle: stamp the answer into the sign of the amplitude

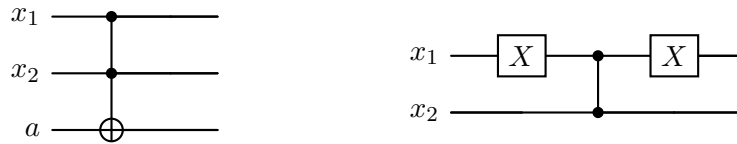
Symbol	Meaning	Unit
$f(x)$	the yes/no function (1 = “marked”)	none
$\oplus$	XOR (add modulo 2)	none
$ a\rangle$	helper (ancilla) qubit	none

Both are reversible, as unitarity demands: apply either twice and everything returns. The phase style leaves no helper to clean up, which is why Grover's algorithm (Chapter 18) prefers it.

WORD TO KNOW

Oracle — A reversible circuit implementing a problem's yes/no function, either into a helper qubit (bit oracle) or into the amplitudes' signs (phase oracle). Built from ordinary controlled gates — never a magic lookup table.

For $f(x_1, x_2) = x_1 \text{ AND } x_2$, the bit oracle is the Toffoli gate: both controls must be 1 to flip the helper. The phase oracle is even smaller: CZ flips the sign of $|11\rangle$ alone. To mark a different pattern, sandwich the gate between X's that temporarily relabel it:



Left: the bit oracle for AND is a Toffoli. Right: a phase oracle marking the pattern $|01\rangle$ — the X's temporarily relabel $|01\rangle$ as $|11\rangle$, CZ stamps the minus sign, and the X's restore the labels.

WORKED EXAMPLE — TRACING THE AND BIT-ORACLE

Feed each basis pattern $|x_1 x_2\rangle |0\rangle$ into the Toffoli oracle and tabulate the outputs.

1. $|00\rangle |0\rangle$: controls not both 1 $\rightarrow$ helper untouched: $|00\rangle |0\rangle$.
2. $|01\rangle |0\rangle$ and $|10\rangle |0\rangle$: still not both 1: unchanged.
3. $|11\rangle |0\rangle$: both controls 1 $\rightarrow$ helper flips: $|11\rangle |1\rangle$.
4. Reversibility check: running the outputs through again restores every input ✓

Answer: the helper ends holding exactly $f(x) = x_1 \wedge x_2$, and the inputs pass through unharmed — the reversible remake of the classical AND from Chapter 1, with no information destroyed.

WORKED EXAMPLE — A PHASE ORACLE FOR "EITHER QUBIT IS 1"

Build a phase oracle for $f = x_1 \text{ OR } x_2$, which must flip the signs of $|01\rangle$, $|10\rangle$, and $|11\rangle$ but not $|00\rangle$.

1. Try the gate menu: $Z \otimes I$ gives signs $(+, +, -, -)$ on $(00, 01, 10, 11)$; $I \otimes Z$ gives $(+, -, +, -)$; CZ gives $(+, +, +, -)$.
2. Multiply the three sign patterns column by column: 00: $+\cdot+\cdot+ = +$; 01: $+\cdot-\cdot+ = -$; 10: $-\cdot+\cdot+ = -$; 11: $-\cdot-\cdot- = -$.
3. That is exactly the OR pattern, so $O_f = (Z \otimes I)(I \otimes Z) \text{ CZ}$ — three gates, order irrelevant (they all commute).

Answer: $O_{\text{OR}} = (Z \otimes I)(I \otimes Z) \text{CZ}$. **Sign patterns multiply, so complicated markings are assembled from simple ones like factors. This “sign algebra” is how real compilers build the oracles inside Grover searches.**

REALITY CHECK

“But if we can build the oracle, don’t we already know the answer?” Careful: we know the *question*, compiled into logic — not the answer. A sudoku checker is easy to build without knowing the solution; a phase oracle marking “assignments that satisfy all clauses” is built clause by clause the same way. The oracle checks; the algorithm searches. Confusing the two is the single most common misreading of Grover’s algorithm.

10.7 Phase Kickback: The Control That Gets Painted

Here is the most productive optical illusion in quantum computing. In a controlled gate we call one qubit “control” and one “target” — but when the target sits in the right state, the effect lands *backwards*, painting a phase onto the control. Algorithms from Deutsch’s to Shor’s are kickback wearing different costumes.

WORKED EXAMPLE — THE CLASSIC KICKBACK

Compute CNOT acting on $|+\rangle \otimes |-\rangle$ (control $|+\rangle$, target $|-\rangle$), and interpret.

1. Expand the control: $|+\rangle |-\rangle = \frac{1}{\sqrt{2}} (|0\rangle |-\rangle + |1\rangle |-\rangle)$.
2. Apply CNOT branchwise: the $|0\rangle$ branch does nothing; the $|1\rangle$ branch applies X to the target: $X|-\rangle = -|-\rangle$.
3. Collect: $\frac{1}{\sqrt{2}} (|0\rangle |-\rangle - |1\rangle |-\rangle) = \left[\frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) \right] |-\rangle = |-\rangle |-\rangle$.
4. Read the miracle: the *target* is exactly as it began; the *control* flipped from $|+\rangle$ to $|-\rangle$. The minus sign, born on the target, factored out and moved into the control’s superposition.

Answer: $\text{CNOT} |+\rangle |-\rangle = |-\rangle |-\rangle$. **Because $|-\rangle$ is an *eigenstate* of X (eigenvalue -1), the target cannot change — only its eigenvalue escapes, as a relative phase on whichever control branches activated the gate. Control and target are labels on the diagram; the physics flows both ways.**

WORD TO KNOW

Phase kickback — When a controlled- U ’s target is an eigenstate of U , the eigenvalue’s phase attaches to the control’s $|1\rangle$ branch instead of disturbing the target. The engine of Deutsch–Jozsa, phase estimation, and Shor’s period-finding.

Why algorithms adore this: prepare the control in $|+\rangle$, let kickback write an unknown phase φ into it, and the control becomes the equator state of longitude φ — then the interferometer of Chapter 5 (a final H) reads the phase as a probability. *Controlled gate + kickback + H = a phase-meter for U ’s eigenvalues.* Phase estimation (Chapter 19) is precisely this meter, run at multiple precisions.

REALITY CHECK

Kickback also explains a CZ curiosity from earlier in this chapter: CZ has no real “target” — it is symmetric, painting the -1 on the joint $|11\rangle$ amplitude — and CNOT is just CZ dressed with H’s on one side. In the H-dressed basis the “obvious” control/target roles swap. Moral for circuit readers: the dot-and- $\oplus$ picture shows *wiring*, not *causation*; amplitudes decide who influences whom.

CHAPTER SUMMARY

- CNOT: flip the target iff the control is 1 — dot and $\oplus$ in circuits; identity-with-last-rows-swapped as a matrix; XOR on basis states; its own inverse.
- On superposed controls, CNOT acts branch by branch — one gate can entangle: $|+\rangle|0\rangle \rightarrow \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.
- CZ flips only the sign of $|11\rangle$: symmetric, histogram-invisible, entangling nonetheless. SWAP exchanges qubits (= 3 CNOTs); compilers spend it on routing.
- Toffoli (two controls) computes AND reversibly onto a fresh target — classical logic imported into quantum circuits, ready to run on superpositions.
- Single-qubit gates alone never entangle; $\{H, T, \text{CNOT}\}$ is universal for all of quantum computing.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher’s Guide.

Problem 1 [Basic] Apply CNOT to each basis state $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ (control first).

Problem 2 [Basic] Draw the circuit symbols for CNOT, CZ, and SWAP from memory, labeling control and target where the words apply.

Problem 3 [Basic] Compute $\text{CNOT}|1\rangle \otimes |+\rangle$, expanding $|1\rangle \otimes |+\rangle = \frac{1}{\sqrt{2}}|10\rangle + \frac{1}{\sqrt{2}}|11\rangle$ first.

Problem 4 [Basic] Toffoli inputs: $a = 1, b = 1, c = 1$. Output? And after applying Toffoli a second time?

Problem 5 [Basic] Using the X-sandwich trick, give the exact gate sequence of a phase oracle that marks $|10\rangle$, and trace all four basis states through it to verify the signs.

Problem 6 [Basic] The XOR bit-oracle for $f = x_1 \oplus x_2$ is two CNOTs, both targeting the helper. Trace $|11\rangle|0\rangle$ and $|10\rangle|0\rangle$ through it and confirm the helper ends as $f(x)$.

Problem 7 [Basic] Compute $\text{CNOT}|+\rangle|+\rangle$ by the branch method and confirm no kickback occurs. Which eigenvalue of X does $|+\rangle$ carry, and why does that answer the question?

Problem 8 [Basic] Compute $\text{CNOT}|-\rangle|-\rangle$: what does the control become, and why?

Problem 9 [Standard] Apply CNOT to $|\psi\rangle = 0.6|00\rangle + 0.8|10\rangle$, and factor the output into (control state) $\otimes$ (target state) if possible. Is the output entangled?

Problem 10 [Standard] Apply CNOT to $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ (the entangled pair). Factor the result. Moral, in one sentence: what does a second CNOT do to entanglement made by a first?

Problem 11 [Standard] Reversed roles: a CNOT with control on qubit B and target on qubit A sends $|01\rangle \rightarrow |11\rangle$. List its action on all four basis states, and write its 4×4 matrix.

Problem 12 [Standard] Verify by tracing basis states that the three-CNOT chain — CNOT(A $\rightarrow$ B), CNOT(B $\rightarrow$ A), CNOT(A $\rightarrow$ B) — maps $|10\rangle \rightarrow |01\rangle$ and $|01\rangle \rightarrow |10\rangle$: SWAP from CNOTs. (Check $|00\rangle$ and $|11\rangle$ too.)

Problem 13 [Standard] Apply CZ to $|+\rangle \otimes |-\rangle$ and identify the output as a product or entangled state (factor it if you can — expand first, stare at signs second).

Problem 14 [Standard] Build the bit oracle for $f = x_1 \text{ OR } x_2$ from De Morgan's law $x_1 \vee x_2 = \neg(\neg x_1 \wedge \neg x_2)$: give the gate list (X's, one Toffoli, X's, and one final X on the helper) and trace the input $|00\rangle|0\rangle$.

Problem 15 [Standard] A three-qubit phase oracle must mark only $|111\rangle$. The gate CCZ does it directly, and CCZ can be built as: H on the third qubit, then Toffoli, then H again. Using $HXH = Z$ from Chapter 7, explain in two sentences why this construction works.

Problem 16 [Standard] Explain when an algorithm needs the bit style and when the phase style, and name one algorithm from this book using each. Why does the phase style need no cleanup?

Problem 17 [Standard] Controlled-Z with target $|1\rangle$ (an eigenstate of Z, eigenvalue -1): show the control in $|+\rangle$ flips to $|-\rangle$, while target $|0\rangle$ leaves the control alone. Then restate BOTH results as facts about CZ painting the $|11\rangle$ amplitude.

Problem 18 [Challenge] Show that (H on target) $\rightarrow$ CZ $\rightarrow$ (H on target) acts on basis states exactly like CNOT: trace all four inputs, using $H|0\rangle = |+\rangle$, $H|1\rangle = |-\rangle$ and CZ's sign rule, then H again. (This is how CZ-native hardware builds CNOT.)

Problem 19 [Challenge] A Toffoli with targets prepared cleverly can compute NAND (= NOT AND), and NAND alone builds all classical logic. Give the input wiring (values of a, b, c) that makes the Toffoli output $\text{NAND}(a, b)$ on some wire, and explain in two sentences why this proves quantum computers can run any classical program.

Problem 20 [Challenge] The “entanglement bookkeeping” claim: a circuit of only single-qubit gates on n qubits keeps the state a product $|\phi_1\rangle \otimes \cdots \otimes |\phi_n\rangle$ at every step. Argue this by induction in three sentences (what does gate U on qubit k do to a product state?), and conclude how many real numbers suffice to track such a circuit classically — versus $2 \cdot 2^n$ for a general state.

Problem 21 [Challenge] Verify the OR phase oracle of the worked example by matrix arithmetic: write the

three 4×4 diagonal matrices, multiply them, and confirm the result is $\text{diag}(1, -1, -1, -1)$. Then show that multiplying it by CZ's matrix instead would produce the oracle for a *different* function — which one?

Problem 22 [Challenge] Build the one-qubit phase-meter: control starts $|+\rangle$; a controlled- U kicks back U 's eigenphase φ (target prepared in the eigenstate); a final H yields $P(0) = (1 + \cos \varphi)/2$. Run it symbolically for $U = T$ (eigenphase 45° on $|1\rangle$) and give the predicted $P(0)$. You have built the core of quantum phase estimation with three gates.

CHAPTER ELEVEN

Entanglement and Bell States

The four Bell states, spooky correlations, and what entanglement can and cannot do

IN THIS CHAPTER YOU WILL LEARN TO

- Create entangled states with the H + CNOT circuit, and verify the calculation by hand.
- Write the four Bell states and convert between them with single-qubit gates.
- Predict the correlated measurement statistics of entangled pairs.
- Explain what makes entanglement genuinely non-classical — and why it still cannot send signals.
- Perform a Bell-basis measurement on paper, the tool behind teleportation and superdense coding.

11.1 Making Entanglement: H + CNOT

Chapter 10 built the machine; now we study what it made. The standard entangling circuit — worth committing to memory, since Parts Four and Five use it dozens of times — is two gates long:

$$\left. \begin{array}{l} q_A = |0\rangle \\ q_B = |0\rangle \end{array} \right\} \begin{array}{c} \boxed{H} \\ \bullet \\ \oplus \end{array} \left. \vphantom{\begin{array}{l} q_A \\ q_B \end{array}} \right\} \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

The Bell-state factory: H puts the control into superposition; CNOT correlates the target with each branch.

WORKED EXAMPLE — THE BELL CIRCUIT, EVERY STEP

Trace the circuit above from $|00\rangle$ to its output.

1. Start: $|00\rangle = |0\rangle_A \otimes |0\rangle_B$.
2. After H on A: $\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) \otimes |0\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|10\rangle$.
3. After CNOT (branch by branch): $|00\rangle$ has control 0 — stays; $|10\rangle$ has control 1 — target flips to give $|11\rangle$.
4. Output: $\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$.

Answer: $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ — the most famous state in quantum information, written $|\Phi^+\rangle$ and called a Bell

state, after John Bell. Chapter 9’s challenge proved it cannot be factored: neither qubit has a state of its own.

WORD TO KNOW

Entangled state — A multi-qubit state that cannot be written as a tensor product of single-qubit states. The qubits have no individual state vectors — only the pair (or group) as a whole has one.

Pause on how strange the last sentence is. For $|\Phi^+\rangle$, there is literally no answer to “what state is qubit A in?” — not “unknown,” but *nonexistent*. Only the pair has a description. Schrödinger, who coined the word *Verschränkung* (entanglement) in 1935, called this “the characteristic trait of quantum mechanics — the one that enforces its entire departure from classical lines of thought.”

11.2 The Four Bell States

$|\Phi^+\rangle$ has three siblings — together they form the **Bell basis**, an orthogonal quartet of maximally entangled two-qubit states.

EQUATION: THE FOUR BELL STATES

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad |\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \quad |\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

In words: Φ states: qubits agree; Ψ states: qubits disagree; the sign records a relative phase

Symbol	Meaning	Unit
$ \Phi^\pm\rangle$	the “agreeing” Bell states (patterns 00 and 11)	none
$ \Psi^\pm\rangle$	the “disagreeing” Bell states (patterns 01 and 10)	none

All four are orthogonal to each other and reachable from $|\Phi^+\rangle$ by one single-qubit gate on qubit A: Z gives Φ^- , X gives Ψ^+ , and XZ (up to phase, Y) gives Ψ^- . Local touches move within the family; only two-qubit gates create or destroy the entanglement itself.

WORKED EXAMPLE — FROM Φ^+ TO Ψ^+ WITH ONE LOCAL GATE

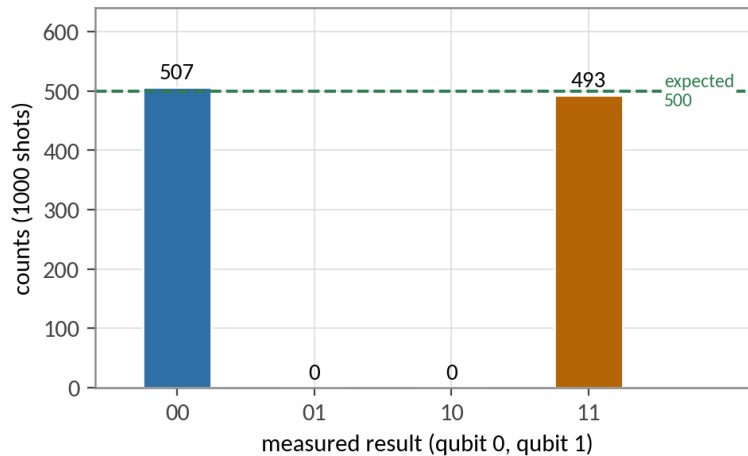
Apply X to qubit A of $|\Phi^+\rangle$ and identify the result.

1. Write the state: $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.
2. Flip the A slot in each term: $|00\rangle \rightarrow |10\rangle$ and $|11\rangle \rightarrow |01\rangle$.
3. Collect: $\frac{1}{\sqrt{2}}(|10\rangle + |01\rangle) = |\Psi^+\rangle$.

Answer: $(X \otimes I)|\Phi^+\rangle = |\Psi^+\rangle$. One local flip turned “agree” into “disagree.” This four-states-one-local-gate structure is precisely the machinery superdense coding will exploit in Chapter 13.

11.3 Correlated Measurements: What Entanglement Does

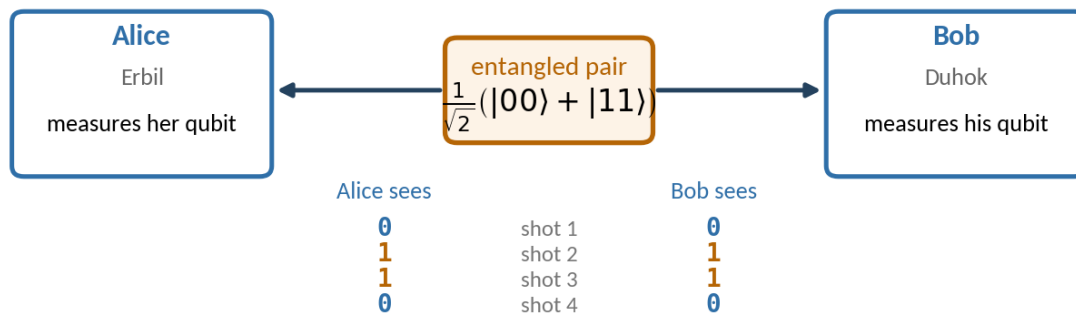
Measure both qubits of $|\Phi^+\rangle$ in the standard basis. The Born rule on the pair reads: $P(00) = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$, $P(11) = \frac{1}{2}$, $P(01) = P(10) = 0$.



A 1000-shot histogram of $|\Phi^+\rangle$: only 00 and 11 appear, each about half the time. The absent middle bars ARE the entanglement, statistically speaking.

Each qubit alone is a perfect coin: $P(A=0) = \frac{1}{2}$, $P(B=0) = \frac{1}{2}$ (marginals, Chapter 9). But the coins are locked together: the results always match. And crucially, this holds *however far apart* the qubits are carried before measuring.

each result is random — but they always match



Individually random, jointly certain — at any distance. Alice's and Bob's results are each fair coins that always agree.

COMMON MISTAKE

X Wrong: “The qubits secretly agreed in advance — like putting one red card and one blue card in two envelopes. No mystery.”

✓ **Right:** That is exactly the classical explanation (“local hidden variables”), it is completely reasonable — and experiment rules it out. Measuring in *rotated* bases produces correlation patterns (violations of Bell’s inequality) stronger than ANY pre-agreed cards could produce. The envelopes story fails; nature really is non-classical. See the case study — this earned the 2022 Nobel Prize.

The card-envelope test deserves one honest paragraph, because it is the deepest experimental fact in this book. In 1964, John Bell proved a theorem: if measurement results were determined by *any* information the particles carried from their common source (however hidden), then a certain combination of correlations, measured along tilted axes, could never exceed the bound 2. Quantum mechanics predicts up to $2\sqrt{2} \approx 2.83$ for Bell states. The experiments — Freedman–Clauser 1972, Aspect 1982, and the loophole-free trio of 2015 — give 2.4 to 2.8, decisively beyond 2, in exact agreement with the quantum calculation. No pre-written script can explain entangled statistics. (The full arithmetic of a Bell test is a Teacher’s-Guide extension; what every student must retain is the verdict.)

11.4 What Entanglement Cannot Do

Alice measures her half of $|\Phi^+\rangle$ in Erbil; instantly, the joint state says Bob’s qubit in Duhok will match. Did something travel faster than light? Can they text through it? A calculation — one every quantum professional should be able to reproduce — says no.

WORKED EXAMPLE — WHY NO SIGNAL CAN BE SENT

Alice tries to send a bit by choosing whether or not to measure her qubit before Bob measures his. Compute what Bob sees in each case.

1. If Alice does nothing: Bob’s marginals (Chapter 9) give $P(0) = P(1) = \frac{1}{2}$.
2. If Alice measures first: she gets 0 or 1, each with $\frac{1}{2}$. If 0, Bob’s qubit collapses to $|0\rangle$ (he sees 0); if 1, he sees 1.
3. Bob’s overall statistics in case 2: $\frac{1}{2} \times (\text{certainly } 0) + \frac{1}{2} \times (\text{certainly } 1) = 50/50$ — *identical* to case 1.

Answer: Bob’s local statistics are 50/50 no matter what Alice does. The correlation is real but only visible when the two result-lists are brought together and compared — by phone, email, or car ride, all limited by light speed. Entanglement is a resource of correlation, never a channel for communication.

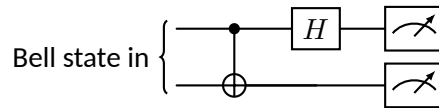
REALITY CHECK

This “no-signalling” fact is not a loophole waiting for a clever engineer — it is a theorem holding for all quantum states and all local operations. So beware two popular myths: entanglement does not enable faster-than-light messaging, and it does not let quantum computers “send answers instantly.” What it *does* enable is quieter and deeper: correlations beyond classical limits (Bell), doubling a channel’s classical capacity (superdense coding, Chapter 13), moving quantum states via classical messages (teleportation, Chapter 13), and device-independent randomness (Chapter 4’s case study). Powerful — and perfectly law-abiding.

11.5 Bell-Basis Measurement

A standard measurement asks a pair “which pattern: 00, 01, 10, 11?” A **Bell measurement** asks instead: “which Bell state: Φ^+ , Φ^- , Ψ^+ , Ψ^- ?” Since the four Bell states form an orthogonal basis, this is a legitimate measurement — and the key move in teleportation (Chapter 13).

The practical recipe is elegant: run the Bell factory *backwards*. The factory circuit (H then CNOT) maps basis states to Bell states; being unitary, its reverse (CNOT then H) maps Bell states back to basis states, where an ordinary measurement can read them.



Bell measurement = disentangle (CNOT, then H on the control), then measure both qubits. Each Bell state announces itself as a distinct two-bit pattern.

WORKED EXAMPLE — DECODING $|\Phi^-\rangle$

Feed $|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$ into the decoder circuit. What do the meters read?

1. After CNOT: $|00\rangle \rightarrow |00\rangle$; $|11\rangle \rightarrow |10\rangle$. State: $\frac{1}{\sqrt{2}}(|00\rangle - |10\rangle) = \left[\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\right] \otimes |0\rangle = |-\rangle \otimes |0\rangle$.
2. After H on qubit A: $H|-\rangle = |1\rangle$. State: $|1\rangle \otimes |0\rangle = |10\rangle$.
3. Measure: qubit A reads 1, qubit B reads 0 — with certainty.

Answer: $|\Phi^-\rangle$ decodes to the pattern 10. The full dictionary (trace the others yourself — Problem 9): $\Phi^+ \rightarrow 00$, $\Psi^+ \rightarrow 01$, $\Phi^- \rightarrow 10$, $\Psi^- \rightarrow 11$. Four entangled siblings, four clean classical labels.

CASE STUDY — THE NOBEL PRIZE FOR PROVING IT REAL • CLAUSER, ASPECT, ZEILINGER, 2022

For decades, “does nature really violate Bell’s inequality?” was philosophy with lab coats. John Clauser built the first test in 1972 (photon pairs, correlations beyond the classical bound); Alain Aspect’s 1982 Paris experiments closed the timing loophole by switching measurement settings while the photons were mid-flight; Anton Zeilinger’s group pushed entanglement to new regimes — multi-photon states, teleportation (Chapter 13), and in 2017 a satellite (Micius) distributing entangled photons across 1200 km of China. In 2015, three “loophole-free” experiments (Delft, Vienna, NIST) closed the remaining escape routes simultaneously. The 2022 Nobel Prize in Physics honored Clauser, Aspect, and Zeilinger — official recognition that entanglement’s strangeness is measured reality, and the foundation of the quantum technologies this book teaches.

ACTIVITY (APP) — BREAK THE ENVELOPES MODEL

1. In the *Entanglement Lab* of the QCFE app, prepare $|\Phi^+\rangle$ with the H+CNOT buttons and run 1000 shots: confirm only 00 and 11.

2. Use the “separate labs” view: watch Alice’s histogram alone (50/50) and Bob’s alone (50/50) — then the joint view (perfectly matched).
3. Switch to $|\Psi^+\rangle$ (X on one qubit) and verify the results now always *disagree*.
4. Open the “Bell test” panel, which measures at tilted angles and reports the correlation score S . Run it: the app shows $S \approx 2.8 > 2$, with a one-line reminder of what the classical bound 2 means.

You need: The QCFE app (Android or Windows)

► App tool: *Entanglement Lab* — build Bell states, split the labs apart, and run a real Bell-inequality test on simulated pairs

11.6 The CHSH Score: Putting a Number on Spookiness

The Bell test of this chapter can be sharpened into a single number, proposed by Clauser, Horne, Shimony, and Holt (CHSH) in 1969 — the version every modern laboratory actually runs. Alice chooses between two measurement settings a and a' ; Bob, far away, between b and b' . Each records $+1$ or -1 per shot, and for every settings pair they estimate the **correlator** E : the average of the product of their two answers.

EQUATION: THE CHSH SCORE

$$S = E(a, b) + E(a', b) + E(a', b') - E(a, b')$$

In words: Add three of the four correlators and subtract the fourth. Any local hidden-answer world obeys $|S| \leq 2$

Symbol	Meaning	Unit
$E(a, b)$	average of (Alice’s ± 1) $\times$ (Bob’s ± 1) at settings a, b	none
a, a'	Alice’s two setting choices	none
b, b'	Bob’s two setting choices	none

Classical ceiling: 2. Quantum ceiling (the Tsirelson bound): $2\sqrt{2} \approx 2.83$. Any experimental $S > 2$ certifies that no local pre-written answer sheet exists — whatever the devices’ inner workings.

WORD TO KNOW

CHSH inequality — The statement $|S| \leq 2$ obeyed by every local hidden-variable theory. Entangled qubits violate it, reaching up to $2\sqrt{2} \approx 2.83$ — the experimentally testable heart of Bell’s theorem.

For the Bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ with all settings in the X-Z plane of the Bloch sphere, the correlator takes the clean form $E = \cos(\theta_A - \theta_B)$. The famous optimal settings are $a = 0^\circ$, $a' = 90^\circ$, $b = 45^\circ$, $b' = 135^\circ$:

Correlators at the optimal CHSH settings (Bell state, ideal)

Settings pair	Angle difference	Correlator E
(a, b)	-45°	$+0.707$
(a', b)	$+45^\circ$	$+0.707$
(a', b')	-45°	$+0.707$
(a, b')	-135°	-0.707

WORKED EXAMPLE — COMPUTING THE QUANTUM SCORE

Use the table to evaluate S for the entangled pair.

1. Insert with signs: $S = 0.707 + 0.707 + 0.707 - (-0.707)$.
2. Add: $S = 4 \times 0.707 = 2.828$.
3. Compare ceilings: $2.828 > 2$ (classical broken) and $2.828 = 2\sqrt{2}$ (quantum ceiling exactly reached) ✓

Answer: $S = 2\sqrt{2} \approx 2.83$. The minus sign in the definition lands on the only negative correlator, so all four terms pull upward — the settings are chosen precisely to engineer that. No arrangement of pre-agreed answers can do better than 2; the qubits do.

WORKED EXAMPLE — HOW AN EAVESDROPPER WRECKS THE SCORE

Suppose every pair is secretly intercepted: a spy measures both qubits in the standard basis and passes on fresh qubits in the observed states. Now the “pair” is really $|00\rangle$ or $|11\rangle$, each half the time. Find the new S .

1. For definite states the correlator factorizes: $E(\theta_A, \theta_B) = \cos \theta_A \cos \theta_B$ (each side’s average is $\pm \cos \theta$ of its own angle, independently).
2. Evaluate: $E(a, b) = \cos 0^\circ \cos 45^\circ = 0.707$; $E(a', b) = \cos 90^\circ \cos 45^\circ = 0$; $E(a', b') = 0$; $E(a, b') = \cos 0^\circ \cos 135^\circ = -0.707$.
3. Score: $S = 0.707 + 0 + 0 - (-0.707) = 1.414 = \sqrt{2}$.

Answer: the spy drags S from 2.83 down to 1.41 — deep below the classical line, let alone the quantum one. A CHSH test is therefore a *tamper seal*: entanglement that has been measured-and-resent can no longer prove it was never touched. Chapter 15 turns exactly this into a key-distribution protocol.

REALITY CHECK

Real experiments never reach 2.83: imperfect detectors, misalignment, and noise shrink every correlator by a visibility factor V , giving $S = 2\sqrt{2}V$. Laboratory values around 2.4-2.7 are typical and still decisively above 2 — the 2015 loophole-free experiments reported violations at overwhelming statistical significance. Beating 2 is the achievement; touching $2\sqrt{2}$ is reserved for textbooks.

11.7 Monogamy: Entanglement Cannot Be Shared Freely

Classical correlation is promiscuous: one weather report correlates perfectly with a million readers. Entanglement obeys the opposite law. If qubits A and B are *maximally* entangled, then neither can be entangled with anything else — not weakly, not secretly, not at all. This is **monogamy of entanglement**, and it is a theorem, not a metaphor.

The GHZ state is the cleanest witness. All three qubits are jointly entangled — yet look at any *pair* inside it. Using the same x-z-plane correlators as the CHSH lesson, a direct computation gives, for qubits A and B of GHZ:

EQUATION: PAIR CORRELATIONS INSIDE GHZ

$$E(\theta_A, \theta_B) = \cos \theta_A \cos \theta_B \implies S_{\text{pair}} \leq \sqrt{2} < 2$$

In words: Any two qubits inside GHZ correlate exactly like classical agreeing coins — their CHSH score cannot even reach the classical ceiling

Symbol	Meaning	Unit
E	pair correlator at settings θ_A, θ_B	none
S_{pair}	the pair's best CHSH score	none

Compare: a true Bell pair gives $E = \cos(\theta_A - \theta_B)$ and $S = 2\sqrt{2}$. The pair inside GHZ produces the same factorized correlations as an intercepted pair in the CHSH lesson — because from the pair's viewpoint, the third qubit is an eavesdropper who already knows everything.

WORKED EXAMPLE — THE PAIR THAT COULD NOT SPOOK

For qubits A,B inside GHZ, evaluate the four CHSH correlators at the optimal settings $a = 0^\circ$, $a' = 90^\circ$, $b = 45^\circ$, $b' = 135^\circ$, and score the pair.

1. $E(a, b) = \cos 0^\circ \cos 45^\circ = 0.707$.
2. $E(a', b) = \cos 90^\circ \cos 45^\circ = 0$; likewise $E(a', b') = 0$.
3. $E(a, b') = \cos 0^\circ \cos 135^\circ = -0.707$.
4. Score: $S = 0.707 + 0 + 0 - (-0.707) = 1.41$ — far below 2, despite the pair living inside a maximally entangled trio.

Answer: $S_{\text{pair}} = \sqrt{2}$. The trio's entanglement is real but *collective*: it shows up in three-way experiments, while every pair, alone, tests classical. Entanglement spent on a third party is entanglement unavailable to the pair — monogamy, measured.

WORD TO KNOW

Monogamy of entanglement — The theorem that maximal entanglement between A and B forbids any entanglement of either with a third system — and that partial sharing is strictly rationed. The physical foundation

of entanglement-based cryptography.

REALITY CHECK

Monogamy is why the E91 protocol of Chapter 15 works: if Alice's and Bob's pairs score $S \approx 2\sqrt{2}$, monogamy leaves zero room for Eve's systems to be correlated with their outcomes — the key is private because arithmetic says nobody else fits. No classical resource behaves this way; a broadcast can always add one more listener. Entanglement is the only correlation in nature with a strict guest list.

ACTIVITY (CLASS) — BEAT THE QUANTUM PAIR (YOU CANNOT)

1. Pairs of students play referee-scored CHSH: each round the referee deals each player a random setting card (a/a' to one, b/b' to the other); players answer $+1$ or -1 with *no communication* after seeing cards.
2. Teams may agree on any strategy beforehand — write it down. Play 40 rounds; the referee computes the four correlators and S .
3. Compare all teams' scores with the classical ceiling 2 and the quantum 2.83. Best classical strategies hit $S = 2$ exactly (the always-agree table); no team beats it.
4. Discuss: what resource would the players need to score 2.83 — and why does sharing written notes, coins, or code words never suffice? (Chapter answer: entangled qubits, and Bell's theorem.)

You need: Setting cards, Paper and pen, A referee with a calculator

CHAPTER SUMMARY

- $H + \text{CNOT}$ on $|00\rangle$ manufactures $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$: entangled — no tensor factorization exists, and neither qubit owns a state.
- The Bell basis: $\Phi^\pm$ (agree), $\Psi^\pm$ (disagree); orthogonal; interconverted by local X and Z on one qubit.
- Measured Bell pairs: each side a fair coin, the pair perfectly correlated, at any distance. Bell-inequality experiments (Nobel 2022) rule out every pre-agreed-answers explanation.
- No-signalling: local statistics never depend on what the distant partner does — entanglement correlates, it does not communicate.
- Bell measurement = inverse factory (CNOT, then H , then measure): $\Phi^+ \rightarrow 00$, $\Psi^+ \rightarrow 01$, $\Phi^- \rightarrow 10$, $\Psi^- \rightarrow 11$. Teleportation runs on it.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] Trace the $H + \text{CNOT}$ circuit starting from $|10\rangle$ (not $|00\rangle$), and name the Bell state you produce.

Problem 2 [Basic] For $|\Psi^+\rangle$: list the four pattern probabilities, and state $P(\text{results agree})$.

Problem 3 [Basic] Which local gate on qubit A converts $|\Phi^+\rangle$ into $|\Phi^-\rangle$? Verify by applying it term by term.

Problem 4 [Basic] A source claims to emit $|\Phi^+\rangle$, but a 1000-shot histogram shows 260 counts of 01. Is the claim credible? What single word describes what the histogram should show at 01?

Problem 5 [Basic] A laboratory reports $E(a, b) = 0.71$, $E(a', b) = 0.69$, $E(a', b') = 0.70$, $E(a, b') = -0.72$. Compute S . Is the result consistent with local hidden answers? With quantum mechanics?

Problem 6 [Basic] Two classical coins are prepared to always agree: every correlator equals +1 regardless of settings. Compute S and confirm that perfect classical correlation exactly *reaches* the classical ceiling without breaking it.

Problem 7 [Basic] Compute $E(45^\circ, 45^\circ)$ for a pair inside GHZ and for a true Bell pair, and state the difference in one sentence.

Problem 8 [Basic] A vendor claims their source gives each of three users a qubit such that *every pair* of users shares a maximally entangled state. Which theorem forbids this, and what is the tell-tale experiment?

Problem 9 [Standard] Alice and Bob share $|\Psi^-\rangle$. Alice measures and gets 0. What is Bob's state now, and what will he measure? Repeat for Alice getting 1.

Problem 10 [Standard] Compute the marginals $P(A=0)$ and $P(B=0)$ for $|\Phi^-\rangle$, and explain why the minus sign is invisible to both marginals *and* to the joint standard-basis histogram. Which measurement from this chapter would see it?

Problem 11 [Standard] Trace the Bell factory on input $|01\rangle$ and on $|11\rangle$, and complete the factory's input-output table: which basis state manufactures which Bell state?

Problem 12 [Standard] Show that $\langle \Phi^+ | \Psi^+ \rangle = 0$ (write both as 4-component vectors and take the inner product). Why does orthogonality matter for the Bell *measurement*?

Problem 13 [Standard] Trace the decoder circuit (CNOT, then H on A, then measure) on $|\Psi^+\rangle$ and on $|\Psi^-\rangle$, confirming the dictionary entries 01 and 11.

Problem 14 [Standard] Noise multiplies every ideal correlator by a visibility $V \in [0, 1]$, so $S = 2\sqrt{2}V$. What is the minimum visibility for which the test still beats the classical bound? Give three decimal places.

Problem 15 [Standard] Keep $a = 0^\circ$, $a' = 90^\circ$, $b = 45^\circ$ but change Bob's second setting to $b' = -45^\circ$. Using $E = \cos(\theta_A - \theta_B)$, recompute all four correlators and S . What does the collapse of the score to zero teach about the role of the settings?

Problem 16 [Standard] A spy intercepts only a fraction f of the pairs (the rest fly untouched), so $S = (1 -$

$f) 2\sqrt{2} + f\sqrt{2}$. Find the largest f for which the measured S still exceeds 2. Why do real protocols nevertheless catch gentler spying, using error rates as well?

Problem 17 [Standard] Show that the GHZ pair correlations $E = \cos \theta_A \cos \theta_B$ are exactly reproducible by a classical model: two coins that always agree, showing $+1$ with probability $\frac{1}{2}$, each reader reporting ± 1 with average $\cos \theta$ of its own angle. Conclude (no computation needed) that no settings can push S_{pair} past 2.

Problem 18 [Challenge] The three-qubit GHZ state is $\frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$. Design a circuit of one H and two CNOTs that builds it from $|000\rangle$, and trace your circuit step by step to verify.

Problem 19 [Challenge] Alice proposes a faster-than-light alarm: “If the dam sensor triggers, I *measure* my qubit; Bob continuously watches his qubit’s statistics for the change.” Using this chapter’s no-signalling worked example, explain in three sentences why Bob’s watching can never detect whether the alarm fired — and identify the exact step where the scheme’s logic breaks.

Problem 20 [Challenge] Entanglement is monogamous: if qubits A and B are in $|\Phi^+\rangle$, qubit C cannot also be in $|\Phi^+\rangle$ with A. Give a plausibility argument in three sentences using the factorization definition (what would A’s “state” have to be in each pairing?), and one sentence on why this matters for quantum cryptography (who fails to get a copy?).

Problem 21 [Challenge] Prove the classical bound: suppose each side’s answers are pre-written numbers $A_a, A_{a'}, B_b, B_{b'} \in \{+1, -1\}$. Write $S = A_a(B_b - B_{b'}) + A_{a'}(B_b + B_{b'})$ and show one bracket is always 0 while the other is ± 2 , so $|S| \leq 2$ for every possible answer sheet — and averaging over sheets cannot help.

Problem 22 [Challenge] Reconcile two facts: pairs inside GHZ test classical, yet measuring qubit C in the X basis leaves A,B in a genuine Bell state (compute the two post-measurement states for C’s outcomes $\pm$, starting from $\frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$ and writing $|0\rangle_C, |1\rangle_C$ in the $\pm$ basis). Why does this “unlocking” not violate monogamy? (Hint: it consumes qubit C.)

CHAPTER TWELVE

Quantum Circuits and Circuit Design

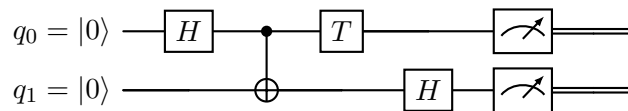
Reading, computing, simplifying, and designing — plus the theorem that forbids copying

IN THIS CHAPTER YOU WILL LEARN TO

- Read every element of a full circuit diagram: wires, gates, controls, meters, classical wires.
- Compute a multi-gate, multi-qubit circuit's output state systematically, column by column.
- Simplify circuits using gate identities, and count depth and gate cost like a compiler.
- State and prove the no-cloning theorem, and draw its practical consequences.
- Design small circuits to specification — including a certified quantum random-number generator.

12.1 Reading a Full Circuit Diagram

You have now met every symbol quantum circuits use. Here is a complete specimen — read it element by element:



Anatomy: qubit wires with initial states at left; time flows rightward; columns are simultaneous steps; dot-and- $\oplus$ is a CNOT; boxed letters are single-qubit gates; the meter boxes are measurements; double lines carry the resulting classical bits.

Conventions this book uses (and most of the field shares): qubits are numbered $q_0, q_1, \dots$ from the top; every wire starts in $|0\rangle$ unless labeled otherwise; gates in the same column act in the same step; measurements are usually last (mid-circuit measurement exists — Chapter 29 uses it — and is marked the same way, just earlier).

WORD TO KNOW

Circuit depth — The number of time-step columns a circuit needs. Depth, not gate count, sets the running time — and on noisy hardware (Chapter 28), the noise exposure. Parallel gates in one column cost one unit

of depth together.

12.2 From Circuit to State, Column by Column

The professional method for hand-computing any circuit: carry the full state from column to column, applying that column's gates term by term. One habit prevents nearly all errors: *write the complete state after every column* — never try to hold two columns in your head.

WORKED EXAMPLE — A THREE-COLUMN, TWO-QUBIT CIRCUIT

Compute the output of: $|00\rangle$, then H on q_0 , then CNOT($q_0 \rightarrow q_1$), then H on q_0 . (This is the specimen circuit above, minus its T gate.)

1. Column 1 (H on q_0): $|00\rangle \rightarrow \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$.
2. Column 2 (CNOT): branch rule — $|00\rangle$ stays, $|10\rangle \rightarrow |11\rangle$: state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ — a Bell state, as Chapter 11 taught.
3. Column 3 (H on q_0): apply $H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ inside each term: $\frac{1}{\sqrt{2}} \left[\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|0\rangle + \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)|1\rangle \right]$.
4. Expand and collect: $\frac{1}{2}(|00\rangle + |10\rangle + |01\rangle - |11\rangle)$.

Answer: $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$: all four patterns at 25%, one negative phase — still entangled (compare Chapter 10's CZ example: same state!). Circuits differing everywhere can meet at the same output; the next lesson turns that into a tool.

COMMON MISTAKE

X Wrong: To compute a circuit, multiply the gate matrices left to right as drawn, then apply to the input at the end.

✓ Right: Two safe options. (1) The column method above — carry the state forward; no big matrices, hardest to get wrong; use it by default. (2) Matrix method: assemble columns into one operator, remembering both rules — same-column gates combine with $\otimes$, successive columns multiply *right to left*: the specimen circuit's operator is $(H \otimes I)(T \otimes H) \text{CNOT} (H \otimes I) \dots$ wait — check yourself against the diagram: which factor stands nearest the ket, and which column does it come from? (First column. If your answer differed, reread Chapter 6, Lesson 1.)

12.3 Circuit Identities and Simplification

Compilers — and good students — shrink circuits using algebraic identities you have already proven along the way:

Identity toolbox (all previously verified in Appendix B and Chapters 3–11)

Identity	Use
$HH = XX = YY = ZZ = I$	cancel adjacent duplicates
$T^2 = S, S^2 = Z, T^4 = Z$	merge phase ladders
$HZH = X$ and $HXH = Z$	change flip-language $\leftrightarrow$ phase-language
$(H\boxtimes H)$ CNOT $(H\boxtimes H)$ reverses the CNOT's direction	re-route control and target
$\text{CNOT} \cdot \text{CNOT} = I$; CZ symmetric; SWAP = 3 CNOTs	untangle and re-route
Gates on different qubits commute	slide columns past each other to expose cancellations

WORKED EXAMPLE — COMPILING A CIRCUIT DOWN

Simplify: q_0 : H, T, T, H — while q_1 (untouched elsewhere): Z, Z. Then state the final circuit's total action.

1. Merge the phase ladder on q_0 : $TT = S$, giving H, S, H on q_0 .
2. Cancel on q_1 : $ZZ = I$ — the wire is empty.
3. q_0 's remaining product (right-to-left as matrices): HSH . No listed identity kills it — but recognize the pattern: an S (90° z -turn) in an H sandwich is the same turn about x : $HSH = R_x(90^\circ)$ up to global phase.

Answer: the two-qubit circuit compiles to a single $R_x(90^\circ)$ on q_0 and nothing on q_1 : six gates became one. Depth 4 became depth 1 — on noisy hardware, that is directly less exposure to error (Chapter 28).

12.4 The No-Cloning Theorem

Chapter 10 warned that CNOT “copies” only basis states. The full truth is stronger, provable in four lines, and one of the most consequential facts in quantum information: *no machine can copy an unknown quantum state*.

EQUATION: THE NO-CLONING THEOREM

No unitary U achieves $U(|\psi\rangle \otimes |0\rangle) = |\psi\rangle \otimes |\psi\rangle$ for every state $|\psi\rangle$.
In words: There is no universal quantum photocopier

Symbol	Meaning	Unit
$ \psi\rangle$	the unknown state to be copied	none
$ 0\rangle$	the fresh “blank page” qubit	none

The proof needs only linearity — gates act term by term — and one honest expansion. It occupies the next worked example, no calculus required.

WORKED EXAMPLE — PROVING NO-CLONING IN FOUR STEPS

Suppose a cloner U exists: it copies $|0\rangle$ and $|1\rangle$ correctly. Show it then fails on $|+\rangle$.

1. Assumed behavior on basis states: $U|0\rangle|0\rangle = |0\rangle|0\rangle$ and $U|1\rangle|0\rangle = |1\rangle|1\rangle$.
2. Feed it $|+\rangle|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle|0\rangle + |1\rangle|0\rangle)$. Linearity forces: $U|+\rangle|0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ — the Bell state.
3. But a true copy would be $|+\rangle|+\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$.
4. Compare: the Bell state has no $|01\rangle$ or $|10\rangle$ terms; $|+\rangle|+\rangle$ does. They are different states — contradiction.

Answer: any machine that copies both basis states is forced by linearity to entangle — not copy — their superpositions. No universal cloner exists. (You proved a theorem today: this argument is essentially Wootters-Zurek 1982, complete.)

No-cloning is not an inconvenience — it is a load-bearing wall of the field. It is why measurement's one-shot collapse (Chapter 4) cannot be sidestepped by “just copy the qubit first and measure the copies.” It is why quantum error correction (Chapter 29) had to invent radically indirect tricks instead of classical backup copies. And it is why quantum cryptography works: an eavesdropper cannot copy the qubits in flight and study them at leisure — as the case study recounts, this “flaw” became a technology.

CASE STUDY — UNFORGEABLE MONEY AND THE BIRTH OF QUANTUM CRYPTOGRAPHY • WIESNER 1970, BB84

Around 1970, a graduate student named Stephen Wiesner had an idea so early it could not even be published for years: banknotes carrying a few qubits in secret bases, unforgeable because copying unknown quantum states is impossible. The paper (finally printed in 1983) planted a seed: in 1984, Charles Bennett and Gilles Brassard turned the same principle into BB84, the first quantum key distribution protocol — eavesdroppers must measure, measurement disturbs, disturbance is detected (you computed exactly this in Chapter 4, Problem 12). Today QKD links run commercially in several countries, and China's Micius satellite has distributed keys intercontinentally. An impossibility theorem, four lines long, became an industry.

12.5 Designing Small Circuits: A Method

Design is where reading becomes fluency. A dependable four-step recipe for small specifications:

1. **Write the target** — the exact final state, or the exact measurement statistics required.
2. **Choose building blocks** — superposition needs H (or R_y for unequal splits); correlation needs CNOT/CZ; phases need Z, S, T; conditional logic needs controls.
3. **Draft and compute** — column method, full state after every column.
4. **Verify and shrink** — check the target exactly; then hunt cancellations with the identity toolbox.

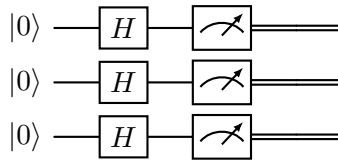
WORKED EXAMPLE — DESIGN: AN 8-SIDED QUANTUM DIE

Specification: one shot must output one of the eight patterns 000–111, each with probability exactly $\frac{1}{8}$, un-

predictable in principle.

1. Target state: the uniform superposition $\frac{1}{\sqrt{8}} \sum_{x=000}^{111} |x\rangle$ — every amplitude $\frac{1}{\sqrt{8}}$, since $\left|\frac{1}{\sqrt{8}}\right|^2 = \frac{1}{8}$.
2. Building block: Chapter 9's rule — H on every qubit of $|000\rangle$ makes exactly this state. No correlations required: no two-qubit gates.
3. Draft: three H gates in one column; then three meters.
4. Verify: $(H|0\rangle)^{\otimes 3} = |+\rangle|+\rangle|+\rangle = \frac{1}{\sqrt{8}} \sum_x |x\rangle$ ✓. Depth 1, three gates — nothing to shrink.

Answer: the circuit below. Read as binary, each shot is a fair roll of 0-7: a random number certified by the Born rule itself, not by any formula. (Compare Chapter 4's case study — and build it for real in Capstone Project 1.)



The 8-sided quantum die: depth 1, three gates, three meters — and randomness no classical machine can manufacture.

REALITY CHECK

Circuit design has a second half this book keeps honest about: real hardware. A textbook circuit assumes any gate anywhere, perfect and instant. A real chip has a limited native gate menu (compilation — Chapter 7's universality — fills the gap), limited connectivity (SWAP insertions — Chapter 10), and noise that grows with depth (Chapter 28). Professionals therefore design twice: once for correctness on paper, once for cost after compilation. The gate counts and depths you practiced measuring in this chapter are exactly the currency of that second design pass.

ACTIVITY (APP) — DESIGN CLINIC

1. In the *Circuit Builder*, reproduce this chapter's specimen circuit and check the app's state display after each column against your own column-method computation.
2. Build the 8-sided die; roll it 40 times; tally your histogram. Does each face appear “about” 5 times? (Recall Chapter 4 on wobble before judging.)
3. Design challenge: build a circuit whose histogram is 50% $|000\rangle$ and 50% $|111\rangle$ (hint: Chapter 11, Problem 10 — the GHZ state). Verify with 1000 shots.
4. Compile: give the app the sequence H, T, T, H, Z, Z on one wire and use the “simplify” button; confirm it reports the same single-gate answer as this chapter's compiling example.

You need: The QCFE app (Android or Windows), Paper and pen

► App tool: *Circuit Builder* — full editor with state inspector after every column, plus a one-press simplifier that names the identities it used

12.6 Scratch Qubits and Uncomputation

Real circuits, like real calculations on paper, need *scratch space*: helper qubits that hold intermediate results. But quantum scratch comes with a trap that classical scratch paper does not have. If a helper still holds information about the data when the algorithm's interference step arrives, it is entangled with the data — and interference between data branches dies, exactly as in the measurement stories of Chapter 4. A dirty blackboard breaks the physics, not just the tidiness.

Charles Bennett's 1973 remedy is the three-beat pattern used by essentially every serious quantum algorithm: **compute** the intermediate result into scratch, **copy** what you need into the output, then **uncompute** — run the first stage backwards — so every scratch qubit returns to $|0\rangle$.

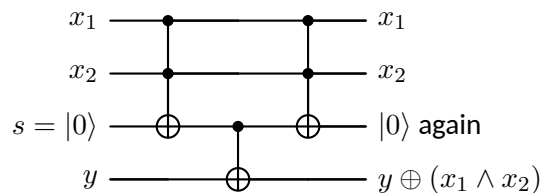
EQUATION: THE UNCOMPUTATION SANDWICH

$$|x\rangle |0\rangle_{\text{scratch}} |y\rangle \xrightarrow{\text{compute, copy, uncompute}} |x\rangle |0\rangle_{\text{scratch}} |y \oplus f(x)\rangle$$

In words: The answer lands in the output register; the scratch returns to zero, provably disentangled from everything

Symbol	Meaning	Unit
$ x\rangle$	data register (any superposition)	none
$ 0\rangle_{\text{scratch}}$	helper qubit(s), clean before and after	none
$y \oplus f(x)$	output, XORed with the function value	none

Because gates are reversible, “run backwards” means: apply the inverse gates in reverse order. For self-inverse gates like Toffoli and CNOT, that is simply the same gates, reversed.



The sandwich for $f = x_1 \text{ AND } x_2$: Toffoli computes the AND into the scratch qubit s , a CNOT copies it to the output y , and a second Toffoli uncomputes s back to $|0\rangle$.

WORD TO KNOW

Uncomputation — Running a computation's gates in reverse to return helper (ancilla) qubits to $|0\rangle$ after their content has been used — mandatory before interference steps, because “dirty” helpers stay entangled with the data and destroy it.

WORKED EXAMPLE — TRACING THE SANDWICH ON A BASIS INPUT

Send $|x_1x_2, s, y\rangle = |11, 0, 0\rangle$ through the three gates.

1. First Toffoli (controls $x_1x_2 = 11$): scratch flips: $|11, 1, 0\rangle$.
2. CNOT (control $s = 1$): output flips: $|11, 1, 1\rangle$.
3. Second Toffoli (controls still 11): scratch flips back: $|11, 0, 1\rangle$.

Answer: $|11, 0, 1\rangle$ — data intact, output holds $f(11) = 1$, scratch restored to $|0\rangle$. Repeat for the other three inputs and the scratch *always* ends clean; that is what makes the next example work.

WORKED EXAMPLE — WHY THE SECOND TOFFOLI IS NOT OPTIONAL

Now feed the superposition $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ into the data register (scratch and output start at 0), and compare the final state with and without the uncompute step.

1. With the full sandwich, each branch is traced as above: $\frac{1}{\sqrt{2}}(|00, 0, 0\rangle + |11, 0, 1\rangle)$. The scratch factor is $|0\rangle$ in *both* branches — it factors out, disentangled.
2. Without the second Toffoli: $\frac{1}{\sqrt{2}}(|00, 0, 0\rangle + |11, 1, 1\rangle)$. Now the scratch differs across branches: it is entangled with the data.
3. Consequence: any later attempt to interfere the data branches (an H, a reverse circuit, a Grover diffusion) now fails, because a “witness” qubit still remembers which branch is which — the interference-killer of Chapter 4, self-inflicted.

Answer: same measured output either way *today*, catastrophically different behavior the moment interference is needed. Uncomputation is not housekeeping; it is what keeps the quantum in quantum computing.

REALITY CHECK

The price is real: the sandwich roughly doubles the gate count (compute forward, compute backward, plus the copy). Compilers fight for every saved uncompute, and clever algorithms share scratch between subroutines. When Chapter 29 multiplies every logical gate by thousands of physical operations, remember that a good fraction of them are circuits diligently cleaning up after themselves.

12.7 The Cost Sheet: Width, Size, and Depth

Engineers reading a circuit ask three numbers before anything else. **Width:** how many qubits (rent). **Size:** how many gates (work). **Depth:** how many time-steps when everything that *can* run in parallel *does* run in

parallel (duration). Depth, not size, meets the coherence clock of Chapter 28 — two gates on different qubits share a time-step, and a wide circuit can be enormous yet fast.

EQUATION: READING DEPTH FROM A DIAGRAM

depth = length of the longest gate-chain along any wire-path $t_{\text{circ}} \approx \text{depth} \times t_{\text{gate}}$
In words: Slice the diagram into columns of simultaneous gates; the number of columns is the depth; multiply by gate time for the duration

Symbol	Meaning	Unit
depth	sequential time-steps	none
t_{gate}	duration of one step (the slowest gate in it)	ns
t_{circ}	total circuit duration	ns- μ s

A gate joins a column only if all its qubits are free that step — two-qubit gates block two wires at once, which is one more reason CNOTs dominate both error and time budgets.

WORKED EXAMPLE — COSTING A GHZ FACTORY

Circuit on five qubits: H on q_0 ; then CNOT $q_0 \rightarrow q_1$; then CNOT $q_1 \rightarrow q_2$ and CNOT $q_0 \rightarrow q_3$ (disjoint qubits!); then CNOT $q_3 \rightarrow q_4$. Compute width, size, depth, and duration at $t_{\text{gate}} = 60$ ns.

1. Width: 5. Size: 5 gates (1 H + 4 CNOTs).
2. Column 1: H. Column 2: CNOT $q_0 q_1$. Column 3: CNOTs $q_1 q_2$ and $q_0 q_3$ *together* (no shared qubit). Column 4: CNOT $q_3 q_4$.
3. Depth 4, though the gate count is 5 — parallelism saved a step. Duration: $4 \times 60 = 240$ ns.
4. Budget check (Chapter 28): at $T_1 = 80 \mu\text{s}$, 240 ns is 0.3% of a lifetime — comfortably inside; the four CNOTs' error toll $(1 - 0.006)^4 \approx 0.976$ dominates instead.

Answer: width 5, size 5, depth 4, duration 0.24 μ s. The habit to build: cost every circuit you design in these three numbers *before* running it — the same one-line professionalism as a chemist balancing an equation.

WORD TO KNOW

Circuit depth — The number of parallel time-steps a circuit needs — its length in time rather than in gates. The quantity that races the decoherence clock.

REALITY CHECK

Compilers fight for depth with two levers you now own: merging runs of single-qubit gates (Euler, Chapter 7) and reordering commuting gates (Chapter 6) to pack columns tighter. But the chip's wiring map fights back: a CNOT between unconnected qubits must be routed through SWAP chains — three CNOTs each — inflating size, depth, and error at once (Chapter 30). "How deep after routing?" is the question that separates paper

circuits from runnable ones.

CHAPTER SUMMARY

- Circuit literacy: wires = qubit timelines; columns = simultaneous steps; dots/ $\oplus$ /meters/double-lines all have exact meanings; depth counts columns.
- Compute circuits by the column method: full state after every column, gates applied term by term. (Matrix alternative: $\otimes$ within a column, right-to-left between columns.)
- Simplify with the identity toolbox: cancellations ($HH=I$), ladders ($TT=S$), sandwiches ($HZH=X$), CNOT re-routing — fewer gates, less depth, less noise.
- No-cloning: linearity forbids any universal copier of unknown states — a four-line proof with heavyweight consequences (no measure-the-copies loophole; error correction must be indirect; eavesdroppers get caught).
- Design method: write the target $\rightarrow$ choose blocks (H/R_y for splits, CNOT for correlation, phase gates for interference) $\rightarrow$ compute $\rightarrow$ verify and shrink. The 8-sided die: three H's, physics-certified randomness.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] In the specimen circuit of Lesson 1, list: the number of columns (depth), every two-qubit gate, and what the double lines carry.

Problem 2 [Basic] Using the column method, compute the output of: $|00\rangle$, X on q_1 , then CNOT (q_0 control).

Problem 3 [Basic] Simplify the one-qubit sequence H, H, T, T, T, T and name the resulting single gate.

Problem 4 [Basic] State the no-cloning theorem in one sentence, and name one everyday computing operation (from your phone or laptop) that it forbids for qubits.

Problem 5 [Basic] Trace $|10, 0, 0\rangle$ and $|10, 0, 1\rangle$ through the AND sandwich, gate by gate. Confirm the output ends as $y \oplus f(x)$ in both and the scratch ends clean.

Problem 6 [Basic] A classmate proposes skipping uncomputation and instead *measuring* the scratch qubit to “reset” it. Using the partial-measurement rule of Chapter 9, explain what this does to a data register that was in superposition.

Problem 7 [Basic] Cost the teleportation core: H on q_1 ; CNOT $q_1 \rightarrow q_2$; CNOT $q_0 \rightarrow q_1$; H on q_0 . Width, size, depth, duration at 60 ns/step?

Problem 8 [Basic] Two circuits have equal size 100. One has depth 100, the other depth 12. Which survives

better on hardware, and by roughly what factor in the time-decay part of the budget ($T_1 = 80 \mu\text{s}$, 60 ns steps)?

Problem 9 [Standard] Column-compute the output of: $|00\rangle$; H on both; CZ; H on both. Compare your answer to the action of a single CNOT ($q_0 \rightarrow q_1$) on $|00\rangle$... then on all four basis states if you dare — which Chapter 10 challenge problem did you just re-derive?

Problem 10 [Standard] Count depth before and after simplifying: q_0 : H, T, T, H; q_1 : Z, X, X, Z — with a CNOT between the two middle columns. Slide commuting gates first; state which identities fire.

Problem 11 [Standard] Compute the full output state of the specimen circuit of Lesson 1 (including its T gate), starting from the worked example's column-2 state. Give all four amplitudes.

Problem 12 [Standard] Design: a two-qubit circuit whose histogram is 25% each of 00, 01, 10, 11, using no two-qubit gates. Then a second design whose histogram is 50% 01 / 50% 10, which *must* use one (build $|\Psi^+\rangle$). Draw both.

Problem 13 [Standard] In the no-cloning proof, point to the exact step that fails if the machine only ever needs to copy the two known states $|0\rangle$ and $|1\rangle$ — and reconcile this with Chapter 10: which gate IS a perfect basis-state copier onto a blank?

Problem 14 [Standard] Build the sandwich for $f = x_1 \text{ OR } x_2$: conjugate the first Toffoli with X gates (De Morgan, Chapter 10) and add one X on the scratch line before the copy. List the full gate sequence and trace $|00, 0, 0\rangle$.

Problem 15 [Standard] For $f = (x_1 \wedge x_2) \wedge x_3$ use two scratch qubits: $s_1 = x_1 \wedge x_2$, then $s_2 = s_1 \wedge x_3$, copy, uncompute. Write the gate list and explain why the uncomputation must run in reverse order (last computed, first uncomputed).

Problem 16 [Standard] A subroutine costs $G = 40$ gates to compute. Count the gates in its full sandwich (compute + one copy CNOT + uncompute). What fraction of the total is cleanup?

Problem 17 [Standard] Draw (or list column by column) the depth-optimal schedule for preparing GHZ on 8 qubits by “doubling”: H, then CNOTs $0 \rightarrow 1$; $\{0 \rightarrow 2, 1 \rightarrow 3\}$; $\{0 \rightarrow 4, 1 \rightarrow 5, 2 \rightarrow 6, 3 \rightarrow 7\}$. Give size and depth, and compare with the naive chain's depth of 8.

Problem 18 [Challenge] Design a three-qubit circuit producing $\frac{1}{\sqrt{2}}|000\rangle + \frac{1}{\sqrt{2}}|110\rangle$, compute its output to verify, then add exactly one more gate to convert the output to $\frac{1}{\sqrt{2}}|000\rangle - \frac{1}{\sqrt{2}}|110\rangle$.

Problem 19 [Challenge] A “quantum backup service” advertises: “send us your qubit; we keep a safety copy and return the original untouched.” Write a three-sentence expert rejection letter citing the no-cloning theorem, and offer the one honest alternative for basis-state-only data.

Problem 20 [Challenge] Depth versus width: Circuit A applies H to each of 8 qubits in one column; Circuit B

applies the same 8 H's one per column on a machine that forbids parallel gates. Same gates, same output state. Give the depth of each; then, using “error risk grows with depth” (Chapter 28 preview), explain in two sentences which machine you would rather run — and why the die of Lesson 5 was deliberately designed at depth 1.

Problem 21 [Challenge] The one-qubit demonstration: data starts in $|+\rangle$, and a CNOT copies it onto a fresh scratch qubit. (a) Write the joint state and show the data alone no longer interferes: applying H to the data now gives $P(0) = \frac{1}{2}$ instead of 1. (b) Show that uncomputing (a second CNOT) before the H restores $P(0) = 1$. This tiny experiment — watchable in any simulator — is decoherence in its purest form.

Problem 22 [Challenge] A chip's wiring is a line: $q_0 - q_1 - q_2 - q_3$. Your circuit needs CNOT $q_0 \rightarrow q_3$. Insert SWAPs (3 CNOTs each) to bring the qubits adjacent, count the true CNOT cost of this one logical gate, and recompute the error toll at $p_2 = 0.6\%$. State the design moral about matching algorithms to connectivity.

PART FIVE

PROTOCOLS, PROGRAMMING, AND CRYPTOGRAPHY

Quantum Teleportation and Superdense Coding

Moving quantum states with classical bits, and packing two bits into one qubit

IN THIS CHAPTER YOU WILL LEARN TO

- State the problem teleportation solves, and why no-cloning and measurement make it hard.
- Trace the complete teleportation protocol, amplitude by amplitude, and verify the four correction cases.
- Explain why teleportation transmits nothing faster than light and never copies the state.
- Run the superdense coding protocol: send two classical bits with one qubit plus shared entanglement.
- Recognize the “resource accounting” view: what one Bell pair plus classical bits can buy.

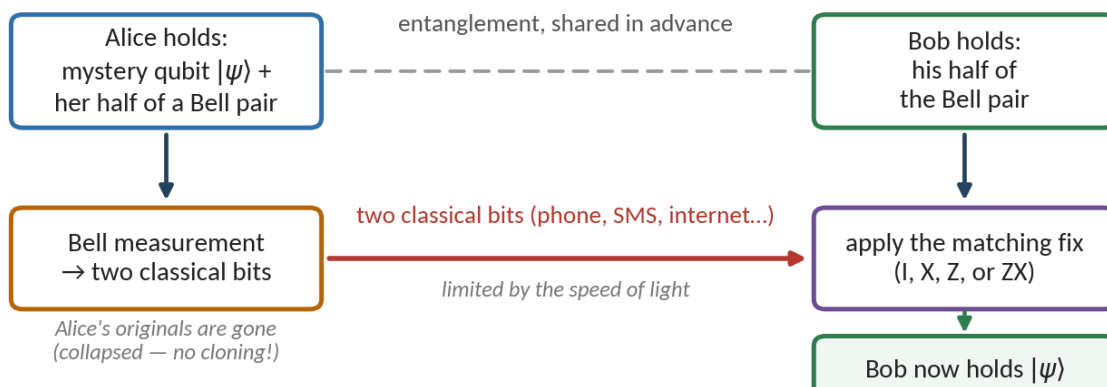
13.1 The Problem: Sending a Qubit Without Sending It

Alice, in Erbil, holds one qubit in an *unknown* state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. She wants Bob, in Duhok, to end up holding exactly that state. The direct options all fail or disappoint. Physically shipping the qubit risks noise destroying it (Chapter 28). Measuring it and phoning the result destroys the superposition and reveals only one bit — α and β are lost (Chapter 4). Measuring many copies would work statistically — but she has *one* copy, and no-cloning (Chapter 12) forbids making more.

The 1993 discovery of Bennett, Brassard, Crépeau, Jozsa, Peres, and Wootters: if Alice and Bob *share a Bell pair in advance*, Alice can transfer $|\psi\rangle$ to Bob exactly, by sending only **two ordinary classical bits**. The protocol was named *teleportation*, and for once the dramatic name is earned — the state vanishes from Alice’s lab and appears in Bob’s, without any quantum object crossing the space between beyond the pre-shared pair.

WORD TO KNOW

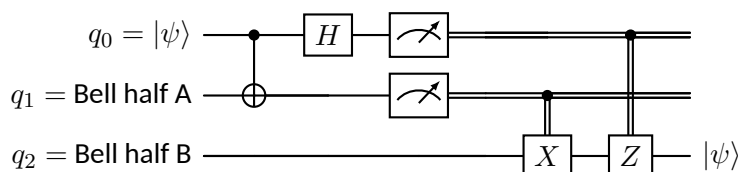
Quantum teleportation — A protocol that transfers an unknown quantum state from Alice to Bob using one pre-shared Bell pair plus two classical bits. The original is necessarily destroyed — teleportation moves states, never copies them.



Teleportation at a glance: pre-shared entanglement (dashed), a Bell measurement on Alice's side, two classical bits across, a corrective gate on Bob's side.

13.2 The Protocol, Step by Step

Three qubits play: q_0 is Alice's mystery qubit $|\psi\rangle$; q_1 and q_2 are the Bell pair $|\Phi^+\rangle$, with q_1 in Alice's lab and q_2 in Bob's. Alice performs a Bell measurement (Chapter 11, Lesson 5: CNOT, then H, then measure both) on her two qubits, and phones the two result bits to Bob.



The teleportation circuit. Alice owns the top two wires; Bob the bottom. Doubled wires carry her two measured bits; Bob applies X if the second bit (m_1) is 1, then Z if the first bit (m_0) is 1.

WORKED EXAMPLE — THE FULL ALGEBRA OF TELEPORTATION

Trace the protocol for a general $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. (This is the longest calculation so far in this book — and the most rewarding. Work with a pencil alongside.)

1. Initial three-qubit state (order $q_0q_1q_2$): $|\psi\rangle \otimes |\Phi^+\rangle = (\alpha|0\rangle + \beta|1\rangle) \otimes \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}[\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle]$.

2. CNOT ($q_0 \rightarrow q_1$): flips the middle bit where the first is 1: $\frac{1}{\sqrt{2}}[\alpha|000\rangle + \alpha|011\rangle + \beta|110\rangle + \beta|101\rangle]$.

3. H on q_0 ($|0\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, $|1\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$), then expand and collect by Alice's two bits:

$$\frac{1}{2} \left[|00\rangle (\alpha|0\rangle + \beta|1\rangle) + |01\rangle (\alpha|1\rangle + \beta|0\rangle) + |10\rangle (\alpha|0\rangle - \beta|1\rangle) + |11\rangle (\alpha|1\rangle - \beta|0\rangle) \right]$$

4. Alice measures her two qubits: each pattern occurs with probability $\left|\frac{1}{2}\right|^2 \times (|\alpha|^2 + |\beta|^2) = \frac{1}{4}$, and Bob's qubit collapses to the matching bracket.

5. Bob's repair table: result 00 — he already holds $\alpha |0\rangle + \beta |1\rangle = |\psi\rangle$ (do nothing). Result 01 — he holds $\alpha |1\rangle + \beta |0\rangle$: apply X to swap amplitudes $\Rightarrow |\psi\rangle$. Result 10 — he holds $\alpha |0\rangle - \beta |1\rangle$: apply Z $\Rightarrow |\psi\rangle$. Result 11 — both fixes: X then Z $\Rightarrow |\psi\rangle$.

Answer: in all four cases Bob finishes with exactly $\alpha |0\rangle + \beta |1\rangle$. Two classical bits selected the right repair from {I, X, Z, ZX} — and the unknown amplitudes crossed from Erbil to Duhok without ever being measured, known, or copied.

13.3 Why Teleportation Breaks No Laws

Three worries, three clean answers — each one a review of an earlier chapter.

Faster than light? No. Until Alice's phone call arrives, Bob's qubit shows the same useless 50/50 statistics regardless of everything (the no-signalling calculation of Chapter 11, Lesson 4, applies verbatim — his marginals are unchanged by her measurement). The quantum state "arrives" only when the classical bits do, and those travel by ordinary channels at light speed or slower.

Cloning? No. Look at step 4: Alice's measurement collapsed her two qubits into plain basis states. The original $|\psi\rangle$ no longer exists on her side — exactly one copy exists in the universe, now in Bob's lab. Teleportation is a *move*, and no-cloning is obeyed to the letter.

Did information really "teleport"? The two classical bits carried by phone are strikingly little: two bits could never describe the continuous amplitudes α, β (that would take infinitely many bits). The missing information traveled through the entanglement — the pre-shared correlation did work that no classical resource could. This is the cleanest demonstration in all of physics that entanglement is a genuine, spendable *resource*: the Bell pair is consumed (it ends as two collapsed classical bits' worth of nothing), one pair per teleported qubit.

COMMON MISTAKE

X Wrong: "Teleportation sends matter, like in the movies — eventually people."

✓ Right: Teleportation transfers a quantum *state* onto a qubit Bob already has; no particle, energy, or matter travels by the protocol. Scaling it to objects is not an engineering roadmap but a category error — though teleporting states between memory qubits and flying photons is real, daily lab work, and the backbone of planned quantum networks (see the case study).

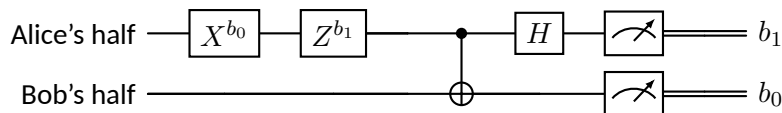
REALITY CHECK

Why teach a protocol that "merely" moves a state? Because moving states is the hard currency of quantum engineering. Future quantum processors will teleport states between chips; quantum repeaters — the only known way to send quantum information across continents through lossy fiber — are chains of teleportations; and in error-corrected machines (Chapter 29), certain logical gates are performed *by* teleportation. The protocol you just verified by hand is infrastructure, not magic.

13.4 Superdense Coding: The Mirror Protocol

Teleportation spends [1 Bell pair + 2 classical bits] to move [1 qubit of state]. Its mirror twin spends [1 Bell pair + 1 sent qubit] to deliver [2 classical bits] — squeezing two bits into a single quantum carrier. Discovered by Bennett and Wiesner in 1992, it is the neatest showcase of Chapter 11’s four Bell states.

The setup: Alice and Bob share $|\Phi^+\rangle$. This time Alice wants to *send data*, not a state: two classical bits, b_1b_0 .



Superdense coding. Left of the gap: Alice encodes two bits by applying X (if $b_0=1$) and Z (if $b_1=1$) to her half, then physically sends that one qubit to Bob. Right: Bob’s Bell measurement reads out both bits.

WORKED EXAMPLE — SUPERDENSE CODING, ALL FOUR MESSAGES

Verify the encoding: starting from $|\Phi^+\rangle$, find the state after Alice’s gates for each two-bit message, and the pattern Bob’s Bell measurement announces.

1. Message 00: apply nothing. State stays $|\Phi^+\rangle$. Bob’s decoder (Chapter 11 dictionary) announces 00.
2. Message 01: apply X to Alice’s qubit. Chapter 11’s worked example computed exactly this: $|\Phi^+\rangle \rightarrow |\Psi^+\rangle$. Decoder announces 01.
3. Message 10: apply Z: flips the sign of the $|11\rangle$ -term’s Alice-part: $|\Phi^+\rangle \rightarrow \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) = |\Phi^-\rangle$. Decoder: 10.
4. Message 11: apply X then Z: $|\Phi^+\rangle \rightarrow \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$, which is $|\Psi^-\rangle$ up to global sign. Decoder: 11.

Answer: Alice’s four local touches steer the pair to the four orthogonal Bell states, and Bob’s Bell measurement distinguishes them perfectly: two full bits recovered from one transmitted qubit. Without the pre-shared pair this is provably impossible — one qubit alone can carry at most one classical bit (Holevo’s theorem, quoted here without proof).

The resource ledger — two protocols, mirror images

Protocol	Spends	Delivers
Teleportation	1 Bell pair + 2 classical bits	1 qubit-state moved
Superdense coding	1 Bell pair + 1 qubit sent	2 classical bits

CASE STUDY — TELEPORTATION FROM LAB BENCH TO SATELLITE • 1997–2017

Anton Zeilinger’s Innsbruck group performed the first experimental quantum teleportation in 1997, moving a photon’s polarization state across a laboratory table. The chase for distance then never stopped: 600 m

across the Danube river (2004), 143 km between the Canary Islands of La Palma and Tenerife (2012), and in 2017 the Chinese satellite Micius teleported photon states from a ground station in Tibet to orbit — more than 500 km straight up, using entangled pairs distributed from space. The same 2017 campaign performed satellite-relayed, entanglement-secured intercontinental communication between Beijing and Vienna. Every one of these headline experiments executed, physically, the identical algebra you traced in this chapter’s worked example — the same four measurement outcomes, the same repair table $\{I, X, Z, ZX\}$.

ACTIVITY (APP) — RUN THE TELEPORTER

1. Open the *Teleportation Lab* in the QCFE app. Set the mystery state with sliders — try $|\psi\rangle = 0.6|0\rangle + 0.8|1\rangle$.
2. Step the protocol column by column, checking the app’s three-qubit state against the worked example’s algebra at each stage.
3. Run 20 single shots. Record Alice’s two bits and the fix Bob applied each time; confirm all four patterns occur (roughly equally) and that Bob’s Bloch arrow always finishes at the original $|\psi\rangle$.
4. Now try to cheat: use the app’s option to skip the classical bits (Bob guesses 00 always). What fraction of runs leaves Bob’s arrow wrong? Explain with the repair table.
5. Switch to the *Superdense panel*, send all four messages, and confirm Bob decodes each perfectly.

You need: The QCFE app (Android or Windows), Paper and pen

► App tool: *Teleportation Lab* — step through the protocol with a live three-qubit state view, then race the superdense channel

13.5 Entanglement Swapping: Linking Qubits That Never Met

Teleportation has a twist ending. Ask: what if the “mystery qubit” Alice teleports is itself *half of another entangled pair*? The answer is the key technology of future quantum networks.

Set the stage with four qubits. Alice holds A, entangled with B; a middle station holds B and C; and C is entangled with D, held by Bob far away. A has never interacted with D — they may sit on different continents. Now the middle station performs a **Bell measurement** on its two qubits, B and C, exactly the joint measurement of this chapter’s protocol.

EQUATION: ENTANGLEMENT SWAPPING

$$\underbrace{|\Phi^+\rangle_{AB}}_{\text{pair 1}} \otimes \underbrace{|\Phi^+\rangle_{CD}}_{\text{pair 2}} \xrightarrow{\text{Bell measurement on } B,C} \text{A and D entangled (one of the four Bell states)}$$

In words: Measuring the two middle qubits jointly throws their entanglement outward onto the two end qubits

Symbol	Meaning	Unit
$ \Phi^+\rangle$	the Bell state $\frac{1}{\sqrt{2}}(00\rangle + 11\rangle)$	none
A, D	end qubits, never in contact	none
B, C	middle qubits, jointly measured	none

Each of the four Bell outcomes occurs with probability $\frac{1}{4}$, and each leaves A-D in the matching Bell state — known exactly once the middle station announces its two classical bits.

The middle station's result tells Bob how to standardize the A-D pair

Bell result on B,C	State of A,D	Bob's correction on D
Φ^+	Φ^+	none
Φ^-	Φ^-	Z
Ψ^+	Ψ^+	X
Ψ^-	Ψ^-	X then Z

WORKED EXAMPLE — ONE BRANCH, FULLY CHECKED

Verify the first row: if the Bell measurement on B,C yields Φ^+ , show A,D end in Φ^+ , and find the outcome's probability.

- Expand the four-qubit state: $|\Phi^+\rangle_{AB} |\Phi^+\rangle_{CD} = \frac{1}{2}(|0000\rangle + |0011\rangle + |1100\rangle + |1111\rangle)$ (order A,B,C,D).
- Project B,C onto $|\Phi^+\rangle_{BC} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)_{BC}$: only terms whose *middle* digits read 00 or 11 survive. Scan the middle digits of the four terms in order: 00, 01, 10, 11 — so only the first and last terms survive.
- Collect the survivors, each picking up the overlap factor $\frac{1}{\sqrt{2}}$: unnormalized result $\frac{1}{2\sqrt{2}}(|00\rangle_{AD} + |11\rangle_{AD})$; its squared length is $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$.
- Renormalize: $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)_{AD} = |\Phi^+\rangle_{AD}$ ✓

Answer: probability $\frac{1}{4}$, and A-D emerge in the perfect Bell state Φ^+ — two qubits, entangled, that have never met. The other three branches work identically (each $\frac{1}{4}$), differing only by the correction in the table.

WORD TO KNOW

Quantum repeater — A chain of entanglement-swapping stations that extends entanglement over long distances segment by segment — storing successful segments in quantum memory until neighbors succeed — rather than hoping one photon survives the whole journey.

REALITY CHECK

Why bother? Optical fiber loses roughly half its photons every 15–20 km, so direct entanglement distribution over 1000 km is astronomically unlikely — and amplifiers are forbidden (no-cloning!). Swapping offers the honest alternative: create entanglement over many *short* segments, store the successes, and stitch neighbors together station by station. The 2017 Micius satellite results and today’s metropolitan test networks are the first rungs of exactly this ladder.

13.6 Teleporting Through Noise: The Fidelity of a Real Link

Real Bell pairs arrive bruised: distributed through fibers and swaps, the shared state is not $|\Phi^+\rangle$ but something close to it. How good is teleportation through a bruised pair? One clean formula answers, and it draws a sharp line between quantum and classical.

EQUATION: TELEPORTATION FIDELITY THROUGH AN IMPERFECT PAIR

$$F_{\text{out}} = \frac{2f + 1}{3}, \quad f = \text{fidelity of the shared pair with } |\Phi^+\rangle$$

In words: The teleported state’s average quality is set by the pair’s quality — linearly, with no mercy and no miracles

Symbol	Meaning	Unit
f	pair quality: $f = 1$ perfect, $f = \frac{1}{4}$ useless	none
F_{out}	average fidelity of the delivered state	none

Landmarks: $f = 1 \Rightarrow F_{\text{out}} = 1$ (the ideal protocol of this chapter). $f = \frac{1}{2}$ (the best any unentangled pair can offer) $\Rightarrow F_{\text{out}} = \frac{2}{3}$ — exactly the ceiling of the best classical strategy (measure the input, phone the result, rebuild). Beating $\frac{2}{3}$ is therefore the proof of genuine quantum teleportation that every experiment must exhibit.

WORKED EXAMPLE — GRADING A LINK

A repeater chain delivers pairs of fidelity $f = 0.85$. Grade the teleportation service, and find the minimum pair quality for a target $F_{\text{out}} \geq 0.95$.

1. Grade: $F_{\text{out}} = (2 \times 0.85 + 1)/3 = 2.7/3 = 0.90$.
2. Context: 90% beats the classical ceiling 66.7% decisively — genuine teleportation — but would wreck a long computation if used repeatedly (Chapter 28’s compounding).
3. Requirement: $F_{\text{out}} \geq 0.95 \Rightarrow 2f + 1 \geq 2.85 \Rightarrow f \geq 0.925$.
4. Engineering consequence: raw long-distance pairs rarely reach 0.925 — which is why real repeater designs add *entanglement distillation*: sacrificing several bruised pairs to forge one better pair before teleporting.

Answer: $F_{\text{out}} = 0.90$ today; the 0.95 service needs pairs at $f \geq 0.925$, hence distillation. The formula turns network engineering into arithmetic: every proposed quantum internet layer must show its f budget the way classical networks show packet-loss budgets.

WORD TO KNOW

Entanglement distillation — Local operations plus classical communication that convert several low-fidelity pairs into fewer higher-fidelity ones — the error correction of entanglement links, and a standard stage in repeater designs.

REALITY CHECK

The classical ceiling $F = \frac{2}{3}$ is the honest referee of the field's headlines. When a 1997 experiment or a 2017 satellite claims teleportation, the reviewer's first question is "average fidelity, over what input states, against $2/3$?" Values published: Bouwmeester 1997, around 0.80; Micius ground-to-satellite 2017, around 0.80 — both cleanly past the line through heroic engineering. Numbers, not adjectives, certify quantum feats — this book's oldest habit, applied at 1400 km altitude.

CHAPTER SUMMARY

- Teleportation moves an unknown state $|\psi\rangle$ from Alice to Bob using one pre-shared Bell pair + two classical bits: Bell-measure Alice's pair, phone two bits, Bob applies I/X/Z/ZX.
- The algebra: rewriting $|\psi\rangle \otimes |\Phi^+\rangle$ groups Bob's qubit into four brackets — each Alice outcome leaves Bob one fixed gate away from $|\psi\rangle$.
- No law is scratched: no signalling before the classical bits arrive; the original is destroyed (no-cloning); the Bell pair is consumed — entanglement is a spendable resource.
- Superdense coding is the mirror: Alice's local $\{I, X, Z, ZX\}$ steers $|\Phi^+\rangle$ among all four Bell states; sending her one qubit lets Bob's Bell measurement read two full bits.
- These two protocols are the foundation stones of quantum networking — repeaters, chip-to-chip links, and satellite experiments all run on them.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] List the three resources present at the start of teleportation and state which are consumed by the end.

Problem 2 [Basic] Alice's measurement gives 01. What state does Bob hold at that instant (for general α, β), and which gate repairs it?

Problem 3 [Basic] In superdense coding, Alice wants to send 10. Which gate(s) does she apply, and which Bell state does Bob receive?

Problem 4 [Basic] Why must Alice's original qubit lose the state $|\psi\rangle$ during teleportation? Name the theorem

that would be violated otherwise.

Problem 5 [Basic] The middle station announces Ψ^+ . What state do A and D share before correction, and what single gate on D standardizes it to Φ^+ ?

Problem 6 [Basic] In a chain A-B, C-D, E-F with swaps at (B,C) and then (D,E), how many classical bits in total must reach the end stations before the final A-F pair is usable? Why can the swaps themselves happen in any order?

Problem 7 [Basic] Compute F_{out} for pair qualities $f = 1, 0.925, 0.85$, and 0.5 , and label each with its meaning.

Problem 8 [Basic] Why is $F_{\text{out}} = \frac{2}{3}$ achievable with no entanglement at all? Describe the classical strategy in two sentences (Chapter 4's best single-shot estimate is the ingredient).

Problem 9 [Standard] Verify the teleportation bracket for outcome 10 explicitly: starting from the step-3 state in the worked example, keep only the terms where Alice's bits are 10, and show Bob's (normalized) qubit is $\alpha|0\rangle - \beta|1\rangle$; then verify Z repairs it.

Problem 10 [Standard] Teleport a known state as a check: run the algebra with $\alpha = 1, \beta = 0$ (so $|\psi\rangle = |0\rangle$). What are Alice's outcome probabilities now, and why does Bob's repair still always work?

Problem 11 [Standard] Eve taps the phone line and learns Alice's two bits (say 11). Show that without Bob's qubit these bits tell her *nothing* about α, β — compare what changes in the four-outcome table when $|\psi\rangle$ changes. What, physically, protects the state's privacy?

Problem 12 [Standard] Superdense in reverse order: Alice applies Z first, then X (instead of X then Z) for message 11. Compute the resulting state and show Bob's decoder still announces 11. What kind of difference did the order make?

Problem 13 [Standard] Chained teleportation: Bob immediately teleports the received $|\psi\rangle$ onward to Charlie. Count the total resources spent moving $|\psi\rangle$ from Alice to Charlie, and state the general rule for a chain of k hops. (This is the arithmetic of a quantum repeater.)

Problem 14 [Standard] A photonic Bell analyzer built from beam splitters can only identify 2 of the 4 Bell states, so each swap succeeds with probability $\frac{1}{2}$ (failures are discarded). For a chain needing $k = 4$ swaps, what fraction of attempts yields an end-to-end pair? Evaluate it.

Problem 15 [Standard] Fiber transmits $T = 10^{-L/50}$ of its photons over L km (a 0.2 dB/km line). Compare (a) sending one photon 100 km directly with (b) two 50-km segments that must *both* succeed in the same attempt. Show the probabilities are identical — and explain why adding quantum *memory* at the middle station (keep a successful half while retrying the other) finally breaks the tie.

Problem 16 [Standard] Explain why entanglement swapping cannot send information by itself: what do A's and D's local statistics look like before the middle station's two bits arrive?

Problem 17 [Standard] A link's pairs pass through two independent degradations, each multiplying the "excess" $f - \frac{1}{4}$ by 0.9. Starting from $f = 1$, compute the delivered f and F_{out} . (Work with the excess: $f = \frac{1}{4} + \frac{3}{4} \times 0.9 \times 0.9$.)

Problem 18 [Challenge] Teleporting half a Bell pair (entanglement swapping): Alice and Bob share $|\Phi^+\rangle$; Bob and Charlie share another $|\Phi^+\rangle$. Bob teleports his half of the FIRST pair to Charlie using the second pair. Argue in three sentences — using only "teleportation moves a state faithfully" — that Alice and Charlie, who never interacted, now share $|\Phi^+\rangle$. Why is this the key to long-distance quantum networks?

Problem 19 [Challenge] A startup claims: "our teleportation compresses data — two classical bits move a whole qubit, so we beat every telecom code." Write a four-sentence referee note: what the two bits do and do not carry, what resource the claim ignores, what Holevo's theorem says a lone qubit can carry, and the correct resource ledger for each protocol.

Problem 20 [Challenge] In the teleportation circuit, suppose Bob impatiently applies his X and Z fixes *conditioned on guessed bits* 00 before Alice's real bits arrive — and applies the true correction later when they do. Show, case by case on the repair table, that "guess-then-correct" still ends with exactly $|\psi\rangle$ (use $XX = ZZ = I$ and the fact that a global sign from reordering Z past X is harmless). What does this teach about the *timing* of corrections versus the timing of information?

Problem 21 [Challenge] Verify the Ψ^- row of the table by the worked example's projection method: expand the four-qubit state, project B,C onto $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$, and show the (renormalized) A-D state is $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$ with probability $\frac{1}{4}$.

Problem 22 [Challenge] Your network delivers $f = 0.80$ pairs. A distillation round consumes two pairs of quality f and outputs (with success probability $\approx \frac{1}{2}$) one pair of quality $f' \approx \frac{f^2}{f^2 + (1-f)^2}$. Compute f' from $f = 0.80$, then F_{out} before and after one round, and the expected number of raw pairs consumed per distilled teleportation. Is the trade worth it for the 0.95 service target?

CHAPTER FOURTEEN

Programming Quantum Computers

Python, NumPy, and Qiskit — from hand calculations to real machines on the cloud

IN THIS CHAPTER YOU WILL LEARN TO

- Explain the difference between a simulator and real quantum hardware, and when to use each.
- Build states and gates in Python with NumPy, and reproduce this book's hand calculations in code.
- Write, draw, and run your first Qiskit circuits.
- Run circuits with shots, read the returned counts, and plot histograms.
- Access real quantum computers over the cloud — from anywhere, including Iraq and the Kurdistan Region.

14.1 Simulators and Real Machines

Everything you have computed by hand — states, gates, probabilities — a computer can compute faster. Two very different kinds of “running a circuit” exist, and professionals use both daily.

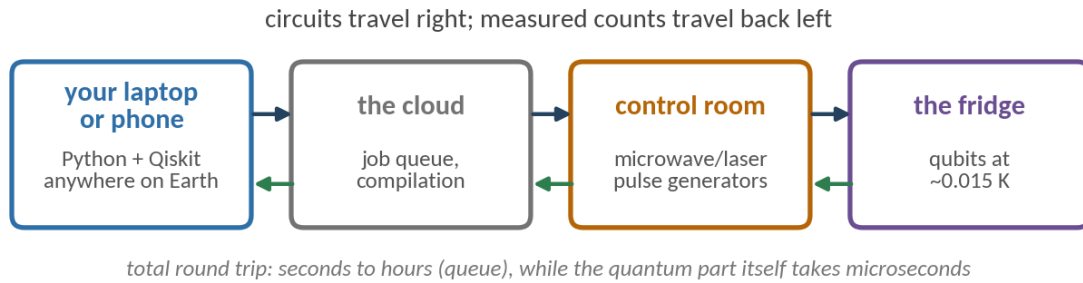
WORD TO KNOW

Simulator — A classical program that stores the state vector (all 2^n amplitudes) and multiplies the matrices exactly — perfect, noiseless results, but limited to roughly 30–40 qubits by the memory wall of Chapter 9.

WORD TO KNOW

Quantum hardware — A physical processor whose qubits undergo the real gates. No memory wall — but every gate and measurement carries noise (Chapter 28). Accessed today almost entirely through the cloud.

The honest division of labor around 2026: simulators are where you *learn, develop, and debug* — they are free, instant, and show you amplitudes directly, which no real machine can (measurement collapse!). Hardware is where you *verify* that a circuit survives real noise, and where genuinely large states live. This book's rule of thumb: every circuit meets a simulator first.



How a quantum program actually runs: your code travels from laptop to cloud queue to control electronics to a refrigerated processor — and the counts travel back.

14.2 Python and NumPy: This Book's Mathematics, Executable

Appendix B and Chapters 3–12 live in about fifteen lines of NumPy. Type this in — there is no better way to make the mathematics permanent:

```
import numpy as np

# --- states as column vectors (Chapter~\ref{ch:qubits}) ---
ket0 = np.array([1, 0], dtype=complex)
ket1 = np.array([0, 1], dtype=complex)

# --- gates as matrices (Chapter~\ref{ch:gates1}-9) ---
X = np.array([[0, 1], [1, 0]], dtype=complex)
Z = np.array([[1, 0], [0, -1]], dtype=complex)
H = np.array([[1, 1], [1, -1]], dtype=complex) / np.sqrt(2)

psi = H @ ket0          # apply H to |0> ("@" = matrix product)
print(psi)              # [0.707+0j 0.707+0j] -> the state |+>

probs = np.abs(psi)**2   # Born rule (Chapter~\ref{ch:measure})
print(probs)            # [0.5 0.5]
```

Eight working lines: states, gates, the Born rule. The @ operator is NumPy's matrix product — gate application is literally one keystroke.

Multi-qubit machinery is one function away — `np.kron` is the tensor product $\otimes$ of Chapter 9:

```
CNOT = np.array([[1,0,0,0],
                 [0,1,0,0],
                 [0,0,0,1],
                 [0,0,1,0]], dtype=complex)
```

```

state = np.kron(ket0, ket0)          # |00> (4-component vector)
state = np.kron(H, np.eye(2)) @ state # H on qubit A: (H x I)|00>
state = CNOT @ state                 # entangle!
print(state)                        # [0.707  0  0  0.707] -> the Bell state (Chapter~\ref{ch:bell})
print(np.abs(state)**2)              # [0.5  0  0  0.5]

```

The Bell-state factory of Chapter 11, executed. Compare each printed vector with your own hand computation — they must match exactly.

WORKED EXAMPLE — VERIFY A BOOK CALCULATION IN CODE

Use NumPy to check Chapter 6's worked example: $HZH = X$.

1. Compute the product in matrix order: $M = H @ Z @ H$.
2. Print it: `print(np.round(M.real, 6))`.
3. Output: `[[0. 1.] [1. 0.]]` — exactly the X matrix.

Answer: the computer confirms $HZH = X$ to numerical precision. Make this a habit: every identity and worked example in this book is three NumPy lines away from an independent check.

COMMON MISTAKE

X Wrong: `H * ket0` — using `*` to apply a gate.

✓ Right: `*` is element-wise multiplication in NumPy; matrix action needs `@` (or `np.dot`). The wrong operator fails silently — shapes may even match — producing quietly nonsensical states. If probabilities stop summing to 1, check your operators first.

14.3 Your First Qiskit Circuit

NumPy scales badly in convenience: hand-building 2^n matrices gets old fast. Quantum programming frameworks — Qiskit (IBM), Cirq (Google), PennyLane (Xanadu), and others — let you write *circuits* instead, then simulate or submit to hardware. This book uses Qiskit, the most widely deployed; install once with `pip install qiskit qiskit-aer`.

```

from qiskit import QuantumCircuit

qc = QuantumCircuit(2, 2)          # 2 qubits, 2 classical bits
qc.h(0)                             # H on qubit 0
qc.cx(0, 1)                         # CNOT: control 0, target 1
qc.measure([0, 1], [0, 1])         # measure both into the classical bits

print(qc.draw())                   # ASCII circuit diagram

```

The Bell circuit in four Qiskit lines. `qc.draw()` prints wires and gates much like this book's diagrams — always draw before running; most bugs are visible.

One convention deserves a red flag. Qiskit numbers qubits from zero and, when printing measurement results, puts **qubit 0 at the right** end of the bit string — the opposite of this book's left-to-right reading. Our $|10\rangle$ ($q_0=1$, $q_1=0$) prints in Qiskit as 01. For the symmetric Bell results (00, 11) nothing visibly changes, which is why the trap stays hidden until it bites. Check the ordering the first time you use *any* framework.

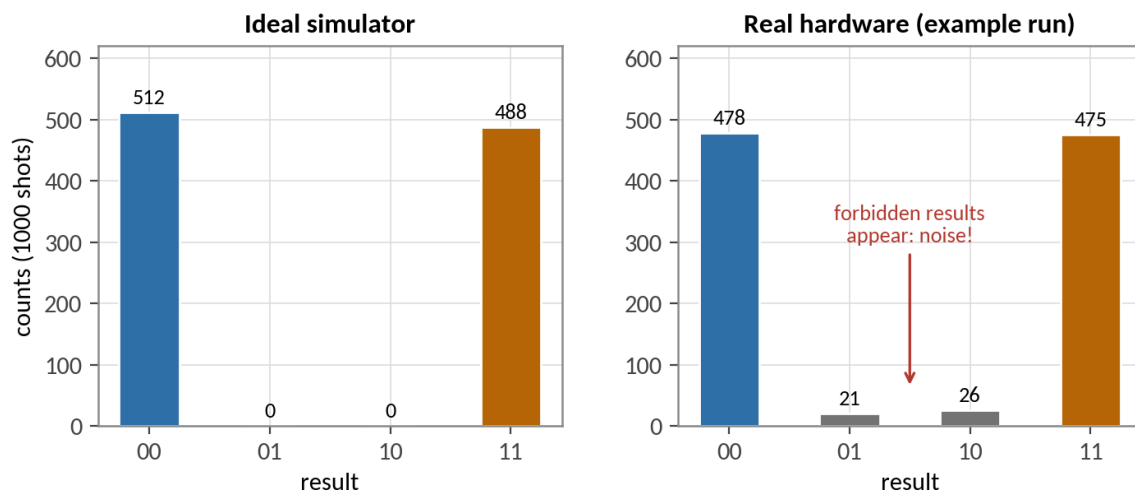
14.4 Running with Shots: Counts and Histograms

```
from qiskit_aer import AerSimulator

sim = AerSimulator()
job = sim.run(qc, shots=1000)          # run the Bell circuit 1000 times
counts = job.result().get_counts()
print(counts)                          # e.g. {'00': 508, '11': 492}
```

Shots on a simulator: prepare, run, measure, repeat — exactly Chapter 4's experimental loop, automated. `plot_histogram(counts)` (from `qiskit.visualization`) draws the bar chart.

The returned counts dictionary is precisely the measurement histogram you learned to read in Chapter 4: keys are result patterns, values are how many of the 1000 shots produced each. All your statistical instincts transfer — expected count = shots $\times$ probability, wobble $\approx \sqrt{N}$, and small-shot histograms lie.



The same Bell circuit on an ideal simulator and on (example) real hardware. The simulator shows textbook physics; the hardware shows physics plus noise — including “impossible” 01 and 10 counts. Chapter 28 explains every imperfection in this picture.

WORKED EXAMPLE — FROM COUNTS BACK TO PHYSICS

A real-hardware run of the Bell circuit returns {00: 478, 01: 21, 10: 26, 11: 475}. Estimate the error level.

1. Total shots: $478 + 21 + 26 + 475 = 1000$.
2. “Forbidden” results: $21 + 26 = 47$ shots $\Rightarrow$ error fraction $\approx 4.7\%$.
3. Compare with statistics: an ideal device would give 0 forbidden counts, and $\sqrt{1000} \approx 32$ cannot explain 47 appearing where probability is exactly zero — noise, not wobble.
4. The legal results split 478 : 475 — comfortably within wobble of 50/50.

Answer: about 5% of shots were corrupted — typical for a two-gate circuit on today’s hardware. Professionals quote such “forbidden-count” fractions as a quick device health check; you now can too.

14.5 To the Cloud: Real Machines from Anywhere

Here is the quiet revolution inside this chapter: the fridge in the figure can be on another continent. Since IBM’s 2016 opening (Chapter 6’s case study), running real quantum hardware requires no visa, no lab, and often no money — only an internet connection and the skills this book teaches.

Cloud access routes (details change fast — verify current terms before relying on them)

Provider (examples)	What you get	Typical cost to students
IBM Quantum	Qiskit access to superconducting processors; open plan with monthly free runtime	free tier + paid plans
Amazon Braket, Microsoft Azure Quantum	one account, many hardware brands (ions, superconducting, neutral atoms)	pay-per-task; academic credits
Google, IonQ, Quantinuum, others	direct or via the platforms above	mostly paid; research programs
Simulators (local)	Aer and friends on your own laptop — this book’s daily workhorse	always free

CASE STUDY — A QUANTUM LAB WITH NO LAB · FROM ERBIL TO THE FRIDGE, TODAY

What does it take for a student at Erbil Polytechnic University — or a self-learner in Duhok, Sulaymaniyah, or Baghdad — to run a real quantum computer? In practice: a laptop, an internet connection, a free cloud account, and the contents of this chapter. The author’s own path in power-systems engineering required laboratories full of expensive hardware; quantum computing is the rare frontier where the actual research instrument is open to anyone on Earth from day one. Regional universities can build a genuine quantum curriculum — circuits homework run on real processors, graduation projects benchmarking hardware noise — with essentially zero equipment budget. Chapter 30 maps this opportunity in detail; every skill it needs, you now have.

REALITY CHECK

Two honest warnings before your first hardware run. *Queues*: free-tier jobs wait behind the world's other students — minutes to many hours; develop on the simulator and submit final versions only. *Noise*: expect your beautiful textbook histogram to come back smudged, as in this chapter's figure — around 1–5% corruption for tiny circuits, worse as circuits deepen. That smudge is not failure; it is the subject of Chapter 28, and learning to anticipate, measure, and reason about it is precisely what separates a quantum professional from a tutorial-follower.

ACTIVITY (HOME) — YOUR FIRST QUANTUM PROGRAM

1. Install Python (python.org), then run `pip install numpy qiskit qiskit-aer`.
2. Type in this chapter's NumPy block and confirm the Bell state's four amplitudes.
3. Type in the Qiskit Bell circuit, draw it, and run 1000 simulator shots. Compare your counts with a classmate's — same physics, different randomness.
4. Change `qc.h(0)` to `qc.x(0)` and rerun. Predict the counts before looking (Chapter 10 knows the answer).
5. Optional, recommended: create a free IBM Quantum account, submit the Bell circuit to a real device, and compute your run's forbidden-count percentage as in the worked example.

You need: A computer with internet, Python 3, about an hour

► *App tool: Code Runner — the QCFE app bundles an offline simulator with editable NumPy/Qiskit-style snippets for every chapter, so all of this works with no installation*

14.6 Debugging Quantum Programs

Classical debugging's favorite tool — print the variables and look — is illegal on quantum hardware: reading the state collapses it, and no-cloning forbids photocopying it for inspection. Debugging quantum programs is therefore a discipline of its own, with three workhorse techniques, in the order a professional reaches for them.

1. **Print the statevector — on the simulator.** Simulators break the laws of physics on purpose: they will happily print all 2^n amplitudes mid-circuit. Develop there first, asserting the exact state at every checkpoint.
2. **Feed deterministic test inputs.** On basis-state inputs, many circuits are supposed to behave classically (the CNOT truth table, the adder's arithmetic). Deterministic in, deterministic out: any randomness in the histogram is already a bug report.
3. **Change the measurement basis.** Phase bugs are invisible in standard-basis counts. Add the right basis rotation before measuring and the invisible becomes a certainty test.

WORD TO KNOW

Assertion — A checkpoint statement — “at this point the state must equal such-and-such” — that a simulator checks exactly and hardware checks statistically. The quantum programmer’s replacement for print-and-stare.

WORKED EXAMPLE — HUNTING A WIRING BUG FROM ITS HISTOGRAM

A student’s two-line program intends a Bell pair: H on q_0 , then CNOT with control q_0 , target q_1 ; expected counts $\approx 50\% 00, 50\% 11$. The run returns $50\% 00, 50\% 10$ (writing q_0q_1). Diagnose.

1. Read the signature: q_0 is behaving (0 half the time, 1 half the time), but q_1 is *never* 1. The correlation is gone and the target never flips.
2. Hypothesis: the CNOT’s control and target are swapped. Then the actual circuit is H on q_0 , CNOT controlled by q_1 .
3. Trace the hypothesis: after H, the state is $\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$; the CNOT’s control q_1 is 0 in both branches, so the gate does nothing. Predicted counts: $50\% 00, 50\% 10$ — exactly the observation ✓

Answer: reversed CNOT arguments. Histogram signatures localize bugs remarkably well: “a qubit that never budes” points at its gate’s wiring, before you read a line of code. (This particular slip — `cx(1, 0)` for `cx(0, 1)` — may be the most-typed bug in quantum computing.)

WORKED EXAMPLE — CATCHING A PHASE BUG THE COUNTS CANNOT SEE

A library routine should output $|+\rangle$. A buggy version outputs $|-\rangle$ (a stray Z crept in). Standard-basis counts are 50/50 for both. Design the test that tells them apart with certainty.

1. Standard basis is blind: $P(0) = \frac{1}{2}$ for both states — 10,000 shots of Z-measurement cannot distinguish them, even in principle.
2. Rotate before measuring: apply H, then measure. Since $H|+\rangle = |0\rangle$ and $H|-\rangle = |1\rangle$, the two candidates become *opposite certainties*.
3. Decision rule: correct routine $\Rightarrow 100\% 0$; buggy $\Rightarrow 100\% 1$. A handful of shots settles it beyond doubt.

Answer: append H, measure, expect all-zeros. Every phase relationship has a basis in which it becomes a plain probability — the practical content of Chapter 8. A test suite that only ever measures in the standard basis is checking half the state.

```
# Assertion-style test for the Bell-pair routine (NumPy simulator)
psi = run_circuit()           # your program returns the statevector
target = np.array([1, 0, 0, 1]) / np.sqrt(2)
assert np.allclose(psi, target), "state check failed"

counts = sample(psi, shots=4000)  # histogram check (hardware-style)
```



```
assert abs(counts['00']/4000 - 0.5) < 0.03 # ~2 sigma tolerance
assert counts.get('01', 0) + counts.get('10', 0) < 40 # forbidden bins
```

REALITY CHECK

On hardware, every assertion becomes statistical: with n shots, a probability estimate wobbles by about $\sigma \approx 0.5/\sqrt{n}$, so distinguishing $P = 0.50$ from $P = 0.55$ reliably needs hundreds of shots, not ten (Chapter 4). Budget shots for your tests the way classical engineers budget CI minutes — and always test the *simulator-verified* circuit on hardware, never a circuit you have only reasoned about.

14.7 From Counts to Truth: Correcting Readout Errors

The last gate fires perfectly — and then the measurement itself lies: real readout misreports a 0 as 1 (and vice versa) a percent or two of the time (Chapter 28). Unlike gate errors, readout errors can be largely *undone in software*, because they act on the final classical statistics in a perfectly modelable way. This is the gentlest, most standard form of error mitigation — and it is pure bookkeeping.

EQUATION: ONE-QUBIT READOUT CORRECTION

$$m_0 = (1 - \epsilon) t_0 + \epsilon t_1 \implies t_0 = \frac{m_0 - \epsilon}{1 - 2\epsilon}$$

In words: Measured frequencies are a known mixing of the true ones; invert the mixing to recover the truth

Symbol	Meaning	Unit
m_0	measured frequency of “0”	none
t_0	true probability of 0 (what you want)	none
ϵ	readout flip rate (from calibration)	none

Assumes symmetric flips for clarity; real devices calibrate $\epsilon_{0 \rightarrow 1} \neq \epsilon_{1 \rightarrow 0}$ and invert a small matrix instead — same idea, two numbers. Calibration is the first measurement: prepare $|0\rangle$, count the lies; prepare $|1\rangle$, count again.

WORKED EXAMPLE — UN-LYING A HISTOGRAM

Calibration says $\epsilon = 2\%$. A 1000-shot run returns 530 zeros. Recover the true probability, with its uncertainty, and judge whether the correction mattered.

1. Raw estimate: $m_0 = 0.530$.
2. Correct: $t_0 = (0.530 - 0.020)/(1 - 0.040) = 0.510/0.960 = 0.531$.
3. Uncertainty survives correction: shot wobble $\sigma \approx 0.016$, slightly *magnified* by the $1/(1 - 2\epsilon)$ factor to ≈ 0.0165 .
4. Judgment: the correction moved the answer by 0.001 — negligible here, because m_0 sits near 0.5 where symmetric flips almost cancel. Repeat for $m_0 = 0.95$: $t_0 = 0.930/0.960 = 0.969$ — a two-point correction, twenty times the size, on exactly the near-certain measurements that specifications care about.

Answer: $t_0 = 0.531 \pm 0.017$ — and the lesson that readout correction matters most at the extremes, where certifications live (Chapter 4’s shot-budget lesson). Every serious toolchain applies this automatically; every serious programmer knows what it did.

WORD TO KNOW

Readout mitigation — Software inversion of the calibrated measurement-error model applied to final counts. The cheapest error mitigation: no extra quantum operations, just linear algebra on histograms.

REALITY CHECK

Two honest limits. First, inversion amplifies statistical noise (the $1/(1 - 2\epsilon)$ factor grows with ϵ) — correcting a terrible readout produces a precise-looking lie with big error bars; budget shots accordingly. Second, multi-qubit readout errors correlate between neighbors on real chips, so the per-qubit model is an approximation — toolchains offer correlated calibration at the cost of 2^n calibration circuits. Mitigation is repair, not resurrection: it cannot recover information the hardware never produced (Chapter 28’s distinction between mitigation and correction).

CHAPTER SUMMARY

- Simulators compute amplitudes exactly (up to ~ 30 – 40 qubits) and are free and instant: learn and debug there. Hardware is noisy, queued, and real: verify there.
- NumPy is this book executable: kets are arrays, gates are matrices, @ applies, `np.kron` is $\otimes$, `np.abs(...)**2` is the Born rule.
- Qiskit writes circuits (`QuantumCircuit`, `.h`, `.cx`, `.measure`), draws them, and runs them on `AerSimulator` with shots; `get_counts()` returns the histogram dictionary.
- Mind the conventions: bit-order differs between this book (q_0 left) and Qiskit’s printed strings (q_0 right).
- Real hardware is a cloud account away — from anywhere, including this book’s region; expect queues and a few percent noise, and treat both as professional data, not disappointment.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher’s Guide.

Problem 1 [Basic] Write the NumPy lines that build $|-\rangle$ two ways: from `ket0`, `ket1` with arithmetic, and as `H @ ket1`. How would you check the two agree?

Problem 2 [Basic] What does `np.kron(ket1, ket0)` return, in this book’s ket notation and as a 4-component vector?

Problem 3 [Basic] List the four Qiskit calls that build and measure the Bell circuit, in order.

Problem 4 [Basic] A simulator returns {0: 245, 1: 755} for 1000 shots on one qubit. Estimate $|\beta|$ for the state that was measured.

Problem 5 [Basic] A one-qubit program should print 50/50 counts but returns 100% 0. Name the single most likely omitted gate, and give the one-line fix.

Problem 6 [Basic] A Bell-pair program returns 50% 01, 50% 10 instead of 00/11. Show that an initialization bug — q_1 starting in $|1\rangle$ — reproduces this histogram exactly.

Problem 7 [Basic] With $\epsilon = 3\%$, correct measured frequencies $m_0 = 0.50$ and $m_1 = 0.97$. Which correction is larger, and why?

Problem 8 [Basic] Describe the two calibration circuits that measure $\epsilon_{0 \rightarrow 1}$ and $\epsilon_{1 \rightarrow 0}$, and why they must be re-run when the machine recalibrates (Chapter 28's drift).

Problem 9 [Standard] Write NumPy code (four lines) verifying that $S^2 = Z$, including the matrix definition of S with a complex entry (1j in Python).

Problem 10 [Standard] Predict `get_counts()` (approximately, for 800 shots) for the circuit: `qc.h(0); qc.h(1); qc.measure(...)` on two qubits — then state which chapter's worked example your prediction reuses.

Problem 11 [Standard] In Qiskit's printed bit-strings, this book's state $|01\rangle$ ($q_0=0, q_1=1$) appears as which string? A Bell-like circuit with control q_1 , target q_0 gives counts {00, 11} in Qiskit's printout — explain why the ordering trap stayed invisible.

Problem 12 [Standard] A 3-qubit GHZ circuit (Chapter 11, Problem 10) is run with 1600 shots on an ideal simulator. Predict the counts dictionary, then write its Qiskit construction (five circuit calls).

Problem 13 [Standard] Hardware returns {000: 389, 111: 371, others: 240 total} for that GHZ circuit. Compute the forbidden fraction, compare it with the Bell circuit's 4.7% from the worked example, and give one sentence on why deeper/wider circuits degrade more (Chapter 12, Problem 12 had the idea).

Problem 14 [Standard] Write the checkpoint (assertion) states for the three stages of the teleportation circuit of Chapter 13 with input $|\psi\rangle = |0\rangle$: after entangling, after Alice's gates, before measurement. (Simulator practice: then verify them.)

Problem 15 [Standard] A routine claims to output $|+\rangle$, but you suspect it outputs a 50/50 mixture of $|0\rangle$ and $|1\rangle$ (a broken, decohered version). Standard-basis counts cannot tell. What histogram does each candidate give after an H, and how many shots make the call convincingly?

Problem 16 [Standard] Using $\sigma \approx 0.5/\sqrt{n}$, how many shots are needed before an observed frequency can reliably (at 2σ) distinguish $P = 0.50$ from $P = 0.55$? Show the arithmetic.

Problem 17 [Standard] Show that for symmetric ϵ , the corrected estimate of $\langle Z \rangle$ is simply $\langle Z \rangle_{\text{true}} = \langle Z \rangle_{\text{meas}} / (1 - 2\epsilon)$, and compute the correction for $\langle Z \rangle_{\text{meas}} = 0.62$, $\epsilon = 2\%$. (This scaling is exactly how measured CHSH scores and VQE energies get un-lied.)

Problem 18 [Challenge] Write a NumPy simulation (pseudo-code accepted, real code better) of Chapter 6's H-Z-H circuit that prints the state after every column, matching the column method. Then add a 20-line loop drawing 1000 Born-rule samples from the final state with `np.random.choice`, and compare your homemade counts to Qiskit's.

Problem 19 [Challenge] Your teleportation homework must run on a simulator that supports mid-circuit measurement and classical control (Aer does). Sketch the Qiskit steps (calls, in order) for Chapter 13's teleportation circuit, using `qc.measure` mid-circuit and with `qc.if_test(...)` (or `c_if`) for the conditional X and Z. What printed evidence would convince a skeptic that $|\psi\rangle$ arrived intact — given that no one can read amplitudes off real hardware?

Problem 20 [Challenge] Design a fair benchmark: you may submit exactly three circuits (any size) and 1000 shots each to an unknown cloud device, and must report a one-number “health grade.” Propose your three circuits and the formula for the grade, justifying each choice in one sentence. (There is no single right answer — but Chapter 28 will show you what the professionals chose.)

Problem 21 [Challenge] Design a three-assertion regression test for a Bell-pair library function: (i) a standard-basis counts check, (ii) an X-basis parity check (H on both qubits before measuring — what counts pattern does Φ^+ give?), and (iii) a marginal check that each qubit alone is 50/50. For each assertion, name one distinct bug it would catch that the others might miss.

Problem 22 [Challenge] Asymmetric device: $\epsilon_{0 \rightarrow 1} = 1\%$, $\epsilon_{1 \rightarrow 0} = 4\%$. Write the 2×2 mixing matrix acting on (t_0, t_1) , invert it, and correct the measured $(0.90, 0.10)$. Then compute what the *symmetric* formula with $\epsilon = 2.5\%$ would have given, and quantify the bias from ignoring asymmetry.

Quantum Cryptography and Communication

BB84, eavesdropper detection, and quantum networks — secrecy guaranteed by physics

IN THIS CHAPTER YOU WILL LEARN TO

- Explain the key-distribution problem that all secret communication rests on.
- Run the BB84 protocol by hand: random bits, random bases, sifting, and key extraction.
- Compute the eavesdropper's fingerprint: why interception forces a 25% error rate, and how quickly checking catches it.
- Describe real QKD systems — fiber links, trusted nodes, satellites — and their honest engineering limits.
- Place quantum cryptography correctly beside post-quantum cryptography: complementary tools, not rivals.

15.1 The Key Problem

Every secret conversation in history reduces to one requirement: the two parties must share a **key** — a random string known to them and no one else. With a shared key, provably unbreakable encryption exists and is almost embarrassingly simple (the one-time pad: XOR each message bit with a fresh key bit — Chapter 1 logic). Without a shared key, no cleverness saves you. So the entire drama of cryptography concentrates in one act: *how do two people who have never met agree on a secret key, over channels an enemy may be watching?*

The classical world has two answers, both with cracks. Couriers with locked briefcases: expensive, slow, and you must trust the courier. Public-key mathematics (RSA and friends): elegant and universal — but resting on assumptions that computers cannot factor fast, assumptions that Chapter 19 showed a quantum computer will someday break, and that “harvest now, decrypt later” attackers exploit already.

Quantum mechanics offers a third answer with a different kind of guarantee. Two principles you have already mastered do the work: *measurement disturbs* (an unknown state cannot be read without risking change — Chapter 4), and *no cloning* (an unknown state cannot be copied for later study — Chapter 12). Together they mean something no classical channel can offer: **an eavesdropper cannot listen without leaving fingerprints**. Quantum key distribution (QKD) turns that into a working protocol.

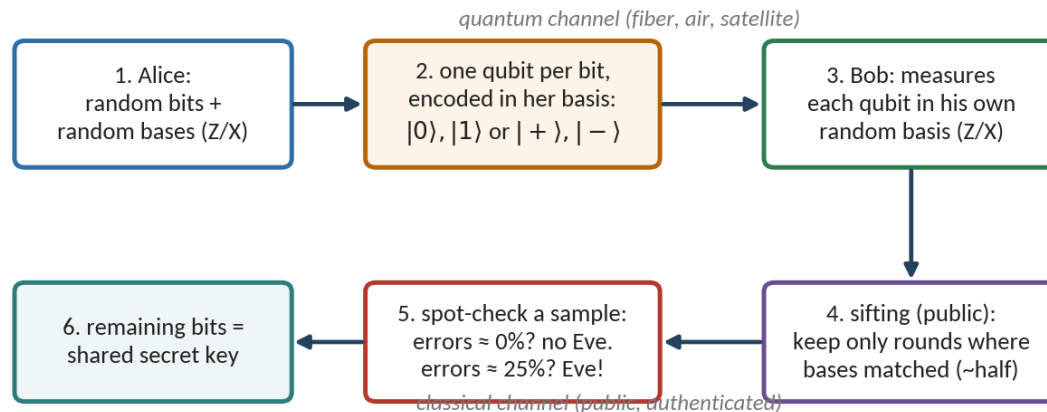
WORD TO KNOW

Quantum key distribution (QKD) — A protocol that uses quantum states to establish a shared secret key between two parties, with eavesdropping made *detectable* by the laws of quantum mechanics. The oldest and most deployed branch of quantum technology.

15.2 Wiesner's Seed: Unforgeable States

The founding idea (Wiesner, around 1970 — Chapter 12's case study) fits in one sentence: *store information in states drawn from two incompatible bases, and nobody who doesn't know the basis can read or copy it cleanly*. Concretely, use the two bases you know best: the Z -basis $\{|0\rangle, |1\rangle\}$ and the X -basis $\{|+\rangle, |-\rangle\}$ (Chapter 4, Lesson 5). A Z -encoded bit measured in the X -basis yields a fair coin — and collapses. A X -encoded bit measured in Z : the same. Guess the wrong basis and you learn nothing while damaging the goods; and no-cloning forbids the safe workaround of copying first. Bennett and Brassard's 1984 insight was that this obstacle course, useless for bank notes in 1970s technology, is *perfect* for distributing keys through an optical fiber.

15.3 BB84, Step by Step



BB84 at a glance: random bits in random bases fly as qubits; bases are compared publicly afterward; matched rounds become key; a spot-check of sacrificed bits reveals any eavesdropper.

The protocol, in five moves. (1) Alice generates a random bit *and* a random basis choice (Z or X) for each round, and sends the matching state: $|0\rangle / |1\rangle$ in the Z rounds, $|+\rangle / |-\rangle$ in the X rounds. (2) Bob, knowing nothing, measures each arriving qubit in his own random basis. (3) **Sifting**: over a public channel, they reveal only their basis choices — never the bits — and discard every round where the bases differed (about half). Where bases matched, Bob's result equals Alice's bit with certainty. (4) **Spot check**: they sacrifice a random sample of the surviving bits, comparing them publicly; any eavesdropping shows up as errors (next lesson). (5) If the error rate passes, the unsacrificed bits are the key — after classical clean-up steps (error correction for channel noise and *privacy amplification*, a hashing step that squeezes out any partial information a subtle

eavesdropper might hold).

WORKED EXAMPLE — A TWELVE-ROUND BB84 RUN, BY HAND

Trace this small run (no eavesdropper, noiseless channel) and extract the key.

1. The run, round by round:

Twelve BB84 rounds (spot-check rounds marked ☆)

Round	1	2	3	4	5	6	7	8	9	10	11	12
Alice's bit	0	1	1	0	1	0	0	1	1	0	1	0
Alice's basis	Z	X	Z	X	X	Z	X	Z	X	Z	Z	X
State sent	$ 0\rangle$	$ -\rangle$	$ 1\rangle$	$ +\rangle$	$ -\rangle$	$ 0\rangle$	$ +\rangle$	$ 1\rangle$	$ -\rangle$	$ 0\rangle$	$ 1\rangle$	$ +\rangle$
Bob's basis	Z	X	X	X	Z	Z	X	Z	Z	X	Z	X
Bases match?	✓	✓	✗	✓	✗	✓	✓	✓	✗	✗	✓	✓
Bob's bit	0	1	?	0	?	0	0	1	?	?	1	0☆

2. Sifting: rounds 3, 5, 9, 10 are discarded (mismatched bases — Bob's results there were fair coins, shown as ?). Eight rounds survive: 1, 2, 4, 6, 7, 8, 11, 12.

3. Spot check: they sacrifice rounds 1 and 12, announce both bits publicly (0 and 0), and find perfect agreement — consistent with no eavesdropper. (Real runs sacrifice a larger sample; Lesson 4 computes how large.)

4. The key: the remaining matched rounds 2, 4, 6, 7, 8, 11 give the shared secret 1, 0, 0, 0, 1, 1.

Answer: Alice and Bob now share the six-bit key 100011, created by randomness and physics, never carried whole across any channel. In real systems the same dance runs millions of rounds per second, and the “key” feeding your encrypted traffic refreshes continuously.

COMMON MISTAKE

✗ Wrong: “BB84 sends the secret message itself through the quantum channel.”

✓ Right: BB84 sends *randomness*, not messages. The output is a shared key; the actual message then travels over ordinary channels, protected by that key (one-time pad or a standard cipher). This division of labor is deliberate: keys are contentless, so a failed, noisy, or intercepted QKD session costs you nothing but time — you discard and rerun. Never put meaning on the quantum channel.

15.4 Catching the Eavesdropper

Now the theorem that makes QKD famous. Suppose Eve intercepts each qubit, measures it (she must pick some basis — say she guesses Z or X at random), and forwards a fresh qubit matching her result. What damage does she cause in the rounds that survive sifting?

WORKED EXAMPLE — THE 25% FINGERPRINT

Compute the error rate Eve’s intercept-and-resend attack forces on sifted rounds.

1. Consider a sifted round: Alice and Bob used the *same* basis — say both Z , with Alice sending $|0\rangle$.
2. Eve guesses right (probability $\frac{1}{2}$): she measures Z , sees 0, resends $|0\rangle$. Bob measures 0. No error.
3. Eve guesses wrong (probability $\frac{1}{2}$): she measures X , gets $|+\rangle$ or $|-\rangle$ (a fair coin — and Alice’s state is destroyed). She resends what she saw — an X -basis state. Bob, measuring Z on $|\pm\rangle$, gets a fair coin: wrong bit with probability $\frac{1}{2}$.
4. Total error probability per sifted bit: $\frac{1}{2} \times 0 + \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

Answer: interception forces 25% errors where an untouched channel gives (ideally) 0%. Eve cannot dodge this by cleverness within this attack: measurement disturbs, and no-cloning forbids keeping a perfect copy while forwarding the original. Her listening is written into the statistics.

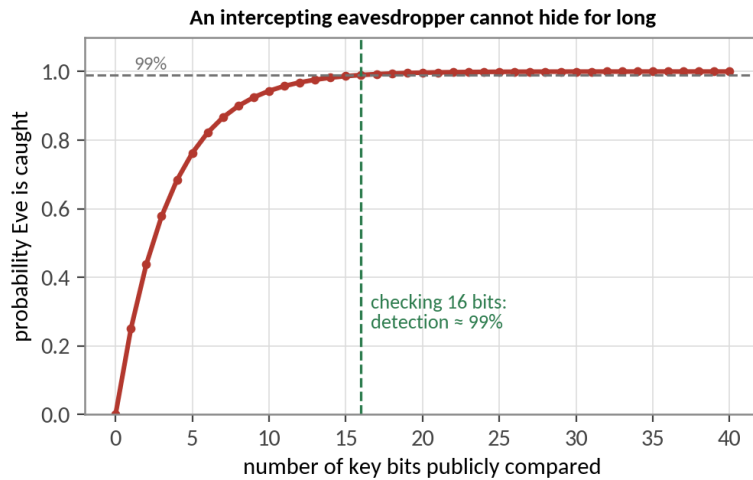
EQUATION: DETECTION PROBABILITY FROM THE SPOT CHECK

$$P_{\text{caught}} = 1 - \left(\frac{3}{4}\right)^n$$

In words: Probability the spot check exposes a full intercept-and-resend attack = 1 minus (3/4) to the power of the number of compared bits

Symbol	Meaning	Unit
n	the number of sifted bits publicly compared in the spot check	bits
$\frac{3}{4}$	the chance one compared bit <i>happens</i> to agree despite Eve	none

Sixteen compared bits already give 99% detection; fifty give 99.9999%. Real systems monitor the error rate continuously rather than in one batch, and abort or discard whenever it climbs.



Certainty by sampling: each compared bit either catches Eve (25%) or not; a handful of checks makes escape essentially impossible.

REALITY CHECK

Honesty about what the guarantee covers. The 25% theorem addresses the quantum channel; real security audits look wider. Channel *noise* also causes errors, so real systems tolerate a small error budget (a few percent) and use privacy amplification to erase whatever partial knowledge that budget might, in the worst case, represent — above a threshold ($\approx 11\%$ for BB84 against general attacks), they abort. The *classical* channel needs authentication (otherwise Eve simply impersonates Bob — a man-in-the-middle), which consumes a small pre-shared or previously-grown key. And the *devices* themselves can leak: real attacks have blinded detectors with bright light, exploited timing, and more — driving the field of device-independent QKD, whose security rests on Bell-test statistics (Chapter 11) rather than trusting hardware. Physics closes the front door; engineering must still lock the windows.

15.5 From Lab to World: Networks, Satellites, and the Road Ahead

QKD left the laboratory decades ago. Commercial fiber systems have run since the 2000s; metropolitan networks have operated in Vienna, Tokyo, Geneva (whose system has protected Swiss election data), and at scale in China, whose Beijing–Shanghai backbone spans some 2000 km. The physics that limits them is loss: photons vanish exponentially with fiber length, and unlike classical light, quantum signals *cannot be amplified* — an amplifier is a copier, and Chapter 12 forbids it. Practical fiber links top out around 100–400 km; longer routes today chain **trusted nodes** (relay stations that must be physically secured — a real weakening of the guarantee), while the clean long-term fix, **quantum repeaters** built from entanglement swapping and teleportation (Chapter 13, Problem 10), remains a laboratory frontier. Above the atmosphere the story leaps ahead: China’s Micius satellite distributed entangled pairs across 1200 km and brokered intercontinental quantum-secured links between Beijing and Vienna in 2017 (Chapter 11’s case study).

How does QKD stand next to the post-quantum cryptography of Chapter 19? As complements, not rivals — and any vendor claiming otherwise fails Chapter 2’s checklist. PQC is software: deployable everywhere instantly, no new hardware, guarding against future quantum computers with *computational* assumptions (believed, not proven, hard). QKD is hardware: point-to-point links with distance limits and real cost, offering *physical* eavesdropping detection and keys whose secrecy does not age — no future computer, quantum or otherwise, can retroactively decrypt a one-time pad. The sober deployment pattern emerging worldwide: PQC everywhere as the general answer; QKD on the highest-value fixed links (interbank backbones, government exchanges, data-center pairs) where its guarantees justify its fibers.

CASE STUDY — FROM A REJECTED MANUSCRIPT TO NATIONAL BACKBONES · BB84’S LONG ROAD, 1984–TODAY

Bennett and Brassard presented BB84 at an obscure IEEE conference in Bangalore in 1984 (the paper, famously, had been hard to publish — reviewers saw no application). The first demonstration, in 1989, sent quantum keys 32 *centimeters* across a lab bench, with a power supply so loud that, the inventors joked, the protocol was secure against any eavesdropper who happened to be deaf. From those 32 cm: kilometers of dark fiber in the 1990s, commercial products by 2003, bank transfers and election data under quantum keys in Geneva by 2007, a 2000-km national backbone and a satellite by 2017, and standards bodies (ETSI,

ITU) writing QKD interoperability specifications today. Forty years from unpublishable idea to infrastructure — driven start to finish by the two theorems you can now prove on one page: measurement disturbs, and nothing quantum can be copied.

ACTIVITY (CLASSROOM) — BB84 WITH CARDS AND COINS

1. Work in threes: Alice, Bob, and (round two) Eve. Alice flips a coin twice per round: one flip for the bit, one for the basis; she writes the state she “sends” on a card, face down.
2. Bob flips his own basis coin. If his basis matches Alice’s, he copies her bit; if not, he flips a coin for his result (that is what measurement in the wrong basis does).
3. Run 20 rounds without Eve: sift publicly, spot-check 4 bits, and extract your key. Error rate should be 0%.
4. Run 20 rounds with Eve between them, following the same rules (guess a basis, “measure,” forward what she saw). Sift, check 4 bits, and compare your observed error rate with the predicted 25%.
5. Discuss: how many check bits would your class demand before trusting a key with real money?

You need: Coins, Index cards, Paper and pen

► App tool: QKD Lab — run BB84 at any length, toggle Eve and channel noise, and watch the sifted-key error rate expose the intruder

15.6 Entanglement-Based Keys: The E91 Idea

BB84 asks Alice to *prepare* states. In 1991 Artur Ekert proposed something bolder: let a central source — possibly owned by the enemy — distribute entangled pairs, and let the CHSH score of Chapter 11 stand guard. If the pairs pass the Bell test, they *cannot* be secretly correlated with anyone else; monogamy of entanglement leaves no room for a third participant.

The recipe: for each incoming pair, Alice measures at an angle chosen randomly from $\{0^\circ, 45^\circ, 90^\circ\}$ and Bob from $\{45^\circ, 90^\circ, 135^\circ\}$. Afterward they publicly announce angles (never results) and sort the rounds.

EQUATION: THE E91 ROUND SORTER

same angle $\Rightarrow$ key bit $\{0^\circ, 90^\circ\} \times \{45^\circ, 135^\circ\} \Rightarrow$ CHSH test else $\Rightarrow$ discard

In words: Matched angles give perfectly correlated bits for the key; the four CHSH combinations measure S ; the rest are thrown away

Symbol	Meaning	Unit
key bit	rounds with equal angles (45° or 90°)	none
S	CHSH score from the test rounds	none
abort rule	proceed only if S is close to $2\sqrt{2}$	none

Security logic: a spy who measures, copies, or replaces pairs necessarily drags S down (Chapter 11); a passed test

therefore certifies the surviving key rounds — even though nobody ever checks the source's hardware.

WORD TO KNOW

Device-independent security — Security argued from measured statistics alone (a CHSH score), without trusting the internals of the source or detectors. E91 is its ancestor; full device-independent QKD was demonstrated in laboratory form in 2022.

WORKED EXAMPLE — SORTING NINE THOUSAND ROUNDS

The source emits 9000 pairs; both sides choose angles uniformly. Estimate the harvest.

1. Nine equally likely angle pairs, so each combination gets about 1000 rounds.
2. Key rounds: $(45^\circ, 45^\circ)$ and $(90^\circ, 90^\circ)$ — 2 of 9 $\rightarrow$ about 2000 bits.
3. Test rounds: $\{0^\circ, 90^\circ\} \times \{45^\circ, 135^\circ\}$ — 4 of 9 $\rightarrow$ about 4000 rounds, 1000 per CHSH correlator.
4. The remaining 3 of 9 (≈ 3000) are discarded.

Answer: about 2000 raw key bits and a CHSH estimate built on 4000 rounds. E91 pays a heavier toll than BB84 (only $\frac{2}{9}$ of rounds become key), and buys the strongest security statement in cryptography with it.

WORKED EXAMPLE — READING THE SCORE

The test rounds yield $S = 2.60$. Should Alice and Bob proceed, and what error rate should they expect in the key?

1. Compare ceilings: $2.60 > 2$ decisively (no local-hidden-answer explanation), though below the ideal $2\sqrt{2} \approx 2.83$ — normal noise.
2. Convert to visibility: $V = S/2\sqrt{2} = 2.60/2.83 \approx 0.92$.
3. Matched-angle rounds agree with probability $(1 + V)/2 \approx 0.96$, so the expected key error rate (QBER) is about 4%.
4. Decision: proceed — 4% is comfortably repaired by error correction and privacy amplification. (Full intercept-and-resend would have shown $S \approx 1.41$: instant abort.)

Answer: proceed, budgeting for a $\approx 4\%$ QBER. One measured number, S , simultaneously certifies quantumness, bounds the eavesdropper, and predicts the repair bill — the elegance that makes E91 the conceptual heart of modern QKD.

REALITY CHECK

The catch is rate: device-independent demonstrations so far produce key at a trickle (bits per second at laboratory distances) because certifying S eats statistics, and detectors must be nearly loophole-free. Deployed

commercial QKD today is BB84-family; E91-style entanglement links are the research frontier — and the design target for the quantum internet of Chapter 13’s repeaters.

15.7 Two Defenses, One Deadline: PQC, QKD, and Mosca’s Clock

Shor’s algorithm created a deadline; this chapter has shown one defense (quantum keys) and mentioned another (post-quantum classical codes). Decision-makers must choose budgets *today* — and there is a clean piece of arithmetic, due to Michele Mosca, that says when “today” is literal.

EQUATION: MOSCA’S INEQUALITY

$$L + M > T \implies \text{you are already late}$$

In words: If (how long secrets must stay secret) plus (how long migration takes) exceeds (time until a code-breaking machine), the deadline has effectively passed

Symbol	Meaning	Unit
L	required secrecy lifetime of the data	years
M	time to migrate systems to safe cryptography	years
T	years until a cryptographically relevant quantum computer	years

The inequality’s teeth come from “harvest now, decrypt later”: traffic recorded today is decrypted the day the machine exists. T is uncertain — but L and M are your numbers, and they are usually shockingly large.

WORKED EXAMPLE — RUNNING THE CLOCK FOR A MINISTRY

A health ministry’s records must stay confidential $L = 25$ years; realistic system-wide migration takes $M = 7$ years. Expert surveys put a cryptographically relevant machine plausibly $T = 15\text{--}30$ years away. Apply Mosca.

1. Compute the threshold: $L + M = 32$ years.
2. Compare across the expert range: $32 > 30 > 15$ — the inequality fires at every value in the surveyed range, optimistic or pessimistic.
3. Translate: records encrypted with RSA today are already exposed to harvest-now collection for their entire mandated lifetime.
4. Decision: begin PQC migration immediately (the standards exist: FIPS 203–205, References); no forecast refinement can change the conclusion, because even $T = 32$ years barely rescues it.

Answer: $32 > T$ for all plausible T : act now. Notice what the arithmetic did — it removed the crystal ball. You never needed to know *when* the machine arrives; you needed one addition and one comparison. This is how honest quantum-risk briefings (capstone Project 12) are actually written.

The two defenses, honestly compared

	Post-quantum cryptography (PQC)	Quantum key distribution (QKD)
What it is	new classical math (lattices, hashes) replacing RSA	keys from qubits (BB84/E91, this chapter)
Deploys on	existing networks, by software update	new optical hardware and links
Security rests on	problems believed hard for quantum computers	physics: no-cloning and Bell tests
Weakness	belief could be wrong (algorithms improve)	range, cost, and trusting the endpoints
Role today	the universal default — migrate everything	special links where physics-level assurance justifies hardware

REALITY CHECK

The professional consensus bears stating plainly: PQC is the seatbelt everyone must wear; QKD is a specialized armored car. They are complements, not rivals — and both are useless against the oldest attacks (bribed insiders, stolen laptops, sloppy endpoints). Quantum-proof mathematics wrapped around unproved institutions protects nothing; Chapter 27's audit instincts apply to security vendors too.

ACTIVITY (CLASS) — BB84 WITH PLAYING CARDS

1. Alice prepares 20 “qubits”: for each, she rolls a die for bit and basis, writing them as cards (red suits = Z basis, black = X basis; number parity = bit).
2. Bob, separately, rolls his measurement basis per card. Matching basis: he reads the bit correctly. Mismatched: he rolls a die — 50/50 random.
3. Sift publicly (compare bases, keep matches ≈ 10 bits), then spot-check 4 bits aloud. Zero errors: keep the rest as the key.
4. Round two with Eve: a third student intercepts every card, measures in her own rolled basis, forwards accordingly. Sift, spot-check — count the errors ($\approx 25\%$ expected). Compute $1 - (3/4)^4 \approx 68\%$: the probability this small check already caught her.

You need: Playing cards, Dice, Three volunteers, Paper and pen

CHAPTER SUMMARY

- All secrecy reduces to sharing random keys; classical solutions rest on couriers or breakable mathematics — and Shor's algorithm dates the latter.
- BB84: random bits in randomly chosen Z/X bases; public sifting keeps the $\approx 50\%$ of rounds with matching bases; a sacrificed sample is compared to detect intrusion; clean-up (error correction + privacy amplification) distills the final key.
- The fingerprint theorem: intercept-and-resend forces 25% errors on sifted bits; n checked bits catch it

with probability $1 - (3/4)^n = 99\%$ by sixteen bits. Measurement disturbance and no-cloning are the load-bearing walls.

- Real QKD: deployed in fibers ($\sim 100\text{--}400$ km per hop), extended by trusted nodes and satellites; loss cannot be amplified away; repeaters via entanglement swapping are the long-term fix; device attacks keep engineers humble.
- QKD and PQC are complements: software for everyone, physics for the highest-value links — and keys distributed today by QKD stay secret against every future computer.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] In BB84, roughly what fraction of transmitted qubits survive sifting, and why?

Problem 2 [Basic] Alice sends $|+\rangle$; Bob measures in the Z basis. What results can he get, with what probabilities, and does this round survive sifting?

Problem 3 [Basic] Using $P = 1 - (3/4)^n$: what is the detection probability after checking 4 bits? After 8?

Problem 4 [Basic] Name the two quantum principles that make eavesdropping detectable, and the chapter where each was established.

Problem 5 [Basic] Re-run the round sorter for 4500 pairs: how many key bits, test rounds, and discards should Alice and Bob expect?

Problem 6 [Basic] A session's test rounds give $S = 1.90$. What must Alice and Bob do, and what two very different situations (one innocent, one hostile) are indistinguishable to them at that moment? Why does the protocol not need to tell them apart?

Problem 7 [Basic] Run Mosca's clock for: (a) a bank with $L = 10$, $M = 5$; (b) a news website with $L = 1$, $M = 2$. For which values of T is each already late?

Problem 8 [Basic] Explain "harvest now, decrypt later" in two sentences, and identify which term of Mosca's inequality it justifies.

Problem 9 [Standard] Extend the worked example: Alice sends 2000 qubits. Estimate the sifted-key length, then the final key length if 200 sifted bits are sacrificed for checking and clean-up costs another 15%.

Problem 10 [Standard] Compute Eve's knowledge: in the intercept-and-resend attack, for what fraction of sifted bits does Eve hold the correct bit value? (Careful: she is right whenever she guessed the basis correctly, and half the time otherwise.) What does your answer say about why privacy amplification is needed even after a passed spot check?

Problem 11 [Standard] A channel has 3% intrinsic noise. Alice and Bob measure a 3.2% sifted error rate one day and 24% the next. Interpret both days, and state the action a real system takes in each case.

Problem 12 [Standard] Explain in three sentences why an optical amplifier — which copies incoming light into stronger outgoing light— cannot rescue QKD over long fibers, naming the theorem involved and the two honest workarounds deployed today.

Problem 13 [Standard] Man-in-the-middle: describe in four sentences what Eve can do if the classical channel is NOT authenticated (she can impersonate each party to the other), and why no quantum principle prevents it. What small classical resource closes the hole?

Problem 14 [Standard] Error correction typically tolerates a QBER up to about 11%. Using $\text{QBER} = (1 - V)/2$ with $V = S/2\sqrt{2}$, find the minimum acceptable S . Give two decimal places.

Problem 15 [Standard] Suppose the untrusted source, trying to cheat cheaply, ships *unentangled* pairs in definite states. Using Chapter 11's result that such pairs obey $|S| \leq 2$, explain why this attack can never survive the test — no matter how cleverly the definite states are chosen.

Problem 16 [Standard] List the trust assumptions of BB84 versus E91 in two short columns: who must be honest, what hardware must be characterized, and what the test statistic is in each. Which assumption of BB84 does E91 remove entirely?

Problem 17 [Standard] A CTO argues: “ T is probably 25 years, our $L + M = 18$ — we can wait 5 more years.” Recompute the margin after waiting (M unchanged, but 5 years of harvested traffic now carry L -year obligations from a later start), and critique the reasoning: what did waiting actually buy and cost?

Problem 18 [Challenge] A lazier Eve intercepts only half the qubits (choosing randomly which). Compute the sifted-bit error rate she causes, the number of check bits needed to catch her with 99% probability (solve $1 - (1 - e)^n \geq 0.99$ with your error rate e per checked bit), and the moral about partial attacks.

Problem 19 [Challenge] Design the briefing: a regional bank asks whether to buy a QKD link to its disaster-recovery site 40 km away, or rely on PQC. Write a six-sentence recommendation covering: what each protects against, hardware and distance feasibility at 40 km, the harvest-now argument, the authentication requirement, and your verdict with one condition attached.

Problem 20 [Challenge] The E91 idea (Ekert, 1991): instead of Alice sending states, a central source distributes Bell pairs, and the two parties measure at angles that double as a CHSH test (Chapter 8, Problem 12). Explain in a paragraph why a passed Bell test certifies the key against eavesdropping *even if the source itself is untrusted* — monogamy of entanglement (Chapter 11, Problem 12) is the key ingredient.

Problem 21 [Challenge] An eavesdropper intercepts a fraction f of pairs (intercept-and-resend), giving $S(f) = (1-f)2\sqrt{2} + f\sqrt{2}$. If Alice and Bob abort whenever $S < 2.70$, find the largest undetectable f , and the number of the 2000 key bits Eve then knows (at most f of them). One sentence: what does privacy amplification do to those bits?

Problem 22 [Challenge] Write the eight-line briefing note (capstone Project 12 style): your organization's L and M (choose and justify), the Mosca verdict across $T \in \{15, 25, 40\}$, the PQC-vs-QKD allocation you recommend from the comparison table, and one sentence on endpoint security. Every claim must carry a number or a chapter reference.

CHAPTER SIXTEEN

The Quantum Internet

Repeaters, distillation, and satellites — engineering entanglement across a planet

IN THIS CHAPTER YOU WILL LEARN TO

- Explain why photon loss plus no-cloning makes long-distance quantum links a genuinely new engineering problem.
- Assemble the repeater chain: short segments, quantum memories, and entanglement swapping.
- Upgrade bruised pairs with entanglement distillation, and price the upgrade in raw pairs.
- Compare fiber and satellite links by their loss arithmetic, and read the Micius results like an engineer.
- Describe what a quantum internet is actually *for* — and what it will never do.

Teleportation moves a qubit anywhere — *if* the two ends already share entanglement. Chapter 15 certifies keys — *if* the pairs arrive intact. Every protocol in Part Five ends with the same unpaid bill: **get high-quality entanglement between distant places**. This chapter pays the bill. It is the newest engineering discipline in this book — the plumbing of a quantum internet — and by now you own every tool it uses.

16.1 The Enemy: Exponential Loss, and No Amplifiers

Optical fiber swallows photons at about 0.2 dB per kilometer — a fixed *fraction* per kilometer, hence an exponential toll with distance:

EQUATION: THE FIBER TOLL

$$T = 10^{-0.2 L/10}$$

In words: Transmission probability falls tenfold every 50 kilometers

Symbol	Meaning	Unit
L	fiber length	km
T	probability one photon survives	none
0.2	typical telecom-fiber loss	dB/km

Landmarks: 50 km $\rightarrow$ 10%; 100 km $\rightarrow$ 1%; 600 km $\rightarrow$ 10^{-12} ; 1000 km $\rightarrow$ 10^{-20} — at a billion attempts per second, one survivor every three thousand years. Classical networks shrug: amplifiers re-copy the signal every 80 km. For qubits, no-cloning (Chapter 12) outlaws exactly that. The exponential must be outsmarted, not amplified.

16.2 The Repeater Chain: Divide, Store, and Swap

The escape uses three ideas this book has already built. **Divide:** split 600 km into six 100-km segments — each segment's success probability is a workable 1%, not 10^{-12} . **Store:** at each station, hold the successful half of a segment's pair in a **quantum memory** while the neighboring segment keeps trying — this is what turns “all segments must succeed *simultaneously*” (probability 10^{-12} again!) into “each segment succeeds *eventually*” (a hundred tries each). **Swap:** once neighboring segments both hold pairs, the station performs the Bell measurement of Chapter 13's swapping lesson, stitching two short pairs into one long one; repeat down the chain until the end stations share entanglement they never transmitted.

WORKED EXAMPLE — BUDGETING A 600-KM LINK

Six 100-km segments ($T = 1\%$ each), attempt rate 1 MHz per segment, matter-based (deterministic) Bell measurements at the five stations. Estimate the entanglement delivery time, with and without memories.

1. Without memories (all six segments lucky in the same attempt): success probability $(0.01)^6 = 10^{-12}$ per attempt — about 10^6 seconds, twelve *days*, per pair. Useless.
2. With memories: each segment independently needs about $1/0.01 = 100$ attempts = 100 μ s; the chain waits for the *slowest* segment — order a few hundred microseconds.
3. Swapping: five deterministic Bell measurements and their classical confirmations (~ 2 ms of light-time over 600 km) finish the stitch.
4. Verdict: milliseconds per end-to-end pair versus twelve days — a 10^9 speedup, bought entirely by *memory*, the unglamorous hero of the architecture. (Real memories must therefore hold qubits faithfully for milliseconds: exactly the T_1, T_2 engineering of Chapter 28, now on the network's critical path.)

Answer: $\sim$ milliseconds with memories, geological time without. The repeater chain converts an exponential problem into a linear chain of solvable ones — divide, store, swap. Every quantum-network roadmap on Earth is a variation on this budget.

WORD TO KNOW

Quantum memory — A device holding half of an entangled pair faithfully while the rest of the network catches up — trapped ions, atomic ensembles, and diamond vacancies are current leaders. Its storage time versus the segment retry time is the make-or-break ratio of repeater designs.

16.3 Distillation: Trading Quantity for Quality

Chapter 13’s fidelity lesson set the bar: teleportation at $F_{\text{out}} \geq 0.95$ needs pairs at $f \geq 0.925$, and long chains deliver worse — each swap and each kilometer bruises the pair. The repair is **entanglement distillation**: consume two bruised pairs, run a small circuit of CNOTs and measurements at both ends (local operations and classical messages only!), and — when the test measurements agree, about half the time — keep one pair, now provably closer to perfect.

EQUATION: ONE DISTILLATION ROUND (RECURRENCE RECIPE)

$$f' \approx \frac{f^2}{f^2 + (1-f)^2} \quad \text{success probability} \approx \frac{1}{2}, \text{ two pairs consumed}$$

In words: Two mediocre pairs make one better pair, about half the time — quality bought with quantity

Symbol	Meaning	Unit
f	input pair quality	none
f'	output quality after a successful round	none
$\frac{1}{2}$	approximate success odds per attempt	none

Only improves what is already decent: for $f > \frac{1}{2}$ the map pushes upward, and rounds can be repeated. The expected bill: about 4 raw pairs per distilled pair per round (2 consumed, half the attempts fail) — multiply through for repeated rounds.

WORKED EXAMPLE — ONE ROUND, PRICED

The 600-km chain delivers $f = 0.80$ pairs. Distill once; check against the 0.95-teleportation bar; price the round.

1. Upgrade: $f' = 0.64 / (0.64 + 0.04) = 0.941$.
2. Check the bar: $0.941 \geq 0.925$ ✓ — teleportation now runs at $F_{\text{out}} = (2 \times 0.941 + 1) / 3 = 0.961$.
3. Price: two pairs per attempt, success $\approx \frac{1}{2}$: about *four* raw end-to-end pairs per distilled pair — the milliseconds-per-pair budget above becomes several milliseconds per *good* pair.
4. Note the beautiful economics: distillation runs on exactly the operations the ends already own (local gates, classical talk) — no new quantum channel needed. It is error correction’s frugal cousin, specialized to entanglement.

Answer: $f : 0.80 \rightarrow 0.941$ at a $4\times$ pair cost — and the network now meets its service target. Real architectures interleave distillation between swap levels (distill short pairs, swap, distill again), a scheduling puzzle that is one of quantum networking’s liveliest research fronts.

16.4 Fiber or Sky: The Satellite Alternative

Fiber’s toll is exponential in distance; a satellite link pays mostly *diffraction* — the beam spreading geometrically through near-empty space — plus a fixed atmosphere fee. For continental distances the comparison is brutal: 1200 km of fiber costs 240 dB (10^{-24}); the Micius satellite’s dual downlinks over the same separation

measured *tens* of dB total. That single contrast is why China's Micius mission (2016–) could distribute entanglement between ground stations 1200 km apart, teleport ground-to-satellite, and relay intercontinental QKD — feats fiber cannot approach at any patience.

CASE STUDY — A NETWORK ACROSS 4,600 KILOMETERS • CHINA, 2021

The first integrated quantum network stitched both technologies together: a 2,000-km fiber backbone from Beijing to Shanghai (32 trusted-relay stations — the pragmatic, physics-lite stopgap where keys are decrypted and re-encrypted at each node) joined to Micius satellite links reaching a far-western station — more than 150 users, banks and grids among them, across 4,600 km. The honest reading cuts both ways. It works, daily, at useful key rates: quantum communication is an *industry*, not a proposal. And its trusted relays are precisely what full repeaters will retire: every relay is a room you must guard, while a true repeater station (memory + swapping, this chapter) never learns the key at all — monogamy and the Bell test guard it instead (Chapters 11, 15). The network of 2021 is the classical-assisted draft; the chain you budgeted above is the final architecture, awaiting better memories.

COMMON MISTAKE

X Wrong: “The quantum internet will replace the ordinary internet and make it faster.”

✓ Right: It will do neither — and runs *on top of* the classical internet, which carries all its control messages (swap announcements, basis sifting, distillation verdicts). What entanglement distribution adds is capabilities classical networks cannot have at any speed: keys certified by physics (Chapter 15), teleportation of quantum states between processors and sensors (Chapter 13), networked quantum computers, and long-baseline quantum sensing. Bandwidth for movies is not on the list — Holevo's bound (Chapter 2) never sleeps.

ACTIVITY (APP) — DESIGN A NATIONAL LINK ON PAPER

1. Pick two cities about 500–700 km apart (Erbil to Baghdad and onward, or any pair you know). Draw the route and divide it into segments of at most 100 km.
2. Budget each segment with the fiber toll ($T = 10^{-0.2L/10}$), choose an attempt rate, and estimate per-segment success time with memories.
3. Place swap stations, count distillation rounds to reach $f \geq 0.925$ from your assumed raw f , and price the good-pair rate.
4. Write the three-line “executive summary”: delivery rate, the component your design depends on most critically (it will be the memory), and what a satellite pass would change.

You need: Paper and pen, Calculator, A map

CHAPTER SUMMARY

- Fiber loss is exponential and amplifiers are forbidden by no-cloning: long-distance entanglement needs architecture, not power.
- The repeater chain: divide the distance, store successes in quantum memories, and stitch segments by entanglement swapping — memory converts “simultaneously lucky” into “eventually ready.”
- Distillation trades pairs for quality ($f' \approx f^2 / (f^2 + (1 - f)^2)$, ~ 4 raw per distilled), letting bruised chains meet teleportation's $f \geq 0.925$ service bar.
- Satellites beat fiber's exponential with diffraction's geometry: Micius distributed entanglement over 1200 km, and the 2021 integrated network spans 4,600 km with trusted relays — the draft version of the repeater future.
- The quantum internet distributes *entanglement and keys*, not bandwidth: physics-certified security, teleported states, networked processors and sensors — running forever alongside, never instead of, the classical net.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] Compute the fiber transmission for 25, 150, and 300 km, and the factor by which each doubling of distance squares the toll.

Problem 2 [Basic] Why exactly does no-cloning forbid optical amplifiers for qubits, while classical repeaters copy freely? Two sentences, citing Chapter 12.

Problem 3 [Basic] In the 600-km budget, what single hardware number must improve for the memory strategy to work at all, and which chapter's protocols measure it?

Problem 4 [Standard] Rebudget the worked example for 800 km as eight 100-km segments: per-segment expected attempts, chain waiting-time scale, and swap count. What grew linearly, and what stayed constant?

Problem 5 [Standard] Apply the distillation map twice starting from $f = 0.80$: compute f'' after the second round, the total raw pairs per twice-distilled pair ($\approx 4^2$), and the resulting teleportation fidelity.

Problem 6 [Standard] Show that distillation cannot rescue $f = 0.5$ pairs: compute f' at $f = 0.5$ and at $f = 0.45$, and connect the fixed point to the separable boundary of Chapter 13's fidelity lesson.

Problem 7 [Standard] A trusted relay learns the key; a repeater station does not. Trace what each station type physically holds during operation, and name the theorem that protects the repeater architecture's keys (Chapter 11).

Problem 8 [Standard] The 2021 network used 32 trusted relays over 2000 km. If full repeaters replaced them

at 100-km spacing, how many stations would remain, and what two hardware capabilities (this chapter) must each possess?

Problem 9 [Standard] Satellite passes are brief and weather-dependent; fiber is always on but exponentially lossy. Propose the hybrid rule a national operator should use for (a) two data centers 80 km apart, (b) two capitals 1200 km apart, with one-line loss arithmetic for each.

Problem 10 [Challenge] Derive the memory-time requirement: with per-segment success q per attempt at rate R , the slowest of k segments finishes in roughly $(\ln k + 1)/(qR)$ (order-of-magnitude reasoning suffices). Evaluate for $q = 0.01$, $R = 1$ MHz, $k = 6$, and compare with the plain $1/(qR)$ single-segment estimate — how much extra memory time does the “slowest of six” cost?

Problem 11 [Challenge] End-to-end service calculation: raw chain delivers pairs at 200 per second at $f = 0.80$; the customer needs teleportations at $F_{\text{out}} \geq 0.95$. Using one distillation round (4 raw per good) and the fidelity formula, compute the deliverable teleportation rate and the margin in F_{out} .

Problem 12 [Challenge] Referee the claim “our satellite makes encryption unbreakable nationwide”: write four questions in the style of Chapter 27’s audit — covering endpoint security, trusted-relay exposure, availability (weather/passes), and the classical-crypto alternative (Chapter 15’s comparison table) — each with the evidence that would settle it.

PART SIX

QUANTUM ALGORITHMS

CHAPTER SEVENTEEN

First Algorithms: Deutsch–Jozsa and Bernstein–Vazirani

The first quantum victories — one question where classical needs two, and a hidden string read in a single query

IN THIS CHAPTER YOU WILL LEARN TO

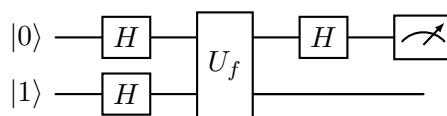
- State the Deutsch problem, and why any classical strategy needs two function calls.
- Trace the Deutsch circuit by hand and show one quantum query settles the question.
- Explain the mechanism — phase kickback plus interference reading a *global* property.
- Scale up to Deutsch–Jozsa, and read a hidden string in one query with Bernstein–Vazirani.
- State honestly what these toy problems do and do not prove — and why they still matter.

Before Grover’s search and Shor’s earthquake came two small, perfect victories. In 1985 David Deutsch found the first task where a quantum computer provably beats a classical one — by a single function call. It is a toy problem, and it launched the field: the interference pattern inside it is the same one working inside every algorithm in this part. Master these twelve or so amplitudes and the big algorithms stop being magic tricks.

17.1 One Question, Two Answers: The Deutsch Problem

A mystery box computes some one-bit function f of one bit. Only four functions exist: the two **constant** ones ($f \equiv 0$ and $f \equiv 1$) and the two **balanced** ones (identity and NOT). The question: is the box constant or balanced?

Classically the answer requires knowing *both* values $f(0)$ and $f(1)$ — learning one value tells you nothing about the category. Two calls, no shortcut, provably. The quantum claim: *one* call to the box (built as the bit-oracle of Chapter 10) suffices.



The Deutsch circuit: query register on top, helper below prepared in $|1\rangle$ then $|-\rangle$. One call to U_f , one final H, one measured bit — and the category is known.

EQUATION: THE DEUTSCH VERDICT

$$\text{measure } 0 \implies f \text{ constant} \qquad \text{measure } 1 \implies f \text{ balanced}$$

In words: One measured bit answers a question about BOTH function values at once — a global property, not a lookup

Symbol	Meaning	Unit
U_f	the bit oracle $ x\rangle a\rangle \rightarrow x\rangle a \oplus f(x)\rangle$	none
constant	$f(0) = f(1)$	none
balanced	$f(0) \neq f(1)$	none

Note what is not learned: the actual values $f(0), f(1)$. One quantum query yields one bit — but it can be a cleverly chosen bit no single classical query could provide. That reframing is the whole trade.

17.2 Why It Works: Kickback Meets Interference

The helper qubit, prepared in $|-\rangle$, is exactly the kickback fuel of Chapter 10: querying the oracle with the helper in $|-\rangle$ converts the answer into a *sign*, $|x\rangle \rightarrow (-1)^{f(x)} |x\rangle$, leaving the helper untouched. From there, the two-path interference of Chapter 6 does the rest.

WORKED EXAMPLE — FULL TRACE: THE BALANCED CASE $f(x) = x$

Trace the circuit for the identity function and confirm the verdict bit reads 1.

1. After the opening H's, the register is $|+\rangle$, and the oracle (with helper $|-\rangle$) stamps signs: $\frac{1}{\sqrt{2}}((-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle) = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.
2. That is $|-\rangle$: the balanced function turned $|+\rangle$ into $|-\rangle$.
3. Final H reads the sign pattern: $H|-\rangle = |1\rangle$.
4. Measure: 1, with certainty — balanced, as promised, from one query.

Answer: the register's phases carried both function values at once; the final H asked “are the phases equal or opposite?” — a question about the *pair*, answered without learning either member. Repeat the trace for $f \equiv 1$: the stamps give $-\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, a global minus on $|+\rangle$, and $H|+\rangle = |0\rangle$: constant. The invisible global phase versus the visible relative phase (Chapter 5) — that distinction IS the algorithm.

COMMON MISTAKE

✗ Wrong: “The quantum computer computed $f(0)$ and $f(1)$ simultaneously and told us both.”

✓ Right: It computed both *amplitudes* simultaneously and told us **one bit**: whether the values agree. The values themselves remain unknown — Holevo's bound (Chapter 2) is never cheated. Quantum speedups come from asking a question whose answer is a pattern across all evaluations, then arranging interference so that pattern lands on a measurable bit. “Try everything, keep everything” is a myth; “try everything, distill one global fact” is the truth — and the template for this whole part.

17.3 Scaling Up: Deutsch–Jozsa

The 1992 upgrade: f now eats n -bit strings, and is promised to be either constant or balanced (exactly half the 2^n inputs give 1). Classically, certainty can cost $2^{n-1} + 1$ queries in the worst case — half a million evaluations for $n = 20$. Quantumly: H everything, one phase-stamping query, H everything, read.

EQUATION: THE DEUTSCH–JOZSA READOUT

$$\text{amplitude of } |00 \cdots 0\rangle \text{ at the end} = \frac{1}{2^n} \sum_x (-1)^{f(x)}$$

In words: The all-zeros amplitude averages every sign stamp: constant averages to ± 1 (certain), balanced cancels to exactly 0 (impossible)

Symbol	Meaning	Unit
$ 00 \cdots 0\rangle$	the all-zeros readout pattern	none
$(-1)^{f(x)}$	the oracle's sign stamp on input x	none

The verdict generalizes cleanly: measure all-zeros $\Rightarrow$ constant; measure anything else $\Rightarrow$ balanced. One query replaces up to $2^{n-1} + 1$ — an exponential separation, for exact answers on promised inputs.

WORKED EXAMPLE — DEUTSCH–JOZSA AT $n = 2$

Take the balanced function $f(x_1x_2) = x_1$. Trace the four amplitudes and find the readout.

1. Uniform start: amplitudes $\frac{1}{2}(+, +, +, +)$ on $(00, 01, 10, 11)$.
2. Stamp signs by x_1 : $\frac{1}{2}(+, +, -, -)$.
3. Final $H \otimes H$, computed by the all-zeros formula first: average of the stamps $= (1 + 1 - 1 - 1)/4 = 0$ — all-zeros is impossible: verdict *balanced*, guaranteed.
4. Full bookkeeping ($H \otimes H$ on the stamped state) puts the entire amplitude on $|10\rangle$ — the readout even names *which* input bit the function depended on. Hold that thought for thirty seconds.

Answer: readout $|10\rangle$ with certainty — not all-zeros, so balanced. The sign pattern $(+, +, -, -)$ is “the x_1 wave,” and the final H’s performed frequency analysis on it — a two-qubit shadow of the Fourier transform that Chapter 19 will scale into an earthquake.

17.4 Reading a Hidden String: Bernstein–Vazirani

That last observation — the readout *named the dependence* — becomes its own algorithm. Promise: $f(x) = s \cdot x \bmod 2$, a parity function built from a secret n -bit string s (the XOR of the input bits where s has a 1). Classically, each query reveals one parity: learning all of s costs n queries. Quantumly, run *exactly the Deutsch–Jozsa circuit*, once:

EQUATION: THE BERNSTEIN–VAZIRANI MIRACLE

$$H^{\otimes n} [\text{stamp } (-1)^{s \cdot x}] H^{\otimes n} |0 \dots 0\rangle = |s\rangle$$

In words: One query, and the measurement prints the entire secret string, bit for bit, with certainty

Symbol	Meaning	Unit
s	the hidden n -bit string	none
$s \cdot x$	parity of the bits of x selected by s	none

n classical queries versus 1 quantum query — a clean, unconditional factor-of- n , no promise gymnastics required. This is also a beloved hardware benchmark: the ideal outcome is a single certain pattern, so any other observed pattern directly measures the machine's error rate.

WORKED EXAMPLE — EXTRACTING $s = 101$

Three qubits, secret $s = 101$. Verify the circuit prints s .

1. Uniform start over eight patterns, amplitudes $\frac{1}{\sqrt{8}}$ each.
2. Stamp $(-1)^{s \cdot x}$: inputs with odd parity on bits 1 and 3 go negative: $-$ on 001, 100, 011, 110 and $+$ on 000, 101, 010, 111. Check two: $s \cdot 001 = 1$ (bit 3 selected, value 1) $\rightarrow -$; $s \cdot 101 = 1 \oplus 1 = 0 \rightarrow +$ ✓
3. The stamped pattern is precisely “the s -wave”: $(-1)^{s \cdot x}$ is the interference fingerprint of the string s .
4. Final $H^{\otimes 3}$ is the matched filter: it maps that fingerprint to the single pattern $|101\rangle$. Measure: 101, probability 1.

Answer: the histogram shows one bar, at the secret itself. Run it on a real machine (many cloud tutorials do) and the height of that bar is your fidelity report — Chapter 30's budgets, benchmarked by an algorithm from 1997 in eight lines of code.

WORD TO KNOW

Query complexity — Counting only calls to the mystery function — the referee's scoreboard on which Deutsch (2 vs 1), Deutsch-Jozsa ($2^{n-1}+1$ vs 1), Bernstein-Vazirani (n vs 1), and Grover (N vs $\sqrt{N}$) are all scored.

CASE STUDY — FROM TOY TO TEMPLATE • 1985 OXFORD → 1994 BELL LABS

No one factors numbers with Deutsch-Jozsa, and no bank fears Bernstein-Vazirani. Their importance is genetic. Deutsch's 1985 paper proved a separation *existed* — one function call where classical logic demands two — and handed the field its blueprint: superpose, kick back phases, interfere, read a global property. Bernstein and Vazirani (1993) sharpened the tool into rigorous quantum complexity theory; Simon's algorithm (1994) used the same skeleton to find a hidden *period*, exponentially faster; and within months Peter Shor, reading Simon, saw that the period of modular exponentiation is a factoring wedge (Chapter 19). From a two-input toy to the collapse of RSA: four papers, nine years, one interference pattern. That is why every course still begins here — these ARE the training wheels of the trillion-dollar bicycle.

REALITY CHECK

Full honesty about the toys: allow classical algorithms a tiny error tolerance (answer “balanced” after a few random agreeing samples would be wrong only rarely) and the Deutsch–Jozsa separation collapses to almost nothing — randomized classical does fine on the promise problem. Bernstein–Vazirani’s factor of n survives all objections but is “only” linear. The permanent lessons are the mechanism (kickback + interference = global questions answered) and the scoreboard (query complexity). The exponential separations that survive every classical counterattack come from period-finding — and that is precisely the road this part now takes.

ACTIVITY (APP) — BE THE ORACLE

1. On paper, run the Deutsch trace for all four one-bit functions: write the stamped state and the final H’s output. Confirm the verdict table: two land on $|0\rangle$, two on $|1\rangle$.
2. In the *Circuit Lab* of the QCFE app, build the Bernstein–Vazirani circuit for a partner’s secret 3-bit string: the oracle is one CNOT from each register qubit where s has a 1, targeting the helper (that computes $s \cdot x!$). Hide your gate choices.
3. Partner predicts, runs once (1000 shots), and reads your secret off the histogram. Swap roles.
4. Break it gently: add one stray T gate inside the oracle and rerun. Which bars appear, and why is the certainty gone?

You need: The QCFE app (Android or Windows), Paper and pen, A partner

CHAPTER SUMMARY

- The Deutsch problem: constant or balanced? Classical: two queries, provably. Quantum: one — the first strict quantum win (1985).
- Mechanism: a $|-\rangle$ helper turns the oracle into sign stamps (kickback); the final H’s ask whether stamps agree (constant $\rightarrow |0 \cdots 0\rangle$) or cancel (balanced $\rightarrow$ anything else).
- One quantum query yields one bit — but a *global* bit no single classical query can provide. No “all answers at once.”
- Deutsch–Jozsa scales the trick (1 vs $2^{n-1}+1$ exact queries); Bernstein–Vazirani reads an n -bit secret in one query — and doubles as a standard hardware benchmark.
- Honest ledger: randomized classical algorithms tame Deutsch–Jozsa’s promise problem; the unconditional exponential wins descend from this template through period-finding (Chapter 19).

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher’s Guide.

Problem 1 [Basic] List the four one-bit functions, classify each as constant or balanced, and explain in one

sentence why one classical query can never classify with certainty.

Problem 2 [Basic] In the Deutsch circuit, why must the helper start in $|1\rangle$ (becoming $|-\rangle$) rather than $|0\rangle$ (becoming $|+\rangle$)? What happens to the stamps with a $|+\rangle$ helper (Chapter 10's kickback rule)?

Problem 3 [Basic] Trace the Deutsch circuit for $f \equiv 0$ and confirm the readout $|0\rangle$.

Problem 4 [Standard] Trace Deutsch for $f(x) = \text{NOT } x$: write the stamped state, factor out the global phase, and confirm the readout $|1\rangle$. Which chapter's global-vs-relative phase distinction did you just use?

Problem 5 [Standard] For Deutsch-Jozsa at $n = 2$ with the balanced function $f = x_2$: write the four stamps and predict the exact readout pattern (mirror the worked example).

Problem 6 [Standard] For Bernstein-Vazirani with $s = 110$: list the eight signs $(-1)^{s \cdot x}$, and state the certain readout.

Problem 7 [Standard] Count queries: for $n = 20$, give the classical worst case for exact Deutsch-Jozsa, the classical cost for Bernstein-Vazirani, and both quantum costs. Which separation would survive a skeptic who tolerates a 10^{-9} error probability?

Problem 8 [Standard] The BV oracle for $s = 101$ is two CNOTs (from register qubits 1 and 3 into the helper). Verify on the input $x = 110$ that the helper ends holding $s \cdot x$, and explain why the phase-oracle version needs no helper cleanup.

Problem 9 [Standard] A hardware run of BV with $s = 101$ returns: 101: 903, 001: 34, 111: 31, 100: 18, others 14. Estimate the circuit's success fidelity, and name two error sources from Chapter 28 consistent with the near-miss patterns (single-bit flips of s).

Problem 10 [Challenge] Prove the classical lower bound for exact Deutsch-Jozsa: with only 2^{n-1} queries all answering 0, show both a constant and a balanced function remain consistent with everything seen — so certainty is impossible.

Problem 11 [Challenge] Prove the all-zeros formula: apply $H^{\otimes n}$ to the stamped uniform state and show the $|0 \cdots 0\rangle$ amplitude is $2^{-n} \sum_x (-1)^{f(x)}$ (every basis state contributes with a $+$ sign to the all-zeros component). Conclude both verdicts.

Problem 12 [Challenge] Bernstein-Vazirani as interference: show for any pattern y that the final amplitude is $2^{-n} \sum_x (-1)^{s \cdot x + y \cdot x}$, and that it vanishes unless $y = s$ (pair up inputs differing in one selected bit). You have proven the one-bar histogram.

CHAPTER EIGHTEEN

Grover's Search Algorithm

Finding the needle by interference — oracles, amplitude amplification, and the square-root speedup

IN THIS CHAPTER YOU WILL LEARN TO

- State the unstructured search problem and its classical cost.
- Explain what a quantum oracle is — and what it honestly does and does not require.
- Perform the two Grover moves — oracle sign-flip and inversion about the mean — by hand.
- Trace the complete algorithm on a 4-entry search, reaching success probability 1 after a single iteration.
- Use $k \approx \frac{\pi}{4} \sqrt{N}$ to choose the iteration count, and explain the overshoot danger and the speedup's honest limits.

18.1 The Search Problem

You face N possibilities — passwords, puzzle states, database entries, parameter settings — and exactly one of them (call it the *target* w) is “correct.” The only tool available is a checker: give it a candidate, it answers yes or no. No sorted order, no clever index, no structure at all: **unstructured search**.

Classically the accounting is grim and simple. Each check clears one candidate; finding w takes $N/2$ checks on average, N in the worst case. A million entries: half a million checks. There is provably no cleverer classical way — without structure, checking is all you can do.

In 1996, Lov Grover found the quantum surprise: a quantum computer needs only about $\sqrt{N}$ quantum checks — roughly a *thousand* for a million entries. Not by checking faster, but by checking *in superposition* and steering interference so the target's amplitude grows while every wrong answer's shrinks. The next three lessons build his machine one move at a time.

WORD TO KNOW

Unstructured search — Finding the unique marked item among N candidates when the only operation is asking “is this one it?” Classical cost: $O(N)$ checks. Grover's quantum cost: $O(\sqrt{N})$.

18.2 The Oracle: Marking Without Revealing

The checker enters the quantum computer as an **oracle**: a unitary that flips the *sign* of the target's amplitude and touches nothing else.

EQUATION: THE GROVER ORACLE

$$O|x\rangle = \begin{cases} -|x\rangle & \text{if } x = w \quad (\text{the target}) \\ |x\rangle & \text{otherwise} \end{cases}$$

In words: The oracle flips the sign of the target's amplitude — an invisible phase mark, planted on every branch of a superposition at once

Symbol	Meaning	Unit
O	the oracle (checker) unitary	none
w	the target bit pattern	none

Fed a superposition, linearity applies the rule to every term simultaneously: one oracle call phase-marks the target inside all N branches. But beware — a sign is not an answer: measuring right after the oracle still gives every pattern with unchanged probability $|\pm a|^2 = |a|^2$. Lesson 3 exists to convert the hidden mark into visible probability.

Where does an oracle come from? Not from magically knowing w . Any efficient checking *circuit* — “does this password unlock the file?”, “does this assignment satisfy the formula?” — compiles into reversible gates (Toffoli logic, Chapter 10) that compute the yes/no into a phase. The oracle knows how to *recognize* the answer, exactly like its classical counterpart; neither knows where the answer *is*. Grover's speedup is honest: it beats classical search using the same checker, called fewer times.

COMMON MISTAKE

✗ Wrong: “The oracle already knows the answer, so Grover's algorithm cheats.”

✓ Right: The classical brute-forcer uses the very same checking subroutine — $N/2$ times on average. Grover uses it $\approx \frac{\pi}{4}\sqrt{N}$ times. Same knowledge in, fewer calls needed: that difference, not the oracle's existence, is the achievement.

18.3 Amplitude Amplification: Inversion About the Mean

Grover's second move — the **diffusion operator** D — is the engine that turns phase marks into probability. Algebraically D reflects every amplitude through the average amplitude:

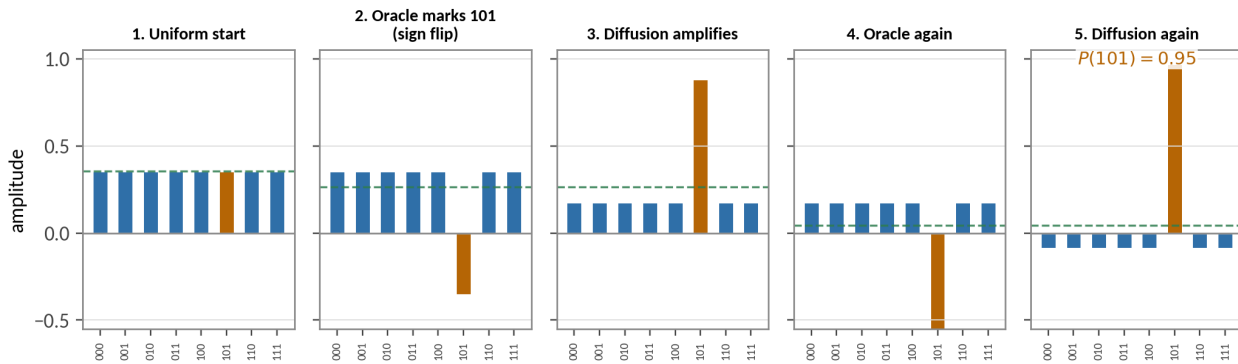
EQUATION: INVERSION ABOUT THE MEAN

$$a_x \longrightarrow 2\bar{a} - a_x \quad \text{where} \quad \bar{a} = \frac{1}{N} \sum_y a_y$$

In words: *New amplitude = twice the mean, minus the old amplitude — every amplitude reflects through the average*

Symbol	Meaning	Unit
a_x	the amplitude of pattern x before the step	none
$\bar{a}$	the mean (average) of all N amplitudes	none

Amplitudes above the mean land below it and vice versa, at equal distance. The target — pushed negative by the oracle — sits far below the mean, so reflection throws it far above: small dip in, large leap out. As a circuit, $D = (H \text{ on all qubits}) \cdot (\text{flip the sign of everything except } |0 \dots 0\rangle) \cdot (H \text{ on all qubits})$ — ordinary gates, no magic.



One picture worth the whole chapter: an 8-entry search. The oracle dips the target below zero; inversion about the (green dashed) mean catapults it upward. After two rounds, $P(101) = 0.95$.

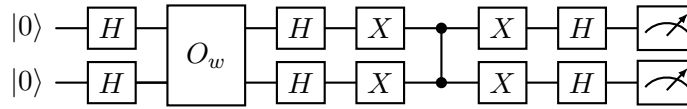
WORKED EXAMPLE — ONE GROVER ROUND ON EIGHT ENTRIES — BY HAND

$N = 8$, uniform start ($a_x = \frac{1}{\sqrt{8}} \approx 0.354$ each), target $w = 101$. Compute the amplitudes after one oracle + one diffusion.

1. Oracle: $a_{101} \rightarrow -0.354$; the other seven stay $+0.354$.
2. Mean: $\bar{a} = \frac{7(0.354) + (-0.354)}{8} = \frac{6 \times 0.354}{8} \approx 0.265$.
3. Reflect the seven: $2(0.265) - 0.354 \approx 0.177$ each.
4. Reflect the target: $2(0.265) - (-0.354) \approx 0.884$.
5. Check normalization: $7(0.177)^2 + (0.884)^2 \approx 0.219 + 0.781 = 1.000 \checkmark$

Answer: after one round, $P(101) = 0.884^2 \approx 78\%$ (from 12.5%). One more round (the figure) lifts it to 95%. Wrong answers did not get “checked off” — their amplitudes were *interfered* downward, all seven at once.

18.4 The Complete Algorithm — and a Perfect Small Case



Grover for $N = 4$ (two qubits): Hadamards make the uniform start; the oracle box marks the target; then the diffusion — written out in gates: H's, X's, a CZ, X's, H's. For $N = 4$ this single iteration suffices.

WORKED EXAMPLE — GROVER ON FOUR ENTRIES: CERTAINTY IN ONE STEP

$N = 4$, target $w = 11$ (oracle = the CZ gate, which flips exactly $|11\rangle$'s sign — Chapter 10). Trace the algorithm.

1. Uniform start (H's on $|00\rangle$): all four amplitudes $\frac{1}{2}$.
2. Oracle: amplitudes $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$ for $(00, 01, 10, 11)$.
3. Mean: $\bar{a} = \frac{3(0.5) + (-0.5)}{4} = \frac{1}{4}$.
4. Diffusion: non-targets $\rightarrow 2(\frac{1}{4}) - \frac{1}{2} = 0$; target $\rightarrow 2(\frac{1}{4}) + \frac{1}{2} = 1$.
5. Final state: $1 \cdot |11\rangle$. Measure: 11, with probability 1.00.

Answer: for $N = 4$, one Grover iteration finds the target with *certainty* — every wrong amplitude interfered to exactly zero. A classical searcher needs 2-3 checks on average and can be unlucky; Grover uses one oracle call and cannot miss. (Verify this on the app or in Qiskit — five minutes, unforgettable.)

18.5 How Many Iterations? $\sqrt{N}$, Overshoot, and Honest Limits

Each Grover round rotates the state a fixed small angle toward the target. Rotation is the right picture — and it warns of a trap: rotate too far and you pass the target and come back down. Success probability oscillates like a wave, and you must *stop at the crest*.

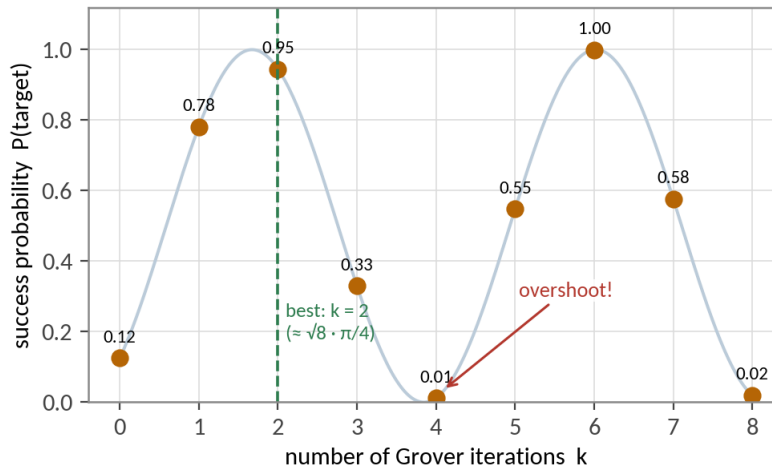
EQUATION: THE OPTIMAL ITERATION COUNT

$$k_{\text{best}} \approx \frac{\pi}{4} \sqrt{N}$$

In words: Best number of Grover rounds $\approx 0.785 \times$ the square root of the number of candidates

Symbol	Meaning	Unit
k_{best}	the iteration count maximizing success probability	rounds
N	the number of candidates	none

Exact machinery for the curious: $P(k) = \sin^2((2k+1)\theta)$ with $\sin \theta = 1/\sqrt{N}$. Check the cases you know: $N=4$ gives $\theta = 30^\circ$ and $P(1) = \sin^2 90^\circ = 1$ — the certainty above; $N=8$ gives $k_{\text{best}} = 2$, $P \approx 0.945$ — the figure.



Success probability versus iterations for $N = 8$: a wave, not a staircase. More is not better — at $k = 4$ the algorithm has rotated past the target back to near-zero. Stop at the crest.

WORKED EXAMPLE — PLANNING A MILLION-ENTRY SEARCH

$N = 10^6$ candidates; one classical check takes $1\ \mu\text{s}$, one quantum oracle round $100\ \mu\text{s}$ (quantum gates are slower — honesty first). Compare total times.

1. Classical average: $N/2 = 500,000$ checks $\times 1\ \mu\text{s} = 0.5\ \text{s}$.
2. Grover rounds: $k \approx 0.785\sqrt{10^6} = 785$ rounds $\times 100\ \mu\text{s} \approx 0.08\ \text{s}$.
3. Speedup despite the $100\times$ slower step: $0.5/0.08 \approx 6\times$. Now rescale ambition: $N = 10^{12}$ gives classical ≈ 139 hours versus Grover ≈ 79 seconds.

Answer: at $N = 10^6$ the win is real but modest; at $N = 10^{12}$ it is transformative. Square-root speedups grow more valuable the bigger the haystack — and are eaten alive by slow hardware on small ones. Both sentences belong in any honest report.

REALITY CHECK

Three honest limits, before anyone sells you a “quantum Google.” (1) Grover speeds up *unstructured* search; real databases have indexes, and classical binary search on sorted data ($\log_2 N \approx 20$ probes for a million entries) demolishes 785 quantum rounds. Grover shines only where no structure exists — brute-forcing, generic constraint problems, black-box subroutines. (2) The quadratic speedup is provably the best possible for this problem — no quantum algorithm searches an unstructured haystack exponentially faster. (3) On noisy hardware, hundreds of coherent rounds are currently out of reach (Chapter 28), so today’s demonstrations live at toy sizes. One concrete consequence you can quote: Grover halves the *effective* key length of symmetric ciphers (2^{128} brute-force $\rightarrow 2^{64}$ quantum steps), which is why AES-256 — not AES-128 — is the post-quantum recommendation. Compare Shor (next chapter), which breaks RSA *exponentially*: different algorithm, entirely different threat level.

CASE STUDY — FROM A PHONE-BOOK PUZZLE TO A CRYPTO STANDARD • BELL LABS 1996 → NIST TODAY

Lov Grover, at Bell Labs in 1996, framed his discovery with a homely image: finding a name in an unsorted phone book of a million entries. The theory community quickly proved his algorithm optimal (the $\sqrt{N}$ cannot be beaten) and endlessly generalizable — “amplitude amplification” now boosts the success of almost any quantum subroutine, and shows up inside algorithms for optimization, machine learning, and Monte-Carlo estimation. Its most concrete worldly impact is defensive: when NIST standardized post-quantum cryptography (2022–2024), symmetric key-length recommendations were set explicitly with Grover in mind — a 1996 interference trick, silently doubling key sizes in the security policies of banks and governments, including every institution modernizing its systems today.

ACTIVITY (APP) — DRIVE THE AMPLIFIER

1. Open the *Grover Lab* in the QCFE app, set $N = 8$ and a secret target.
2. Step one oracle call; watch one bar dip negative in the amplitude view — and confirm the probability view has not changed at all.
3. Step the diffusion; watch the catapult. Read $P(\text{target})$ and compare with the worked example's 78%.
4. Continue stepping to $k = 4$ and watch success *collapse* (overshoot!); then use the app's “optimal k ” button and run 1000 shots.
5. Set $N = 64$: predict k_{best} with the formula, then verify against the app.

You need: The QCFE app (Android or Windows), A calculator

► App tool: *Grover Lab* — amplitude bars, probability bars, and the success-vs-iterations wave, with oracle and diffusion as single buttons

18.6 Several Needles at Once: Grover with Multiple Targets

Real searches rarely have exactly one answer: a sudoku may have several completions, a database several matching records. Grover's machinery handles M marked items out of N with one elegant change: the initial angle grows from $\sin \theta = \sqrt{1/N}$ to $\sin \theta = \sqrt{M/N}$, and everything else — oracle, diffusion, rotation picture — runs unchanged.

EQUATION: GROVER WITH M MARKED ITEMS

$$\sin \theta = \sqrt{\frac{M}{N}}, \quad P_{\text{success}}(k) = \sin^2((2k+1)\theta), \quad k_{\text{opt}} \approx \frac{\pi}{4} \sqrt{\frac{N}{M}}$$

In words: More needles = bigger starting angle = fewer iterations; success lands on any one of the marked items

Symbol	Meaning	Unit
M	number of marked (solution) items	none
θ	rotation per half-iteration	rad
k_{opt}	iteration count nearest the peak	none

The speedup becomes $\sqrt{N/M}$ versus the classical expected N/M tries. On success the measurement returns one marked item uniformly at random — rerun if you want more of them.

WORKED EXAMPLE — TWO NEEDLES IN EIGHT BOXES — PERFECTION AGAIN

$N = 8$, with items 101 and 110 both marked. Analyze one Grover iteration.

1. Starting angle: $\sin \theta = \sqrt{2/8} = \frac{1}{2}$, so $\theta = 30^\circ$ — exactly the lucky angle of the $N = 4$, single-target story.
2. After $k = 1$ iteration: $P = \sin^2(3 \times 30^\circ) = \sin^2 90^\circ = 1$: certainty.
3. Where the amplitude sits: entirely on the marked pair, 0.5 probability each — the unmarked six patterns have amplitude exactly zero.
4. Optimal count check: $k_{\text{opt}} \approx \frac{\pi}{4}\sqrt{4} = 1.57 \rightarrow$ the integer 1 is closest, and here it is not merely close but perfect.

Answer: one iteration, then measure: you get 101 or 110, a fair coin between them, never anything else. The $M/N = \frac{1}{4}$ ratio is quantum search's sweet spot — any (N, M) pair with that ratio reaches certainty in one step.

WORKED EXAMPLE — THE MERRY-GO-ROUND DOES NOT STOP

Continue the same search for a second iteration.

1. The rotation continues past the top: after $k = 2$, $P = \sin^2(5 \times 30^\circ) = \sin^2 150^\circ = 0.25$.
2. Interpretation: the state vector rotated *past* the marked subspace; three quarters of the probability has flowed back onto unmarked items.

Answer: P falls from 100% to 25%. Overshoot is not a rounding nuisance — it is the geometry. Every extra iteration after the peak actively *destroys* success probability, which is why iteration counting (and knowing M !) matters.

WORD TO KNOW

Quantum counting — A phase-estimation routine (Chapter 19's QPE applied to the Grover rotation) that estimates the number of marked items M before searching — the standard cure when M is unknown.

REALITY CHECK

Unknown M is the common case, and guessing k badly can land you at a trough. Two honest cures: run quantum counting first, or use the randomized schedule (try k chosen randomly below an increasing bound), which finds a marked item in $O(\sqrt{N/M})$ expected iterations without ever learning M . At scale, remember Grover's other honesty clause: a big instance like $N = 2^{20}$, $M = 4$ still needs $k_{\text{opt}} \approx \frac{\pi}{4} \sqrt{2^{20}/4} \approx 402$ *sequential, error-free* oracle calls — the quadratic speedup is real and modest at once.

18.7 Amplitude Amplification: Grover Beyond Search

Grover's machinery is more general than finding marked database entries. Any procedure that *sometimes* succeeds — a randomized algorithm, a heuristic, a physics simulation post-selecting a rare event — can have its success amplified quadratically. The recipe never changes: reflect about failure, reflect about the start, repeat.

EQUATION: AMPLIFYING ANY LUCKY PROCEDURE

$$p \rightarrow \sin^2((2k+1)\theta), \quad \sin \theta = \sqrt{p}, \quad k_{\text{opt}} \approx \frac{\pi}{4\sqrt{p}} \quad \text{vs.} \quad \text{classical} \approx \frac{1}{p} \text{ retries}$$

In words: A one-in- N -shots subroutine becomes near-certain after about $(\pi/4) \sqrt{N}$ amplification rounds — the square-root speedup, detached from databases

Symbol	Meaning	Unit
p	the subroutine's success probability	none
k	amplification rounds	none
θ	the rotation angle per half-round	rad

Requirements with teeth: the subroutine must run coherently (as a unitary circuit — no measurements inside!), and you must be able to flag success with an oracle-style phase flip. Chapter 12's uncomputation discipline is what makes subroutines coherent.

WORKED EXAMPLE — FROM 1% TO 98% IN EIGHT ROUNDS

A coherent subroutine succeeds with $p = 1\%$. Compare the classical and amplified costs of getting one success.

1. Classical: expected retries $1/p = 100$ (and a 50% chance of needing about 69 or more).
2. Amplified: $\theta = \arcsin \sqrt{0.01} = 5.74^\circ$; $k_{\text{opt}} \approx \pi/(4 \times 0.1) = 7.85 \rightarrow 8$ rounds.
3. Success after 8 rounds: $\sin^2(17 \times 5.74^\circ) = \sin^2(97.6^\circ) = 0.983$.
4. Cost accounting: each round runs the subroutine *twice* (forward and inverted, inside the reflection), so about 16 coherent subroutine calls replace about 100 classical ones — a six-fold saving that grows as $\sqrt{1/p}$ for rarer successes.

Answer: eight rounds, 98.3% success, versus a hundred classical tries. At $p = 10^{-6}$ the comparison becomes 785 rounds versus a million retries. This is Grover's true legacy: not a database gadget but a universal $\sqrt{\cdot}$ discount on luck — provided the luck can be run as a circuit.

WORD TO KNOW

Amplitude amplification — The generalization of Grover's iteration to any coherent success-flagging procedure: success probability p needs only $O(1/\sqrt{p})$ amplification rounds instead of $O(1/p)$ retries.

REALITY CHECK

The fine print carries the usual two clauses. *Coherence*: a subroutine that measures internally (or leaks scratch qubits, Chapter 12) breaks the rotation picture and amplifies nothing. *Depth*: k_{opt} rounds run *sequentially*, so the noise budgets of Chapter 28 bite exactly as in plain Grover — amplifying a 10^{-6} event needs coherence across ~ 1600 subroutine executions, far beyond unprotected near-term machines. The mathematics is a permanent asset; the deployment waits for error correction.

ACTIVITY (CLASS) — HUMAN AMPLITUDE AMPLIFICATION

1. Eight students stand in a row holding equal paper amplitudes (0.354 each); one secretly holds the marked card.
2. Oracle round: the marked student flips their sign (turns the paper upside down).
3. Diffusion round: the class computes the mean aloud, and every student replaces their value with $(2 \times \text{mean} - \text{own value})$. Recompute and re-post the eight numbers.
4. Repeat oracle+diffusion; after round two, compare the marked student's probability ($\approx 94.5\%$) with the start (12.5%). Run one round too many and watch it *shrink* — the overshoot, embodied.

You need: Nine large paper cards, A calculator, Eight volunteers

CHAPTER SUMMARY

- Unstructured search: N candidates, one checker, classically $N/2$ checks on average. Grover: about $\frac{\pi}{4}\sqrt{N}$ quantum oracle calls.
- The oracle phase-flips the target's amplitude inside the full superposition — same knowledge as a classical checker, deployed on all branches at once; alone it changes no probabilities.
- Diffusion reflects every amplitude through the mean ($a \rightarrow 2\bar{a} - a$): the negative-marked target leaps above the crowd while all wrong amplitudes shrink together.
- Worked flagships: $N = 4$ — one round, certainty; $N = 8$ — two rounds, 95%; success follows

$\sin^2((2k+1)\theta)$ — a wave with an overshoot: stop at the crest.

- Honest verdict: quadratic, optimal, structure-blind, noise-limited today; its lasting worldly footprint is doubled symmetric-key lengths (AES-256).

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] A search has $N = 100$ candidates. How many checks does a classical searcher need on average, and roughly how many Grover rounds ($\frac{\pi}{4}\sqrt{N}$) suffice?

Problem 2 [Basic] After the oracle acts on the uniform 4-entry state with target 10, write all four amplitudes. What are the four measurement probabilities at this instant?

Problem 3 [Basic] Amplitudes are $(0.5, 0.5, -0.5, 0.5)$. Compute the mean, then apply $a \rightarrow 2\bar{a} - a$ to all four.

Problem 4 [Basic] For $N = 16$: compute k_{best} and round sensibly.

Problem 5 [Basic] $N = 4$ with $M = 2$ marked items: compute θ and the success probability after one iteration. Compare with simply guessing at random, and state the uncomfortable conclusion for this special ratio.

Problem 6 [Basic] For $N = 16$: compute k_{opt} for $M = 1$ and for $M = 4$. For the $M = 4$ case, evaluate the success probability at $k = 1$ exactly.

Problem 7 [Basic] For $p = 4\%$: compute θ , k_{opt} , and the amplified success probability at that k .

Problem 8 [Basic] State the two requirements a subroutine must satisfy before amplitude amplification applies, and give one concrete violation of each.

Problem 9 [Standard] Redo the complete $N = 4$ trace with target $w = 01$ instead of 11 (the oracle now flips $|01\rangle$'s sign), and confirm certainty is reached in one round — nothing about the mean-reflection arithmetic depended on which entry was marked.

Problem 10 [Standard] Show by direct computation that a second Grover round on the finished $N=4$ state (amplitudes $0, 0, 0, 1$) makes things worse: oracle, mean, reflection — what is $P(11)$ after round two? Which figure of this chapter predicted this?

Problem 11 [Standard] Using $P(k) = \sin^2((2k+1)\theta)$ with $\sin \theta = 1/\sqrt{8}$ ($\theta \approx 20.7^\circ$): compute $P(0), P(1), P(2)$ and check against the worked example and the figure (12.5%, 78%, 94.5%).

Problem 12 [Standard] A password space has $N = 2^{20}$ ($\approx 10^6$) entries; the checker circuit uses 50 gates, and the diffusion 30 gates. Estimate the total gate count of an optimal Grover search, and name (Chapter 28 preview) the resource this number collides with on today's machines.

Problem 13 [Standard] Sorted versus unsorted: a telephone directory of 10^6 entries is alphabetically sorted. Compare three costs of finding a name: classical linear scan, classical binary search ($\log_2 N$ probes), and Grover. Rank them, and state the moral in one sentence.

Problem 14 [Standard] $N = 64$, $M = 2$: find θ in degrees, k_{opt} , and P at that k (three decimals). How many classical random draws (with replacement) would be needed for a 50% chance of hitting a marked item?

Problem 15 [Standard] Show from $k_{\text{opt}} \approx \frac{\pi}{4} \sqrt{N/M}$ that quadrupling the number of solutions halves the iteration count. What happens to the count if M grows all the way to $N/4$?

Problem 16 [Standard] A colleague runs $N = 8$, $M = 2$ for three iterations, reasoning “more amplification is safer.” Compute P at $k = 3$ and write the one-sentence correction you would send them.

Problem 17 [Standard] Compare total subroutine executions (classical $1/p$ vs. amplified $2k_{\text{opt}}$) for $p = 10^{-2}, 10^{-4}, 10^{-6}$, and verify the ratio scales as $\sqrt{1/p}/(\pi/2)$.

Problem 18 [Challenge] Two targets among $N = 8$ (say 011 and 101): redo one full round by hand — oracle flips both signs; compute the new mean ($\bar{a} = \frac{4 \times 0.354}{8}$); reflect all eight. What is the total probability of hitting *either* target after one round? (General fact worth stating after you compute: with M targets, θ obeys $\sin \theta = \sqrt{M/N}$ — more needles, fewer rounds.)

Problem 19 [Challenge] Overshoot forensics: with $N = 8$, a student runs $k = 4$ rounds and gets $P(\text{target}) \approx 0.01$ — worse than random guessing. Explain, using the rotation picture, how “too much amplification” can bury the target below its starting probability, and give the two professional fixes (stop at k_{best} ; or measure-and-repeat, computing the expected total oracle calls for the repeat strategy at $P = 0.945$ per attempt).

Problem 20 [Challenge] Grover halves effective key bits: brute-forcing a b -bit key is search with $N = 2^b$, so Grover needs about $2^{b/2}$ rounds. Compute the rounds for $b = 128$; assuming (very generously) 10^9 rounds per second, convert to years; then write the two-sentence security-policy conclusion that led real standards bodies to prefer AES-256 while still not panicking about AES-128 today.

Problem 21 [Challenge] Using the rotation picture, derive $P(k) = \sin^2((2k+1)\theta)$ for general M : define the normalized marked and unmarked superpositions as the two axes, show the initial state sits at angle θ with $\sin \theta = \sqrt{M/N}$, and argue each oracle-plus-diffusion pair is a rotation by 2θ .

Problem 22 [Challenge] Amplification overshoots like all Grover rotations: for $p = 25\%$, compute the success probability after $k = 1$ and $k = 2$ rounds, find the exact k giving certainty (hint: $\theta = 30^\circ$), and explain why “round up k_{opt} ” is sometimes the wrong rule.

The QFT, Phase Estimation, and Shor's Algorithm

The quantum Fourier transform, phase estimation, and the algorithm that alarmed the world

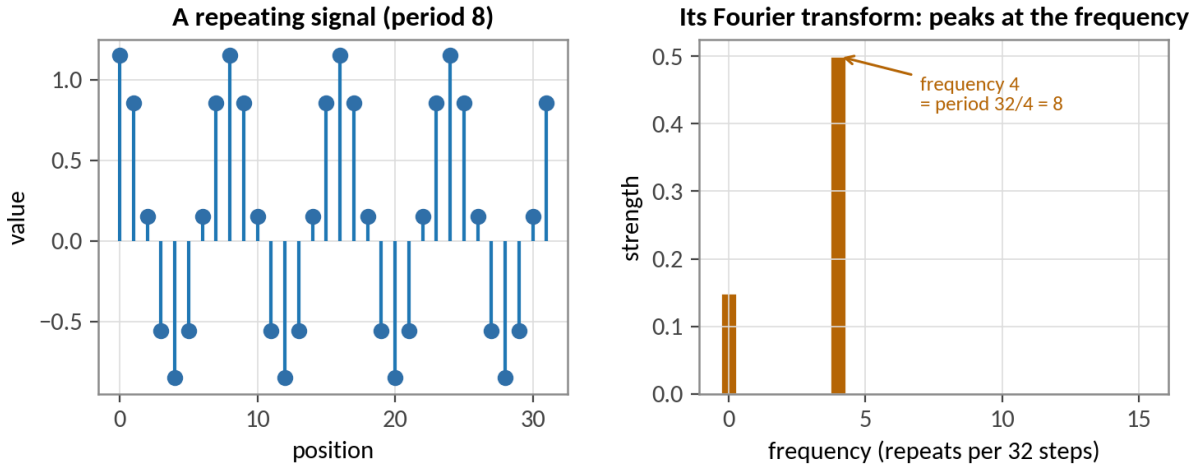
IN THIS CHAPTER YOU WILL LEARN TO

- Explain what a Fourier transform does: turn repetition in a signal into peaks at its frequency.
- Apply the quantum Fourier transform (QFT) on two qubits by hand, and read its circuit.
- Describe quantum phase estimation (QPE): how phase kickback plus inverse QFT reads out an unknown phase as binary digits.
- Trace Shor's algorithm end to end on $N = 15$: reduction to period finding, the quantum period measurement, and the classical finish.
- Assess soberly what Shor means for today's cryptography and the migration to post-quantum standards.

19.1 Waves, Periods, and Frequencies

Grover needed only signs. The deeper quantum algorithms run on the full machinery of *phases* — and the master tool for phases is the Fourier transform, the same mathematics that lets audio software find the pitch of a note.

The core idea needs no calculus. A signal that *repeats* every r steps is built from oscillations whose frequencies are multiples of $1/r$. The Fourier transform re-describes the signal in frequency language: feed in a repeating list of numbers, and the output is nearly zero everywhere except sharp **peaks at the repetition frequency** and its multiples. Repetition hidden in position becomes peaks visible in frequency.



Fourier's bargain: a signal repeating with period 8 (left) becomes a clean spike at frequency $4 = 32/8$ (right). Find the peak, read off the period.

WORD TO KNOW

Period — The number of steps after which a repeating pattern returns to its start: the pattern 3, 7, 1, 3, 7, 1, ... has period $r = 3$. Shor's algorithm reduces factoring to finding such an r .

Keep one number in view all chapter: *period times frequency = total length*. Peaks at frequency f in a length- N signal mean period $r = N/f$.

19.2 The Quantum Fourier Transform

The QFT is the Fourier transform performed on *amplitudes*. It is a unitary gate on n qubits ($N = 2^n$ patterns), defined by what it does to each basis state:

EQUATION: THE QUANTUM FOURIER TRANSFORM

$$\text{QFT } |x\rangle = \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} \omega^{xy} |y\rangle, \quad \omega = e^{2\pi i/N}$$

In words: Each input number x becomes an equal-weight superposition whose phases rotate y -by- y at a speed set by x

Symbol	Meaning	Unit
$N = 2^n$	the number of patterns of the n qubits	none
ω	the basic phase step: one N -th of a full turn on the unit circle (Appendix A)	none
ω^{xy}	the phase attached to output $ y\rangle$: xy steps around the circle	none

For $N=4$: $\omega = i$ (a quarter turn). Fed a superposition whose amplitudes repeat with period r , linearity makes the outputs interfere: only y near multiples of N/r survive — the peak-finding of Lesson 1, executed by interference

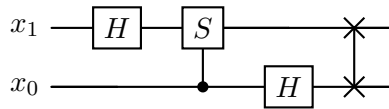
on 2^n amplitudes at once, in about n^2 gates.

WORKED EXAMPLE — QFT ON TWO QUBITS, BY HAND

Compute QFT $|10\rangle$ for $N = 4$ (so $x = 2$, $\omega = i$).

1. Amplitudes of output $|y\rangle$: $\frac{1}{2} \omega^{2y} = \frac{1}{2} i^{2y}$ for $y = 0, 1, 2, 3$.
2. Evaluate with the i -cycle (Appendix A): $i^0 = 1, i^2 = -1, i^4 = 1, i^6 = -1$.
3. Assemble: QFT $|10\rangle = \frac{1}{2} (|00\rangle - |01\rangle + |10\rangle - |11\rangle)$.
4. Sanity: four amplitudes of size $\frac{1}{2}$ — normalized ✓; and the phases alternate with “frequency” $x=2$, exactly as the definition promises.

Answer: $\frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$ — an equal superposition whose *sign pattern* encodes the input number. The QFT stores numbers as phase speeds; the inverse QFT reads them back.



The complete 2-qubit QFT: one H, one controlled-S (a quarter-turn phase, conditioned on the other qubit), another H, and a final SWAP. For n qubits the same recipe uses n Hadamards and about $n^2/2$ controlled phase rotations — polynomial cost for an exponential-size transform.

19.3 Quantum Phase Estimation

Many quantum problems reduce to: a unitary U “stamps” a phase $e^{2\pi i \varphi}$ on a special state, and we need the number φ . Phases are invisible to direct measurement (Chapter 4) — QPE is the standard machine that converts them into readable bits. Two ingredients:

Phase kickback. Apply *controlled- U* with the control in superposition: the branch where the control is $|1\rangle$ acquires the phase $e^{2\pi i \varphi}$ — and since the target state is unchanged, the phase belongs to the *control’s* relative phase (Chapter 10’s warning that “influence runs both ways,” now working for us). Using controlled- U^1 , controlled- U^2 , controlled- U^4 , ... onto a register of m control qubits writes the phases $e^{2\pi i \varphi k}$ across all its 2^m branches $|k\rangle$.

Inverse QFT. The control register now holds exactly a Fourier pattern with “speed” $\varphi \cdot 2^m$. Run the QFT backwards and the register collapses to the *binary digits* of φ .

WORKED EXAMPLE — QPE READS A QUARTER TURN

U stamps the phase $\varphi = 0.25$ (a quarter turn). Run QPE with $m = 2$ counting qubits and predict the measurement.

1. Hadamards put the counting register into $\frac{1}{2} \sum_{k=0}^3 |k\rangle$; controlled- U^k kickback attaches $e^{2\pi i (0.25)k} = i^k$:

state = $\frac{1}{2} (|00\rangle + i|01\rangle - |10\rangle - i|11\rangle)$ (counting register shown; k in binary).

2. Compare with the QFT definition: these are precisely the amplitudes of QFT $|01\rangle$ ($x = 1$: amplitudes $\frac{1}{2}i^y$).

3. Apply the *inverse* QFT: it maps QFT $|01\rangle \mapsto |01\rangle$.

4. Measure: 01 with certainty. Read as a binary fraction: $01 \rightarrow 0.01_2 = \frac{0}{2} + \frac{1}{4} = 0.25$ ✓

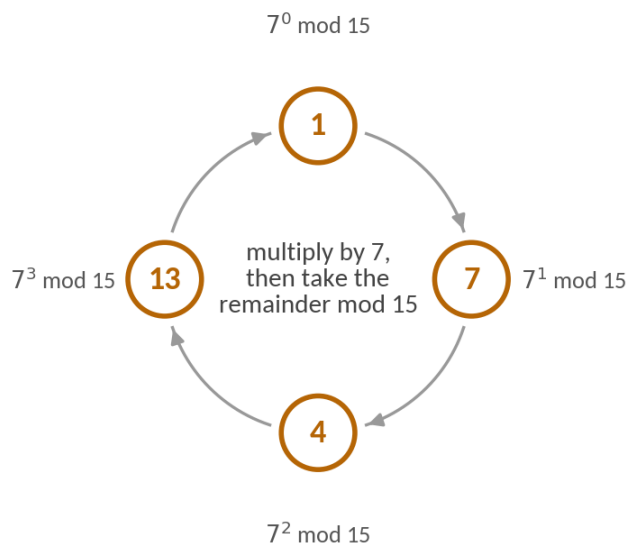
Answer: the register prints φ 's binary digits: $0.25 \rightarrow "01"$. With m counting qubits, QPE delivers φ to m binary digits (exactly when φ is an exact m -bit fraction; to high probability otherwise). This one primitive powers Shor's algorithm below, quantum chemistry's energy solvers, and more.

19.4 Shor's Algorithm: Factoring by Finding a Period

Now the famous application. Multiplying two large primes is easy; recovering them from the product N is believed classically hopeless — the best known classical algorithms are super-polynomial, and 2048-bit numbers would take them longer than the age of the universe. On that asymmetry rests RSA, the encryption protecting most of the world's online commerce.

Shor's 1994 discovery has two halves. The first is *classical number theory*: factoring reduces to period finding. Pick any number a sharing no factor with N , and list the powers $a^1, a^2, a^3, \dots$ each taken *mod* N (keep only remainders). The list always cycles with some period r — and knowing r (if it cooperates: r even, and $a^{r/2} \not\equiv -1 \pmod{N}$) cracks N via a greatest-common-divisor computation. The second half is quantum: *finding r is exactly a period-finding problem*, and QPE/QFT machinery solves it in polynomial time — the exponential speedup.

The powers of 7 (mod 15) repeat with period $r = 4$



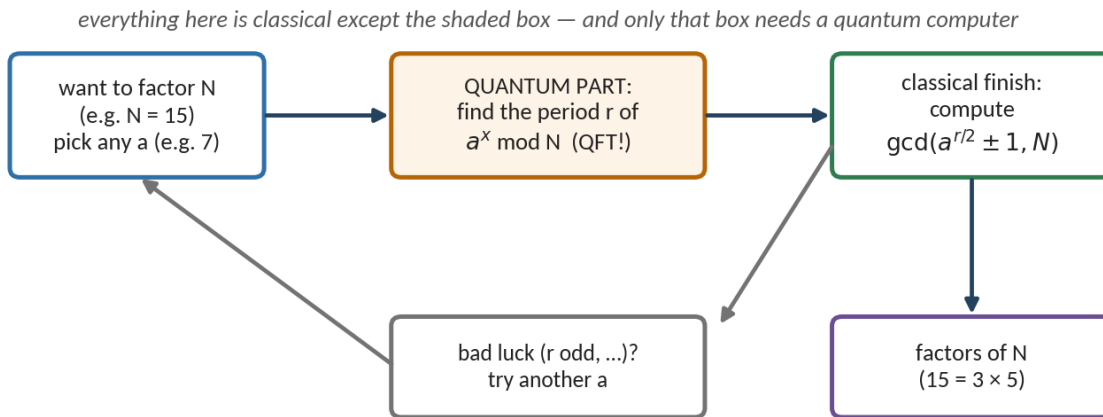
The whole secret of factoring 15: the powers of 7 (mod 15) cycle through 1, 7, 4, 13 — period $r = 4$. Classically you find r by walking the cycle (hopeless when the cycle has 10^{300} steps); quantumly, a Fourier transform reads r from a superposition over the whole cycle at once.

WORKED EXAMPLE — FACTORING 15, END TO END

Run Shor's recipe on $N = 15$ with $a = 7$.

1. Quantum subroutine (conceptually): prepare a superposition over x , compute $7^x \bmod 15$ into a second register (reversible Toffoli arithmetic, Chapter 10), and apply the inverse QFT to the x -register. Interference produces peaks; measuring returns a multiple of N_{states}/r , from which classical post-processing extracts the period $r = 4$ — matching the cycle in the figure.
2. Check r 's cooperation: $r = 4$ is even ✓, and $a^{r/2} = 7^2 = 49 \equiv 4 \pmod{15}$, which is not $-1 \pmod{15}$ (≡ 14) ✓.
3. Classical finish: compute $\gcd(a^{r/2} - 1, N) = \gcd(3, 15) = 3$ and $\gcd(a^{r/2} + 1, N) = \gcd(5, 15) = 5$.
4. Verify: $3 \times 5 = 15$ ✓

Answer: $15 = 3 \times 5$, found via the period of modular powers. Every step scales polynomially — the same recipe on a 2048-bit N needs a bigger machine, not a new idea. (If step 2's checks fail, pick another a and repeat — random a cooperates at least half the time.)



Shor's algorithm as an honest flowchart: classical preparation, one quantum period-finding subroutine (the only part needing a quantum computer), classical GCD finish, and a retry loop for unlucky choices of a .

19.5 What Shor Means for Codes and Security

REALITY CHECK

Where does the threat actually stand (as of this book's writing, 2026)? The largest numbers honestly factored by Shor's algorithm on real hardware remain tiny — demonstrations like $15 = 3 \times 5$ and $21 = 3 \times 7$, using handfuls of qubits and considerable "compiler foresight." Breaking RSA-2048 is estimated to require on the order of a *million* good physical qubits running error-corrected for days (recent careful estimates have fallen from tens of millions to under a million — progress worth watching). Today's machines: thousands of noisy qubits, no large-scale error correction. So: no, Shor is not reading your messages this year. And yet the

migration must happen *now* — because of “harvest now, decrypt later”: encrypted traffic recorded today can be stored cheaply and decrypted the day a machine exists. Data whose secrecy must survive 20+ years is already at risk, retroactively.

The world’s response is **post-quantum cryptography** (PQC): new public-key systems built on mathematical problems (chiefly lattice problems) that neither Shor’s period finding nor any known quantum algorithm cracks. NIST published the first PQC standards in 2024 (ML-KEM for key exchange, ML-DSA and SLH-DSA for signatures), and operating systems, browsers, and messaging apps began shipping hybrid classical+PQC encryption in 2024–2025. Symmetric ciphers, remember, face only Grover’s quadratic dent (Chapter 18): AES-256 stays comfortable.

Quantum threats to cryptography — the honest scoreboard

System	Quantum attack	Verdict and remedy
RSA, Diffie–Hellman, elliptic curves	Shor: exponential break	migrate to PQC standards (underway now)
AES and other symmetric ciphers	Grover: quadratic only	use 256-bit keys; no migration crisis
Hash functions	Grover-type searches	prefer longer outputs (SHA-256 and up)
PQC (lattice-based, hash-based)	no known quantum break	the new default; standardized 2024

CASE STUDY — THE TALK THAT MOVED GOVERNMENTS • PETER SHOR, 1994 — AND THE MIGRATION, 2016–TODAY

In April 1994, Peter Shor of Bell Labs presented his factoring algorithm; within weeks it had transformed quantum computing from professors’ curiosity into a matter of national policy — here was a concrete, world-relevant task with proven exponential quantum advantage. The long echo is just as instructive: in 2016, NIST opened a worldwide competition for quantum-resistant cryptography; 82 candidate systems from 25 countries were attacked publicly for eight years (several fell to *classical* attacks along the way — healthy scientific brutality); the survivors became the 2024 standards now rolling out across the internet. A single mathematical insight, thirty years ahead of the hardware able to run it, rewired global security planning — the strongest argument this book knows for learning quantum computing before the machines mature, not after.

ACTIVITY (APP) — PERIODS AND PEAKS

1. Open the *Shor Lab* in the QCFE app with $N = 15$, $a = 7$. Watch the app build the powers cycle (1, 7, 4, 13) and confirm $r = 4$ by eye.
2. Press “quantum view”: the app shows the superposition’s amplitudes and then the QFT spectrum — count the peaks and check peak spacing = total/ r .
3. Let the app measure; take its outcome through the GCD finish yourself on paper, reproducing 3×5 .

4. Rerun with $a = 2$ ($r = 4$ again) and with $a = 4$ ($r = 2$ — check whether the cooperation tests pass or the app retries).
5. Try $N = 21$: find a working a , and report r and the factors.

You need: The QCFE app (Android or Windows), Paper and pen

► App tool: *Shor Lab* — modular-power cycles, the QFT spectrum with its peaks, and the full factoring pipeline step by step

19.6 A Three-Qubit QFT, Gate by Gate

The QFT stops feeling like magic the moment you compute one, completely, by hand. Take $N = 8$ (three qubits) and the input $|101\rangle$ — the number 5.

EQUATION: QFT ON A BASIS STATE

$$\text{QFT}_N |x\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i xk/N} |k\rangle$$

In words: Input the number x ; output every pattern with equal loudness, phases climbing by x/N of a turn per step — a pure musical note of frequency x

Symbol	Meaning	Unit
x	the input number (0 to $N-1$)	none
k	output pattern index	none
$e^{2\pi i xk/N}$	phase of pattern k : xk/N of a full turn	none

All magnitudes equal $1/\sqrt{N}$; the information sits entirely in the phases. That is why the QFT is only useful inside interference sandwiches like phase estimation — measuring its output directly would show a featureless uniform histogram.

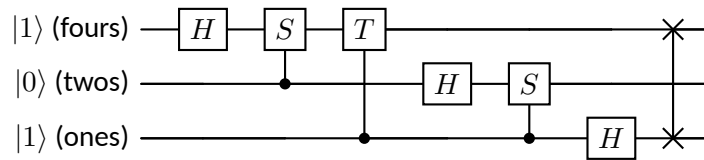
WORKED EXAMPLE — THE PHASE STAIRCASE OF $|101\rangle$

Compute all eight amplitudes of $\text{QFT}_8 |5\rangle$.

1. Magnitudes: all $1/\sqrt{8} \approx 0.354$.
2. Phase step: $x/N = 5/8$ of a turn = 225° per unit of k .
3. Climb the staircase, reducing mod 360° : $k = 0: 0^\circ; 1: 225^\circ; 2: 450^\circ \rightarrow 90^\circ; 3: 315^\circ; 4: 180^\circ; 5: 45^\circ; 6: 270^\circ; 7: 135^\circ$.
4. Sanity: after 8 steps the phase returns to $8 \times 225^\circ = 1800^\circ \equiv 0^\circ$ ✓

Answer: $\text{QFT}_8 |5\rangle = \frac{1}{\sqrt{8}} (|0\rangle + e^{i225^\circ} |1\rangle + e^{i90^\circ} |2\rangle + \dots + e^{i135^\circ} |7\rangle)$. **Eight equal bars, phases spinning 225° per step. The inverse QFT is simply the staircase read backwards — feed it this state and $|101\rangle$ comes out with certainty.**

The circuit that builds this uses only Hadamards and two controlled phase gates — S (a quarter turn) and T (an eighth turn):



The three-qubit QFT: H on the top digit, then finer and finer controlled turns ($S = 90^\circ$, $T = 45^\circ$) conditioned on the lower digits; repeat one level down; finish with a swap. Six gates plus a swap — versus 8×8 matrix arithmetic classically.

WORD TO KNOW

Controlled phase gate — A two-qubit gate applying a phase turn to $|11\rangle$ only — the QFT’s workhorse. The pattern continues: n -qubit QFTs need turns of $90^\circ, 45^\circ, 22.5^\circ, \dots$, each half the last.

WORKED EXAMPLE — COUNTING THE QFT’S COST

How many gates does an n -qubit QFT need, and what does that mean for $n = 20$?

1. Each qubit gets one H plus one controlled turn per qubit below it: n Hadamards and $n(n-1)/2$ controlled phases.
2. Add about $n/2$ final swaps. For $n = 3$: $3 + 3 + 1 = 7$ operations — the circuit above.
3. For $n = 20$: $20 + 190 + 10 = 220$ gates — while the matrix it implements is $2^{20} \times 2^{20}$, over a trillion entries.

Answer: $O(n^2)$ gates to apply a 2^n -dimensional Fourier transform — the exponential compression at Shor’s core. The honest fine print stays: the result lives in amplitudes’ phases, retrievable only through interference tricks like QPE, never by direct printout.

REALITY CHECK

In period-finding, this staircase is run in reverse: the modular-arithmetic register hides a period r , and the *inverse* QFT converts “hidden repetition every r steps” into “measurable spike near multiples of N/r ” — turning rhythm into a readable number, exactly as Fourier analysis turns a heard chord into named notes. If you can hand-compute the example above, the engine room of Shor’s algorithm holds no further mysteries — only bigger staircases.

19.7 How Big Is the Machine That Breaks RSA?

“Shor’s algorithm breaks RSA” is a mathematical fact; *when* is an engineering question with a moving answer. The honest way to track it is resource estimation: published, refereed counts of exactly how much machine the deed requires. Reading two landmark estimates — and how the number moved between them — teaches

the skill.

WORKED EXAMPLE — READING THE RESOURCE ESTIMATES

The 2021 benchmark (Gidney-Ekerå) priced factoring a 2048-bit RSA key at about 20 million physical qubits running for 8 hours. A 2025 estimate cut it below one million. Unpack the numbers and the lesson in their motion.

1. Separate the layers: the *logical* algorithm needs only thousands of logical qubits — but each logical qubit costs $\sim 10^3$ physical qubits of error correction (Chapter 29's ledger), plus factories for the expensive T-gates (Chapter 7's T-count).
2. 2021 arithmetic: $\sim$ few thousand logical $\times$ $\sim$ thousand-to-one overhead + factories $\approx 2 \times 10^7$ physical qubits, 8 hours of below-threshold operation.
3. 2025 arithmetic: better algorithms (approximate modular arithmetic), denser codes, and smarter factories shaved the same job under 10^6 qubits — a twenty-fold cut with *no new hardware*, purely from cleverness.
4. Compare with machines: current devices field 10^2 – 10^3 physical qubits at error rates near threshold — three orders of magnitude of scaling still missing, but the target moved twenty-fold closer in four years.

Answer: thousands of logical qubits dressed as a million physical ones, for hours — versus today's hundreds. The gap is real and so is its shrinkage from *both* sides (hardware up, requirements down). This two-sided motion is why serious risk models (Mosca's clock, Chapter 15) refuse to bet secrets on the gap's persistence.

WORD TO KNOW

Resource estimation — The engineering discipline of counting, honestly and in public, the qubits, gates, factories, and hours a target algorithm requires on specified hardware — the field's exchange rate between mathematical threat and physical timeline.

REALITY CHECK

Resource estimates are also the antidote to two opposite hypes. Against “RSA falls next year”: no published estimate fits any existing or announced machine — the missing factor is $\sim 10^3$ in qubit count *at* error rates most devices haven't reached. Against “never in our lifetime”: the requirement curve has fallen twenty-fold in four years while hardware error rates crossed the surface-code threshold (Chapter 29) — both curves point the same direction. The professional position is neither panic nor comfort: it is Mosca's inequality, evaluated annually with the newest numbers from the References.

CHAPTER SUMMARY

- Fourier transforms turn repetition into peaks: period r in a length- N signal $\Rightarrow$ spikes spaced N/r in frequency.

- The QFT does this to amplitudes: $|x\rangle \rightarrow \frac{1}{\sqrt{N}} \sum_y \omega^{xy} |y\rangle$, $\omega = e^{2\pi i/N}$ — an exponential-size transform in $\sim n^2$ gates (H's and controlled phase turns).
- QPE = phase kickback (controlled- U^{2^j} writes $e^{2\pi i \varphi k}$ across a counting register) + inverse QFT (reads φ as binary digits).
- Shor: factoring $\rightarrow$ period of $a^x \bmod N \rightarrow$ quantum period finding $\rightarrow$ classical GCD finish; fully traced here for $15 = 3 \times 5$ via $a=7, r=4$.
- Consequences: RSA-class systems fall to a future fault-tolerant machine (thousands of logical, $\sim$ million physical qubits); PQC standards (2024) are the migration path; harvest-now-decrypt-later makes the timeline urgent despite the hardware gap.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] A signal of length 20 repeats every 5 steps. At which frequencies does its Fourier transform peak?

Problem 2 [Basic] List the powers of 2 $\bmod 15$ until they cycle, and state the period r .

Problem 3 [Basic] For the 2-qubit QFT ($\omega = i$): compute QFT $|00\rangle$ (take $x = 0$), and name the familiar state you obtain.

Problem 4 [Basic] In QPE with $m = 3$ counting qubits, the register measures 101. What phase φ does this announce (binary fraction 0.101_2)?

Problem 5 [Basic] Redo the staircase for QFT₈ $|3\rangle$: give all eight phases in degrees. What is the phase step per unit of k ?

Problem 6 [Basic] Compute QFT₈ $|0\rangle$ from the formula. Which familiar state is it, and why does “frequency zero” explain the answer?

Problem 7 [Basic] Using the 1000-physical-per-logical rule of thumb, how many logical qubits does a million-physical-qubit machine offer? Roughly what fraction of the 2025 RSA-2048 requirement is that?

Problem 8 [Basic] Name the three layers whose costs multiply into a resource estimate (algorithm, error correction, magic-state factories), and state which chapters of this book price each.

Problem 9 [Standard] Compute QFT $|01\rangle$ ($x = 1, N = 4$) by hand and verify your amplitudes match the kickback state in the QPE worked example.

Problem 10 [Standard] Verify the other cooperative choice for 15: with $a = 2$ and its period from Problem 2, run the classical finish ($\gcd(2^{r/2} \pm 1, 15)$) and confirm the factors.

Problem 11 [Standard] An unlucky draw: for $N = 15$, $a = 14$, compute the powers cycle, find r , and show why the cooperation test rejects it (what is $14^{r/2} \bmod 15$?).

Problem 12 [Standard] QPE with $m = 2$ on a phase $\varphi = 0.3$ (not an exact 2-bit fraction): which two readouts are nearest to the truth, and why does increasing m help? (Answer in words; no computation of the full distribution required.)

Problem 13 [Standard] The QFT circuit for n qubits uses about $n^2/2$ controlled-phase gates. Count them for $n = 10$ and $n = 2048$, and compare the latter with Grover's $\sqrt{N}$ oracle calls for the same 2048-bit search space — which scaling is polynomial in n , and which exponential?

Problem 14 [Standard] Hand-compute the four amplitudes of $\text{QFT}_4 |2\rangle$ (phase step 180°), then show the result equals the product state $|+\rangle \otimes |-\rangle$. (QFT outputs of basis states are always product states — the deep reason the circuit is so short.)

Problem 15 [Standard] For a five-qubit QFT, count the Hadamards, the controlled phase gates, and the swaps. What is the smallest turn angle used, in degrees?

Problem 16 [Standard] In the circuit above, the controlled-T couples the top and bottom qubits. Using the staircase, explain in one sentence what goes wrong with the output phases if that single gate is omitted — and why the smallest turns matter less and less (the insight behind the “approximate QFT” used on real hardware).

Problem 17 [Standard] The requirement fell from 2×10^7 to $< 10^6$ qubits in four years while hardware grew from $\sim 10^2$ to $\sim 10^3$. If both trends continued geometrically, estimate the crossing time — then write two sentences on why such extrapolation deserves distrust in both directions.

Problem 18 [Challenge] Prove the classical reduction's punchline for the worked example's numbers: from $7^4 \equiv 1 \pmod{15}$, show 15 divides $(7^2 - 1)(7^2 + 1) = 48 \times 50$, and explain in two sentences why each factor of 15 must then hide inside $\gcd(7^2 \pm 1, 15)$ — barring the two unlucky cases the algorithm screens out.

Problem 19 [Challenge] Estimate the “harvest-now” horizon: a hospital's patient records must stay confidential for 50 years; encrypted backups leak today. Using this chapter's honest hardware picture (RSA-2048 breakable at $\sim 10^6$ good qubits; today $\sim 10^3$ – 10^4 noisy ones; capability doubling optimistically every 2–3 years), argue in four sentences whether classical RSA protects those records, and write the one-line policy recommendation.

Problem 20 [Challenge] A vendor demo “factors 15 on 4 qubits” — but inspection shows the circuit hard-codes the answer's period into its gates. Using the claim-checking checklist of Chapter 2 plus this chapter, write a five-line review distinguishing an honest small-scale Shor demonstration from a “compiled” one, and state which single scaling question exposes the difference.

Problem 21 [Challenge] Verify the two-qubit case fully by matrices: compute the 4×4 product $H(\text{top}) \cdot \text{controlled-S} \cdot H(\text{bottom}) \cdot \text{SWAP}$ and confirm it equals QFT_4 with entries $\frac{1}{2}i^{jk}$. (Patient bookkeeping, big payoff: you will have proven a Fourier transform equals five elementary gates.)

Problem 22 [Challenge] A headline reads “Researchers factor a 48-bit number on 10 qubits — RSA in danger.” Using this lesson plus Chapter 19’s scaling, write a four-sentence referee note: what was plausibly done, why 48 bits on 10 qubits does not extrapolate (which resource explodes?), and what single measured quantity would change your assessment.

Variational Algorithms, Optimization, and Quantum Machine Learning

VQE, QAOA, and quantum machine learning — hybrid algorithms for imperfect machines

IN THIS CHAPTER YOU WILL LEARN TO

- Explain the hybrid quantum–classical loop, and why it suits today’s noisy hardware.
- Run a one-qubit VQE by hand: parameterized circuit, measured energy, classical dial-turning.
- Describe QAOA and work a small MaxCut example’s cost function.
- Give a balanced account of quantum machine learning: the ideas, the hopes, and the caveats.
- Recognize the failure modes — barren plateaus, noise floors, and classical competition.

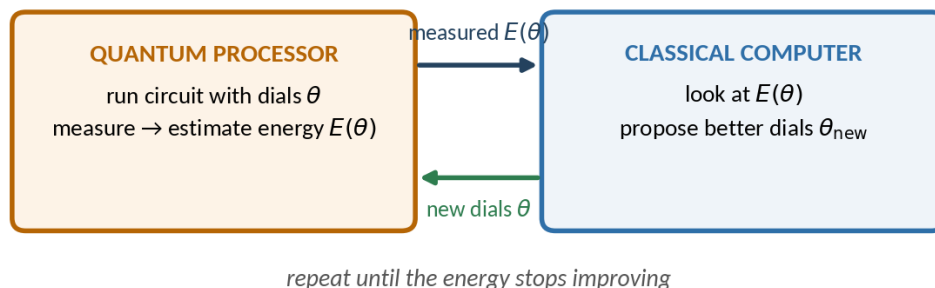
20.1 The Hybrid Idea: Quantum + Classical Loops

Grover and Shor assume long, clean circuits — a luxury today’s noisy machines (Chapter 28) cannot afford. The 2014-era response was a family of algorithms built around a different bargain: keep the quantum circuit *short* and adjustable, and let a classical computer do the strategizing.

WORD TO KNOW

Variational algorithm — A hybrid method in which a short quantum circuit with adjustable dials (parameters θ) is run and measured; a classical optimizer then adjusts the dials to improve the measured score, and the loop repeats.

The template never changes. A **parameterized circuit** (an *ansatz* — rotation gates whose angles are the dials, entangled by some CNOTs) prepares a trial state $|\psi(\theta)\rangle$. Measurements estimate a **score** — typically an energy. The **classical optimizer** proposes new dials. Around and around, until the score stops improving.



The variational loop: a short quantum circuit proposes, measurement scores, a classical optimizer disposes. Each partner does what it is best at.

Why is this noise-friendly? Three reasons, all pragmatic. Short circuits finish before decoherence erases them. The output is an *average* over many shots, and averages forgive scattered errors better than single delicate amplitudes. And the optimizer, chasing whatever landscape it is given, automatically works around some systematic hardware quirks — it optimizes the machine that exists, not the ideal one.

20.2 VQE: The Variational Quantum Eigensolver

The flagship variational algorithm hunts the **ground-state energy** — the lowest possible energy — of a physical system such as a molecule. Chemistry cares intensely: reaction rates, binding strengths, and material properties all flow from such energies, and exact classical computation hits the 2^n wall for even modest molecules (Chapter 1’s case study).

The mathematical guarantee behind VQE is the *variational principle*: every trial state’s energy lies *at or above* the true ground energy. So the search is safe — you can only ever be too high, and lower is always truer.

EQUATION: THE VQE OBJECTIVE

$$E(\theta) = \langle \psi(\theta) | H_{\text{sys}} | \psi(\theta) \rangle \geq E_{\text{ground}}$$

In words: The measured average energy of the trial state is never below the true lowest energy — so: turn the dials to push it down

Symbol	Meaning	Unit
$E(\theta)$	the average energy measured for dial settings θ	energy (chemists: hartree)
H_{sys}	the Hamiltonian: the observable “energy” of the system, a sum of simple measurable pieces	—
E_{ground}	the true minimum energy being hunted	energy

In practice H_{sys} is a recipe of Pauli measurements (X’s and Z’s — Chapters 6–7): run the circuit, measure each piece over many shots, and combine the averages into $E(\theta)$.

WORKED EXAMPLE — A COMPLETE ONE-QUBIT VQE, BY HAND

Toy system with energy recipe $E = \langle Z \rangle + 0.35 \langle X \rangle$ (averages of Z and X measurements). Ansatz: $|\psi(\theta)\rangle = R_y(\theta) |0\rangle = \cos \frac{\theta}{2} |0\rangle + \sin \frac{\theta}{2} |1\rangle$. Find the best dial.

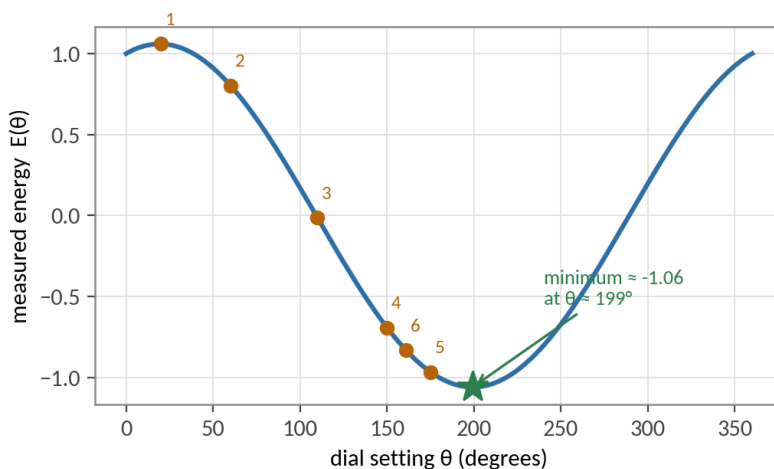
1. Averages for this ansatz (from the Born rule; both are standard results you can verify): $\langle Z \rangle = \cos \theta$ and $\langle X \rangle = \sin \theta$.

2. Objective: $E(\theta) = \cos \theta + 0.35 \sin \theta$.

3. Optimizer at work (measured points, like the figure): $E(20^\circ) \approx 1.06$, $E(110^\circ) \approx -0.01$, $E(175^\circ) \approx -0.97$, $E(199^\circ) \approx -1.06$ — and probing nearby shows no improvement.

4. Exact check (the luxury of a toy): minimum of $\cos \theta + 0.35 \sin \theta$ is $-\sqrt{1 + 0.35^2} \approx -1.060$ at $\theta \approx 199.3^\circ$ ✓

Answer: the loop settles at $\theta \approx 199^\circ$, $E \approx -1.06$ — matching the exact ground energy. Real VQE is this, scaled: dozens of dials, entangled ansatz circuits, thousands of shots per energy estimate — but the same loop you just turned by hand.



The energy landscape of the worked example, with the optimizer's visits numbered. The classical partner only ever sees sampled points — it walks downhill on a curve it never fully sees.

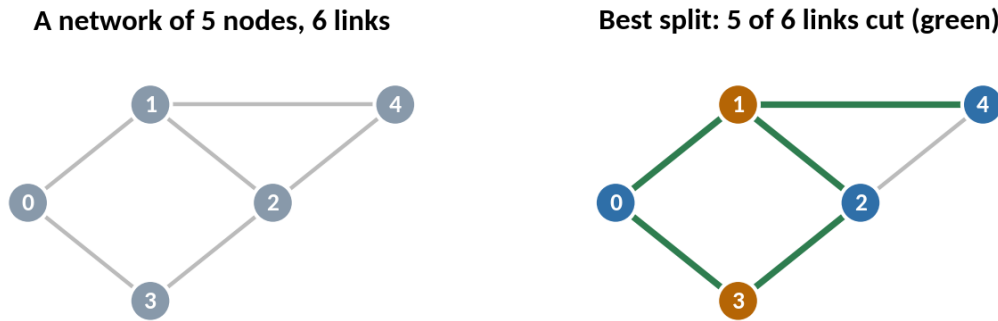
COMMON MISTAKE

✗ Wrong: "VQE found an energy, so that is the molecule's true ground energy."

✓ Right: VQE reports the lowest energy *reachable* by its *ansatz*, estimated with *shot noise*, on *noisy hardware*. All three can lie upward (and noise can even lie downward past the bound). Professional VQE reports come with ansatz descriptions, shot budgets, error-mitigation methods, and comparisons against classical references — never a bare number.

20.3 Quantum Optimization and QAOA

Swap physics energy for a *cost function* and the same loop attacks optimization problems. The standard-bearer is QAOA (Quantum Approximate Optimization Algorithm), and its favorite demonstration problem is **MaxCut**: divide a network's nodes into two camps so that as many links as possible run *between* the camps.



MaxCut: split the nodes into two teams (blue/orange) to cut as many links as possible. For this 5-node network the best split cuts 5 of 6.

WORKED EXAMPLE — SCORING MAXCUT ASSIGNMENTS

For the figure's network (links: 0-1, 0-3, 1-2, 2-3, 2-4, 1-4), score the split $\{0, 2, 4\}$ vs $\{1, 3\}$, and the split $\{0, 1\}$ vs $\{2, 3, 4\}$.

1. Split A — check each link: 0-1 crosses ✓; 0-3 crosses ✓; 1-2 crosses ✓; 2-3 crosses ✓; 2-4 stays inside ✗; 1-4 crosses ✓. Cut = 5.
2. Split B: 0-1 inside ✗; 0-3 crosses ✓; 1-2 crosses ✓; 2-3 inside ✗; 2-4 inside ✗; 1-4 crosses ✓. Cut = 3.
3. Encoding for a quantum computer: one qubit per node, $|0\rangle/|1\rangle$ = camp; each assignment is a basis pattern; the cut count becomes an energy-style cost read from measurements.

Answer: split A scores 5 (optimal here), split B scores 3. QAOA prepares a parameterized superposition over all 2^5 assignments, measures the average cut, and lets the classical loop tune dials so that high-cut patterns dominate the histogram.

The honest state of play: for small demonstrations QAOA works charmingly; for industrial sizes, classical heuristics (which are excellent at MaxCut-type problems) remain ahead. Whether QAOA-family methods will beat them at useful scale is a genuinely open research question — this book flags it as such, and Chapter 31's capstone lets you form your own judgment on small instances.

20.4 Quantum Machine Learning: Promise and Caution

Quantum machine learning (QML) is the field's most searched — and most oversold — frontier. The sober map has three territories.

Variational models. Use a parameterized circuit as a trainable model: encode data into rotation angles, train

the dials like network weights. These run on today's machines at small scale. The open problem is advantage: for datasets of practical size, well-tuned classical networks so far match or beat them, and encoding large classical datasets into qubits is itself expensive.

Linear-algebra speedups. Quantum algorithms exist that solve certain huge linear systems exponentially faster — with heavy fine print: data must arrive in quantum form, solutions come out as quantum states (readable only by sampling), and several early headline applications were later “dequantized” when classically inspired algorithms caught up (the Tang results, 2018–2019). A humbling, healthy episode: quantum claims sharpen classical ingenuity.

Quantum data. The least glamorous and best-founded territory: when the data is *itself* quantum — states from a quantum sensor, a chemistry simulation, a quantum network — a quantum processor is the natural learner, and provable separations exist. Many researchers expect QML's first real wins here.

REALITY CHECK

A checklist for any “quantum AI” headline, in the spirit of Chapter 2: Was the comparison against a *tuned* classical baseline of similar size? Does the input-loading cost (encoding classical data into states) appear in the accounting? Is the claimed speedup theoretical-asymptotic or measured end-to-end? Could the circuit be simulated classically (too small to matter)? As of this writing, no production machine-learning workload runs faster on quantum hardware end to end — and saying so plainly is fully compatible with taking QML research seriously.

CASE STUDY — A MOLECULE IN A PROCESSOR · VQE ON REAL HARDWARE, 2017–TODAY

In 2017, an IBM team used VQE on a six-qubit superconducting processor to compute the ground-state energy curve of small molecules — hydrogen (H_2), lithium hydride (LiH), and beryllium hydride (BeH_2) — directly on noisy hardware, no error correction, agreeing with exact answers to chemical usefulness for these tiny cases. Every molecule computed was, admittedly, classically trivial; the point was existence proof: the full hybrid loop — ansatz, Pauli measurements, optimizer — running through a real refrigerator. Since then VQE-family experiments have grown steadily in size and sophistication (better ansätze, error mitigation, smarter measurement grouping), while an honest scoreboard still reads: classical quantum-chemistry methods remain ahead at useful scale. The race's outcome will likely be decided by the error-corrected machines of Chapter 29 — carrying the algorithms you learned here.

ACTIVITY (APP) — TURN THE DIALS YOURSELF

1. Open the *VQE Lab* in the QCFE app — it presents the worked example's system with a θ slider and a live measured $E(\theta)$ (with realistic shot noise).
2. Be the optimizer: starting from $\theta = 20^\circ$, adjust the dial by hand, always seeking lower energy, and log your visits like the figure's numbered points. How many evaluations did you need?
3. Switch on “auto-optimize” and watch the machine's strategy; compare its evaluation count to yours.

4. Raise the shot noise from 1000 to 50 shots per estimate and repeat: feel how noisy scores mislead the descent — the daily reality of variational work.
5. In the QAOA *panel*, load the 5-node MaxCut, run the loop, and check the final histogram concentrates on split A's patterns (each optimum appears as two mirror patterns — why?).

You need: The QCFE app (Android or Windows), Paper and pen

► App tool: VQE Lab — energy landscapes with shot noise, manual and automatic optimizers, and a QAOA MaxCut playground

20.5 Measuring a Many-Term Energy — and Trusting the Bound

VQE's cost function is an energy $\langle H \rangle$, but no machine measures “ H ” in one go. A molecular Hamiltonian arrives as a shopping list of Pauli terms, each with a weight, and the machine measures the terms — cleverly batched.

EQUATION: ENERGY, TERM BY TERM

$$H = \sum_i c_i P_i \quad \Rightarrow \quad \langle H \rangle = \sum_i c_i \langle P_i \rangle$$

In words: The energy is the weighted sum of simple Pauli measurements — each estimated from its own histogram

Symbol	Meaning	Unit
P_i	a Pauli string, e.g. $Z \otimes Z$ or $X \otimes X$	none
c_i	its real weight from the chemistry	varies
$\langle P_i \rangle$	its measured average, between -1 and $+1$	none

Batching rule: terms that are **qubit-wise commuting** — on every qubit their letters agree or one is I — share one measurement setting. $\{ZZ, ZI, IZ\}$ is one batch; XX needs its own (measure after H on each qubit).

WORD TO KNOW

Qubit-wise commuting group — A set of Pauli terms that agree (or defer to I) qubit by qubit, so a single choice of per-qubit measurement basis serves them all — the standard trick for slashing VQE's measurement bill.

WORKED EXAMPLE — ONE TRIAL STATE, TWO SETTINGS, ONE ENERGY

Let $H = 0.5 ZZ + 0.3 XX + 0.2 ZI$ and take the trial state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. Organize the measurements and compute the energy.

1. Batch: $\{ZZ, ZI\}$ commute qubit-wise (setting: plain standard-basis readout); $\{XX\}$ needs the X-setting (H on both, then read).
2. Setting 1 statistics on $|\Phi^+\rangle$: outcomes 00 and 11 only, so $\langle ZZ \rangle = +1$ (digits always agree) and $\langle ZI \rangle = 0$ (first digit is a fair coin).
3. Setting 2: applying H to both maps $|\Phi^+\rangle$ to itself, so again perfect agreement: $\langle XX \rangle = +1$.
4. Combine: $E = 0.5(1) + 0.3(1) + 0.2(0) = 0.8$.

Answer: $E = 0.8$, from two histogram settings instead of three — a 33% saving that becomes a lifesaver when molecules bring thousands of terms. Every VQE iteration repeats exactly this bookkeeping at its current dial settings.

WORKED EXAMPLE — THE VARIATIONAL BOUND, CAUGHT IN THE ACT

For the same H , hunt the ground energy with Bell-state trials, then compare with the exact answer.

1. Try the singlet $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$: $\langle ZZ \rangle = -1$, $\langle XX \rangle = -1$, $\langle ZI \rangle = 0$, so $E = -0.5 - 0.3 + 0 = -0.8$.
2. Exact diagonalization of the 4×4 matrix gives ground energy $E_0 \approx -0.861$.
3. Verdict: even the *best* Bell state sits 0.061 above truth — and the variational principle guaranteed in advance it could not sit below.

Answer: $E_{\text{trial}} = -0.8 \geq E_0 \approx -0.861$. **Trials never lie low — so VQE progress is monotone truth: every lower energy found is a genuinely better state. Closing the last 0.061 needs an ansatz richer than Bell states (a tilted entangler), which is precisely the ansatz-design game of this chapter.**

REALITY CHECK

Measurement, not optimization, is VQE's practical bottleneck. Each $\langle P_i \rangle$ carries shot noise $\sim 1/\sqrt{n}$, and chemistry demands “chemical accuracy” (≈ 1.6 milli-hartree), so realistic molecules need millions of shots per energy evaluation, repeated every optimizer step. Grouping, weight-proportional shot allocation (spend shots where $|c_i|$ is large), and smarter estimators are an active research front — the unglamorous plumbing on which quantum chemistry's hopes rest.

20.6 Optimizer Traps: Local Minima, Noise, and the Restart Discipline

The variational bound guarantees trial energies never lie *low* — but nothing guarantees the optimizer finds the lowest trial. Real VQE landscapes have valleys that are not the valley: local minima, plus shot-noise fog over every reading. Professionals answer with a discipline as old as optimization itself, sharpened by the bound's one-sided honesty.

WORKED EXAMPLE — THREE STARTS, ONE ANSWER

A two-parameter ansatz is trained three times from random starts on the Hamiltonian of the grouping lesson (exact ground energy -0.861). Runs converge to $E_1 = -0.72$, $E_2 = -0.86$, $E_3 = -0.73$. Interpret and decide.

1. Apply the bound: all three are legal upper bounds — the true ground energy lies at or below -0.86 .
2. Diagnose the spread: two runs found one valley (≈ -0.72), one found a deeper valley — a classic local-minimum landscape, not noise (the repeats agree too well).
3. Decide: report $E \leq -0.86$; restart several more times around run 2's parameters to test whether -0.86 itself is the floor.
4. Cross-check with theory: exact diagonalization (possible for toy sizes) gives -0.861 — run 2 landed within 0.001 , and the “keep the best, trust the bound” rule delivered the right answer despite two-thirds of runs failing.

Answer: the best of n restarts is the estimate, and the variational bound makes “best” meaningful — lower is *provably* better, never a statistical fluke of lying low. Contrast classical fitting, where the lowest loss can be overfitting; VQE's floor is load-bearing.

WORD TO KNOW

Restart discipline — Running a variational optimization from multiple independent initializations and keeping the lowest converged energy — the standard defense against local minima, made rigorous by the variational bound's one-sided guarantee.

REALITY CHECK

Budget the discipline honestly: restarts multiply the shot bill of Chapter 25's pricing lesson, so practitioners restart *small* ansätze freely and large ones grudgingly, warm-starting from smaller solved instances (layerwise growth) instead. And keep the two failure modes distinct: a *spread* of converged energies signals local minima (restart more); converged runs that *wander* without settling signal shot noise or barren flatness (raise shots, shrink the circuit). The cure for one is waste on the other — diagnose before spending.

CHAPTER SUMMARY

- Variational algorithms marry short parameterized quantum circuits with classical optimizers — built for noisy, shallow hardware.
- VQE hunts ground energies under the variational safety net $E(\theta) \geq E_{\text{ground}}$: energies are measured as Pauli-average recipes; you ran the full loop by hand on $E = \langle Z \rangle + 0.35 \langle X \rangle$.
- QAOA turns optimization problems (MaxCut) into the same loop: qubits = decisions, cost = measured average, dials steer probability toward good assignments; classical heuristics remain the industrial benchmark.
- QML: variational models (runnable now; advantage unproven), linear-algebra speedups (heavy caveats;

partially dequantized), and quantum-data learning (best-founded hopes).

- Professional habits: report ansatz/shots/mitigation with every number; distrust bare comparisons; watch for barren plateaus and noise floors; treat open questions as open.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] Name the two partners of the hybrid loop and what each contributes per iteration.

Problem 2 [Basic] For the worked example's $E(\theta) = \cos \theta + 0.35 \sin \theta$: evaluate $E(0^\circ)$, $E(90^\circ)$, and $E(180^\circ)$.

Problem 3 [Basic] Why can a VQE energy never (ideally) fall below the true ground energy? Name the principle.

Problem 4 [Basic] In the MaxCut figure's network, score the split $\{0\}$ vs $\{1, 2, 3, 4\}$.

Problem 5 [Basic] Which of these fit into a single measurement setting: $\{ZI, IZ, ZZ\}$? $\{ZZ, XX\}$? $\{XI, IX, XX\}$? Answer for each with a one-clause reason.

Problem 6 [Basic] A standard-basis run of 1000 shots on two qubits returns: 00: 420, 01: 80, 10: 90, 11: 410. Estimate $\langle ZZ \rangle$ (agreements minus disagreements, over shots) and $\langle ZI \rangle$.

Problem 7 [Basic] Five restarts give $-0.41, -0.55, -0.54, -0.55, -0.41$. What is your energy estimate, what does the bound let you claim, and what do the repeated values suggest about the landscape?

Problem 8 [Basic] Why can a lower converged VQE energy never be "overfitting"? One sentence, naming the theorem.

Problem 9 [Standard] Using $\langle Z \rangle = \cos \theta$, $\langle X \rangle = \sin \theta$ for the R_y ansatz: write $E(\theta)$ for the recipe $E = 0.5 \langle Z \rangle - \langle X \rangle$, evaluate it at $\theta = 90^\circ, 120^\circ, 150^\circ$, and bracket where its minimum lies.

Problem 10 [Standard] A 1000-shot estimate of $\langle Z \rangle$ from counts $\{0: 620, 1: 380\}$ gives what value? ($\langle Z \rangle = P(0) - P(1)$.) Give the shot-noise scale ($\approx 1/\sqrt{N_{\text{shots}}}$) and decide whether E differences of 0.01 are resolvable at this budget.

Problem 11 [Standard] MaxCut mirror symmetry: show that any split and its swap (exchange the two camps) cut the same links, and state the consequence for the measurement histogram of a successful QAOA run.

Problem 12 [Standard] Your optimizer reports these visits: $E = -0.71, -0.96, -1.02, -1.05, -1.058, -1.059$. Recommend stopping or continuing, with a one-sentence justification referencing both the trend and the worked example's known floor.

Problem 13 [Standard] List the three QML territories with one honest sentence each, and mark which one runs on today's hardware, which was partially dequantized, and which most experts consider the safest bet.

Problem 14 [Standard] Group $H = 0.4 ZZ + 0.4 ZI + 0.2 IX$ into the fewest settings, and describe each setting's readout recipe explicitly.

Problem 15 [Standard] Compute $\langle H \rangle$ for the worked examples' H in the trial state $|01\rangle$. Where does this trial rank among the four you now know ($|\Phi^+\rangle$, singlet, $|01\rangle$, and the exact ground state)?

Problem 16 [Standard] You have 1000 shots and two groups with total weights 0.7 ($\{ZZ, ZI\}$) and 0.3 ($\{XX\}$). Allocate shots proportionally to weight, and explain in one sentence why equal splitting wastes precision.

Problem 17 [Standard] Each optimizer step costs 60 evaluations $\times$ 2000 shots (Chapter 25); convergence takes 150 steps. Price one run and a 12-restart campaign. If budget caps you at 10^8 shots, how many restarts fit?

Problem 18 [Challenge] Ansatz limitation in one qubit: suppose the recipe is $E = \langle Y \rangle$ but your ansatz remains $R_y(\theta) |0\rangle$ (real amplitudes only, so $\langle Y \rangle = 0$ for every θ). What minimum does your VQE report, what is the true minimum (-1 , at $|-i\rangle$), and which single gate added to the ansatz fixes the miss? Moral, in one sentence, about "the lowest energy reachable by the ansatz."

Problem 19 [Challenge] Barren plateaus: in large random ansätze, gradients of $E(\theta)$ shrink exponentially with qubit count, so measured differences drown in shot noise. Using Problem 6's noise scale, estimate how many shots per evaluation you would need to resolve a gradient of size 10^{-4} , and draw the strategic conclusion (two sentences) about "just make the ansatz bigger."

Problem 20 [Challenge] Design a fair fight: specify a complete protocol (five bullet-style sentences are fine — but write them as prose) for comparing QAOA against a classical heuristic on 200-node MaxCut instances without stacking the deck: instance selection, tuning budget for BOTH sides, end-to-end timing, success metric, and repetitions. Which single omission most often flatters the quantum side in published comparisons?

Problem 21 [Challenge] A press release: "Our quantum AI analyzed hospital X-rays with 98% accuracy." Draft four referee questions using this chapter's checklist (baseline, encoding cost, end-to-end timing, simulability), and explain in one closing sentence which answer would most change your assessment.

Problem 22 [Challenge] Prove the X-setting recipe: show $(H \otimes H) |\Phi^+\rangle = |\Phi^+\rangle$ by direct expansion, and explain why standard-basis agreement after the H's measures exactly $\langle X \otimes X \rangle$ (hint: $HZH = X$ from Chapter 7).

Problem 23 [Challenge] Design the decision tree for a stuck VQE: converged-but-spread energies vs. non-converging noisy traces vs. flat gradients from step one. For each branch, name the diagnosis, the cheapest test confirming it, and the correct spend (restarts / more shots / smaller ansatz) — one paragraph, citing this chapter and Chapter 25.

Quantum Simulation: The Original Killer App

Why Feynman wanted this machine — molecules and materials, simulated one small time-step at a time

IN THIS CHAPTER YOU WILL LEARN TO

- State the simulation problem that started the whole field, and why classical computers hit the 2^n wall on it.
- Read e^{-iHt} as “the circuit nature runs,” built from the same Pauli-term shopping lists as VQE.
- Slice time with Trotter’s recipe, compute its error, and buy accuracy with more steps.
- Estimate the gate bill for simulating a small material, and check it against hardware budgets.
- Locate simulation’s real prizes — chemistry, materials, nuclear and particle physics — and its honest competition.

Shor’s algorithm gets the headlines, but it is the field’s adopted child. The *founding* application — the one in Feynman’s 1982 paper that Chapter 2 began with — is humbler and vaster: use a controllable quantum system to imitate an uncontrollable one. Molecules computing chemistry, electrons computing materials — nature runs these “programs” effortlessly while our best supercomputers choke on thirty particles. This chapter turns that founding dream into circuits you can count.

21.1 The Problem Nature Solves Daily

A drug molecule docking, a catalyst cracking nitrogen, electrons pairing into a superconductor — each is a quantum state evolving under known physical laws. The laws fit on a postcard; the *bookkeeping* explodes: n interacting particles need 2^n amplitudes (Chapter 9), and the exact dynamics of even 50 spins outruns every computer on Earth. Chemists survive on brilliant approximations (they earn Nobel Prizes for them), but the approximations fail exactly where the chemistry gets interesting — strongly correlated electrons, transition states, exotic magnetism.

Feynman’s proposal, in this book’s language: build a machine whose qubits obey *our* chosen Hamiltonian, map the target system onto it, press play, and measure. The bookkeeping never happens — the amplitudes live in the hardware, all 2^n of them, rent-free.

21.2 Time Evolution as a Circuit Target

EQUATION: SCHRÖDINGER, IN ONE LINE

$$|\psi(t)\rangle = e^{-iHt} |\psi(0)\rangle, \quad H = \sum_j c_j P_j$$

In words: Waiting for time t applies one specific unitary, generated by the energy operator — which arrives, as in VQE, as a weighted list of simple Pauli terms

Symbol	Meaning	Unit
H	the Hamiltonian (energy operator) of the target system	varies
e^{-iHt}	the time-evolution unitary — our compilation target	none
$c_j P_j$	one local term: a weighted Pauli string	varies

Two familiar faces: H 's shopping-list form is exactly Chapter 20's, and for a single term the evolution is just a rotation gate — $e^{-i\theta Z/2} = R_z(\theta)$, Chapter 7. Simulation = compiling the sum when the terms refuse to commute.

If all terms commuted, the compile would be trivial: $e^{-i(A+B)t} = e^{-iAt}e^{-iBt}$, one rotation per term, done. Nature declines: position-like and momentum-like terms, or X -type and Z -type couplings, do not commute (Chapter 6's field guide) — and then the exponential of a sum is *not* the product of exponentials. Trotter's recipe tames the gap.

21.3 Trotter's Recipe: Slice Time Thin

EQUATION: THE TROTTER STEP

$$e^{-i(A+B)t} \approx \left(e^{-iAt/n} e^{-iBt/n} \right)^n, \quad \text{error} \lesssim \frac{t^2 \|[A, B]\|}{2n}$$

In words: Alternate small doses of each term, n rounds; the mistake shrinks in proportion to $1/n$, priced by how badly the terms refuse to commute

Symbol	Meaning	Unit
n	number of time slices (circuit repetitions)	none
$[A, B]$	the commutator $AB - BA$ — the “refusal to commute”	varies
$\ \cdot\ $	its size (largest effect on any state)	none

Intuition: over a thin slice t/n , the terms barely notice each other's motion — alternating tiny doses almost equals dosing together. Halve the slices' width (double n), halve the error: accuracy is bought by circuit depth, the currency Chapter 28 taught you to spend carefully.

WORKED EXAMPLE — TROTTER ON ONE QUBIT, EXACTLY CHECKED

Take $H = 0.5 X + 0.8 Z$ (two stubbornly non-commuting terms) and $t = 1$. Compare the Trotter circuit against the exact evolution for $n = 1, 4, 16$ slices.

1. Each slice is two rotations from Chapter 7: $R_x(2 \times 0.5/n)$ then $R_z(2 \times 0.8/n)$; the whole circuit is that pair repeated n times.
2. Predicted error from the bound: $\|[A, B]\| = 0.8$ here, so error $\lesssim 0.4/n$: about 0.40, 0.10, 0.025 for $n = 1, 4, 16$.
3. Exact comparison (matrix arithmetic, Appendix B): true operator error 0.362, 0.086, 0.0215 — the bound tracks the truth within 15%.
4. Cash it out as fidelity on the state $e^{-iHt} |0\rangle$: 91.2%, 99.4%, 99.96%. Sixteen thin slices make a nearly perfect movie of the dynamics.

Answer: $n = 1$ is a caricature, $n = 4$ a decent sketch, $n = 16$ a portrait — with cost growing linearly ($2n$ rotations) while error falls as $1/n$. Every serious simulation paper is this worked example scaled up: choose target accuracy, invert the bound for n , multiply out the gate bill.

WORD TO KNOW

Trotterization — Compiling e^{-iHt} by alternating small evolutions of H 's individual terms. The error per unit time is set by the terms' commutators and shrinks as slices thin — accuracy traded for depth.

WORKED EXAMPLE — THE GATE BILL FOR A SMALL MAGNET

A 50-spin magnetic chain: 49 neighbor couplings (ZZ -type) and 50 field terms (X -type). Price a simulation of $t = 10$ time units at accuracy demanding $n = 100$ slices.

1. Per slice: each coupling term costs one two-qubit interaction (a ZZ -rotation compiles to CNOT, R_z , CNOT — Chapter 10); each field term one R_x . About $49 \times 3 + 50 = 197$ gates per slice.
2. Parallelism (Chapter 12's cost sheet): odd and even couplings run in alternating columns — depth per slice ≈ 7 , not 197.
3. Total: 100 slices $\times$ 197 gates $\approx$ 19,700 gates, of which $\approx$ 14,700 are CNOTs; depth $\approx$ 700.
4. Budget check (Chapter 30): at $p_2 = 0.6\%$, survival $(1 - 0.006)^{14700} \approx 10^{-38}$ — hopeless raw. At error-corrected $p_L = 10^{-6}$ per logical gate: survival $\approx 98.5\%$ — comfortable.

Answer: $\sim 20k$ gates: trivial for an error-corrected machine, unreachable for a raw one — the recurring arithmetic of Part Eight, now derived by you from a physical model. Real research squeezes the bill with commuting-term grouping (Chapter 20's trick!), higher-order slicing, and smarter encodings — factors of 10, fighting for feasibility.

21.4 The Prize List, Honestly Ranked

CASE STUDY — NITROGEN'S SECRET • FROM HABER-BOSCH FURNACES TO FeMoco, 2017–

About 1–2% of world energy feeds Haber–Bosch reactors, cracking N_2 at hundreds of degrees and atmospheres to make fertilizer — while soil bacteria do the same chemistry at ambient conditions, using an enzyme core called **FeMoco** whose strongly correlated electrons defeat every classical method. In 2017, Reiher and colleagues published the founding resource estimate: simulating FeMoco's electronic structure to chemical accuracy on an error-corrected quantum computer — millions of qubit-hours in their original count, slashed repeatedly since by better algorithms (the same falling-requirements story as Chapter 19's factoring estimates). The prize: understanding, and perhaps industrializing, nature's room-temperature route to fertilizer. It is the field's favorite example because it is concrete, humanitarian, measurable — and honest: nobody claims it for this decade; everybody can now count what it costs.

Beyond FeMoco, the serious list: battery and catalyst materials (strongly correlated candidates first), high-temperature superconductivity's mechanism, nuclear and particle physics on lattices — and, nearer term, **analog simulators**: cold atoms and trapped ions arranged to *be* the model system, already probing magnetism and exotic phases with hundreds of particles, special-purpose but real. Meanwhile the honest competition never sleeps: classical methods (DFT, tensor networks, quantum Monte Carlo) keep improving, and they own every weakly correlated molecule. The professional forecast: quantum simulation wins first, and specifically, where entanglement in the *target* is dense — exactly where classical bookkeeping dies.

COMMON MISTAKE

✗ Wrong: “Quantum computers will soon replace chemistry labs and classical simulation software.”

✓ Right: Neither, on any near horizon. Most industrial chemistry is weakly correlated — classical methods are excellent and staying. The quantum case is the *strongly correlated* minority: FeMoco-class enzymes, exotic materials, lattice theories — problems where classical errors are qualitative, not cosmetic. And the machines that decide it are the error-corrected ones of Chapter 29, priced by ledgers you can now audit. “Original killer app” names the oldest, best-founded target — not the earliest deadline.

ACTIVITY (APP) — SLICE TIME YOURSELF

1. In the *Circuit Lab* of the QCFE app (or the Appendix E simulator), build the worked example: alternate $R_x(1/n)$ and $R_z(1.6/n)$, repeated n times, on $|0\rangle$.
2. For $n = 1, 2, 4, 8, 16$: record the final $P(0)$ and compare against the exact target ($P(0) = 0.816$ for these numbers — computable with the matrix tools of Appendix B).
3. Plot error versus n on paper: confirm the $1/n$ shrinkage of the Trotter bound.
4. Sabotage: swap the order (R_z first). Which n notices, and what does that teach about error *direction* versus error *size*?

You need: The QCFE app (Android or Windows) or Python, Paper and pen

CHAPTER SUMMARY

- Simulation is the founding application: nature's 2^n bookkeeping, hosted on hardware that pays no rent for amplitudes.
- The target is one unitary, e^{-iHt} , with H the same Pauli shopping list VQE measures — simulation evolves it instead.
- Trotter slices time: alternate small doses of non-commuting terms; error $\sim t^2 \| [A, B] \| / 2n$ falls linearly in the slice count, paid in depth.
- Gate bills are countable: terms $\times$ slices, compressed by parallel columns and commuting groups — feasible on error-corrected machines, fatal raw.
- The honest prize list leads with strongly correlated chemistry and materials (FeMoco the flagship), analog simulators deliver physics today, and classical methods keep the rest — the founding dream, on an engineering schedule.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] Why does $e^{-i(A+B)t} = e^{-iAt}e^{-iBt}$ hold when $[A, B] = 0$, and what familiar single-term gates are $e^{-i\theta Z/2}$ and $e^{-i\theta X/2}$?

Problem 2 [Basic] For the worked example's H , how many slices does the bound $0.4/n$ suggest for target error 0.01? And for 0.001?

Problem 3 [Basic] A Hamiltonian has terms $\{ZZ \text{ on } (1, 2), ZZ \text{ on } (3, 4), XI \text{ on } 5\}$. Which terms share a Trotter slice's parallel column (Chapter 12)? Does this set even need Trotterization?

Problem 4 [Standard] Doubling the simulated time t at fixed slice count n multiplies the Trotter error bound by what factor? Restore the original error: what must n become, and what happened to the gate bill?

Problem 5 [Standard] Recompute the small-magnet bill for 100 spins (99 couplings, 100 fields), $n = 200$ slices: gates, CNOTs, and depth (odd/even parallel columns). At logical error 10^{-6} per gate, estimate survival.

Problem 6 [Standard] The ZZ -rotation compiles as CNOT, $R_z(2ct/n)$, CNOT. Verify the construction's logic: why does conjugating R_z on the target by CNOTs make the phase depend on the parity of the two qubits (Chapter 10)?

Problem 7 [Standard] An analog cold-atom simulator realizes the magnet directly with 300 atoms but offers no gates and limited measurement. List one question it answers better than the digital circuit, and one question

it cannot answer at all — and the general moral about special-purpose machines.

Problem 8 [Standard] FeMoco resource estimates fell by orders of magnitude between 2017 and the mid-2020s with no new hardware. Name the two ingredients of such falls (better algorithms, tighter error analysis), and connect to the same phenomenon in Chapter 19's factoring estimates.

Problem 9 [Challenge] Second-order slicing uses the symmetric step $e^{-iAt/2n}e^{-iBt/n}e^{-iAt/2n}$, with error falling as $1/n^2$. For the worked example, estimate the slices needed for error 0.001 under first order ($0.4/n$) versus second order ($\approx 0.4/n^2$ as a crude model), and compare gate bills (the symmetric step costs $\frac{3}{2}$ rotations per slice on average).

Problem 10 [Challenge] Compute $[A, B]$ for $A = 0.5X$, $B = 0.8Z$ explicitly (use $XZ = -iY$, $ZX = +iY$), verify $\|[A, B]\| = 0.8$, and explain why two *commuting* Z -type terms would have made the whole chapter's problem disappear.

Problem 11 [Challenge] Design the measurement stage: after simulating the magnet for time t , you must estimate the magnetization $\langle Z_1 + \dots + Z_{50} \rangle / 50$ and one correlation $\langle Z_1 Z_2 \rangle$. Using Chapters 8 and 20, give the measurement settings, the shot budget for ± 0.01 precision, and whether both quantities share one setting.

Problem 12 [Challenge] Referee this claim: "Our 40-qubit processor simulated a 40-spin magnet beyond classical reach." Using Chapter 9 (what classical simulators actually reach), Chapter 28 (what raw hardware fidelity survives the required depth), and this chapter's bills, write the four questions whose answers decide the claim.

PART SEVEN

QUANTUM AI AND MACHINE LEARNING

Encoding Data into Quantum States

Basis, angle, and amplitude encoding — how classical data enters a quantum computer

IN THIS CHAPTER YOU WILL LEARN TO

- Explain why every quantum machine-learning model begins with an encoding decision.
- Use basis encoding for bit patterns and angle encoding for numbers, by hand.
- Build amplitude encodings, count their remarkable compression — and their loading cost.
- Describe feature maps: how encoding itself creates the nonlinearity models need.
- Judge the encoding bottleneck honestly: when loading data erases a quantum advantage.

22.1 The First Step of Every Quantum Model

Part Six asks the question this book has been building toward: can quantum computers *learn*? Before any learning, though, comes a humbler step that decides more than beginners expect. Machine-learning data is classical — columns of numbers: exam scores, pixel brightnesses, sensor readings. Qubits speak amplitudes and phases. Something must translate, and that something is the **encoding**: a fixed recipe turning each data point x into a quantum state $|\phi(x)\rangle$.

WORD TO KNOW

Data encoding — The fixed circuit recipe that prepares a quantum state $|\phi(x)\rangle$ from a classical data point x . Every quantum model's first stage — and often the stage that decides its cost and its power.

Why does the choice matter so much? Three reasons this chapter unpacks. *Capacity*: different encodings pack wildly different amounts of data per qubit. *Cost*: some encodings load in one circuit layer, others need depth that grows with the data size — and depth is the currency noise taxes (Chapter 28). *Geometry*: encoding decides how data points are arranged as states — which points end up close, which far — and that arrangement is what any classifier downstream must work with. In classical machine learning, feature engineering shapes success; in quantum machine learning, the encoding *is* the feature engineering.

	the practical trade-offs		
	basis	angle	amplitude
BASIS ENCODING bits become the pattern $1,0 \rightarrow 10\rangle$	qubits needed	d	d (or fewer)
			$\log_2 d$
ANGLE ENCODING numbers become rotation angles $x \rightarrow R_y(x) 0\rangle$	loading depth	1	up to $\sim d$
	carries	bit patterns	a few numbers
			whole vectors
AMPLITUDE ENCODING numbers become amplitudes $(0.6, 0.8) \rightarrow 0.6 0\rangle + 0.8 1\rangle$			

The three standard encodings and their trade-offs. This book’s worked examples use angle encoding — one rotation per feature, one layer deep, honest and hardware-friendly.

22.2 Basis and Angle Encoding

Basis encoding is the translation you have used since Chapter 9 without naming it: a bit pattern becomes the matching basis state. The customer record (paid: yes, late: no, renewed: yes) becomes $|101\rangle$. It is exact, free, and shallow — and it stores just one classical bit per qubit, with no use made of superposition at all. Its true role in algorithms is indexing: a superposition over basis-encoded records is how Grover’s oracle (Chapter 18) “sees” a whole database.

Angle encoding is the workhorse of near-term quantum ML: each numerical feature becomes a rotation angle.

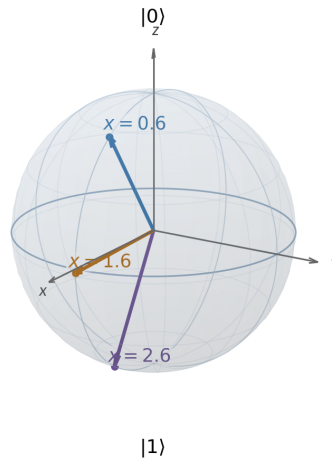
EQUATION: ANGLE ENCODING

$$|\phi(x)\rangle = R_y(x) |0\rangle = \cos\frac{x}{2} |0\rangle + \sin\frac{x}{2} |1\rangle$$

In words: The feature value becomes a rotation angle: the data point is written into the qubit’s latitude

Symbol	Meaning	Unit
x	one numerical feature, typically rescaled into a range like $[0, \pi]$	radians after rescaling
R_y	the rotation gate of Chapter 7	none

Several features use several rotations: this book’s standard map for a 2-feature point is $R_z(x_2)R_y(x_1) |0\rangle$ — latitude from x_1 , longitude from x_2 : the whole data point becomes one Bloch arrow. One layer deep, one qubit per couple of features, and every gate is a calibrated hardware native.



Angle encoding in a picture: three values of a feature x become three latitudes of the Bloch arrow. Nearby data lands on nearby states — similarity in, similarity preserved.

WORKED EXAMPLE — ENCODING A DATA POINT, START TO FINISH

A weather record has two features, already rescaled: $x_1 = 1.0472$ (that is 60°) and $x_2 = 1.5708$ (90°). Encode it with this book's standard map and describe the resulting state.

1. Latitude first: $R_y(60^\circ) |0\rangle = \cos 30^\circ |0\rangle + \sin 30^\circ |1\rangle = 0.866 |0\rangle + 0.5 |1\rangle$ (Chapter 7's worked example, reused verbatim).

2. Longitude second: $R_z(90^\circ)$ attaches phases $e^{\mp i 45^\circ}$ to the two amplitudes; up to global phase the state is $0.866 |0\rangle + 0.5 e^{i 90^\circ} |1\rangle = 0.866 |0\rangle + 0.5i |1\rangle$.

3. Locate it: Bloch angles $\theta = 60^\circ$, $\varphi = 90^\circ$ — exactly the state of Chapter 5's from-angles worked example.

Answer: the record now lives at one definite point on the sphere, $(\theta, \varphi) = (60^\circ, 90^\circ)$. Every downstream model in this part — classifiers, kernels, networks — starts from states prepared exactly like this.

COMMON MISTAKE

X Wrong: "Feed the raw numbers in — the encoder will cope with any range."

✓ Right: Angles live on a circle: x and $x + 2\pi$ encode to the *same* state, so unscaled features (incomes in millions, pixel values 0–255) wrap around and scramble similarity. Always rescale features into a chosen window — $[0, \pi]$ is standard, keeping distinct values on distinct latitudes. Data preparation discipline transfers from classical ML fully intact.

22.3 Amplitude Encoding: Maximum Compression, Maximum Cost

The third strategy is the most quantum of all: store the data *in the amplitudes themselves*. A vector of 2^n numbers, normalized to length 1 (Chapter 3), is a legal state of n qubits.

EQUATION: AMPLITUDE ENCODING

$$x = (x_0, x_1, \dots, x_{N-1}) \longrightarrow |\phi(x)\rangle = \frac{1}{\|x\|} \sum_{k=0}^{N-1} x_k |k\rangle, \quad N = 2^n$$

In words: Each data value becomes the amplitude of one basis pattern; n qubits carry 2^n values

Symbol	Meaning	Unit
x_k	the k -th data value, indexing basis pattern $ k\rangle$	none
$\ x\ $	the vector's length, dividing out for normalization	none
$N = 2^n$	values stored by n qubits	none

The compression is breathtaking: 10 qubits hold a 1024-value vector; 30 qubits, a billion. The catch follows in the worked example and the Reality Check — read all three together.

WORKED EXAMPLE — AMPLITUDE-ENCODING A FOUR-VALUE RECORD

Encode the vector $x = (3, 1, 1, 5)$ into two qubits.

1. Length: $\|x\| = \sqrt{9 + 1 + 1 + 25} = \sqrt{36} = 6$.
2. Divide: amplitudes $(0.5, 0.1\bar{6}, 0.1\bar{6}, 0.8\bar{3})$ — precisely $(3, 1, 1, 5)/6$.
3. Write the state: $|\phi(x)\rangle = 0.5|00\rangle + 0.167|01\rangle + 0.167|10\rangle + 0.833|11\rangle$; check $0.25 + 0.028 + 0.028 + 0.694 = 1.000$ ✓
4. Now the honesty step — the loading circuit: preparing an *arbitrary* 4-amplitude state needs a small cascade of conditioned rotations (Chapter 10 machinery); for 2^n arbitrary amplitudes the required circuit generally grows to order 2^n gates. The exponential compression is paid for by exponential loading depth — unless the data has special structure.

Answer: $|\phi(x)\rangle = \frac{1}{6}(3|00\rangle + |01\rangle + |10\rangle + 5|11\rangle)$. **Two qubits, four values — and a loading bill that this part will keep in view every time amplitude encoding is proposed.**

22.4 Feature Maps: Nonlinearity for Free

Classical machine learning long ago learned a trick: problems unsolvable in the raw features often become easy after a clever nonlinear re-description — a *feature map*. The rings dataset coming in Chapter 23 is the classic example: no straight line separates the classes, but add the feature $x_1^2 + x_2^2$ (the radius) and a mere threshold solves it.

Here is the beautiful observation powering quantum ML: **quantum encodings are nonlinear feature maps automatically**. The state $R_y(x)|0\rangle$ carries amplitudes $\cos(x/2)$ and $\sin(x/2)$ — already trigonometric, not linear, in the data. Measured quantities inherit this: the simple score $\langle Z \rangle$ of an angle-encoded point is $\cos x$;

richer circuits build richer combinations — products of sines and cosines of all the features, which is exactly the raw material of Fourier analysis (Chapter 19’s toolbox, resurfacing in learning theory). Repeating the encoding between trainable layers (*data re-uploading*) stacks these harmonics higher, the way Chapter 23’s worked classifier will exploit; entangling encoded qubits (as in the ZZ-style feature maps of Chapter 24) mixes features together into correlation terms no single qubit could hold.

WORD TO KNOW

Feature map — A re-description of data intended to make its structure easier to separate. In quantum ML, the encoding circuit IS the feature map: data point in, point on a high-dimensional quantum state space out — with trigonometric nonlinearity built in by the physics of rotations.

WORKED EXAMPLE — A NONLINEAR FEATURE, MEASURED

Show that a plain $\langle Z \rangle$ readout of an angle-encoded feature computes a nonlinear function of the data — with no explicit “nonlinearity” programmed anywhere.

1. Encode: $|\phi(x)\rangle = \cos \frac{x}{2} |0\rangle + \sin \frac{x}{2} |1\rangle$.
2. Read out: $\langle Z \rangle = P(0) - P(1) = \cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}$.
3. Simplify with the half-angle identity: $\langle Z \rangle = \cos x$.

Answer: the measurement returns $\cos x$ — a curve, not a line, conjured purely by encoding-then-measuring. Stack three re-uploads with trainable weights w_k and the model’s output becomes a sum of harmonics $\cos(w_1 x), \cos(w_2 x), \dots$ — a trainable Fourier series. That is the machinery behind the decision boundaries you will watch being learned in the next chapter.

22.5 The Encoding Bottleneck

REALITY CHECK

The most important honesty checkpoint in quantum machine learning lives right here, at the doorstep. Several famous quantum speedups (Chapter 2’s map; the linear-algebra family discussed in Chapter 27) assume the data is *already* loaded — amplitude-encoded, or served by a hypothetical quantum memory (“qRAM”) that no one has built at scale. For a dataset of N numbers, if loading alone costs order- N operations, then any algorithm claiming to beat classical $O(N)$ after loading has spent its advantage at the door. This is the encoding bottleneck, and it disciplines the whole field: near-term quantum ML sticks to shallow encodings of *small* feature sets (angle encoding — this book’s choice); claimed exponential speedups must state their data-access assumptions in writing; and the cleanest escape — data that is *born quantum* and needs no loading at all — is exactly where Chapter 25 will locate quantum learning’s best-founded hopes.

CASE STUDY — WHEN THE DOORWAY DECIDED THE RACE • RECOMMENDATION SYSTEMS, 2016–2019

In 2016, Kerenidis and Prakash published a quantum recommendation algorithm — the engine behind “users who liked this also liked...” — exponentially faster than the best classical methods known, *assuming* the preference data sat pre-loaded in a qRAM-style structure. It became a flagship example of quantum ML’s promise. Then in 2018, Ewin Tang — an eighteen-year-old undergraduate — asked the referee’s question: if the quantum algorithm is allowed such generous data access, what could a classical algorithm do with a comparable structure? The answer: essentially match it. Tang’s “dequantized” algorithm, followed by a wave of similar results for other linear-algebra speedups, redrew the field’s map overnight. The lesson was not that quantum ML is empty — it was that *advantage claims live or die at the encoding step*, and every serious paper since states its input assumptions up front. You now read the field’s literature with exactly the right first question: *how does the data get in?*

ACTIVITY (APP) — THE ENCODING PLAYGROUND

1. Open the *Encoding Lab* in the QCFE app. Type the two features $(60^\circ, 90^\circ)$ into the angle encoder and confirm the Bloch arrow lands where the worked example predicted.
2. Drag feature x_1 smoothly from 0° to 180° and watch the arrow sweep pole to pole — then push past 360° and watch the wrap-around that the Common Mistake box warned about.
3. Switch to the amplitude encoder, enter $(3, 1, 1, 5)$, and check the app’s normalized amplitudes against your hand computation.
4. Open the “loading cost” readout and compare the circuit depth the app reports for the angle map (constant) versus amplitude maps of length 4, 16, and 64 (growing). You are watching the encoding bottleneck form in real time.

You need: The QCFE app (Android or Windows), A calculator

► App tool: *Encoding Lab* — angle and amplitude encoders with live Bloch view, normalization check, and loading-depth meter

22.6 How Many Qubits to Hold Your Data?

Before any training begins, an engineer must answer a bookkeeping question: *how many qubits does my dataset need?* The three encodings of this chapter give three very different invoices — and the differences drive real design decisions.

EQUATION: THE ENCODING INVOICE

$$\text{basis: } n_q = \text{bits}(x) \quad \text{angle: } n_q = d \quad \text{amplitude: } n_q = \lceil \log_2 d \rceil$$

In words: Basis encoding pays one qubit per bit; angle encoding one qubit per feature; amplitude encoding only the logarithm of the feature count

Symbol	Meaning	Unit
d	number of features in one data point	none
n_q	qubits required	none
$\lceil \cdot \rceil$	round up	none

Amplitude encoding's exponential compression carries fine print: the amplitudes must be normalized ($\sum_i |x_i|^2 = 1$), and loading an arbitrary vector generally needs a circuit whose depth grows like 2^{n_q} — the compression is in storage, not necessarily in loading time.

WORKED EXAMPLE — INVOICING A 28×28 IMAGE

A grayscale image has $28 \times 28 = 784$ pixel values. Price all three encodings (8 bits per pixel for the basis option).

1. Basis: $784 \times 8 = 6272$ qubits — absurd today; this style suits short bit strings, not images.
2. Angle: 784 qubits, one rotation each — beyond near-term machines, though shallow and gentle.
3. Amplitude: $\lceil \log_2 784 \rceil = 10$ qubits — ten! But the loading circuit for an arbitrary 784-entry vector needs on the order of $2^{10} \approx 10^3$ gates, and the pixel vector must first be normalized.

Answer: 6272 vs. 784 vs. 10 qubits. The exponential storage win of amplitude encoding is real, and so is its loading bill — there is no free lunch, only a choice of which currency to pay in. Practical 2026 QML mostly angles-encodes a few informative features instead (often after classical compression such as PCA).

WORD TO KNOW

State-preparation cost — The gate count needed to load data into amplitudes. For arbitrary vectors it grows exponentially with qubit count — the “input problem” that quietly dominates many claimed QML speedups.

REALITY CHECK

When a paper announces “we processed 1024-dimensional data on 10 qubits,” the professional's first question is: *what did the loading circuit cost, and was it included in the runtime?* If data loading takes exponential time, an exponential speedup downstream is an illusion — the checklist point D of Chapter 27. Honest QML papers now report end-to-end costs, loading included.

22.7 Feature Scaling: The Forgotten First Layer

Between the raw dataset and the encoding gates sits an unglamorous step that decides more accuracy than any circuit choice: **scaling**. Angle encodings feed features into rotation gates, and rotations live on a circle — feed them numbers of the wrong size and the geometry silently betrays you (the wrap-around trap of Chapter 24).

EQUATION: MIN-MAX SCALING INTO THE SAFE WINDOW

$$x_{\text{scaled}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \times \pi$$

In words: Slide and stretch each feature so its whole dataset range spans exactly zero to π — half the circle, no wrap-around possible

Symbol	Meaning	Unit
$x_{\min}, x_{\max}$	the feature's smallest and largest <i>training</i> values	varies
x_{scaled}	the angle fed to the rotation gate	rad
π	the safe span: distinct inputs stay distinguishable	none

Compute $x_{\min}, x_{\max}$ on training data only, and clip test points that land outside — letting test data redefine the scaling is a classic silent leak of information between sets.

WORKED EXAMPLE — SCALING A TEMPERATURE COLUMN

A sensor feature takes training values between 12 and 47 (degrees). Scale the readings 12, 30, and 47, and check the geometry.

1. Range: $47 - 12 = 35$.
2. Scale: $12 \rightarrow 0$; $47 \rightarrow \pi$ (180°); $30 \rightarrow \frac{18}{35}\pi \approx 0.51\pi \approx 92.6^\circ$.
3. Geometry check with Chapter 24's ruler: the extremes land 180° apart — kernel similarity $\cos^2(90^\circ) = 0$, perfectly distinguishable; no two distinct readings can collide.
4. Contrast the unscaled disaster: raw values 12 and $12 + 2\pi \approx 18.3$ fed directly as radians would encode to the *same state* — two different temperatures the model constitutionally cannot tell apart.

Answer: $(0^\circ, 92.6^\circ, 180^\circ)$. **One line of preprocessing buys what no amount of training can recover afterward. The habit: scale first, encode second, and log the $(x_{\min}, x_{\max})$ pair as part of the model — it is part of the model.**

WORD TO KNOW

Data leakage — Any information flow from test data into training-time choices — including computing scaling ranges on the full dataset. A leading cause of accuracy claims that evaporate on truly fresh data; auditors (Chapter 27) check for it first.

REALITY CHECK

Scaling choices are honest hyperparameters: $[0, \pi]$ is the conservative default, $[0, 2\pi)$ deliberately wraps (occasionally right for genuinely periodic features like wind direction or time-of-day!), and standardization (mean 0, spread 1, then a gain factor) suits bell-shaped data. The QML-specific rule is only this: *know your encoding's period, and place your data inside it on purpose*. Periodicity is a feature of the hardware — make

it a design decision, never an accident.

CHAPTER SUMMARY

- Encoding — the fixed recipe $x \rightarrow |\phi(x)\rangle$ — is stage one of every quantum model, and it sets capacity, cost, and the geometry all learning inherits.
- Basis encoding: bit patterns to basis states; one bit per qubit; the indexing language of oracles. Angle encoding: rescaled features to rotation angles ($R_z(x_2)R_y(x_1)|0\rangle$); shallow, native, this book's standard.
- Amplitude encoding: 2^n normalized values in n qubits — exponential compression bought with loading circuits that generally grow like 2^n .
- Encodings are built-in nonlinear feature maps: rotations make trigonometric features ($\langle Z \rangle = \cos x$), re-uploading stacks harmonics, entangled encodings mix features — the raw material of quantum models.
- The encoding bottleneck is the field's honesty gate: speedups quoted *after* free loading can vanish once loading is priced (the dequantization story) — and quantum-born data sidesteps the gate entirely.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] Basis-encode the record (1, 1, 0, 1) and state how many qubits it needs.

Problem 2 [Basic] Angle-encode $x = 120^\circ$ with $R_y(x)|0\rangle$: write the state's amplitudes ($\cos 60^\circ = 0.5$) and its measurement probabilities.

Problem 3 [Basic] How many qubits does amplitude encoding need for a 256-value vector? For a million values (use $2^{20} \approx 10^6$)?

Problem 4 [Basic] Rescaling drill: incomes range from 0 to 5,000,000 dinars. Give the formula mapping income v into the window $[0, \pi]$, and the angle for $v = 2,500,000$.

Problem 5 [Basic] Price all three encodings for a data point with 64 features (use 8 bits per feature for basis encoding).

Problem 6 [Basic] Amplitude-encode the vector (3, 4): normalize it, write the resulting qubit state, and give the probabilities a measurement would show.

Problem 7 [Basic] Scale the training values 5, 20, 35 (range learned from a column spanning 5 to 35) into $[0, \pi]$, in degrees.

Problem 8 [Basic] A test point arrives at 41 on the 5–35 column. What angle would naive scaling give, why is that dangerous, and what does the clipping rule do instead?

Problem 9 [Standard] Amplitude-encode $x = (1, 2, 2, 4)$: compute the normalization, all four amplitudes, and verify the probabilities sum to 1.

Problem 10 [Standard] Two records $a = (30^\circ, 0^\circ)$ and $b = (150^\circ, 0^\circ)$ are angle-encoded with $R_z(x_2)R_y(x_1)|0\rangle$. Compute both states and their overlap $|\langle\phi(a)|\phi(b)\rangle|^2 = \cos^2$ of half the latitude difference — verify!). Which downstream chapter turns this overlap into a similarity score?

Problem 11 [Standard] Show that features x and $x + 2\pi$ produce physically identical angle-encoded states (recall global phase, Chapter 5), and explain the practical rescaling rule this forces.

Problem 12 [Standard] A hospital dataset has 8 features per patient. Compare three designs: (a) angle encoding, one feature per qubit; (b) angle encoding, two features per qubit (R_y then R_z); (c) amplitude encoding. Give qubit counts and one-sentence loading-cost verdicts for each.

Problem 13 [Standard] Verify the feature-map identity of Lesson 4 numerically at $x = 60^\circ$: compute $P(0)$, $P(1)$, and $\langle Z \rangle$ from the amplitudes, and check against $\cos 60^\circ = 0.5$.

Problem 14 [Standard] A sensor stream has 12 features. Compare angle encoding (12 qubits, depth 1) against amplitude encoding (4 qubits, depth $\approx 2^4$): which fits better on a machine with 27 noisy qubits and a two-qubit error rate of 0.6% (Chapter 30's budget)? Show your survival estimates.

Problem 15 [Standard] Why must an amplitude-encoded vector be normalized first, and what information is lost when two data points differ only by overall scale? Name one dataset where that loss is harmless and one where it is not.

Problem 16 [Standard] For d features, angle encoding costs d qubits and $O(d)$ gates; amplitude costs $\lceil \log_2 d \rceil$ qubits and $O(d)$ gates. At what feature count does each resource (qubits, gates) become the binding constraint on a 100-qubit, depth-200 machine?

Problem 17 [Standard] A feature is time-of-day in hours $[0, 24)$. Argue for scaling into $[0, 2\pi)$ with wrap-around, compute the encoded angles of 23:00 and 01:00, and show the kernel now correctly rates them as near neighbors.

Problem 18 [Challenge] Design a two-qubit *entangling* feature map for features (x_1, x_2) : angle-encode each feature on its own qubit, then apply CNOT, then $R_z(x_1x_2)$ on the target, then CNOT again. Trace the circuit on the input (π, π) far enough to show the product-term phase $e^{\pm ix_1x_2/2}$ appearing — a feature (the *product* of inputs) that no single-qubit encoding of the same features can produce. (This is the heart of the ZZ feature maps used in Chapter 24.)

Problem 19 [Challenge] The bottleneck, quantified: a claimed quantum ML routine runs in $\sqrt{N}$ time on amplitude-encoded data of size $N = 10^{12}$, but loading costs N operations at 10^9 per second. Compute the loading time, the post-loading quantum time (at the same rate), and the end-to-end total; compare with a classical $N \log N$ scan; and write the two-sentence referee verdict.

Problem 20 [Challenge] Amplitude encoding stores values in amplitudes — but Chapter 4 says amplitudes cannot be read back from one copy. Reconcile the two facts in a short paragraph: what CAN a quantum algorithm usefully do with an amplitude-encoded state, if not read it? (Hint: think interference between whole vectors, and overlaps — the operations Chapters 24 and 27 are built on.)

Problem 21 [Challenge] A colleague proposes: “compress 1024 features to 8 by classical PCA, then angle-encode 8 qubits.” Estimate the full pipeline’s qubit and depth budget, and argue in a short paragraph whether any quantum advantage can survive the classical compression step — and what experiment would settle it.

Problem 22 [Challenge] Design the scaling audit for a published QML result: the three questions you would ask (computed on which split? period matched to encoding? test clipping policy?), and for each, the specific failure it would expose — tying one to the leakage definition and one to the wrap-around trap.

Variational Quantum Classifiers

A circuit that learns — training a real quantum model on a dataset no line can split

IN THIS CHAPTER YOU WILL LEARN TO

- Define supervised learning in one page: features, labels, training, and testing.
- Build a variational quantum classifier: encoding, trainable layers, and a measured score.
- Train it: loss functions, gradients, and the hybrid loop — the VQE engine repurposed.
- Follow a complete, real example: a one-qubit model learning the rings dataset to 100% test accuracy.
- Evaluate models honestly: train/test splits, baselines, and the questions that keep results trustworthy.

23.1 Learning Machines in One Page

Machine learning, stripped to essentials: a **model** is a function with adjustable dials; **training** is turning the dials until the function fits examples; the hope is **generalization** — that the fitted function keeps working on examples it never saw. In *supervised classification*, each example is a feature vector x with a label y (spam or not; disease or not; in this chapter, $+1$ or -1), and the model must output the right label for new x .

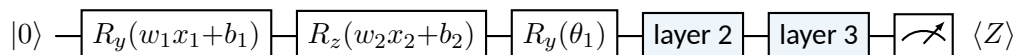
You already possess every ingredient. A parameterized circuit is a function with dials — Chapter 20 trained one to minimize energy. An encoding turns x into a state — Chapter 22. A measured average turns the circuit into a number — Chapter 8. Classification simply rewires the same loop: instead of “turn the dials to lower the energy,” the command becomes “turn the dials until the score agrees with the labels.”

WORD TO KNOW

Variational quantum classifier (VQC) — A supervised model whose prediction is a measured expectation value of a parameterized quantum circuit: encode the data point, apply trainable gates, measure a score, threshold it into a label. Trained by the classical-quantum hybrid loop of Chapter 20.

23.2 The Model: A Circuit as a Function

This book's demonstration model is deliberately minimal — one qubit, three repeating layers — yet it will conquer a dataset that defeats every straight-line classifier. Each layer re-encodes the two features (with trainable scales and offsets) and adds a trainable twist:



The one-qubit re-uploading classifier. Each layer: encode x_1 into a y -rotation and x_2 into a z -rotation — with trainable weights w and biases b — then apply a trainable rotation θ . Layers 2 and 3 repeat the pattern with their own dials: fifteen trainable numbers in all.

EQUATION: THE CLASSIFIER'S PREDICTION RULE

$$s(x; \theta) = \langle Z \rangle \text{ of the circuit's output, } \hat{y} = \text{sign}(s(x; \theta))$$

In words: Run the circuit on the encoded point, measure the average of Z — a score between -1 and $+1$ — and predict the label by its sign

Symbol	Meaning	Unit
$s(x; \theta)$	the model's score for point x at dial settings θ	between -1 and $+1$
$\hat{y}$	the predicted label	$+1$ or -1
θ	all trainable dials (w 's, b 's, θ 's — 15 here)	radians

On hardware, $\langle Z \rangle$ is estimated with shots (Chapter 8); on a simulator it is exact. The score's magnitude doubles as confidence: $s = +0.9$ is a confident $+1$; $s = +0.05$ is a shrug.

Why re-upload the data in every layer? Chapter 22's Fourier picture: one encoding pass gives the model curves like $\cos(wx + b)$; each repetition multiplies the available harmonics, letting the trained function bend into rings, stripes, and pockets. One pass could never fence in a circle; three passes, as you are about to watch, can.

23.3 Loss and the Training Loop

Training needs a target to descend. The standard choice for measured scores is the **squared loss**:

EQUATION: THE TRAINING LOSS

$$L(\theta) = \frac{1}{N_{\text{train}}} \sum_i \left(s(x_i; \theta) - 0.9 y_i \right)^2$$

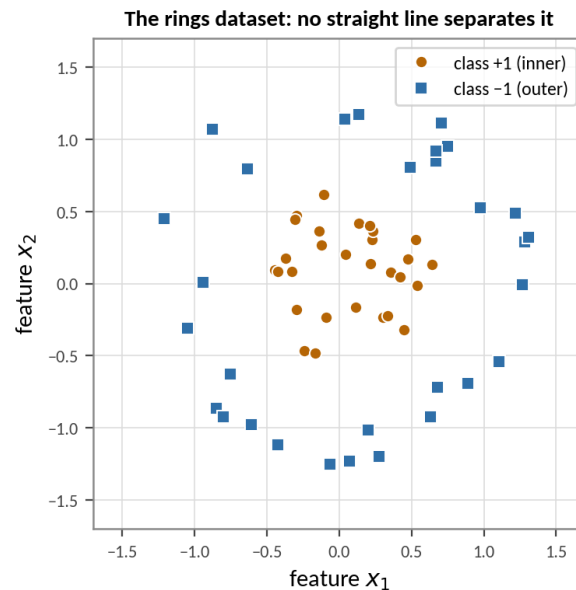
In words: Average, over the training examples, of (score minus target)² — pushing each score toward $+0.9$ for class $+1$ and -0.9 for class -1

Symbol	Meaning	Unit
L	the loss: 0 would mean every score sits exactly on its target	none
x_i, y_i	training example i : features and true label	—
0.9	a softened target: asking for ± 1 exactly would demand infinite confidence at the boundary of reachable scores	none

The loop is Chapter 20's, verbatim: quantum circuit scores the batch, classical optimizer proposes new dials, repeat. Gradients come either from finite differences or — more elegantly — from the parameter-shift rule of Chapter 25.

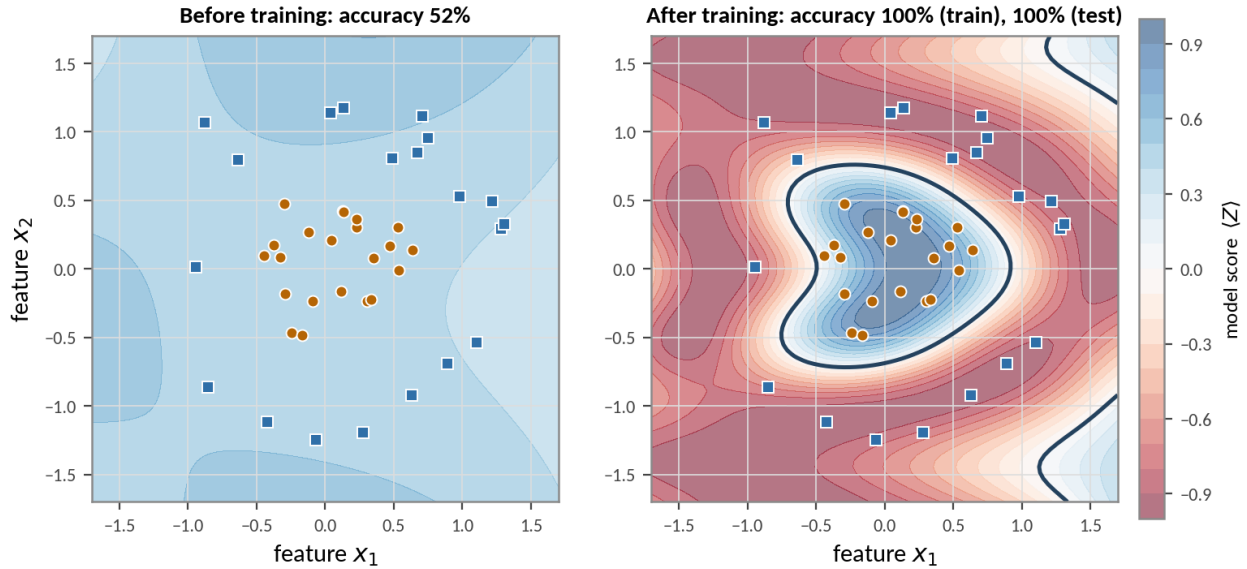
23.4 A Complete, Real Training Run

Everything in this lesson is a genuine computation — the shipped script `fig_defs_ml.py` reproduces every number and figure. The dataset: 60 points in two features, an inner disk (class +1) surrounded by a ring (class -1), split 42 for training and 18 held out for testing.

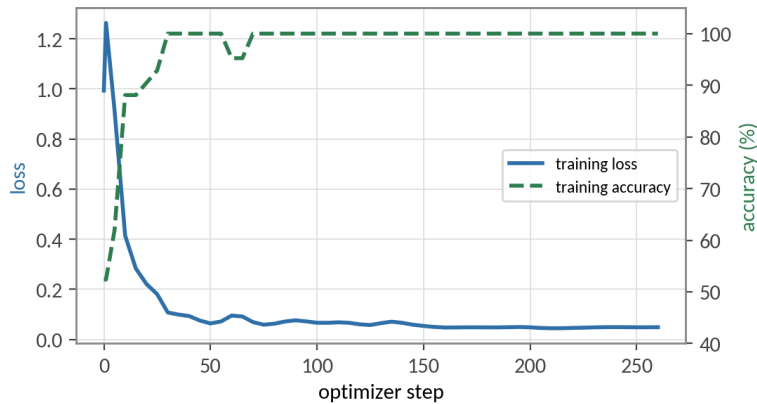


The rings dataset. Radius, not any straight line, separates the classes — the classic test that forces a model to be nonlinear.

The run: dials start random; the optimizer (Adam, a standard adaptive gradient method) takes 260 steps; gradients are computed by parameter shifts. Watch the whole story in two pictures:



The model's score $\langle Z \rangle$ across the whole feature plane, before and after training. Random dials (left): a meaningless tilt, 52% accuracy — coin flipping. Trained dials (right): the blue high-score region has wrapped itself around the inner class, and the dark decision line $s = 0$ encircles it. Accuracy: 100% on training points, and 100% on the 18 test points the model never saw.



The loop at work: loss falls, accuracy climbs to 100% within roughly 60 optimizer steps, and the remaining steps polish confidence. Every point on these curves cost one full sweep of parameter-shift circuit evaluations.

WORKED EXAMPLE — ONE PREDICTION BY HAND, ONE BY MACHINE

Sanity-check the trained machine's logic on the point $x = (0, 0)$ — the dead center, which plainly belongs to the inner class (+1).

1. At $x = (0, 0)$ every encoding rotation collapses to its bias: layer k applies $R_y(b_1^{(k)})$, $R_z(b_2^{(k)})$, $R_y(\theta^{(k)})$ — six fixed y/z -rotations in a row.
2. Composed rotations of one qubit are one net rotation (Chapter 7): the origin is sent to one definite Bloch point, and training has chosen the dials so that point sits far up the northern hemisphere.

3. Northern latitude means $P(0) > P(1)$, so $s = \langle Z \rangle > 0$: predicted label +1 ✓. (The script reports $s(0, 0) = +0.89$ — a confident, correct call.)

4. Contrast a genuine outer-ring point like $(-1.2, 0.3)$: its encoded rotations land the arrow deep in the southern hemisphere; the script reports $s = -0.95$: label -1 ✓.

Answer: the trained function routes inner points north and outer points south of the Bloch equator — the “decision boundary” in feature space is the equator’s shadow. Fifteen numbers, one qubit, a fully nonlinear classifier: quantum machine learning at its most honest scale.

COMMON MISTAKE

X Wrong: “100% accuracy — so quantum classifiers beat classical machine learning.”

✓ Right: A classical network with fifteen parameters ALSO solves the rings dataset perfectly — so does the one-line feature trick $x_1^2 + x_2^2$. This demonstration establishes that quantum circuits can *learn* — mechanism, not advantage. Whether quantum models beat strong classical baselines on problems anyone cares about is precisely the open question Chapter 27 treats with the respect open questions deserve. Never let a toy victory write a headline.

23.5 Honest Evaluation

The habits that keep machine-learning claims trustworthy transfer to quantum models unchanged — and this book asks you to treat them as non-negotiable as normalization.

Hold out a test set. Dials are fitted to training data; only performance on *unseen* points measures learning rather than memorizing. Our 42/18 split is the miniature of the universal practice. **Fear the gap.** Training accuracy far above test accuracy is **overfitting** — the model memorized noise. Cures mirror classical ones: fewer layers, more data, earlier stopping. **Mind the shots.** On hardware, each score carries $\pm 1/\sqrt{N_{\text{shots}}}$ noise (Chapter 8); scores near zero flip labels under noise, so hardware runs of this chapter’s model would want thousands of shots per prediction near the boundary. **Always run the baseline.** A quantum result without a tuned classical comparison at equal parameter count is an anecdote, not an experiment. **And trust no extrapolation.** Every model — quantum or classical — is only calibrated where its training data lives; probe this chapter’s machine at a far corner like $(1.5, 1.5)$, outside both rings, and its score $(+0.15)$, per the script) is a low-confidence guess about a region it has never seen. Deployed models must either see training data covering their whole input region or refuse to answer outside it.

WORD TO KNOW

Overfitting — Fitting the training examples so closely that performance on new data degrades — memorization masquerading as learning. Diagnosed by the train-versus-test gap; treated with simpler models, more data, or earlier stopping.

REALITY CHECK

Where does this scale? Adding qubits and entangling gates grows the model family fast — and awakens the two dragons of Chapter 25: barren plateaus (gradients vanishing exponentially in deep random circuits) and the shot tax (every gradient component bought at $1/\epsilon^2$ shots). Meanwhile classical hardware trains billion-parameter networks between your two coffee sips. The near-term frontier is therefore not “replace deep learning” but the sharper questions: which *structured* circuits train well at scale, and on which data — especially quantum-born data — do they see things classical models cannot? Hold both the working demo and the open frontier in mind at once; that double vision is this part’s whole discipline.

CASE STUDY — A CIRCUIT LEARNS ON REAL HARDWARE • IBM RESEARCH, NATURE, 2019

The experiment that moved variational classification from whiteboards to refrigerators was run by Havlíček and colleagues at IBM (published in *Nature*, 2019). On a five-qubit superconducting chip, they trained exactly this chapter’s species of model — entangling feature map, shallow trainable layers, measured score, hybrid loop — on small synthetic datasets, achieving clean classification despite 2019-era noise, and introduced alongside it the quantum-kernel method starring in the next chapter. The datasets were deliberately artificial, chosen to suit the feature map — the authors said so plainly, modeling the honesty this book keeps preaching. The paper’s real cargo was existence and method: learning loops genuinely close around quantum processors. Its thousand successors — and this chapter’s rings demo on your own laptop — all run in the groove it cut.

ACTIVITY (APP) — TRAIN IT YOURSELF

1. Open the *Classifier Lab* in the QCFE app and load the rings dataset. Inspect the untrained boundary — a useless tilt, like the book’s left panel.
2. Press “train” and watch boundary, loss, and accuracy evolve live; pause around step 60 to catch the moment the boundary closes around the inner class.
3. Drag three new points onto the map — one deep inside, one far outside, one right on the boundary — and read their scores. Relate magnitude to confidence.
4. Turn on shot noise (500 shots per score) and re-classify your boundary point ten times. How often does its label flip? Which lesson of this chapter did you just experience?
5. Switch the dataset to “stripes” and retrain. Same fifteen dials, different function — that flexibility is the whole point of trainable circuits.

You need: The QCFE app (Android or Windows)

► *App tool: Classifier Lab — live decision-boundary training on toy datasets, with shot-noise and layer-count controls*

23.6 Decision Boundaries You Can Draw by Hand

A classifier's soul is its **decision boundary** — the line where its score crosses zero. For one qubit and one feature the whole story fits on paper, and drawing it by hand vaccinates you against treating trained models as unreadable oracles.

Take the tiny model: encode feature $x \in [0, 1]$ as a rotation $R_y(180^\circ x)$ on $|0\rangle$ and score with $s(x) = \langle Z \rangle$.

EQUATION: SCORE AND BOUNDARY OF THE ONE-QUBIT MODEL

$$s(x) = \cos(180^\circ x), \quad \text{boundary where } s(x) = 0$$

In words: The score is a cosine of the encoding angle; the class flips where the cosine crosses zero

Symbol	Meaning	Unit
x	the (scaled) input feature	none
$s(x)$	the score $\langle Z \rangle$, between -1 and $+1$	none
boundary	input(s) where the score changes sign	none

Trainable weights simply reshape the cosine: $R_y(w 180^\circ x + b)$ gives $s = \cos(w 180^\circ x + b)$ — w stretches the wave, b slides it. Training is sliding and stretching until the sign pattern matches the labels.

WORKED EXAMPLE — ONE MODEL, READ END TO END

For $s(x) = \cos(180^\circ x)$: score the points $x = 0.25$ and $x = 0.75$, find the boundary, and describe the classifier in plain words.

1. $s(0.25) = \cos 45^\circ = +0.707$: confidently class +.
2. $s(0.75) = \cos 135^\circ = -0.707$: confidently class −.
3. Boundary: $\cos(180^\circ x) = 0$ at $180^\circ x = 90^\circ$, i.e. $x = 0.5$ — a single threshold, dead center.

Answer: this qubit is a threshold detector: “below one half $\Rightarrow$ class +, above $\Rightarrow$ class −,” with confidence falling smoothly to zero at the threshold. Every giant neural classifier is philosophically this picture — iterated, stacked, and no longer drawable by hand. Here you can still see the whole machine at once.

WORD TO KNOW

Decision boundary — The set of inputs where a classifier's score crosses zero, separating the regions it labels differently. For quantum classifiers the boundary's possible shapes are set by the encoding's frequencies (Chapter 25).

REALITY CHECK

The one-boundary cosine also shows the model's limits: a single R_y layer cannot separate a label pattern like $-, +, -$ along one feature (two boundaries needed), no matter how training tunes w and b within one

half-period. Cures: re-upload the feature (more wiggles per Chapter 25), or add qubits and entanglement. Knowing *why* a model must fail before you train it is a superpower this tiny example teaches better than any large one.

23.7 From Scores to Decisions: Thresholds and Imbalance

The quantum circuit outputs a score $s(x) \in [-1, +1]$; reality demands a decision. The bridge — *where to cut* — is classical decision theory, and skipping it is how technically sound models fail their users. The default cut at $s = 0$ silently assumes both classes are equally common and both mistakes equally costly. Fraud detection, medical screening, and defect inspection violate both assumptions before breakfast.

WORKED EXAMPLE — THE ACCURACY TRAP, DEFUSED

A dataset has 1% fraud (positive) and 99% honest transactions. Model A ignores its input and declares “honest” always. Model B, a trained classifier cut at $s = 0$, catches 80% of fraud while flagging 5% of honest traffic. Score both on accuracy, then judge properly.

1. Model A's accuracy: 99% — spectacular, useless: it catches nothing.
2. Model B's accuracy: $0.99 \times 0.95 + 0.01 \times 0.80 = 0.9485$ — *lower* than the useless model. Accuracy has voted for the fraudster.
3. Judge with per-class rates instead: recall (fraud caught) = 80%; false-alarm rate = 5%. Model B does real work; Model A scores 0% recall.
4. Tune the cut to the costs: if missing a fraud costs $100\times$ a false alarm, lower the threshold (say, flag when $s < 0.3$ instead of $s < 0$): recall rises, false alarms rise, and the *cost-weighted* sum falls. The dial belongs to the deployment, not the training.

Answer: accuracy 99% vs 94.85% — and B is enormously better. On imbalanced problems, quote recall and false-alarm rate (or precision), never bare accuracy; and treat the threshold as a business decision made *after* training. Auditors of Chapter 27: add “accuracy on imbalanced data” to your red-flag list.

WORD TO KNOW

Recall and precision — Recall: the fraction of true positives the model catches. Precision: the fraction of its alarms that are real. The pair replaces accuracy whenever classes are rare or mistakes asymmetric — which is most deployed reality.

REALITY CHECK

The quantum twist on thresholds: near the cut, scores need *shots*. A hardware estimate of $s(x)$ carries wobble $\approx 1/\sqrt{n}$ (Chapter 23's boundary lesson), so a transaction scoring $s = 0.05$ against a threshold of 0 needs hundreds of shots before the decision is statistics rather than coin-flipping — while a confident

$s = 0.9$ needs a dozen. Adaptive shot allocation — spend measurements where the decision is close — is the quantum-native version of a very old idea: think longest about the borderline cases.

CHAPTER SUMMARY

- Supervised learning: models with dials, fitted on labeled examples, judged on unseen ones. A VQC is Chapter 20's loop pointed at labels: encode $\rightarrow$ trainable layers $\rightarrow$ measured score $s = \langle Z \rangle \rightarrow$ predict $\text{sign}(s)$.
- Re-uploading the data between trainable layers stacks trigonometric harmonics — the expressive power that lets even one qubit bend its boundary around a circle.
- Training minimizes the squared loss toward softened targets ± 0.9 ; gradients by parameter shift; optimizer classical (Adam here).
- The real run: rings dataset, 42/18 split, 15 dials, 260 steps — from 52% (random) to 100% train AND 100% test accuracy, fully reproducible from the shipped script.
- Honesty kit: hold-out testing, overfitting vigilance, shot-noise awareness near the boundary, and tuned classical baselines — a toy win demonstrates mechanism, never advantage.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] Name the three stages of a VQC and the chapter that supplied each.

Problem 2 [Basic] A model outputs scores $+0.72$, -0.05 , -0.61 for three points. Give the predicted labels and rank the predictions by confidence.

Problem 3 [Basic] With targets ± 0.9 : compute the squared-loss contribution of a point with label $+1$ and score $+0.5$; then with score -0.5 . Which error costs more, and by what factor?

Problem 4 [Basic] The chapter's model has 3 layers with 5 dials each. A 5-layer version of the same design has how many dials?

Problem 5 [Basic] For $s(x) = \cos(180^\circ x)$, score $x = 0.1$ and $x = 0.9$ and give each point's class and confidence.

Problem 6 [Basic] Slide the wave: for $s(x) = \cos(180^\circ x - 45^\circ)$, find the new boundary. Which direction did the -45° move it, and by how much?

Problem 7 [Basic] A screening model on 2% disease prevalence catches 90% of cases with a 8% false-alarm rate. Compute its accuracy, and the accuracy of the do-nothing model. Which numbers should the clinic quote?

Problem 8 [Basic] Define recall and precision for the fraud Model B (80% catch, 5% false alarms, 1% prevalence): compute both. (Precision needs Bayes-style counting: of 1000 transactions, how many alarms, how many real?)

Problem 9 [Standard] Why must the data be re-encoded in every layer rather than once at the start? Answer in three sentences using the harmonic/Fourier picture of Chapter 22.

Problem 10 [Standard] A run reports 98% training accuracy and 71% test accuracy. Diagnose, name the phenomenon, and prescribe two treatments from this chapter.

Problem 11 [Standard] Shot arithmetic at the boundary: a prediction's true score is $s = +0.06$ and hardware estimates it with 400 shots (noise $\approx 1/\sqrt{400} = 0.05$). Is a wrong-sign estimate plausible? Roughly how many shots would make the noise bar three times smaller than the score?

Problem 12 [Standard] Design the baseline: specify, in four sentences, the classical comparison you would run against this chapter's rings result to make the comparison fair (model size, tuning budget, same split, same metric) — and predict its outcome honestly.

Problem 13 [Standard] At $x = (0, 0)$ the encoding rotations reduce to biases (worked example). What dataset property would make the ORIGIN a bad diagnostic point for this check, and which alternative probe point would you use instead?

Problem 14 [Standard] Stretch the wave: for $s(x) = \cos(360^\circ x)$ on $x \in [0, 1]$, find *all* boundaries and the resulting class pattern. What label sequence along x can this model fit that the 180° model cannot?

Problem 15 [Standard] A trained model reports $w = 1.5$, $b = -60^\circ$ in $s = \cos(w 180^\circ x + b)$. Find its boundary locations in $[0, 1]$ and sketch the sign pattern.

Problem 16 [Standard] Show that with hard thresholding at score zero, the model's confidence $|s(x)|$ near a boundary shrinks linearly, and explain (shot noise $\sigma \approx 1/\sqrt{n}$, Chapter 4) why points near boundaries need more shots to classify reliably.

Problem 17 [Standard] Moving the threshold from $s = 0$ to $s = 0.3$ raised recall from 80% to 92% and false alarms from 5% to 11%. At cost ratio 100:1 (miss : false alarm) and 1% prevalence, compute the expected cost per 1000 transactions before and after, and pick the better cut.

Problem 18 [Challenge] One-layer ceiling: with a single layer, the model's score along the x_1 -axis has the form $s = A \cos(w_1 x_1 + c) + B$ (one harmonic). Argue in a short paragraph why no dial setting of this form can score + inside the ring and – on BOTH sides outside it along a line through the center — and why adding a second re-uploading layer removes the obstruction. (This is the expressivity argument behind the three-layer choice.)

Problem 19 [Challenge] Multi-class by tournament: extend the ± 1 machine to classify three classes using three one-vs-rest classifiers. Specify the training labels each sub-model sees, the prediction rule at inference

(argmax of scores), and one failure mode when two scores tie — then propose the tie-breaker.

Problem 20 [Challenge] Hardware pilot plan: write a six-sentence protocol for running this chapter's trained model (dials fixed) as an *inference-only* service on a real cloud device: circuit per prediction, shots per prediction (justify with Problem 7's arithmetic), expected corrupted-shot correction (Chapter 28), throughput per hour at 10,000 shots/second, and the two numbers you would report to certify the service's accuracy honestly.

Problem 21 [Challenge] Prove the counting rule for this family: $s(x) = \cos(w 180^\circ x + b)$ with $|w| \leq L$ has at most L boundaries in $[0, 1]$, and connect it to the frequency budget of re-uploading circuits in Chapter 25 — what does adding one re-upload layer buy, in boundary count?

Problem 22 [Challenge] Design the shot-allocation policy: scores arrive with wobble $0.5/\sqrt{n}$; decisions at threshold 0 must be wrong (statistically) less than 1% of the time. Give the shot rule as a function of the first-look score from 50 shots, and estimate the average shots per transaction if first-look scores are uniform on $[-1, 1]$.

Quantum Kernels

Similarity as an inner product — measuring overlaps and letting classical mathematics do the learning

IN THIS CHAPTER YOU WILL LEARN TO

- Explain the kernel idea: learning from pairwise similarities instead of raw features.
- Define the quantum kernel $K(x, y) = |\langle \phi(x) | \phi(y) \rangle|^2$ and compute small cases by hand.
- Measure a kernel on hardware with the overlap circuit, and budget its shots.
- Assemble and read a kernel matrix, and use it to classify — the division of labor between quantum similarity and classical optimization.
- State where quantum kernels could genuinely win, and the honest caveats attached.

24.1 The Similarity Trick

Chapter 23 trained the quantum circuit itself. This chapter takes the other royal road of machine learning: leave the heavy optimization to trusted *classical* algorithms, and use the quantum computer for exactly one job — **judging similarity**.

The insight, one of classical ML's crown jewels (the “kernel trick” behind support-vector machines): many learning algorithms never need the feature vectors themselves — only the table of *pairwise similarities* between training points. Hand such an algorithm a good similarity function $K(x, y)$ and it builds a classifier as a weighted vote of similarities to memorized examples: new point close to inner-ring examples → inner ring. The similarity function carries all the geometry; the classical optimizer (an SVM solver — polished, convex, reliable since the 1990s) carries all the learning.

WORD TO KNOW

Kernel — A similarity function $K(x, y)$ between data points, equal to an inner product of the points' images in some feature space. Kernel methods learn using only the table of pairwise similarities — never touching the feature space directly.

The quantum opening is immediate. Chapter 22 maps data into quantum states — a feature space of exponential dimension. Inner products of states are exactly what quantum mechanics computes natively (Appendix B; Chapter 4). So: let the encoding be the feature map, let the measured overlap be the similarity, and plug the resulting table into a stock classical SVM. No trainable quantum dials, no barren plateaus — the quantum processor is a *similarity meter*.

24.2 The Quantum Kernel

EQUATION: THE QUANTUM KERNEL

$$K(x, y) = |\langle \phi(x) | \phi(y) \rangle|^2$$

In words: Similarity of two data points = squared overlap of their encoded states

Symbol	Meaning	Unit
$ \phi(x)\rangle$	the encoded state of point x (Chapter 22's feature map)	none
$K(x, y)$	the kernel value	between 0 and 1

Landmarks: $K(x, x) = 1$ (everything is fully similar to itself); $K = 0$ for orthogonally encoded points; and K is symmetric. For this book's one-qubit angle map, geometry gives a closed form: $K = \cos^2$ (half the Bloch angle between the two encoded arrows).

WORKED EXAMPLE — THREE KERNEL VALUES BY HAND

Use the one-feature map $|\phi(x)\rangle = R_y(x) |0\rangle$. Compute $K(0^\circ, 0^\circ)$, $K(0^\circ, 90^\circ)$, and $K(0^\circ, 180^\circ)$.

1. Encode: $\phi(0^\circ) = |0\rangle$; $\phi(90^\circ) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$; $\phi(180^\circ) = |1\rangle$.
2. $K(0^\circ, 0^\circ) = |\langle 0|0\rangle|^2 = 1$.
3. $K(0^\circ, 90^\circ) = |\langle 0|+\rangle|^2 = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$ (Chapter 3's overlap, recycled).
4. $K(0^\circ, 180^\circ) = |\langle 0|1\rangle|^2 = 0$.

Answer: $1, \frac{1}{2}, 0$ — similarity decaying smoothly as the encoded arrows swing apart, reaching zero at opposite poles. The formula $\cos^2\left(\frac{\Delta x}{2}\right)$ reproduces all three: check $\cos^2 0 = 1$, $\cos^2 45^\circ = \frac{1}{2}$, $\cos^2 90^\circ = 0$ ✓

24.3 Measuring a Kernel on Hardware

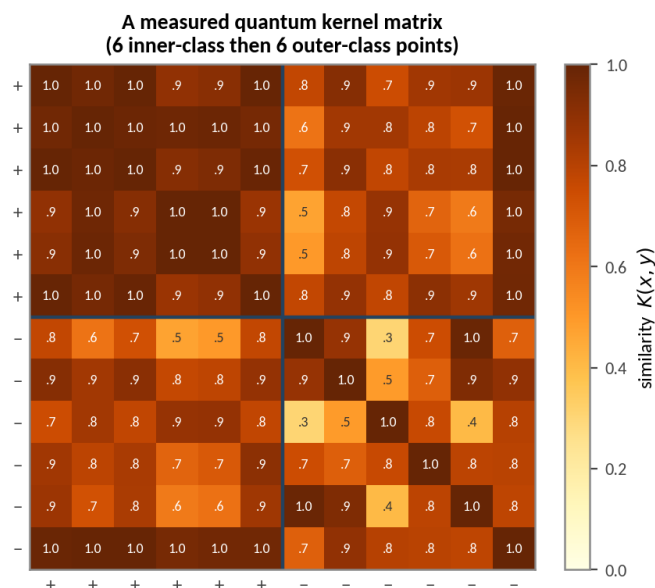
On paper, overlaps are inner products. On hardware, amplitudes are unreadable (Chapter 4) — yet the *squared* overlap has a beautifully direct meter. Encoding circuits are unitaries: $|\phi(x)\rangle = U(x) |0\rangle$. Run $U(y)$'s inverse after $U(x)$ and measure:

$$|0\rangle \longrightarrow \boxed{U(x)} \longrightarrow \boxed{U(y)^\dagger} \longrightarrow \boxed{\text{meter}} \quad P(\text{result } 0) = K(x, y)$$

The overlap test. Encode x , un-encode y , and ask: did we land back on $|0\rangle$? The probability of “yes” is exactly the kernel value. (Inverses are free in quantum computing: run the gates backward, daggered — Appendix B.)

Why it works, in two lines of Chapter-4 algebra: the final state is $U(y)^\dagger U(x) |0\rangle$, and the Born probability of reading 0 is $|\langle 0 | U(y)^\dagger U(x) | 0 \rangle|^2 = |\langle \phi(y) | \phi(x) \rangle|^2 = K(x, y)$. Estimating each entry to precision ε costs the standard $1/\varepsilon^2$ shots (Chapter 8) — and a training set of m points needs $m(m-1)/2$ distinct pairs: the kernel's shot bill scales with the *square* of the dataset, a number worth computing before promising anyone a large-scale run (Problem 9 makes you do exactly that).

24.4 A Working Kernel Machine



A real computed kernel matrix for twelve points of Chapter 23's rings dataset (six inner-class, then six outer-class), using the two-feature angle map. The block structure is the learning signal: within-class pairs (diagonal blocks) skew similar; cross-class pairs (off-diagonal blocks) skew dissimilar.

WORKED EXAMPLE — CLASSIFYING WITH THE MATRIX ALONE

Using only similarity votes — no features — classify a new point z whose measured kernel values against the twelve stored examples average 0.81 to the six inner-class points and 0.44 to the six outer-class points.

1. The simplest kernel classifier (a nearest-class-mean vote): $\text{score} = \bar{K}_{\text{inner}} - \bar{K}_{\text{outer}} = 0.81 - 0.44 = +0.37$.
2. Positive score $\Rightarrow$ the new point resembles the inner class more $\Rightarrow$ predict +1.
3. A full SVM refines exactly this vote: it learns *weights* for each stored example (boundary examples matter more), but consumes the same table of measured similarities and nothing else.

Answer: class +1, by similarity vote +0.37. Notice what just happened: the quantum device only ever answered “how alike are these two points?”, and a completely classical rule did the classifying. Convex, no local minima, no plateau — the whole training difficulty of Chapter 23 traded for a measurement budget.

COMMON MISTAKE

X Wrong: “Kernel methods avoid quantum noise — the classical SVM does the learning.”

✓ Right: The SVM is noiseless, but every kernel *entry* it consumes is a hardware estimate with shot noise and device errors (Chapter 28), and SVM solutions can be sensitive to a noisy matrix (it may even lose the mathematical property — positive semidefiniteness — solvers expect, requiring a repair step). Error bars do not disappear by changing method; they move from the gradient loop into the similarity table.

24.5 When Quantum Kernels Could Win — and the Fine Print

A kernel method is only as good as the geometry its feature map induces. So the decisive question is: *are there useful similarity functions that quantum circuits compute efficiently and classical computers cannot?* Three honest findings frame the answer.

The existence proof. Yes, such kernels exist. A 2021 result (Liu, Arunachalam, Temme) constructed a classification problem — built on the discrete-logarithm problem, cousin of Chapter 19’s factoring — where a quantum kernel provably classifies efficiently while *any* classical learner needs superpolynomial time (assuming the same hardness that protects classical cryptography). Advantage is possible in principle; the field is not chasing a phantom.

The relevance gap. That constructed problem is cryptographic, not commercial. For natural datasets — images, medical records, sensor logs — no quantum kernel has yet beaten well-tuned classical kernels. Simple angle maps like this chapter’s induce similarities that classical formulas reproduce exactly (you derived one: $\cos^2(\Delta x/2)$); genuinely hard-to-simulate feature maps (deep, entangling — the ZZ family of Chapter 22, Problem 10) may help or may drown structure in dimensionality — an effect (“exponential concentration”) where all similarities crowd toward the same value and the matrix carries no signal.

The design frontier. The productive research stance treats the feature map as a hypothesis about the data: match the encoding’s structure to the problem’s structure (symmetries, physics, periodicities), test against strong baselines, and report shot budgets. Quantum kernels are a genuinely new family of similarity geometries — the open question is which real data speaks their language.

CASE STUDY — SIMILARITY, MEASURED • SUPERVISED LEARNING WITH QUANTUM-ENHANCED FEATURE SPACES, 2019

The same IBM *Nature* paper that launched hardware variational classification (Chapter 23’s case study) carried a second protocol that named this chapter’s field: estimate kernels on the processor, learn with a classical SVM. Havlíček and colleagues measured overlap circuits pair by pair on five superconducting qubits, assembled the matrix, solved the SVM classically, and classified their test data cleanly — while being explicit that the dataset was tailored to the feature map. The paper crystallized the division of labor (quantum similarity, classical optimization), set off the theory race that produced both the provable-advantage construction and the concentration warnings of Lesson 5, and made “quantum kernel” a standard entry in the machine-learning lexicon. Its method section is, almost line for line, the workflow you can now execute on twelve points with the mathematics of Chapters 3, 4, and 8.

ACTIVITY (APP) — BE THE SIMILARITY METER

1. Open the *Kernel Lab* in the QCFE app. Place two points on the feature plane and read the app's $K(x, y)$; drag one point around and watch similarity fall as the encoded arrows separate.
2. Verify the worked example: set the points to 0° and 90° (one feature) and confirm $K = 0.50$.
3. Load the twelve-point rings sample and press “build matrix”: watch the heatmap fill pair by pair, then compare with the book's figure.
4. Switch on shot noise at 200 shots per entry and rebuild: the blocks blur. Raise to 5000 and watch them sharpen — the mistake box, live.
5. Press “classify” to run the SVM on your measured matrix, then drop a fresh point and watch its similarity votes decide its class.

You need: The QCFE app (Android or Windows)

► App tool: *Kernel Lab* — pairwise overlap meter, live kernel-matrix builder with shot noise, and a one-press SVM on top

24.6 The Kernel Matrix Health Check

Before feeding a quantum kernel to a classifier, a professional *looks* at the matrix — five minutes of inspection that catches most broken feature maps. For the one-feature map $|\phi(x)\rangle = R_y(x)|0\rangle$, the kernel has a closed form you can check by hand:

EQUATION: THE ONE-FEATURE KERNEL, EXACTLY

$$K(x, y) = |\langle \phi(x) | \phi(y) \rangle|^2 = \cos^2\left(\frac{x-y}{2}\right)$$

In words: Similarity depends only on the feature gap, falling from 1 (identical) toward 0 (opposite poles) as the gap grows

Symbol	Meaning	Unit
x, y	two data points' (scaled) features	rad
K	their kernel similarity, 0 to 1	none
$(x - y)/2$	half the feature gap = half the Bloch angle	rad

Check the extremes: $K(x, x) = 1$ always; $K = 0$ exactly when the two encoded states are perpendicular ($|x - y| = \pi$). And note the trap: $K(x, x + 2\pi) = 1$ — the circle wraps, and badly scaled features alias like wheels in old westerns.

WORKED EXAMPLE — THREE SIMILARITIES BY HAND

With features already scaled to radians, compute K for the gaps $\Delta = 0$, $\Delta = 1.6$, and $\Delta = 2\pi$.

1. $\Delta = 0$: $K = \cos^2 0 = 1$ — identical points, perfect similarity.

2. $\Delta = 1.6$: $K = \cos^2(0.8) = (0.697)^2 \approx 0.485$ — half-similar, healthily in between.

3. $\Delta = 2\pi$: $K = \cos^2(\pi) = 1$ — maximal similarity for very *different* features: the wrap-around trap, sprung.

Answer: 1, ≈ 0.49 , and a treacherous 1. **Rule of practice:** scale features so the data's full range spans well under 2π (a common choice: into $[0, \pi]$), so distinct points can never masquerade as identical.

WORD TO KNOW

Kernel concentration — The failure mode of over-rich feature maps: as qubits and depth grow, off-diagonal kernel values crowd toward one tiny value, the matrix approaches the identity, and the model memorizes instead of generalizing — the kernel cousin of barren plateaus.

The health check itself — four glances at the computed matrix:

Kernel matrix inspection checklist

Glance	Healthy	Sick (and diagnosis)
Diagonal	exactly 1	anything else: a bug, full stop
Same-class blocks	visibly bright	no structure: map too weak or features unscaled
Off-diagonal spread	broad range of values	all ≈ 1 : encoding barely moves the state
Off-diagonal size	moderate	all ≈ 0 : concentration — simplify the map

REALITY CHECK

Estimating each kernel entry on hardware costs shots: K is a probability, so its estimate wobbles by $\approx \sqrt{K(1-K)/n}$. A 60-point dataset needs $60 \times 59/2 = 1770$ distinct entries; at 1000 shots each, 1.77 million shots *before any learning happens*. Kernel methods buy their elegance (convex training, no plateaus — this chapter's selling points) with a measurement bill that grows quadratically in dataset size — honest accounting required.

24.7 What the Classifier Remembers: Support Vectors

The kernel matrix feeds a classical optimizer (the support-vector machine), and what comes out deserves demystifying: the trained classifier is nothing but a *weighted comparison with remembered training points*.

EQUATION: THE KERNEL CLASSIFIER'S DECISION

$$f(x) = \sum \alpha_i y_i K(x_i, x) + b \quad \text{decide by the sign of } f$$

In words: Score a new point by its ⁱkernel similarity to each remembered training point, weighted by importance and label, plus an offset

Symbol	Meaning	Unit
$\alpha_i \geq 0$	the learned importance of training point i	none
$y_i = \pm 1$	that point's label	none
$K(x_i, x)$	quantum-estimated similarity to the new point	none

Training discovers that most $\alpha_i = 0$: only the points nearest the boundary — the **support vectors** — matter. The model is that short list. Inference on hardware costs one kernel-circuit run per support vector per new point: a countable, billable number.

WORKED EXAMPLE — SCORING A NEW POINT BY HAND

A trained toy model kept two support vectors: $(x_1, y_1 = +1, \alpha_1 = 0.8)$ and $(x_2, y_2 = -1, \alpha_2 = 0.5)$, with $b = -0.1$. For a new point x , the quantum kernel circuits report $K(x_1, x) = 0.9$ and $K(x_2, x) = 0.4$. Classify.

1. Plus-side pull: $0.8 \times (+1) \times 0.9 = +0.72$.
2. Minus-side pull: $0.5 \times (-1) \times 0.4 = -0.20$.
3. Total: $f = 0.72 - 0.20 - 0.1 = +0.42 > 0$: class +1, with a healthy margin.
4. Shot-noise sanity (Chapter 24's bill): each K came from a histogram; at 1000 shots each entry wobbles by ≈ 0.015 , so f wobbles by $\approx 0.8 \times 0.015 + 0.5 \times 0.015 \approx 0.02$ — tiny against the 0.42 margin. Borderline points deserve more shots; this one is safe.

Answer: class +1, $f = 0.42 \pm 0.02$. The entire trained “quantum AI” is two remembered points, three multiplications, and an addition — plus quantum circuits doing the one thing classical hardware cannot: computing the similarities in the quantum feature space. Knowing exactly where the quantum ends and the classical begins is the audit skill of Chapter 27, made concrete.

WORD TO KNOW

Support vector — A training point with nonzero learned weight — one the decision boundary actually leans on. Sparse support = a compact, cheap-to-deploy model; every-point-a-support = memorization worth investigating.

REALITY CHECK

The support count is a deployment invoice: classifying one transaction costs (support vectors) $\times$ (shots per kernel estimate) circuit runs. Thirty supports at 500 shots = 15,000 runs *per decision* — fine for screening rare events, absurd for a packet filter. When a QML vendor quotes training accuracy, the operations question is always inference cost; now you can compute it from two numbers they must provide.

CHAPTER SUMMARY

- Kernel methods learn from pairwise similarities only; a trusted classical SVM does the optimizing — convex, plateau-free.
- The quantum kernel is the squared overlap of encoded states: $K = |\langle \phi(x) | \phi(y) \rangle|^2$; for the angle map it reduces to $\cos^2$ of half the Bloch separation.
- Hardware meters it with the overlap circuit — $U(x)$ then $U(y)^\dagger$, probability of all-zeros — at $1/\varepsilon^2$ shots per entry and $m(m-1)/2$ entries per dataset.
- The measured matrix's block structure is the signal; a similarity vote (or full SVM) turns it into predictions — quantum similarity, classical learning.
- Status, honestly: provable advantage exists on cryptography-flavored constructions; natural-data wins remain open; deep feature maps risk exponential concentration; the craft is matching encodings to data structure, against real baselines.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] State the two properties $K(x, x) = 1$ and $K(x, y) = K(y, x)$ in plain words, and verify both from the definition.

Problem 2 [Basic] With the one-feature map $R_y(x) |0\rangle$: compute $K(30^\circ, 90^\circ)$ using the $\cos^2(\Delta x/2)$ formula.

Problem 3 [Basic] In the overlap circuit, what measured event's probability equals the kernel value, and why must the second unitary be daggered?

Problem 4 [Basic] A 20-point training set: how many distinct kernel pairs must be measured?

Problem 5 [Basic] Compute K for feature gaps $\Delta = \pi/2$ and $\Delta = \pi$. What geometric relationship between the two encoded states does $K = 0$ signal?

Problem 6 [Basic] A computed kernel matrix has diagonal entries 0.97–0.99. Diagnose using the checklist, and name the most likely software cause on hardware (hint: what error type taxes every estimate, per Chapter 28?).

Problem 7 [Basic] Re-score the worked example's point if $K(x_2, x)$ were 0.95 instead of 0.4. Class and margin?

Problem 8 [Basic] Why do points far from the boundary end with $\alpha = 0$, intuitively? What would it suggest if training returned nonzero α for every one of 200 training points (Chapter 24's concentration lesson may help)?

Problem 9 [Standard] Compute K by explicit inner product for the two-feature map $R_z(x_2)R_y(x_1)|0\rangle$ between $a = (90^\circ, 0^\circ)$ and $b = (90^\circ, 90^\circ)$ — same latitude, quarter-turn phase apart. (Answer first with the Bloch picture: what is the angle between two equator arrows 90° of longitude apart, and what does $\cos^2$ of half of it give?)

Problem 10 [Standard] Shot budget: each entry to precision 0.02 costs how many shots? For the 20-point set of Problem 4, what is the total, and the wall-clock time at 10^4 shots per second?

Problem 11 [Standard] A measured matrix shows within-class averages 0.55 and cross-class averages 0.53. Diagnose in two sentences (name the Lesson-5 phenomenon this resembles) and propose two changes to the feature map worth testing.

Problem 12 [Standard] Similarity vote: a new point averages $K = 0.62$ to class-A examples and 0.71 to class-B examples. Predict, give the vote margin, and state one reason a full SVM might still disagree with the plain vote.

Problem 13 [Standard] Kernel versus VQC bookkeeping: for a 100-point training set, compare what each method asks of the quantum computer — number of distinct circuits, presence or absence of a training loop, and where each pays its shot bill. Two sentences per method.

Problem 14 [Standard] Features in $[0, 100]$ are encoded without scaling (x used directly as radians). Using the wrap-around trap, exhibit two feature values 6.28 apart that the kernel calls identical, and give the scaling recipe that prevents this.

Problem 15 [Standard] Count the kernel entries and total shots needed for a 200-point dataset at 500 shots per entry (symmetric matrix, unit diagonal free). Compare against the 60-point figure and state the scaling law.

Problem 16 [Standard] For the two-feature map $R_z(x_2)R_y(x_1)|0\rangle$ of this chapter, the nearby points $(0.2, 0.1)$ and $(0.3, 0.4)$ give $K \approx 0.996$ while the distant pair $(0.2, 0.1)$ and $(1.9, 1.4)$ gives $K \approx 0.367$. Are these values healthy by the checklist? What would $K \approx 0.001$ everywhere off-diagonal have suggested instead?

Problem 17 [Standard] A deployed model has 25 support vectors and needs decision wobble on f below 0.02. With weights summing to $\sum \alpha_i = 4$ and per-entry wobble $\approx 0.5\sqrt{K(1-K)}/\sqrt{n}$ bounded by $0.5/\sqrt{n}$, derive a sufficient per-entry shot count, and the total circuit runs per decision.

Problem 18 [Challenge] Prove the overlap circuit's correctness in general: show $P(0 \cdots 0) = |\langle 0|U(y)^\dagger U(x)|0\rangle|^2 = K(x, y)$ for any number of qubits, citing the Born rule in its inner-product form (Chapter 4, Lesson 5).

Problem 19 [Challenge] Concentration by dimension: suppose a deep feature map scatters encoded states essentially at random over an N -dimensional state space, so typical overlaps obey $K \sim 1/N$. For 10 qubits ($N = 1024$), estimate a typical kernel entry, compare it with shot noise at 10^4 shots, and write the two-sentence conclusion about deep random feature maps.

Problem 20 [Challenge] Design a feature map with a symmetry: your data is known to satisfy “label depends only on $x_1^2 + x_2^2$ ” (like the rings). Propose an encoding whose kernel automatically respects rotational symmetry of the feature plane (hint: encode the radius — or engineer angles so longitude cancels in overlaps), verify its invariance on one example pair, and explain in two sentences why building known symmetries into the map should improve learning from few examples.

Problem 21 [Challenge] Derive the closed form: show $|\langle 0 | R_y(-x) R_y(y) | 0 \rangle|^2 = \cos^2(\frac{y-x}{2})$ using the rotation matrix of Chapter 7, and explain why *any* single-qubit feature map’s kernel depends only on the angle between the two encoded Bloch vectors.

Problem 22 [Challenge] Hand-build the smallest SVM: training points $x = 0.3$ ($y = +1$) and $x = 1.1$ ($y = -1$) with the $\cos^2$ kernel of this chapter, symmetric weights $\alpha_1 = \alpha_2 = \alpha$, and b chosen so $f = 0$ exactly midway at $x = 0.7$. Compute K to each training point from a test point at $x = 0.5$, show the sign of f there for any $\alpha > 0$, and interpret: what did the kernel’s distance-decay just guarantee about points left of center?

Quantum Neural Networks and Training

Layers, exact gradients, barren plateaus — and the quantum data where these networks shine

IN THIS CHAPTER YOU WILL LEARN TO

- Translate the vocabulary of neural networks — layers, weights, activations — into circuit language, and note where the analogy breaks.
- Compute exact gradients with the parameter-shift rule, and verify it by hand on a known curve.
- Explain barren plateaus: why deep random circuits starve their own training signal, and the design responses.
- Describe data re-uploading and expressivity: what makes a circuit family rich enough to learn.
- Identify the arena where quantum networks have the principled edge: data that is born quantum.

25.1 From Classical Networks to Quantum Layers

The deep networks behind modern AI are stacks of simple pieces: layers of weighted sums followed by nonlinear activations, trained end to end by gradient descent on millions of dials. “Quantum neural network” (QNN) is the field’s name for the quantum-native counterpart — and at this point in the book you can define it in one breath: *a parameterized circuit organized into repeating layers, trained by gradients on a measured loss*. Chapter 23’s classifier was already a small QNN; this chapter studies the species properly.

The dictionary — and its honest gaps

Classical networks	Quantum networks	Faithful translation?
weights and biases	rotation angles θ	yes — the trainable dials
layer of neurons	block of rotations + entanglers	yes — the repeating unit
nonlinear activation	none inside (unitaries are linear!) — nonlinearity enters via encoding and measurement	the sharpest break: Chapters 22 and 8 carry what ReLU carries classically
network output	measured expectation values	yes — with shot noise attached
backpropagation	parameter-shift gradients (Lesson 2)	exact but priced per parameter, not per pass
millions of parameters	tens to thousands today	the scale gap — and Lesson 3's plateau polices it

One row deserves a highlight before anything else: *there is no ReLU inside a quantum circuit*. Every gate is unitary — linear on amplitudes (Appendix B) — so all nonlinearity in a QNN's input-output function comes from the two ends: the trigonometric encoding of data (Chapter 22) and the squared-modulus of measurement (Chapter 4). Quantum networks are sandwiches: nonlinear bread, linear filling.

25.2 Gradients: The Parameter-Shift Rule

Classical backpropagation reads gradients from stored intermediate values — but measurement collapse (Chapter 4) forbids peeking inside a quantum computation. The field's elegant workaround gives *exact* derivatives from ordinary runs:

EQUATION: THE PARAMETER-SHIFT RULE

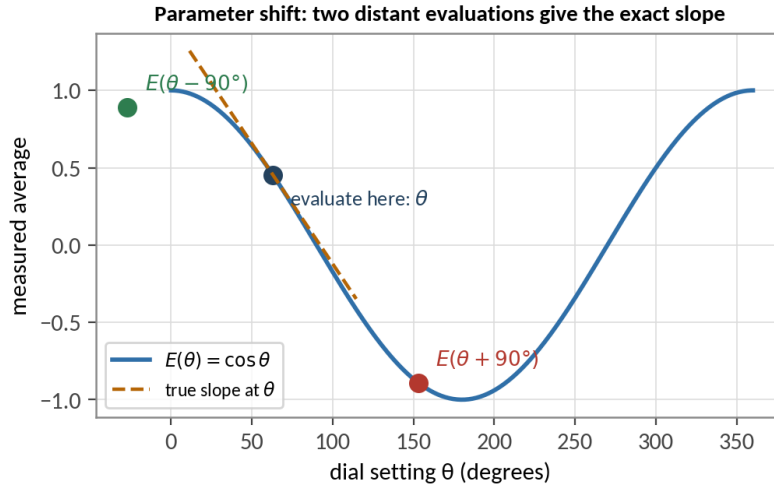
$$\frac{\partial E}{\partial \theta} = \frac{E(\theta + 90^\circ) - E(\theta - 90^\circ)}{2}$$

In words: The exact slope at θ = half the difference of the measured values a quarter-turn above and below θ

Symbol	Meaning	Unit
$E(\theta)$	the measured expectation value as a function of one dial (others held fixed)	none
$\pm 90^\circ$	the two shifted settings — full quarter-turns, not tiny nudges	degrees ($\pi/2$)

Valid for the workhorse gates R_x, R_y, R_z (any gate generated by a Pauli), because their measured averages are always sinusoids in each dial — and for a sinusoid, this two-point formula is not an approximation but an identity.

Cost: two circuit evaluations per parameter per gradient.



No tiny nudges: the rule samples a quarter-turn away on each side, yet returns the exact tangent slope at θ — because expectation curves of Pauli-rotation dials are pure sinusoids.

WORKED EXAMPLE — VERIFYING THE RULE ON A KNOWN CURVE

Chapter 8 showed the $R_y(\theta) |0\rangle$ family has $E(\theta) = \langle Z \rangle = \cos \theta$. Check the parameter-shift rule at $\theta = 60^\circ$.

1. True slope, from the calculus fact slope of $\cos$ is $-\sin$: $-\sin 60^\circ \approx -0.866$. (Take this on trust if calculus is new — the point is the comparison.)
2. Shifted evaluations: $E(150^\circ) = \cos 150^\circ = -0.866$ and $E(-30^\circ) = \cos(-30^\circ) = +0.866$.
3. The rule: $(E(150^\circ) - E(-30^\circ))/2 = (-0.866 - 0.866)/2 = -0.866$.

Answer: -0.866 from the rule, -0.866 from the truth — exact agreement, using evaluations 90° away from the point of interest. On hardware each evaluation carries shot noise, so gradient precision is bought at Chapter 8's $1/\varepsilon^2$ rate — times two circuits, times the number of dials: the full price list of quantum training.

COMMON MISTAKE

X Wrong: “Approximate the gradient with a tiny finite difference, $(E(\theta + 0.001) - E(\theta))/0.001$ — standard numerics.”

✓ Right: On hardware this is the worst of both worlds: the true difference between two settings 0.001 apart is microscopic, while each estimate carries shot noise $\sim 1/\sqrt{N}$ — the noise *divided by* 0.001 explodes. The parameter shift’s genius is using *large* separations (where signals differ by order one) while remaining exact. Quantum training without parameter shifts is like weighing sand grains on a truck scale.

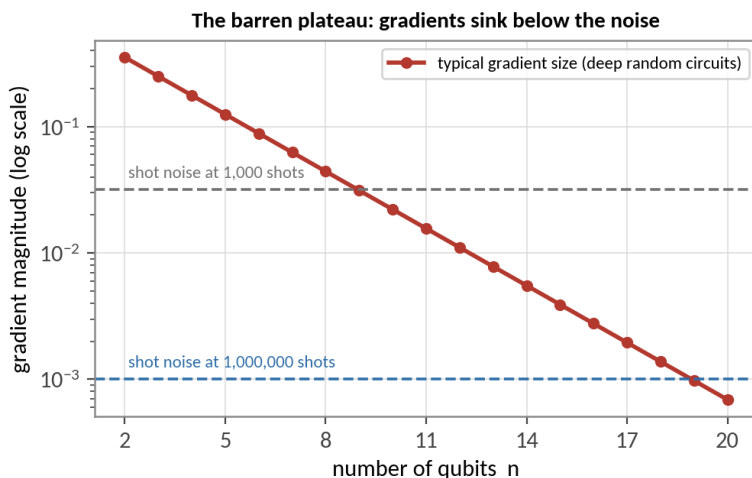
25.3 Barren Plateaus: The Desert of Deep Random Circuits

Scaling QNNs meets a phenomenon with no true classical twin. Make the circuit deep and random and wide, and the loss landscape flattens *exponentially*: gradients of typical parameters shrink like 2^{-n} in the qubit

number n . The intuition follows from this book's own arithmetic: a deep random circuit scrambles the state across all 2^n amplitudes (Chapter 9); any single dial then influences the measured average only through an exponentially diluted share of that space; and detecting a 2^{-n} slope through shot noise $1/\sqrt{N_{\text{shots}}}$ costs $N_{\text{shots}} \sim 4^n$ — the exponential wall of Chapter 1, now billed to the optimizer.

WORD TO KNOW

Barren plateau — A training landscape whose gradients vanish exponentially with system size, leaving the optimizer wandering a nearly flat desert where every direction reads as zero through shot noise. The central scaling obstacle of variational quantum computing, identified by McClean and coworkers in 2018.



The desert forms: typical gradient size in deep random circuits (red) falls below the shot-noise floors (dashed) — already by ~ 10 qubits at 1000 shots, by ~ 20 qubits even at a million.

The response is not despair but *design* — randomness is the culprit, structure the cure. Proven or promising escapes: keep circuits **shallow** relative to qubit count (Chapter 23's three layers); use **local** cost functions (measure few-qubit observables rather than global all-qubit ones — provably milder plateaus); **initialize** near identity or warm-start from smaller trained models rather than randomly; and build **symmetry-matched** circuit families whose parameters all do meaningful, non-diluted work (the geometric QML program). Every serious QNN proposal is now judged first by its answer to the plateau question.

25.4 Expressivity and Data Re-uploading

Between too-simple and too-random lies the working zone. A circuit family's **expressivity** is the richness of input-output functions it can realize — and for the re-uploading models of this part, the picture is beautifully concrete: each encoding pass adds harmonics, so the model's output in each feature is a Fourier series whose depth of repetition sets the number of available frequencies (Chapter 23, Problem 10 made you feel the one-layer ceiling). Expressivity is a budget, not a virtue: too little and the model cannot bend around the data (underfitting); too much and it memorizes noise (overfitting, Chapter 23) — or wanders onto the plateau. The craft of QNN architecture — like all network architecture — is spending the budget where the data's structure

lives.

WORKED EXAMPLE — COUNTING A MODEL'S FREQUENCIES

How many distinct frequencies (harmonics) can Chapter 23's three-layer re-uploading model express in feature x_1 — and what would five layers buy?

1. Each pass encodes x_1 once through a rotation: the state's amplitudes gain factors oscillating at the encoding frequency (scaled by that layer's trainable weight $w^{(k)}$).
2. Measured averages combine amplitudes in conjugate pairs (Chapter 8): passes interfere, and L encoding passes generate output terms at sums and differences of up to L single-pass frequencies — with trainable weights, up to L independent harmonics (plus a constant).
3. So three layers: up to 3 harmonics per feature — enough to carve one closed ring, as the trained boundary figure showed. Five layers: up to 5 — enough for double rings or finer stripes.

Answer: layers = frequency budget: L re-uploads buy up to L harmonics per feature. This single counting rule lets you size a re-uploading model to a dataset the way Chapter 1 sized bit-widths to alphabets — structure first, dials second.

25.5 Quantum Data: Where the Edge Is Principled

Every chapter of this part has repeated one honest refrain: on *classical* data, quantum learners fight uphill — loading costs at the door (Chapter 22), plateaus in the middle, and formidably tuned classical rivals at the finish line. This final lesson states the arena where the fight inverts.

Some data is *born quantum*: output states of quantum simulations and chemistry (Chapter 20), signals from entangled-sensor networks, states arriving over quantum links (Chapter 15), snapshots of exotic materials. For such data the encoding bottleneck vanishes — there is nothing to load; the state IS the input — while any *classical* learner must first measure the state into classical numbers, and Chapter 4's collapse guarantees that step discards information. Theory has sharpened this intuition into proof: for identifying phases of quantum matter and related tasks, quantum-processing learners provably need *exponentially fewer* experiment repetitions than any measure-first classical pipeline (the “quantum advantage in learning from experiments” results, demonstrated on Google hardware in 2022). Architectures built for this arena — notably the quantum convolutional network (QCNN), a layered pattern of local unitaries and mid-circuit measurements that also provably dodges barren plateaus — read quantum inputs the way image networks read pixels.

REALITY CHECK

So the field's map, drawn without decoration: on everyday classical datasets, QNNs are a research program — mechanisms proven, advantage unproven, plateaus policing scale. On quantum data, the advantage is not marketing but theorem-grade, with hardware demonstrations — yet “quantum data at industrial volume” mostly awaits the mature quantum sensors, simulators, and networks of the coming decades. Part Six's honest summary in one sentence: *quantum machine learning's surest customers are quantum machines themselves* — and the students who understand both sides of that sentence are the ones who will build it.

CASE STUDY — THE PLATEAU PAPER AND THE MAP IT REDREW • McCLEAN ET AL., 2018

In 2018, Jarrod McClean and colleagues at Google published a result nobody in the young variational-circuits community wanted: for wide, deep, randomly initialized circuit families, gradients concentrate exponentially close to zero — training signals invisible beneath any realistic shot budget. The “barren plateaus” paper could have read as an obituary; instead it became the field’s zoning law. Within five years, the literature it provoked mapped which ingredients cause deserts (depth, randomness, global costs, too much entanglement in the wrong places) and which build oases (locality, symmetry, shallow structure, clever initialization, QCNN-style architectures — later *proven* plateau-free). The episode is this book’s favorite specimen of healthy science: a hard negative result, published loudly by the very community it inconvenienced, that turned a naive gold rush into an engineering discipline. When you design your first circuit model, you will consult its descendants before you spend a single shot.

ACTIVITY (APP) — GRADIENTS AND DESERTS

1. In the *QNN Lab* of the QCFE app, load the one-dial demo ($E(\theta) = \cos \theta$). Pick $\theta = 60^\circ$ and press “parameter shift”: the app runs the two shifted circuits and reports the slope. Check against the worked example.
2. Switch on shots (1000) and repeat ten times: watch the gradient estimate scatter. Now try the “tiny finite difference” button and watch the same budget produce garbage — the mistake box, measured.
3. Open the plateau explorer: at 4, 8, 12 qubits (random deep ansatz), read the typical gradient size and compare with the shot-noise line. At which width does training become hopeless at your budget?
4. Switch the ansatz to “local cost + shallow” and repeat: the desert recedes. Structure is the cure.

You need: The QCFE app (Android or Windows)

► App tool: *QNN Lab — parameter-shift demonstrations with shot noise, and a barren-plateau explorer with architecture switches*

25.6 Initialization, Depth, and Staying Out of the Desert

The barren-plateau figure of this chapter is a warning sign; this lesson is the driving manual. Three practical decisions — circuit depth, parameter initialization, and shot budget — decide whether training ever leaves the parking lot.

EQUATION: THE GRADIENT’S SHRINKING BUDGET

typical gradient size $\sim \frac{1}{2^n}$ (deep, scrambled circuits) vs. $\sigma_{\text{shot}} \approx \frac{1}{\sqrt{n_{\text{shots}}}}$
In words: Deep random circuits on n qubits have exponentially tiny gradients; shot noise only shrinks as the square root of the shot count — the race the desert always wins at scale

Symbol	Meaning	Unit
n	number of qubits	none
n_{shots}	shots per gradient estimate	none
σ_{shot}	statistical wobble of the estimate	none

To see a gradient it must beat the wobble: $n_{\text{shots}} \gtrsim 2^{2n}$ for fully scrambled circuits — a million shots per parameter step at just $n = 10$. The cure is never more shots; it is refusing to scramble.

WORKED EXAMPLE — PRICING ONE TRAINING RUN

A re-uploading classifier has 30 parameters and trains for 200 optimizer steps with parameter-shift gradients at 2000 shots per circuit evaluation. Price the run, then check whether 2000 shots can even resolve the gradients if the model were a scrambled 10-qubit circuit.

1. Parameter-shift needs 2 evaluations per parameter (Chapter 25's $\pm 90^\circ$ rule): $30 \times 2 = 60$ evaluations per step.
2. Total: $60 \times 200 \times 2000 = 24,000,000$ shots — twenty-four million, for a toy.
3. Desert check: scrambled at $n = 10$ means gradients $\sim 2^{-10} \approx 0.001$, but $\sigma = 1/\sqrt{2000} \approx 0.022$ — the signal would sit *twenty times below* the noise, and the 24 million shots would optimize pure static.

Answer: 24M shots if the landscape is healthy; effectively ∞ if it is barren. The billable lesson: budget shots *after* confirming trainability — with shallow depth, few qubits, structured (not random) circuits, and the initialization tricks below.

WORD TO KNOW

Identity-block initialization — Starting a layered circuit at parameters where consecutive blocks cancel to the identity, so training begins at an unscrambled, gradient-rich point and grows complexity only as needed — the standard anti-plateau opening move.

REALITY CHECK

Working habits that keep 2026 practitioners out of the desert: start shallow and small (the Part-Seven demos use 1–2 qubits and 3 layers for a reason); initialize near identity or from a trained shallower model (layerwise growth); prefer problem-structured circuits over expressive random ones; monitor the gradient's *size* from step one — a healthy run shows gradients orders of magnitude above σ , and a flatlining gradient norm is grounds to stop, shrink, and restructure rather than to buy more shots. None of these tricks *provably* scales to large quantum advantage — that open question is exactly where QML research stands.

25.7 Overfitting: The Model That Memorized the Noise

A model can be too rich for its data: with enough frequencies (Chapter 25's budget), a re-uploading circuit can thread its boundary between every training point — including the mislabeled and the unlucky. It then aces training and fails the only exam that matters: fresh data. The disease is overfitting; the symptom is a **generalization gap**.

WORKED EXAMPLE — READING THE GAP

Three models train on 40 points (2 features) and are tested on 20 held-out points. Results — Model 1 (1 layer): train 82%, test 80%. Model 2 (3 layers): train 97%, test 93%. Model 3 (8 layers): train 100%, test 72%. Diagnose each and pick the winner.

1. Model 1: gap 2 points — honest but possibly *underfitted*: too few frequencies to trace the true boundary. Check: would more capacity raise *test* accuracy?
2. Model 2: gap 4 points, best test score — the sweet spot: enough frequencies for the pattern, not enough to memorize the noise.
3. Model 3: gap 28 points — textbook overfitting: those 8 layers bought wiggles that fit the training set's accidents. Its perfect training score is the *symptom*, not an achievement.
4. Decision: deploy Model 2. And report *test* numbers with their shot-and-split error bars ($\pm$ several points on 20 test samples — Chapter 27's statistics lesson).

Answer: Model 2. The ranking rule — judge on held-out data, watch the gap, distrust perfection — is universal machine learning; the quantum-specific lever is that *layers = frequencies*, so capacity control is architectural and exact: remove a re-upload, lose a harmonic, smooth the boundary. Few classical models offer so clean a dial.

WORD TO KNOW

Generalization gap — Training accuracy minus test accuracy. Small: the model learned the pattern. Large: it memorized the sample — reduce capacity (layers/frequencies), add data, or regularize.

REALITY CHECK

Two quantum wrinkles worth knowing. First, shot noise acts as accidental regularization — a model trained on noisy estimates cannot chase razor-thin wiggles, so modest hardware noise sometimes *improves* test accuracy (a documented, slightly comic effect). Second, tiny QML datasets (dozens of points, as in most demos) make the gap estimate itself noisy: a 20-point test set measures accuracy to ± 10 points at best — which is how “100% test accuracy” headlines are born and why Chapter 27 demands seeds and error bars before applause.

CHAPTER SUMMARY

- QNNs are layered parameterized circuits trained on measured losses; the network dictionary maps cleanly except its most important word — nonlinearity lives only in encoding and measurement, never inside the unitary filling.
- Parameter shift: exact gradients from two quarter-turn-shifted evaluations per dial — large separations, exact slopes; tiny finite differences drown in shot noise.
- Barren plateaus: deep random circuits dilute every dial's influence across 2^n amplitudes; gradients $\sim 2^{-n}$ sink under shot noise. Cures are structural: shallow depth, local costs, smart initialization, symmetry-matched families.
- Expressivity is a spendable budget: L re-uploads buy L harmonics per feature — size the model to the data's structure.
- The principled edge is quantum data: no loading door, provable exponential savings over measure-first pipelines, plateau-free architectures (QCNN) — quantum learners for quantum machines.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] Name the two places nonlinearity enters a QNN, and the chapter that owns each.

Problem 2 [Basic] Apply the parameter-shift rule to $E(\theta) = \cos \theta$ at $\theta = 0^\circ$ and at $\theta = 90^\circ$, and interpret both slopes on the curve.

Problem 3 [Basic] A model has 40 dials. How many circuit evaluations does one full parameter-shift gradient cost?

Problem 4 [Basic] From the dictionary table: which classical concept has NO faithful quantum translation inside the circuit, and what stands in for it?

Problem 5 [Basic] Recompute the worked example's total for a 12-parameter, 100-step, 1000-shot run. Which of the three factors is cheapest to cut in practice, and why?

Problem 6 [Basic] At $n = 6$ qubits, scrambled gradients are $\sim 2^{-6}$. How many shots per evaluation are needed for the shot wobble to shrink to half that size?

Problem 7 [Basic] Train 99%, test 97% — diagnose. Train 100%, test 60% — diagnose and prescribe two remedies with their frequency-budget rationale.

Problem 8 [Basic] Why is a perfect training score on noisy labels a warning rather than a success? One sentence.

Problem 9 [Standard] Gradient budget: each evaluation uses 2000 shots (noise ≈ 0.022), and the model has 40 dials. Compute the shots per gradient; then, for a plateau gradient of true size 0.003, explain with numbers why this budget cannot see it, and the shots that could.

Problem 10 [Standard] Verify the parameter-shift rule at $\theta = 30^\circ$ for $E = \cos \theta$ (true slope $-\sin 30^\circ = -0.5$), then show the rule ALSO works for $E(\theta) = 0.7 \cos \theta + 0.2$ — and state which feature of these curves the proof rides on.

Problem 11 [Standard] Plateau arithmetic: with typical gradient $2^{-n}/2$ and shot noise $1/\sqrt{N}$, derive the shot count N needed for signal = 3× noise at $n = 8$ and $n = 16$. What scaling law connects the two answers?

Problem 12 [Standard] Frequency budgeting: a periodic dataset needs 4 harmonics in each of 2 features. Using Lesson 4’s counting rule, specify the minimum re-uploading depth, the resulting dial count at 5 dials per layer, and one overfitting risk of doubling the depth “for safety.”

Problem 13 [Standard] Sort these tasks into “classical data (uphill)” vs “quantum data (principled edge),” one sentence each: (a) classifying customer churn from spreadsheets; (b) recognizing the phase of matter produced by a quantum simulator; (c) reading handwritten digits; (d) certifying states arriving over the network of Chapter 15.

Problem 14 [Standard] Make the desert table: for $n = 4, 8, 12, 16$, tabulate scrambled gradient size 2^{-n} and the shots $\approx 2^{2n}$ needed to resolve it. At today’s $\sim 10^4$ shots per second, how long is one gradient evaluation at $n = 16$?

Problem 15 [Standard] Explain why identity-block initialization helps, in the language of this chapter’s frequency picture: what does starting at the identity do to the circuit’s effective depth and its initial function?

Problem 16 [Standard] A colleague’s 8-qubit, 40-layer random ansatz trains beautifully in a *noiseless statevector simulator* but flatlines on hardware at 4000 shots. Reconcile the two observations quantitatively.

Problem 17 [Standard] A 20-point test set reports 90% accuracy. Using binomial wobble $\sqrt{p(1-p)/n}$, give the two-sigma interval, and judge whether a competitor’s 85% on the same set is meaningfully worse.

Problem 18 [Challenge] Prove the parameter-shift rule for sinusoids: assume $E(\theta) = A \cos \theta + B \sin \theta + C$ (the general form for Pauli-rotation dials), evaluate $(E(\theta + 90^\circ) - E(\theta - 90^\circ))/2$ using the angle-addition identities, and show it equals the calculus derivative $-A \sin \theta + B \cos \theta$ exactly.

Problem 19 [Challenge] Global versus local costs: to score “all n qubits read 0,” compare observable A = the projector onto $|0 \cdots 0\rangle$ (global: one term, value 1 on a single pattern out of 2^n) with observable B = the average of the n single-qubit checks “qubit k reads 0” (local). For a typical scrambled state, estimate each observable’s size and argue in three sentences which one leaves a detectable training signal — the intuition behind “local costs soften plateaus.”

Problem 20 [Challenge] Measure-first information loss: a source emits one of two known single-qubit states,

$|0\rangle$ or $|+\rangle$, each shot. A classical pipeline must pick ONE fixed measurement basis and record bits; a quantum pipeline may store and process the states coherently. Using Chapter 4's tables, show every fixed classical basis confuses the two sources on some fraction of shots, compute the best single-basis distinguishing probability ($\cos^2 22.5^\circ \approx 0.85$ — the optimal tilted basis), and explain in two sentences how this baby example scales into Lesson 5's exponential separations.

Problem 21 [Challenge] Design a trainability pre-flight check costing under 100,000 shots: choose a handful of random parameter points and one parameter's shift-rule gradient at each, specify shots per estimate, and give the decision rule (gradient norm vs. σ) that clears the run for full training or grounds it. Justify each number.

Problem 22 [Challenge] Design the capacity sweep for the rings dataset: layers $L = 1, 2, 3, 5, 8$, fixed budget of shots (allocate it), train/validation/test splits from 60 points, and the decision rule for the deployed L . Predict the qualitative shape of the train and validation curves versus L , and mark where the gap opens.

Quantum Generative Models and the Question of Advantage

Born machines, quantum GANs, and an honest map of where quantum advantage in AI might live

IN THIS CHAPTER YOU WILL LEARN TO

- Explain the difference between discriminative and generative models, and why quantum circuits are *natively* generative.
- Define the Born machine, and compute the distribution it samples from a given circuit.
- Measure the gap between two distributions with the total variation distance, and use it as a training score.
- Describe the quantum GAN training loop, and its known failure modes.
- Audit “quantum advantage in AI” claims: sampling hardness, quantum data, hard kernels — and the de-quantization counterattack.

The classifiers of the last four chapters answer questions: *which class is this point?* A second family of machine-learning models does something stranger and often more valuable: it *creates* — new images, new molecules, new plausible data. These are generative models, and they are the arena where quantum computing’s native talent is most obvious. A quantum circuit, measured, does not merely compute: it *samples* from a probability distribution that the circuit itself defines. Randomness is not bolted on — it is the Born rule, working for a living.

This chapter builds the smallest honest versions of these ideas: the Born machine, trained by shrinking a distance between histograms; the quantum GAN, where two models duel; and then the question every practitioner must face with open eyes — *where, exactly, could quantum advantage in AI come from?* The answer, honestly mapped, is this book’s last technical gift before the capstones.

26.1 Discriminate or Generate: Two Jobs for a Model

A discriminative model draws boundaries: given an input, output a score (Chapter 23). A generative model holds a whole *distribution* inside itself: ask it for a sample and it produces one, with the right relative frequencies. Classically this is hard in a specific way — neural generators must convert uniform random numbers

through learned transformations, and evaluating or normalizing the resulting distribution can be painful. A quantum circuit skips the conversion. Prepare $|\psi(\theta)\rangle$, measure, and nature itself hands you a sample:

EQUATION: THE BORN MACHINE

$$P_{\theta}(x) = |\langle x | \psi(\theta) \rangle|^2$$

In words: The circuit's output state defines a probability for every bit pattern; measuring IS sampling from that distribution

Symbol	Meaning	Unit
$ \psi(\theta)\rangle$	the state prepared by the circuit at dial settings θ	none
x	a bit pattern (a sample)	none
$P_{\theta}(x)$	the machine's probability of producing x	none

Named for the Born rule of Chapter 4. An n -qubit Born machine carries a distribution over 2^n patterns using 2^n amplitudes — classically describing the same distribution needs the same exponential list, which is the storage argument of Chapter 9 wearing a machine-learning hat.

WORD TO KNOW

Generative model — A model that represents a probability distribution and can produce fresh samples from it — as opposed to a discriminative model, which only scores inputs. Quantum circuits are generative by construction: measurement is sampling.

Training a generative model means reshaping its distribution until it matches the data's. For that we need a ruler between distributions.

26.2 A Ruler Between Distributions

EQUATION: TOTAL VARIATION (TV) DISTANCE

$$\text{TV}(p, q) = \frac{1}{2} \sum_x |p(x) - q(x)|$$

In words: Add up all the histogram disagreements and $\frac{1}{2}$ halve — 0 means identical distributions, 1 means no overlap at all

Symbol	Meaning	Unit
p, q	the two distributions being compared	none
$p(x)$	probability distribution p gives pattern x	none
TV	their distance, between 0 and 1	none

Interpretation with teeth: TV is the best possible probability of telling the two distributions apart from one sample, minus one half, doubled. $\text{TV} = 0.5$ means a perfect judge guesses right 75% of the time. Real training pipelines

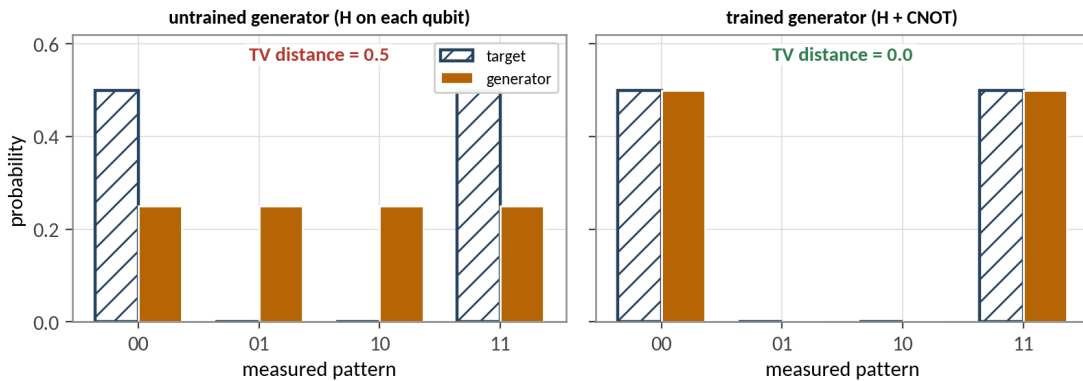
often use smoother rulers (MMD, KL); TV is the honest hand-computable one.

WORKED EXAMPLE — TRAINING A TWO-QUBIT BORN MACHINE, START TO FINISH

Target distribution: $q(00) = q(11) = \frac{1}{2}$ (perfectly correlated bits, patterns 01 and 10 never occur). Untrained generator: H on each qubit. Score the untrained machine, then find dial settings that finish the job.

1. Untrained machine: $H \otimes H$ gives the uniform state, so $p = (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ over (00, 01, 10, 11).
2. Score it: $TV = \frac{1}{2}(|\frac{1}{4} - \frac{1}{2}| + |\frac{1}{4} - \frac{1}{4}| + |\frac{1}{4} - \frac{1}{2}|) = \frac{1}{2} \times 1 = 0.5$ — badly wrong, as the left panel of the figure shows.
3. Think like Chapter 11: the target is exactly the statistics of the Bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. So the “trained” circuit is: H on the first qubit, CNOT to the second.
4. Rescore: the Bell circuit gives $p = (\frac{1}{2}, 0, 0, \frac{1}{2})$, so $TV = 0$ — a perfect generator, right panel.

Answer: from $TV = 0.5$ to $TV = 0$ by adding one CNOT. The example is tiny but the moral is structural: correlated data wants entangling gates — a generator without two-qubit gates can only ever produce independent bits, no matter how its single-qubit dials are tuned.



Training as histogram-matching: the untrained generator produces all four patterns equally (left, $TV = 0.5$); adding the entangling gate matches the correlated target exactly (right, $TV = 0$).

WORD TO KNOW

Total variation distance — The histogram-disagreement ruler between two distributions: half the sum of absolute probability differences. The Born machine’s training loss in this chapter.

In real Born-machine training the target is an unknown data distribution, the TV is *estimated* from finite samples on both sides, and the dials are adjusted by the parameter-shift gradients of Chapter 25 — the same loop as Part Seven’s classifiers, chasing histograms instead of labels.

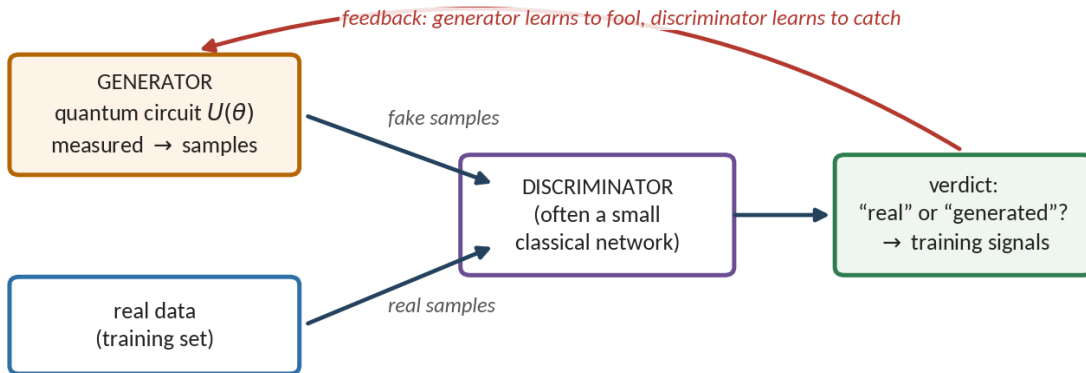
COMMON MISTAKE

X Wrong: “A quantum computer tries all possible weights in superposition, so it trains any model exponentially faster.”

✓ Right: No known result says anything like this. Training remains an iterative, classical-feedback loop (Chapter 20); superposition lives *inside* each circuit run, and measurement returns one sample. Where quantum generative models genuinely differ is in *what distributions they can represent and sample natively* — a representational claim, not a training-speed claim. Confusing the two is the fastest way to fail the audit of Chapter 27.

26.3 Quantum GANs: Two Models, One Duel

Generative adversarial networks (GANs) train a generator not with a fixed ruler but with a living opponent: a **discriminator** that learns to tell real samples from generated ones, while the generator learns to fool it. When the duel balances, the generator’s distribution has become indistinguishable from the data’s. The quantum version simply casts a parameterized circuit as the generator — and, usually, keeps the discriminator classical (a small neural network reading measured bit patterns):



The quantum GAN loop: a circuit generates samples by measurement, a (typically classical) discriminator judges them against real data, and both models update from the verdicts — the generator by parameter-shift gradients, the discriminator by ordinary backpropagation.

Why bother with the duel? Because for high-dimensional data the fixed rulers fail first: estimating TV over 2^n patterns needs astronomically many samples, while a discriminator learns *which* disagreements matter and focuses the training signal there. The 2019 finance experiments (loading market distributions into few-qubit states for risk analysis) used exactly this trick, and quantum GANs remain a lively engineering line.

The failure modes are inherited from classical GANs, plus one of our own. **Mode collapse:** the generator finds one pattern the discriminator likes and produces only that — diversity dies. **Instability:** the duel oscillates instead of settling. And the quantum addition, by now an old friend: **shot cost** — every gradient step needs fresh histograms on both fronts, priced exactly as in Chapter 25’s budgeting lesson.

WORD TO KNOW

Quantum GAN — An adversarial training scheme whose generator is a parameterized quantum circuit sampling by measurement, dueling a (usually classical) discriminator. The equilibrium goal: generated samples statistically indistinguishable from data.

26.4 Where Could Quantum Advantage in AI Actually Live?

Time to assemble this book's most requested map. Four candidate territories, with the evidence honestly graded:

The advantage map, as of the mid-2020s

Candidate	The claim and its evidence	The catch
Sampling hardness	Random-circuit sampling is provably hard to imitate classically (the supremacy experiments, Chapter 30); Born machines live next door to this hardness	Hard-to-imitate $\neq$ useful-to-have: no one has tied the hard distributions to data anyone wants to model
Quantum data	For learning from <i>quantum</i> experiments (states from sensors, simulators), quantum memory + two-copy measurements give proven exponential advantages — demonstrated on real hardware (2022)	The input must genuinely be quantum states; your customer database is not
Hard kernels	A feature map built on a believed-hard number-theory problem gives a provable kernel advantage (2021)	Provable, but for a cryptographic toy task — no natural dataset speaks that language yet
Heuristic QML on classical data	Small demos run; occasionally competitive	No proven separation; every claimed speedup so far has fallen to better classical baselines or dequantization

WORD TO KNOW

Dequantization — The counterattack: finding a classical algorithm that matches a claimed quantum speedup once the quantum algorithm's assumptions (like cheap data loading) are granted to the classical side too. Ewin Tang's 2018 demolition of the quantum recommendation-system speedup is the founding example — done as an undergraduate.

The dequantization story deserves its moral spelled out. The quantum recommendation algorithm assumed data pre-loaded into quantum-friendly form; Tang showed that granting classical algorithms an analogous data structure lets them match the performance. The general lesson, now a reflex in serious QML reviews: *an advantage claimed over a handicapped classical opponent is not an advantage*. It is the same fairness principle as the tuned-baseline rule of Chapter 27 — applied to theory.

CASE STUDY — LEARNING FROM QUANTUM EXPERIMENTS • GOOGLE QUANTUM AI AND CALTECH, 2022

The strongest advantage evidence in quantum machine learning did not involve classifying flowers or fraud. Huang and collaborators asked: how many runs of a quantum *experiment* does a learner need to predict its properties? They proved a stark separation — a learner with **quantum memory**, storing two copies of each experimental state and measuring them *jointly*, needs exponentially fewer experiments than any learner that measures each copy immediately and keeps only classical notes. Then they demonstrated it on hardware, with tasks like identifying an unknown process’s symmetries in orders of magnitude fewer runs. The punchline for this book: the proven quantum advantage in learning, so far, is about learning from *quantum* data — nature’s experiments, quantum sensors, quantum simulators — where the learner’s quantum memory speaks the data’s native language. For spreadsheets and photographs, the jury is still out; for the coming era of quantum sensors and simulators, the verdict is already in.

REALITY CHECK

Keep the map’s proportions: territories two and three hold *proofs*; territory one holds hardness without use; territory four — the one in most press releases — holds neither, yet. A working scientist treats that as an agenda, not an insult: the open problem “find a natural classical-data task with proven quantum learning advantage” is one of the field’s great prizes, and it is entirely possible the honest answer will be “there is none.” Both outcomes are discoveries.

26.5 The Engineer’s Compass

Condensing this chapter into working rules: reach for a **Born machine** when the data is few-variable but strangely correlated, and let entangling gates carry the correlations. Reach for a **quantum GAN** when a fixed ruler is unaffordable and you can pay the duel’s instability tax. Reach for **quantum-data methods** the moment your input is genuinely quantum — sensor states, simulation outputs — because there the advantage is proven. And when the input is a plain spreadsheet, run the classical baseline *first*, audit any quantum gain with Chapter 27’s five questions, and remember that “no advantage found” is a publishable, honorable, and common result.

ACTIVITY (APP) — TRAIN A BORN MACHINE BY HAND

1. Open the *Two-Qubit Lab* in the QCFE app. Build the untrained generator: H on each qubit. Run 1000 shots and record the four frequencies.
2. Compute the TV distance to the target $\{00: 50\%, 11: 50\%\}$ from your measured histogram (expect ≈ 0.5).
3. Now “train” in one move: replace the second H with a CNOT from qubit A. Run 1000 shots, recompute TV (expect ≈ 0 , up to shot wobble).
4. Sabotage the trained machine: add a small $R_y(20^\circ)$ on qubit B after the CNOT. Predict qualitatively, then measure: which forbidden patterns appear, and how big is TV now?

You need: The QCFE app (Android or Windows), Paper and pen, Calculator

26.6 Judging a Generator: Beyond Eyeballing

Classifiers grade themselves — accuracy against labels. A generator has no labels, only the claim “my samples look like the data.” Turning *look like* into a number is the evaluation problem, and sloppy answers to it inflate half the generative-model hype in every field, quantum included.

WORKED EXAMPLE — THREE HONEST CHECKS ON A TRAINED BORN MACHINE

A 3-qubit Born machine was trained on a dataset over 8 patterns. Design the evaluation, with numbers.

1. Marginal check (cheap): compare each qubit’s 1-frequency, generated vs. real. Real: (0.50, 0.30, 0.20); generated: (0.35, 0.35, 0.30). Per-qubit TV = $\frac{1}{2}(|0.15| + |0.05| + |0.10|)$ applied coordinate-wise: gaps of 15, 5, and 10 points — qubit 1’s marginal is badly off; the training under-weighted it.

2. Joint check (the real exam): TV over all 8 pattern frequencies, from equal-sized sample histograms; report with its shot wobble. Marginals can match while the joint fails — correlations are the whole point of entangling generators.

3. Novelty check (the memorization alarm): what fraction of generated samples are verbatim copies of training points? A generator matching the joint distribution *and* producing only copies has learned the list, not the law — the generative cousin of overfitting.

Answer: marginals flag which qubit to fix, the joint TV grades the model, and the novelty rate catches memorization. Report all three with error bars, and you are already more rigorous than most demos. (Chapter 27’s question H applies verbatim: how many samples, from hardware or simulator, with what spread across seeds?)

WORD TO KNOW

Memorization (generative) — A generator that reproduces training examples rather than the distribution behind them — perfect scores on naive checks, zero value in deployment, and a privacy hazard when training data is sensitive. Detected by novelty-rate checks.

REALITY CHECK

The deep honesty problem: for n qubits, certifying the *joint* distribution needs samples exponential in n — the same wall that makes Born machines potentially powerful makes them practically ungradable at scale. Working compromises: grade marginals and low-order correlations (they need only polynomial samples), grade downstream utility (does a classifier trained on synthetic data work on real data?), and state plainly which you did. “Our 20-qubit generator matches the data distribution” is a claim no one can verify; “matches all two-qubit correlations within 0.02” is science.

CHAPTER SUMMARY

- Quantum circuits are natively generative: the Born rule makes every measurement a sample, giving the Born machine $P_\theta(x) = |\langle x|\psi(\theta)\rangle|^2$.
- Generative training means shrinking a distance between histograms; the total variation distance $TV = \frac{1}{2} \sum_x |p - q|$ is the hand-computable ruler, and correlated targets demand entangling gates.
- Quantum GANs replace the fixed ruler with a learned discriminator — more scalable, less stable, and priced in shots like all variational training.
- The advantage map: proven wins exist for quantum data (exponential, demonstrated 2022) and for one cryptography-flavored kernel; sampling hardness lacks a use; heuristic QML on classical data has no proven separation.
- Dequantization is the fairness police: grant the classical opponent the same assumptions, and many claimed speedups evaporate. Audit before believing — including your own results.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] A generator produces the uniform distribution over two bits; the target is {00: 100%}. Compute the TV distance, and give the perfect generator circuit (it needs zero gates — explain).

Problem 2 [Basic] A one-qubit Born machine is the circuit $R_y(\theta)$ on $|0\rangle$. Find θ so the machine outputs 1 with probability 0.30, using $P(1) = \sin^2(\theta/2)$.

Problem 3 [Basic] In one sentence each: why does sampling from a trained Born machine cost nothing beyond running the circuit, and what is the classical generative modeler's corresponding headache?

Problem 4 [Basic] Compute the joint TV between generated (0.40, 0.10, 0.10, 0.40) and real (0.45, 0.05, 0.05, 0.45) over four patterns.

Problem 5 [Basic] A generator's novelty rate is 3% (97% of outputs are verbatim training points) with excellent TV scores. Verdict, and the one-sentence reason.

Problem 6 [Standard] An imperfectly trained generator gives $p = (0.45, 0.05, 0.05, 0.45)$ against the target $q = (0.5, 0, 0, 0.5)$. Compute TV, and interpret it as a best-judge distinguishing probability.

Problem 7 [Standard] Give the three-qubit generator circuit for the target {000: 50%, 111: 50%}, and name the state it prepares (Chapter 9).

Problem 8 [Standard] Estimating each of 4 pattern probabilities to ± 0.01 needs roughly $n \approx p(1-p)/0.01^2$ shots per pattern's estimate. Budget the shots to certify $TV < 0.05$ against the Bell target, and state why certifying small TV over 2^{10} patterns is a different world.

Problem 9 [Standard] In the quantum GAN loop, list what flows along each arrow of the pipeline figure, and mark which updates use parameter-shift gradients and which use ordinary backpropagation.

Problem 10 [Standard] Mode collapse strikes: the generator outputs $\{00: 100\%$ against the Bell target. Compute TV, and explain in two sentences how the discriminator's response eventually punishes the collapse.

Problem 11 [Standard] With 1000 samples per side, each pattern frequency carries wobble $\approx \sqrt{p(1-p)/1000}$. For the four-pattern example above, propagate a rough error bar onto the TV estimate and decide whether a measured TV of 0.10 is distinguishable from 0.

Problem 12 [Challenge] “Random-circuit sampling is classically hard, and our Born machine uses random circuits — therefore our product has quantum advantage.” Write a five-sentence referee response separating the true premise from the false conclusion, using the trainability results of Chapter 25 as your second witness.

Problem 13 [Challenge] Reconstruct the dequantization argument's shape: state the quantum recommendation claim's hidden assumption, what the classical algorithm was granted in response, and the resulting fairness rule — then apply the same rule to amplitude-encoding speedup claims (Chapter 22).

Problem 14 [Challenge] Explain, in a paragraph for a lab director, why learning from quantum sensors may need exponentially fewer experiments with quantum memory: what can joint measurements on *two stored copies* see that immediate single-copy measurements statistically cannot? (Chapter 8's incompatible-measurement trade-offs are the seed of the answer.)

Problem 15 [Challenge] Design the audit (Chapter 27's five questions) for a startup claiming “our 12-qubit quantum GAN generates synthetic patient records indistinguishable from real ones.” For each question, name the specific evidence you would demand and the red flag you expect.

Problem 16 [Challenge] Explain why full joint certification needs exponentially many samples (how many patterns must be populated at n qubits?), then design the polynomial-cost report card: which marginal and correlation checks, at what sample sizes, plus one downstream-utility test — and state what your card can and cannot certify.

Quantum AI in Practice

The honest scoreboard, hybrid workflows, generative models — and how to evaluate every claim you will ever hear

IN THIS CHAPTER YOU WILL LEARN TO

- Summarize the state of quantum machine learning on one honest scoreboard.
- Tell the dequantization story and extract its permanent lessons.
- Describe generative quantum models and their special relationship to sampling.
- Place quantum components correctly inside today's real ML pipelines.
- Apply the QML claim-checking checklist — and chart realistic entry points for students and researchers in our region.

27.1 The Honest Scoreboard

Part Six has built every working part of quantum machine learning: encodings (Chapter 22), trainable classifiers (Chapter 23), kernels (Chapter 24), and networks with their gradients and plateaus (Chapter 25). This closing chapter assembles the parts into a professional's view of the field — starting with the scoreboard a colleague, minister, or investor deserves to see.

Quantum machine learning, scored honestly (state of play around 2026)

Claim	Verdict	Evidence
Quantum circuits can learn real decision boundaries	established	trained models on simulators and hardware since 2019 — including this book's rings demo
Exact gradients are measurable on hardware	established	parameter-shift rule (Chapter 25)
Provable advantage exists on some constructed problems	established (theory)	discrete-log kernel separation; learning-from-experiments theorems
Advantage on natural classical data (images, records, text)	open — none shown	tuned classical baselines match or win to date
Exponential speedups for classical big-data linear algebra	largely withdrawn	the dequantization results (Lesson 2)
Advantage on quantum-born data	established (theory + demos)	exponential sample savings, hardware demonstration 2022
Scaling of deep variational training	obstructed	barren plateaus; structural cures under study

Notice the scoreboard's shape: solid mechanisms, honest obstacles, one arena of principled advantage — and the biggest commercial question still open. A field like this rewards exactly two postures, both taught by this book: rigorous participation, and calibrated skepticism. It punishes a third: cheerleading.

27.2 The Dequantization Story — and Its Permanent Lessons

Chapter 22's case study introduced the young researcher; here is the fuller arc, because no episode teaches QML's epistemics better. Through the 2010s, a family of quantum algorithms promised exponential speedups for the linear algebra at machine learning's core — solving huge linear systems, principal-component analysis, recommendation systems — all assuming data pre-loaded in quantum-friendly form. In 2018–2019, Ewin Tang and collaborators showed that granting classical algorithms a *comparable* data-access assumption (efficient random sampling from the stored vectors) lets them match the quantum runtimes up to polynomial factors, for one flagship after another.

Three permanent lessons crystallized. **Compare like with like:** an advantage claim is only as strong as the fairness of its input-access accounting — quantum with magic memory versus classical with a filing cabinet is not a race. **Advantage must be architectural, not assumed:** the surviving quantum advantages (Shor's, the quantum-data results) run on structure classical machines provably lack, not on generous bookkeeping. **Negative results are gifts:** dequantization sharpened the field, redirecting effort toward the arenas — quantum data, constructed-hard kernels, sampling (next lesson) — where the quantum-classical boundary genuinely lies. Science that cannot celebrate its own corrections cannot be trusted with its own claims.

27.3 Generative Models: Learning to Sample

Classification answers questions; **generative** models produce things — the family behind modern image and text AI. Their core operation is *sampling*: draw new examples from the distribution the training data sketches.

And here quantum machines hold a native card this book has kept face-up since Chapter 4: *a measured quantum circuit IS a sampler*. Every shot draws a fresh pattern from the Born distribution $P(x) = |a_x|^2$ — no algorithmic trick required; nature does the drawing.

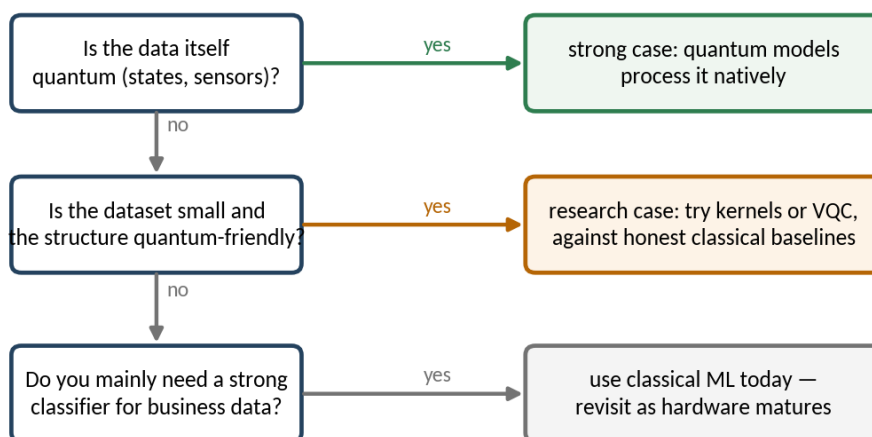
WORD TO KNOW

Born machine — A generative model whose output distribution is the Born-rule distribution of a parameterized circuit: train the dials so the circuit’s measurement statistics imitate the data’s statistics. Sampling is then a single shot per example.

The intriguing asymmetry: sampling is precisely where believed quantum-classical separations live. The circuit families behind the quantum-advantage experiments of Chapter 2 are hard for classical computers to *sample from* — suggesting Born machines can represent distributions no efficient classical sampler expresses. The sober counterweights: training them meets the same plateau and shot-cost dragons (Chapter 25); “expressible” has not yet met “useful” on a commercial dataset; and small hardware demonstrations (including finance-flavored pilots) remain proofs of mechanism. Verdict: the most theoretically motivated corner of classical-data QML — and still a research corner.

27.4 Hybrid Workflows: Where Quantum Sits in a Real Pipeline

No serious 2026 deployment replaces an ML stack with a quantum computer. The realistic pattern — visible in every industrial pilot worth reading — embeds a *quantum component* inside a classical pipeline at one of a few slots: a **kernel service** (Chapter 24) feeding an ordinary SVM; a **feature transformer** whose measured statistics become extra columns for a classical model; a **sampler** proposing candidates (molecules, portfolios, configurations) that classical models screen; or an **optimizer subroutine** (QAOA, Chapter 20) attacking a combinatorial sub-problem. Around the slot, everything stays boringly classical: data cleaning, baselines, deployment, monitoring. This modesty is engineering wisdom — small quantum surface area means testable claims, honest ablations (“remove the quantum piece; did accuracy actually drop?”), and graceful failure.



The deployment decision, honestly drawn. The strong case today is quantum-born data; classical business data earns a research posture at best — always against real baselines.

COMMON MISTAKE

X Wrong: “Our pipeline includes a quantum step, so the product is quantum-accelerated.”

✓ Right: A pipeline is quantum-*accelerated* only if removing the quantum step degrades a metric a customer cares about, at matched cost — the ablation test. Pilots that skip it are marketing with extra refrigeration. Demand the ablation, the tuned classical baseline, and the shot bill; Lesson 5 packages these into a checklist you can wield in any meeting.

27.5 The Checklist — and the Road for Our Region

EQUATION: THE QML CLAIM-CHECKING CHECKLIST

D — A — S — H — B

In words: Five questions that sort every quantum-AI claim: Data access · Ablation · Scaling · Hardware honesty · Baselines

Symbol	Meaning	Unit
D	how does the data get in — and is loading priced? (Chapter 22)	—
A	does removing the quantum piece change the result?	—
S	do gradients, shots, and depth survive the plateau arithmetic at the promised size? (Chapter 25)	—
H	simulator or hardware — and with what noise and shot budgets? (Chapters 8, 28)	—
B	is the classical baseline tuned as hard as the quantum model?	—

Extends Chapter 2’s general checklist with the four QML-specific traps this part has armed you against. Any paper, pitch, or press release that answers all five cleanly deserves your serious attention — most do not survive question D.

WORKED EXAMPLE — RUNNING THE CHECKLIST ON A PITCH

A startup’s deck claims: “Our 12-qubit quantum AI classifies medical scans 40% faster than deep learning.” Apply DASHB.

1. D: scans are megapixel classical images — how are they encoded into 12 qubits? (Either catastrophic compression or a classical front-end doing the real work.)
2. A: was the identical pipeline run with the quantum block deleted? Not reported.
3. S: 12 qubits trains fine — but the deck’s roadmap says “1000 qubits next year”: the plateau and shot arithmetic of Chapter 25 goes unmentioned.

4. H: “faster” turns out to compare a simulator against an *untuned* network on one GPU; no shot budgets appear.

5. B: the deep-learning baseline is three years old and un-optimized.

Answer: the claim fails D, A, and B outright and dodges S and H — verdict: decline, with a written request for the five answers. Ten minutes of this book’s checklist is worth more than any demo day. (And when a claim passes? Then you have found something genuinely worth funding, publishing, or joining — the checklist cuts both ways.)

REALITY CHECK

For Iraq and the Kurdistan Region, quantum machine learning is — surprisingly — one of the field’s most accessible research frontiers. Every result in Part Six runs on a laptop simulator or free cloud tier (Chapter 30); the datasets are small by design; and the field’s most valuable current products are exactly the things a well-trained group can make anywhere: honest benchmarks, reproductions, ablation studies, and symmetry-matched model designs. A graduation project reproducing this book’s rings classifier on real hardware with full DASHB reporting is publishable-adjacent work; a master’s thesis benchmarking three feature maps against tuned classical baselines on a regional dataset (agriculture, energy-grid, medical) is a genuine contribution; and the quantum-data arena will need exactly the simulation skills Chapter 20 teaches. The entry ticket is this book plus discipline — not a refrigerator.

CASE STUDY — LEARNING FROM QUANTUM EXPERIMENTS • GOOGLE AND CALTECH, SCIENCE, 2022

The cleanest demonstrated quantum-learning advantage to date closed Part Six’s central loop. A Caltech–Google collaboration (Huang, McClean, and colleagues, *Science* 2022) compared two ways of learning about quantum systems: the conventional pipeline — measure everything into classical data, then analyze — versus a quantum pipeline that stores and processes the quantum states directly before measuring. Theory predicted, and 40-qubit Sycamore experiments confirmed, an *exponential* gap: tasks (like sensing certain noise properties) where the quantum learner needed hundreds of experiment runs while the best classical pipeline would need more runs than atoms-in-planets budgets allow. No classical dataset was involved — the data was quantum from birth, exactly Lesson 5 of Chapter 25. The result set QML’s compass for the decade: the surest advantages sit where quantum computers learn about the quantum world — simulators, sensors, materials, networks — an arena that will only widen as the machines of Chapters 28–30 mature.

ACTIVITY (CLASSROOM) — THE DASHB TRIBUNAL

1. Split into groups; each group finds one public quantum-AI claim (press release, paper abstract, vendor page).
2. Each group prepares a five-minute DASHB audit: one slide-equivalent per letter, with the exact quotes being judged.

3. Present; the class votes each claim onto a scoreboard from “fund it” to “file it.”
4. Finale, in the spirit of fairness: each group must also propose the *minimal experiment* that would make their audited claim pass DASHB — criticism that cannot state its own success condition is just noise.

You need: Internet access, Paper and pen, This chapter’s checklist

► *App tool: QML Studio — the app’s capstone workspace bundling the encoding, classifier, kernel, and QNN labs into one pipeline with an ablation toggle and a DASHB report card*

27.6 A Worked Audit, End to End

The checklist of this chapter earns its keep only when exercised. Here is a fictional-but-typical claim, audited in the open. The vendor’s page reads:

“Our 16-qubit quantum classifier achieved 94% accuracy on a financial fraud dataset, outperforming classical machine learning. Quantum advantage is here for banking.”

WORKED EXAMPLE — RUNNING THE FIVE QUESTIONS

Apply the audit checklist, question by question, listing what to ask for and what would count as a pass.

1. **D — Data access:** How many samples and features — and, critically, how was 16 qubits’ worth of data loaded, at what priced cost? If 784 features were compressed classically to 16, the heavy lifting may be classical (Chapter 22’s invoice). Demand: logistic regression, random forest, and a tuned SVM on the *same* features and splits. (Typical finding: a tuned random forest matches or beats 94% — the claim’s “outperforming” often means “outperformed our own untuned baseline.”)
2. **A — Ablation:** Swap the quantum kernel/model for a classical one inside the identical pipeline: does the 94% survive? “We used entangling layers” is a circuit description, not evidence — only the ablation shows the quantum piece did anything (Chapter 24).
3. **S — Scaling:** Do gradients, shots, and depth survive the plateau arithmetic at the promised size (Chapter 25)? A 16-qubit demo with a random deep ansatz already sits near the desert’s edge; “enterprise scale” claims need the 4ⁿ shot table answered.
4. **H — Hardware honesty:** Simulator or hardware — with what shot budgets, noise, and error bars (Chapters 8, 28)? Demand hardware numbers with spread over seeds and calibrations, not one lucky simulator run.
5. **B — Baselines:** Which classical models, tuned how hard? Demand logistic regression, random forest, and a tuned SVM on the same features and splits. (Typical finding: a tuned random forest matches or beats 94% — “outperforming classical ML” usually means “outperformed our own untuned baseline.”)

Answer: as stated, the claim fails D (loading unpriced), A (no ablation), and B (no serious baseline), while dodging S and H — the standard result for 2026-era marketing. The polite, professional verdict template: *“promising engineering demo; no evidence of quantum advantage; re-evaluate against tuned classical baselines with error bars.”*

WORD TO KNOW

Ablation — Re-running an experiment with one ingredient removed or replaced (e.g., the quantum kernel swapped for a classical one) to expose which ingredient actually produced the performance. The single most clarifying experiment to demand from any QML claim.

REALITY CHECK

Auditing is not cynicism — it is how the field matures. The same five questions applied to the strongest current work (quantum-data experiments in the style of Huang et al. 2022, learning from *quantum* sensors and states) return quite different answers: a provable separation, stated scaling, hardware error bars. Knowing what a *pass* looks like keeps the audit fair — and keeps you hireable in both directions: able to puncture hype, and able to recognize the real thing.

27.7 The Project Template: Six Steps from Idea to Report

Everything in Part Seven compresses into one reusable checklist — the shape of every capstone (Projects 9–11) and of every credible small QML study you will read or write.

The six-step QML project template

Step	Do	The trap it prevents
1	Freeze the splits; train the tuned classical baseline first	“beat classical” claims with no real opponent
2	Scale features to the encoding’s period; log the ranges	wrap-around aliasing; leakage via test-informed scaling
3	Start with the smallest circuit (1–2 qubits, 1–3 layers)	barren deserts; untraceable failures
4	Train with tracked gradients and restarts; budget shots in advance	silent non-convergence; surprise invoices
5	Evaluate on held-out data with error bars over seeds; run the ablation	20-point “100%” headlines; quantum ingredients that did nothing
6	Write the audit of your own study (the five questions)	shipping hype you would have caught in others

WORKED EXAMPLE — THE TEMPLATE, FILLED IN FOR THE RINGS STUDY

Fill the template for this book's running example, one line per step.

1. (1) 60 points frozen 42/18; tuned RBF-SVM baseline: 100% test — the bar is *parity at smaller model size*, stated before training.
2. (2) Both features scaled into $[0, \pi]$; ranges logged in the model card.
3. (3) One qubit, three re-uploading layers — 15 parameters, drawable by hand.
4. (4) Parameter-shift training, 3 restarts, 260 steps; shot budget priced by Chapter 25's lesson *before launch*.
5. (5) Test 100% ($\pm$ error bars over seeds and shots); ablation: classical kernel in the same pipeline also strong — so the claim is *parity-at-tiny-size*, not supremacy.
6. (6) Self-audit passes B, D, S; A honestly answered “no advantage claimed”; H states the scaling wall. Publishable — *because* of what it does not claim.

Answer: the same study that a vendor might headline as “quantum AI achieves 100%” becomes, through the template, a modest true sentence: *a 15-parameter quantum model matches a tuned classical baseline on a toy dataset*. Learning to prefer the second sentence is this part's real graduation exam.

REALITY CHECK

Keep the template when you leave this book: it is field-agnostic. Steps 1, 2, 5, 6 are simply good machine learning; steps 3 and 4 are the quantum-specific survival rules. And the report it produces — claims sized to evidence, costs stated, ablations run — is precisely what reviewers, employers, and funding committees now ask of quantum machine learning. The field's credibility is rebuilt one honestly-reported small study at a time; yours can be among them.

CHAPTER SUMMARY

- The scoreboard: mechanisms established (training, gradients, kernels), constructed-problem advantages proven, natural-data advantage open, big-data linear-algebra claims dequantized, quantum-data advantage proven and demonstrated, deep-training scaling obstructed by plateaus.
- Dequantization's lessons are permanent: price the data access, demand architectural (not assumed) advantage, and treat corrections as the field's immune system.
- Born machines make sampling native — one shot, one sample — sitting atop believed sampling separations; training and usefulness remain research-grade.
- Real deployments are hybrid: a small quantum slot (kernel, features, sampler, optimizer) inside a classical pipeline, validated by ablation.
- DASHB — Data access, Ablation, Scaling, Hardware honesty, Baselines — is your permanent audit tool; and for our region, QML's low hardware bar makes honest benchmarking and design genuine, publishable entry points.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] From the scoreboard: name one “established,” one “open,” and one “withdrawn” claim, each in your own words.

Problem 2 [Basic] What single question defeated the quantum recommendation-system speedup, and who asked it?

Problem 3 [Basic] Why is a measured quantum circuit automatically a sampler? Cite the rule and chapter.

Problem 4 [Basic] Expand DASHB from memory, one clause each.

Problem 5 [Basic] Rewrite the vendor claim so that it would pass question H: list the three experimental details the new sentence must contain.

Problem 6 [Basic] Name the ablation experiment for the vendor’s system, and state the two possible outcomes with what each would prove.

Problem 7 [Basic] Order these correctly and name the trap each prevents: train the quantum model; tune the classical baseline; freeze the splits.

Problem 8 [Basic] Which two template steps are quantum-specific, and which chapters of this part justify them?

Problem 9 [Standard] Ablation design: a hybrid pipeline uses a quantum kernel plus classical SVM for crop-disease detection. Specify the exact ablation run, the metric to compare, and the two additional DASHB letters most at risk for this setup.

Problem 10 [Standard] A paper reports a Born machine matching a classical generative model on 20-dimensional synthetic data, trained on a simulator. Which scoreboard rows does this move, and which does it not touch? Three sentences.

Problem 11 [Standard] The access-fairness principle: restate Lesson 2’s first lesson as a rule for referees, then apply it to this claim: “our quantum PCA is exponentially faster (data assumed amplitude-encoded).”

Problem 12 [Standard] Sort into hybrid slots (kernel / features / sampler / optimizer): (a) proposing candidate battery molecules for classical screening; (b) adding ten measured circuit statistics as extra spreadsheet columns; (c) computing similarity tables for a legal-document SVM; (d) attacking the routing sub-problem of a logistics stack.

Problem 13 [Standard] Regional project scoping: draft a one-paragraph master’s-thesis proposal following the Reality Check’s pattern — dataset, three feature maps, tuned baseline, DASHB reporting — for a dataset plausible at a university in Erbil or Sulaymaniyah.

Problem 14 [Standard] A paper reports a quantum kernel beating a linear SVM 88% to 81% on 40 test points. Using shot-noise reasoning ($\sigma \approx \sqrt{p(1-p)/n}$ with $n = 40$), decide whether the 7-point gap is even statistically meaningful. Show the arithmetic.

Problem 15 [Standard] Audit this stronger claim with the five questions, marking pass/fail/needs-info for each: “On 60 samples of 2 features, our 1-qubit re-uploading classifier reached 100% test accuracy (500 hardware shots per point, 5 seeds, $\pm 2\%$); a tuned RBF-SVM also reached 100%; we claim only parity at drastically smaller model size.”

Problem 16 [Standard] Question D hides a subtlety: explain how classical preprocessing (PCA, feature selection) can smuggle the real learning out of the quantum model, and design the control experiment that measures the preprocessing’s own accuracy.

Problem 17 [Standard] A study skipped step 5’s ablation and claims its quantum kernel drove the 94% result. Write the two-sentence reviewer request, and the two possible ablation outcomes with their verdicts.

Problem 18 [Challenge] The sampling asymmetry: explain in a short paragraph why “hard to sample from classically” (the advantage-experiment circuits) does NOT immediately yield a useful generative model — naming the two missing ingredients (trainability toward a *target* distribution; usefulness of that distribution) and the chapter-dragons guarding the first.

Problem 19 [Challenge] Write the referee report: three paragraphs (strengths, concerns, verdict) for a submission claiming “quantum feature maps improve heart-disease prediction from 12 clinical features; accuracy 84% vs 81% classical; 6 qubits; simulator; 5000 shots per kernel entry.” Use at least four DASHB letters explicitly, praise what deserves praise, and state the single revision most likely to change your verdict.

Problem 20 [Challenge] The decade bet: state, in five sentences, where you expect QML’s first commercially decisive win — classical data, quantum data, or sampling — defending your pick with one theorem-grade fact, one hardware-trend fact (Chapter 30), and one economics fact (shot bills or baseline strength) from this book. There is no wrong answer — only undefended ones.

Problem 21 [Challenge] Write a one-page referee report (five short paragraphs, one per checklist letter) on the vendor claim, ending with a constructive experiment list that would change your verdict. Tone requirement: every criticism paired with the concrete evidence that would answer it.

Problem 22 [Challenge] Apply the full template, on paper, to a proposed study: “classify 100 patient records (8 features) with a 4-qubit VQC.” For each step, one concrete decision with numbers (splits, scaling window, architecture, shot budget via Chapter 25, evaluation plan, audit risks) — a one-page project plan ready for capstone Project 9’s rubric.

PART EIGHT

REAL MACHINES, REAL LIMITS, REAL PROJECTS

Noise, Decoherence, and Errors

Why qubits fail — T1, T2, gate errors, readout errors, and the fight to compute anyway

IN THIS CHAPTER YOU WILL LEARN TO

- Explain why qubits are fragile: the same couplings that control them also disturb them.
- Use the lifetimes T1 (energy relaxation) and T2 (phase memory) to predict how long a qubit stays useful.
- Read an error budget: gate errors, readout errors, and idle errors, and combine them across a circuit.
- Estimate a circuit's survival probability from its depth, and judge what today's machines can and cannot run.
- Describe error mitigation — squeezing better answers from noisy data — and how it differs from error correction.

28.1 Why Qubits Are Fragile

Chapter 1 praised the classical bit's ruggedness: a stored “1” in your phone survives years of heat, vibration, and cosmic rays. A quantum bit enjoys no such peace. The reason is a genuine dilemma, not sloppy engineering. To compute, we must *couple* to the qubit — shine lasers, send microwave pulses, read signals out. But any channel through which we can disturb the qubit is a channel through which the environment can too. A perfectly isolated qubit would keep its state forever — and be perfectly useless, since no gate could reach it.

WORD TO KNOW

Decoherence — The gradual loss of a qubit's quantum character — its amplitudes and phases — through unwanted interaction with its environment. The environment “measures” the qubit continuously, ruining superpositions the way a mid-circuit measurement did in Chapter 6.

Recall from Chapter 6 what an accidental measurement costs: the HH interference experiment collapses to a coin flip. Decoherence is that catastrophe in slow motion, running constantly, on every qubit. The entire engineering of quantum hardware — dilution refrigerators near absolute zero, vacuum chambers, magnetic shielding (Chapter 30) — is one long fight to slow it down.

28.2 T1 and T2: The Two Lifetimes

Two standard clocks quantify how fast a qubit forgets. Hardware datasheets quote both for every qubit, and you should read them the way readers of this book's sister project, *Solar Energy for Everyone*, learn to read a solar panel's datasheet.

WORD TO KNOW

T1 (relaxation time) — The lifetime of the qubit's energy: prepared in $|1\rangle$, the qubit decays toward $|0\rangle$, surviving with probability e^{-t/T_1} . After one T1, only 37% remains.

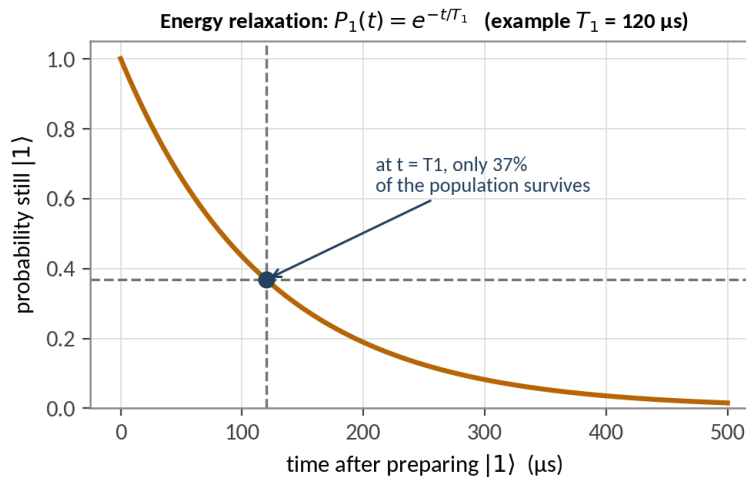
EQUATION: ENERGY RELAXATION

$$P_1(t) = e^{-t/T_1}$$

In words: The probability of still finding $|1\rangle$ decays exponentially, with time constant T1

Symbol	Meaning	Unit
$P_1(t)$	probability the qubit is still $ 1\rangle$ after waiting time t	none
t	the waiting (idle) time	μs
T_1	the relaxation time of this particular qubit	μs

Example scales around 2026: superconducting qubits $T_1 \sim 0.1\text{--}1\text{ ms}$; trapped ions: seconds to minutes. Numbers improve steadily — always read the current datasheet.

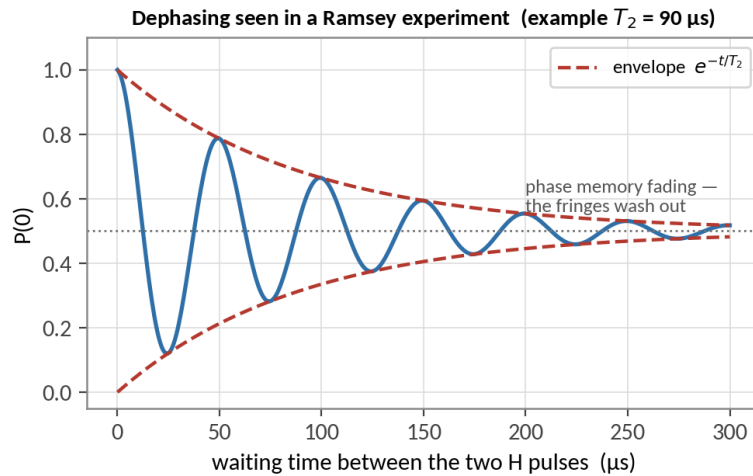


Relaxation: an excited qubit's slow fall to $|0\rangle$. At $t = T_1$ the survival probability has dropped to $1/e \approx 37\%$.

WORD TO KNOW

T2 (dephasing time) — The lifetime of phase memory: how long the relative phase in $\alpha|0\rangle + e^{i\varphi}\beta|1\rangle$ stays trustworthy. Since phases power all interference, T2 is the deadline for any quantum algorithm. Always $T_2 \leq 2T_1$; in practice often $T_2 < T_1$.

Phase is measured by the Ramsey experiment you already understand from Chapter 7's atomic-clock case study: H, wait a time t , H, measure. With perfect phase memory the interference fringes persist forever; with dephasing they wash out with envelope e^{-t/T_2} :



A Ramsey fringe decaying: the oscillation shows the accumulating phase; the shrinking envelope shows the phase becoming unreliable — superpositions aging into classical coin flips.

WORKED EXAMPLE — HOW LONG IS A QUBIT USEFUL?

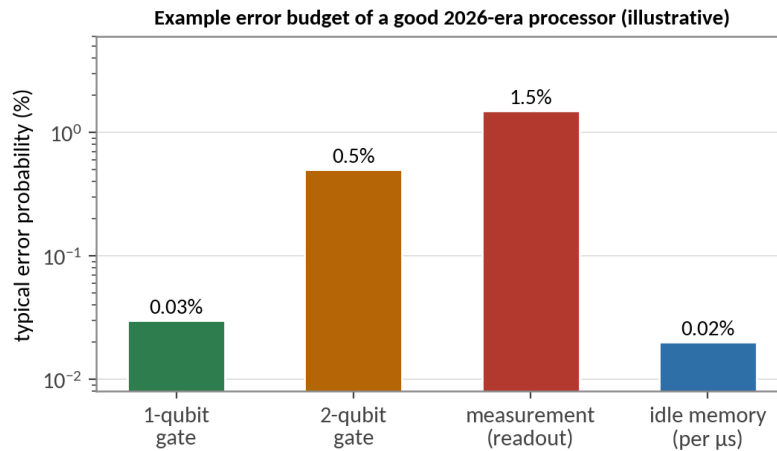
A qubit has $T_1 = 120 \mu\text{s}$ and $T_2 = 90 \mu\text{s}$. Its two-qubit gates take $0.3 \mu\text{s}$. Roughly how many sequential gates fit inside the phase-memory budget?

1. The tighter clock governs interference: $T_2 = 90 \mu\text{s}$.
2. Naive fit: $90 \div 0.3 = 300$ gate slots before one T_2 has passed.
3. But at $t = T_2$ the phase is already badly degraded ($e^{-1} \approx 37\%$ fidelity envelope); a practical budget keeps total time under $\sim 0.1 T_2$: about 30 gate slots for high-quality work.

Answer: around 30 clean sequential two-qubit gates — a few hundred at degraded quality. This single estimate explains the shape of near-term quantum computing: shallow circuits (Chapter 20's variational algorithms) fit; textbook Shor at scale (Chapter 19) does not. (Fortunately gates also fail for their own reasons — next lesson — so the real budget is usually even tighter.)

28.3 Gate Errors and Readout Errors

Even within the coherence window, each operation is slightly imperfect — a pulse fractionally too long, a stray coupling to a neighbor. Three error types dominate the budget:



An example error budget (illustrative, but realistically shaped): two-qubit gates cost roughly ten times more error than single-qubit gates, and readout is often the most error-prone single step of all.

WORD TO KNOW

Gate fidelity — The probability a gate does exactly its job; the error rate is $1 - \text{fidelity}$. A “99.5% two-qubit gate” errs once per 200 uses. Two-qubit gates are consistently the weakest link — which is why compilers count CNOTs (Chapters 10, 12).

WORD TO KNOW

Readout error — The probability a measurement reports the wrong classical bit — flipping a real $|1\rangle$ result into a reported 0 or vice versa. Typically 0.5–3% per qubit around 2026.

Errors compound across a circuit. If each of d steps independently succeeds with probability $(1 - p)$, the whole run survives with probability $(1 - p)^d$ — the same exponential arithmetic as Chapter 1’s doubling, now working against us:

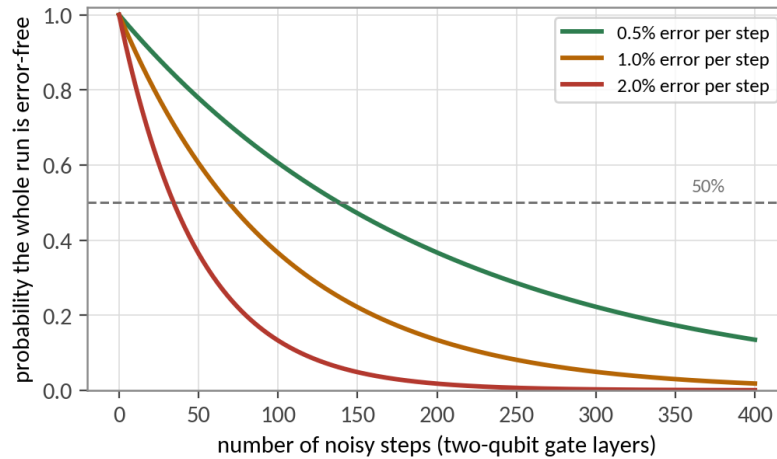
EQUATION: CIRCUIT SURVIVAL PROBABILITY

$$P_{\text{clean}} \approx (1 - p)^d$$

In words: Probability the whole circuit runs error-free $\approx$ (per-step success) multiplied over all steps

Symbol	Meaning	Unit
p	the dominant per-step error probability (usually the two-qubit gate)	none
d	the number of such steps (roughly, circuit depth)	none

Rule of thumb: trouble begins near $d \approx 1/p$. At $p = 1\%$: about 100 steps. This is why “how many CNOT layers?” is the first question a hardware engineer asks about your algorithm.



The depth wall: survival probability versus depth for three per-step error rates. Halving the error rate roughly doubles the usable depth — which is why fidelity, not qubit count, is the headline metric professionals watch.

WORKED EXAMPLE — BUDGETING A REAL RUN

Your Bell-state circuit compiles to 1 two-qubit gate ($p_2 = 0.6\%$), 2 single-qubit gates ($p_1 = 0.05\%$ each), and 2 readouts ($p_r = 1.5\%$ each). Predict the corrupted-shot fraction and compare with Chapter 14’s observed 4.7%.

1. Success probability: $(1 - 0.006) \times (1 - 0.0005)^2 \times (1 - 0.015)^2$.
2. Evaluate: $0.994 \times 0.999 \times 0.9702 \approx 0.9634$.
3. Corruption estimate: $1 - 0.963 \approx 3.7\%$, dominated by readout ($\sim 3\%$) and the CNOT (0.6%).

Answer: about 4% corrupted shots — matching the 4.7% measured in Chapter 14 within statistical wobble. Reading a datasheet forward to a prediction, then checking it against counts, is exactly how professionals qualify a device.

COMMON MISTAKE

X Wrong: “Our chip has 1000 qubits, so it is more powerful than their 100-qubit chip.”

✓ Right: Ask for error rates before counting qubits. A useful figure of merit multiplies both: 100 qubits at 0.1% two-qubit error can run depth- ~ 1000 circuits; 1000 qubits at 2% error die by depth 50. (Composite

benchmarks like “quantum volume” exist precisely to punish this confusion — Chapter 30.) Count and quality must grow *together*; either alone is marketing.

28.4 Error Mitigation: Better Answers from Noisy Data

Until full error correction arrives (Chapter 29), practitioners use **error mitigation**: classical statistical tricks that trade extra runs for better estimates — without any extra qubits.

The mitigation toolbox (all in production use today)

Technique	Idea in one sentence
Readout-error correction	Measure each qubit’s flip rates on known states, then statistically un-flip your real histograms.
Zero-noise extrapolation	Deliberately run the circuit at 1×, 2×, 3× the noise (stretching pulses), then extrapolate the results back to zero noise.
Probabilistic error cancellation	Learn the device’s noise model, then run an ensemble of modified circuits whose weighted average cancels it.
Dynamical decoupling	Insert echo pulses (X–wait–X) into idle periods so slow phase drifts cancel themselves.

Mitigation’s honest price tag: it reduces *bias*, not uncertainty — and its sampling cost grows exponentially with circuit size. It extends the reach of today’s machines; it does not repeal the depth wall. That repeal requires error *correction*, the subject of the next chapter.

REALITY CHECK

Keep the two words straight forever: *mitigation* is statistics on top of a noisy computation (no extra qubits; exponential sampling cost; today’s tool). *Correction* is redundancy inside the computation (many extra qubits; exponentially *suppressed* errors; tomorrow’s foundation — and Chapter 29’s subject). Vendors blur them; a professional never does.

CASE STUDY — THE DECOHERENCE CENTURY • FROM SCHRÖDINGER’S CAT TO DATASHEETS

In 1935, Schrödinger’s famous cat argument mocked the idea of macroscopic superpositions: why do we never see a cat both alive and dead? The modern answer — developed by Zeh, Zurek, and others from the 1970s on — is decoherence: big objects interact with so many environmental degrees of freedom that their superpositions “leak” into the world almost instantly; a dust grain decoheres in nanoseconds, a cat unimaginably faster. The same physics, measured rather than mocked, became the engineering discipline of this chapter: T1 and T2 are decoherence, tamed into datasheet numbers. Between 1999 and today, superconducting-qubit coherence improved from about one *nanosecond* to around a *millisecond* — six orders of magnitude in twenty-five years, won layer by layer (better materials, better shielding, better geometry). The fragility is fun-

damental; the progress is relentless; both facts belong in every honest sentence about quantum hardware.

ACTIVITY (APP) — BREAK A CIRCUIT ON PURPOSE

1. In the *Noise Lab* of the QCFE app, load the Bell circuit with noise OFF: run 1000 shots and admire the clean histogram.
2. Switch on realistic noise (defaults: $p_2 = 0.6\%$, $p_r = 1.5\%$). Run, and compare the corrupted fraction with this chapter's budgeting example.
3. Drag the T2 slider down from 90 μs to 10 μs and watch the Ramsey demo's fringes collapse.
4. Load the depth-test circuit (repeated CNOT pairs) and increase depth 10 $\rightarrow$ 50 $\rightarrow$ 200, recording the ideal-answer fraction. Plot your three points; do they follow $(1 - p)^d$?
5. Switch on readout mitigation and re-run step 2: watch the histogram straighten — and note the shot count the app spent to do it.

You need: The QCFE app (Android or Windows), Paper and pen

► App tool: *Noise Lab* — dials for T1, T2, gate and readout errors on any circuit, plus one-press mitigation to see the cost and the benefit

28.5 Measuring T_1 and T_2 Yourself

Datasheet lifetimes are not looked up — they are *measured*, by two protocols so standard that every quantum cloud dashboard runs them hourly. Both fit in five circuit lines, and both are excellent first experiments on real hardware.

EQUATION: THE TWO LIFETIME PROTOCOLS

$$\mathbf{T}_1: \quad P_1(t) = e^{-t/T_1} \qquad \mathbf{T}_2 \text{ (Ramsey)}: \quad P_1(t) = \frac{1}{2} \left(1 + \cos(2\pi\delta t) e^{-t/T_2} \right)$$

In words: T1: prepare 1, wait, ask if it survived. T2: prepare +, wait, un-rotate, ask if the phase survived — a fading interference fringe

Symbol	Meaning	Unit
t	the deliberately inserted waiting time	μs
δ	small intentional detuning (sets the fringe rate)	MHz
e^{-t/T_2}	the fringe's shrinking envelope	none

Physics guarantees $T_2 \leq 2T_1$: energy loss also scrambles phase. Datasheets distinguish T_2^* (plain Ramsey, pessimistic) from T_2 echo (with a refocusing X pulse mid-wait, kinder) — read the footnote before comparing machines.

WORD TO KNOW

Ramsey experiment — The phase-lifetime protocol: H, wait t , H, measure, swept over t . Its oscillating signal decays with envelope e^{-t/T_2} ; the oscillation's frequency also reveals calibration errors — two diagnostics for the price of one.

WORKED EXAMPLE — AN INVERSION-RECOVERY RUN

Protocol: X gate (prepare $|1\rangle$), wait t , measure; 1000 shots per point. On a qubit with $T_1 = 100\ \mu\text{s}$, predict $t = 50\ \mu\text{s}$ and $t = 100\ \mu\text{s}$, then invert a measurement: a different qubit shows $P_1 = 0.50$ at $t = 50\ \mu\text{s}$ — find its T_1 .

1. Predictions: $P_1(50) = e^{-0.5} \approx 0.607$ (about 607 of 1000 shots); $P_1(100) = e^{-1} \approx 0.368$.
2. Inversion: $0.50 = e^{-50/T_1} \Rightarrow T_1 = 50/\ln 2 \approx 72\ \mu\text{s}$.
3. Professional habit: fit *all* points to one exponential rather than trusting one; a single misbehaving point then shows itself.

Answer: ≈ 607 and ≈ 368 counts; the second qubit's $T_1 \approx 72\ \mu\text{s}$. The half-life-to-lifetime conversion $T_1 = t_{1/2}/\ln 2$ is worth memorizing — it turns any survival plot into a lifetime at sight.

WORKED EXAMPLE — READING A RAMSEY FRINGE

A Ramsey run uses detuning $\delta = 0.25\ \text{MHz}$ on a qubit with $T_2 = 200\ \mu\text{s}$. Find the fringe period, and the fringe amplitude remaining at $t = 50\ \mu\text{s}$.

1. Period: $1/\delta = 4\ \mu\text{s}$ per full oscillation — the histogram swings 1-heavy, 0-heavy, 1-heavy every $4\ \mu\text{s}$ of waiting.
2. Envelope at $50\ \mu\text{s}$: $e^{-50/200} = e^{-0.25} \approx 0.78$; the swing still covers 78% of its full range.
3. Extracting T_2 in practice: record the swing amplitude at several t , fit the shrinking envelope; the oscillation frequency separately confirms δ (any surprise there means the qubit's frequency calibration has drifted).

Answer: $4\ \mu\text{s}$ fringes inside a $e^{-t/200\ \mu\text{s}}$ envelope, at 78% strength by $t = 50\ \mu\text{s}$. A dead-flat Ramsey signal at short times is diagnostic gold: it means dephasing so fast the fringe dies before the first swing — check the wiring, not the qubit.

REALITY CHECK

Lifetimes drift — between days, between cooldowns, even between hours, as stray two-level defects in the chip wander in and out of resonance. Serious labs (and serious cloud dashboards) republish T_1/T_2 maps continuously, and serious users check the morning's numbers before believing yesterday's circuit budget. Treat every lifetime as a weather report, not a constant of nature.

28.6 Zero-Noise Extrapolation, By Hand

Error mitigation sounded abstract earlier in this chapter; one flagship technique is so concrete you can run it with a ruler. **Zero-noise extrapolation (ZNE)**: measure your quantity at the machine’s natural noise level, measure it again with the noise *deliberately made worse* by a known factor, then extend the trend line backwards to the noise level you cannot have — zero.

EQUATION: ZNE IN TWO POINTS

$$E(\lambda) \approx a + b\lambda \quad \implies \quad E(0) \approx E(1) + \frac{E(1) - E(3)}{2}$$

In words: Measure at noise scale 1 and (folded) scale 3; the zero-noise estimate is the straight line’s intercept

Symbol	Meaning	Unit
λ	noise scale factor (1 = the machine as-is)	none
$E(\lambda)$	the measured expectation value at scale λ	varies
$E(0)$	the extrapolated noiseless estimate	varies

How to worsen noise on purpose: gate folding — replace each gate G by $G G^\dagger G$ (same unitary, three times the exposure, $\lambda = 3$). Richer versions use several λ values and curved fits; two points and a line already capture the idea.

WORKED EXAMPLE — EXTRAPOLATING AN ENERGY BACK TO HONESTY

A VQE energy term reads $E(1) = 0.82$ on the machine, and $E(3) = 0.58$ with every gate folded. Estimate the noiseless value, and audit the method’s assumptions.

1. Slope of the damage: $(0.58 - 0.82)/(3 - 1) = -0.12$ per unit of noise scale.
2. Extrapolate to zero: $E(0) = 0.82 + 0.12 = 0.94$.
3. Sanity-check the direction: noise shrinks expectation values toward zero (Chapter 28’s depolarizing picture), so the corrected value moving away from zero is expected.
4. Audit: the estimate trusts (i) linearity over $\lambda \in [0, 3]$ — dubious if the circuit is deep enough that E has mostly collapsed already — and (ii) shot noise: each E carries wobble, and extrapolation amplifies it by the lever-arm geometry.

Answer: $E(0) \approx 0.94$, up from a raw 0.82 — a 15% correction bought with one extra (folded) experiment and one line of arithmetic. ZNE and readout inversion (Chapter 14) are the mitigation pair that rescued most published near-term results — including the error-bar honesty that must accompany them.

WORD TO KNOW

Zero-noise extrapolation — Estimating a noiseless expectation value by measuring at amplified noise levels and extrapolating the trend to zero noise. Costs extra runs, not extra qubits; trusts a smoothness assumption that must be checked.

REALITY CHECK

Mitigation's economics deserve one honest sentence each. ZNE multiplies runtime (each λ is a full experiment) and cannot help once noise has flattened the signal into the floor — extrapolating from mush yields confident mush. Probabilistic error cancellation, its stronger sibling, needs exponentially many samples as circuits grow. This is why mitigation is a bridge technology for the NISQ era, and error *correction* (Chapter 29) — costlier but scalable — is the destination. Use the bridge; do not build a city on it.

CHAPTER SUMMARY

- Qubits are fragile by dilemma: control channels are disturbance channels; decoherence is continuous accidental measurement by the environment.
- T_1 clocks energy decay ($P_1 = e^{-t/T_1}$); T_2 clocks phase memory (Ramsey envelope e^{-t/T_2} , $T_2 \leq 2T_1$) — and phase is where algorithms live.
- Error budget: single-qubit gates ~ 0.01 – 0.1% , two-qubit gates ~ 0.1 – 1% , readout ~ 0.5 – 3% (2026-era, improving). Survival $\approx (1 - p)^d$: trouble at depth $\sim 1/p$.
- Predict-then-measure: datasheet numbers forecast corrupted-shot fractions — and Chapter 14's real histogram confirmed the forecast.
- Mitigation (readout un-flipping, zero-noise extrapolation, echoes) buys accuracy with extra shots, not extra qubits; it stretches the depth wall but only correction (next chapter) removes it.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] A qubit with $T_1 = 200 \mu\text{s}$ is prepared in $|1\rangle$. What is the probability it is still $|1\rangle$ after $200 \mu\text{s}$? After $400 \mu\text{s}$?

Problem 2 [Basic] Which lifetime — T_1 or T_2 — limits an interference experiment, and why (one sentence)?

Problem 3 [Basic] A two-qubit gate has fidelity 99.2%. What is its error rate, and how many uses until the expected number of errors reaches 1?

Problem 4 [Basic] Using $P \approx (1 - p)^d$ with $p = 1\%$: estimate the survival probability at depth 50 (use $(0.99)^{50} \approx 0.605$).

Problem 5 [Basic] A qubit has $T_1 = 80 \mu\text{s}$. Predict the surviving-1 fraction at $t = 80 \mu\text{s}$ and at $t = 160 \mu\text{s}$, and state the general rule for whole multiples of T_1 .

Problem 6 [Basic] Write the two five-line protocols (gate list per shot) for T_1 and for Ramsey T_2 , marking where the waiting time goes in each.

Problem 7 [Basic] $E(1) = 0.40$, $E(3) = 0.28$: compute the ZNE estimate $E(0)$, and state the slope's sign check.

Problem 8 [Basic] Explain gate folding $G \rightarrow GG^\dagger G$: why is the ideal circuit unchanged, and why does the noise triple (Chapter 28's per-gate error picture)?

Problem 9 [Standard] A datasheet says $T_2 = 60 \mu\text{s}$ and single-qubit gates take 25 ns. How many sequential single-qubit gates fit within 0.1 T_2 ? Why do compilers still worry mainly about the (slower, noisier) two-qubit gates?

Problem 10 [Standard] Your GHZ circuit uses 2 CNOTs ($p_2 = 0.8\%$ each), 1 Hadamard (0.05%), and 3 readouts (1.2% each). Predict the corrupted-shot fraction, and identify the single largest contributor.

Problem 11 [Standard] Hardware A: 0.5% two-qubit error. Hardware B: 0.25%. Using the $d \approx 1/p$ rule, compare usable depths, and state in one sentence why “2× better gates” can matter more than “10× more qubits.”

Problem 12 [Standard] Readout un-flipping: calibration shows this qubit reports a true 1 as “0” 4% of the time (and true 0 as “1” 1%). A run returns 300 zeros in 1000 shots on a state you believe is $|1\rangle$ -heavy. Is the raw $P(0) = 0.30$ an over- or under-estimate of the true value, and roughly what correction direction does mitigation apply?

Problem 13 [Standard] A Ramsey experiment's fringes vanish into the 0.5 line after about 150 μs of waiting. Estimate this qubit's T_2 to one significant figure, stating the reasoning from the envelope e^{-t/T_2} .

Problem 14 [Standard] An inversion-recovery run gives $P_1 = 0.37$ at $t = 100 \mu\text{s}$ and $P_1 = 0.135$ at $t = 200 \mu\text{s}$. Show both points are consistent with one T_1 , and give it.

Problem 15 [Standard] With 1000 shots per point and wobble $\sigma \approx 0.5/\sqrt{n}$ (Chapter 4), can you reliably distinguish $P_1 = 0.61$ from $P_1 = 0.55$ at one time point? Show the arithmetic, and name the cheap fix.

Problem 16 [Standard] Explain why T_2^* (plain Ramsey) is usually shorter than T_2 echo on the same qubit, and which one better predicts a long algorithm's phase coherence.

Problem 17 [Standard] Each measurement carries wobble ± 0.02 . Propagate through the two-point formula to find the wobble of $E(0)$ (the coefficients are $\frac{3}{2}$ and $-\frac{1}{2}$; add variances). What does the answer say about mitigation's shot bill?

Problem 18 [Challenge] Two error channels compete on an idle qubit: relaxation ($T_1 = 100 \mu\text{s}$) and pure dephasing. The measured T_2 is 70 μs . Using the standard relation $\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\varphi}$, extract the pure-dephasing time T_φ . Which improvement — doubling T_1 or doubling T_φ — would raise T_2 more? Compute both.

Problem 19 [Challenge] Grover meets the depth wall: Chapter 18's million-entry search needed ≈ 785 rounds; suppose each round compiles to 40 two-qubit steps. At $p_2 = 0.1\%$, estimate the survival probability of the

full run (use $(1 - p)^d \approx e^{-pd}$), and state the smallest p_2 (order of magnitude) at which the search becomes plausible. What technology from the next chapter is this problem begging for?

Problem 20 [Challenge] Design a coherence experiment: with only the app's (or Qiskit's) tools — state preparation, waits, H gates, measurement, histograms — write two short numbered protocols, one that measures T_1 and one that measures T_2 , four sentences each, including what you plot and how you read the lifetime off the plot. (You have re-invented two of the standard calibration experiments every quantum lab runs daily.)

Problem 21 [Challenge] You suspect a “ T_2 problem” is actually a miscalibrated gate over-rotating by a fixed small angle each shot. Design measurements that separate the two hypotheses (hint: one produces a *decaying random* spread; the other a *repeatable deterministic* offset that flips sign under pulse reversal), and state the telltale signature of each.

Problem 22 [Challenge] A colleague reports $E(1) = 0.10$, $E(3) = 0.09$, extrapolates $E(0) = 0.105$, and celebrates precision. Using the “mush floor” warning, explain the more likely reality (both readings sit at the depolarized floor ≈ 0 plus noise), design the control experiment that exposes it (hint: fold to $\lambda = 5$, or ZNE a circuit whose true answer is known), and restate the general rule about extrapolating flat lines.

CHAPTER TWENTY-NINE

Quantum Error Correction

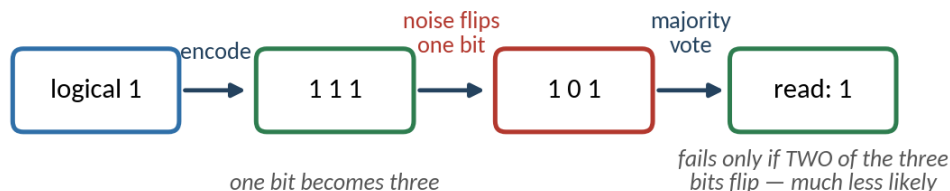
Protecting qubits without copying or looking — repetition codes, syndromes, and the surface code

IN THIS CHAPTER YOU WILL LEARN TO

- Explain why quantum error correction seems impossible — no cloning, no looking — and how both barriers fall.
- Encode one logical qubit into the three-qubit bit-flip code and trace an error through detection and repair.
- Measure syndromes: extract *where* an error struck without learning *what* the data is.
- Describe the surface code, code distance, and the threshold theorem — why below-threshold hardware can be scaled to any reliability.
- Count the overhead honestly: physical qubits per logical qubit, and what that means for timelines.

29.1 Why We Cannot Just Copy — and What We Can Do

Classical machines conquered noise with redundancy: store a bit three times, take majority votes. Chapter 28's depth wall says quantum computing needs something equally strong — but the obvious plan hits two walls this book has already built. **No cloning** (Chapter 12): an unknown state $\alpha|0\rangle + \beta|1\rangle$ cannot be copied three times. **No looking** (Chapter 4): checking a qubit “for errors” by measuring collapses the very superposition being protected.



The classical cure: triplicate and vote. One flipped bit is outvoted; failure needs two simultaneous flips. Quantum computing needs this power — without copying and without looking.

The escape (Shor 1995, Steane 1996) is one of the cleverest ideas in modern science, and it fits in one sentence: *spread the information into entanglement, and measure only questions whose answers are the same for $|0\rangle$ -ish and $|1\rangle$ -ish data — parity questions — so that errors are revealed while data stays secret.*

WORD TO KNOW

Logical qubit — One protected qubit of information, stored non-locally in the entangled state of many physical qubits. The algorithm sees one clean qubit; the hardware runs many noisy ones beneath it.

29.2 The Three-Qubit Bit-Flip Code

The training-wheels code protects against one error type — bit flips (X errors) — and exhibits every idea the full machinery uses.

Encoding. Not copying, but entangling: two CNOTs from the data qubit onto two fresh $|0\rangle$ s.

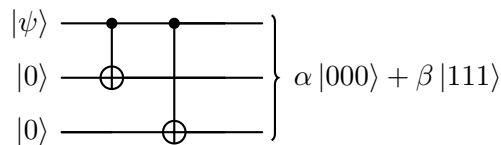
EQUATION: THE BIT-FLIP CODE

$$\alpha |0\rangle + \beta |1\rangle \xrightarrow{\text{encode}} \alpha |000\rangle + \beta |111\rangle$$

In words: The logical state lives across three qubits: "all zeros" carries the α amplitude, "all ones" carries β

Symbol	Meaning	Unit
$ 000\rangle$	the code word carrying the α amplitude	none
$ 111\rangle$	the code word carrying the β amplitude	none

Compare $(\alpha |0\rangle + \beta |1\rangle)^{\otimes 3}$ — the forbidden triple copy — which expands into eight terms. The code state has only two: it is a GHZ-style entangled state (Chapter 11, Problem 10), fully legal, cloning nothing.



Encoding: two CNOTs entangle the data across three qubits. (Exactly the basis-state "copier" of Chapter 10 — put to lawful work.)

WORKED EXAMPLE — AN ERROR STRIKES — AND IS CAUGHT

Noise applies X to the middle qubit of $\alpha |000\rangle + \beta |111\rangle$. Show the state before and after, and check the damage is repairable.

1. Apply X to slot 2 in each term: $|000\rangle \rightarrow |010\rangle$ and $|111\rangle \rightarrow |101\rangle$.

2. Corrupted state: $\alpha |010\rangle + \beta |101\rangle$.

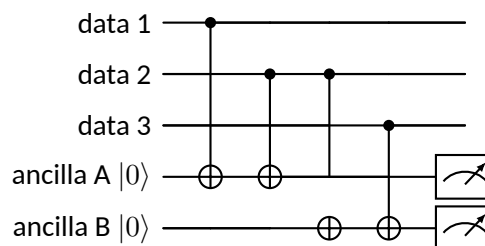
3. Observe the fingerprint: in *both* terms, the middle qubit now disagrees with its neighbors — the disagreement pattern is the same whatever α, β are.

4. Repair (once located — next lesson): apply X to the middle qubit; both terms return simultaneously: $\alpha |000\rangle + \beta |111\rangle$.

Answer: the error moved the state to a recognizable “wrong ward” of the state space without touching α or β . One X undoes it exactly — for every superposition at once. The amplitudes never appeared in the diagnosis.

29.3 Syndromes: Finding Errors Without Looking at Data

How do we *locate* the flip without measuring the data? Ask only **parity questions**: “are qubits 1 and 2 equal?” and “are qubits 2 and 3 equal?” Each is measured by a helper (*ancilla*) qubit: two CNOTs write the parity into the ancilla, and measuring the *ancilla* reveals agree/disagree — nothing more.



Syndrome extraction: ancilla A collects the parity of data qubits 1,2; ancilla B the parity of 2,3. Only the ancillas are measured — the data qubits are never looked at.

WORD TO KNOW

Syndrome — The pattern of parity-check results. It identifies which error occurred and where — while carrying zero information about the encoded amplitudes, so measuring it collapses nothing that matters.

The bit-flip code's complete diagnosis chart

Parity (1,2)	Parity (2,3)	Diagnosis and repair
agree	agree	no error — do nothing
disagree	agree	X struck qubit 1 — apply X to qubit 1
disagree	disagree	X struck qubit 2 — apply X to qubit 2
agree	disagree	X struck qubit 3 — apply X to qubit 3

WORKED EXAMPLE — RUNNING THE DIAGNOSIS

Continue the previous example: the corrupted state is $\alpha |010\rangle + \beta |101\rangle$. What do the two parity checks report?

1. Parity (1,2): in the term $|010\rangle$, qubits 1,2 read 0,1 — disagree. In $|101\rangle$: 1,0 — disagree. Same answer in both branches: the ancilla measures “disagree” with certainty, no collapse of the superposition.
2. Parity (2,3): $|010\rangle$: 1,0 disagree; $|101\rangle$: 0,1 disagree. Certain “disagree.”
3. Chart lookup: (disagree, disagree) $\Rightarrow$ X on qubit 2. Apply it: state restored.

Answer: syndrome = (disagree, disagree) $\rightarrow$ repair qubit 2 — matching where we secretly injected the error. Notice the miracle in step 1: because both branches answer every parity question identically, the measurement is *safe* — this is the design principle of every quantum code.

COMMON MISTAKE

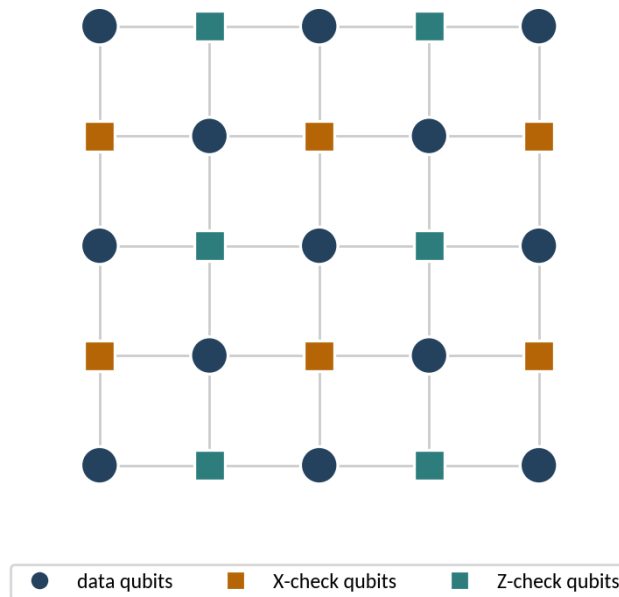
X Wrong: “Error correction reads the qubits, fixes them, and writes them back — like fixing a corrupted file.”

✓ Right: The data qubits are never read. Only ancillas are measured, and only about parities — questions whose answers are identical across the superposition’s branches. A code that measured data would destroy exactly what it guards. (This is also why the bit-flip code alone is incomplete: it says nothing about *phase* flips — Z errors — which need parity checks in the $|+\rangle / |-\rangle$ language. Full codes check both; see Problem 10.)

29.4 Toward the Surface Code: Distance and Threshold

Real codes must catch both X and Z errors (a Z on any code qubit attacks the phase between α and β — invisible to bit-parities), survive errors *during* the checking itself, and run forever in repeated cycles. The workhorse meeting all three demands is the **surface code**: data qubits on a grid, guarded by a checkerboard of ancillas measuring local X-parities and Z-parities, cycle after cycle, with a classical *decoder* watching the syndrome stream and inferring repairs in real time.

A surface-code patch: data qubits guarded by a checkerboard of check qubits



A surface-code patch. Every check involves only a qubit's nearest neighbors — exactly what flat chip hardware can provide. Grow the patch, and protection grows with it.

WORD TO KNOW

Code distance — The size d of the smallest error pattern that can fool the code (defeat needs about $d/2$ simultaneous well-placed errors). A distance- d surface-code patch uses roughly $2d^2$ physical qubits: $d=3 \rightarrow 17$, $d=11 \rightarrow$ about 250, $d=25 \rightarrow$ about 1250.

The payoff is the most important scaling law in quantum computing:

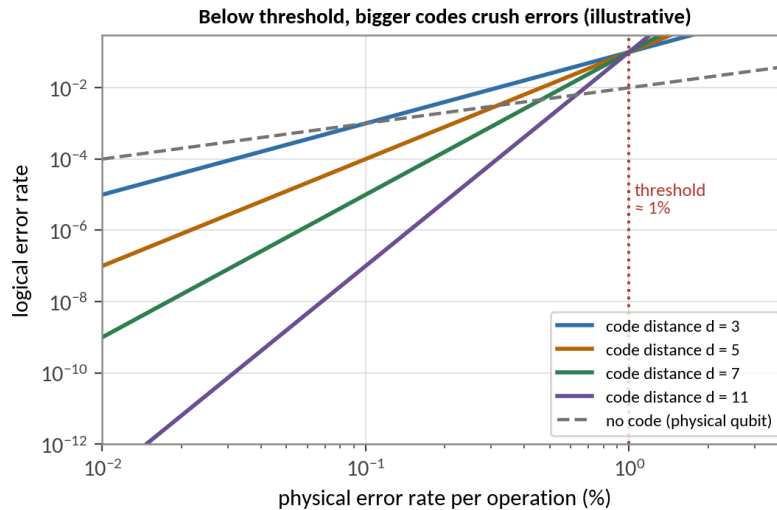
EQUATION: THE THRESHOLD LAW (SIMPLIFIED)

$$p_{\text{logical}} \sim A \left(\frac{p}{p_{\text{th}}} \right)^{(d+1)/2}$$

In words: If the physical error rate p is below the threshold p_{th} , every increase of the code distance multiplies the protection — logical errors shrink exponentially with d

Symbol	Meaning	Unit
p	the physical error rate per operation	none
p_{th}	the threshold ($\approx 1\%$ for the surface code)	none
d	the code distance	none
A	a constant of order 0.1	none

Below threshold, reliability is bought with size: each step $d \rightarrow d + 2$ multiplies the suppression by the factor p_{th}/p . Above threshold, bigger codes make things WORSE — the checking machinery injects errors faster than it removes them. Crossing this line is the defining milestone of the field (see the case study).



The threshold law drawn (illustrative): below $\sim 1\%$ physical error, each larger code crushes the logical rate further. At $p = 0.1\%$ with $d = 25$ -scale codes, logical rates reach the 10^{-12} regime that algorithms like full Shor demand.

WORKED EXAMPLE — PRICING A FAULT-TOLERANT FACTORING MACHINE

A future machine has $p = 0.1\%$ physical error. Shor on RSA-2048 needs roughly 4000 logical qubits at logical error $\sim 10^{-12}$. Estimate the physical qubit bill.

1. Suppression per distance step: $p_{th}/p = 10$. Need $(1/10)^{(d+1)/2} \cdot 0.1 \lesssim 10^{-12}$, so $(d+1)/2 \approx 11$, giving $d \approx 21$ — call it 25 with margin.
2. Physical qubits per logical: $\approx 2d^2 = 2 \times 625 = 1250$.
3. Total: $4000 \times 1250 = 5$ million physical qubits — plus overheads for the T gates Chapter 7 flagged as expensive (magic-state factories), which published designs fold into totals of about 1–20 million depending on cleverness; recent optimized estimates have pushed below one million.

Answer: order of millions of below-threshold physical qubits — versus today's thousands. This 1000× gap, not any mystery of principle, is the honest distance between the present and cryptographically relevant quantum computing. It is a factory-scaling problem now — which is precisely why progress became strikingly fast once the threshold was crossed.

CASE STUDY — CROSSING THE THRESHOLD • GOOGLE WILLOW AND THE ERROR-CORRECTION RACE, 2023–2025

For twenty-five years, the threshold theorem was a promise on paper: no experiment had shown a larger

code beating a smaller one. Google’s milestones changed that. In 2023, a distance-5 surface code finally edged out distance-3. In December 2024, the Willow chip (105 qubits, p comfortably below threshold) demonstrated the full scaling law: distance-3, -5, and -7 patches each cut the logical error rate roughly in half at every step, with logical qubits outliving the chip’s best physical qubit — the “below threshold, at scale” moment the field had chased since 1995. Rival platforms kept pace in their own styles: Harvard/QuEra’s neutral-atom system ran dozens of logical qubits in 2023–24, and trapped-ion machines demonstrated high-fidelity logical operations. None of these machines yet does useful protected computation — but the curve you studied in this chapter’s figure is now measured physics, not theory. Every serious roadmap dates its promises from that crossing.

REALITY CHECK

Error correction’s costs buy something subtle: not a noiseless machine, but an *arbitrarily reliable* one — reliability on demand, priced in qubits. The classical world made this exact trade invisibly: every phone call, disk, and satellite link runs on classical error correction you never see. Quantum computing is simply paying the same toll at a far harsher exchange rate. When headlines shout “1000 qubits!”, your first professional reflexes should be: below threshold or not? distance of what size demonstrated? logical error rate measured how? Those three questions — all from this chapter — separate roadmap from reality.

ACTIVITY (APP) — RUN THE REPAIR SHOP

1. In the *Error-Correction Lab* of the QCFE app, encode $|\psi\rangle = 0.6|0\rangle + 0.8|1\rangle$ into three qubits and confirm the state display shows two terms, not eight.
2. Inject an X error on a qubit of your choice; run syndrome extraction; check the ancilla readouts against the diagnosis chart; repair, and verify the Bloch display of the decoded qubit returns to the original.
3. Turn on random errors at 5% per qubit per cycle and run 50 cycles with auto-repair: log the survival. Then raise the rate to 60% and watch the code *fail* (double flips now beat the vote — the toy threshold sits at 50%).
4. Open the surface-code panel: watch syndrome flashes on a $d = 5$ patch, then grow to $d = 7$ at the same physical rate and compare logical failure counters — the threshold law, live.

You need: The QCFE app (Android or Windows), Paper and pen

► App tool: *Error-Correction Lab* — encode, inject, diagnose, repair; then scale surface-code patches and watch the threshold law emerge

29.5 Nine Qubits Against Everything: Shor’s Code

The three-qubit code repairs bit flips; a mirror-image code (the same trick in the $|+\rangle / |-\rangle$ basis) repairs phase flips. Peter Shor’s 1995 masterstroke: nest one inside the other, and *both* enemies — in fact, any error on any single qubit — become repairable. It was the result that convinced physicists quantum computing could

survive contact with reality.

EQUATION: THE SHOR CODE WORDS

$$|0_L\rangle = \frac{1}{2\sqrt{2}} (|000\rangle + |111\rangle) (|000\rangle + |111\rangle) (|000\rangle + |111\rangle)$$

$$|1_L\rangle = \frac{1}{2\sqrt{2}} (|000\rangle - |111\rangle) (|000\rangle - |111\rangle) (|000\rangle - |111\rangle)$$

In words: Three blocks of three: repetition inside each block fights bit flips; the pattern of signs across blocks fights phase flips

Symbol	Meaning	Unit
block	one factor of three qubits	none
$\pm$ sign	the block's phase — the outer code's "digit"	none
$ 0_L\rangle, 1_L\rangle$	the two logical code words	none

Eight stabilizer checks guard the nine qubits: six ZZ parities (two per block) locate bit flips to the qubit, and two six-qubit $XXXXXX$ comparisons (blocks 1-2 and 2-3) locate phase flips to the block. $9 - 8 = 1$ logical qubit survives the bookkeeping.

WORKED EXAMPLE — CATCHING A BIT FLIP ON QUBIT 4

An X error strikes qubit 4 (the first qubit of block 2). Read the syndrome and prescribe the repair.

1. Within block 2 the checks are Z_4Z_5 and Z_5Z_6 . An X_4 anticommutes with Z_4Z_5 (they clash on qubit 4) but commutes with Z_5Z_6 .
2. Syndrome: $Z_4Z_5 \rightarrow -1$, $Z_5Z_6 \rightarrow +1$; all other blocks' checks read $+1$; the two X -type checks also read $+1$ (X errors commute with X checks).
3. Decode with the three-qubit code's logic, block-locally: "first pair disagrees, second agrees" $\Rightarrow$ qubit 4. Apply X_4 .

Answer: syndrome $(-1, +1)$ inside block 2 fingers qubit 4 exactly; one X repairs it. The inner machinery is precisely the three-qubit code of this chapter, running independently in each block — concatenation means never learning new arithmetic, just stacking the old.

WORKED EXAMPLE — CATCHING A PHASE FLIP — TO THE BLOCK, AND WHY THAT SUFFICES

A Z error strikes qubit 4. Trace the syndrome, and explain the odd-looking repair rule.

1. All six ZZ checks still read $+1$: Z errors are invisible to Z parities.
2. The X -type comparisons: $X_1 \cdots X_6$ (blocks 1-2) anticommutes with $Z_4 \rightarrow -1$; $X_4 \cdots X_9$ (blocks 2-3) also contains qubit 4 $\rightarrow -1$.
3. Syndrome $(-1, -1)$ says only: "the phase flip is in block 2" — it cannot name the qubit. (Z_4, Z_5, Z_6 all give the identical syndrome.)

4. Repair anyway: apply Z_4 (or Z_5 , or Z_6 — any one). If the error was Z_5 and we apply Z_4 , the residue is $Z_4 Z_5$, which acts as the *identity* on both code words: each block factor $|000\rangle \pm |111\rangle$ picks up $(+1)$ from $|000\rangle$ and $(-1)(-1)$ from $|111\rangle$.

Answer: the code hears the block, not the qubit — and heals perfectly anyway, because within a block, single phase flips differ only by an operator the code words cannot feel. This graceful bonus is called *degeneracy*, and it is a first taste of how error correction exploits structure rather than brute-force lookup.

WORD TO KNOW

Degenerate code — A code in which different physical errors produce the same syndrome AND the same effect on the code space, so one repair fixes them all — Shor's code's phase sector is the classic example.

REALITY CHECK

Why does a Y error (bit and phase at once) also fall? Because $Y = iXZ$: its X part trips the Z checks (naming the qubit) and its Z part trips the X comparisons (naming the block) — both repairs fire, and the error is undone up to a harmless overall phase. One code word, every single-qubit error: that 1995 proof of principle is the direct ancestor of the surface-code machinery later in this chapter, which trades Shor's 9 qubits for hundreds in exchange for tolerating *realistic* error rates.

29.6 The Logical Qubit Ledger: Overhead, Honestly Priced

“One logical qubit costs a thousand physical ones” — this chapter's headline overhead deserves its derivation, because the arithmetic is short, and because *which* number drives it holds the field's whole strategy.

EQUATION: THE SURFACE-CODE LEDGER

$$p_L \approx \Lambda^{-(d+1)/2}, \quad \text{qubits per logical} \approx 2d^2$$

In words: Each step up in code distance divides the logical error rate by Lambda and costs quadratically more qubits

Symbol	Meaning	Unit
p_L	logical error rate per operation	none
d	code distance (odd: 3, 5, 7, ...)	none
Λ	error-suppression factor per distance step — set by how far <i>below threshold</i> the hardware operates	none

$\Lambda \approx (\text{threshold error rate})/(\text{physical error rate})$, roughly. At threshold $\Lambda = 1$: growing the code buys nothing. The 2023–2024 Google results measured $\Lambda > 2$ for the first time — the moment “below threshold” became data instead of hope.

WORKED EXAMPLE — PRICING A SHOR-CLASS LOGICAL QUBIT

Algorithms like RSA-2048 factoring need $p_L \sim 10^{-12}$ per operation. Price the logical qubit at $\Lambda = 10$ (excellent hardware) and $\Lambda = 2$ (barely below threshold).

1. Solve for distance: $\Lambda^{(d+1)/2} = 10^{12}$, so $(d+1)/2 = 12/\log_{10} \Lambda$.
2. $\Lambda = 10$: $(d+1)/2 = 12 \Rightarrow d = 23 \Rightarrow \text{qubits} \approx 2 \times 23^2 = 1058$.
3. $\Lambda = 2$: $(d+1)/2 = 12/0.301 = 39.9 \Rightarrow d = 79 \Rightarrow \text{qubits} \approx 2 \times 79^2 = 12,482$.
4. Read the lever: a $5\times$ better Λ cut the bill $12\times$. Physical error rate improvements act *exponentially* through the ledger; adding raw qubits acts only linearly.

Answer: ~ 1000 physical per logical on excellent hardware, $\sim 12,000$ on marginal hardware — same target, same code, one order of magnitude apart on a single quality number. This is why every serious roadmap leads with error rates, not qubit counts, and why Chapter 30's datasheet lesson taught you to read them first.

WORD TO KNOW

Below threshold — The regime where physical error rates are low enough ($\Lambda > 1$) that enlarging the code makes logical qubits *better*. Above threshold, more qubits make things worse — scale is only useful after quality.

REALITY CHECK

The ledger also explains the industry's two-track race. Track one: push Λ up (better gates, better calibration — exponential payoff). Track two: manufacture enough qubits to afford $2d^2$ per logical, times thousands of logical (linear grind). The 2024 demonstrations of logical qubits *beating* their physical components — Google's below-threshold surface code, Harvard/QuEra's atom arrays — were small- d down-payments on this arithmetic. Between those results and a million-qubit machine stands no new physics, “only” a decade-class engineering campaign — the honest sentence on which this book's hardware story closes.

ACTIVITY (CLASS) — THE SYNDROME GAME

1. Three students are data qubits holding a secret shared bit (all 0 or all 1, agreed secretly); the class must protect it without ever asking a student's value directly.
2. A saboteur (eyes-closed draw) may flip one student — or none.
3. Two referees measure parities only: referee 1 asks students 1,2 to raise hands if they *differ*; referee 2 the same for students 2,3. (No individual value is ever revealed!)
4. The class decodes the syndrome table of Chapter 29, names the flipped student (or “nobody”), repairs, and only then reveals values to check. Play several rounds, including the no-error round — and discuss why never-looking-at-data is the entire quantum point.

You need: Five volunteers, Paper signs

CHAPTER SUMMARY

- The two quantum barriers — no cloning, no looking — fall to one design: store data in entanglement ($\alpha |000\rangle + \beta |111\rangle$), and measure only parity questions whose answers are branch-independent.
- Bit-flip code: encode with two CNOTs; two ancilla-measured parities locate any single X error (four-row diagnosis chart); one X repairs it — amplitudes untouched throughout.
- Full protection needs phase checks too; the surface code arranges X- and Z-parities on a nearest-neighbor grid, cycling forever with a classical decoder.
- Threshold law: below $p_{\text{th}} \approx 1\%$, logical error $\sim (p/p_{\text{th}})^{(d+1)/2}$ — reliability grows exponentially with code distance; above threshold, codes backfire. Crossed experimentally in 2023–2024.
- Honest price: $\sim 2d^2$ physical qubits per logical ($\sim 1000\times$ at useful distances); cryptography-scale machines need millions of below-threshold qubits — an industrial gap, no longer a conceptual one.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] Encode $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ with the bit-flip code: write the encoded state, and explain in one sentence why it is not the forbidden state $|+\rangle|+\rangle|+\rangle$.

Problem 2 [Basic] An X error strikes qubit 3 of $\alpha|000\rangle + \beta|111\rangle$. Write the corrupted state and predict both parity readings from the chart.

Problem 3 [Basic] The syndrome reads (disagree, agree). Which repair does the chart order?

Problem 4 [Basic] A distance-3 surface-code patch uses 17 qubits ($2d^2$ is approximate). How many for $d = 5$ (use $2d^2$)? For $d = 9$?

Problem 5 [Basic] An X error strikes qubit 7. List the values of all six ZZ checks and both $XXXXXX$ checks, and give the repair.

Problem 6 [Basic] Every one of the eight checks reads +1. What are the two possible situations, and why is the code's response ("do nothing") correct for both?

Problem 7 [Basic] At $\Lambda = 4$, what distance reaches $p_L = 10^{-12}$, and how many physical qubits per logical is that? (Use $(d+1)/2 = 12/\log_{10} 4$; round d up to odd.)

Problem 8 [Basic] Why does operating exactly at threshold ($\Lambda = 1$) make code growth useless? Answer from the ledger formula.

Problem 9 [Standard] Failure analysis: with independent flip probability $q = 10\%$ per data qubit per cycle, compute the probability the three-qubit code fails (two or three flips): $3q^2(1 - q) + q^3$. Compare with the unprotected single-qubit failure rate q . Now redo both at $q = 60\%$ — what reversed, and which concept from Lesson 4 did you just discover numerically? (For a cleaner view, also check the break-even point $q = 50\%$.)

Problem 10 [Standard] Show that the parity measurement is safe on a *clean* code state: for $\alpha|000\rangle + \beta|111\rangle$, verify both branches answer “agree” to both checks — so the ancillas read (agree, agree) with certainty and the state is undisturbed.

Problem 11 [Standard] Why does the bit-flip code fail against a Z error on any one qubit? Apply Z to qubit 1 of $\alpha|000\rangle + \beta|111\rangle$, write the result, and check what both parity readings report. What damage slipped through?

Problem 12 [Standard] Using the threshold law with $p_{\text{th}} = 1\%$, $A = 0.1$: compute the logical error rate at $p = 0.5\%$ for $d = 3$ and $d = 7$. Then repeat at $p = 2\%$ — and describe what the numbers do instead.

Problem 13 [Standard] A vendor announces “48 logical qubits.” Using this chapter, write the three technical questions you would ask before applauding, one line each.

Problem 14 [Standard] A Y error strikes qubit 4. Using $Y = iXZ$, give the full eight-check syndrome, the two-part repair, and explain why the leftover factor of i is harmless.

Problem 15 [Standard] Explain why two X-type comparisons suffice to locate a phase flip among three blocks, by writing the three blocks’ syndrome patterns $(\pm 1, \pm 1)$ and checking they are all distinct — the same trick as the three-qubit code’s two parities.

Problem 16 [Standard] Do the stabilizer bookkeeping: nine physical qubits, eight independent checks — how many logical qubits? Generalize: what does a code with n physical qubits and s independent checks encode?

Problem 17 [Standard] A vendor doubles qubit count yearly at fixed $\Lambda = 1.5$; a rival halves physical error rates yearly at fixed 1000 qubits. Using the ledger, argue which reaches a $p_L = 10^{-9}$ logical qubit first, stating your assumptions.

Problem 18 [Challenge] The phase-flip code: show that encoding $\alpha|+++ \rangle + \beta|--- \rangle$ (the bit-flip code conjugated by Hadamards) turns a Z error on any qubit into a detectable event: use $Z|+\rangle = |-\rangle$ (Chapter 6) to show a single Z now flips one “+” to “−,” caught by parity checks in the $\pm$ basis. What error type is this code now blind to — and why does nesting both codes (Shor’s 9-qubit construction) catch everything?

Problem 19 [Challenge] Threshold engineering: a lab improves physical error from $p = 0.8\%$ to $p = 0.4\%$ ($p_{\text{th}} = 1\%$, $A = 0.1$). For a target logical rate of 10^{-9} , estimate the required distance d in both cases (solve $0.1 \cdot (p/p_{\text{th}})^{(d+1)/2} = 10^{-9}$ by taking logs), then the qubit count $2d^2$ per logical qubit. Summarize in one sentence what halving the physical error rate bought.

Problem 20 [Challenge] Decoding under pressure: syndromes stream out every microsecond, and repairs

must keep up or errors pile up. Estimate the classical decoding throughput needed for 1000 logical qubits at $d = 25$ (each patch: $\sim 2d^2$ checks per cycle, one cycle per μs) in checks per second. Compare with a CPU doing 10^9 simple operations per second, and state the engineering conclusion (one sentence) about why decoder hardware is itself a frontier research topic.

Problem 21 [Challenge] Show by direct calculation on one block that $Z_1 Z_2$ acts as the identity on both $|000\rangle + |111\rangle$ and $|000\rangle - |111\rangle$, and conclude (degeneracy) that repairing a block phase flip with the “wrong” qubit’s Z still restores the code word exactly.

Problem 22 [Challenge] Full campaign estimate: RSA-2048 needs roughly 1500 logical qubits of algorithm plus factory overhead tripling the count, at $p_L = 10^{-12}$. Price the machine at $\Lambda = 10$ and $\Lambda = 2$, compare with the published estimates of Chapter 19’s resource lesson, and identify which ingredient of your estimate is crudest.

CHAPTER THIRTY

Quantum Hardware and Our Region

Five ways to build a qubit, how to compare machines honestly — and the quantum opportunity for Iraq and the Kurdistan Region

IN THIS CHAPTER YOU WILL LEARN TO

- Describe the five leading qubit technologies and each one's signature strengths and weaknesses.
- Compare machines like a professional: coherence, fidelity, speed, connectivity, and composite benchmarks — not qubit counts alone.
- Explain what lives inside a dilution refrigerator, and why most quantum computers are cold.
- Map the practical routes by which students and universities in Iraq and the Kurdistan Region can do real quantum computing today.
- Sketch a realistic regional roadmap: education first, cloud research second, specialized niches third.

30.1 How to Build a Qubit: Five Technologies

Chapter 3 promised that any two-level quantum system can be a qubit. Thirty years of engineering have narrowed the field to a handful of serious contenders — each a different bargain among speed, quality, and scalability. (All numbers below are example figures for the mid-2020s, marked for orientation, not procurement.)

The five main qubit platforms (example figures, mid-2020s — always check current data)

Platform	The qubit is...	Signature strength	Signature weakness	Championed by
Superconducting circuits	current states of a tiny cold circuit	fast gates (~ 10 – 100 ns); chip fabrication	short coherence; needs dilution fridge	Google, IBM, Rigetti
Trapped ions	energy levels of single ions in electric traps	superb fidelity and coherence (seconds+); identical qubits	slow gates ($\sim \mu$ s–ms); hard to grow beyond 100s	IonQ, Quantinuum
Neutral atoms	atoms held in laser “tweezer” arrays	huge arrays (100s–1000s); reconfigurable layout	newer gate technology; mid-pack fidelity	QuEra, Pasqal
Photonics	states of single light particles	room temperature; natural for networking	photons vanish (loss); two-qubit gates are probabilistic	PsiQuantum, Xanadu
Spins in silicon	electron/nuclear spins in chips	rides the trillion-dollar silicon industry	few-qubit demos; uniformity challenges	Intel, academic labs

No platform wins everywhere, and hedged bets are rational: the machine that first factors a real RSA key may not be built from today’s leader. A wise student learns the *physics-independent* layer — everything in the appendices and Chapters 3–29 — which transfers untouched across all five.

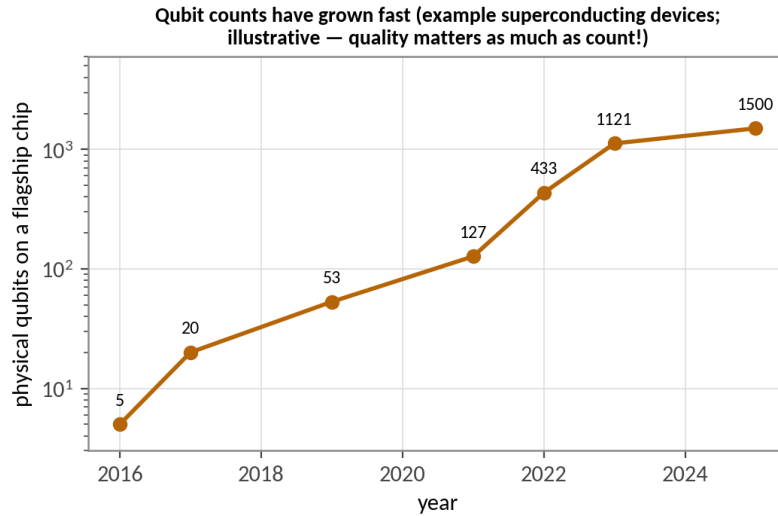
30.2 Comparing Machines Honestly

Chapter 28’s mistake box warned against counting qubits. The professional’s scorecard has at least five lines:

The honest scorecard

Metric	The question it answers
Qubit count	how wide can circuits be?
Two-qubit fidelity	how deep can circuits go? (Chapter 28: depth $\sim 1/p$)
Coherence (T1, T2)	how long does memory last relative to gate time?
Connectivity	which qubit pairs can interact directly? (sparse grids pay SWAP taxes — Chapter 10)
Speed (gates/s, shots/s)	how much data per hour of real experiments?

Because vendors quote whichever line flatters them, composite benchmarks exist. **Quantum volume** runs randomized square circuits (n qubits, depth n) and reports the largest size the machine passes — punishing weak links automatically. Newer measures track error-per-layered-gate (EPLG) or application-level scores. None is perfect; all beat qubit-counting.



Growth in headline qubit counts (example superconducting flagships; illustrative). The professional reads this together with fidelity data: the industry's true turning point was crossing the error-correction threshold (Chapter 29), not any single count.

WORKED EXAMPLE — TWO MACHINES, ONE JOB

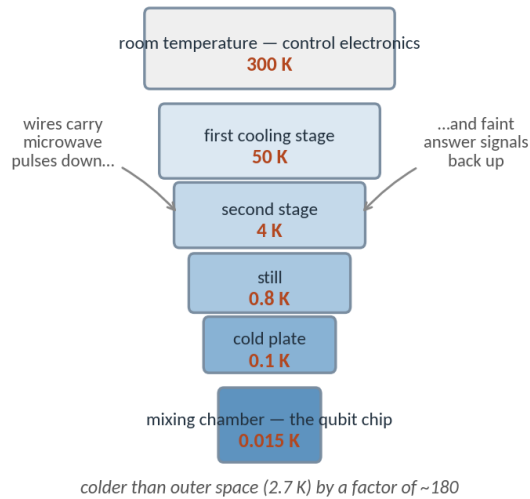
Machine A: 400 qubits, two-qubit error 1.2%, grid connectivity. Machine B: 56 qubits, error 0.15%, all-to-all connectivity. Your algorithm needs 40 qubits and depth 300 (after compilation on all-to-all hardware). Which machine?

1. Depth capacity (Chapter 28 rule $d \sim 1/p$): A manages $\sim 1/0.012 \approx 83$ layers; B manages $\sim 1/0.0015 \approx 667$.
2. Compilation tax: on A's grid, distant pairs need SWAP chains — realistic depth might grow from 300 toward 500+; on B, 300 stands.
3. Verdict: A fails twice over ($83 \ll 500$); B passes with margin ($667 > 300$) and has enough qubits ($56 > 40$).

Answer: Machine B — with one-seventh the qubits. Width you don't need is worthless; depth you do need is everything. This is the calculation behind every serious "which hardware?" decision — and why ion/atom platforms with modest counts remain fierce competitors.

30.3 Cold Machines: Inside the Refrigerator

Why do most quantum computer photographs show a golden chandelier? That is a **dilution refrigerator** with its covers off — a nested set of ever-colder stages, hung from the ceiling, with the qubit chip at the bottom.



The cold stack: from room temperature to about 15 millikelvin — roughly 180 times colder than deep space. Each stage anchors away heat carried by the control wiring.

The cold serves Chapter 28's masters. Thermal energy flips qubits and scrambles phases; at 15 mK, thermal disturbance at superconducting-qubit frequencies is essentially frozen out. The environment is quieted — decoherence slowed — by brute refrigeration, layered shielding, and careful filtering of every wire. (Trapped ions and neutral atoms relocate the battle: their qubits sit in vacuum chambers at room temperature, but their *lasers* must be fantastically stable — every platform pays the isolation bill somewhere.)

WORD TO KNOW

Dilution refrigerator — A cryostat reaching millikelvin temperatures by circulating a mixture of helium isotopes. The standard home of superconducting and spin qubits — and the reason those quantum computers look like golden chandeliers.

REALITY CHECK

A quantum computer is not just its qubits. Around the fridge stand racks of microwave electronics (one or more control lines per qubit), FPGA decoders for error correction (Chapter 29, Problem 12), vacuum and laser systems, and software stacks doing compilation, calibration, and queueing. Scaling to millions of qubits (Chapter 29's bill) is largely a *systems* engineering problem: wiring density, heat budgets, control multiplexing, real-time classical processing. Students with backgrounds in electrical engineering, control systems, cryogenics, and software are exactly who this industry hires — a directly relevant note for readers of this book's region, where such engineering skills are strong.

30.4 Cloud Quantum Computing from Anywhere

Now the good news this book has been building toward. Because every serious machine sits behind a cloud interface (Chapter 14), *geography has almost vanished as a barrier to entry*. What a student in Erbil, Duhok, Sulaymaniyah, Baghdad, or Basra needs to run real quantum hardware:

1. A computer and an internet connection — modest bandwidth suffices; jobs are kilobytes.
2. Free software: Python, Qiskit or a peer framework (all open-source).
3. A free cloud account (IBM Quantum’s open plan and periodic academic programs of other providers).
4. The mathematics and circuit skills of this book — the only truly scarce ingredient.

Every experiment referenced in this book at ten qubits or fewer — Bell tests, teleportation, Grover at toy scale, VQE on tiny molecules, the three-qubit repetition code — has been run by students worldwide through exactly this route. Nothing in the workflow distinguishes a submission from Kurdistan from one sent from Boston, except the timezone stamp.

CASE STUDY — QUANTUM PROGRAMS WITHOUT QUANTUM MACHINES · A WORLDWIDE PATTERN

Most of the world’s quantum-active universities own no quantum computer. The pattern that works, visible from Africa to Southeast Asia to Latin America: begin with a course built on simulators (cost: zero); add cloud-hardware labs (cost: near zero); join a vendor academic network or regional hub for larger allocations; grow a research group around algorithms, error mitigation, benchmarking, or applications — all publishable through the cloud; and only much later, if ever, consider hardware of one’s own. National quantum initiatives now exist on every continent, and several countries whose first quantum publication is younger than this decade have active groups today. The lesson for Iraq and the Kurdistan Region is direct: the entry ticket is educational, not industrial — and this book is designed to be that ticket.

30.5 Building Quantum Skills in Iraq and the Kurdistan Region

This final lesson is written as a practical roadmap — the author’s recommendations, offered to students, teachers, and university planners in the region. (These are proposals to act on, not descriptions of existing programs; verify current offerings at your institution.)

For the student (this semester). Work through this book’s practice problems honestly. Install Python and reproduce Chapter 14’s runs. Create a free cloud account and complete your first hardware run — Chapter 31’s Project 1 is designed as that milestone. Join the global online communities (Qiskit’s forums and events, open courseware from major universities) where working quantum scientists answer beginners daily. None of this requires anyone’s permission.

For the teacher or department (this year). A one-semester elective needs: this book (Student Edition + Teacher’s Guide), a computer lab with Python — machines that run a browser comfortably will do for simulators at this book’s scale — and one instructor willing to stay a chapter ahead. Pair it with existing strengths: electrical engineering departments can slot quantum hardware topics beside microwave and control courses;

computer science beside algorithms and cryptography; physics beside quantum mechanics. Capstone projects (Chapter 31) double as student conference material.

For the region (this decade). The realistic sequence mirrors the worldwide pattern above: (1) education at scale — graduates who read hardware datasheets and algorithm papers fluently; (2) cloud-based research groups — benchmarking, error mitigation, applications in optimization and chemistry relevant to local industry (energy networks, logistics, materials); (3) niche specialization — the region need not build fridges to contribute: quantum-safe cryptography migration (Chapter 19's PQC — every bank and ministry will need it), quantum-curriculum development in Kurdish and Arabic, and remote-hardware benchmarking services are all export-grade specialties with low capital cost.

WORKED EXAMPLE — COSTING A UNIVERSITY QUANTUM LAB (CLOUD MODEL)

Estimate the setup cost for a 25-student quantum computing elective using the cloud model, assuming an existing computer lab. (Example prices for planning only.)

1. Software: Python, Qiskit, simulators — \$0 (open source).
2. Hardware access: free-tier cloud accounts for coursework; one shared paid allocation for capstones, say \$500–\$2000/year depending on provider promotions and academic credits.
3. Materials: this book and Teacher's Guide are provided free of charge; printing costs only.
4. Staff: one trained instructor — the real investment; a semester of preparation time using freely available graduate courseware.

Answer: on the order of \$1000–\$2000 per year beyond ordinary lab operation — the cost of a single conventional laboratory instrument, for a course at the frontier of computing. Compare: a starter superconducting lab begins around \$1 million per fridge before staff. The cloud model is not a compromise; at educational scale it is simply the correct engineering choice.

COMMON MISTAKE

X Wrong: “Serious quantum work is only possible in countries that own quantum computers.”

✓ Right: The overwhelming majority of published quantum-algorithm, error-mitigation, and benchmarking research runs on cloud hardware and simulators the authors do not own. Owning fridges matters for hardware physics; for everything else in this book, the laboratory is the internet plus mathematics. The scarce resource is trained people — which is exactly the resource a determined region can build fastest.

ACTIVITY (CLASSROOM) — DRAFT YOUR DEPARTMENT'S QUANTUM PLAN

1. In groups: pick a real department you know (engineering, CS, physics) at a university in the region.
2. Draft a one-page elective proposal: prerequisites (which chapters of this book cover them?), weekly topics for 14 weeks, lab sessions (simulator/cloud), and assessment.
3. Cost it with the worked example's method, replacing assumptions with your local numbers.

4. Identify the single biggest obstacle (instructor training? lab access? approval process?) and one concrete first step against it.
5. Present in five minutes; class votes on the most implementable plan.

You need: Paper and pen, This chapter, Optionally a real course catalogue

► App tool: Hardware Explorer — browse the five platforms with datasheet-style cards, and a scorecard tool for comparing example machines

30.6 Reading a Real Datasheet

Sooner or later a datasheet lands on your desk — a cloud dashboard, a vendor PDF, a conference slide — and someone asks: *can our circuit run on this?* Here is a representative 2026 superconducting machine’s summary table, and the four-step read that turns it into a yes or no.

A representative datasheet (median values; illustrative numbers)

Quantity	Value	First reaction
Qubits	27	capacity, not quality
T_1 median	80 μ s	the survival clock
T_2 echo median	60 μ s	the phase clock
1-qubit gate error	0.03%	rarely the bottleneck
2-qubit gate error	0.6%	the bottleneck
Readout error	1.2%	taxes every qubit once
2-qubit gate time	60 ns	sets the time budget

EQUATION: THE SURVIVAL BUDGET

$$F \approx (1 - p_1)^{n_1} (1 - p_2)^{n_2} (1 - p_r)^{n_q} \times e^{-t_{\text{circ}}/T_1}$$

In words: Multiply the survival odds of every gate, every readout, and the waiting itself — the honest estimate of the run’s fidelity

Symbol	Meaning	Unit
p_1, p_2, p_r	1-qubit, 2-qubit, readout error rates	none
n_1, n_2, n_q	counts of each in your circuit	none
t_{circ}	total circuit duration	μ s

Rules of thumb fall straight out: keep $n_2 \lesssim 0.1/p_2$ for a $\lesssim 10\%$ gate-error toll, and $t_{\text{circ}} \lesssim 0.1 T_1$ for a $\lesssim 10\%$ decay toll.

WORKED EXAMPLE — FOUR-STEP DATASHEET READ FOR A 100-GATE JOB

Your circuit: 12 qubits, 100 two-qubit gates, 90 one-qubit gates, everything read out. Can the machine above run it?

1. Two-qubit toll: $(1 - 0.006)^{100} \approx 0.55$ — the run keeps only 55% fidelity from its worst enemy alone. (At 20 such gates the toll would be a friendly 0.89.)
2. One-qubit toll: $(1 - 0.0003)^{90} \approx 0.97$ — confirming rule one: single-qubit gates are cheap.
3. Readout toll: $(1 - 0.012)^{12} \approx 0.87$.
4. Time toll: about $100 \times 60 \text{ ns} = 6 \mu\text{s} \ll T_1 = 80 \mu\text{s}$: $e^{-6/80} \approx 0.93$. Combined estimate: $0.55 \times 0.97 \times 0.87 \times 0.93 \approx 0.43$.

Answer: roughly 43% of shots survive un-corrupted — marginal but workable *with* the error-mitigation tools of Chapter 28, hopeless without them. The verdict came from four multiplications; that is the entire craft of datasheet reading.

WORKED EXAMPLE — MORE QUBITS OR BETTER QUBITS?

Machine A boasts 200 qubits at $p_2 = 1.5\%$; machine B offers 27 qubits at $p_2 = 0.3\%$. Which runs a 50-two-qubit-gate circuit better?

1. Machine A: $(1 - 0.015)^{50} \approx 0.47$.
2. Machine B: $(1 - 0.003)^{50} \approx 0.86$.
3. The 200-qubit machine loses badly *for this job* — unless the circuit simply needs more than 27 qubits, in which case B cannot enter the race at all.

Answer: B by a landslide (86% vs 47%). Qubit count is capacity; error rate is capability. Marketing leads with the first; your survival budget runs on the second. The professional's question is never “how many qubits?” but “how many gates before the signal dies?”

WORD TO KNOW

Median vs. best qubit — Datasheets quote medians because chips are uneven — a few hero qubits beat the median several-fold, a few duds miss it badly. Serious toolchains read the per-qubit calibration map and route circuits around the duds.

REALITY CHECK

Datasheet numbers are *isolated* benchmarks: one gate at a time, freshly calibrated. Real circuits suffer crosstalk (neighbors firing together), drift (this morning's calibration aging by afternoon), and routing overhead (SWAP gates inserted when your circuit's connections don't match the chip's wiring diagram — each SWAP is three CNOTs of budget!). Treat the budget F as an optimistic ceiling, subtract mentally for the un-quoted taxes, and always benchmark with a small canary circuit before the real run.

30.7 The Roadmap Game: Reading Vendor Promises

Every quantum company publishes a roadmap — glossy slides of qubit counts doubling into the confident future. After this book, you can read them the way an engineer does: by asking which *budget line* each promised number actually improves.

WORKED EXAMPLE — AUDITING A ROADMAP SLIDE

A (fictional but typical) slide promises: 2026: 400 qubits. 2027: 1000 qubits. 2028: 4000 qubits, “enabling enterprise advantage.” Error rates appear nowhere. Audit it.

1. Apply the survival budget (this chapter): usable circuit depth is set by p_2 , not qubit count. At an unimproved $p_2 = 0.6\%$, all three machines run out of fidelity near the same ~ 100 –150 two-qubit gates — the 4000-qubit machine mostly adds *width* it cannot use at depth.
2. Apply the ledger (Chapter 29): at 0.6% ($\Lambda \approx 1.7$ against a $\sim 1\%$ threshold), error correction needs $d \approx$ large and $\sim 10^4$ physical per logical — 4000 qubits buy *zero* good logical qubits. Halving p_2 would do more than the entire tripling of count.
3. Check the missing lines: no T_1/T_2 trend, no Λ measurement, no connectivity map, no calibration-stability data — the numbers that decide everything are the numbers not shown.
4. Verdict, in referee language: “Roadmap demonstrates manufacturing scale-up; provides no evidence of the error-rate trajectory on which claimed applications depend. Request: measured p_2 and Λ per generation.”

Answer: the slide promises the linear ingredient (count) while silent on the exponential one (quality). The one-line audit you now own: *find the error rate; if it is absent, the roadmap is marketing; if it is falling, the roadmap is serious* — everything else in this book’s Part Eight follows from that check.

WORD TO KNOW

Roadmap audit — Reading a hardware announcement by budget lines: qubit count (width), error rates (depth and Λ), coherence times, connectivity, and calibration stability — and refusing conclusions from the lines not shown.

REALITY CHECK

The same audit, run in reverse, is how to spot *genuine* progress that headlines undersell: a modest “we improved two-qubit fidelity from 99.4% to 99.7%” quietly halves the effective error rate, doubles usable depth, and slashes the error-correction ledger — worth more than a qubit-count doubling, while generating a tenth of the press. The professionals’ rule of thumb by now is yours: *in quantum hardware, quality compounds; quantity only adds*.

ACTIVITY (APP) — DATASHEET SCAVENGER HUNT

1. In pairs, open a real public dashboard or datasheet (any cloud quantum provider — Chapter 14 lists starting points; your instructor may print snapshots if the lab is offline).
2. Fill the datasheet table of this chapter for the real machine: qubits, T_1 , T_2 , one- and two-qubit errors, readout error, gate time. Note which numbers were *missing* — that absence is data too.
3. Run the four-step read of the worked example for a 50-CNOT, 10-qubit circuit on the real numbers: survival estimate, verdict.
4. Compare machines across pairs: rank by survival on the *same* circuit, not by qubit count — then check the ranking against each vendor's own headline claim.

You need: Internet access or printed snapshots, Calculator, Paper and pen

CHAPTER SUMMARY

- Five platforms lead: superconducting (fast, cold), trapped ions (precise, slow), neutral atoms (many, reconfigurable), photonics (room-temperature, lossy), silicon spins (industrial promise, early). The mathematics of this book runs on all of them.
- Compare machines by fidelity, coherence, connectivity, and speed together — composite benchmarks like quantum volume exist because single numbers mislead; depth capacity ($\sim 1/p$) beats qubit count for most jobs.
- Superconducting and spin qubits live at ~ 15 mK in dilution refrigerators; every platform pays its isolation bill somewhere (fridges, vacuums, or laser stability); the machine is a full classical-quantum system, hungry for exactly the engineering skills this region trains.
- The cloud has erased geography: laptop + internet + free accounts + this book's skills = real quantum experiments from anywhere, including every city in Iraq and the Kurdistan Region.
- Regional roadmap: education now (near-zero cost), cloud research groups next, niche specialties after (PQC migration, curriculum, benchmarking) — people before hardware, always.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] Match each platform to its signature strength: superconducting, trapped ions, neutral atoms, photonics.

Problem 2 [Basic] Name the five scorecard metrics, and state which one Chapter 28's depth rule connects to.

Problem 3 [Basic] Why are superconducting qubits kept at millikelvin temperatures? (One sentence, using the word “decoherence.”)

Problem 4 [Basic] List the four ingredients a regional student needs for a first real-hardware run.

Problem 5 [Basic] Using the datasheet above, compute the readout toll for reading out 5 qubits, and the two-qubit toll for a 20-gate circuit. Which costs more fidelity?

Problem 6 [Basic] A vendor slide says “433 qubits!” and nothing else. List the three numbers you would demand before believing any circuit of yours can run, and give each number’s role in the budget formula.

Problem 7 [Basic] Machine A: 4000 qubits, $p_2 = 0.6\%$. Machine B: 400 qubits, $p_2 = 0.15\%$. Compute each machine’s $\approx 0.1/p_2$ clean-depth budget, and say which runs a 200-CNOT algorithm better.

Problem 8 [Basic] List the five budget lines of a roadmap audit, and mark which two compound exponentially through the error-correction ledger.

Problem 9 [Standard] Machine A: 1000 qubits at 2% two-qubit error. Machine B: 100 qubits at 0.2%. For a 60-qubit, depth-200 algorithm, repeat the worked example’s analysis and pick a machine.

Problem 10 [Standard] A trapped-ion machine has $T_2 = 1$ s and gate time 100 μ s; a superconducting machine has $T_2 = 100$ μ s and gate time 50 ns. Compute the rough “gates within T_2 ” figure for each, and comment on why both platforms remain competitive despite the 10,000 $\times$ coherence gap.

Problem 11 [Standard] Quantum volume doubles from 512 to 1024. What, roughly, does that claim mean in circuit terms, and name two different underlying improvements that could each have caused it.

Problem 12 [Standard] Your department has \$3000/year and 20 students for a quantum elective. Allocate the budget using the worked example’s categories, and state the one item you would fund first and why.

Problem 13 [Standard] A ministry asks: “should we buy a quantum computer?” Write a four-sentence professional answer using this chapter: what to do first, what the purchase would and would not enable, the realistic cost scale, and the people-first alternative.

Problem 14 [Standard] On the datasheet machine, how many two-qubit gates can a circuit contain before gate errors alone cut fidelity to 50%? Solve $(1 - 0.006)^n = 0.5$.

Problem 15 [Standard] A compiler must insert 12 SWAP gates (3 CNOTs each) to fit your circuit onto the chip’s wiring. Recompute the two-qubit toll of the 100-gate worked example with these extra CNOTs included, and state the moral about connectivity.

Problem 16 [Standard] The worked example ignored T_2 . With $T_2 = 60$ μ s and a phase-sensitive circuit idling 30 μ s on its most delicate qubit, estimate the extra dephasing factor e^{-t/T_2} and revise the overall budget.

Problem 17 [Standard] A press release: “Our new chip reaches 99.9% two-qubit fidelity ($p_2 = 0.1\%$), up from 99.5%.” Quantify the improvement three ways: clean-depth budget, survival of a 300-gate circuit, and the ledger’s Λ (threshold $\approx 1\%$). Which framing best conveys the gain?

Problem 18 [Challenge] SWAP-tax estimate: on a square-grid machine, two qubits sit 6 steps apart, and your circuit needs 20 CNOTs between them. Each routing SWAP = 3 CNOTs, and moving adjacent takes 5 SWAPs each way (there and back per interaction round, but assume clever compilation amortizes to 2 SWAP-chains total). Estimate total extra CNOTs from routing, the resulting extra error at 0.5% per CNOT, and compare with an all-to-all machine.

Problem 19 [Challenge] Design a fair national benchmark: a region will grant cloud-credit funding to the best of three student teams. Propose the evaluation — three tasks of increasing difficulty drawn from this book (name chapters), the metrics per task, and an anti-gaming rule — in about eight sentences of prose.

Problem 20 [Challenge] The “leapfrog” question: some argue regions without legacy computing industries can jump directly into quantum-era specialties (PQC migration services, quantum-literate energy optimization, curriculum export), as some countries leapt to mobile banking without wired telephony. Argue for or against, in one paragraph, using at least three specific facts from Chapters 19–20 and 28–30 — and end with the single investment you would make first.

Problem 21 [Challenge] Write the full budget for this job on the datasheet machine: 12 qubits, 40 two-qubit gates, 90 one-qubit gates, full readout, 6 μs duration. Then find which single datasheet improvement — halving p_2 , halving p_r , or doubling T_1 — buys the most fidelity, showing the three revised budgets.

Problem 22 [Challenge] Write the two-paragraph “roadmap referee report” for a vendor of your choice (or the fictional one above): paragraph one, what the shown numbers genuinely support; paragraph two, the specific missing measurements and the experiment you would request — ending with the one-sentence quality-vs-quantity verdict.

CHAPTER THIRTY-ONE

Capstone Projects

Twelve complete projects — apply the whole course, from randomness to a professional briefing

IN THIS CHAPTER YOU WILL LEARN TO

- Apply the full toolkit — states, gates, circuits, algorithms, noise, and honest judgment — to twelve self-contained projects.
- Plan an experiment: hypothesis, circuit, shot budget, and pass/fail criteria stated *before* running.
- Analyze results like a professional: statistics with wobble bars, comparisons against prediction, and honest anomaly hunting.
- Write engineering reports that a teacher, colleague, or decision-maker can trust and check.
- Communicate quantum computing responsibly to a non-technical audience without hype or false modesty.

Welcome to the capstone chapter. The appendices and Chapters 3–30 taught the ideas one piece at a time; each project here is a complete, realistic exercise — specified by hand, checked by simulation, and (for those with cloud access, Chapter 14) runnable on real hardware. Every project ends in a written deliverable, because in professional life *the calculation that is not communicated does not exist*. Use the same marking rubric for all twelve:

Marking rubric for every project report

Component	What earns the marks	Weight
Plan	goal, circuit, shot budget, and success criteria stated before results	20%
Correct execution	circuits and calculations match the plan; arithmetic verifiable	25%
Analysis	histograms read with wobble ($\sqrt{N}$) in mind; prediction vs. result compared honestly	25%
Judgment	anomalies investigated, limitations stated, claims sized to evidence	20%
Clarity	a reader can reproduce the work from the report alone	10%

31.1 Project 1 — A Certified Quantum Coin

Mission: build, run, and statistically certify the fairest coin in the world. **Uses:** Chapters 4, 6, 12, 14.

Tasks. (1) Design the one-qubit coin circuit (H + measure) and the three-qubit die (Chapter 12, Lesson 5). (2) Before running, write your acceptance test: for 1000 shots, what count range counts as “fair”? (3) Run 10, 100, 1000, 10,000 shots on the simulator; tabulate counts, deviations, and deviation/ $\sqrt{N}$. (4) If cloud access is available, repeat 1000 shots on hardware and compare. (5) Deliverable: a two-page certification report.

WORKED EXAMPLE — THE ACCEPTANCE TEST, DONE RIGHT

Set a professional pass window for “fair coin, 1000 shots.”

1. Expectation: 500 heads; wobble $\sigma \approx \sqrt{1000 \times 0.5 \times 0.5} \approx 15.8$.
2. A “two-sigma” window (passes $\sim 95\%$ of honest coins): 500 ± 32 , i.e. 468–532 heads.
3. Add a second check the window misses: split the run into ten 100-shot blocks and confirm no drift trend (early blocks vs late blocks), guarding against a device warming up or drifting.

Answer: pass if heads $\in [468, 532]$ AND no block trend. Writing the window *before* running is what turns a demo into an experiment — and this two-line habit is transferable to every project below.

31.2 Project 2 — Build and Verify a Bell Pair

Mission: manufacture entanglement and prove it — to a skeptic who suspects your “entangled” pair is just classical correlated coins. **Uses:** Chapters 9–11, 14.

Tasks. (1) Build H+CNOT; predict the four-pattern table. (2) Run 2000 shots; verify only 00/11 within wobble. (3) The skeptic says: “two hidden coins glued together would match too.” Answer experimentally: measure both qubits in the $|+\rangle / |-\rangle$ basis (H on each before measuring — Chapter 4, Lesson 5) and show the results *still* correlate perfectly, which glued classical coins cannot do in both bases at once. (4) Compute the four-basis-combination correlation table. (5) Deliverable: “Certificate of Entanglement” report with both histograms.

WORKED EXAMPLE — THE SECOND-BASIS CHECK

Predict the $|+\rangle / |-\rangle$ -basis statistics of $|\Phi^+\rangle$.

1. Algebraic identity (verify by expanding the right side): $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}(|++\rangle + |--\rangle)$.
2. So in the rotated basis the pair reads ++ or --, each 50% — perfect agreement again.
3. Classical glued coins set in the 0/1 basis, measured diagonally, would agree only 50% of the time (each coin answers a diagonal question randomly) — prediction: correlation collapses to coin-flipping.

Answer: quantum pair — agreement in BOTH bases; classical mimic — agreement in at most one. Your two histograms together are the certificate. (The full courtroom version is the Bell test of Chapter 11 — this two-basis check is its undergraduate sibling, and catches every classical forger that keeps a definite answer per basis.)

31.3 Project 3 — The Teleportation Machine

Mission: implement Chapter 13's protocol and demonstrate, with evidence, that an arbitrary state crossed the circuit. **Uses:** Chapters 11–14.

Tasks. (1) Pick three test states: $|1\rangle$, $|+\rangle$, and $0.6|0\rangle + 0.8|1\rangle$. (2) For each, trace the algebra by hand (the four outcome brackets and fixes). (3) Simulate with mid-circuit measurement and conditional gates (Chapter 14, Problem 11); verify all four outcomes occur $\approx 25\%$ each. (4) Evidence of arrival: since amplitudes cannot be read directly, run each teleportation *followed by* the appropriate verification measurement on Bob's qubit — standard basis for $|1\rangle$ (expect all 1s), $\pm$ basis for $|+\rangle$ (expect all +), and a 1000-shot standard-basis histogram for the third state (expect 36%/64%). (5) Deliverable: protocol report including the “cheating Bob” control run (skip the classical fixes) showing verification then fails — and explain the failure rate you observe.

31.4 Project 4 — A Secret Message with Superdense Coding

Mission: transmit the four-symbol message 00, 01, 10, 11 — one qubit per symbol — and measure the channel's reliability. **Uses:** Chapters 11, 13, 14, 28.

Tasks. (1) Implement encode (I/X/Z/XZ) and decode (CNOT+H+measure) circuits. (2) Simulate each symbol 500 shots, ideal: confirm 100% decoding. (3) Turn on simulated noise (Chapter 28's Noise Lab settings, or Qiskit's noise models): measure the symbol error rate at two-qubit gate error 0.5%, 2%, 5%. (4) Plot symbol error vs. gate error. (5) Deliverable: a channel datasheet — capacity two bits/qubit, measured error curve, and a one-paragraph recommendation of the maximum gate error at which you would trust this channel with, say, exam grades.

31.5 Project 5 — Grover Search of an 8-Entry Database

Mission: find the marked entry among eight with two oracle calls, and map the overshoot. **Uses:** Chapters 12, 18, 14.

Tasks. (1) Build the 3-qubit Grover circuit for a target of your choice (oracle: a controlled-controlled-Z pattern with X-conjugations selecting your pattern; diffusion: Chapter 18's H/X/CCZ/X/H). (2) Hand-compute the amplitude table after each of $k = 1, 2, 3, 4$ iterations (extend the worked example of Chapter 18). (3) Simulate 1000 shots at each k ; overlay measured success frequency on your predicted $\sin^2((2k+1)\theta)$ curve. (4) Change the target; confirm nothing in the performance changes. (5) Deliverable: a lab report whose central figure is your measured-vs-predicted success curve, with the overshoot at $k = 4$ clearly demonstrated and explained.

31.6 Project 6 — Period Finding: Factor 15 End to End

Mission: reproduce Shor's factoring of 15 with every step either computed by hand or simulated. **Uses:** Appendix A and Chapters 10 and 19.

Tasks. (1) By hand: build the powers table of $a = 7, 2, 4, 8, 11, 13 \bmod 15$; find each period; mark which values pass the cooperation tests. (2) By hand: run the classical finish for two passing choices, obtaining 3×5 twice. (3) Simulate (or use the app's Shor Lab): the QFT spectrum for the $a = 7$ superposition; verify peaks

at spacing total/r . (4) Study the failure branch: for $a = 14$, show $r = 2$ but $14^{r/2} \equiv -1$, and the algorithm retries. (5) Deliverable: a complete worked dossier on “how 15 falls,” written so a Chapter-1–17 reader can follow — including one honest paragraph on why 15 is easy and RSA-2048 is a different world (Chapter 29’s overhead numbers belong here).

31.7 Project 7 — Ideal vs. Noisy: A Benchmark Report

Mission: characterize a (simulated or real) device the way a lab qualifies an instrument. **Uses:** Chapters 14, 28, and everything before.

Tasks. (1) Choose your benchmark suite — justify it: this book suggests (a) single-qubit coin (readout + 1q gates), (b) Bell pair (adds one CNOT), (c) GHZ-5 (adds depth and width), (d) depth-ladder: repeated CNOT pairs at depth 10/50/200. (2) Predict every histogram, with corruption estimates from an assumed error budget. (3) Run all, 1000+ shots each, at two noise settings (or simulator vs real hardware). (4) Extract: forbidden-count fractions, survival vs depth fit to $(1-p)^d$, and your fitted effective p . (5) Deliverable: a four-page device qualification report with a one-line “health grade” and the evidence behind it — your professional answer to Chapter 14, Problem 12.

31.8 Project 8 — Protect One Qubit: A Repetition-Code Study

Mission: demonstrate error correction beating no error correction — and find where it stops helping. **Uses:** Chapters 10, 12, 28, 29.

Tasks. (1) Implement encode-noise-syndrome-repair-decode for the three-qubit bit-flip code (app Error-Correction Lab, or Qiskit with resets/conditionals). (2) Sweep injected flip probability $q = 1\%, 5\%, 10\%, 25\%, 40\%, 50\%, 60\%$ per qubit per round; run 2000 trials each; record logical failure rates. (3) Plot measured failures against the theory curve $3q^2(1-q) + q^3$ (Chapter 29, Problem 5) and against the unprotected line q . (4) Locate the crossover experimentally — the toy threshold. (5) Deliverable: a report titled “When does protection pay?,” its centerpiece your measured crossover, plus one paragraph connecting it to the real surface-code threshold story of Chapter 29.

31.9 Project 9 — Train a Quantum Classifier, Simulator to Hardware

Mission: take Part Seven’s rings classifier from clean mathematics to noisy reality, and report what survives the trip. **Uses:** Chapters 22, 23, 25, 28, 30.

Tasks. (1) Reproduce the rings-dataset training of Chapter 27 in a statevector simulator: one qubit, three re-uploading layers, parameter-shift gradients; confirm you reach the book’s reported accuracies. (2) Re-evaluate the *trained* model with finite shots (100, 500, 2000 per point): plot accuracy vs. shots and explain the trend with $\sigma \approx 0.5/\sqrt{n}$ near-boundary logic (Chapter 23). (3) Add a hardware-style error budget (Chapter 30’s datasheet numbers) to every inference circuit and predict the accuracy loss before measuring it. (4) Run the sabotage study: retrain from three random seeds and report the spread — how repeatable is your model? (5) Deliverable: a two-page *model card* — architecture, training budget in shots, accuracies with error bars at each noise level, and an honest “intended use / known limits” paragraph.

31.10 Project 10 — Quantum Kernel vs. Classical Baseline: The Fair Fight

Mission: stage the comparison most QML papers dodge — your quantum kernel against a *tuned* classical opponent, refereed by yourself. **Uses:** Chapters 4, 24, 27.

Tasks. (1) Compute the full quantum kernel matrix for the 60-point rings dataset with the $R_z(x_2)R_y(x_1)$ map: all $60 \times 59/2 = 1770$ entries; run the health check of Chapter 24 and include the matrix picture. (2) Budget and simulate shot noise: recompute the matrix at 200 and at 2000 shots per entry; quantify how much the entries move. (3) Train a kernel classifier on each matrix; train a *tuned* classical RBF-kernel SVM on the same splits as your baseline. (4) Run the ablation of Chapter 27: swap the quantum kernel for the classical one inside the same pipeline — does performance actually depend on the quantum ingredient? (5) Deliverable: a comparison report with accuracies $\pm$ error bars, the total shot invoice, and a verdict written in the audit template's language — *parity, advantage, or disadvantage, and the evidence*.

31.11 Project 11 — The QML Claim Audit, in the Wild

Mission: graduate from auditing textbook examples to auditing the real literature. **Uses:** Chapters 2, 26, 27.

Tasks. (1) Find one real, recent quantum machine-learning claim: a preprint, a vendor page, or a news story (your instructor may assign one; harvest candidates from the References section). (2) Extract its checkable claims into a one-page summary table: dataset, size, encoding, hardware vs. simulator, baseline, headline number. (3) Run the full five-question audit of Chapter 27, in writing, question by question, citing page or figure numbers from the source for every judgment. (4) Draft the two experiments you would request as a referee — one ablation, one baseline upgrade — each with its expected-outcome fork ("if X, the claim strengthens; if Y, it dissolves"). (5) Deliverable: a referee report of at most two pages, written so its subject's authors would call it tough and fair — plus a three-sentence plain-language version for a newspaper reader.

31.12 Project 12 — A Realistic Briefing for Decision-Makers

Mission: the communication capstone. A university president, ministry director, or company board in the region asks: "*Should we invest in quantum computing — and how?*" Prepare the honest briefing. **Uses:** Chapter 2, Chapters 19 and 28–30 — and every honesty habit in this book.

Tasks. (1) Write a two-page executive brief: what quantum computers do and do not do (Chapter 2's map), the real threat timeline to cryptography (Chapter 19's harvest-now argument, Chapter 29's overhead numbers), the state of hardware (Chapters 28–30), and the education-first regional roadmap (Chapter 30, Lesson 5) with the cloud-lab costing. (2) Prepare five slides' worth of content — one figure each, chosen from this book's figure styles, each with an honest caption. (3) Anticipate and answer, in writing, the three questions decision-makers always ask: "When will it break our encryption?", "Should we buy a machine?", "What would you do with \$50,000?". (4) Have a classmate play hostile reviewer using Chapter 2's claim-checking checklist against *your own* brief. (5) Deliverable: brief + Q&A sheet + one-paragraph reflection on the hardest claim to state honestly.

REALITY CHECK

Project 12 is deliberately last: it is the one every reader will actually perform in life — at a family dinner, in a staff meeting, or before a minister. The quantum century will be navigated well or badly depending on whether the people explaining it can hold three truths at once: the mathematics is solid and spectacular; the engineering is real but young; and the timeline is uncertain in both directions. If this book has done its work, you can now hold all three — with worked examples to back every sentence.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] For Project 1: compute the two-sigma pass window for 10,000 shots.

Problem 2 [Basic] For Project 2: expand $\frac{1}{\sqrt{2}}(|++\rangle + |--\rangle)$ back into the standard basis and confirm it equals $|\Phi^+\rangle$.

Problem 3 [Standard] For Project 4: with two-qubit gate error 2% and three CNOT-equivalents per symbol round-trip, estimate the symbol error rate and check against the trend your simulation should show.

Problem 4 [Standard] For Project 5: predict the success probability at $k = 3$ for $N = 8$ and explain in one sentence why running *fewer* iterations than optimal can still be the right engineering choice on noisy hardware (Chapter 28 logic).

Problem 5 [Standard] For Project 8: at $q = 25\%$, compute the theoretical protected failure rate and compare it with the unprotected rate. Which side of the crossover is this?

Problem 6 [Challenge] Design Project 13 yourself: a one-page specification (mission, chapters used, five tasks, deliverable) for a capstone this chapter does not cover — candidates: a QAOA MaxCut study (Chapter 20), a phase-estimation precision ladder (Chapter 19), or a T1/T2 measurement campaign (Chapter 28, Problem 12). Include one worked-example-style calculation kernel.

APPENDICES

THE MATHEMATICS AND TOOLS YOU NEED

APPENDIX A

Complex Numbers

The number i , the complex plane, and the circle where quantum phases live

IN THIS CHAPTER YOU WILL LEARN TO

- Explain why quantum computing needs numbers beyond the ordinary real numbers.
- Calculate with the imaginary unit i , using $i^2 = -1$.
- Add, subtract, and multiply complex numbers, and draw them on the complex plane.
- Find the conjugate z^* , the modulus $|z|$, and the squared modulus $|z|^2$ of any complex number — the quantity that becomes probability in quantum mechanics.
- Use the polar form $e^{i\theta}$ and read it as a point on the unit circle — the home of quantum phases.

A.1 Why Quantum Computing Needs New Numbers

In Chapter 2 you learned that quantum amplitudes can be negative, so that computational paths can cancel. Nature goes one step further: amplitudes can also be *complex* — numbers built from the mysterious quantity $i = \sqrt{-1}$. This is not decoration. The phases of complex amplitudes are physical: they decide how paths interfere, they are what quantum gates like S and T (Chapter 7) manipulate, and they carry the “hidden” information that algorithms like Shor’s extract (Chapter 19).

Here is the good news: complex numbers are *easier* than their name suggests. They follow the ordinary rules of algebra plus exactly one new rule. If you can expand $(a + b)(c + d)$, you can master this chapter.

WORD TO KNOW

Complex number — A number of the form $z = a + bi$, where a and b are ordinary real numbers and i is the imaginary unit with $i^2 = -1$. Example: $3 + 2i$.

A brief note for engineering students: in electrical engineering the same number is often written j instead of i (because i is reserved for current). This book uses i , the standard in physics, mathematics, and quantum computing. If you have used phasors to analyze AC circuits, you already know complex numbers — quantum amplitudes will feel familiar.

CASE STUDY — THE ENGINEER WHO TAMED AC WITH COMPLEX NUMBERS • CHARLES STEINMETZ, USA, 1893

Before quantum mechanics existed, complex numbers proved themselves in power engineering. Alternating currents oscillate like waves, and adding two oscillations with different timing (phase) was a nightmare of trigonometry. In 1893 Charles Steinmetz showed that representing each oscillation as a single complex number (a phasor) turns that nightmare into simple multiplication and addition. Every electrical grid on Earth — including the ones this book's sister project, *Solar Energy for Everyone*, designs solar systems for — is analyzed with complex numbers to this day. Quantum mechanics later adopted the same tool for the same reason: complex numbers are the natural language of anything that oscillates or has a phase.

A.2 The Imaginary Unit *i*

For centuries, mathematicians insisted equations like $x^2 = -1$ have no solution, because any real number squared is positive or zero. The breakthrough was to stop insisting and *define* a new number that solves it.

WORD TO KNOW

Imaginary unit — The number i , defined by the single rule $i^2 = -1$. “Imaginary” is a historical nickname, not an insult — i is exactly as legitimate as -1 or $\sqrt{2}$, and physics uses it to describe real, measurable phenomena.

EQUATION: THE DEFINING RULE OF *i*

$$i^2 = -1$$

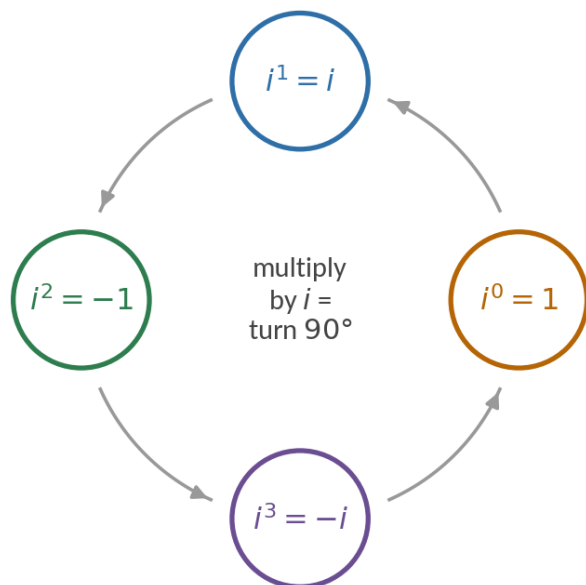
In words: i is the number whose square is minus one

Symbol	Meaning	Unit
i	the imaginary unit	none — a number

Everything else about complex numbers follows from this one rule plus ordinary algebra.

Powers of i follow a beautiful four-step cycle, worth knowing by heart:

$$i^0 = 1, \quad i^1 = i, \quad i^2 = -1, \quad i^3 = i^2 \cdot i = -i, \quad i^4 = (i^2)^2 = 1, \quad \text{and repeat.}$$



The powers of i cycle with period four. Geometrically (Lesson 5), each multiplication by i is a quarter turn.

WORKED EXAMPLE — COMPUTING A BIG POWER OF i

Find i^{23} .

1. The cycle repeats every 4 steps, so divide: $23 \div 4 = 5$ remainder 3.
2. Therefore $i^{23} = i^3$ (five full cycles bring you back to the start, then take 3 steps).
3. From the cycle: $i^3 = -i$.

Answer: $i^{23} = -i$

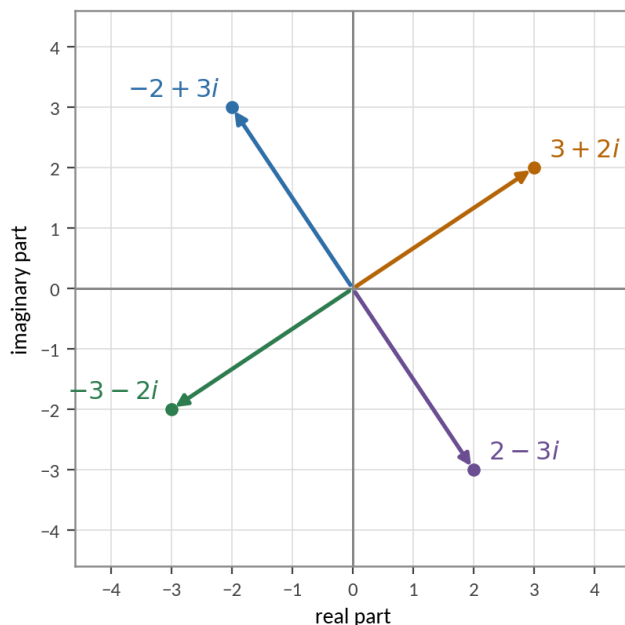
COMMON MISTAKE

✗ Wrong: $\sqrt{-9} = -3$, because the minus “comes out.”

✓ Right: $\sqrt{-9} = \sqrt{9}\sqrt{-1} = 3i$. Check: $(3i)^2 = 9i^2 = -9$ ✓. A real number can never square to a negative — only imaginary numbers do that.

A.3 Complex Numbers and the Complex Plane

A general complex number mixes a real part and an imaginary part: $z = a + bi$. The real part is $\text{Re}(z) = a$; the imaginary part is $\text{Im}(z) = b$ (note: b , not bi). Since z is built from two real numbers, the natural picture of complex numbers is a *plane*: real part on the horizontal axis, imaginary part on the vertical axis.



Every complex number is a point (or an arrow from the origin) in the complex plane: $a + bi$ sits a steps across and b steps up.

Adding and subtracting is done part by part — exactly like adding the arrows head-to-tail:

$$(a + bi) + (c + di) = (a + c) + (b + d)i.$$

Multiplying uses ordinary expansion of brackets plus the rule $i^2 = -1$:

EQUATION: MULTIPLYING COMPLEX NUMBERS

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i$$

In words: Expand the brackets as usual; wherever i^2 appears, replace it by -1 ; then collect real and imaginary parts

Symbol	Meaning	Unit
a, c	the real parts of the two numbers	none
b, d	the imaginary parts of the two numbers	none

You do not need to memorize the result — just expand and use $i^2 = -1$ every time.

WORKED EXAMPLE — ADDING AND MULTIPLYING

Let $z = 2 + 3i$ and $w = 1 - 2i$. Find $z + w$ and $z \times w$.

1. Add part by part: $z + w = (2 + 1) + (3 - 2)i = 3 + i$.
2. Multiply by expanding: $zw = (2)(1) + (2)(-2i) + (3i)(1) + (3i)(-2i)$.
3. Compute each piece: $= 2 - 4i + 3i - 6i^2$.

4. Use $i^2 = -1$: $-6i^2 = +6$, so $zw = (2 + 6) + (-4 + 3)i = 8 - i$.

Answer: $z + w = 3 + i$ and $zw = 8 - i$

WORKED EXAMPLE — A PRODUCT YOU WILL MEET AGAIN

Compute $(1 + i)(1 - i)$.

1. Expand: $(1)(1) + (1)(-i) + (i)(1) + (i)(-i) = 1 - i + i - i^2$.

2. The $\pm i$ terms cancel; $-i^2 = +1$.

3. Total: $1 + 1 = 2$.

Answer: $(1 + i)(1 - i) = 2$ — a real number. Multiplying a number by its “mirror” always gives a real result, as the next lesson explains.

ACTIVITY (CLASSROOM) — COMPLEX-NUMBER RACE

- Each student writes one complex number with small whole parts, like $2 - i$.
- In pairs, swap numbers and compute: the sum, the product, and each number plotted on a shared complex-plane drawing on the board.
- Check each other’s products by substituting back (does the expansion really give your answer?).
- Race round two: everyone computes $i \times$ (their own number) and plots it. What did multiplication by i do to every point on the board?

You need: Paper and pen, A board for the shared plane

► *App tool: Complex Calculator — type two complex numbers and see the sum and product appear on the plane*

A.4 Modulus, Conjugate, and $|z|^2$

Quantum mechanics constantly asks one question about a complex amplitude: *how big is it?* The answer is the modulus — the length of the arrow in the complex plane. By Pythagoras:

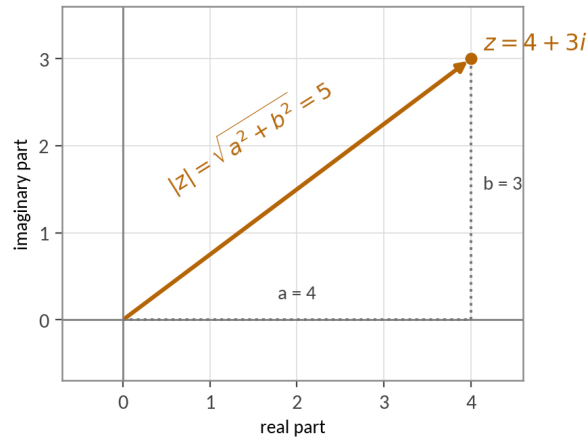
EQUATION: MODULUS (SIZE) OF A COMPLEX NUMBER

$$|z| = \sqrt{a^2 + b^2}$$

In words: Size of z = square root of (real part squared + imaginary part squared)

Symbol	Meaning	Unit
$ z $	the modulus (length of the arrow for z)	none — a size, never negative
a	the real part of z	none
b	the imaginary part of z	none

For a real number ($b = 0$) the modulus is just the absolute value — the same bars $| |$ mean the same idea.



The modulus is the length of the arrow: for $z = 4 + 3i$, Pythagoras gives $|z| = \sqrt{16 + 9} = 5$.

WORD TO KNOW

Conjugate — The mirror image of a complex number across the real axis: if $z = a + bi$, then $z^* = a - bi$ (flip the sign of the imaginary part). Read z^* as “z star.”

The conjugate looks like a small thing, but it is the key that unlocks $|z|^2$ without any square roots. Multiply a number by its own conjugate and watch:

$$z z^* = (a + bi)(a - bi) = a^2 - abi + abi - b^2 i^2 = a^2 + b^2.$$

EQUATION: THE SQUARED MODULUS — THE QUANTITY THAT BECOMES PROBABILITY

$$|z|^2 = z z^* = a^2 + b^2$$

In words: Size squared = the number times its conjugate = real part² + imaginary part²

Symbol	Meaning	Unit
$ z ^2$	the squared modulus of z	none — never negative
z^*	the conjugate of z : flip the sign of the imaginary part	none

Preview of Chapter 4: if an amplitude is z , the probability of its outcome is exactly $|z|^2$. This equation is used on almost every page of quantum computing.

WORKED EXAMPLE — FROM AMPLITUDE TO (FUTURE) PROBABILITY

A quantum amplitude will turn out to be $z = \frac{1}{2} + \frac{1}{2}i$. Find z^* , then $|z|^2$ two ways.

1. Conjugate: flip the imaginary sign: $z^* = \frac{1}{2} - \frac{1}{2}i$.
2. Way 1 — parts: $|z|^2 = a^2 + b^2 = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$.
3. Way 2 — product: $zz^* = \left(\frac{1}{2} + \frac{1}{2}i\right)\left(\frac{1}{2} - \frac{1}{2}i\right) = \frac{1}{4} - \frac{1}{4}i^2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$ ✓

Answer: $|z|^2 = 0.5$. In Chapter 4 language: this amplitude's outcome would occur with probability 50%.

COMMON MISTAKE

X Wrong: $|z|^2 = z^2$ — “just square the number.”

✓ Right: $|z|^2 = zz^*$, with the *conjugate*. Try $z = i$: squaring gives $z^2 = -1$ (negative — impossible for a probability!), while $zz^* = i \times (-i) = -i^2 = +1$. The conjugate is what guarantees the answer is real and never negative.

A.5 The Unit Circle and $e^{i\theta}$ (Polar Form)

Instead of asking “how far across and how far up?” (a and b), we can describe a complex number by “how long is the arrow, and at what angle?” This is the **polar form**, and it turns multiplication into geometry.

WORD TO KNOW

Phase — The angle θ of a complex number's arrow, measured counter-clockwise from the positive real axis. In quantum computing, relative phases between amplitudes are exactly what interference feeds on.

The bridge between angle language and $a + bi$ language is one of the most celebrated equations in mathematics:

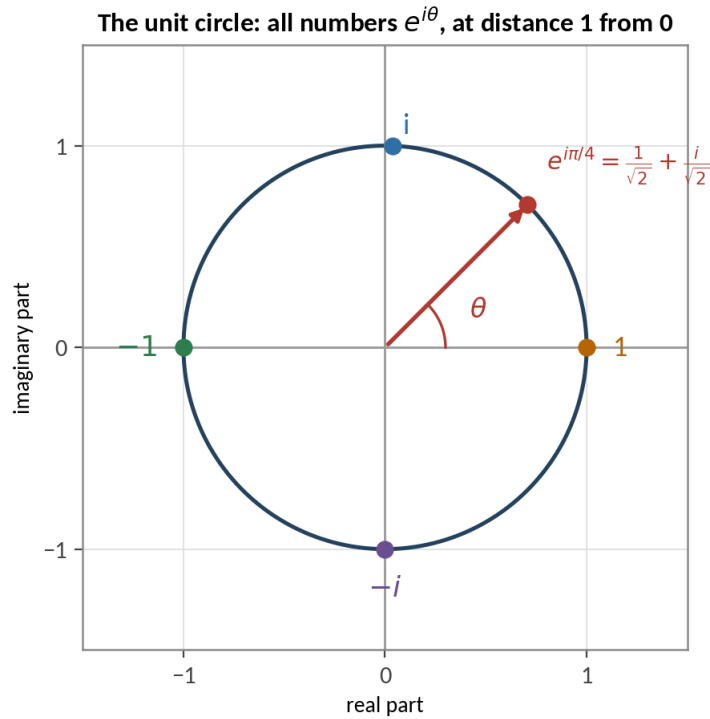
EQUATION: EULER'S FORMULA

$$e^{i\theta} = \cos \theta + i \sin \theta$$

In words: The number $e^{i\theta}$ is the point on the unit circle at angle θ

Symbol	Meaning	Unit
$e^{i\theta}$	a complex number of modulus 1 (arrow of length 1) at angle θ	none
θ	the angle (phase), measured counter-clockwise from the positive real axis	radians or degrees
$\cos \theta, \sin \theta$	its real and imaginary parts	none

You may take this formula as a given tool — this book never needs its calculus proof. Check the landmarks: $\theta = 0$ gives 1; $\theta = 90^\circ$ gives i ; $\theta = 180^\circ$ gives -1 ; $\theta = 270^\circ$ gives $-i$.

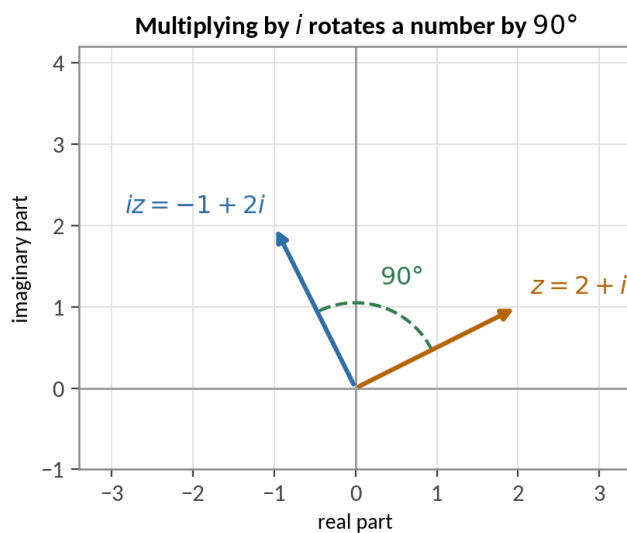


The unit circle. Every number $e^{i\theta}$ has modulus 1; the angle θ picks the point. The four landmarks $1, i, -1, -i$ sit at $0^\circ, 90^\circ, 180^\circ, 270^\circ$.

Any complex number can now be written as *length times direction*: $z = r e^{i\theta}$, with $r = |z|$. The payoff is the multiplication rule: to multiply complex numbers, **multiply the lengths and add the angles**:

$$(r_1 e^{i\theta_1}) (r_2 e^{i\theta_2}) = r_1 r_2 e^{i(\theta_1 + \theta_2)}.$$

In particular, multiplying by $e^{i\theta}$ *rotates* a number by θ without changing its size. That is why quantum phase gates (Chapter 7) are described as rotations.



Multiplying by $i = e^{i90^\circ}$ rotates any complex number a quarter turn: $2 + i$ becomes $-1 + 2i$, same length, turned 90° .

WORKED EXAMPLE — USING EULER'S FORMULA

Write $e^{i\pi/4}$ (that is, $\theta = 45^\circ$) in the form $a + bi$, and verify its modulus is 1.

1. Euler: $e^{i\pi/4} = \cos 45^\circ + i \sin 45^\circ$.
2. Standard values: $\cos 45^\circ = \sin 45^\circ = \frac{1}{\sqrt{2}} \approx 0.7071$.
3. So $e^{i\pi/4} = \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} \approx 0.7071 + 0.7071 i$.
4. Modulus: $\sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2} = \sqrt{\frac{1}{2} + \frac{1}{2}} = 1 \checkmark$

Answer: $e^{i\pi/4} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} i$, a unit-length arrow at 45° . You will meet this exact number as the phase of the T gate in Chapter 7.

WORKED EXAMPLE — MULTIPLY BY ADDING ANGLES

Compute $(e^{i90^\circ})(e^{i90^\circ})$ two ways.

1. Angle way: multiply lengths ($1 \times 1 = 1$), add angles ($90^\circ + 90^\circ = 180^\circ$): result $e^{i180^\circ} = -1$.
2. Algebra way: $e^{i90^\circ} = i$, and $i \times i = i^2 = -1 \checkmark$

Answer: -1 both ways. Two quarter-turns make a half-turn — Euler's formula turns algebra into geometry.

REALITY CHECK

Do quantum phases matter in real life, or are they mathematical decoration? They are as real as voltage. Interference of phases is how noise-cancelling headphones silence engines, how MRI machines image your body, and — in Chapters 18 and 19 — how Grover's and Shor's algorithms concentrate probability on correct answers. A quantum computer that could not control phases would be no computer at all.

ACTIVITY (APP) — WALK AROUND THE UNIT CIRCLE

1. Open the *Phase Explorer* in the QCFE app.
2. Drag the angle slider from 0° to 360° and watch the point $e^{i\theta}$ travel the circle while its $a + bi$ form updates live.
3. Stop at 60° . Write down a and b from the screen, then verify with a calculator: $a = \cos 60^\circ = 0.5$, $b = \sin 60^\circ \approx 0.866$.
4. Set two phases, $\theta_1 = 30^\circ$ and $\theta_2 = 60^\circ$, and use the multiply button. Confirm the product lands at 90° — exactly i .

You need: The QCFE app (Android or Windows), A calculator

► App tool: *Phase Explorer* — see $e^{i\theta}$ move on the unit circle as you turn the phase dial

A.6 The Clock Numbers: Roots of Unity

One family of complex numbers deserves its own farewell lesson, because a famous quantum algorithm runs on it. Fix a whole number N and look at the numbers whose N -th power is 1 — the **N -th roots of unity**. On the unit circle they are simply the N hour-marks of a clock:

EQUATION: THE N -TH ROOTS OF UNITY

$$\omega = e^{i360^\circ/N}, \quad \text{roots} = 1, \omega, \omega^2, \dots, \omega^{N-1}, \quad 1 + \omega + \omega^2 + \dots + \omega^{N-1} = 0$$

In words: N equally spaced points around the unit circle; step by multiplying, wrap after N steps — and all N of them add to zero

Symbol	Meaning	Unit
ω	one clock-step: $360^\circ/N$ of a turn	none
ω^k	k steps around the circle	none
sum = 0	perfectly spread arrows cancel	none

The sum-to-zero fact is pure symmetry: rotate the whole bundle of arrows by one step and it maps to itself, yet the rotation must also turn their sum — only the zero arrow survives both demands. This cancellation is the engine of the quantum Fourier transform (Chapter 19).

WORKED EXAMPLE — THE EIGHT HOURS OF $N = 8$

For $N = 8$: locate the roots, compute ω^3 in $a+bi$ form, and verify the cancellation for the four “odd corners.”

1. One step is $360^\circ/8 = 45^\circ$: the roots sit at $0^\circ, 45^\circ, 90^\circ, \dots, 315^\circ$ — compass points of the unit circle.
2. $\omega^3 = e^{i135^\circ} = \cos 135^\circ + i \sin 135^\circ = -0.707 + 0.707i$.
3. Opposite pairs cancel exactly: $\omega^1 + \omega^5 = 0$ and $\omega^3 + \omega^7 = 0$ (each pair points in opposite directions), and likewise for the even corners — hence the full sum is 0 ✓

Answer: eight arrows at 45° spacing, summing to zero by opposite-pair cancellation. When Chapter 19 computes $\text{QFT}_8 |5\rangle$ and its “phase staircase” of 225° steps, every amplitude will be one of these eight clock numbers — and the algorithm’s power is precisely their organized cancellation.

CHAPTER SUMMARY

- Quantum amplitudes are complex numbers; their phases power interference. The one new rule is $i^2 = -1$.
- A complex number $z = a + bi$ is a point (or arrow) in the complex plane. Add part by part; multiply by expanding brackets and replacing i^2 with -1 .
- Powers of i cycle with period four: $1, i, -1, -i$ — each multiplication by i is a quarter turn.

- The conjugate flips the imaginary sign: $z^* = a - bi$. The modulus is the arrow length: $|z| = \sqrt{a^2 + b^2}$.
- The workhorse of quantum probability: $|z|^2 = zz^* = a^2 + b^2$ — always real, never negative.
- Euler's formula $e^{i\theta} = \cos \theta + i \sin \theta$ describes the unit circle. Multiplying complex numbers multiplies lengths and adds angles; multiplying by $e^{i\theta}$ is a pure rotation.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] Simplify: (a) i^2 , (b) i^4 , (c) i^{10} , (d) $\sqrt{-25}$.

Problem 2 [Basic] Let $z = 4 + 2i$ and $w = 1 + 3i$. Compute $z + w$ and $z - w$, and plot all four numbers on one complex plane.

Problem 3 [Basic] Compute $(2 + i)(3 + 2i)$, showing the i^2 step explicitly.

Problem 4 [Basic] Write the conjugate of: (a) $5 + 4i$, (b) $-2 - 7i$, (c) $6i$, (d) 9 .

Problem 5 [Standard] Find $|z|$ and $|z|^2$ for $z = 6 - 8i$, using both formulas for $|z|^2$ (parts, and zz^*) and checking they agree.

Problem 6 [Standard] An amplitude is $z = 0.6i$. Compute $|z|^2$. Why would z^2 have been a physically impossible “probability”?

Problem 7 [Standard] Compute i^{101} using the remainder trick, and explain the trick in one sentence.

Problem 8 [Standard] Using Euler's formula, write in $a + bi$ form: (a) e^{i0° , (b) e^{i180° , (c) e^{i30° (use $\cos 30^\circ \approx 0.866$).

Problem 9 [Standard] Verify that $z = \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i$ lies on the unit circle, and give its phase angle in degrees (measure clockwise angles as negative, or equivalently give the angle between 0° and 360°).

Problem 10 [Challenge] Show algebraically that $|zw| = |z||w|$ for $z = 1 + i$ and $w = 2 - i$: compute zw first, then all three moduli.

Problem 11 [Challenge] The number $\omega = e^{i120^\circ}$ satisfies $\omega^3 = 1$ (a “cube root of 1”). Show this using the add-the-angles rule, then find ω in $a + bi$ form ($\cos 120^\circ = -0.5$, $\sin 120^\circ \approx 0.866$). Numbers like ω run the quantum Fourier transform in Chapter 19.

Problem 12 [Challenge] Two amplitudes $z_1 = \frac{1}{2}e^{i0^\circ}$ and $z_2 = \frac{1}{2}e^{i\theta}$ are added. Compute $|z_1 + z_2|^2$ for $\theta = 0^\circ$, 90° , and 180° . (For 90° : $z_2 = \frac{1}{2}i$.) Which value of θ gives constructive interference, which destructive, and which is in between?

Problem 13 [Basic] For $N = 4$: write the four roots in $a + bi$ form and verify their sum is zero.

Problem 14 [Standard] Show that $\omega^k \cdot \omega^m = \omega^{k+m}$ wraps around ($\omega^{k+N} = \omega^k$) using the add-the-angles rule — clock arithmetic on the circle.

Problem 15 [Challenge] Prove the sum-to-zero fact algebraically: multiply $S = 1 + \omega + \cdots + \omega^{N-1}$ by ω , observe $\omega S = S$, and conclude $S = 0$ whenever $\omega \neq 1$. Where does the argument fail for $N = 1$?

APPENDIX B

Vectors and Matrices

Column vectors, matrices, and the special matrices quantum mechanics allows

IN THIS CHAPTER YOU WILL LEARN TO

- Read and write column vectors, and picture them as arrows.
- Add vectors, multiply them by numbers, and compute inner products.
- Multiply a matrix by a vector, and multiply matrices together — in the right order.
- Find the dagger $M^\dagger$ (conjugate transpose) of a matrix.
- Test whether a matrix is unitary ($U^\dagger U = I$) — the entrance exam every quantum gate must pass.

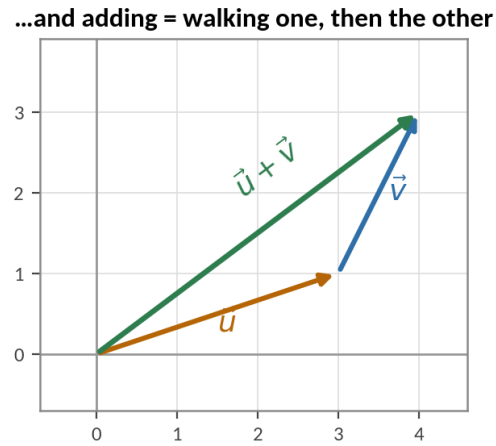
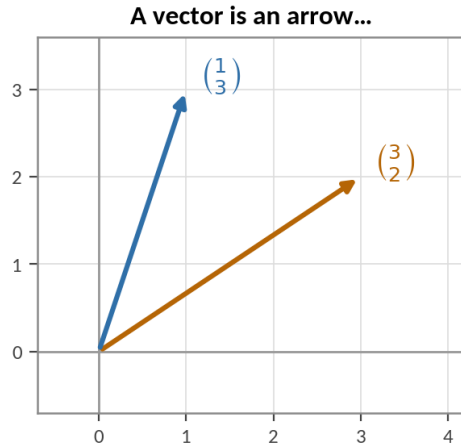
B.1 Vectors: Lists of Numbers with Meaning

Chapter 3 will represent every qubit state as a small list of amplitudes, and Chapter 6 will represent every quantum gate as a small grid of numbers acting on that list. The lists are **vectors**, the grids are **matrices**, and this chapter teaches you to handle both fluently. Nothing here is harder than careful multiplication and addition — but everything later stands on it, so work every example by hand.

WORD TO KNOW

Vector — An ordered list of numbers, written as a column. The vector $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ is the pair “3, then 2.” The numbers in it are called its components, and they may be complex.

A vector of two components can be drawn as an arrow in the plane: $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ is the arrow from the origin to the point 3 across, 2 up. This picture builds intuition, but remember the components are the definition — especially once they become complex numbers, which no flat drawing fully captures.



Left: vectors as arrows. Right: adding vectors means walking along one, then the other — component-wise addition does exactly this.

WORD TO KNOW

Dimension — The number of components in a vector. Qubit states live in dimension 2; two-qubit states in dimension 4; n -qubit states in dimension 2^n .

B.2 Adding, Scaling, and the Inner Product

Vectors combine by three basic operations, each computed component by component.

Addition adds matching components. **Scaling** multiplies every component by one number (called a *scalar*). These two are exactly how superpositions are built: the state $\alpha |0\rangle + \beta |1\rangle$ of Chapter 3 is literally “scale one vector by α , scale the other by β , add.”

EQUATION: VECTOR ADDITION AND SCALING

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} + \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \end{pmatrix} \qquad c \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} c u_1 \\ c u_2 \end{pmatrix}$$

In words: Add matching components; scaling multiplies every component by the same number

Symbol	Meaning	Unit
u_1, u_2, v_1, v_2	the components of the vectors (may be complex)	none
c	a scalar — a single (possibly complex) number	none

Vectors of different dimensions cannot be added.

WORKED EXAMPLE — BUILDING A COMBINATION

Compute $0.6 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 0.8 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ — the exact arithmetic behind the state $0.6 |0\rangle + 0.8 |1\rangle$.

1. Scale the first: $0.6 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.6 \\ 0 \end{pmatrix}$.

2. Scale the second: $0.8 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0.8 \end{pmatrix}$.

3. Add: $\begin{pmatrix} 0.6 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0.8 \end{pmatrix} = \begin{pmatrix} 0.6 \\ 0.8 \end{pmatrix}$.

Answer: $\begin{pmatrix} 0.6 \\ 0.8 \end{pmatrix}$ — the top component is the amount of the first building block, the bottom the amount of the second.

The third operation compares two vectors and returns a single number.

WORD TO KNOW

Inner product — A way to multiply two vectors into one number: conjugate the components of the first vector, multiply matching components, add. It measures overlap: how much the vectors point the same way.

EQUATION: THE INNER PRODUCT

$$\langle u|v \rangle = u_1^* v_1 + u_2^* v_2$$

In words: Inner product = (conjugate of first's components) $\times$ (second's components), summed

Symbol	Meaning	Unit
$\langle u v \rangle$	the inner product of u with v (Dirac writes it this way; Chapter 3)	none — one number
u_i^*	the complex conjugates of the first vector's components	none
v_i	the second vector's components	none

Why the conjugate? So that a vector's inner product with itself, $\langle u|u \rangle = |u_1|^2 + |u_2|^2$, is real and never negative — it is the squared length.

WORKED EXAMPLE — OVERLAP AND LENGTH

Let $u = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $v = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Find $\langle u|v \rangle$, $\langle u|u \rangle$, and $\langle v|v \rangle$.

1. $\langle u|v \rangle = 1^* \cdot \frac{1}{\sqrt{2}} + 0^* \cdot \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \approx 0.707$. Partial overlap.

2. $\langle u|u \rangle = 1 \cdot 1 + 0 \cdot 0 = 1$. Length $\sqrt{1} = 1$.

3. $\langle v|v \rangle = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2} + \frac{1}{2} = 1$. Also length 1.

Answer: $\langle u|v \rangle = 1/\sqrt{2}$; both vectors have length 1. Preview: length-1 vectors are exactly the legal quantum states, and $|\langle u|v \rangle|^2 = 1/2$ will be a measurement probability in Chapter 4.

WORD TO KNOW

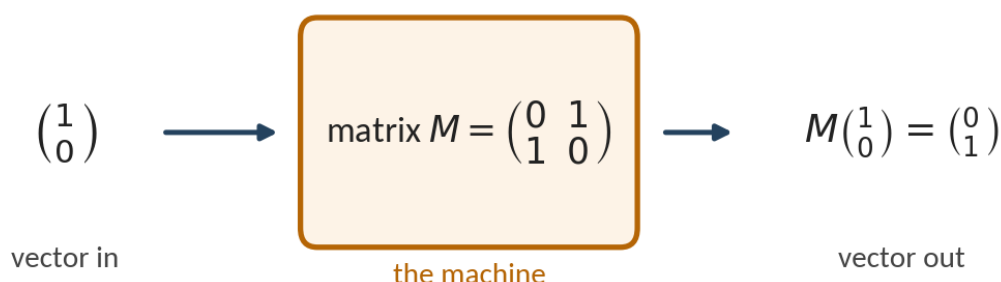
Orthogonal — Two vectors with inner product zero — they point in completely independent directions, with no overlap. The building blocks $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are orthogonal.

B.3 Matrices and Matrix-Vector Multiplication

WORD TO KNOW

Matrix — A rectangular grid of numbers. A 2×2 matrix has 2 rows and 2 columns. In quantum computing, matrices are the machines that transform state vectors — every quantum gate is a matrix.

Think of a matrix as a *machine*: a vector goes in, a vector comes out. The machine's behavior is fully described by its grid of numbers, and “running the machine” is a fixed arithmetic recipe: each output component is the inner-product-style combination of one *row* of the matrix with the input *column*.



A matrix is a machine that transforms vectors. This particular machine (you will meet it in Chapter 6 as the X gate) swaps the two components: in goes $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, out comes $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

EQUATION: MATRIX $\times$ VECTOR

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} av_1 + bv_2 \\ cv_1 + dv_2 \end{pmatrix}$$

In words: Top output = top row $\cdot$ vector; bottom output = bottom row $\cdot$ vector (row times column, then add)

Symbol	Meaning	Unit
a, b, c, d	the four entries of the matrix (row by row)	none
v_1, v_2	the components of the input vector	none

Output row i uses matrix row i . Say it aloud while computing: “row times column.”

WORKED EXAMPLE — RUNNING A MACHINE YOU WILL MEET AGAIN

Apply $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ to the vector $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$. (This is the Hadamard gate acting on $|0\rangle$ — the most important single calculation in this book.)

1. Ignore the scalar for a moment; compute $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

2. Top row: $1 \cdot 1 + 1 \cdot 0 = 1$. Bottom row: $1 \cdot 1 + (-1) \cdot 0 = 1$.

3. Result so far: $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Now apply the scalar $\frac{1}{\sqrt{2}}$: multiply every component.

Answer: $H \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} \approx \begin{pmatrix} 0.707 \\ 0.707 \end{pmatrix}$ — the equal superposition. Remember this result; Chapter 6 builds a whole lesson on it.

COMMON MISTAKE

✗ Wrong: Multiplying a matrix by a vector multiplies matching entries: top output = av_1 , bottom output = dv_2 .

✓ Right: Each output component uses a *whole row*: top = $av_1 + bv_2$, bottom = $cv_1 + dv_2$. The off-diagonal entries b and c are how machines mix components — without them, no gate could ever create a superposition.

B.4 Matrix $\times$ Matrix, the Identity, and the Dagger

Composing machines. Feeding a vector through machine A and then machine B is the same as feeding it once through a single combined machine. That combined machine is the *matrix product* BA — and the order matters enormously. Because the vector meets A first, A stands *closest to the vector*: first A , then B , is written BAv . Quantum circuits (Chapter 6) are read left to right, but their matrix products are written right to left — practice this now and it will never confuse you again.

EQUATION: MATRIX $\times$ MATRIX

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{pmatrix}$$

In words: Entry in row i , column j of the product = (row i of the left matrix) $\cdot$ (column j of the right matrix)

Symbol	Meaning	Unit
$a \dots d$	entries of the left matrix	none
$e \dots h$	entries of the right matrix	none

In general $AB \neq BA$ — test any two random matrices and see. Quantum gates inherit this: order of operations matters physically.

WORD TO KNOW

Identity matrix — The do-nothing machine $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$: for every vector v , $Iv = v$. It plays the role of “1” in matrix arithmetic.

WORKED EXAMPLE — TWO FLIPS MAKE NOTHING

Let $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (the swap-components machine). Compute XX .

1. Row 1 of left times column 1 of right: $0 \cdot 0 + 1 \cdot 1 = 1$.
2. Row 1 times column 2: $0 \cdot 1 + 1 \cdot 0 = 0$.
3. Row 2 times column 1: $1 \cdot 0 + 0 \cdot 1 = 0$. Row 2 times column 2: $1 \cdot 1 + 0 \cdot 0 = 1$.

Answer: $XX = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$. **Swapping twice returns everything — the algebra knows that flipping a bit twice does nothing.**

WORD TO KNOW

Dagger — The conjugate transpose $M^\dagger$ of a matrix: flip it across its main diagonal (transpose), then conjugate every entry. Read “M dagger.” For vectors, the dagger turns a column into a conjugated row.

EQUATION: THE DAGGER (CONJUGATE TRANSPOSE)

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^\dagger = \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix}$$

In words: Swap across the diagonal, then conjugate every entry

Symbol	Meaning	Unit
$M^\dagger$	the dagger (conjugate transpose) of M	none
a^* etc.	complex conjugates of the entries	none

For real matrices the dagger is just the transpose. Note how b and c trade places while a and d stay.

WORKED EXAMPLE — A DAGGER WITH COMPLEX ENTRIES

Find $M^\dagger$ for $M = \begin{pmatrix} 1 & i \\ 0 & 2 \end{pmatrix}$, and compute $M^\dagger M$.

1. Transpose (swap across the diagonal): $\begin{pmatrix} 1 & 0 \\ i & 2 \end{pmatrix}$.
2. Conjugate every entry: $M^\dagger = \begin{pmatrix} 1 & 0 \\ -i & 2 \end{pmatrix}$.
3. Multiply: $M^\dagger M = \begin{pmatrix} 1 & 0 \\ -i & 2 \end{pmatrix} \begin{pmatrix} 1 & i \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 \cdot 1 + 0 \cdot 0 & 1 \cdot i + 0 \cdot 2 \\ -i \cdot 1 + 2 \cdot 0 & -i \cdot i + 2 \cdot 2 \end{pmatrix} = \begin{pmatrix} 1 & i \\ -i & 5 \end{pmatrix}$.

Answer: $M^\dagger = \begin{pmatrix} 1 & 0 \\ -i & 2 \end{pmatrix}$ and $M^\dagger M = \begin{pmatrix} 1 & i \\ -i & 5 \end{pmatrix}$. **Since $M^\dagger M \neq I$, this M is not unitary — it could never be a quantum gate, as the final lesson now explains.**

B.5 Unitary Matrices: The Entrance Exam for Quantum Gates

Quantum mechanics puts one non-negotiable condition on every machine that transforms quantum states: *it must preserve total probability*. States are length-1 vectors (Chapter 3), and probabilities are squared amplitudes, so the machine must preserve vector length. Matrices that preserve length have a name and a beautifully simple test.

WORD TO KNOW

Unitary matrix — A matrix U satisfying $U^\dagger U = I$. Unitary machines preserve lengths and inner products — they rotate and re-phase vectors but never stretch, shrink, or destroy them. Every quantum gate is unitary.

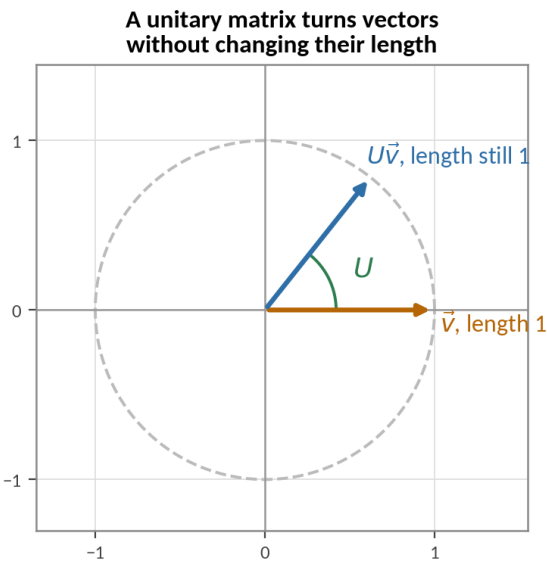
EQUATION: THE UNITARY CONDITION

$$U^\dagger U = I$$

In words: A matrix is unitary if its dagger times itself gives the identity

Symbol	Meaning	Unit
U	the matrix being tested	none
$U^\dagger$	its conjugate transpose	none
I	the identity matrix	none

Equivalent statement: $U^\dagger = U^{-1}$ — the dagger undoes U . Every quantum gate is therefore reversible, in the strongest possible sense: its inverse is free, just take the dagger.



What unitaries do: turn and re-phase, never stretch. A length-1 state vector stays length 1 — total probability stays 100%.

WORKED EXAMPLE — H PASSES THE ENTRANCE EXAM

Show that the Hadamard matrix $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ is unitary.

1. H is real, so $H^\dagger = \text{transpose}$. Transposing swaps the off-diagonal entries — but they are both 1, so $H^\dagger = H$.
2. Compute $H^\dagger H = HH = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \cdot 1 + 1 \cdot 1 & 1 \cdot 1 + 1 \cdot (-1) \\ 1 \cdot 1 + (-1) \cdot 1 & 1 \cdot 1 + (-1)(-1) \end{pmatrix}$.
3. Evaluate: $\frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \checkmark$

Answer: $H^\dagger H = I$, so H is unitary — and since $H^\dagger = H$, the Hadamard gate is its own inverse: applying it twice does nothing. Chapter 6 uses this fact constantly.

REALITY CHECK

Unitarity is not bureaucracy — it is physics enforcing three deep facts at once. (1) Probabilities always total 100%: no state can leak away. (2) Quantum computation is reversible: unlike the AND gate of Chapter 1, no unitary gate ever destroys information. (3) There is no legal “copy” machine or “delete” machine — both would violate unitarity, as Chapter 12 proves. When you check $U^\dagger U = I$, you are checking the laws of nature.

ACTIVITY (HOME) — THE UNITARY ENTRANCE EXAM

1. Test whether $M_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is unitary: form the dagger, multiply, compare with I .
2. Test $M_2 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ the same way.
3. Test $M_3 = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$ (careful: the dagger conjugates i to $-i$).
4. For each matrix that failed, apply it to $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and find a vector whose length it changed.

You need: Paper and pen

► App tool: *Matrix Lab* — enter any 2×2 matrix; the app applies it to states, checks unitarity, and shows lengths before and after

B.6 Eigenvectors: The Directions a Matrix Keeps

One last matrix idea completes your toolkit — the one behind measurement outcomes, phase kickback, and phase estimation. Most vectors, hit by a matrix, change direction. A few special ones do not: the matrix only *stretches* them by a number. Those directions are the matrix’s **eigenvectors**; the stretch factors are its

eigenvalues.

EQUATION: THE EIGENVECTOR EQUATION

$$A \vec{v} = \lambda \vec{v}$$

In words: The matrix acts on its eigenvector as a pure number — direction kept, length scaled by the eigenvalue

Symbol	Meaning	Unit
$\vec{v} \neq 0$	an eigenvector of A	none
λ	its eigenvalue (the stretch factor)	none

Quantum dictionary: an observable's eigenvectors are the states with a definite answer, and its eigenvalues are the answers (Chapter 8). For unitary gates every eigenvalue is a pure phase $e^{i\varphi}$ — length 1, as unitarity demands — and extracting such φ 's is exactly what phase kickback (Chapter 10) and phase estimation (Chapter 19) do.

WORKED EXAMPLE — FINDING X'S SPECIAL DIRECTIONS

Find the eigenvectors and eigenvalues of $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

1. Write the demand $X \begin{pmatrix} a \\ b \end{pmatrix} = \lambda \begin{pmatrix} a \\ b \end{pmatrix}$: components give $b = \lambda a$ and $a = \lambda b$.
2. Substitute: $a = \lambda^2 a$, so $\lambda^2 = 1$: eigenvalues $\lambda = +1$ and $\lambda = -1$.
3. For $\lambda = +1$: $b = a$: direction $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$, normalized $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ — the state $|+\rangle$. For $\lambda = -1$: $b = -a$: $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = |-\rangle$.
4. Check one: $X |-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = -|-\rangle$ ✓

Answer: X keeps exactly two directions, $|+\rangle$ (stretch +1) and $|-\rangle$ (stretch −1). This is why $|\pm\rangle$ are the X -measurement's answer states, and why a CNOT with target $|-\rangle$ kicks a −1 back onto its control — the eigen-machinery you have now derived by hand.

CHAPTER SUMMARY

- Vectors are columns of (possibly complex) numbers; add component-wise, scale by multiplying every component.
- The inner product $\langle u|v \rangle = u_1^* v_1 + u_2^* v_2$ measures overlap; $\langle u|u \rangle$ is the squared length; inner product 0 means orthogonal.
- A matrix is a machine on vectors: output row i = row i times the input column. Matrix products compose machines and are written right to left: first A then B is BA .
- The identity I does nothing. The dagger $M^\dagger$ transposes and conjugates.
- Quantum gates must be unitary: $U^\dagger U = I$. Unitaries preserve length (total probability), are perfectly reversible, and their inverse is simply $U^\dagger$.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher's Guide.

Problem 1 [Basic] Compute $\begin{pmatrix} 2 \\ 5 \end{pmatrix} + \begin{pmatrix} -1 \\ 3 \end{pmatrix}$ and $4\begin{pmatrix} 1 \\ -2 \end{pmatrix}$.

Problem 2 [Basic] Compute the inner product of $u = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ with itself, and hence the length of u .

Problem 3 [Basic] Apply $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ to $\begin{pmatrix} 0.6 \\ 0.8 \end{pmatrix}$.

Problem 4 [Basic] Write the dagger of $\begin{pmatrix} 2 & 3i \\ 1-i & 4 \end{pmatrix}$.

Problem 5 [Standard] Compute $\langle u|v \rangle$ for $u = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $v = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$. What word describes this pair of vectors?

Problem 6 [Standard] Let $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Compute XZ and ZX . Are they equal? What does your result mean for the order of quantum operations?

Problem 7 [Standard] Apply $H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ to $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, and compare the result with $H\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ from the lesson. What differs, and in which component?

Problem 8 [Standard] Verify that Z (from Problem 6) is unitary.

Problem 9 [Standard] A machine is $M = \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$. Apply it to $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and to $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, compute the output lengths, and explain in one sentence why no quantum computer may contain this machine.

Problem 10 [Challenge] Show that the product of two unitaries is unitary, using the rule $(AB)^\dagger = B^\dagger A^\dagger$: expand $(UV)^\dagger(UV)$ step by step. Why is this closure property essential for building circuits from gates?

Problem 11 [Challenge] Find all values of the complex number c that make $M = \begin{pmatrix} c & 0 \\ 0 & c \end{pmatrix}$ unitary. (Hint: the condition becomes $|c|^2 = 1$; describe the answer as a set of points in the complex plane.)

Problem 12 [Challenge] Compute HZH (multiply right to left: first ZH , then $H \cdot (ZH)$), where H and Z are as above. Compare your result with X . This identity — “ X is Z dressed in Hadamards” — returns with a starring role in Chapter 6.

Problem 13 [Basic] Read off (no computation) the eigenvectors and eigenvalues of $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, and connect them to the standard measurement.

Problem 14 [Standard] Verify that $\vec{v} = (\cos 22.5^\circ, \sin 22.5^\circ)$ is an eigenvector of H with eigenvalue $+1$ by direct multiplication ($\cos 22.5^\circ = 0.924$, $\sin 22.5^\circ = 0.383$). Where does this direction sit on the Bloch sphere relative to the z - and x -axes?

Problem 15 [Challenge] Show that a unitary matrix's eigenvalues must have $|\lambda| = 1$: apply U to a normalized eigenvector, use the length-preservation property $\|U\vec{v}\| = \|\vec{v}\|$, and conclude. What does this say about the “answers” a phase-estimation algorithm can ever return?

APPENDIX C

The Probability You Need

Chance, averages, spread, and the square-root law — the statistics this book runs on

Quantum mechanics speaks probability as its native language: the Born rule turns amplitudes into chances, histograms into evidence, and shot counts into error bars. This appendix teaches, from zero, exactly the probability the main text uses — no more, no less. If Chapter 4’s wobbles or Chapter 28’s budgets ever feel slippery, this is the ground to stand on.

C.1 Probability: The Rules of Chance

A **probability** is a number between 0 (impossible) and 1 (certain) attached to an outcome, with one book-keeping law: over all mutually exclusive outcomes of an experiment, the probabilities **add to 1**. From that, two everyday rules. *Either-or* (exclusive outcomes): add. *Not*: $P(\text{not } A) = 1 - P(A)$ — the complement trick that repeatedly rescues hard counts in this book (“the probability at least one error occurs” = $1 -$ “none occur”).

WORKED EXAMPLE — THE COMPLEMENT TRICK, TWICE

(a) A die shows a number: what is $P(\text{at least } 3)$? (b) A check catches an eavesdropper with probability $\frac{1}{4}$ per checked bit; what is $P(\text{caught})$ after 12 checks?

1. (a) Direct add: $P(3) + P(4) + P(5) + P(6) = \frac{4}{6} = \frac{2}{3}$. Or complement: $1 - P(1) - P(2) = 1 - \frac{2}{6} = \frac{2}{3}$

✓

2. (b) Complement is the only sane route: $P(\text{missed every time}) = (\frac{3}{4})^{12} = 0.0317$.

3. So $P(\text{caught}) = 1 - 0.0317 = 96.8\%$.

Answer: $\frac{2}{3}$; and 96.8% — which the reader of Chapter 15 will recognize as BB84’s eavesdropper-detection arithmetic, verbatim. “One minus the probability nothing happens” is the single most reused formula in this book.

C.2 Independence: When Chances Multiply

Two events are **independent** when neither influences the other’s odds; then, and only then, $P(A \text{ and } B) = P(A) P(B)$. Chains of independent steps multiply: survive step one AND step two AND step three...

EQUATION: THE SURVIVAL CHAIN

$$P(\text{all } d \text{ steps succeed}) = (1 - p)^d$$

In words: Independent per-step success probabilities multiply down the chain

Symbol	Meaning	Unit
p	per-step failure probability	none
d	number of steps	none

This one formula runs Chapter 28's circuit budgets, Chapter 30's datasheet reads, and Chapter 16's fiber tolls. Handy approximation: for small p , $(1 - p)^d \approx 1 - pd$ until pd nears 1 — “errors add before they compound.”

C.3 Expectation and Spread: One Number Each

A random quantity is summarized by its **expectation** (probability-weighted average) and its **spread** (standard deviation — the typical distance from that average):

EQUATION: EXPECTATION AND STANDARD DEVIATION

$$\mathbb{E}[X] = \sum x P(x), \quad \sigma = \sqrt{\mathbb{E}[X^2] - \mathbb{E}[X]^2}$$

In words: Average = outcomes weighted by their chances; spread = square root of (average of squares minus square of average)

Symbol	Meaning	Unit
$\mathbb{E}[X]$	the expectation (long-run mean)	varies
σ	the standard deviation (spread)	varies

These are precisely Chapter 8's $\langle A \rangle$ and ΔA — expectation values and uncertainties are ordinary statistics applied to quantum outcomes.

WORKED EXAMPLE — A DIE, FULLY PROCESSED

Compute the expectation and spread of one fair die roll.

1. Expectation: $\mathbb{E}[X] = \frac{1+2+\dots+6}{6} = 3.5$.
2. Average of squares: $\mathbb{E}[X^2] = \frac{1+4+9+16+25+36}{6} = \frac{91}{6} = 15.17$.
3. Spread: $\sigma = \sqrt{15.17 - 3.5^2} = \sqrt{2.92} = 1.71$.

Answer: 3.5 ± 1.71 : rolls average 3.5, typically landing within about 1.7 of it. The same three-step recipe computes every $\langle Z \rangle$ and ΔZ in Chapter 8 — with outcomes ± 1 instead of 1–6.

C.4 Repetition and the Square-Root Law

Repeat an experiment n times and count successes: the count's expectation is np , and its spread is $\sqrt{np(1-p)}$ — growing only like $\sqrt{n}$. Divide by n for the observed frequency, and the spread shrinks like

$1/\sqrt{n}$:

EQUATION: THE SQUARE-ROOT LAW (SHOT NOISE)

$$\text{frequency} = \frac{\text{count}}{n} \approx p \pm \sqrt{\frac{p(1-p)}{n}} \quad \text{worst case } \pm \frac{0.5}{\sqrt{n}}$$

In words: Estimates sharpen with the square root of the repetitions: 4 times the shots, half the wobble

Symbol	Meaning	Unit
n	number of repetitions (shots)	none
p	the true probability being estimated	none
$\pm$	the one-sigma wobble of the estimate	none

The origin of every “ $\sigma \approx 0.5/\sqrt{n}$ ” in this book, from Chapter 4’s histograms to Chapter 20’s energy estimates. Rules of thumb: within one sigma about 68% of the time, within two about 95% — “two sigma” is this book’s default standard of evidence.

WORKED EXAMPLE — HOW WOBBLY IS A THOUSAND-SHOT HISTOGRAM?

A fair-coin qubit ($p = 0.5$) is measured 1000 times. What counts are normal, and when should you suspect the device?

1. Expected count: $np = 500$.
2. Spread of the count: $\sqrt{1000 \times 0.25} = 15.8$.
3. Normal range (two sigma): 500 ± 32 , i.e. roughly 468–532 zeros.
4. A run showing 561 zeros sits $61/15.8 = 3.9$ sigma out: chance alone does that about once in ten thousand runs — investigate the device, not your luck (Chapter 4’s certification logic).

Answer: 500 ± 32 is business as usual; 561 is evidence. The square-root law is what lets finite histograms certify infinite claims — and its $1/\sqrt{n}$ stinginess is why precision always costs quadratically in shots.

C.5 Conditional Probability: Updating on Evidence

$P(A \mid B)$ — the probability of A given that B happened — is computed by shrinking the world to the outcomes where B holds and renormalizing: $P(A \mid B) = P(A \text{ and } B)/P(B)$. The main text uses exactly this shape twice: the partial-measurement rule of Chapter 9 (keep consistent amplitudes, renormalize) and the readout un-lying of Chapter 14.

WORKED EXAMPLE — THE RARE-DISEASE SURPRISE (BAYES IN ONE TABLE)

A condition affects 1% of a population. A test catches 90% of true cases but also flags 8% of healthy people. A random person tests positive: how worried should they be?

1. Imagine 10,000 people: 100 have the condition, 9900 do not.
2. Positives among the sick: $0.90 \times 100 = 90$. Positives among the healthy: $0.08 \times 9900 = 792$.

3. Condition given positive: $90/(90 + 792) = 10.2\%$.

Answer: about 10% — the flag is nine-to-one a false alarm, because the condition is rare and the test imperfect. This “base-rate” arithmetic is Chapter 23’s precision computation for imbalanced classifiers, and the sober lens for every rare-event detector, quantum or classical.

REALITY CHECK

Everything above is *classical* probability — and quantum mechanics obeys all of it at the level of measurement outcomes. What quantum theory changes is where the probabilities *come from*: amplitudes that add and cancel *before* squaring (Chapter 2’s double slit). Keep the two layers straight and the subject stays honest: amplitudes interfere; probabilities, once born, add, multiply, and update exactly as this appendix taught.

Practice Problems

Try each problem on paper first. Full step-by-step solutions are in the companion Teacher’s Guide.

Problem 1 [Basic] A qubit gives 0 with probability 0.36. Compute $P(1)$, and then the probability of seeing at least one 1 across three independent shots (use the complement trick).

Problem 2 [Basic] Compute the expectation and spread of a ± 1 coin with $P(+1) = 0.8$ — and connect the two numbers to $\langle Z \rangle$ and ΔZ of Chapter 8.

Problem 3 [Basic] A circuit has 60 independent steps at $p = 0.5\%$ failure each. Compute the survival probability exactly and by the $1 - pd$ approximation, and compare.

Problem 4 [Standard] How many shots reduce the worst-case wobble to ± 0.01 ? To ± 0.001 ? State the general scaling rule these two answers demonstrate.

Problem 5 [Standard] A 400-shot histogram shows 236 zeros. Give the frequency estimate with its one-sigma wobble, and test the hypothesis $p = 0.5$ at the two-sigma standard.

Problem 6 [Standard] Rework the rare-disease table for a better test (98% catch rate, 2% false alarms) at the same 1% prevalence. What is $P(\text{condition} \mid \text{positive})$ now, and what single factor helped most?

Problem 7 [Challenge] Derive the worst-case wobble: show $p(1 - p) \leq \frac{1}{4}$ for $p \in [0, 1]$ (complete the square), so the frequency spread never exceeds $0.5/\sqrt{n}$ — the bound this book quotes everywhere.

Problem 8 [Challenge] The 3-qubit code fails when 2 or 3 qubits flip (per-qubit probability q , independent). Derive $P_{\text{fail}} = 3q^2(1 - q) + q^3$ by counting the ways, evaluate at $q = 10\%$, and compare with the unprotected failure q — reproducing Chapter 29’s central formula from pure probability.

APPENDIX D

Formula Sheet and Gate Reference

Every formula, gate, and rule of thumb in this book — on six pages, for the exam and the lab

This appendix is the book folded into a reference card. Each entry names the chapter that teaches it — if a line is not obvious, that chapter is your remedy. Angles are written both ways where useful: $90^\circ = \pi/2$.

D.1 States and Probability

One qubit, many qubits, and the Born rule

Rule	Formula	Where taught
General qubit state	$ \psi\rangle = \alpha 0\rangle + \beta 1\rangle$, α, β complex	Chapter 3
Normalization	$ \alpha ^2 + \beta ^2 = 1$	Chapter 3
Born rule	$P(0) = \alpha ^2$, $P(1) = \beta ^2$	Chapter 4
Measurement in any basis	$P(\phi) = \langle\phi \psi\rangle ^2$	Chapter 4
Fidelity (state overlap)	$F = \langle\phi \psi\rangle ^2$	Chapter 3
Collapse	after result a : state $\rightarrow$ the a -eigenstate	Chapter 4
Tensor product	$(a 0\rangle + b 1\rangle) \otimes (c 0\rangle + d 1\rangle) = ac 00\rangle + ad 01\rangle + bc 10\rangle + bd 11\rangle$	Chapter 9
Dimension	n qubits $\Rightarrow 2^n$ amplitudes	Chapter 9
Marginal	$P(A=1) = \sum_y a_{1y} ^2$	Chapter 9
Partial measurement	keep consistent amplitudes, divide by $\sqrt{P}$	Chapter 9
Bloch vector	$r = (\langle X \rangle, \langle Y \rangle, \langle Z \rangle)$, $ r = 1$ pure, $ r < 1$ mixed	Chapters 5, 8
Expectation value	$\langle A \rangle = \sum_a a P(a)$; for Pauli: $\langle Z \rangle = P(0) - P(1)$	Chapter 8
Spread and uncertainty	$\Delta P = \sqrt{1 - \langle P \rangle^2}$; $\Delta Z \Delta X \geq \langle Y \rangle $	Chapter 8
Shot wobble	frequency estimate $\pm \approx 0.5/\sqrt{n}$	Chapter 4

D.2 The Gate Menu

Single-qubit gates (matrices in the $|0\rangle, |1\rangle$ basis)

Gate	Matrix	Action / Bloch move
X	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	bit flip; 180° turn about x-axis
Y	$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$	bit+phase flip; 180° about y-axis
Z	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	phase flip $ 1\rangle \rightarrow - 1\rangle$; 180° about z-axis
H	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	$ 0\rangle \leftrightarrow +\rangle, 1\rangle \leftrightarrow -\rangle$; swaps x- and z-axes
S	$\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$	quarter turn (90°) about z; $S^2 = Z$
T	$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$	eighth turn (45°) about z; $T^2 = S$; the expensive gate of fault tolerance
$R_y(\theta)$	$\begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}$	smooth turn by θ about y; the workhorse of encodings and ansätze
$R_z(\theta)$	$\begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}$	smooth turn by θ about z; phase dial

Multi-qubit gates

Gate	Action on basis states	Notes
CNOT	target $\oplus$ control	entangler-in-chief; self-inverse
CZ	sign flip on $ 11\rangle$ only	symmetric between the two qubits
SWAP	exchange the two qubits	three alternating CNOTs
Toffoli (CCX)	target $\oplus$ (control ₁ AND control ₂)	reversible AND; universal for classical logic
controlled- U	apply U on target iff control is 1	phases kick back onto the control

Useful identities: $HZH = X$, $HXH = Z$, $XZ = -ZX$, $S^2 = Z$, $T^2 = S$, $(H \otimes H) \text{CNOT}_{1 \rightarrow 2} (H \otimes H) = \text{CNOT}_{2 \rightarrow 1}$ (Chapters 6, 7, 10).

D.3 The Famous States and Protocol Scorecards

States worth knowing by heart

State	Definition	Signature
$ \pm\rangle$	$\frac{1}{\sqrt{2}}(0\rangle \pm 1\rangle)$	equator of the Bloch sphere
Bell $\Phi^\pm$	$\frac{1}{\sqrt{2}}(00\rangle \pm 11\rangle)$	always agree
Bell $\Psi^\pm$	$\frac{1}{\sqrt{2}}(01\rangle \pm 10\rangle)$	always disagree
GHZ	$\frac{1}{\sqrt{2}}(000\rangle + 111\rangle)$	three-way all-or-nothing
Uniform	$H^{\otimes n} 0 \dots 0\rangle = \frac{1}{\sqrt{2^n}} \sum_x x\rangle$	every algorithm's opening move

Protocols and algorithms at a glance

Item	The numbers
Teleportation	1 Bell pair + 2 classical bits per qubit; corrections $\Phi^+ : I, \Psi^+ : X, \Phi^- : Z, \Psi^- : ZX$ (Chapter 13)
Superdense coding	1 Bell pair + 1 qubit carries 2 classical bits (Chapter 13)
BB84	$\approx$ half the rounds survive sifting; intercept-resend leaves 25% errors in checked bits; detection $P = 1 - (3/4)^n$ (Chapter 15)
CHSH / E91	classical $ S \leq 2$; quantum $\max 2\sqrt{2} \approx 2.83$; full intercept-resend $\rightarrow S = \sqrt{2} \approx 1.41$ (Chapters 11, 15)
Grover	$\sin \theta = \sqrt{M/N}$; $P(k) = \sin^2((2k+1)\theta)$; $k_{\text{opt}} \approx \frac{\pi}{4} \sqrt{N/M}$; overshoot is real (Chapter 18)
QFT	$ x\rangle \rightarrow \frac{1}{\sqrt{N}} \sum_k e^{2\pi i x k / N} k\rangle$; circuit cost $O(n^2)$ (Chapter 19)
Shor	period r of $a^x \bmod N \Rightarrow$ factors from $\gcd(a^{r/2} \pm 1, N)$ (Chapter 19)
VQE	$\langle H \rangle = \sum_i c_i \langle P_i \rangle \geq E_0$; batch qubit-wise-commuting terms (Chapter 20)
Parameter shift	$\partial E / \partial \theta = \frac{1}{2} [E(\theta + 90^\circ) - E(\theta - 90^\circ)]$ (Chapter 25)
Kernel (1 feature)	$K(x, y) = \cos^2(\frac{x-y}{2})$ for the R_y map; scale features into $[0, \pi]$ (Chapter 24)
TV distance	$\text{TV}(p, q) = \frac{1}{2} \sum_x p(x) - q(x) $ (Chapter 26)

D.4 Noise, Errors, and Budgets

The reliability formulas

Rule	Formula	Where taught
Energy relaxation	$P_1(t) = e^{-t/T_1}$; half-life = $T_1 \ln 2$	Chapter 28
Ramsey dephasing	fringe envelope e^{-t/T_2} ; $T_2 \leq 2T_1$	Chapter 28
Circuit survival	$P \approx (1 - p)^d$; trouble near $d \approx 1/p$	Chapter 28
Full budget	$F \approx (1 - p_1)^{n_1} (1 - p_2)^{n_2} (1 - p_r)^{n_q} e^{-t/T_1}$	Chapter 30
Rules of thumb	$n_2 \lesssim 0.1/p_2$; $t_{\text{circ}} \lesssim 0.1 T_1$	Chapter 30
3-qubit code failure	$P_{\text{fail}} = 3q^2(1 - q) + q^3$; beats no protection for $q < 50\%$	Chapter 29
Code distance	distance d corrects $\lfloor (d - 1)/2 \rfloor$ errors	Chapter 29
Logical error scaling	below threshold, error $\sim \Lambda^{-(d+1)/2}$: bigger code, better qubit	Chapter 29

D.5 Quick Machine-Learning Reference

Encoding invoice (Chapter 22): basis $n_q = \text{bits}$; angle $n_q = d$; amplitude $n_q = \lceil \log_2 d \rceil$ (but loading may cost $\sim 2^{n_q}$ gates). Re-uploading (Chapter 25): L layers buy up to L frequencies per feature. Barren-plateau flag (Chapter 25): deep scrambled circuits have gradients $\sim 2^{-n}$; resolve needs $\sim 2^{2n}$ shots — restructure instead. Audit questions (Chapter 27): Baseline, Data, Advantage-source, Shots, Honest-scaling.

REALITY CHECK

A formula sheet is a map, not the territory: every line above was *earned* somewhere in this book by a worked example you can redo. If a formula here ever surprises you, treat that as the book's invitation — the chapter reference beside it is where the surprise dissolves.

APPENDIX E

Python and NumPy Quick Start

From zero to simulating qubits — the ten-minute path into the language this book’s code speaks

Chapter 14 simulates qubits in Python with the NumPy library, and the capstones assume you can run such code. This appendix is the on-ramp for readers who have never touched Python: enough of the language to read, run, and modify every script in this book — honestly, that is less Python than most tutorials teach, because quantum simulation needs only arrays and arithmetic.

E.1 Installing Python (Once)

Install Python 3 from python.org (Windows/macOS; on Linux it is usually pre-installed), ticking “Add Python to PATH” during Windows installation. Then, in a terminal or command prompt, install the two libraries this book uses:

```
pip install numpy matplotlib
```

No internet in the lab? Both libraries ship inside the free Anaconda distribution, which can be downloaded once and installed offline — a practical route for university computer rooms in the region. Everything below also runs, with zero installation, in free cloud notebooks (search: Google Colab, or the IBM Quantum platform of Chapter 14).

E.2 Python in Ten Minutes

Four ideas cover this book’s needs: variables, lists, loops, and functions.

```
shots = 1000                                # a variable (no type declarations needed)
patterns = ['00', '01', '10', '11']         # a list

for p in patterns:                          # a loop: note the colon and the indentation
    print(p)                                # indentation IS the grammar in Python

def prob(amplitude):                        # a function
```



```
return abs(amplitude)**2 # ** is "to the power of"
```

Two habits prevent ninety percent of beginner errors: indent consistently (four spaces), and read error messages bottom line first — Python names the problem and the line number.

E.3 NumPy Is the Quantum Part

NumPy gives Python fast arrays and matrix arithmetic — exactly the mathematics of Appendices A and B. The dictionary between the book’s notation and NumPy’s:

Book notation → **NumPy**

Mathematics	NumPy
i (imaginary unit)	1j
state vector $ 0\rangle$	<code>np.array([1, 0], dtype=complex)</code>
matrix-vector product $H \psi\rangle$	<code>H @ psi</code>
tensor product $\otimes$	<code>np.kron(a, b)</code>
$ \alpha ^2$	<code>np.abs(alpha)**2</code>
norm $\sqrt{\langle\psi \psi\rangle}$	<code>np.linalg.norm(psi)</code>
inner product $\langle\phi \psi\rangle$	<code>np.vdot(phi, psi)</code>

E.4 A Complete First Simulation: The Bell Pair

The following script is this book’s entire simulation method in twenty lines — type it, run it, and you have reproduced Chapter 14’s core experiment. (Every line matches a formula from the main text.)

```
import numpy as np

# gates and starting state (Appendix C's tables, verbatim)
ket0 = np.array([1, 0], dtype=complex)
H     = np.array([[1, 1], [1, -1]], dtype=complex) / np.sqrt(2)
I2    = np.eye(2, dtype=complex)
CNOT  = np.array([[1,0,0,0],[0,1,0,0],[0,0,0,1],[0,0,1,0]], dtype=complex)

state = np.kron(ket0, ket0)      # |00>
state = np.kron(H, I2) @ state  # H on the first qubit
state = CNOT @ state            # entangle
print("state =", np.round(state, 4)) # [0.7071 0 0 0.7071]

# measurement = sampling from the Born rule (Chapter 6)
rng = np.random.default_rng(seed=7)
```

```

probs = np.abs(state)**2
patterns = ['00', '01', '10', '11']
shots = rng.choice(patterns, size=1000, p=probs)
for p in patterns:
    n = int(np.sum(shots == p))
    if n: print(p, n)

```

Run it (python3 bell.py) and the output is the book’s Bell signature: the state (0.7071, 0, 0, 0.7071) and counts of about half-and-half — with seed=7 exactly 00: 502, 11: 498, and never a 01 or 10. Every worked example in this book can be checked by the same three moves: build the state with @ and np.kron, square amplitudes for probabilities, sample with rng.choice.

E.5 Seeing Your Results

```

import matplotlib.pyplot as plt
plt.bar(['00', '11'], [502, 498])
plt.xlabel('pattern'); plt.ylabel('counts (1000 shots)')
plt.savefig('bell_counts.png', dpi=150)

```

Four lines turn counts into the histogram pictures used throughout this book — the file lands beside your script.

E.6 From NumPy to Qiskit

NumPy simulation hits the 2^n wall of Chapter 9 around 25–30 qubits, and it never touches real hardware. Both limits fall to the free Qiskit library (pip install qiskit), where the same Bell pair reads:

```

from qiskit import QuantumCircuit
qc = QuantumCircuit(2)
qc.h(0)
qc.cx(0, 1)
qc.measure_all()
print(qc.draw())

```

Chapter 14 takes the story from here — simulators, cloud access to real machines, and the debugging discipline that keeps you honest on both.

REALITY CHECK

Learning exactly this much Python — arrays, @, np.kron, sampling — is a genuine professional credential: it is the difference between *reading about* every result in this book and *checking* it. Every number in the worked

examples of Parts Five through Eight was verified with scripts no fancier than this appendix. You now hold the same tools as the author.

References and Further Reading

The original papers and books behind this book’s stories, case studies, and results — grouped by part, each with a plain-language note on why it matters. Everything here is findable free of charge (published versions via any university library; nearly all papers also on arXiv.org).

A Quantum Century: Milestones

From quanta to quantum computers — the road this book travels

Year	Milestone	In this book
1900–26	Planck’s quanta; Born makes amplitudes into probabilities	Chapters 2, 4
1935	EPR paradox; Schrödinger names entanglement	Chapter 11
1964–69	Bell’s theorem; CHSH make it testable	Chapter 11
1982	Feynman’s simulation proposal; no-cloning theorem; Aspect’s Bell tests	Chapters 2, 12, 21
1984	Bennett and Brassard: BB84 quantum key distribution	Chapter 15
1985	Deutsch: universal quantum computer, first quantum algorithm	Chapter 17
1991–93	Ekert’s E91; superdense coding; teleportation protocol	Chapters 15, 13
1994	Shor’s factoring algorithm — the field acquires a budget	Chapter 19
1995–96	First quantum error-correcting codes (Shor, Steane); trapped-ion gate; Grover’s search; Lloyd’s universal simulator	Chapters 29, 30, 18, 21
1997–99	First teleportation experiments; first superconducting qubit	Chapters 13, 30
2014	VQE and QAOA: the variational era opens	Chapter 20
2015	Loophole-free Bell tests (three at once)	Chapter 11
2016–17	Micius satellite: entanglement at 1200 km; FeMoco resource estimate; VQE on real molecules	Chapters 16, 21, 20
2018	“NISQ” era named; Bell-certified randomness; barren plateaus discovered	Chapters 30, 4, 25
2019	Sycamore sampling milestone; quantum kernels on hardware; Tang’s dequantization	Chapters 9, 24, 26
2021	4,600-km integrated quantum network; provable kernel advantage	Chapters 16, 26
2022	Quantum advantage in learning from experiments; device-independent QKD; Nobel Prize for Bell tests	Chapters 26, 15, 11
2023–24	Surface code beats its parts; logical atom-array processor; below-threshold error correction; post-quantum standards finalized	Chapters 29, 15
2026	You, reading this book — the era of engineering	Everywhere

Foundations: Quantum Mechanics and the Bell Story

M. Born. *Zur Quantenmechanik der Stoßvorgänge*. Zeitschrift für Physik 37, 863–867 (1926).

The footnote that earned a Nobel Prize: probabilities are the squared magnitudes of amplitudes — the Born rule of Chapter 4.

A. Einstein, B. Podolsky, and N. Rosen. *Can quantum-mechanical description of physical reality be considered complete?*. Physical Review 47, 777–780 (1935).

The EPR paper: entanglement's strangeness stated as an objection — the challenge Bell answered.

E. Schrödinger. *Die gegenwärtige Situation in der Quantenmechanik*. Naturwissenschaften 23, 807–812 (1935).

Names “entanglement” (Verschränkung) and the famous cat; declares entanglement the characteristic trait of quantum mechanics.

F. Bloch. *Nuclear induction*. Physical Review 70, 460–474 (1946).

The physics behind the Bloch sphere of Chapter 5.

N. F. Ramsey. *A molecular beam resonance method with separated oscillating fields*. Physical Review 78, 695–699 (1950).

The Ramsey interference method — today's atomic clocks and the T_2 measurements of Chapter 28.

J. S. Bell. *On the Einstein Podolsky Rosen paradox*. Physics Physique Fizika 1, 195–200 (1964).

The theorem of Chapter 11: no local hidden answers can reproduce quantum correlations.

J. F. Clauser, M. A. Horne, A. Shimony, and R. A. Holt. *Proposed experiment to test local hidden-variable theories*. Physical Review Letters 23, 880–884 (1969).

The CHSH inequality — the testable $|S| \leq 2$ form used throughout Chapter 11.

A. Aspect, J. Dalibard, and G. Roger. *Experimental test of Bell's inequalities using time-varying analyzers*. Physical Review Letters 49, 1804–1807 (1982).

The landmark Bell-test experiments (2022 Nobel Prize, with Clauser and Zeilinger).

B. Hensen et al.. *Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres*. Nature 526, 682–686 (2015).

The first loophole-free Bell test — with Giustina et al. and Shalm et al. (both Physical Review Letters 115, 2015), the closing of the case.

W. H. Zurek. *Decoherence, einselection, and the quantum origins of the classical*. Reviews of Modern Physics 75, 715–775 (2003).

Why the everyday world looks classical: the environment as a relentless measuring device (Chapter 28).

The Birth of Quantum Computing

R. Landauer. *Irreversibility and heat generation in the computing process*. IBM Journal of Research and Development 5, 183–191 (1961).

“Information is physical”: erasing a bit costs energy — the thermodynamic seed of reversible computing (Chapter 12).

C. H. Bennett. *Logical reversibility of computation*. IBM Journal of Research and Development 17, 525–532 (1973).

Compute-copy-uncompute: the reversible-computing blueprint behind Chapter 12's scratch-qubit discipline.

A. S. Holevo. *Bounds for the quantity of information transmitted by a quantum communication channel*. Problems of Information Transmission 9, 177–183 (1973).

Why n qubits can never deliver more than n classical bits — the honesty theorem quoted in Chapters 2 and 9.

S. Wiesner. *Conjugate coding*. ACM SIGACT News 15(1), 78–88 (1983; written c. 1970).

Quantum money: the unforgeable-states idea, a decade ahead of its time, that seeded BB84 (Chapter 15).

- R. P. Feynman. *Simulating physics with computers*. International Journal of Theoretical Physics 21, 467–488 (1982).
The founding argument of Chapter 2: classical machines choke on quantum physics; build quantum machines.
- D. Deutsch. *Quantum theory, the Church–Turing principle and the universal quantum computer*. Proceedings of the Royal Society of London A 400, 97–117 (1985).
The universal quantum computer defined — and the first glimpse of quantum parallelism.
- W. K. Wootters and W. H. Zurek. *A single quantum cannot be cloned*. Nature 299, 802–803 (1982).
The no-cloning theorem of Chapter 12 — one page, enormous consequences.
- A. Barenco et al.. *Elementary gates for quantum computation*. Physical Review A 52, 3457–3467 (1995).
Every unitary from single-qubit gates plus CNOT: the universality result behind Chapter 7’s gate menu.

Protocols and Quantum Cryptography

- C. H. Bennett and G. Brassard. *Quantum cryptography: public key distribution and coin tossing*. Proceedings of the IEEE International Conference on Computers, Systems and Signal Processing, Bangalore, 175–179 (1984).
BB84 itself (Chapter 15) — arguably the most consequential conference paper in the field’s history.
- A. K. Ekert. *Quantum cryptography based on Bell’s theorem*. Physical Review Letters 67, 661–663 (1991).
E91: keys certified by a CHSH score, the entanglement-based protocol of Chapter 15.
- C. H. Bennett and S. J. Wiesner. *Communication via one- and two-particle operators on Einstein-Podolsky-Rosen states*. Physical Review Letters 69, 2881–2884 (1992).
Superdense coding: two classical bits on one entangled qubit (Chapter 13).
- C. H. Bennett, G. Brassard, C. Crépeau, R. Jozsa, A. Peres, and W. K. Wootters. *Teleporting an unknown quantum state via dual classical and Einstein-Podolsky-Rosen channels*. Physical Review Letters 70, 1895–1899 (1993).
The teleportation protocol of Chapter 13, exactly as this book computes it.
- D. Bouwmeester et al.. *Experimental quantum teleportation*. Nature 390, 575–579 (1997).
Teleportation leaves the blackboard: the first photonic demonstration.
- J.-G. Ren et al.. *Ground-to-satellite quantum teleportation*. Nature 549, 70–73 (2017).
The Micius satellite era: teleportation and entanglement distribution over 1200 km (with Yin et al., Science 356, 1140, 2017).
- P. Bierhorst et al.. *Experimentally generated randomness certified by the impossibility of superluminal signals*. Nature 556, 223–226 (2018).
Certified randomness from a Bell test — Chapter 4’s case study.
- D. P. Nadlinger et al.. *Experimental quantum key distribution certified by Bell’s theorem*. Nature 607, 682–686 (2022).
Device-independent QKD demonstrated — E91’s dream, in hardware.
- H. J. Kimble. *The quantum internet*. Nature 453, 1023–1030 (2008).
The vision paper that named the field of Chapter 16.
- S. Wehner, D. Elkouss, and R. Hanson. *Quantum internet: a vision for the road ahead*. Science 362, eaam9288 (2018).
The staged roadmap — from trusted relays to full repeaters — that Chapter 16 walks.
- Y.-A. Chen et al.. *An integrated space-to-ground quantum communication network over 4,600 kilometres*. Nature 589, 214–219 (2021).
The Beijing–Shanghai backbone plus Micius: quantum communication as infrastructure (Chapter 16’s case study).

NIST. *Module-lattice-based key-encapsulation mechanism standard (FIPS 203); ML-DSA (FIPS 204); SLH-DSA (FIPS 205)*. U.S. National Institute of Standards and Technology (2024).
The post-quantum cryptography standards — the classical countermeasure to Chapter 19, already rolling out.

The Famous Algorithms

D. Deutsch and R. Jozsa. *Rapid solution of problems by quantum computation*. Proceedings of the Royal Society of London A 439, 553–558 (1992).

The first clean quantum-beats-classical example — Chapter 17's centerpiece.

E. Bernstein and U. Vazirani. *Quantum complexity theory*. SIAM Journal on Computing 26, 1411–1473 (1997).

Rigorous foundations — and the hidden-string algorithm of Chapter 17 that doubles as today's favorite hardware benchmark.

D. R. Simon. *On the power of quantum computation*. SIAM Journal on Computing 26, 1474–1483 (1997).

The hidden-period algorithm that showed Shor the way — the missing link in Chapter 17's lineage story.

S. Lloyd. *Universal quantum simulators*. Science 273, 1073–1078 (1996).

Feynman's conjecture proven: quantum computers can efficiently simulate local quantum systems — Chapter 21's charter.

I. M. Georgescu, S. Ashhab, and F. Nori. *Quantum simulation*. Reviews of Modern Physics 86, 153–185 (2014).

The standard survey of digital and analog simulation — the wide-angle view of Chapter 21.

M. Reiher, N. Wiebe, K. M. Svore, D. Wecker, and M. Troyer. *Elucidating reaction mechanisms on quantum computers*. Proceedings of the National Academy of Sciences 114, 7555–7560 (2017).

The FeMoco resource estimate — quantum chemistry's flagship target, priced in public (Chapter 21's case study).

P. W. Shor. *Algorithms for quantum computation: discrete logarithms and factoring*. Proceedings of the 35th Annual Symposium on Foundations of Computer Science, 124–134 (1994); journal version SIAM Journal on Computing 26, 1484–1509 (1997).

Chapter 19: period-finding breaks RSA, and quantum computing acquires a budget line.

L. K. Grover. *A fast quantum mechanical algorithm for database search*. Proceedings of the 28th Annual ACM Symposium on Theory of Computing, 212–219 (1996).

Chapter 18's $\sqrt{N}$ search — optimal, modest, and honest.

C. Gidney and M. Ekerå. *How to factor 2048 bit RSA integers in 8 hours using 20 million noisy qubits*. Quantum 5, 433 (2021).

The resource-estimate benchmark behind Chapter 19's honesty tables.

C. Gidney. *How to factor 2048 bit RSA integers with less than a million noisy qubits*. arXiv:2505.15917 (2025).

The estimate falling twenty-fold in four years — why “harvest now, decrypt later” is taken seriously (Chapter 15).

Variational Algorithms and Quantum Machine Learning

A. Peruzzo et al.. *A variational eigenvalue solver on a photonic quantum processor*. Nature Communications 5, 4213 (2014).

VQE is born (Chapter 20).

E. Farhi, J. Goldstone, and S. Gutmann. *A quantum approximate optimization algorithm*. arXiv:1411.4028 (2014).

QAOA — Chapter 20's optimization branch.

A. Kandala et al.. *Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets*. Nature 549, 242–246 (2017).

Real molecules on real superconducting hardware — and the noise realities that motivated error mitigation.

M. Cerezo et al.. *Variational quantum algorithms*. Nature Reviews Physics 3, 625–644 (2021).

The standard survey of the whole variational family — the wide-angle photograph of Part Six's territory.

J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven. *Barren plateaus in quantum neural network training landscapes*. Nature Communications 9, 4812 (2018).

The desert of Chapter 25: why deep random circuits cannot be trained.

K. Mitarai, M. Negoro, M. Kitagawa, and K. Fujii. *Quantum circuit learning*. Physical Review A 98, 032309 (2018).

Trainable quantum circuits, and the parameter-shift idea in its first form.

M. Schuld, V. Bergholm, C. Gogolin, J. Izaac, and N. Killoran. *Evaluating analytic gradients on quantum hardware*. Physical Review A 99, 032331 (2019).

The $\pm 90^\circ$ parameter-shift rule exactly as Chapter 25 teaches it.

V. Havlíček et al.. *Supervised learning with quantum-enhanced feature spaces*. Nature 567, 209–212 (2019).

Quantum kernels and variational classifiers on hardware — the founding experiment of Chapter 24.

A. Pérez-Salinas, A. Cervera-Lierta, E. Gil-Fuster, and J. I. Latorre. *Data re-uploading for a universal quantum classifier*. Quantum 4, 226 (2020).

Chapter 23's architecture: re-encoding data between layers buys frequencies.

Y. Liu, S. Arunachalam, and K. Temme. *A rigorous and robust quantum speed-up in supervised machine learning*. Nature Physics 17, 1013–1017 (2021).

The provable kernel advantage (for a cryptography-flavored problem) on Chapter 26's advantage map.

E. Tang. *A quantum-inspired classical algorithm for recommendation systems*. Proceedings of the 51st Annual ACM Symposium on Theory of Computing, 217–228 (2019).

Dequantization's founding strike (Chapter 26) — fairness enforced on speedup claims.

H.-Y. Huang et al.. *Quantum advantage in learning from experiments*. Science 376, 1182–1186 (2022).

Proven, demonstrated exponential advantage — for quantum data. Chapter 26's case study.

C. Zoufal, A. Lucchi, and S. Woerner. *Quantum generative adversarial networks for learning and loading random distributions*. npj Quantum Information 5, 103 (2019).

Quantum GANs put to work loading finance distributions (Chapter 26).

Hardware, Noise, and Error Correction

J. I. Cirac and P. Zoller. *Quantum computations with cold trapped ions*. Physical Review Letters 74, 4091–4094 (1995).

The trapped-ion gate proposal — realized within months by Monroe et al. (Physical Review Letters 75, 4714, 1995).

Y. Nakamura, Yu. A. Pashkin, and J. S. Tsai. *Coherent control of macroscopic quantum states in a single-Cooper-pair box*. Nature 398, 786–788 (1999).

The first superconducting qubit — ancestor of Chapter 30's refrigerated processors.

P. W. Shor. *Scheme for reducing decoherence in quantum computer memory*. Physical Review A 52, R2493–R2496 (1995).

The nine-qubit code of Chapter 29: error correction is possible after all.

A. M. Steane. *Error correcting codes in quantum theory*. Physical Review Letters 77, 793–797 (1996).

The independent second pillar of quantum error correction.

A. G. Fowler, M. Mariantoni, J. M. Martinis, and A. N. Cleland. *Surface codes: towards practical large-scale quantum computation*. Physical Review A 86, 032324 (2012).

The surface-code engineering manual — source of Chapter 29's overhead arithmetic.

J. Preskill. *Quantum computing in the NISQ era and beyond*. Quantum 2, 79 (2018).

Names our era — noisy, intermediate-scale — and frames its honest possibilities; the backdrop of Part Eight.

F. Arute et al.. *Quantum supremacy using a programmable superconducting processor*. Nature 574, 505–510 (2019).

The Sycamore sampling milestone of Chapter 9's case study — and the classical-rebuttal saga that followed.

A. W. Cross, L. S. Bishop, S. Sheldon, P. D. Nation, and J. M. Gambetta. *Validating quantum computers using randomized model circuits*. Physical Review A 100, 032328 (2019).

Quantum volume: the whole-machine benchmark of Chapter 30.

Google Quantum AI. *Suppressing quantum errors by scaling a surface code logical qubit*. Nature 614, 676–681 (2023).

Bigger code, fewer errors — the first experimental step of Chapter 29's scaling story.

Google Quantum AI and collaborators. *Quantum error correction below the surface code threshold*. Nature 638, 920–926 (2025); arXiv:2408.13687.

The Willow result: logical qubits improving as the code grows — below threshold at last.

D. Bluvstein et al.. *Logical quantum processor based on reconfigurable atom arrays*. Nature 626, 58–65 (2024).

Dozens of logical qubits in neutral atoms — the platform race of Chapter 30 in full stride.

Books and Free Online Resources

M. A. Nielsen and I. L. Chuang. *Quantum Computation and Quantum Information*. Cambridge University Press, 10th anniversary edition (2010).

"Mike and Ike": the field's standard reference — the natural next book after this one.

N. D. Mermin. *Quantum Computer Science: An Introduction*. Cambridge University Press (2007).

Short, sharp, and computer-science flavored; a beautiful second telling of Parts One–Five.

S. Aaronson. *Quantum Computing Since Democritus*. Cambridge University Press (2013).

Ideas, complexity theory, and wit — the book to read for why it all matters.

M. Schuld and F. Petruccione. *Machine Learning with Quantum Computers*. Springer, 2nd edition (2021).

The standard textbook for Part Seven's subject, by two of its architects.

IBM Quantum Learning (learning.quantum.ibm.com). *Interactive courses and the open-source Qiskit textbook*. online, free.

Runnable notebooks for nearly every algorithm in this book — and free real-hardware access (Chapter 14).

J. Preskill. *Lecture notes for Ph219: Quantum Computation*. California Institute of Technology, online, free.

The graduate-level path: rigorous notes that pick up exactly where this book stops.

Glossary

Every important term used in this book, in one place.

Ablation — Re-running an experiment with one ingredient removed or swapped (e.g., quantum kernel replaced by classical) to expose which ingredient produced the performance. The decisive test to demand of any QML claim.

Amplitude — The complex number attached to each basis state in a quantum state. Gates transform amplitudes; interference adds and cancels them; squared moduli give probabilities.

Ancilla — A helper qubit used alongside the data — for example, to collect a parity in error correction — and then measured or reset.

Ansatz — The adjustable trial circuit of a variational algorithm; its rotation-gate dials are the parameters an optimizer tunes.

Assertion (quantum debugging) — A checkpoint statement — “here the state must equal such-and-such” — checked exactly on a simulator and statistically on hardware. The quantum replacement for print-and-stare debugging.

Barren plateau — A training desert: for deep, unstructured parameterized circuits, gradients shrink exponentially with qubit count, so no realistic number of shots can resolve a training direction. Avoided by shallow, structured circuits and identity-style initialization.

Basis — A set of orthogonal reference states used to describe or measure a system, such as $|0\rangle, |1\rangle$ (standard) or $|+\rangle, |-\rangle$.

Bell measurement — A joint measurement asking a pair of qubits “which Bell state?”; performed by CNOT, then H, then measuring both. Central to teleportation.

Bell states — The four maximally entangled two-qubit states $\Phi^\pm = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle)$ and $\Psi^\pm = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)$.

Binary number — A number written with digits 0 and 1, each place worth twice the one to its right. Measurement results of qubit registers are binary patterns.

Bit — The smallest unit of classical information: one choice between two states, 0 or 1.

Bloch sphere — The unit sphere on which every pure single-qubit state is one point: $|0\rangle$ north, $|1\rangle$ south, equal superpositions on the equator; latitude sets probabilities, longitude is relative phase.

Born machine — A generative model whose samples are measurement outcomes of a parameterized circuit: $P_\theta(x) = |\langle x|\psi(\theta)\rangle|^2$. Sampling is native; training reshapes the circuit’s histogram toward the data’s.

Born rule — Probability = squared modulus of amplitude: $P = |\alpha|^2$; in general $P(\phi) = |\langle \phi | \psi \rangle|^2$.

Bra — The row-vector partner $\langle \psi |$ of a ket — its conjugate transpose. Bra times ket, $\langle \phi | \psi \rangle$, is the inner product.

Byte — Eight bits; holds $2^8 = 256$ patterns.

CHSH score — The Bell-test statistic $S = E(a, b) + E(a', b) + E(a', b') - E(a, b')$. Local hidden-variable worlds obey $|S| \leq 2$; entangled qubits reach $2\sqrt{2} \approx 2.83$ (the Tsirelson bound).

Circuit depth — The number of time-step columns in a circuit; the quantity noise punishes most (survival $\approx (1 - p)^{\text{depth}}$).

Classical bit/wire — In circuit diagrams, the doubled line carrying a measurement's 0/1 result; can control later gates (as in teleportation).

CNOT — Controlled-NOT: flips the target qubit iff the control is $|1\rangle$; the standard entangling gate; XOR on basis states; its own inverse.

Code distance — The size d of the smallest error pattern that defeats an error-correcting code; larger distance = exponentially better protection below threshold.

Collapse — The state change caused by measurement: the state becomes the outcome's basis state, and the previous amplitudes are gone.

Complex number — A number $a + bi$ with $i^2 = -1$; the number system of quantum amplitudes.

Computational basis — The standard reference states $|0\rangle$ and $|1\rangle$ (and their multi-qubit patterns $|x\rangle$).

Conjugate — $z^* = a - bi$, the mirror of $z = a + bi$ across the real axis; $zz^* = |z|^2$.

Controlled gate — A gate applied to a target qubit only when control qubit(s) are $|1\rangle$: CNOT, CZ, Toffoli, controlled- U .

CZ (controlled-Z) — The symmetric two-qubit gate flipping only the sign of $|11\rangle$; native on several hardware platforms.

Dagger — The conjugate transpose $M^\dagger$: transpose the matrix, conjugate every entry. For unitaries, $U^\dagger = U^{-1}$.

Data re-uploading — Encoding the input repeatedly, between trainable layers, in a variational classifier. Each re-upload adds a frequency to the model's function family, enriching the decision boundaries it can draw.

Decision boundary — The set of inputs where a classifier's score crosses zero, separating differently-labeled regions. For quantum models, its possible shapes are set by the encoding's frequency budget.

Decoder (QEC) — The classical algorithm that watches syndrome streams and infers which repairs to apply in an error-corrected computer.

Decoherence — The loss of amplitudes' phase relationships through unwanted interaction with the environment — continuous accidental measurement; quantified by T1 and T2.

Degenerate code — An error-correcting code in which different physical errors share a syndrome AND act identically on the code space, so one repair heals them all — the phase sector of Shor's nine-qubit code is the classic example.

Dequantization — Matching a claimed quantum speedup with a classical algorithm once the classical side is granted analogous assumptions (e.g., comparable data access). Tang's 2019 recommendation-system result founded the practice.

Device-independent security — Cryptographic security argued from measured statistics alone (a CHSH score), without trusting the devices' internals. E91 is the ancestor protocol; hardware demonstrations arrived in 2022.

Diffusion operator — Grover's "inversion about the mean": every amplitude $a \rightarrow 2\bar{a} - a$; amplifies the oracle-marked target.

Dilution refrigerator — The millikelvin cryostat housing superconducting and spin qubits — the "golden chandelier" of quantum labs.

Dirac notation — The bra-ket language: kets $|\psi\rangle$ for states (columns), bras $\langle\psi|$ for their daggers (rows), $\langle\phi|\psi\rangle$ for inner products.

E91 protocol — Ekert's 1991 entanglement-based key distribution: matched measurement angles yield key bits; the remaining rounds fund a CHSH test that certifies the pairs untouched — even if the source is hostile.

Entanglement — A correlation between qubits such that the group has a definite state but the individuals do not; cannot be written as a tensor product; verified experimentally by Bell tests.

Entanglement swapping — A Bell measurement on the middle two of four qubits (A-B, C-D pairs) that leaves the never-met outer qubits A and D entangled. The elementary step of quantum repeaters.

Error budget — The itemized error rates of a device — one- and two-qubit gates, readout, idling — combined to predict a circuit's corruption fraction.

Error correction (quantum) — Protecting logical information by spreading it into entangled states of many physical qubits and repairing errors located via syndrome measurements.

Error mitigation — Statistical techniques (readout un-flipping, zero-noise extrapolation, echoes) that improve noisy results using extra runs rather than extra qubits.

Euler's formula — $e^{i\theta} = \cos \theta + i \sin \theta$: the unit-circle description of phases.

Expectation value — The average of an observable's outcomes, weighted by their probabilities: $\langle A \rangle = \sum_a a P(a)$. Estimated on hardware as a histogram average; the quantity variational algorithms optimize.

Exponential growth — Growth by repeated multiplication, like 2^n ; the reason brute-force search, exact simulation of quantum systems, and unprotected error rates all explode.

Fidelity (gate) — The probability an operation does exactly its job; error rate = $1 - \text{fidelity}$. Two-qubit gate fidelity is the headline hardware metric.

Fidelity (state overlap) — The closeness of two states, $F = |\langle \phi | \psi \rangle|^2$ for pure states: 1 for identical, 0 for perpendicular. The ruler for comparing a prepared state against its target.

Gate (quantum) — A unitary matrix applied to one or more qubits; the building block of circuits.

Generative model — A model representing a whole probability distribution, able to produce fresh samples — versus a discriminative model, which only scores inputs. Quantum circuits are natively generative: measurement is sampling.

GHZ state — The three-(or more-)qubit entangled state $\frac{1}{\sqrt{2}}(|00 \cdots 0\rangle + |11 \cdots 1\rangle)$.

Global phase — An overall factor $e^{i\gamma}$ on a whole state; physically undetectable. Contrast relative phase.

Grover's algorithm — Quantum search of N unstructured possibilities in about $\frac{\pi}{4}\sqrt{N}$ oracle calls, via sign-marking and amplitude amplification; quadratic speedup, provably optimal.

Hadamard gate (H) — $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$: turns $|0\rangle, |1\rangle$ into $|+\rangle, |-\rangle$ and back; the standard superposition maker; its own inverse.

Hamiltonian — The observable “energy” of a system, written as a sum of simple measurable pieces; VQE hunts its lowest average value.

Identity matrix (I) — The do-nothing gate/matrix: $Iv = v$.

Identity-block initialization — Starting a layered variational circuit at parameters where blocks cancel to the identity, so training begins unscrambled and gradient-rich — the standard opening move against barren plateaus.

Imaginary unit — i , defined by $i^2 = -1$; powers cycle $1, i, -1, -i$.

Inner product — $\langle u | v \rangle = \sum_k u_k^* v_k$: the overlap of two states; its squared modulus is a transition probability.

Interference — The adding (constructive) and cancelling (destructive) of amplitudes; the working principle of every quantum algorithm.

Kernel concentration — The failure mode of over-expressive feature maps: off-diagonal kernel values crowd toward one tiny number, the matrix nears the identity, and the model memorizes rather than generalizes.

Ket — The symbol $|\rangle$ for a state vector; $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

Logic gate (classical) — A device computing a fixed truth table (NOT, AND, OR, XOR); most classical gates erase information — quantum gates never do.

Logical qubit — One protected qubit of information encoded across many physical qubits by an error-correcting code.

Marginal probability — The probability of a part-question (“qubit A = 1, regardless of B”): add the probabilities of all matching patterns.

Matrix — A grid of numbers acting as a machine on vectors; row-times-column gives each output component.

MaxCut — The optimization problem of splitting a network’s nodes into two camps cutting as many links as possible; QAOA’s standard demonstration.

Measurement — Reading a quantum system: returns one classical outcome with Born-rule probabilities and collapses the state to match.

Mixed state — A state of genuine ignorance — a probabilistic mixture of pure states rather than a superposition of them. Lives strictly inside the Bloch sphere; distinguished from superposition by measuring in a second basis.

Mode collapse — A GAN failure where the generator produces only one (or few) patterns the discriminator currently accepts, abandoning the target distribution’s diversity.

Modulus — $|z| = \sqrt{a^2 + b^2}$: the length of a complex number’s arrow; $|z|^2 = zz^*$.

No-cloning theorem — No machine can copy an unknown quantum state; proven from linearity alone; underlies quantum cryptography and shapes error correction.

Noise — Every unwanted influence on qubits — decay, dephasing, gate imperfection, readout error; the central obstacle of current hardware.

Normalization — The requirement $\sum |a_x|^2 = 1$: state vectors have length 1, so probabilities total 100%.

Oracle — A subroutine packaged as a unitary that marks answers (for example by a sign flip) without revealing them; the “checker” Grover’s algorithm calls.

Orthogonal — Having inner product zero; orthogonal states are perfectly distinguishable and sit at antipodal Bloch points.

Parameter-shift rule — Exact gradients from hardware: for standard rotation gates, $\partial E / \partial \theta = \frac{1}{2}[E(\theta + 90^\circ) - E(\theta - 90^\circ)]$ — two extra circuit runs per parameter, no approximation.

Partial measurement — Measuring some qubits of a register: outcome probabilities add over consistent patterns, and the survivors' amplitudes, renormalized, form the state of the rest.

Pauli gates — X (bit flip), Y (bit+phase flip), Z (phase flip): 180° rotations about the three Bloch axes; every single-qubit error can be expressed through them.

Pauli group (single qubit) — The gate family $\{\pm I, \pm iI, \pm X, \pm iX, \dots\}$ generated by X, Y, Z under multiplication: any two Pauli products multiply to another, up to a phase $\pm 1, \pm i$.

Phase — The angle of a complex amplitude on the unit circle. Relative phases between amplitudes are physical; global phase is not.

Phase estimation (QPE) — The algorithm reading an unknown phase φ as binary digits: kickback onto a counting register, then inverse QFT.

Phase kickback — The mechanism by which a controlled- U writes U 's phase onto the *control* qubit's superposition; the engine of phase estimation.

Physical qubit — An actual hardware two-level system, noise and all — as opposed to a logical qubit.

Post-quantum cryptography (PQC) — Classical encryption designed to resist quantum attack (lattice-based and related); standardized by NIST in 2024; the migration answer to Shor.

Probability table — The professional bookkeeping of a state: each pattern, its amplitude, its probability.

Product state — A multi-qubit state that factors into single-qubit states — no entanglement.

QAOA — The Quantum Approximate Optimization Algorithm: a variational circuit whose dials steer superpositions toward high-scoring assignments of an optimization problem.

Quantum advantage — A demonstrated computation on quantum hardware materially beyond the reach of the best classical methods; claims require careful verification.

Quantum counting — Phase estimation applied to the Grover rotation to learn the number of marked items M before searching — the standard cure for unknown M .

Quantum Fourier transform (QFT) — The unitary $|x\rangle \rightarrow \frac{1}{\sqrt{N}} \sum_y \omega^{xy} |y\rangle$: converts hidden periods into measurable peaks in about n^2 gates.

Quantum GAN — Adversarial training with a parameterized quantum circuit as generator (sampling by measurement) and usually a classical network as discriminator, converging when generated samples become indistinguishable from data.

Quantum repeater — A chain of entanglement-swapping stations with quantum memories, extending entanglement segment by segment — the architecture for continental quantum links, since amplifiers are forbidden by no-cloning.

Quantum volume — A composite benchmark: the largest random square circuit (width = depth) a machine runs acceptably; punishes weak-link metrics.

Quantum Zeno effect — Freezing of evolution by frequent measurement: each check collapses the state back, erasing accumulated change — “a watched quantum pot never boils.”

Qubit — A quantum two-level system: measured, it yields 0 or 1; between measurements it carries two complex amplitudes and can superpose and entangle.

Qubit-wise commuting group — Pauli terms whose letters agree (or defer to I) on every qubit, hence measurable in one shared setting — the standard batching trick that slashes VQE’s measurement bill.

Ramsey experiment — H – wait – H: converts accumulated phase into populations; measures T2 and runs the world’s atomic clocks.

Readout error — The probability a measurement reports the wrong classical bit; often the largest single entry in an error budget.

Relative phase — The phase difference between amplitudes — the angle φ on the Bloch equator; physical, gate-controllable, interference-deciding.

Repetition code — The three-qubit training code: $\alpha|0\rangle + \beta|1\rangle \rightarrow \alpha|000\rangle + \beta|111\rangle$; parity checks locate one bit flip.

Reversible computing — Logic that never erases information; forced on quantum computers by unitarity; realized classically by Toffoli-style gates.

Rotation gates — $R_x(\theta), R_y(\theta), R_z(\theta)$: turns of the Bloch sphere by any angle about the named axis; the dials of variational circuits.

S gate — The quarter-turn phase gate: $|1\rangle \rightarrow i|1\rangle$; $S = T^2, S^2 = Z$.

Shor code — The 1995 nine-qubit code — three blocks of three, repetition inside blocks and sign-agreement across them — correcting any error on any single qubit; the first proof that full quantum error correction is possible.

Shor’s algorithm — Factoring via quantum period finding (QFT/QPE) plus classical number theory; exponentially faster than known classical methods; the reason PQC exists.

Shot — One prepare-run-measure cycle; quantum data is gathered over many shots.

Simulator — A classical program computing a circuit’s exact amplitudes (up to ~ 30 – 40 qubits); the free, noiseless companion of every quantum programmer.

State-preparation cost — The gate count to load data into amplitudes; exponential in qubit number for arbitrary vectors. The “input problem” that quietly dominates many claimed QML speedups.

Superdense coding — Sending two classical bits with one transmitted qubit plus one pre-shared Bell pair; the mirror protocol of teleportation.

Superposition — A definite quantum state combining basis states with amplitudes; not “both at once,” not ignorance — a third thing, with measurable interference consequences.

Surface code — The leading error-correcting code: data qubits on a grid guarded by nearest-neighbor parity checks; threshold $\approx 1\%$, overhead $\approx 2d^2$ physical qubits per logical.

SWAP gate — Exchanges two qubits' states; buildable from three CNOTs; the routing currency of limited-connectivity chips.

Syndrome — The pattern of parity-check outcomes locating an error without revealing the protected data.

T gate — The eighth-turn phase gate: $|1\rangle \rightarrow e^{i\pi/4} |1\rangle$; with H it generates all single-qubit gates; the expensive gate of fault-tolerant computing.

T1 (relaxation time) — The energy lifetime: prepared $|1\rangle$ survives as e^{-t/T_1} .

T2 (dephasing time) — The phase-memory lifetime, read from decaying Ramsey fringes; $T_2 \leq 2T_1$; the deadline for interference.

T-count — The number of T gates in a fault-tolerant circuit — the dominant cost metric, since error-corrected T gates are vastly more expensive than Clifford gates.

Teleportation — Transferring an unknown state using one Bell pair plus two classical bits: Bell measurement, phone call, corrective gate. Moves, never copies.

Tensor product — The $\otimes$ combination of systems: amplitudes multiply, labels concatenate; n qubits live in 2^n dimensions.

Threshold theorem — Below a critical physical error rate ($\approx 1\%$ for the surface code), increasing code distance suppresses logical errors exponentially — scalable quantum computing is possible in principle.

Toffoli gate (CCX) — Two controls, one target: flips the target iff both controls are 1; reversible AND; imports classical logic into quantum circuits.

Total variation distance — The histogram-disagreement ruler between distributions: $TV = \frac{1}{2} \sum_x |p(x) - q(x)|$, from 0 (identical) to 1 (disjoint). A hand-computable generative-model loss.

Truth table — The complete input-output listing of a classical gate or circuit.

Tsirelson bound — Quantum mechanics' own ceiling on the CHSH score: $|S| \leq 2\sqrt{2} \approx 2.83$. Entanglement breaks the classical bound of 2 but cannot break this one.

Uncertainty relation — A provable floor under the product of two observables' spreads (for qubits: $\Delta Z \Delta X \geq |\langle Y \rangle|$), holding for every state — a property of states, not of clumsy devices.

Uncomputation — Running a computation's gates in reverse to return scratch qubits to $|0\rangle$ after use — mandatory before interference, since dirty scratch stays entangled with the data and kills it.

Unitary matrix — $U^\dagger U = I$: length-preserving, perfectly reversible; the entrance requirement for every quantum gate.

Universal gate set — A small gate menu (e.g. $\{H, T, \text{CNOT}\}$) whose combinations approximate any quantum computation to any accuracy.

Variational algorithm — The hybrid loop: parameterized quantum circuit, measured score, classical optimizer; VQE and QAOA are the flagships.

Vector — An ordered column of numbers; quantum states are complex vectors of length 1.

VQE — The Variational Quantum Eigensolver: hunts a Hamiltonian's ground energy under the guarantee $E(\theta) \geq E_{\text{ground}}$.

XOR — The “different?” logic gate; addition of bits without carry; the classical shadow of CNOT.

Zero-noise extrapolation — Error mitigation by running at amplified noise levels and extrapolating measurements back to zero noise.